

MATHEMATICS SUBJECT TEST

Ordered Glossary and Worked Problem Companion

A systematic guide to the uploaded
Princeton Review, fourth edition

316 concept entries • 1,647 solved problems

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Prepared for Robert

Readability edition • 6 October 2026

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Three ways to move around

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Page numbers match the PDF. Entry numbers identify concepts; page numbers locate pages. Source references labeled “Book pp.” refer to the uploaded book. [Source corrections](#) and [scope, notation, and exam notes](#) are at the back. All [worked solutions](#) are in a separate section at the end.

Compact glossary

Book order, grouped by section and subsection. Click a title to open its explanation and questions. Gray text lists additional terms covered in that entry. “p.” identifies the destination page.

1. Precalculus

Introduction

Prerequisite algebra

001 Exponents, radicals, and complex arithmetic p. 26

principal square root; complex number; imaginary unit; complex conjugate; modulus; absolute value.

Functions

Definition and domain

002 Function, domain, codomain, range, and fixed point p. 28

image; natural domain.

Composition of Functions

003 Composition and iteration of functions p. 30

composite function; functional recurrence; identity function.

Inverse Functions

004 Injectivity, surjectivity, bijectivity, and inverse functions p. 31

injective; one-to-one; surjective; onto; bijective; reciprocal; restriction; rationalization.

Graphs in the xy -Plane

Graphs and symmetry

005 Graphs, symmetry, coordinates, and intercepts p. 33

vertical-line test; horizontal-line test; abscissa; even function; odd function; reflection.

Piecewise graphs

006 Piecewise functions, absolute value, floor, and graph transformations p. 35

greatest integer function; step function; translation; scaling.

Analytic Geometry

General conic equation

007 Conic sections, completing squares, rotations, and degeneracy p. 37

analytic geometry; general second-degree equation; rotation of axes; degenerate conic; completing the square.

Lines

008 Lines, slopes, distances, and perpendicular bisectors p. 39

point-slope formula; parallel; distance formula; midpoint; locus.

Parabolas

009 Parabola, focus, directrix, vertex, axis, and latus rectum p. 40

focal width.

Circles

010 Circle, center, radius, diameter, chord, and tangent p. 42

tangent line.

Ellipses

011 Ellipse, foci, axes, eccentricity, and area scaling p. 43

major axis; minor axis; semiaxis; focal distance.

Hyperbolas

012 Hyperbola, branches, transverse axis, vertices, and asymptotes p. 44

focal axis; hyperbola eccentricity.

Polynomial Equations

Polynomials and quadratics

013 Polynomial degree, coefficients, quadratic formula, and discriminant p. 46

leading coefficient; constant coefficient; quadratic polynomial; monic polynomial.

*Division Algorithm, Remainder Theorem, and Factor Theorem***014** Polynomial division, remainders, and factors p. 48

division algorithm; quotient; Remainder theorem; Factor theorem; polynomial divisibility.

*Fundamental Theorem of Algebra and Roots***015** Fundamental Theorem of Algebra, multiplicity, and polynomial identity p. 49

root; zero; linear factor; distinct roots.

*Root Location Theorem***016** Root location and sign changes p. 51

Root Location theorem; Intermediate Value theorem; root isolation.

*Rational Roots Theorem***017** Rational root theorem and root testing p. 53

Rational Roots theorem; integer coefficient; lowest terms.

*Conjugate Root Theorems***018** Radical and complex conjugate roots p. 55

Conjugate Radical Roots theorem; Complex Conjugate Roots theorem; quadratic conjugate; real coefficient; rational coefficient.

*Sum and Product of the Roots***019** Vieta formulas and symmetric expressions in roots p. 56

sum of roots; product of roots; elementary symmetric sum; power sum.

Logarithms*Exponential and logarithmic functions***020** Logarithms, exponentials, laws, and change of base p. 57

exponential function; natural logarithm; Euler number; change-of-base formula; logarithmic equation; exponential equation.

Trigonometry*Trig Functions of Acute Angles***021** Right-triangle trigonometry and special angles p. 59

sine; cosine; tangent; cosecant; secant; cotangent; hypotenuse; complementary angles; cofunctions; special right triangle.

*Arbitrary Angles and Real Numbers***022** Unit circle, radian measure, quadrants, and directed angles p. 61

degree measure; initial side; terminal side; positive angle; negative angle; arc length.

*Trig Identities and Formulas***023** Reciprocal, opposite-angle, and Pythagorean identities p. 62

trigonometric identity; reciprocal identity; opposite-angle identity; Pythagorean identity; trigonometric parity; reduction formula.

024 Addition, subtraction, double-angle, and half-angle formulas p. 64

addition formula; subtraction formula; double-angle formula; angle sum; angle difference.

*Periodicity and Graphs of the Trig Functions***025** Periodicity, fundamental period, and trigonometric domains and ranges p. 65

periodic function; amplitude; phase shift; trigonometric asymptote.

*The Inverse Trig Functions***026** Inverse trigonometric functions and principal branches p. 67

arcsine; arccosine; arctangent; arccosecant; arcsecant; arccotangent.

Chapter 1 Review Questions*Radical equations***027** Radical equations and extraneous roots p. 69

squaring an equation; equivalent transformation.

*Rational inequalities***028** Rational inequalities and sign charts p. 70

rational inequality; critical endpoint; strict inequality; interval notation.

*Hyperbolic functions***029** Hyperbolic sine, cosine, tangent, and their inverses p. 72

hyperbolic cosine; hyperbolic tangent; inverse hyperbolic sine; inverse hyperbolic tangent; inverse hyperbolic cosine.

2. Calculus I**Limits of Sequences***Definitions***030** Sequences, convergence, and divergence p. 73

limit of a sequence; epsilon–N definition; convergent sequence; divergent sequence; divergence to infinity; oscillation.

*Monotone convergence***031** Boundedness, monotonicity, and recursive limits p. 75

bounded sequence; monotone sequence; eventually monotone; monotone convergence theorem; recursive sequence.

*Algebra and estimates***032** Limit laws and the squeeze theorem p. 77

sum law; product law; quotient law; sandwich theorem; bounded factor times null sequence; growth comparison.

Limits of Functions*Definitions and calculations***033** Function limits, one-sided limits, and epsilon–delta reasoning p. 79

limit of a function; epsilon–delta definition; punctured neighborhood; sequential criterion; left-hand limit; right-hand limit.

Limits of Functions as $x \rightarrow \pm\infty$ *End behavior***034** Limits at infinity and asymptotes p. 81

limit at infinity; horizontal asymptote; vertical asymptote; slant asymptote; oblique asymptote.

Continuous Functions*Definition and closure properties***035** Continuity and types of discontinuity p. 83

continuity at a point; continuous function; one-sided endpoint continuity; removable discontinuity; jump discontinuity; infinite discontinuity; continuity of a composition.

*A nontrivial continuity example***036** Rational–irrational constructions and Thomae’s function p. 85

Thomae function; rational–irrational function; dense set; continuity via bounded denominators; Dirichlet function.

Theorems Concerning Continuous Functions*Extreme value theorem***037** Extreme value theorem and attainment p. 87

absolute maximum; absolute minimum; closed bounded interval.

*Bolzano and intermediate value theorems***038** Intermediate values, zeros, and fixed points p. 89

Bolzano theorem; intermediate value theorem; fixed-point theorem on an interval; sign change.

The Derivative*Definition and geometry***039** Derivative, differentiability, tangent, and normal p. 91

difference quotient; secant line; tangent line; normal line; corner; cusp; vertical tangent; Leibniz notation.

Rules of differentiation

040 Differentiation rules and the chain rule p. 93

sum rule; constant multiple rule; product rule; quotient rule; composite derivative.

Inverse-function rule

041 Derivative of an inverse function p. 95

inverse-function derivative; local inverse; reciprocal slope.

Table of differentials

042 Elementary derivatives and differentials p. 97

power rule; exponential derivative; logarithmic derivative; trigonometric derivatives; inverse trigonometric derivatives.

Linear Approximations Using Differentials*Local approximation*

043 Linearization, differentials, and approximation error p. 99

linear approximation; differential approximation; relative error; tangent approximation.

Implicit Differentiation*Local implicit functions*

044 Implicit differentiation and the implicit function theorem p. 101

implicit equation; dependent variable; independent variable.

Higher-Order Derivatives*Repeated differentiation*

045 Higher-order derivatives p. 103

second derivative; n th derivative; Leibniz product formula.

Curve Sketching*Properties of the first derivative*

046 Critical points, stationary points, and monotonicity p. 104

increasing function; decreasing function; first-derivative test.

Properties of the second derivative

047 Concavity, convexity, and inflection points p. 106

concave up; convex function; concave down; second-derivative sign.

Classifying critical points

048 Second-derivative and higher-derivative tests p. 108

second-derivative test; degenerate critical point; local maximum; local minimum.

Theorems Concerning Differentiable Functions*Rolle's theorem*

049 Rolle's theorem and repeated zeros p. 110

Rolle theorem; horizontal tangent; zero counting.

Mean value theorem

050 Mean value theorem and derivative bounds p. 112

secant slope; constant derivative theorem; Lipschitz bound; monotonicity from derivative.

Max/Min Problems*Finding extrema*

051 Optimization and constrained extrema p. 114

max/min problem; local extremum; absolute extremum; endpoint test; constraint reduction.

Related Rates*Differentiating a geometric constraint*

052 Related rates and time differentiation p. 116

instantaneous rate; time derivative; rate of change.

Indefinite Integration (Antidifferentiation)*Definitions and elementary formulas*

053 Antiderivatives and indefinite integration p. 118

indefinite integral; integrand; constant of integration; variable of integration; power integration rule.

Techniques of Integration*Integration by Substitution***054** Integration by substitution p. 120

u-substitution; change of variable; reverse chain rule; transformed limits.

*Integration by Parts***055** Integration by parts p. 122

reverse product rule; reduction formula; cyclic integral.

*Trig Substitutions***056** Trigonometric substitutions and radical integrals p. 124

Pythagorean substitution; power-reduction identity.

*The Method of Partial Fractions***057** Partial fractions and rational integration p. 126

partial-fraction decomposition; proper rational function; polynomial division; repeated linear factor; irreducible quadratic factor.

Definite Integration*Area as a limit of sums***058** Riemann sums and definite integrals p. 128

partition; mesh; tagged partition; Riemann integrability; sum of squares.

The Fundamental Theorem of Calculus*Integration and differentiation***059** Fundamental theorem of calculus and variable limits p. 130

accumulation function; evaluation theorem; dummy variable; Leibniz rule for endpoints.

*Properties of definite integrals***060** Integral properties, signed area, and piecewise integration p. 132

linearity of the integral; reversal of limits; additivity over intervals; integral comparison; geometric area; piecewise integral; floor-function integral.

The Average Value of a Function*Integral averages***061** Average value and the integral mean value theorem p. 134

average value of a function; mean value theorem for integrals; average rate.

Finding the Area Between Two Curves*Choosing the integration variable***062** Area between curves p. 136

upper minus lower; right minus left; intersection bounds; horizontal slicing.

Polar Coordinates*Coordinates and curves***063** Polar coordinates and standard polar curves p. 137

Cartesian coordinates; negative radius; polar circle; cardioid; polar-to-Cartesian conversion.

*Area in polar coordinates***064** Polar area and overlap regions p. 139

polar area formula; sector area; radial bounds; overlap of polar regions; rose curve; lemniscate.

Volumes of Solids of Revolution*Disks and washers***065** Volumes of revolution: disks and washers p. 141

solid of revolution; disk method; washer method; outer radius; inner radius; cross-sectional area.

Arc Length*Length by integration***066** Arc length p. 142

smooth curve; line element; rectifiable curve; parametric arc length.

The Natural Exponential and Logarithm Functions

Definitions and calculus

067 The natural exponential, logarithm, and the number e p. 144

natural logarithm; Euler number e ; exponential limit; integral definition of logarithm; transcendental number.

Variable exponents

068 Logarithmic differentiation and variable powers p. 146

variable exponent; logarithmic derivative; exponential rewriting.

L'Hopital's Rule*Limits by differentiation*

069 L'Hopital's rule and indeterminate forms p. 148

L'Hopital rule; zero over zero; infinity over infinity; indeterminate product; indeterminate power; growth hierarchy.

Improper Integrals*Definition and evaluation*

070 Improper integrals and independent limiting operations p. 150

unbounded interval; endpoint singularity; interior singularity; convergent improper integral; divergent improper integral; Cauchy principal value; improper integral of the first kind; improper integral of the second kind; absolute convergence of an improper integral; conditional convergence of an improper integral.

Convergence criteria

071 Improper p -integrals and comparison near singularities p. 152

comparison test for integrals; limit comparison for integrals; logarithmic singularity; endpoint convergence.

Infinite Series*Definitions and exact sums*

072 Infinite series, partial sums, geometric and telescoping sums p. 154

sum of a series; geometric series; common ratio; telescoping series.

Basic convergence tests

073 The term test, harmonic series, and p -series p. 156

n th-term test; necessary condition for series convergence.

The comparison test

074 Direct and limit comparison for series p. 158

comparison test; direct comparison; nonnegative series; eventual inequalities.

The ratio test

075 Ratio test p. 160

factorial series; absolute ratio; inconclusive ratio.

The root test

076 Root test and limsup formulation p. 162

n th root; limsup root test; absolute convergence test.

The integral test

077 Integral test and remainder bounds p. 164

eventually decreasing terms; integral remainder estimate; logarithmic series.

Alternating Series*Alternating-series test*

078 Alternating series and their error estimates p. 166

Leibniz alternating-series test; alternating remainder bound; eventual monotonicity.

Absolute versus conditional convergence

079 Absolute and conditional convergence p. 168

absolute convergence; absolute convergence implies convergence; signed series.

Power Series*Convergence sets*

080 Power series, radius, and interval of convergence p. 170

center of a power series; radius of convergence; endpoint testing; Cauchy–Hadamard formula.

Functions Defined by Power Series

081 Termwise differentiation and integration of power series p. 172

termwise integration; unchanged radius; endpoint caution; power-series regularity.

Operations on series

082 Power-series algebra, substitution, and coefficient extraction p. 174

power-series substitution; Cauchy product; power-series multiplication; power-series division; geometric-series transformation.

Taylor Series*Coefficients and centers*

083 Taylor coefficients, Taylor series, and analyticity p. 176

Maclaurin series; expansion center; analytic function; smooth nonanalytic function.

Standard expansions

084 Standard Maclaurin expansions and leading-order limits p. 178

exponential series; sine series; cosine series; logarithm series; arctangent series; leading nonzero term; order of vanishing.

Taylor Polynomials*Approximation and error*

085 Taylor polynomials and rigorous remainder bounds p. 180

Taylor theorem; Lagrange remainder; truncation error; error bound; degree versus number of terms.

3. Calculus II**Analytic Geometry of $\mathbb{R}^3$** *Coordinates and Vectors*

086 Euclidean space, vectors, norm, and distance p. 182

Cartesian coordinates; octant; first octant; standard unit vectors; vector components; scalar multiplication; unit vector.

The Dot Product

087 Dot product, angle, and orthogonality p. 183

scalar product; angle between vectors; orthogonal vectors; perpendicular vectors.

088 Scalar and vector projection p. 185

scalar projection; orthogonal decomposition.

The Cross Product

089 Cross product, orientation, and area p. 187

vector product; right-hand rule; normal vector; parallelogram area; triangle area; vector triple-product identity.

The Triple Scalar Product

090 Scalar triple product and volume p. 189

parallelepiped; coplanarity; oriented volume; tetrahedron volume; Gram determinant.

Lines in 3-Space

091 Parametric and symmetric equations of lines p. 190

line in space; direction vector; parametric equations; skew lines.

Planes in 3-Space

092 Planes and normal equations p. 192

coordinate plane; parallel planes; angle between planes.

Cylinders

093 Cylindrical surfaces and generators p. 194

cylinder; right circular cylinder; elliptic cylinder; parabolic cylinder; element of a cylinder.

*Surfaces of Revolution***094** Surfaces of revolution p. 196

surface of revolution; axis of revolution; radial distance; ellipsoid of revolution.

*Level Curves and Level Surfaces***095** Level sets, contours, and quadric surfaces p. 198

level curve; level surface; contour curve; cross-section; circular paraboloid; elliptic paraboloid; hyperboloid of one sheet.

*Cylindrical Coordinates***096** Cylindrical coordinates p. 200

cylindrical radius; azimuthal angle; coordinate surface.

*Spherical Coordinates***097** Spherical coordinates p. 202

radial distance; polar angle; azimuth; sphere; cone.

Partial Derivatives*Definition and Geometric Interpretation***098** Partial derivatives and coordinate traces p. 204

independent variable; partial derivative as slope.

*Higher-Order Partial Derivatives***099** Mixed partials and equality of differentiation orders p. 206

higher-order partial derivative; mixed partial derivative; Clairaut theorem; Schwarz theorem.

*The Tangent Plane to a Surface***100** Tangent planes and implicit partial differentiation p. 208

implicit differentiation; implicit function condition; regular level surface.

*Linear Approximations***101** Total differential and linear approximation p. 210

linearization; differentiability; first-order error.

*The Chain Rule for Partial Derivatives***102** Multivariable chain rule and held-fixed variables p. 212

dependent variable; intermediate variable; independent variable; total derivative; cyclic partial derivative identity.

Directional Derivatives and the Gradient*Rates of Change***103** Directional derivative and steepest ascent p. 214

gradient; del operator; steepest descent; maximum rate of change.

*Normals to Level Sets***104** Gradient normals and tangent directions p. 216

normal to a level set; tangent vector; normal line; intersection of level surfaces.

Max/Min Problems*Critical Points and the Second-Derivative Test***105** Critical points, Hessian, and saddle points p. 218

stationary point; local maximum; local minimum; Hessian matrix; second-derivative test.

*Absolute Extrema on a Closed Region***106** Absolute extrema and boundary analysis p. 220

absolute maximum; absolute minimum; closed and bounded region; compact region; extreme value theorem; boundary critical point.

*Max/Min with a Constraint; Lagrange Multipliers***107** Constrained extrema and Lagrange multipliers p. 222

constraint; constrained extremum; regular constraint; elimination method.

Line Integrals

Line Integrals with Respect to Arc Length

108 Smooth curves, arc length, and scalar line integrals p. 224

parameterized curve; piecewise smooth curve; arc length element; curve integral; contour integral; path integral; linear density; mass of a wire.

The Line Integral of a Vector Field

109 Vector fields, work, and oriented line integrals p. 226

vector-valued function; work integral; orientation; differential one-form; circulation.

The Fundamental Theorem of Calculus for Line Integrals

110 Potential functions and conservative fields p. 228

scalar field; gradient field; path independence; fundamental theorem for line integrals; closed curve.

Double Integrals*Iterated Integrals and Reversing Order*

111 Double integrals, Fubini, and region bounds p. 230

area element; Riemann sum; signed volume; iterated integral; order of integration; Fubini theorem.

Double Integrals in Polar Coordinates

112 Polar area element and radial integration p. 232

polar double integral; radial bound; polar substitution.

Green's Theorem*Circulation and Area*

113 Green's theorem, positive orientation, and area p. 234

simple closed curve; boundary orientation; Green theorem; area from a line integral.

Path Independence and Gradient Fields

114 Convexity, simple connectivity, and the hole obstruction p. 236

convex region; simply connected region; curl-free planar field; closed-path criterion.

Chapter 3 Review Questions*Vector Identities*

115 Vector triple product and cross-product equations p. 238

BAC-CAB identity.

Distance to a Plane

116 Perpendicular distance to a plane p. 240

point-to-plane distance; orthogonal foot; distance between parallel planes.

Continuity at the Origin

117 Multivariable limits and continuity p. 241

path test; continuity in several variables; polar estimate; squeeze theorem.

4. Differential Equations**Introduction***Differential equations and initial conditions*

118 Differential equations, order, solutions, integral curves, and IVPs p. 243

ordinary differential equation; partial differential equation; general solution; particular solution; initial condition; initial value problem.

Separable Equations*Separation of variables*

119 Separable ODEs, equilibrium solutions, and maximal intervals p. 245

separable equation; separation of variables; constant solution; finite-time blow-up.

Homogeneous Equations*Scaling and the substitution $y=xv$*

120 Homogeneous functions and first-order homogeneous equations p. 247

homogeneous degree; homogeneous first-order equation; substitution $y=xv$; scale invariance; Euler identity for homogeneous functions.

Exact Equations*Total differentials and potentials***121** Exact equations, total differentials, and potential functions p. 249

exact differential; mixed partial derivatives; level curve; simply connected domain.

Nonexact Equations and Integrating Factors*Factors depending on one variable***122** Integrating factors for nonexact equations p. 251

one-variable integrating factor; equivalent differential equation.

First-Order Linear Equations*Standard form and integrating factor***123** First-order linear ODEs and the product-derivative method p. 252

first-order linear equation; standard form; linear integrating factor; inhomogeneous term; variation of constants.

Higher-Order Linear Equations with Constant Coefficients*Linear structure and general solutions***124** Linear differential operators, superposition, and solution spaces p. 254

linear differential equation; constant coefficient; homogeneous linear equation; nonhomogeneous linear equation; complementary solution; particular solution; linear independence; fundamental set.

*Auxiliary equation: distinct real roots***125** Characteristic polynomial and distinct exponential modes p. 256

auxiliary equation; characteristic equation; distinct characteristic roots.

*Auxiliary equation: repeated roots***126** Repeated characteristic roots and polynomial factors p. 257

root multiplicity; polynomial-exponential solution; zero characteristic root.

*Auxiliary equation: nonreal roots***127** Complex characteristic roots, oscillation, and damping p. 259

oscillatory solution; angular frequency; damped oscillation; complex exponential; real solution basis.

*Particular solutions of nonhomogeneous equations***128** Particular-solution guessing, undetermined coefficients, and resonance p. 261

forcing term; particular-solution ansatz; verification by substitution.

Chapter 4 Review Questions*Growth proportional to population***129** Exponential growth, decay, doubling time, and half-life p. 263

exponential decay; proportional rate; growth constant.

*Area conditions and differential modeling***130** Converting integral or geometric conditions into ODEs p. 265

integral equation; differential modeling; area condition; Fundamental Theorem of Calculus; boundary consistency.

*Power-product integrating factors***131** Monomial integrating factors and coefficient matching p. 267

power-product integrating factor; singular axis.

*Orthogonal trajectories***132** Orthogonal trajectories and eliminating a family parameter p. 269

orthogonal trajectory; orthogonal family; negative reciprocal slope; parameter elimination; tangent direction.

*First-order partial differential equations***133** Linear first-order PDEs, characteristic directions, and level curves p. 271

transport equation; characteristic curve; directional derivative; gradient.

5. Linear Algebra

Solutions of Linear Systems

Consistency and Number of Solutions

134 Linear equations and consistent systems p. 273

linear system; coefficient; unknown; solution; inconsistent system.

Matrices and Matrix Algebra

Dimensions and Matrix Operations

135 Matrices, entries, and triangular form p. 275

matrix; matrix dimensions; matrix entry; row; column; diagonal entry; main diagonal; square matrix; upper triangular matrix; lower triangular matrix; zero matrix; matrix addition; scalar multiple; diagonal matrix.

The Transpose

136 Transpose and row-column duality p. 277

row vector; column vector; transpose of a product.

Matrix Multiplication

137 Matrix multiplication and noncommutativity p. 279

matrix product; compatible dimensions; commuting matrices; associativity; distributivity.

Identity Matrices and Inverses

138 Identity matrices and one-sided inverses p. 281

identity matrix; Kronecker delta; left inverse; right inverse; inverse matrix; invertible matrix; singular matrix; nonsingular matrix.

Matrix Powers and Polynomial Identities

139 Matrix powers and polynomial identities p. 283

matrix polynomial; polynomial identity; inverse from a polynomial identity.

Gaussian Elimination

Row Operations and Echelon Form

140 Gaussian elimination and echelon form p. 285

coefficient matrix; augmented matrix; elementary row operation; row equivalence; pivot; back-substitution; reduced row echelon form.

Free Variables and Homogeneous Systems

141 Homogeneous systems and affine solution spaces p. 287

free variable; parameter; nonhomogeneous system; trivial solution; particular solution; general solution.

Solving Matrix Equations Using an Inverse

The Two-by-Two Formula

142 Inverse solution formula and the two-by-two inverse p. 289

matrix equation; unique solvability.

Vector Spaces

Vector Spaces and Subspaces

143 Vector space axioms and the subspace test p. 291

real vector space; scalar field; closure; zero vector; trivial subspace; proper subspace.

The Nullspace

144 Nullspace and homogeneous directions p. 293

null space; kernel of a matrix; nullity of a matrix.

Linear Combinations

145 Linear combinations and span p. 295

spanning set; trivial linear combination.

Linear Independence

146 Linear independence and dependence p. 297

linearly independent; linearly dependent; dependence relation; redundant vector.

Bases and Dimension

147 Basis, dimension, and minimal spanning sets p. 299

finite-dimensional space; infinite-dimensional space; basis extension.

The Rank, Column Space, and Row Space

148 Rank, column space, and row space p. 301

row rank; column rank; pivot column; full rank.

Other Vector Spaces

149 Polynomial, function, and matrix spaces p. 303

polynomial space; function space; bounded functions; continuous functions; differentiable functions; matrix vector space; matrix units.

Determinants*Permutations and the Defining Formula*

150 Permutation signs and determinant definition p. 305

transposition; even permutation; odd permutation; sign of a permutation; Leibniz determinant formula.

Properties and Elimination

151 Determinant laws and row operations p. 307

determinant multilinearity; alternating determinant; determinant row operations; triangular determinant; determinant of a product; determinant scaling; determinant of a transpose.

Laplace Expansions

152 Minors, cofactors, and Laplace expansion p. 309

cofactor expansion; determinant test for collinearity.

The Adjugate Matrix

153 Adjugate and the invertibility criterion p. 311

classical adjoint; cofactor matrix; adjugate inverse formula; determinant criterion for invertibility.

Cramer's Rule

154 Cramer's rule and a selected unknown p. 313

Cramer rule; column replacement determinant.

Linear Transformations*Linearity and Component Vectors*

155 Linear transformations and coordinates in a basis p. 315

linear map; linear operator; component vector; coordinate vector; image of a basis.

Standard Matrix Representative

156 Matrix representation of a linear map p. 317

standard basis; standard matrix; matrix representative; column image rule.

The Rank Plus Nullity Theorem

157 Kernel, image, and rank-nullity p. 319

range; rank-nullity theorem; injective linear map; surjective linear map; isomorphism.

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158 Composition and inverse transformations p. 321

composition of linear maps; inverse linear transformation; matrix order in composition.

Eigenvalues and Eigenvectors*Characteristic Equation*

159 Eigenvalues, eigenvectors, and characteristic polynomial p. 323

characteristic value; characteristic vector; characteristic equation.

Trace and Determinant

160 Trace and eigenvalue sums and products p. 325

sum of eigenvalues; product of eigenvalues; algebraic multiplicity; two-by-two characteristic polynomial.

Eigenspaces

161 Eigenspaces and geometric multiplicity p. 327

zero eigenspace; independent eigenvectors.

The Cayley-Hamilton Theorem*Reducing Matrix Powers*

162 Cayley-Hamilton theorem p. 329

Cayley-Hamilton theorem; characteristic identity; reduction of matrix powers.

Chapter 5 Review Questions*Counterexamples to Scalar Rules*

163 Nilpotent and involutory matrices p. 331

nilpotent matrix; nilpotency index; involutory matrix; matrix cancellation failure; zero divisor.

Symmetric Matrix Spaces

164 Symmetric and skew-symmetric matrices p. 333

symmetric matrix; skew-symmetric matrix; symmetric matrix space; dimension of symmetric matrices.

Rotations and Reflections

165 Rotation and reflection operators p. 335

rotation matrix; reflection matrix; orthogonal transformation; orientation-preserving map; orientation-reversing map.

The Cross Product as a Linear Map

166 Cross-product linear operators p. 337

cross-product matrix; cross-product linear map; skew-symmetric operator.

6. Number Theory and Abstract Algebra**Part A: Number Theory***Divisibility*

167 Divisibility, parity, primes, and coprimality p. 339

divides; divisor; factor; multiple; even; odd; composite; relatively prime; pairwise relatively prime; divisibility tests; square-free integer.

The Division Algorithm

168 Quotient and remainder p. 341
division algorithm; floor function.

Primes

169 Prime factorization and the distribution of primes p. 342

fundamental theorem of arithmetic; prime-counting function; prime number theorem; Bertrand postulate; prime gaps; prime-adic valuation; factorial trailing zeros.

The Greatest Common Divisor and the Least Common Multiple; The Euclidean Algorithm

170 GCD, LCM, Euclid, and Bezout p. 344

greatest common divisor; least common multiple; Euclidean algorithm; Bezout identity; extended Euclidean algorithm.

The Diophantine Equation $ax + by = c$

171 Linear Diophantine equations p. 346

integer solution; Fermat last theorem.

Congruences

172 Congruence arithmetic and valid cancellation p. 348

modulus; residue class; modular arithmetic; cancellation modulo n ; Chinese remainder theorem.

173 Fermat, Euler, and modular powers p. 350

Fermat little theorem; Euler totient function; Euler theorem; modular exponentiation.

The Congruence Equation $ax = b \pmod{n}$

174 Linear congruences and solution counts p. 351

modular inverse; number of congruence solutions.

Part B: Abstract Algebra*Binary Structures and the Definition of a Group*

175 Binary operations, semigroups, and monoids p. 352

closure; associativity; binary structure; identity.

176 Groups, inverses, and commutativity p. 354

Abelian group; commutative operation; finite group; infinite group; group order; cancellation law.

Examples of Groups

177 General and special linear groups p. 356
 general linear group; matrix group; $GL(n, F)$; $SL(n, F)$; unimodular matrix.

178 Roots of unity as a finite group p. 358
 root of unity; primitive root of unity; multiplicative roots-of-unity group.

179 Permutations and cycle decompositions p. 359
 symmetric group; cycle notation; disjoint cycles; transposition; permutation order.

180 Dihedral groups and geometric symmetry p. 361
 rotation; reflection; symmetry group.

181 Residue groups and multiplication tables p. 362
 additive group modulo n ; unit group modulo n ; group table; Cayley table.

Cyclic Groups

182 Cyclic groups and generators p. 364
 infinite cyclic group; Klein four-group.

Subgroups

183 Subgroups and the center p. 366
 proper subgroup; trivial subgroup; subgroup test; center of a group.

Cyclic Subgroups

184 Element order and generated subgroups p. 368
 order of an element; cyclic subgroup; finitely generated group; group exponent.

Generators and Relations

185 Group presentations and word reduction p. 370
 generators and relations; defining relation; word in generators.

Some Theorems Concerning Subgroups

186 Lagrange, Cauchy, and Sylow existence p. 371
 Lagrange theorem; Cauchy theorem; Sylow first theorem; Sylow subgroup; subgroup order; index of a subgroup.

The Concept of Isomorphism

187 Isomorphism and structural invariants p. 373
 isomorphic groups; nonisomorphic groups.

The Classification of Finite Abelian Groups

188 Direct products, direct sums, and element orders p. 375
 direct product of groups; finite direct sum; coordinatewise operation; order in a direct product; cyclic product criterion.

189 Elementary divisors and invariant factors p. 377
 fundamental theorem of finite Abelian groups; integer partitions; finite Abelian group classification.

Group Homomorphisms

190 Homomorphisms and maps from cyclic groups p. 379
 group homomorphism; monomorphism; epimorphism; endomorphism; automorphism; homomorphic image.

191 Kernels and normal subgroups p. 381
 kernel of a group homomorphism; conjugation; trivial kernel; fiber of a homomorphism.

192 Inner automorphisms and automorphism groups p. 383
 inner automorphism group; conjugation map.

*Rings***193** Rings, unity, and subrings p. 385

commutative ring; ring with unity; distributive laws; additive identity; multiplicative identity.

194 Ring characteristic p. 387

characteristic of a ring; prime characteristic; characteristic zero; additive order of unity.

195 Polynomial, Gaussian-integer, and function rings p. 389

polynomial ring; indeterminate; Gaussian integer; quadratic integer ring; pointwise addition; pointwise multiplication.

196 Nilpotent elements p. 391

nilpotency index; sum of commuting nilpotents.

*Ring Homomorphisms***197** Ring homomorphisms and evaluation p. 393

ring endomorphism; evaluation homomorphism; substitution homomorphism; kernel of a ring homomorphism; unital homomorphism.

198 Frobenius endomorphism p. 395

freshman dream in prime characteristic.

199 Ideals and absorption p. 397

two-sided ideal; left ideal; right ideal; principal ideal; ideal generated by elements.

*Integral Domains***200** Zero divisors, cancellation, and integral domains p. 399

left zero divisor; right zero divisor; cancellation in a ring.

*Fields***201** Units, division rings, fields, and finite domains p. 401

subfield; finite field; quadratic field; quaternion division ring; strictly skew field.

202 Idempotents and Boolean rings p. 403

idempotent element; orthogonal idempotents.

7. Additional Topics**Logic***Sentential Calculus***203** Propositions and logical connectives p. 405

sentential calculus; propositional calculus; sentence; negation; conjunction; disjunction; conditional; biconditional; if and only if.

*Logical Equivalences and Truth Tables***204** Truth tables, tautologies, and contraposition p. 407

logical equivalence; tautology; contradiction; contrapositive; converse; inverse of an implication; De Morgan laws; distributive laws; excluded middle; quantifier negation.

Set Theory*Subsets and Complements***205** Sets, subsets, and complements p. 409

element; membership; set-builder notation; family of sets; empty set; proper subset; superset; universal set; relative complement; set difference.

*Union and Intersection; Venn Diagrams***206** Unions, intersections, and symmetric difference p. 410

disjoint sets; indexed family; indexed union; indexed intersection; Venn diagram; set De Morgan laws.

*Cartesian Products; Intervals of the Real Line***207** Cartesian products and real intervals p. 412

ordered pair; open interval; closed interval; half-open interval; unbounded interval; extended real numbers.

*Cardinality***208** Cardinality, countability, and algebraic numbers p. 414

equinumerous sets; bijection; finite set; infinite set; countably infinite; countable set; aleph-null; countable union.

209 Power sets, uncountability, and the continuum p. 416

Cantor theorem; uncountable set; transcendental number; cardinal inequality; diagonal argument.

Graph Theory*Vertices, Edges, and Degree***210** Simple graphs and vertex degrees p. 418

edge; undirected graph; adjacent vertices; odd vertex; even vertex; complete graph; graph order; graph size; handshaking lemma; loop; graph complement; regular graph; graphical degree sequence.

*Graph Isomorphism; Review Matrices***211** Graph isomorphisms and incidence matrices p. 420

adjacency preservation; adjacency matrix; incidence matrix; unoriented incidence matrix; graph invariant.

*Paths, Trees, and Subgraphs***212** Walks, circuits, trees, and spanning subgraphs p. 422

trail; path; closed walk; cycle; connected graph; connected component; forest; spanning tree.

Algorithms*Tracing and Termination***213** Algorithms, loops, and invariants p. 424

input; output; assignment; conditional branch; while loop; iteration; termination; deterministic algorithm; loop invariant; Collatz iteration.

Combinatorics*The Multiplication Principle***214** Sum and product counting principles p. 426

combinatorics; addition principle; multiplication principle; fundamental counting principle; complementary counting.

*Permutations and Combinations***215** Permutations, repeated symbols, and circular orders p. 428

permutation count; factorial; ordered selection; arrangement; permutation with repeated symbols; circular permutation.

216 Combinations, binomial coefficients, and partitions into teams p. 430

binomial theorem; Pascal identity; multinomial coefficient; labeled groups; unlabeled groups; multinomial theorem; hockey-stick identity.

*With Repetitions Allowed***217** Repetition and stars and bars p. 432

ordered selection with repetition; combination with repetition; multiset selection; weak composition; positive composition.

*The Pigeonhole Principle***218** Pigeonhole and averaging arguments p. 434

pigeonhole principle; generalized pigeonhole principle; residue-class pigeonholes; averaging principle.

Probability and Statistics*Elementary Probability***219** Equally likely outcomes and geometric probability p. 435

favorable outcomes; sampling without replacement; uniform distribution.

*Probability Spaces***220** Sample spaces, events, and probability measures p. 436

outcome; algebra of sets; Boolean algebra of sets; sigma-algebra; probability space; countable additivity.

221 Independence, exclusion, and inclusion-exclusion p. 438

independent events; mutually independent events; pairwise independence; mutually exclusive events; conditional probability; Bayes theorem; law of total probability.

Bernoulli Trials

222 Bernoulli trials and the binomial distribution p. 440

success probability; failure probability; binomial probability mass function; binomial mode.

Random Variables

223 Random variables and cumulative distribution functions p. 442

measurability; CDF; discrete random variable; probability mass; atom; right continuity.

224 Densities and transformations of continuous variables p. 444

probability density function; absolutely continuous distribution; density normalization; exponential distribution; survival probability; transformed random variable.

Expectation, Variance, and Standard Deviation

225 Expectation, indicators, and moments p. 446

mean; expected value; linearity of expectation; indicator random variable; law of the unconscious statistician.

226 Variance, standard deviation, covariance, and correlation p. 448

second moment; variance under scaling.

The Normal Distribution

227 Normal distributions and standardization p. 450

Gaussian distribution; normal density; standard normal distribution; z-score; standard normal CDF.

The Normal Approximation to the Binomial Distribution

228 Normal approximation and continuity correction p. 452

normal approximation to binomial; binomial standard deviation; approximation regime.

Point-Set Topology*Introduction*

229 Topology, open and closed sets, and neighborhoods p. 454

topological space; open set; standard topology.

230 Hausdorff separation p. 456

Hausdorff space; separation of points; uniqueness of limits; T1 separation; cofinite topology; cocountable topology.

231 Discrete, indiscrete, finer, and coarser topologies p. 458

discrete topology; indiscrete topology; trivial topology; finer topology; coarser topology.

The Subspace Topology

232 Subspace or relative topology p. 460

subspace topology; relatively open; relatively closed.

The Interior, Exterior, Boundary, Limit Points, and Closure of a Set

233 Interior, exterior, and boundary p. 462

clopen set.

234 Limit points, derived set, closure, and density p. 464

accumulation point; cluster point; isolated point; dense subset.

Basis for a Topology

235 Topological bases and the lower-limit topology p. 465

basis; basis element; generated topology; Sorgenfrey line; second countable space.

The Product Topology

236 Product topology and open rectangles p. 467

coordinate projection.

Connectedness

237 Connectedness, separations, and connected components p. 469

connected space; disconnected space.

238 Paths, path connectedness, and convexity p. 471

path-connected space; convex set; star-shaped set.

Compactness

239 Open covers and compactness p. 473

covering; finite subcover; compact space; finite intersection property; sequential compactness.

240 Boundedness, norm, and the Heine–Borel theorem p. 475

Euclidean norm; bounded set; closed and bounded.

Metric Spaces

241 Metrics, open balls, and equivalent metric topologies p. 477

metric space; triangle inequality; metric topology; Euclidean metric; square metric; sup metric; sup norm; bounded function; sup metric on bounded functions.

Continuous Functions

242 Topological and metric continuity p. 479

continuous map; inverse image criterion; continuity at a point; epsilon-delta continuity; uniform continuity; sequential criterion for continuity.

243 Continuous images and the extreme value theorem p. 481

continuous image of compact set; continuous image of connected set; absolute maximum; absolute minimum.

Open Maps and Homeomorphisms

244 Open maps, homeomorphisms, and topological invariants p. 483

compact-to-Hausdorff theorem.

Real Analysis*The Completeness of the Real Numbers*

245 Bounds, supremum, infimum, and order completeness p. 485

upper bound; lower bound; bounded above; bounded below; least upper bound; greatest lower bound; least upper bound axiom.

246 Cauchy sequences and complete metric spaces p. 487

complete space; metric completeness; closed subspace; total boundedness.

Lebesgue Measure

247 Outer measure and measurable sets p. 489

Lebesgue outer measure; Lebesgue measurable set; Caratheodory criterion; sigma-algebra.

248 Measure, null sets, and countable additivity p. 491

Lebesgue measure; measure zero; almost everywhere; monotonicity of measure; Cantor set; continuity of measure.

Lebesgue Measurable Functions

249 Measurable functions and characteristic functions p. 493

indicator function; pointwise limit; almost-everywhere limit.

250 Simple functions and approximation from below p. 495

step function; finite-valued function; simple-function approximation.

Lebesgue Integrable Functions

251 Lebesgue integral and positive and negative parts p. 497

Lebesgue integrable function; positive part; absolute integrability.

252 Riemann versus Lebesgue integration p. 499

Riemann integrability criterion; Lebesgue criterion; Dirichlet function; conditional improper integral.

253 Monotone and dominated convergence p. 501

monotone convergence theorem; Levi theorem; dominated convergence theorem; interchange of limit and integral.

254 Lebesgue spaces and completeness p. 503

L_p space; L_p norm; almost-everywhere equivalence; Banach space; Holder inequality; conjugate exponents.

Complex Variables*Introduction*

255 Complex arithmetic and conjugation p. 505

complex number; complex field; imaginary unit; real part; imaginary part; complex conjugate; complex plane; real axis; imaginary axis; pure imaginary number.

The Polar Form

256 Modulus, arguments, and polar geometry p. 507

absolute value; principal argument; polar form; triangle inequality in $\mathbb{C}$.

The Exponential Form

257 Euler formula, complex exponential, and de Moivre formula p. 509

exponential form; exponential periodicity.

Complex Roots

258 Complex roots and roots of unity p. 511

n th root; principal n th root; primitive root of unity.

Complex Logarithms

259 Multivalued logarithm and principal logarithm p. 512

complex logarithm; logarithm branch; branch cut.

Complex Powers

260 Complex powers and branch dependence p. 514

principal value; multivalued power; failure of radical identities.

The Trigonometric Functions

261 Complex trigonometric functions and equations p. 516

complex sine; complex cosine; complex tangent; complex cotangent; complex secant; complex cosecant; trigonometric identities over $\mathbb{C}$.

The Hyperbolic Functions

262 Hyperbolic functions and complex trigonometric modulus p. 518

hyperbolic sine; hyperbolic cosine; $\sinh$; $\cosh$; hyperbolic identity; hyperbolic tangent; hyperbolic cotangent.

The Derivative of a Function of a Complex Variable

263 Complex differentiability p. 520

complex derivative; difference quotient; direction-independent limit.

The Cauchy–Riemann Equations

264 Cauchy–Riemann equations p. 522

necessary condition for complex differentiability; sufficient C^1 criterion.

265 Harmonic functions and harmonic conjugates p. 524

Laplace equation.

Analytic Functions

266 Analytic, holomorphic, and entire functions p. 526

analytic function; holomorphic function; analytic rational function; higher complex derivatives; identity theorem; isolated zeros.

Complex Line Integrals

267 Complex line integrals and primitives p. 528

contour integral; parametrized path; antiderivative; path independence; orientation.

Theorems Concerning Analytic Functions

268 Cauchy theorem and simply connected domains p. 530

closed contour; primitive existence; contour deformation.

269 Morera theorem p. 532

vanishing triangle integrals; analyticity from integration; local uniform convergence.

270 Cauchy integral and derivative formulas p. 534

Cauchy integral formula; Cauchy derivative formula; simple closed contour; positive orientation.

271 Cauchy derivative estimates and growth bounds p. 536

coefficient estimate; polynomial growth.

272 Liouville theorem p. 538

bounded entire function; fundamental theorem of algebra via analysis.

273 Maximum modulus principle p. 540

maximum principle; boundary maximum; minimum modulus principle.

Taylor Series for Functions of a Complex Variable

274 Complex Taylor series and disks of convergence p. 542

power series; disk of convergence; radius of convergence; Cauchy–Hadamard formula.

Singularities, Poles, and Laurent Series

275 Isolated singularities and their classification p. 544

singularity; isolated singularity; punctured disk; removable singularity; pole; order of a pole; simple pole; double pole; essential singularity.

276 Laurent series, annuli, and principal parts p. 546

Laurent coefficient; annulus; singular part; analytic part.

The Residue Theorem

277 Residues and residue computation p. 548

coefficient of reciprocal term; simple-pole residue; higher-order-pole residue; logarithmic derivative.

278 Residue theorem and contour evaluation p. 550

sum of residues; enclosed singularity; winding number; argument principle.

Numerical Analysis*Iterative Methods*

279 Iterations, error, tolerance, and convergence rate p. 552

iterative method; initial guess; approximation error; error tolerance; rate of convergence; linear convergence; quadratic convergence; relative error; super-linear convergence.

Bisection Method

280 Bisection and certified root brackets p. 554

bisection method; sign-changing bracket; intermediate value theorem; bisection error bound.

Newton Method

281 Newton iteration and multiple roots p. 556

Newton method; Newton–Raphson method; tangent-line iteration; simple root; multiple-root convergence; Newton failure; modified Newton method.

8. Solutions to the Chapter Review Questions

Chapter 5 Review Questions

Question 20: complex matrices

282 Hermitian matrices, conjugate transpose, and the spectral theorem p. 558

Hermitian matrix; unitary matrix; Rayleigh quotient.

Chapter 6 Review Questions

Question 8: a subgroup-order counterexample

283 Alternating groups and permutation parity p. 560

even permutation; odd permutation; sign homomorphism; double transposition.

9. Practice Test

Practice Test

Question 26: surfaces of revolution

284 Torus and circular cross sections p. 562

major radius; minor radius; solid torus; annular cross section.

Question 38: a quantitative continuity condition

285 Lipschitz and Holder conditions; constancy from a strong difference bound p. 564

Lipschitz continuity; Holder continuity; Holder exponent; uniform continuity.

Question 57: Cauchy mean-value formula

286 Cauchy mean-value theorem and simultaneous derivative ratios p. 566

generalized mean-value theorem; common intermediate point.

Question 58: a topology with finite complements

287 Cofinite topology and nonunique limits p. 568

finite-complement topology; T1 separation; nonunique sequence limit.

10. Practice Test Answers and Explanations

Practice Test Answers and Explanations

Question 46: Pascal triangle

288 Pascal triangle and binomial sum identities p. 570

Pascal identity; hockey-stick identity; Vandermonde identity; weighted binomial sum.

Supplement. Breadth beyond the book

Multivariable calculus

289 Jacobians and multivariable change of variables p. 572

Jacobian matrix; Jacobian determinant; inverse function theorem.

290 Triple integrals and solid coordinate systems p. 574

volume element; spherical volume element; cylindrical volume element.

291 Surface area and flux integrals p. 575

regular surface parametrization; surface element; surface integral; orientation.

292 Divergence, curl, and differential identities p. 577

solenoidal field; irrotational field; Laplacian; vector Laplacian.

293 The divergence theorem p. 579

Gauss theorem; outward flux; closed surface.

294 Stokes's theorem p. 581

Stokes theorem; boundary orientation; circulation.

Differential equations

295 Existence, uniqueness, and maximal solution intervals p. 583

initial-value problem; local existence; local uniqueness; Picard-Lindelof theorem; maximal interval of existence.

296 Autonomous equations and phase-line stability p. 585

autonomous differential equation; equilibrium solution; stable equilibrium; asymptotic stability; unstable equilibrium; phase line.

297 Euler-Cauchy differential equations p. 587

Euler-Cauchy equation; equidimensional equation; indicial equation.

298 Linear differential systems and matrix exponentials p. 588

fundamental matrix; Jordan-block solution.

Linear algebra

299 Inner products, orthogonality, and Gram-Schmidt p. 590

norm induced by an inner product; orthonormal basis; Gram-Schmidt process; Cauchy-Schwarz inequality.

300 Orthogonal projections and least squares p. 592

orthogonal projection onto a subspace; normal equations; projection matrix.

301 Change of basis, similarity, and diagonalization p. 594

coordinate vector; change-of-basis matrix; similar matrices; diagonalizable operator; algebraic multiplicity; geometric multiplicity.

302 Rank inequalities and dimensions of intersections p. 596

Sylvester rank inequality; dimension formula; rank of a composition.

303 Quadratic forms and positive definiteness p. 598

positive definite matrix; positive semidefinite matrix; Sylvester criterion; Sylvester law of inertia.

304 Minimal polynomials and nilpotent Jordan structure p. 600

annihilating polynomial; Jordan block; Jordan normal form; nilpotency index.

Abstract algebra

305 Quotient groups and the first isomorphism theorem p. 602

coset multiplication; factorization through a quotient.

306 Ideals, quotient rings, and polynomial quotients p. 604

prime ideal; maximal ideal; irreducible polynomial; comaximal ideals.

307 Modules, free modules, and annihilators p. 606

submodule; module homomorphism; torsion module; torsion-free module.

308 Group actions, conjugacy, and orbit counting p. 608

stabilizer; orbit-stabilizer theorem; conjugacy class; centralizer; Burnside counting lemma.

309 Sylow subgroup counts and normality p. 610

Sylow conjugacy theorem; Sylow counting theorem.

310 Polynomial irreducibility and finite field extensions p. 612

irreducible polynomial; minimal polynomial of an algebraic element; extension degree; tower law; algebraic element; algebraic closure; compositum of fields; Eisenstein criterion; Gauss lemma.

Real analysis

311 Pointwise and uniform convergence of functions p. 614

pointwise convergence; supremum norm; uniform limit theorem; Weierstrass M-test.

312 Interchanging limits, derivatives, and integrals p. 616

termwise differentiation; uniform convergence of derivatives; differentiation under the integral sign.

Probability and statistics

313 Sample means, standard error, and the central limit theorem p. 618
sampling distribution.

314 Geometric, Poisson, and exponential distributions p. 620
geometric distribution; Poisson distribution; memoryless property; Poisson thinning; rate parameter; homogeneous Poisson process.

315 Conditional expectation and total expectation and variance p. 622

law of total expectation; law of total variance; conditional variance; random sum.

Geometry

316 Triangle geometry and circle power p. 624

law of cosines; law of sines; Heron formula; inradius; circumradius; power of a point.

1. Precalculus

Introduction ▸ Prerequisite algebra

001 Exponents, radicals, and complex arithmeticBook pp. 5. [Find in compact glossary](#)

In this entry: exponent; radical; principal square root; complex number; imaginary unit; complex conjugate; modulus; absolute value.

Definition and key ideas

For positive real a , real powers satisfy $a^{u+v} = a^u a^v$, $(a^u)^v = a^{uv}$, and $a^{-u} = 1/a^u$. For real x , $|x| = \max\{x, -x\}$. An even real root denotes its nonnegative value, so $\sqrt{x^2} = |x|$. Negative bases require restrictions on real exponents; applying familiar exponent laws without those restrictions can change a problem's meaning. Rationalize a denominator by multiplying by a conjugate expression.

A complex number is $z = a + bi$, where $i^2 = -1$; its conjugate is $\bar{z} = a - bi$ and its modulus is $|z| = \sqrt{a^2 + b^2}$. Thus $z\bar{z} = |z|^2$, and division by nonzero z can be performed using its conjugate. Real and imaginary parts must agree separately in a complex equation.

Problem 001.1[See solution](#)

How many real numbers x satisfy $\sqrt{(x-1)^2} + \sqrt{(x+2)^2} = 5$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinitely many

Problem 001.2[See solution](#)

A complex number z satisfies $z + 2\bar{z} = 6 + 3i$. What is $|z^2 + 1|$?

- (A) 5 (B) $2\sqrt{10}$ (C) $4\sqrt{10}$ (D) 13 (E) 20

Problem 001.3[See solution](#)

If $u = \sqrt{5 + 2\sqrt{6}} - \sqrt{5 - 2\sqrt{6}}$, what is u^4 ?

- (A) 64 (B) 32 (C) 144 (D) 96 (E) 16

Problem 001.4[See solution](#)

A nonzero complex number z satisfies $z/\bar{z} = i$ and $|z| = \sqrt{10}$. What is $|z^2 - 6|$?

- (A) 16 (B) 4 (C) $2\sqrt{10}$ (D) $2\sqrt{34}$ (E) 8

Problem 001.5[See solution](#)

For $a > 1$, suppose $a^x = 2$ and $a^y = 3$. What is $a^{(x+y)/2} - a^{(x-y)/2}$?

- (A) $2\sqrt{6}/3$ (B) 1 (C) $\sqrt{6}/3$ (D) $\sqrt{3/2}$ (E) $\sqrt{6}$

Problem 001.6[See solution](#)

For real x , let $f(x) = (x^2)^{1/2}$ and $g(x) = (x^{1/2})^2$, using real principal square roots. Which statement is correct?

- (A) Both functions have domain $\mathbb{R}$ and equal x .
- (B) Both functions have domain $[0, \infty)$ and equal x .
- (C) f has domain $\mathbb{R}$ and equals $|x|$; g has domain $[0, \infty)$ and equals x .
- (D) f has domain $\mathbb{R}$ and equals x ; g has domain $[0, \infty)$ and equals $|x|$.
- (E) f and g have different values at every positive x .

1. Precalculus

Functions ▸ Definition and domain

002 Function, domain, codomain, range, and fixed point

Book pp. 5-6. [Find in compact glossary](#)**In this entry:** function; domain; codomain; range; image; fixed point; natural domain.

Definition and key ideas

A function $f : A \rightarrow B$ assigns exactly one element of B to each element of its domain A . The target set B is the codomain; the actual range, or image, is $f(A) = \{f(x) : x \in A\} \subseteq B$. The book sometimes uses “range” for the target set; distinguishing the two is essential when testing surjectivity. The natural real domain of a formula excludes zero denominators, negative radicands for even roots, and nonpositive logarithm arguments.

A fixed point is an input x with $f(x) = x$. It must belong to the domain and also be a possible output. Algebraic manipulation of this equation can introduce extraneous roots, especially after squaring or cancelling denominators. Domain analysis should precede algebra, and candidate solutions should be checked in the original expression.

Problem 002.1

[See solution](#)

What is the natural real domain of $f(x) = \sqrt{(x-1)/(x+2)}/\log(x^2)$?

- (A) $(-\infty, -2) \cup [1, \infty)$ (B) $(-\infty, -2) \cup (1, \infty)$
 (C) $(-\infty, -2) \cup [-1, 1]$ (D) $(-2, 1) \setminus \{0\}$
 (E) $(-\infty, -2) \cup (0, 1)$

Problem 002.2

[See solution](#)

For $f(x) = \sqrt{3x+4}$ on its natural real domain, how many fixed points does $f \circ f$ have?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 002.3

[See solution](#)

What is the range of $f(x) = (x^2 + 1)/(x^2 - 1)$ on its natural real domain?

- (A) $\mathbb{R} \setminus \{1\}$ (B) $[-1, 1]$ (C) $(-\infty, 1)$ (D) $(-\infty, -1] \cup (1, \infty)$ (E) $(-\infty, -1) \cup (1, \infty)$

Problem 002.4

[See solution](#)

Let $f(x) = \log(4 - x^2)$. What is the natural real domain of $f(f(x))$?

- (A) $(-\sqrt{3}, \sqrt{3})$ (B) $(-\sqrt{4 - e^{-2}}, \sqrt{4 - e^{-2}})$
 (C) $(-2, 2)$ (D) $(-\sqrt{4 - e^{-1}}, \sqrt{4 - e^{-1}})$
 (E) $(-1, 1)$

Problem 002.5[See solution](#)

For how many real values of a does $f(x) = x^2 + a$ have exactly one real fixed point?

- (A) 1 (B) Infinitely many (C) 2 (D) 3 (E) 0

Problem 002.6[See solution](#)

A function $f : \mathbb{R} \rightarrow [0, \infty)$ is defined by $f(x) = (x^2 - 1)^2 + 2$. What is the largest set B for which the same rule defines a surjection $\mathbb{R} \rightarrow B$?

- (A) $[0, \infty)$ (B) $[1, \infty)$ (C) $[2, \infty)$ (D) $(2, \infty)$ (E) $[2, 3]$

1. Precalculus

Functions ▸ Composition of Functions

003 Composition and iteration of functions

Book pp. 6-7; 30. [Find in compact glossary](#)**In this entry:** composition; composite function; iteration; functional recurrence; identity function.**Definition and key ideas**

For compatible functions, $(g \circ f)(x) = g(f(x))$: apply the rightmost function first. The exact domain is $\{x \in \text{dom } f : f(x) \in \text{dom } g\}$. Composition is associative but usually not commutative. The identity map $\text{id}_A(x) = x$ is neutral for composition on A .

Iteration means repeated composition, denoted $f^{\circ n}$, and is different from the numerical power $[f(x)]^n$. A recurrence $u_{n+1} = F(u_n)$ is a sequence of iterates. When an equation determines f only through $f(g(x))$, it determines f on the image of g , not automatically everywhere in its stated domain.

Problem 003.1[See solution](#)

Let $f(x) = 1/(1-x)$, with the natural domains used in all compositions. Which describes $f^{\circ 3}$?

- (A) x on $\mathbb{R} \setminus \{1\}$ (B) x on $\mathbb{R} \setminus \{0, 1\}$
 (C) $-x$ on $\mathbb{R} \setminus \{0, 1\}$ (D) $1/x$ on $\mathbb{R} \setminus \{0\}$
 (E) 1 on $\mathbb{R} \setminus \{0, 1\}$

Problem 003.2[See solution](#)

The sequence $u_0 = t$, $u_{n+1} = 2 - u_n^2$ satisfies $u_2 = u_0$. What is the product of all distinct real values of t with this property?

- (A) -4 (B) -2 (C) 1 (D) 2 (E) 4

Problem 003.3[See solution](#)

Let $f(x) = 2x + 3$. What is $f^{\circ 10}(1)$?

- (A) 3073 (B) 2045 (C) 4096 (D) 1027 (E) 4093

Problem 003.4[See solution](#)

Functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $g(x) = x^2$ and $(f \circ g)(x) = |x|$ for every real x . Which expression is determined by these conditions?

- (A) $f(-4)$ (B) $f(4) - f(-4)$ (C) $f(-1) + f(1)$ (D) $f(9)$ (E) $(g \circ f)(-9)$

Problem 003.5[See solution](#)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(2x - 1) = x^2 + 1$ for every real x . What is $f(f(1))$?

- (A) $5/2$ (B) 2 (C) 5 (D) $13/4$ (E) 4

1. Precalculus

Functions ▸ Inverse Functions

004 Injectivity, surjectivity, bijectivity, and inverse functions

Book pp. 7-9. [Find in compact glossary](#)

In this entry: injective; one-to-one; surjective; onto; bijective; inverse function; reciprocal; restriction; rationalization.

Definition and key ideas

A map $f : A \rightarrow B$ is injective if $f(x) = f(y)$ implies $x = y$, and surjective if $f(A) = B$. It is bijective if both hold, precisely when a two-sided inverse $f^{-1} : B \rightarrow A$ exists. Then $f^{-1} \circ f = \text{id}_A$ and $f \circ f^{-1} = \text{id}_B$. The reciprocal $1/f$ is unrelated to the inverse map.

To find an inverse formula, solve $y = f(x)$ for x , selecting the branch whose values lie in the original domain. A restriction may make a noninjective function invertible. Rationalizing a derived formula can reveal a removable apparent singularity. To evaluate $f^{-1}(b)$, solve $f(a) = b$; an explicit inverse formula is often unnecessary.

Problem 004.1

[See solution](#)

The function $f : (-1, 1) \rightarrow \mathbb{R}$ is $f(x) = 2x/(1 - x^2)$. What is $f^{-1}(3/2)$?

- (A) $1/3$ (B) $(\sqrt{13} - 2)/3$ (C) $1/2$ (D) $2/3$ (E) $(\sqrt{13} + 2)/3$

Problem 004.2

[See solution](#)

Let $f : A \rightarrow B$ and $g : B \rightarrow A$ satisfy $g \circ f = \text{id}_A$. Which must be true?

- (A) f and g are bijective
(B) f is surjective and g is injective
(C) f is injective and g is surjective
(D) $g \circ f = f \circ g$
(E) $A = B$

Problem 004.3

[See solution](#)

For $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3 - 3x$, which is true?

- (A) One-to-one but not onto
(B) Onto but not one-to-one
(C) Neither one-to-one nor onto
(D) Both one-to-one and onto
(E) Its range is $[-2, 2]$

Problem 004.4[See solution](#)

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be bijections. For $h = g \circ f$, which is necessarily the inverse of h ?

- (A) $g^{-1} \circ f^{-1}$ (B) $f^{-1} \circ g^{-1}$ (C) $f \circ g$ (D) $g \circ f^{-1}$ (E) $f^{-1} \circ g$

Problem 004.5[See solution](#)

Let $f(x) = x^2 - 4x + 7$. Which restriction has an inverse $f^{-1}(y) = 2 - \sqrt{y - 3}$?

- (A) $f : [2, \infty) \rightarrow [3, \infty)$ (B) $f : (-\infty, 2] \rightarrow [3, \infty)$
(C) $f : \mathbb{R} \rightarrow [3, \infty)$ (D) $f : (-\infty, 3] \rightarrow [2, \infty)$
(E) $f : [3, \infty) \rightarrow [2, \infty)$

1. Precalculus

Graphs in the xy -Plane ▸ Graphs and symmetry

005 Graphs, symmetry, coordinates, and intercepts

Book pp. 9. [Find in compact glossary](#)

In this entry: graph; vertical-line test; horizontal-line test; abscissa; ordinate; intercept; even function; odd function; reflection.

Definition and key ideas

The graph of f is $\{(x, f(x)) : x \in \text{dom } f\}$. A plane relation is a function of x exactly when each vertical line meets it at most once; an injective function additionally passes the horizontal-line test. An x -intercept has ordinate 0; a y -intercept has abscissa 0. Intercepts need not be crossings: a graph can merely touch an axis.

Reflection across the y -axis replaces x by $-x$. On a domain symmetric about 0, an even function satisfies $f(-x) = f(x)$ and an odd function satisfies $f(-x) = -f(x)$. Reflection across $y = x$ interchanges a bijection's graph with its inverse's graph. Symmetry of a relation about the x -axis generally conflicts with being a function unless all its ordinates are zero.

Problem 005.1

[See solution](#)

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is even and satisfies $f(x+2) = -f(x)$ for every x . Which value must be zero?

- (A) $f(0)$ (B) $f(1)$ (C) $f(2)$ (D) $f(4)$ (E) $f(1/2)$

Problem 005.2

[See solution](#)

For how many real values of a is the relation $y^2 + 2y = x^2 + a$ the graph of a real-valued function whose domain is all of $\mathbb{R}$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinitely many

Problem 005.3

[See solution](#)

A function f is symmetric about both lines $x = 1$ and $x = 4$. Which positive number must be a period of f ?

- (A) 6 (B) 1 (C) 3 (D) 2 (E) 4

Problem 005.4

[See solution](#)

If f is odd and $g(x) = f(x-2) + 3$, about which point is the graph of g centrally symmetric?

- (A) $(-2, -3)$ (B) $(0, 0)$ (C) $(2, -3)$ (D) $(-2, 3)$ (E) $(2, 3)$

Problem 005.5[See solution](#)

How many intercept points does the graph $y = (x^2 - 1)/(x - 1)$ have with the two coordinate axes combined?

- (A) 1 (B) 4 (C) 3 (D) 0 (E) 2

Problem 005.6[See solution](#)

An odd function $f : \mathbb{R} \rightarrow \mathbb{R}$ is bijective. Which property must the graph of f^{-1} have?

- (A) Symmetry about the x -axis
- (B) Symmetry about the y -axis
- (C) Symmetry about the origin
- (D) Symmetry about the line $y = x$
- (E) A horizontal tangent at the origin

1. Precalculus

Graphs in the xy -Plane ▸ Piecewise graphs

006 Piecewise functions, absolute value, floor, and graph transformations

Book pp. 10; 30. [Find in compact glossary](#)

In this entry: piecewise function; absolute value; greatest integer function; floor; step function; translation; scaling.

Definition and key ideas

A piecewise function uses different formulas on specified parts of its domain; endpoint inequalities determine which value is assigned at a boundary. Absolute value is $|x| = x$ for $x \geq 0$ and $-x$ for $x < 0$. The floor $\lfloor x \rfloor$ is the greatest integer no larger than x , so $\lfloor -1.2 \rfloor = -2$. A step function is constant on intervals but can jump between them.

The graph of $y = af(b(x - h)) + k$ shifts by (h, k) , scales vertically by $|a|$ and horizontally by $1/|b|$, and reflects when a multiplier is negative. For absolute-value equations, split at zeros of the enclosed expressions; retain each algebraic solution only if it lies in the interval used to derive it. Floor equations are solved by inequalities $n \leq x < n + 1$.

Problem 006.1

[See solution](#)

What is the total length of the set of real numbers satisfying $\lfloor x \rfloor + \lfloor 2x \rfloor = 7$?

- (A) 0 (B) $1/4$ (C) $1/2$ (D) 1 (E) $3/2$

Problem 006.2

[See solution](#)

What is the sum of all real solutions of $|x - 2| + 2|x + 1| = 7$?

- (A) -4 (B) -2 (C) 0 (D) 2 (E) 4

Problem 006.3

[See solution](#)

What is $\sum_{k=1}^{10} \lfloor 3k/4 \rfloor$?

- (A) 38 (B) 41 (C) 36 (D) 37 (E) 35

Problem 006.4

[See solution](#)

What is the minimum of $|x - 1| + |x - 4| + |x - 9|$ over real x ?

- (A) 6 (B) 7 (C) 8 (D) 5 (E) 9

Problem 006.5

[See solution](#)

A function f has a unique global maximum $f(-2) = 5$. Define $g(x) = 7 - 3f(4 - 2x)$. At which point does g attain its global minimum?

- (A) $(-2, -8)$ (B) $(3, -8)$ (C) $(3, 22)$ (D) $(0, -8)$ (E) $(-3, -8)$

Problem 006.6[See solution](#)

Let $f(x) = x + 2$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$. How many real numbers satisfy $f(f(x)) = 2$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

1. Precalculus

Analytic Geometry ▸ General conic equation

007 Conic sections, completing squares, rotations, and degeneracy

Book pp. 11. [Find in compact glossary](#)

In this entry: analytic geometry; conic section; general second-degree equation; rotation of axes; degenerate conic; completing the square.

Definition and key ideas

A real conic is described by a quadratic equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. A rotation diagonalizes the symmetric quadratic-part matrix $\begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}$; a translation often removes linear terms. Completing squares exposes the center or vertex and the real locus.

For a nondegenerate conic, $B^2 - 4AC < 0$ gives ellipse type, $= 0$ parabola type, and > 0 hyperbola type. The discriminant alone does not rule out an empty locus, a point, or pairs of lines. Check the constant after normalization. Coordinate changes must be inverted when substituting: replacing points by $(u, v) = (2x, y)$ means substitute $x = u/2$ in the old equation.

For $Ax^2 + Bxy + Cy^2$, the rotation $x = u \cos \theta - v \sin \theta$, $y = u \sin \theta + v \cos \theta$ removes the mixed term when $(C - A) \sin 2\theta + B \cos 2\theta = 0$. A point map must be inverted before substituting into the original equation. Degenerate conics include pairs of lines, a point, and the empty set.

Problem 007.1

[See solution](#)

Which best describes $5x^2 + 6xy + 5y^2 - 8 = 0$?

- (A) A circle of radius 2
- (B) An ellipse with semiaxes 1 and 2
- (C) An ellipse with semiaxes $\sqrt{2}$ and 2
- (D) A hyperbola
- (E) A pair of lines

Problem 007.2

[See solution](#)

For which value of k does $x^2 + 4y^2 - 6x + 8y + k = 0$ have exactly one real point?

- (A) 5 (B) 9 (C) 12 (D) 13 (E) 17

Problem 007.3

[See solution](#)

Which equation represents two distinct parallel lines?

- (A) $x^2 + 2xy + y^2 - 4 = 0$
- (B) $x^2 + 2xy + y^2 - 4x = 0$
- (C) $x^2 + y^2 - 4 = 0$
- (D) $x^2 - y^2 - 4 = 0$
- (E) $x^2 + 2xy + y^2 + 4 = 0$

Problem 007.4[See solution](#)

Every point (x, y) on $x^2 + 4y^2 = 4$ is mapped to $(u, v) = (x + y, y)$. Which equation describes the image?

(A) $u^2 + 4v^2 = 4$

(B) $u^2 + 2uv + 5v^2 = 4$

(C) $u^2 - 2uv + 5v^2 = 4$

(D) $u^2 - 2uv + 3v^2 = 4$

(E) $u^2 + v^2 = 4$

Problem 007.5[See solution](#)

Under the rotation $x = u \cos \theta - v \sin \theta$, $y = u \sin \theta + v \cos \theta$, the mixed term in $3x^2 + 4xy + 6y^2$ vanishes. What is $\tan(2\theta)$?

(A) $-4/3$ **(B)** $4/3$ **(C)** $-3/4$ **(D)** $3/4$ **(E)** $2/3$

1. Precalculus

Analytic Geometry ▸ Lines

008 Lines, slopes, distances, and perpendicular bisectors

Book pp. 11; 31. [Find in compact glossary](#)

In this entry: line; slope; point-slope formula; parallel; perpendicular; distance formula; midpoint; perpendicular bisector; locus.

Definition and key ideas

A line has equation $ax + by + c = 0$, with $(a, b) \neq (0, 0)$. For a nonvertical line its slope is $(y_2 - y_1)/(x_2 - x_1)$, and point-slope form is $y - y_0 = m(x - x_0)$. Parallel nonvertical lines have equal slopes; perpendicular ones have slopes whose product is -1 . Vertical and horizontal lines form the missing perpendicular case.

The distance between two points is the Euclidean norm of their difference; their midpoint is their coordinatewise average. The points equidistant from two distinct fixed points form the perpendicular bisector of their joining segment. The distance from (u, v) to $ax + by + c = 0$ is $|au + bv + c|/\sqrt{a^2 + b^2}$, useful for tangency and conic definitions.

Problem 008.1

[See solution](#)

The point on $x + 2y = 8$ equidistant from $(0, 0)$ and $(4, 2)$ has x -coordinate

- (A) $-2/3$ (B) 0 (C) $2/3$ (D) $4/3$ (E) 2

Problem 008.2

[See solution](#)

A line of slope 2 is tangent to the circle $x^2 + y^2 = 5$. What is the product of the two possible y -intercepts?

- (A) -25 (B) -10 (C) -5 (D) 5 (E) 25

Problem 008.3

[See solution](#)

The lines $2x - y = 1$ and $x + 2y = 7$ meet at P . What is the distance from P to $3x + 4y = 0$?

- (A) $14/5$ (B) $79/25$ (C) 3 (D) $12/5$ (E) 2

Problem 008.4

[See solution](#)

What is the area of the triangle formed by $x + y = 6$, $x - y = 2$, and the y -axis?

- (A) 12 (B) 8 (C) 24 (D) 18 (E) 16

Problem 008.5

[See solution](#)

Points $A = (1, 2)$ and $B = (5, 4)$ lie above the x -axis. What is the minimum of $|AP| + |PB|$ over points P on the x -axis?

- (A) $2\sqrt{5}$ (B) $2\sqrt{10}$ (C) $2\sqrt{13}$ (D) 8 (E) 10

1. Precalculus

Analytic Geometry ▸ Parabolas

009 Parabola, focus, directrix, vertex, axis, and latus rectum

Book pp. 11-13. [Find in compact glossary](#)**In this entry:** parabola; focus; directrix; vertex; axis; latus rectum; focal width.**Definition and key ideas**

A parabola is the locus of points equally distant from a fixed focus and a fixed line, the directrix, which does not contain the focus. In $(x - h)^2 = 4p(y - k)$ with $p \neq 0$, the vertex is (h, k) , the focus is $(h, k + p)$, the directrix is $y = k - p$, and the axis is $x = h$. The sign of p determines the opening direction.

For a horizontal axis, interchange x and y : $(y - k)^2 = 4p(x - h)$. The latus rectum is the chord through the focus perpendicular to the axis; its length is $4|p|$. Do not confuse the coefficient of the squared coordinate with $4p$: in $y = ax^2$, the focal parameter is $p = 1/(4a)$.

Problem 009.1[See solution](#)

A parabola has focus $(2, 3)$ and directrix $y = -1$. What is the sum of the x -coordinates where it meets $y = 9$?

- (A) 0 (B) 2 (C) 4 (D) 8 (E) 16

Problem 009.2[See solution](#)

The latus rectum of $y = ax^2 + bx + c$, with $a \neq 0$, has length 3. Which must equal $1/3$?

- (A) a (B) $|a|$ (C) $b^2 - 4ac$ (D) $|b|$ (E) $|c|$

Problem 009.3[See solution](#)

The parabola $y^2 - 6y - 8x + 1 = 0$ has focus (a, b) . What is $a + b$?

- (A) 6 (B) 2 (C) 0 (D) 4 (E) 3

Problem 009.4[See solution](#)

A parabola has focus $(1, 1)$ and directrix $x + y = -2$. Which equation defines it?

- (A) $2[(x - 1)^2 + (y - 1)^2] = (x + y + 2)^2$
 (B) $(x - 1)^2 + (y - 1)^2 = (x + y + 2)^2$
 (C) $2[(x + 1)^2 + (y + 1)^2] = (x + y - 2)^2$
 (D) $(x - 1)^2 + (y - 1)^2 = 2(x + y + 2)^2$
 (E) $x^2 + y^2 = 2(x + y + 2)^2$

Problem 009.5[See solution](#)

The line $y = x - 1$ passes through the focus of $y^2 = 4x$. What is the distance between its two intersections with the parabola?

- (A) 4 (B) $4\sqrt{2}$ (C) 6 (D) 8 (E) $8\sqrt{2}$

1. Precalculus

Analytic Geometry ▸ Circles

010 Circle, center, radius, diameter, chord, and tangent

Book pp. 13–14. [Find in compact glossary](#)**In this entry:** circle; center; radius; diameter; chord; tangent line.**Definition and key ideas**

A circle consists of all points at a fixed positive distance r from a center (h, k) ; its equation is $(x - h)^2 + (y - k)^2 = r^2$. A chord is a segment joining two points of the circle. A diameter is a chord through the center, and its midpoint is the center. The circumference is $2\pi r$ and the enclosed area is πr^2 .

A tangent at a point P on a circle is perpendicular to the radius joining the center to P . If the center is the origin and $P = (u, v)$ lies on the circle of radius r , the tangent is $ux + vy = r^2$. A line is tangent exactly when its distance from the center equals r ; this avoids slope exceptions and is often faster than solving a quadratic intersection equation.

Problem 010.1[See solution](#)

A circle has diameter endpoints $(-3, 1)$ and $(5, 5)$. The tangent at $(5, 5)$ meets the x -axis at which point?

- (A) $(-5/2, 0)$ (B) $(5/2, 0)$ (C) $(15/2, 0)$ (D) $(10, 0)$ (E) $(15, 0)$

Problem 010.2[See solution](#)

The circles $x^2 + y^2 = 25$ and $(x - 6)^2 + y^2 = 25$ intersect. What is the length of their common chord?

- (A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Problem 010.3[See solution](#)

A circle has center $(2, -1)$ and is tangent to $3x + 4y = 12$. What is its area?

- (A) 4π (B) 2π (C) 5π (D) 16π (E) π

Problem 010.4[See solution](#)

A chord of a circle of radius 5 has length 6. The two tangents at its endpoints meet at P . What is the distance from P to the center?

- (A) $20/3$ (B) 5 (C) $15/4$ (D) $25/4$ (E) $25/3$

Problem 010.5[See solution](#)

The circle $x^2 + y^2 - 6x - 8y = 0$ meets $y = x$ at the origin and at P . What is the distance between these points?

- (A) $7\sqrt{2}$ (B) $5\sqrt{2}$ (C) $6\sqrt{2}$ (D) 7 (E) 14

1. Precalculus

Analytic Geometry ▸ Ellipses

011 Ellipse, foci, axes, eccentricity, and area scaling

Book pp. 14–15; 31. [Find in compact glossary](#)**In this entry:** ellipse; major axis; minor axis; semiaxis; eccentricity; focal distance; area scaling.**Definition and key ideas**

An ellipse is the locus where the sum of distances to two foci is a constant exceeding the distance between them. In $x^2/a^2 + y^2/b^2 = 1$ with $a \geq b > 0$, the semimajor axis is a , the semiminor axis is b , the foci are $(\pm c, 0)$ where $c^2 = a^2 - b^2$, and the distance sum is $2a$. Its eccentricity is $e = c/a \in [0, 1)$, with $e = 0$ the circle case.

Vertices lie at the ends of the major axis. If the larger denominator is beneath y^2 , interchange the axis roles. An ellipse has area πab . A transformation $(x, y) \mapsto (sx, ty)$ scales all areas by $|st|$, so changing a coordinate in a point map and replacing a variable in an equation have reciprocal effects.

Problem 011.1[See solution](#)An ellipse has foci $(\pm 3, 0)$ and eccentricity $3/5$. What is its area?

- (A)
- 12π
- (B)
- 15π
- (C)
- 16π
- (D)
- 20π
- (E)
- 25π

Problem 011.2[See solution](#)Every point (x, y) on $4x^2 + 9y^2 = 36$ is sent to $(2x, y/3)$. What is the eccentricity of the resulting ellipse?

- (A)
- $\sqrt{35}/6$
- (B)
- $\sqrt{77}/9$
- (C)
- $\sqrt{80}/9$
- (D)
- $\sqrt{8}/3$
- (E)
- $\sqrt{5}/3$

Problem 011.3[See solution](#)

An ellipse has major axis length 10 and the distance between its foci is 8. What is the length of a chord through one focus perpendicular to the major axis?

- (A) 3 (B)
- $32/5$
- (C)
- $18/5$
- (D)
- $24/5$
- (E) 6

Problem 011.4[See solution](#)What is the largest possible value of $x + y$ on $x^2/9 + y^2/4 = 1$?

- (A)
- $\sqrt{13}$
- (B) 13 (C)
- $\sqrt{5}$
- (D) 3 (E) 5

Problem 011.5[See solution](#)An ellipse has foci $F_1 = (-4, 0)$ and $F_2 = (4, 0)$ and major-axis length 10. A point P on it satisfies $|PF_1| = 2|PF_2|$. What is $|PF_1||PF_2|$?

- (A) 16 (B) 20 (C)
- $200/9$
- (D) 25 (E) 32

1. Precalculus

Analytic Geometry ▸ Hyperbolas

012 Hyperbola, branches, transverse axis, vertices, and asymptotes

Book pp. 15-17. [Find in compact glossary](#)**In this entry:** hyperbola; branch; focal axis; transverse axis; asymptote; hyperbola eccentricity.**Definition and key ideas**

A hyperbola is the locus where the absolute difference of distances to two foci is a positive constant smaller than their separation. For $x^2/a^2 - y^2/b^2 = 1$, $a, b > 0$, the vertices are $(\pm a, 0)$, foci $(\pm c, 0)$ with $c^2 = a^2 + b^2$, and the distance difference is $2a$. The two connected pieces are branches; their focal or transverse axis joins the vertices.

Its asymptotes are $y = \pm(b/a)x$, and its eccentricity is $c/a > 1$. For $y^2/a^2 - x^2/b^2 = 1$, the transverse axis is vertical and the asymptotes are $y = \pm(a/b)x$. Translations shift the center and both asymptotes. The plus sign in $c^2 = a^2 + b^2$ distinguishes the hyperbola from the ellipse.

A hyperbola has eccentricity $e = c/a > 1$. For $x^2/a^2 - y^2/b^2 = 1$, $c^2 = a^2 + b^2$ and the directrices are $x = \pm a^2/c$. Relative to either focus and its corresponding directrix, the distance to the focus is e times the perpendicular distance to the directrix.

Problem 012.1[See solution](#)

A hyperbola centered at the origin has vertices $(\pm 2, 0)$ and asymptotes $y = \pm 3x/2$. What is the distance between its foci?

- (A) $2\sqrt{5}$ (B) $2\sqrt{13}$ (C) $\sqrt{13}$ (D) 10 (E) 13

Problem 012.2[See solution](#)

The absolute difference of distances from a point on a hyperbola to $(1, 7)$ and $(1, -3)$ is 6. Which is an asymptote?

- (A) $y - 2 = \frac{3}{4}(x - 1)$ (B) $y - 2 = \frac{4}{3}(x - 1)$
 (C) $y + 2 = \frac{3}{4}(x - 1)$ (D) $y - 2 = \frac{5}{3}(x - 1)$
 (E) $y - 2 = \frac{3}{5}(x - 1)$

Problem 012.3[See solution](#)

For the hyperbola $x^2/9 - y^2/16 = 1$, the right-hand branch is also described by $|PF| = e d(P, L)$, with $F = (5, 0)$ and a vertical line L . Which pair (e, L) is correct?

- (A) $(3/5, x = 5/3)$ (B) $(5/3, x = 9/5)$
 (C) $(5/3, x = 5)$ (D) $(4/3, x = 9/5)$
 (E) $(5/4, x = 16/5)$

Problem 012.4[See solution](#)

The line $y = mx$ meets $x^2/4 - y^2/9 = 1$ in two real points exactly when

- (A) $|m| \leq 3/2$ (B) $|m| < 2/3$ (C) $m \neq 0$ (D) $|m| > 3/2$ (E) $|m| < 3/2$

Problem 012.5[See solution](#)

The curve $xy = 12$ is rotated to its principal axes. What is the distance between its vertices?

- (A) $8\sqrt{3}$ (B) $4\sqrt{3}$ (C) $2\sqrt{6}$ (D) $4\sqrt{6}$ (E) 12

1. Precalculus

Polynomial Equations > Polynomials and quadratics

013 Polynomial degree, coefficients, quadratic formula, and discriminant

Book pp. 17. [Find in compact glossary](#)

In this entry: polynomial; degree; leading coefficient; constant coefficient; quadratic polynomial; quadratic formula; discriminant; monic polynomial.

Definition and key ideas

A polynomial is $p(x) = a_n x^n + \cdots + a_0$ with nonnegative integer powers. If $a_n \neq 0$, its degree is n and a_n is the leading coefficient; it is monic when $a_n = 1$. The zero polynomial is exceptional: it has no finite degree under the convention used here. Polynomial equations can often be obtained by substituting a new variable into an expression involving radicals, exponentials, or logarithms.

For $ax^2 + bx + c = 0$ with $a \neq 0$, the roots are $(-b \pm \sqrt{b^2 - 4ac})/(2a)$. With real coefficients, the discriminant $b^2 - 4ac$ is positive for two distinct real roots, zero for a repeated real root, and negative for a nonreal conjugate pair. Parameter problems must also consider when the leading coefficient vanishes and the equation ceases to be quadratic.

Problem 013.1

[See solution](#)

For how many real values of k does $(k - 1)x^2 + 2kx + k + 1 = 0$ have exactly one distinct real solution?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinitely many

Problem 013.2

[See solution](#)

The quadratic $x^2 - 2tx + t + 2$ is strictly positive for every real x . Which is the set of possible t ?

- (A) $(-\infty, -1) \cup (2, \infty)$ (B) $[-1, 2]$ (C) $(-1, 2)$ (D) $(0, 2)$ (E) $\mathbb{R}$

Problem 013.3

[See solution](#)

A nonzero real polynomial p satisfies $p(x)^2 = p(x^2)$ for every real x , and $p(0) = 0$. Which must be true?

- (A) p has even degree.
(B) p has odd degree.
(C) The leading coefficient of p is 1.
(D) $p(1) = 0$.
(E) p has a nonzero real root.

Problem 013.4[See solution](#)

If p and q are nonzero polynomials of degrees 4 and 7, respectively, what is the degree of $p(q(x)) - q(p(x))$ necessarily at most?

- (A) 7 (B) 27 (C) 28 (D) 21 (E) 11

Problem 013.5[See solution](#)

For what real c is $x^2 + cx + c$ a factor of $x^3 - 3x - 2$?

- (A) -2 only (B) 1 only (C) 2 only (D) -2 and 2 (E) No real c

1. Precalculus

Polynomial Equations > Division Algorithm, Remainder Theorem, and Factor Theorem

014 Polynomial division, remainders, and factors

Book pp. 17; 32. [Find in compact glossary](#)**In this entry:** division algorithm; quotient; remainder; Remainder theorem; Factor theorem; polynomial divisibility.**Definition and key ideas**

Over a field, division by a nonzero polynomial d gives unique polynomials q, r with $p = qd + r$, where $r = 0$ or $\deg r < \deg d$. Dividing by $x - a$ leaves remainder $p(a)$, the Remainder theorem. Consequently $x - a$ divides p exactly when $p(a) = 0$, the Factor theorem.

For a divisor with distinct roots, evaluating at those roots determines the low-degree remainder by interpolation. For a repeated factor $(x - a)^m$, use derivative data: the remainder modulo $(x - a)^m$ is the Taylor polynomial of p at a through degree $m - 1$. These methods avoid a long division when only the remainder is needed.

Problem 014.1[See solution](#)

A polynomial p satisfies $p(1) = 3$ and $p(-2) = -6$. What is its remainder on division by $x^2 + x - 2$?

- (A) $x + 2$ (B) $2x + 1$ (C) $3x$ (D) $3x - 1$ (E) $-3x$

Problem 014.2[See solution](#)

What is the remainder when x^{20} is divided by $(x - 1)^2$?

- (A) $20x$ (B) $20x - 19$ (C) $19x - 18$ (D) $x + 19$ (E) 1

Problem 014.3[See solution](#)

What is the remainder when x^{101} is divided by $x^2 + x + 1$?

- (A) 1 (B) $-x - 1$ (C) $x + 1$ (D) $-x$ (E) x

Problem 014.4[See solution](#)

When a polynomial p is divided by $x^2 + 1$, the quotient is $x - 2$ and the remainder is $3x + 4$. What is the quotient when p is divided by $x - 1$?

- (A) $x^2 - x + 3$ (B) $x^2 - 2x + 4$ (C) $x^2 - x + 4$ (D) $x^2 + x + 3$ (E) $x^2 - 3x + 1$

Problem 014.5[See solution](#)

A polynomial leaves remainder $x + 1$ on division by $(x - 1)^2$ and remainder 2 on division by $x + 1$. What is its remainder on division by $(x - 1)^2(x + 1)$?

- (A) $x + 1$ (B) $\frac{1}{2}x^2 + \frac{3}{2}$ (C) $x^2 + 1$ (D) $\frac{1}{2}x^2 + x + \frac{1}{2}$ (E) $2x^2$

1. Precalculus

Polynomial Equations > Fundamental Theorem of Algebra and Roots

015 Fundamental Theorem of Algebra, multiplicity, and polynomial identity

Book pp. 18; 32. [Find in compact glossary](#)

In this entry: Fundamental Theorem of Algebra; root; zero; linear factor; multiplicity; polynomial identity; distinct roots.

Definition and key ideas

Every nonconstant complex polynomial has a complex root. Repeated division therefore expresses a degree- n polynomial as $a_n \prod_{j=1}^n (x - r_j)$, counting repetitions; it has exactly n complex roots with multiplicity. A root r has multiplicity m when $(x - r)^m$ divides the polynomial but $(x - r)^{m+1}$ does not. Factorization is unique up to order.

A nonzero degree- n polynomial has at most n distinct roots. Thus two polynomials of degree at most n agreeing at $n + 1$ distinct points are identical. Equivalently, a polynomial with more distinct roots than its degree is the zero polynomial. Multiplicity can be detected by derivatives: $p(r) = \cdots = p^{(m-1)}(r) = 0$ and $p^{(m)}(r) \neq 0$.

Over $\mathbb{C}$, a root of multiplicity $m \geq 1$ in p has multiplicity $m - 1$ in p' . Thus $\deg \gcd(p, p') = \deg p - r$, where r is the number of distinct roots.

Problem 015.1

[See solution](#)

A real polynomial p of degree at most 5 satisfies $p(j) = j^2$ for $j = 0, 1, 2, 3, 4, 5$. What is $p(6)$?

(A) 0 (B) 6 (C) 25 (D) 36 (E) Not determined

Problem 015.2

[See solution](#)

For $p(x) = x^6 - 3x^4 + 3x^2 - 1$, how many distinct complex roots does p have?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Problem 015.3

[See solution](#)

A degree-7 real polynomial has a root $2 + i$ of multiplicity 2 and a real root 3 of multiplicity 2. How many additional roots, counted with multiplicity, remain?

(A) 1 (B) 3 (C) 5 (D) 2 (E) 0

Problem 015.4

[See solution](#)

If a polynomial p satisfies $p(x + 1) = p(x) + 2x + 1$ for all real x and $p(0) = 3$, what is $p(10)$?

(A) 101 (B) 103 (C) 110 (D) 100 (E) 123

Problem 015.5[See solution](#)

A polynomial p of degree 7 has exactly three distinct complex roots. What is the degree of $\gcd(p, p')$?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

1. Precalculus

Polynomial Equations > Root Location Theorem

016 Root location and sign changes

Book pp. 18. [Find in compact glossary](#)**In this entry:** Root Location theorem; Intermediate Value theorem; sign change; root isolation.**Definition and key ideas**

A real polynomial is continuous. If $p(a)$ and $p(b)$ have opposite signs, the Intermediate Value theorem guarantees at least one root strictly between a and b . Disjoint sign-changing intervals give distinct roots. Combined with the degree bound, enough such intervals can establish the exact number of roots without solving for them.

The absence of a sign change does not imply absence of a root: an even-multiplicity root can merely touch the axis. A sign change alone gives existence, not uniqueness; monotonicity or a degree count supplies additional information. Every real polynomial of odd degree has a real root because its two tails have opposite signs.

Problem 016.1[See solution](#)

A quartic real polynomial satisfies $p(-2) > 0$, $p(-1) < 0$, $p(0) > 0$, $p(1) < 0$, and $p(2) > 0$. How many distinct real roots does it have?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) Cannot be determined

Problem 016.2[See solution](#)

A monic cubic polynomial p has $p(-1) = 2$, $p(0) = -1$, and $p(1) = 2$. In which set must one of its roots lie?

- (A) $(-\infty, -1)$ (B) $(1, 2)$ (C) $(2, \infty)$ (D) $\{-1, 0, 1\}$ (E) $\{2\}$

Problem 016.3[See solution](#)

For which real a does $x^3 - 3x + a = 0$ have three distinct real roots?

- (A) $a < -2$ (B) $-2 < a < 2$ (C) $-2 \leq a \leq 2$ (D) $a > 2$ (E) $a \neq \pm 2$

Problem 016.4[See solution](#)

For $p(x) = x^4 - 4x^2 - 1$, how many positive real roots are there?

- (A) 2 (B) 3 (C) 0 (D) 4 (E) 1

Problem 016.5[See solution](#)

A real polynomial p satisfies $p(0) < 0 < p(1)$ and has no zeros at either endpoint. Which must hold for its roots in $(0, 1)$?

- (A) There is exactly one distinct root.
- (B) Every root has odd multiplicity.
- (C) The sum of the multiplicities is odd.
- (D) There are at least three roots counting multiplicity.
- (E) The derivative is nonzero at every root.

1. Precalculus

Polynomial Equations > Rational Roots Theorem

017 Rational root theorem and root testing

Book pp. 18; 20. [Find in compact glossary](#)

In this entry: Rational Roots theorem; integer coefficient; lowest terms; rational root.

Definition and key ideas

If $p(x) = a_nx^n + \cdots + a_0$ has integer coefficients and s/t is a rational root in lowest terms with $t > 0$, then s divides a_0 and t divides a_n . For a monic integer polynomial, every rational root is an integer. If $a_0 = 0$, factor out powers of x before listing candidates.

The theorem supplies a finite candidate list, not a guarantee that any candidate is a root. Evaluate efficiently or divide after discovering a root. A quadratic with integer coefficients can have rational roots only when its discriminant is a square of an integer; this often finishes a factorization after rational roots have been removed.

Problem 017.1

[See solution](#)

How many rational roots does $2x^3 - 3x^2 - 8x + 12$ have, counted without multiplicity?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Cannot be determined

Problem 017.2

[See solution](#)

If a rational number r satisfies $r^4 - 5r^2 + 1 = 0$, which conclusion follows?

- (A) $r = 1$
(B) $r = -1$
(C) $r = 1/5$
(D) $r = 5$
(E) No such rational number exists

Problem 017.3

[See solution](#)

An integer-coefficient polynomial has roots $2/3$ and $3/5$. What is the least possible positive leading coefficient?

- (A) 3 (B) 5 (C) 8 (D) 15 (E) 30

Problem 017.4

[See solution](#)

How many integers k make both roots of $x^2 - kx + 6$ rational?

- (A) 2 (B) 3 (C) 4 (D) 6 (E) 8

Problem 017.5[See solution](#)

The polynomial $2x^3 + kx^2 - 5x + 2$, with integer k , has root $1/2$. What is the sum of its other two roots?

- (A) 2 (B) $-3/2$ (C) -2 (D) -1 (E) -3

1. Precalculus

Polynomial Equations > Conjugate Root Theorems

018 Radical and complex conjugate roots

Book pp. 18-19; 31-32. [Find in compact glossary](#)

In this entry: Conjugate Radical Roots theorem; Complex Conjugate Roots theorem; quadratic conjugate; real coefficient; rational coefficient.

Definition and key ideas

For a polynomial with real coefficients, $p(\bar{z}) = \overline{p(z)}$. Consequently nonreal roots occur in complex-conjugate pairs, with equal multiplicities. This need not hold for polynomials with genuinely complex coefficients. An odd-degree real polynomial therefore has at least one real root.

For rational coefficients, if $a + b\sqrt{d}$ is a root with $a, b \in \mathbb{Q}$, $b \neq 0$, and $d \in \mathbb{Q}$ nonsquare, then $a - b\sqrt{d}$ is also a root. These hypotheses matter: not every irrational number belongs to a quadratic radical pair. With several independent square roots, more conjugates may be forced; eliminating the radicals can produce a polynomial and indicate the required degree.

Problem 018.1

[See solution](#)

What is the least possible degree of a nonzero polynomial in $\mathbb{Q}[x]$ having both $1 + \sqrt{2}$ and $2 + i$ as roots?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 018.2

[See solution](#)

Which monic polynomial of least degree in $\mathbb{Q}[x]$ has $\sqrt{2} + \sqrt{3}$ as a root?

- (A) $x^2 - 5$ (B) $x^4 - 10x^2 + 1$ (C) $x^4 - 5x^2 + 6$ (D) $x^4 - 10x^2 - 1$ (E) $x^2 - 2x - 5$

Problem 018.3

[See solution](#)

A polynomial with coefficients in $\mathbb{Q}(\sqrt{2})$ has $\sqrt{2} + i$ as a root. Which additional number must be a root?

- (A) $-\sqrt{2} + i$ (B) $-\sqrt{2} - i$ (C) $\sqrt{2} - i$ (D) i (E) $-\sqrt{2}$

Problem 018.4

[See solution](#)

A monic real quartic has roots $1 + i$ and $2 + 3i$. What is its constant term?

- (A) 52 (B) 25 (C) 26 (D) 20 (E) 13

Problem 018.5

[See solution](#)

A rational-coefficient polynomial satisfies $p(\sqrt{2}) = 3 + 2\sqrt{2}$. What is $p(\sqrt{2})p(-\sqrt{2})$?

- (A) -1 (B) 1 (C) 7 (D) 9 (E) 17

1. Precalculus

Polynomial Equations > Sum and Product of the Roots

019 Vieta formulas and symmetric expressions in roots

Book pp. 19-20; 32. [Find in compact glossary](#)**In this entry:** Vieta formulas; sum of roots; product of roots; elementary symmetric sum; power sum.**Definition and key ideas**

For $a_n \prod_{j=1}^n (x - r_j) = a_n x^n + \cdots + a_0$, comparison of coefficients gives $\sum r_j = -a_{n-1}/a_n$ and $\prod r_j = (-1)^n a_0/a_n$, with multiplicities included. More generally, the elementary symmetric sum of products of k distinct root positions is $(-1)^k a_{n-k}/a_n$.

Symmetric expressions can often be computed without finding individual roots. For two roots, $r^2 + s^2 = (r + s)^2 - 2rs$ and $r^3 + s^3 = (r + s)^3 - 3rs(r + s)$. When transforming roots, compute their new sum and product, or substitute into the original relation. For nonzero roots, sums of reciprocals come from the coefficients near the constant term.

If $p(z) = a \prod_{j=1}^n (z - r_j)$ with roots counted by multiplicity, then $p'(z)/p(z) = \sum_j 1/(z - r_j)$ whenever $p(z) \neq 0$. This computes shifted reciprocal sums directly from the coefficients.

Problem 019.1[See solution](#)

The roots of $x^3 - 4x^2 + 5x - 7$ are r, s, t . What is $r^2 + s^2 + t^2$?

- (A) 3 (B) 6 (C) 9 (D) 11 (E) 16

Problem 019.2[See solution](#)

The roots of $x^2 - 3x + 1 = 0$ are r, s . Which polynomial has roots r^3, s^3 ?

- (A) $x^2 - 9x + 1$ (B) $x^2 - 18x + 1$ (C) $x^2 - 27x + 1$ (D) $x^2 - 18x - 1$ (E) $x^2 - 6x + 1$

Problem 019.3[See solution](#)

The roots of $x^3 - 4x^2 + 5x - 7$ are r, s, t , counted with multiplicity. What is $(r + s)(s + t)(t + r)$?

- (A) 7 (B) 13 (C) 20 (D) 27 (E) 35

Problem 019.4[See solution](#)

If r_1, r_2, r_3 are the roots of $p(x) = x^3 - 2x + 3$, what is $\sum_{j=1}^3 (2 - r_j)^{-1}$?

- (A) 7/10 (B) 10/7 (C) 3/7 (D) 2 (E) 7/3

Problem 019.5[See solution](#)

The roots of a monic cubic are nonzero and have sum 3, pairwise-product sum -2 , and product 5. What is the sum of their reciprocals?

- (A) 5/3 (B) $-2/5$ (C) $-5/2$ (D) 2/5 (E) 3/5

1. Precalculus

Logarithms ▸ Exponential and logarithmic functions

020 Logarithms, exponentials, laws, and change of base

Book pp. 20–22. [Find in compact glossary](#)

In this entry: logarithm; base; exponential function; natural logarithm; Euler number; change-of-base formula; logarithmic equation; exponential equation.

Definition and key ideas

For $b > 0$, $b \neq 1$, $\log_b x = y$ means $b^y = x$, with real domain $x > 0$. It is the inverse of b^x . Both functions increase when $b > 1$ and decrease when $0 < b < 1$. The natural base is e , and this guide uses $\log x = \ln x = \log_e x$.

For positive arguments, $\log_b(xy) = \log_b x + \log_b y$, $\log_b(x/y) = \log_b x - \log_b y$, and $\log_b(x^r) = r \log_b x$. Change of base is $\log_b x = \log_a x / \log_a b$. These laws do not turn sums into products: generally $\log(x + y) \neq \log x + \log y$. Substitutions such as $u = b^x > 0$ turn many exponential equations into polynomials, with positivity then eliminating spurious algebraic roots.

Problem 020.1

[See solution](#)

What is the sum of the real solutions of $\log_2 x + \log_x 2 = 5/2$?

- (A) $2 + \sqrt{2}$ (B) $4 + \sqrt{2}$ (C) $4 + \sqrt[4]{2}$ (D) 6 (E) 8

Problem 020.2

[See solution](#)

What is the unique real solution of $9^x - 4 \cdot 3^x - 5 = 0$?

- (A) 1 (B) $\log_3 5$ (C) $\log_3 4$ (D) $\log_3 9$ (E) -1

Problem 020.3

[See solution](#)

How many real solutions does $\log_2(x - 1) + \log_2(x + 1) = 3$ have?

- (A) 3 (B) 2 (C) 0 (D) 1 (E) 4

Problem 020.4

[See solution](#)

If $\log_a b = 2$ and $\log_b c = 3$, for positive bases $a, b \neq 1$, what is $\log_{ac}(b^2)$?

- (A) $2/7$ (B) $4/5$ (C) $3/7$ (D) $4/7$ (E) $6/7$

Problem 020.5[See solution](#)

What is the real solution set of $\log_{1/2}(x^2 - 3x + 2) > -2$?

- (A) $(1, 2)$
- (B) $(-\infty, 1) \cup (2, \infty)$
- (C) $((3 - \sqrt{17})/2, 1) \cup (2, (3 + \sqrt{17})/2)$
- (D) $((3 - \sqrt{17})/2, (3 + \sqrt{17})/2)$
- (E) $(-\infty, (3 - \sqrt{17})/2) \cup ((3 + \sqrt{17})/2, \infty)$

1. Precalculus

Trigonometry ▸ Trig Functions of Acute Angles

021 Right-triangle trigonometry and special angles

Book pp. 22-23. [Find in compact glossary](#)

In this entry: sine; cosine; tangent; cosecant; secant; cotangent; hypotenuse; complementary angles; cofunctions; special right triangle.

Definition and key ideas

For an acute angle in a right triangle, sine is opposite over hypotenuse, cosine adjacent over hypotenuse, and tangent opposite over adjacent. Cosecant, secant, and cotangent are the reciprocals of sine, cosine, and tangent, respectively. Complementary angles add to $\pi/2$; their sine and cosine values are exchanged, as are tangent and cotangent, secant and cosecant.

The 45-45-90 triangle has side ratio $1 : 1 : \sqrt{2}$, and the 30-60-90 triangle has ratio $1 : \sqrt{3} : 2$. These give exact values at $\pi/6$, $\pi/4$, and $\pi/3$. Always interpret “opposite” relative to the chosen angle, and use a unitless ratio rather than an angle measure as the value of a trig function.

Problem 021.1

[See solution](#)

In a right triangle, an acute angle θ satisfies $\tan \theta + \cot \theta = 5$. What is $\sin \theta \cos \theta$?

- (A) $1/10$ (B) $1/5$ (C) $2/5$ (D) $1/2$ (E) $\sqrt{5}/5$

Problem 021.2

[See solution](#)

The altitude to the hypotenuse of a right triangle divides that hypotenuse into segments of lengths 1 and 3. What is the sine of the smaller acute angle?

- (A) $1/4$ (B) $1/2$ (C) $\sqrt{3}/2$ (D) $\sqrt{3}/4$ (E) $1/\sqrt{3}$

Problem 021.3

[See solution](#)

A right triangle has hypotenuse 13 and inradius 2. What is its area?

- (A) 30 (B) 36 (C) 26 (D) 32 (E) 24

Problem 021.4

[See solution](#)

A vertical tower stands on level ground. From a point P its top has elevation 30° ; after walking 20 units directly toward its base to Q , the elevation is 60° . What is the tower height?

- (A) $5\sqrt{3}$ (B) 10 (C) $10\sqrt{3}$ (D) 20 (E) $20\sqrt{3}$

Problem 021.5

[See solution](#)

An acute angle satisfies $\tan \theta = 2$. What is $\sin \theta \cos \theta / (\sin \theta + \cos \theta)^2$?

- (A) $4/9$ (B) $1/5$ (C) $1/3$ (D) $2/9$ (E) $2/5$

Problem 021.6[See solution](#)

The acute angles α, β of a right triangle satisfy $\sec \alpha = 3 \csc \alpha$. What is $\sec \beta + \csc \beta$?

- (A) $\sqrt{10}$ (B) $4\sqrt{10}/3$ (C) $3\sqrt{10}/4$ (D) $4\sqrt{3}/3$ (E) $10/3$

1. Precalculus

Trigonometry ▸ Arbitrary Angles and Real Numbers

022 Unit circle, radian measure, quadrants, and directed angles

Book pp. 23–24. [Find in compact glossary](#)

In this entry: unit circle; radian; degree measure; initial side; terminal side; positive angle; negative angle; quadrant; arc length.

Definition and key ideas

An angle in standard position begins on the positive x -axis; counterclockwise rotation is positive. If $(x, y) \neq (0, 0)$ lies on its terminal side and $r = \sqrt{x^2 + y^2}$, then $\cos \theta = x/r$ and $\sin \theta = y/r$. The coordinate signs determine the signs of all six functions, while zero denominators identify undefined values.

Radian measure is arc length divided by radius, $\theta = s/r$, so π radians equals 180 degrees. On the unit circle the terminal point is $(\cos \theta, \sin \theta)$. Angles differing by $2\pi k$ have the same terminal side. This definition treats sine and cosine as functions of arbitrary real numbers, including negative numbers and numbers larger than one revolution.

Problem 022.1

[See solution](#)

The terminal side of θ contains $(-5, 12)$ and $\theta \in (\pi/2, \pi)$. What is $\sec \theta + \csc \theta$?

- (A) $-91/60$ (B) $91/60$ (C) $-13/60$ (D) $-17/13$ (E) $17/13$

Problem 022.2

[See solution](#)

A wheel of radius 3 rotates clockwise through an arc length 17π . Starting at $(3, 0)$ with center at the origin, where does the marked point end?

- (A) $(3/2, 3\sqrt{3}/2)$ (B) $(3/2, -3\sqrt{3}/2)$ (C) $(-3/2, 3\sqrt{3}/2)$ (D) $(-3/2, -3\sqrt{3}/2)$ (E) $(0, 3)$

Problem 022.3

[See solution](#)

For how many angles $\theta \in [0, 2\pi)$ does $2\cos^2 \theta + \sin \theta = 1$ hold?

- (A) 3 (B) 1 (C) 2 (D) 6 (E) 4

Problem 022.4

[See solution](#)

An angle θ has measure $90 + 180k$ degrees, where k is an integer. Which pair of functions is undefined for every such θ ?

- (A) $\sin \theta, \cos \theta$ (B) $\tan \theta, \sec \theta$ (C) $\cot \theta, \csc \theta$ (D) $\tan \theta, \csc \theta$ (E) $\sec \theta, \cot \theta$

Problem 022.5

[See solution](#)

If θ is in quadrant III and $\cos(2\theta) = 7/25$, what is $\sin \theta$?

- (A) $4/5$ (B) $-\sqrt{7}/5$ (C) $-3/5$ (D) $-4/5$ (E) $3/5$

1. Precalculus

Trigonometry ▸ Trig Identities and Formulas

023 Reciprocal, opposite-angle, and Pythagorean identities

Book pp. 24–25. [Find in compact glossary](#)

In this entry: trigonometric identity; reciprocal identity; opposite-angle identity; Pythagorean identity; trigonometric parity; reduction formula.

Definition and key ideas

An identity holds for every argument where both sides are defined. The basic identities are $\tan \theta = \sin \theta / \cos \theta$, $\cot \theta = \cos \theta / \sin \theta$, and the reciprocal definitions of secant and cosecant. Sine and tangent are odd, while cosine is even; the remaining reciprocal functions inherit the corresponding parity.

The Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$ yields $1 + \tan^2 \theta = \sec^2 \theta$ and $1 + \cot^2 \theta = \csc^2 \theta$ when denominators are nonzero. Complementary-angle formulas include $\sin(\pi/2 - \theta) = \cos \theta$. Squared identities determine magnitudes, not signs; determine the quadrant before extracting a square root.

Problem 023.1

[See solution](#)

If $\sec \theta - \tan \theta = 3$, what is $\sin \theta$?

- (A) $-4/5$ (B) $-3/5$ (C) $3/5$ (D) $4/5$ (E) $5/4$

Problem 023.2

[See solution](#)

For every x where it is defined, $\frac{1 - \cos x}{\sin x} + \frac{\sin x}{1 - \cos x}$ equals

- (A) $2 \sin x$ (B) $2 \cos x$ (C) $2 \csc x$ (D) $2 \sec x$ (E) $2 \cot x$

Problem 023.3

[See solution](#)

If $\tan \theta + \cot \theta = 4$, what is $\tan^4 \theta + \cot^4 \theta$?

- (A) 192 (B) 98 (C) 194 (D) 196 (E) 254

Problem 023.4

[See solution](#)

On its natural domain, $F(x) = \sin(-x) \sec x + \cos(-x) \csc x$. Which statement is true?

- (A) F is even and equals $\tan x + \cot x$.
(B) F is odd and equals $\cot x - \tan x$.
(C) F is even and equals $\cot x - \tan x$.
(D) F is odd and equals $\tan x - \cot x$.
(E) F is defined at $x = 0$.

Problem 023.5[See solution](#)

For $0 < x < \pi/2$, which is equal to $\sqrt{1 - \sin^2(\pi/2 + x)}$?

- (A) $-\sin x$ (B) $\sin x$ (C) $\cos x$ (D) $-\cos x$ (E) $\tan x$

1. Precalculus

Trigonometry ▸ Trig Identities and Formulas

024 Addition, subtraction, double-angle, and half-angle formulas

Book pp. 25–26. [Find in compact glossary](#)

In this entry: addition formula; subtraction formula; double-angle formula; half-angle formula; angle sum; angle difference.

Definition and key ideas

The addition formulas are $\sin(a + b) = \sin a \cos b + \cos a \sin b$, $\cos(a + b) = \cos a \cos b - \sin a \sin b$, and $\tan(a + b) = (\tan a + \tan b)/(1 - \tan a \tan b)$ where all terms are defined. Replacing b by $-b$ gives subtraction formulas. Setting $a = b$ gives $\sin 2a = 2 \sin a \cos a$ and $\cos 2a = 1 - 2 \sin^2 a = 2 \cos^2 a - 1$.

Half-angle formulas are $\sin(a/2) = \pm \sqrt{(1 - \cos a)/2}$ and $\cos(a/2) = \pm \sqrt{(1 + \cos a)/2}$, with signs determined by the half-angle quadrant. Also $\tan(a/2) = \sin a/(1 + \cos a)$ where defined. Angle formulas are especially useful for exact values, eliminating inverse trig functions, and reducing multiple-angle equations to algebra.

Problem 024.1

[See solution](#)

If $\sin \alpha = 3/5$ with $\alpha \in (\pi/2, \pi)$ and $\cos \beta = 12/13$ with $\beta \in (0, \pi/2)$, what is $\sin(\alpha + \beta)$?

- (A) 16/65 (B) 33/65 (C) 56/65 (D) -16/65 (E) -56/65

Problem 024.2

[See solution](#)

If $\cos \theta = -7/25$ and $\pi < \theta < 2\pi$, what is $\tan(\theta/2)$?

- (A) -4/3 (B) -3/4 (C) 3/4 (D) 4/3 (E) 7/24

Problem 024.3

[See solution](#)

How many $x \in [0, 2\pi)$ satisfy $\sin(2x) = \cos x$?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 024.4

[See solution](#)

What is $\cos(\pi/12) \cos(5\pi/12)$?

- (A) $\sqrt{3}/4$ (B) $-1/4$ (C) $1/2$ (D) $1/4$ (E) $\sqrt{2}/4$

Problem 024.5

[See solution](#)

Let $\alpha, \beta \in (0, \pi/2)$ satisfy $\alpha + \beta = \pi/4$. What is $(1 + \tan \alpha)(1 + \tan \beta)$?

- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $1 + \sqrt{2}$ (E) It depends on α .

1. Precalculus

Trigonometry ▷ Periodicity and Graphs of the Trig Functions

025 Periodicity, fundamental period, and trigonometric domains and ranges

Book pp. 27. [Find in compact glossary](#)

In this entry: periodic function; period; fundamental period; amplitude; phase shift; trigonometric asymptote.

Definition and key ideas

A positive number T is a period when translation by T preserves the domain and $f(x + T) = f(x)$ throughout it. If a smallest positive period exists, it is the fundamental period. Sine and cosine have fundamental period 2π ; tangent and cotangent have period π . A constant has every positive period and no smallest one. For nonzero b , replacing x by bx divides a fundamental period by $|b|$.

Sine and cosine have domain $\mathbb{R}$ and range $[-1, 1]$. Tangent and secant exclude $\pi/2 + k\pi$; cotangent and cosecant exclude $k\pi$. Tangent and cotangent have range $\mathbb{R}$, while secant and cosecant have range $(-\infty, -1] \cup [1, \infty)$. In $A \sin(bx + c) + d$, $|A|$ is amplitude, $-c/b$ the horizontal phase shift, and d the midline.

The excluded inputs of tangent, secant, cotangent, and cosecant are vertical asymptotes: the function becomes unbounded on approach from at least one side. Phase shifts translate these asymptotes; a nonzero vertical scale does not remove them.

Problem 025.1

[See solution](#)

What is the smallest positive period of $f(x) = \sin(2x) + \cos(3x)$?

- (A) $\pi/3$ (B) $\pi/2$ (C) π (D) 2π (E) 6π

Problem 025.2

[See solution](#)

How many points in $[0, 2\pi)$ are excluded from the natural domain of $\tan(2x) + \csc(3x)$?

- (A) 6 (B) 8 (C) 9 (D) 10 (E) 12

Problem 025.3

[See solution](#)

The function $f(x) = 3 - 2 \sin(4x - \pi/3)$ first attains its maximum at a positive $x = a$. What is a ?

- (A) $\pi/24$ (B) $\pi/12$ (C) $5\pi/24$ (D) $11\pi/24$ (E) $7\pi/12$

Problem 025.4

[See solution](#)

What is the range of $2 \sin x + \cos(2x)$?

- (A) $[-1, 3/2]$ (B) $[-2, 2]$ (C) $[-3, 3]$ (D) $[-3, 1]$ (E) $[-3, 3/2]$

Problem 025.5[See solution](#)

For $g(x) = 2 \sec x - 3$, which pair gives its range and all its vertical asymptotes?

- (A) $(-\infty, -5] \cup [-1, \infty)$; $x = \pi/2 + k\pi$ (B) $[-1, \infty)$; $x = k\pi$
(C) $(-\infty, -1] \cup [1, \infty)$; $x = \pi/2 + k\pi$ (D) $(-\infty, -5] \cup [-1, \infty)$; $x = k\pi$
(E) $\mathbb{R}$; $x = \pi/2 + k\pi$

1. Precalculus

Trigonometry ▸ The Inverse Trig Functions

026 Inverse trigonometric functions and principal branches

Book pp. 28-29; 33-34. [Find in compact glossary](#)

In this entry: arcsine; arccosine; arctangent; arccosecant; arcsecant; arccotangent; principal branch; inverse trigonometric function.

Definition and key ideas

Inverse trig functions invert injective restrictions of the periodic functions. The principal ranges used here are $\arcsin x \in [-\pi/2, \pi/2]$, $\arccos x \in [0, \pi]$, and $\arctan x \in (-\pi/2, \pi/2)$. Their domains are $[-1, 1]$, $[-1, 1]$, and $\mathbb{R}$, respectively. Thus $\arcsin(\sin x)$ is not always x ; it is the unique principal angle with the same sine.

We use $\operatorname{arccot} x \in (0, \pi)$, $\operatorname{arcsec} x \in [0, \pi] \setminus \{\pi/2\}$, and $\operatorname{arccsc} x \in [-\pi/2, \pi/2] \setminus \{0\}$. The last two require $|x| \geq 1$. Conventions can vary, especially for arccotangent, so specify them. To simplify compositions, set the inverse expression equal to an angle, recover its other trig values, and track its quadrant before identifying any sum of angles.

Problem 026.1

[See solution](#)

What is $\arcsin(\sin(7\pi/6)) + \arccos(\cos(5\pi/3))$?

- (A) $-\pi/2$ (B) $-\pi/6$ (C) $\pi/6$ (D) $\pi/2$ (E) $3\pi/2$

Problem 026.2

[See solution](#)

What is $\arctan 2 + \arctan 3 - \arctan(1/5)$?

- (A) $\arctan(4/5)$ (B) $\pi/2$ (C) $\pi - \arctan(3/2)$ (D) $\pi + \arctan(3/2)$ (E) $3\pi/4$

Problem 026.3

[See solution](#)

With the principal branches stated above, what is $\operatorname{arcsec}(-2) + \operatorname{arccsc}(-2)$?

- (A) $-\pi/2$ (B) $\pi/6$ (C) $\pi/2$ (D) $5\pi/6$ (E) $7\pi/6$

Problem 026.4

[See solution](#)

For real x , which identity holds under the convention $\operatorname{arccot} x \in (0, \pi)$?

- (A) $\operatorname{arccot} x = \arctan(1/x)$ for every $x \neq 0$
 (B) $\operatorname{arccot} x = \pi/2 - \arctan x$
 (C) $\operatorname{arccot}(-x) = -\operatorname{arccot} x$
 (D) $\operatorname{arccot} x = \pi/2 + \arctan x$
 (E) $\operatorname{arccot} x$ is undefined at $x = 0$

Problem 026.5[See solution](#)

What is $\cos(2 \arcsin(2/3))$?

- (A) $5/9$ (B) $1/9$ (C) $\sqrt{5}/3$ (D) $-1/9$ (E) $4/9$

1. Precalculus

Chapter 1 Review Questions ▸ Radical equations

027 Radical equations and extraneous roots

Book pp. 30; 332. [Find in compact glossary](#)**In this entry:** radical equation; extraneous root; squaring an equation; equivalent transformation.**Definition and key ideas**

A radical equation contains the unknown beneath a root. Begin by requiring all even-root radicands to be nonnegative. Isolate a radical and square to eliminate it, repeating if necessary. Squaring preserves every original solution but can add solutions because $u^2 = v^2$ permits $u = -v$ as well as $u = v$.

A reliable solution records sign restrictions before each squaring and substitutes every final candidate into the original equation. Dividing by an expression can instead lose solutions where that expression is zero. Solving a transformed polynomial is therefore an intermediate step; its full root set is not automatically the answer to the original radical equation.

Problem 027.1[See solution](#)What is the sum of all real solutions of $\sqrt{x+3} = x-3$?

- (A) 1 (B) 3 (C) 6 (D) 7 (E) 8

Problem 027.2[See solution](#)How many real solutions does $\sqrt{x+1} + \sqrt{9-x} = 4$ have?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 027.3[See solution](#)How many real solutions does $x\sqrt{x+2} = x$ have?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 027.4[See solution](#)For what real a does $\sqrt{x^2+1} = x+a$ have a real solution?

- (A)
- $a \geq 0$
- (B)
- $a > 0$
- (C)
- $a \neq 0$
- (D)
- $|a| \geq 1$
- (E) Every real
- a

Problem 027.5[See solution](#)For which $a \geq 0$ does $\sqrt{x} + \sqrt{1-x} = a$ have exactly one real solution?

- (A)
- $a = 0$
- (B)
- $a = 3/2$
- (C)
- $a = \sqrt{2}$
- (D)
- $a = 2$
- (E)
- $a = 1$

1. Precalculus

Chapter 1 Review Questions > Rational inequalities

028 Rational inequalities and sign charts

Book pp. 30; 332. [Find in compact glossary](#)**In this entry:** rational inequality; sign chart; critical endpoint; strict inequality; interval notation.**Definition and key ideas**

To solve $P(x)/Q(x) > 0$, first exclude every zero of Q . Factor numerator and denominator and mark all real zeros on the number line; the sign is constant on each intervening interval. A factor of odd multiplicity changes sign when crossed, while one of even multiplicity does not.

Test one point in each interval or track signs systematically. For a nonstrict inequality, include numerator zeros that are in the domain; never include denominator zeros. Cancelling a common factor simplifies signs but does not restore an excluded point. Multiplying an inequality by an expression of unknown sign requires splitting into cases or retaining its sign information.

Problem 028.1[See solution](#)Which is the solution set of $(x - 1)^2(x + 2)/(x - 3) \leq 0$?

- (A) $[-2, 3)$ (B) $(-\infty, -2] \cup (3, \infty)$ (C) $[-2, 1] \cup (1, 3]$ (D) $(-2, 3)$ (E) $[-2, 3]$

Problem 028.2[See solution](#)What is the sum of the integer solutions of $1/(x - 1) > 1/(x + 2)$ with $-5 \leq x \leq 5$?

- (A) -5 (B) -3 (C) 0 (D) 2 (E) 5

Problem 028.3[See solution](#)What is the solution set of $\frac{(x - 1)^2(x + 2)}{(x - 1)(x - 3)} \geq 0$?

- (A) $[-2, 1] \cup (3, \infty)$ (B) $[-2, 1) \cup (3, \infty)$
 (C) $(-\infty, -2] \cup (1, 3)$ (D) $[-2, 3)$
 (E) $[-2, 1) \cup [3, \infty)$

Problem 028.4[See solution](#)For which real a does $\frac{x^2 + ax + 1}{(x - 2)^2} > 0$ hold for every real $x \neq 2$?

- (A) $-2 < a < 2$ (B) $-2 \leq a \leq 2$ (C) $a < 2$ (D) $a > -2$ (E) $a = 2$ only

Problem 028.5[See solution](#)

What is the solution set of $(x^2 - 4)/(x^2 - 1) \geq 0$?

(A) $\mathbb{R} \setminus \{-1, 1\}$

(B) $(-\infty, -1) \cup (1, \infty)$

(C) $(-\infty, -2] \cup (-1, 1) \cup [2, \infty)$

(D) $[-2, -1) \cup (1, 2]$

(E) $[-2, 2]$

1. Precalculus

Chapter 1 Review Questions ▸ Hyperbolic functions

029 Hyperbolic sine, cosine, tangent, and their inverses

Book pp. 33; 336-337. [Find in compact glossary](#)

In this entry: hyperbolic sine; hyperbolic cosine; hyperbolic tangent; inverse hyperbolic sine; inverse hyperbolic tangent; inverse hyperbolic cosine.

Definition and key ideas

Define $\sinh x = (e^x - e^{-x})/2$, $\cosh x = (e^x + e^{-x})/2$, and $\tanh x = \sinh x / \cosh x$. They satisfy $\cosh^2 x - \sinh^2 x = 1$. Hyperbolic sine is an odd increasing bijection $\mathbb{R} \rightarrow \mathbb{R}$; hyperbolic cosine is even with range $[1, \infty)$; hyperbolic tangent is an increasing bijection $\mathbb{R} \rightarrow (-1, 1)$.

Solving the exponential equations gives $\operatorname{arsinh} x = \log(x + \sqrt{x^2 + 1})$ for every real x , and $\operatorname{artanh} x = \frac{1}{2} \log((1+x)/(1-x))$ for $|x| < 1$. The inverse of $\cosh$ requires the restriction $x \geq 0$ and is $\operatorname{arcosh} u = \log(u + \sqrt{u^2 - 1})$, $u \geq 1$. In inverse formulas a negative candidate for e^x must be discarded.

The addition identities are $\sinh(u+v) = \sinh u \cosh v + \cosh u \sinh v$ and $\cosh(u+v) = \cosh u \cosh v + \sinh u \sinh v$. Thus $\sinh 2u = 2 \sinh u \cosh u$ and $\cosh 2u = 1 + 2 \sinh^2 u$.

Problem 029.1

[See solution](#)

If $\tanh x = 3/5$, what is e^{2x} ?

- (A) $1/4$ (B) $3/2$ (C) 2 (D) 4 (E) 8

Problem 029.2

[See solution](#)

What is $\operatorname{arsinh}(3/4) + \operatorname{artanh}(3/5)$?

- (A) $\log 2$ (B) $\log 3$ (C) $\log 4$ (D) $\log 5$ (E) $\log 8$

Problem 029.3

[See solution](#)

If $\sinh x = 2$, what is $\tanh(2x)$?

- (A) $\sqrt{5}/3$ (B) $4\sqrt{5}/9$ (C) $2/\sqrt{5}$ (D) $8/9$ (E) $4/9$

Problem 029.4

[See solution](#)

For real x , what is the minimum of $3 \cosh x - 2 \sinh x$?

- (A) 5 (B) $\sqrt{5}$ (C) 3 (D) 2 (E) 1

Problem 029.5

[See solution](#)

For real x , which expression equals $\operatorname{arcosh}(\cosh x) + \operatorname{arsinh}(\sinh x)$?

- (A) $2x$ (B) 0 (C) $2|x|$ (D) $|x| + x$ (E) $|x| - x$

2. Calculus I

Limits of Sequences ▸ Definitions

030 Sequences, convergence, and divergenceBook pp. 35-36. [Find in compact glossary](#)

In this entry: sequence; limit of a sequence; epsilon–N definition; convergent sequence; divergent sequence; divergence to infinity; oscillation.

Definition and key ideas

A real sequence is a function $a : \mathbb{N} \rightarrow \mathbb{R}$, written (a_n) . It converges to $L \in \mathbb{R}$ if for every $\varepsilon > 0$ there is N such that $n \geq N$ implies $|a_n - L| < \varepsilon$. The quantifier order matters: N may depend on ε , but must work for every subsequent index. A finite limit is unique, and altering finitely many terms changes neither convergence nor the limit.

Divergence means failure to converge to a finite real number. The notation $a_n \rightarrow +\infty$ means every real threshold is eventually exceeded; it is a special kind of divergence. Oscillation need not prevent convergence if its amplitude tends to zero. Conversely, two subsequences with different limiting values disprove convergence. Identify the size of a term before treating its alternating sign.

Problem 030.1[See solution](#)

What is $\lim_{n \rightarrow \infty} n(\sqrt{n^2 + 3} - n)$?

- (A) 0 (B) 1 (C) $3/2$ (D) 3 (E) $+\infty$

Problem 030.2[See solution](#)

Let $a_n = n \sin(n\pi/2)$. Which statement is true?

- (A) (a_n) converges to 0.
 (B) (a_n) is bounded but divergent.
 (C) $a_n \rightarrow +\infty$.
 (D) $a_n \rightarrow -\infty$.
 (E) (a_n) is unbounded and has a subsequence converging to 0.

Problem 030.3[See solution](#)

For $a_n = 2 + (-1)^n/n$, what is the smallest positive integer N such that $|a_n - 2| < 1/100$ for every $n \geq N$?

- (A) 99 (B) 100 (C) 101 (D) 199 (E) 200

Problem 030.4[See solution](#)

How many subsequential limits does $a_n = (-1)^n + 1/n$ have?

- (A) 2 (B) 0 (C) 1 (D) Infinitely many (E) 3

Problem 030.5[See solution](#)

If $a_n \rightarrow 2$ and $b_n \rightarrow -2$, which sequence must converge?

- (A) $a_n^2 - b_n^2$ (B) $1/(a_n + b_n)$ (C) $n(a_n + b_n)$ (D) $(a_n - 2)/(b_n + 2)$ (E) $(-1)^n a_n$

2. Calculus I

Limits of Sequences ▸ Monotone convergence

031 Boundedness, monotonicity, and recursive limits

Book pp. 36-38. [Find in compact glossary](#)

In this entry: bounded sequence; monotone sequence; eventually monotone; monotone convergence theorem; recursive sequence.

Definition and key ideas

A sequence is bounded if $|a_n| \leq M$ for some fixed M and every n . It is nondecreasing if $a_{n+1} \geq a_n$ and nonincreasing if the inequality is reversed. The monotone convergence theorem says that a nondecreasing sequence bounded above converges to its supremum; the decreasing version uses a lower bound and infimum. Eventual monotonicity suffices. Every convergent sequence is bounded, but bounded sequences may oscillate without converging.

For $a_{n+1} = F(a_n)$, first prove an invariant interval and monotonicity or another convergence criterion. Only then may continuity of F justify $L = F(L)$. A solution of this fixed-point equation is merely a candidate until convergence and the admissible branch are checked. Some recurrences simplify after taking reciprocals or differences.

Problem 031.1

[See solution](#)

Let $a_1 = 3$ and $a_{n+1} = \frac{1}{2}(a_n + 6/a_n)$. Which description is correct?

- (A) It increases without bound.
- (B) It decreases to $\sqrt{6}$.
- (C) It decreases to 0.
- (D) It oscillates between two distinct limits.
- (E) It increases to 3.

Problem 031.2

[See solution](#)

If $a_1 = 2$ and $a_{n+1} = a_n/(1 + a_n)$, what is $\lim_{n \rightarrow \infty} na_n$?

- (A) 0
- (B) $1/2$
- (C) 1
- (D) 2
- (E) The limit does not exist.

Problem 031.3[See solution](#)

Let $a_1 = 0$ and $a_{n+1} = 1 - a_n$. The equation $L = 1 - L$ has the unique solution $L = 1/2$. Which describes (a_n) ?

- (A) It increases to $1/2$.
- (B) It decreases to $1/2$.
- (C) It converges to 0.
- (D) It alternates between 0 and 1 and does not converge.
- (E) It is unbounded.

Problem 031.4[See solution](#)

Let $b_n = \prod_{k=1}^n (1 - 1/(k+1)^2)$. Which is its limit?

- (A) 0 (B) $1/4$ (C) $1/2$ (D) 1 (E) It does not exist.

Problem 031.5[See solution](#)

Let $a_1 = 0$ and $a_{n+1} = 1 - a_n/2$. What is $a_n - 2/3$?

- (A) $-\frac{2}{3}(\frac{1}{2})^{n-1}$ (B) $-\frac{2}{3}(-\frac{1}{2})^{n-1}$ (C) $\frac{2}{3}(-\frac{1}{2})^n$ (D) $(-\frac{1}{2})^{n-1}$ (E) $\frac{2}{3}(\frac{1}{2})^{n-1}$

2. Calculus I

Limits of Sequences ▸ Algebra and estimates

032 Limit laws and the squeeze theorem

Book pp. 36-38. [Find in compact glossary](#)

In this entry: sum law; product law; quotient law; squeeze theorem; sandwich theorem; bounded factor times null sequence; growth comparison.

Definition and key ideas

For convergent real sequences, limits commute with sums and products, and with quotients when the denominator limit is nonzero. If $u_n \leq a_n \leq v_n$ eventually and both outer sequences tend to L , then $a_n \rightarrow L$. In particular, a bounded sequence times a sequence tending to zero tends to zero. Powers n^{-p} tend to zero for $p > 0$, and $r^n \rightarrow 0$ when $|r| < 1$.

Divide rational expressions by the dominant power and rationalize differences of nearly equal radicals. Do not use quotient laws with a zero denominator limit, or subtract two divergent infinities as if they were numbers. A function limit $f(x) \rightarrow L$ as $x \rightarrow \infty$ implies $f(n) \rightarrow L$, but the converse is false: integer samples do not control intervening values.

Problem 032.1

[See solution](#)

What is $\lim_{n \rightarrow \infty} \left[n \left(\sqrt{1 + \frac{2}{n}} - 1 \right) + \frac{\cos(n^2)}{\sqrt{n}} \right]$?

- (A) 0
- (B) $1/2$
- (C) 1
- (D) 2
- (E) The limit does not exist.

Problem 032.2

[See solution](#)

Real sequences satisfy $a_n \rightarrow 0$ and $a_n b_n \rightarrow 1$. Which conclusion must hold?

- (A) $b_n \rightarrow +\infty$. (B) $b_n \rightarrow -\infty$. (C) (b_n) is bounded. (D) $|b_n| \rightarrow +\infty$. (E) $b_n/n \rightarrow 0$.

Problem 032.3

[See solution](#)

What is $\lim_{x \rightarrow 0} x^2 \lfloor 1/|x| \rfloor$?

- (A) ∞
- (B) 1
- (C) The limit does not exist
- (D) 0
- (E) -1

Problem 032.4[See solution](#)

A continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(n) = 0$ for every positive integer n . Which conclusion follows?

- (A) $f(x) \rightarrow 0$ as $x \rightarrow \infty$.
- (B) f is bounded.
- (C) $f(x)/x \rightarrow 0$ as $x \rightarrow \infty$.
- (D) $\lim_{n \rightarrow \infty} f(n) = 0$, but none of the preceding three conclusions is forced.
- (E) f must be identically zero.

Problem 032.5[See solution](#)

What is $\lim_{n \rightarrow \infty} (2^n + 3^n)^{1/n}$?

- (A) 5 (B) 3 (C) $\sqrt{6}$ (D) 6 (E) 2

Problem 032.6[See solution](#)

For any fixed positive integer m , what is $\lim_{n \rightarrow \infty} n^m/2^n$?

- (A) 0 (B) $1/2$ (C) 1 (D) $m!$ (E) $+\infty$

2. Calculus I

Limits of Functions > Definitions and calculations

033 Function limits, one-sided limits, and epsilon–delta reasoning

Book pp. 38–40. [Find in compact glossary](#)

In this entry: limit of a function; epsilon–delta definition; punctured neighborhood; sequential criterion; left-hand limit; right-hand limit.

Definition and key ideas

The statement $\lim_{x \rightarrow a} f(x) = L$ means that for each $\varepsilon > 0$ some $\delta > 0$ makes $0 < |x - a| < \delta$ imply $|f(x) - L| < \varepsilon$ for domain points. The value at a is excluded and may be undefined. Equivalently, every sequence of domain points $x_n \neq a$ with $x_n \rightarrow a$ must satisfy $f(x_n) \rightarrow L$. Testing only one sequence is insufficient.

Left and right limits restrict to $x < a$ and $x > a$. At a two-sided accumulation point, the two-sided finite limit exists exactly when both one-sided limits exist and agree. Algebraic limit laws and squeezing apply. For floor functions and absolute values, determine their formula on each side before substituting; cancellation is legitimate on a punctured neighborhood even if the original expression is undefined at the limit point.

Problem 033.1

[See solution](#)

For $x \neq 0$, let $f(x) = (\sqrt{1 + |x|} - 1)/x$. What are its left and right limits at zero, respectively?

- (A) $(1/2, 1/2)$
- (B) $(-1/2, 1/2)$
- (C) $(-1, 1)$
- (D) $(0, 0)$
- (E) Neither one-sided limit exists.

Problem 033.2

[See solution](#)

Let $f(x) = [x] + [-x]$. Which statement holds at every integer m ?

- (A) $\lim_{x \rightarrow m} f(x) = f(m) = 0$.
- (B) The left limit is -1 and the right limit is 0 .
- (C) The two-sided limit fails to exist.
- (D) $\lim_{x \rightarrow m} f(x) = -1$, whereas $f(m) = 0$.
- (E) $\lim_{x \rightarrow m} f(x) = 1$.

Problem 033.3[See solution](#)

What is $\lim_{x \rightarrow 2} (x^3 - 8)/(x^2 - 4)$?

- (A) 6
- (B) The limit does not exist
- (C) 3
- (D) 2
- (E) 4

Problem 033.4[See solution](#)

For $0 < \varepsilon < 1$, which choice of δ guarantees $|x^2 - 4| < \varepsilon$ whenever $|x - 2| < \delta$?

- (A) $\delta = 2\varepsilon$ (B) $\delta = 1$ (C) $\delta = \sqrt{\varepsilon}$ (D) $\delta = \varepsilon/5$ (E) $\delta = \varepsilon$

Problem 033.5[See solution](#)

For a function defined on a punctured interval about a , consider: I. $f(x_n) \rightarrow L$ for every sequence $x_n \neq a$ tending to a ; II. this holds for one sequence from each side; III. this holds for every monotone sequence $x_n \neq a$ tending to a . Which conditions are each sufficient for $\lim_{x \rightarrow a} f(x) = L$?

- (A) I only (B) II only (C) I and III only (D) II and III only (E) I, II, and III

2. Calculus I

Limits of Functions as $x \rightarrow \pm\infty$ ▷ End behavior

034 Limits at infinity and asymptotes

Book pp. 40-41, 95. [Find in compact glossary](#)**In this entry:** limit at infinity; horizontal asymptote; vertical asymptote; slant asymptote; oblique asymptote.**Definition and key ideas**

A finite limit $f(x) \rightarrow L$ as $x \rightarrow +\infty$ or $-\infty$ gives a horizontal asymptote $y = L$ at that end. A vertical line $x = a$ is an asymptote if at least one one-sided limit is infinite. A slant asymptote $y = mx + b$ satisfies $f(x) - (mx + b) \rightarrow 0$ at the relevant infinite end. Polynomial division is especially effective for rational functions.

A zero of a denominator is not automatically a vertical asymptote: a common factor can produce a removable hole instead. For radicals, $\sqrt{x^2} = |x|$, so positive and negative infinity often produce different end behavior. Infinite limits describe growth rather than finite real limits; matching signs on two sides must be checked separately.

The condition $f(x) - (mx + b) \rightarrow 0$ does not imply $f'(x) \rightarrow m$: a small-amplitude term can have a rapidly oscillating derivative. Differentiate limits only when a theorem justifies doing so.

Problem 034.1[See solution](#)The graph of $f(x) = (x^2 + 3x + 1)/(x + 1)$ has which asymptotes?

- (A) $x = -1$ and $y = x + 2$ (B) $x = -1$ and $y = x + 3$
 (C) $x = 1$ and $y = x + 2$ (D) $y = 1$ only
 (E) $y = x + 2$ only

Problem 034.2[See solution](#)What is $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 4x + 5} + x)$?

- (A) $-\infty$ (B) -2 (C) 0 (D) 2 (E) $+\infty$

Problem 034.3[See solution](#)What are the horizontal asymptotes of $f(x) = x/\sqrt{x^2 + 1}$ at $+\infty$ and $-\infty$, respectively?

- (A) $y = 1$ and $y = 1$ (B) $y = 1$ and $y = -1$
 (C) $y = -1$ and $y = 1$ (D) $y = 0$ and $y = 0$
 (E) Neither exists.

Problem 034.4[See solution](#)

For $x > 0$, define $f(x) = x + \sin(x^2)/x$. Which statement is correct as $x \rightarrow \infty$?

- (A) $y = x$ is an asymptote and $f'(x) \rightarrow 1$.
- (B) $y = x$ is an asymptote but $f'(x)$ has no limit.
- (C) $y = 0$ is an asymptote.
- (D) No slant asymptote exists because f' oscillates.
- (E) $f(x) - x$ has no limit.

Problem 034.5[See solution](#)

For $f(x) = \frac{(x-1)^2}{(x-1)(x+2)}$ on its natural domain, which statement is correct?

- (A) Both $x = 1$ and $x = -2$ are vertical asymptotes.
- (B) Only $x = 1$ is a vertical asymptote.
- (C) Only $x = -2$ is a vertical asymptote; $x = 1$ is a removable hole.
- (D) Neither excluded point admits a finite or infinite one-sided limit.
- (E) The horizontal asymptote is $y = 0$.

Problem 035.4[See solution](#)

At how many real points is $f(x) = (x^2 - 1)/(x^2 - x)$, on its natural domain, discontinuous relative to that domain?

- (A) 1 (B) Infinitely many (C) 2 (D) 0 (E) 3

Problem 035.5[See solution](#)

Define $f(0) = 0$ and $f(x) = x \sin(1/x)$ otherwise. Which is true at zero?

- (A) Unbounded near zero
(B) Continuous but not differentiable
(C) Differentiable with derivative 1
(D) Differentiable with derivative 0
(E) A jump discontinuity occurs

Problem 035.6[See solution](#)

Define $f : [0, 1] \rightarrow \mathbb{R}$ by $f(0) = a$, $f(1) = b$, and $f(x) = x/(1 + \sqrt{x})$ for $0 < x < 1$. Which (a, b) makes f continuous on its entire domain?

- (A) $(0, 0)$
(B) $(0, 1/2)$
(C) $(1, 1/2)$
(D) $(0, 1)$
(E) No pair, because two-sided limits at the endpoints are unavailable.

2. Calculus I

Continuous Functions > A nontrivial continuity example

036 Rational–irrational constructions and Thomae’s function

Book pp. 43–44. [Find in compact glossary](#)

In this entry: Thomae function; rational–irrational function; dense set; continuity via bounded denominators; Dirichlet function.

Definition and key ideas

Thomae’s function on $(0, 1)$ is $T(x) = 0$ at irrational x , and $T(p/q) = 1/q$ for a reduced rational p/q with $q > 0$. It is continuous at every irrational point and discontinuous at every rational point. For a prescribed ε , only finitely many rationals in a bounded interval have denominator below $1/\varepsilon$; avoid those finitely many points to prove continuity at an irrational point.

Rational and irrational numbers are both dense, so a function defined by different formulas on those two sets can be continuous only where the two limiting behaviors agree. The Dirichlet function, equal to one on rationals and zero on irrationals, is nowhere continuous. Density and the sequential criterion replace ordinary graph intuition for these examples.

Problem 036.1

[See solution](#)

Define $g(x) = x^2$ for rational x and $g(x) = -x^2$ for irrational x . At which points is g differentiable?

- (A) No points
- (B) Exactly the rational points
- (C) Exactly the irrational points
- (D) Exactly $x = 0$
- (E) Every real point

Problem 036.2

[See solution](#)

For Thomae’s function T on $(0, 1)$, what is $\lim_{x \rightarrow 2/5} T(x)$?

- (A) 0
- (B) $1/5$
- (C) $2/5$
- (D) 1
- (E) The limit does not exist.

Problem 036.3[See solution](#)

Let $f(x) = x$ for rational x and $f(x) = 1 - x$ for irrational x . Where is f continuous?

- (A) Only at zero
- (B) Only at $x = 1/2$
- (C) Everywhere
- (D) At every rational point
- (E) Nowhere

Problem 036.4[See solution](#)

Let $D(x) = 1$ for rational x and $D(x) = 0$ for irrational x . Which function is continuous at every real number?

- (A) $D(x)$ (B) $xD(x)$ (C) $D(x)^2$ (D) $D(D(x))$ (E) $D(x) + x$

Problem 036.5[See solution](#)

For Thomae's function T on $[0, 1]$, with $T(p/q) = 1/q$ in lowest terms and $T = 0$ at irrationals, what is $\int_0^1 T(x) dx$?

- (A) $\pi^2/6$
- (B) 1
- (C) 0
- (D) The Riemann integral does not exist
- (E) $1/2$

2. Calculus I

Theorems Concerning Continuous Functions ▷ Extreme value theorem

037 Extreme value theorem and attainment

Book pp. 44. [Find in compact glossary](#)

In this entry: extreme value theorem; absolute maximum; absolute minimum; attainment; closed bounded interval.

Definition and key ideas

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous and $a \leq b$, then it attains both an absolute maximum and an absolute minimum. Thus there are $c, d \in [a, b]$ with $f(c) \leq f(x) \leq f(d)$ for all x in the interval. The hypotheses combine continuity with compactness: a closed bounded interval supplies both endpoints and prevents escape to infinity.

Attainment is stronger than boundedness. For example, a function may have finite supremum on an open interval without taking that value. The theorem locates no particular maximizer and guarantees no uniqueness. In calculations, examine endpoints as well as interior critical points; for a continuous function whose minimum is strictly positive, taking reciprocals preserves continuity and boundedness.

Problem 037.1

[See solution](#)

If f is continuous and strictly positive on $[0, 1]$, which statement must hold?

- (A) f is increasing.
- (B) $\inf_{[0,1]} f = 0$.
- (C) There is $M > 0$ such that $1/f(x) \leq M$ for every x .
- (D) f has exactly one minimum point.
- (E) f' is bounded.

Problem 037.2

[See solution](#)

Which continuous function has a finite supremum on its stated domain but no maximum?

- (A) $f(x) = x^2$ on $[-1, 1]$
- (B) $f(x) = x/(1+x)$ on $[0, \infty)$
- (C) $f(x) = 1/x$ on $(0, 1]$
- (D) $f(x) = \sin x$ on $\mathbb{R}$
- (E) $f(x) = e^{-x^2}$ on $\mathbb{R}$

Problem 037.3[See solution](#)

A continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x) \rightarrow \infty$ as $|x| \rightarrow \infty$. Which must hold?

- (A) f attains a global minimum
- (B) f has only finitely many critical points
- (C) f attains a global maximum
- (D) f is one-to-one
- (E) f has a zero

Problem 037.4[See solution](#)

What are the infimum and supremum of $x(1 - x)$ on $(0, 1)$?

- (A) Neither is finite (B) $0, 1/4$ (C) $-1, 1/4$ (D) $0, 1$ (E) $1/4, 1/4$

Problem 037.5[See solution](#)

Which hypothesis guarantees that a continuous real function on a nonempty subset $S \subset \mathbb{R}$ attains both extrema?

- (A) S is connected
- (B) S is bounded
- (C) S contains infinitely many points
- (D) S is closed
- (E) S is closed and bounded

2. Calculus I

Theorems Concerning Continuous Functions ▸ Bolzano and intermediate value theorems

038 Intermediate values, zeros, and fixed points

Book pp. 44-45. [Find in compact glossary](#)

In this entry: Bolzano theorem; intermediate value theorem; fixed-point theorem on an interval; sign change.

Definition and key ideas

If f is continuous on $[a, b]$, every value between $f(a)$ and $f(b)$ is attained. In particular, $f(a)f(b) < 0$ guarantees a zero in (a, b) ; this special case is often called Bolzano's theorem. Continuity is essential, and existence alone gives no uniqueness. A strictly monotone function can supply uniqueness independently.

The interval fixed-point theorem follows by applying the sign-change argument to $g(x) = f(x) - x$: every continuous map $f : [a, b] \rightarrow [a, b]$ has a point c with $f(c) = c$. Similar auxiliary functions turn equations such as $f(c) = \lambda c$ into zero problems. Do not confuse this elementary interval statement with higher-dimensional fixed-point theorems.

Having the intermediate value property does not by itself imply continuity. Continuity is a sufficient hypothesis for the theorem, not a necessary characterization of all functions whose images of intervals are intervals.

Problem 038.1

[See solution](#)

How many real roots does $x^3 + x - 1 = 0$ have?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinitely many

Problem 038.2

[See solution](#)

Every continuous $f : [0, 1] \rightarrow [0, 1]$ must satisfy which assertion?

- (A) $f(c) = c^2$ for some $c \in [0, 1]$
(B) $f(c) = 2$ for some $c \in [0, 1]$
(C) $f(c) = 0$ for some $c \in [0, 1]$
(D) $f(c) = 1$ for some $c \in [0, 1]$
(E) $f(c) = c$ for exactly one $c \in [0, 1]$

Problem 038.3

[See solution](#)

A continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + 1) = f(x) + 2$ for every x . Which must be its range?

- (A) A bounded interval (B) $[f(0), \infty)$ (C) $(-\infty, f(0)]$ (D) $\mathbb{R}$ (E) $\mathbb{R} \setminus \{0\}$

Problem 038.4[See solution](#)

Which function, with the stated value at zero, has the intermediate value property on every real interval but is not continuous at zero?

- (A) $f(x) = \sin(1/x)$ for $x \neq 0$, $f(0) = 0$
- (B) $f(x) = 1$ for $x > 0$, $f(x) = 0$ for $x \leq 0$
- (C) $f(x) = 1/x$ for $x \neq 0$, $f(0) = 0$
- (D) $f(x) = 1$ for rational x , $f(x) = 0$ otherwise
- (E) $f(x) = x \sin(1/x)$ for $x \neq 0$, $f(0) = 0$

Problem 038.5[See solution](#)

A continuous $f : [0, 1] \rightarrow \mathbb{R}$ satisfies $f(0) = f(1)$. Which must hold for some $c \in [0, 1/2]$?

- (A) $f(c) = f(c + 1/2)$
- (B) f is constant on $[c, c + 1/2]$
- (C) $f(c) = f(0)/2$
- (D) $f(c) = c$
- (E) $f(c) = 0$

2. Calculus I

The Derivative ▸ Definition and geometry

039 Derivative, differentiability, tangent, and normal

Book pp. 45-47, 49-50. [Find in compact glossary](#)

In this entry: derivative; difference quotient; secant line; tangent line; normal line; differentiability; corner; cusp; vertical tangent; Leibniz notation.

Definition and key ideas

The derivative is $f'(a) = \lim_{h \rightarrow 0} [f(a+h) - f(a)]/h$ when this finite limit exists. It gives instantaneous rate of change and tangent slope. The tangent line is $y - f(a) = f'(a)(x - a)$; for nonzero slope, the normal has slope $-1/f'(a)$. A horizontal tangent has a vertical normal. Secant slopes use finite increments; differentiation takes their limit.

Differentiability implies continuity, but continuity does not imply differentiability. Corners have unequal one-sided slopes; vertical tangents have unbounded slopes and do not give finite derivatives. A derivative may exist without being continuous. Prime notation, df/dx , and the operator d/dx describe the same first derivative. Piecewise differentiability requires matching both values and one-sided difference quotients.

Problem 039.1

[See solution](#)

Assume only that f is differentiable at a . What is $\lim_{h \rightarrow 0} \frac{f(a+5h^2) - f(a-2h^2)}{h^2}$?

- (A) $3f'(a)$
- (B) $5f'(a)$
- (C) $7f'(a)$
- (D) $10f'(a)$
- (E) The limit need not exist.

Problem 039.2

[See solution](#)

For $f(x) = x^3 - 3x$, what is the equation of the normal line at the positive x -intercept?

- | | |
|--------------------------------------|--------------------------------------|
| (A) $y = 6(x - \sqrt{3})$ | (B) $y = -\frac{1}{6}(x - \sqrt{3})$ |
| (C) $y = -\frac{1}{3}(x - \sqrt{3})$ | (D) $y = \frac{1}{6}(x + \sqrt{3})$ |
| (E) $x = \sqrt{3}$ | |

Problem 039.3[See solution](#)

Let $f(0) = 0$ and $f(x) = x^2 \sin(1/x)$ for $x \neq 0$. Which statement holds?

- (A) f is discontinuous at zero.
- (B) f is continuous but not differentiable at zero.
- (C) $f'(0) = 0$, but f' is not continuous at zero.
- (D) $f'(0) = 1$.
- (E) f' is continuous at zero.

Problem 039.4[See solution](#)

The tangent to $y = x^3$ at a positive point intersects the y -axis at -16 . What is its slope?

- (A) 8 (B) 6 (C) 16 (D) 24 (E) 12

Problem 039.5[See solution](#)

At which real points is $|x^2 - 1|$ not differentiable?

- (A) $x = -1, 0, 1$ only (B) Nowhere (C) $x = -1, 1$ only (D) $x = 0$ only (E) $x = 1$ only

Problem 039.6[See solution](#)

The curve $y = x^{1/3}$ passes through the origin. Which description is correct there?

- (A) Finite derivative 0; horizontal tangent
- (B) Finite derivative 1; tangent $y = x$
- (C) No finite derivative; vertical tangent $x = 0$ and horizontal normal $y = 0$
- (D) No tangent because the function is discontinuous
- (E) A corner with two different finite slopes

2. Calculus I

The Derivative ▸ Rules of differentiation

040 Differentiation rules and the chain rule

Book pp. 47-49. [Find in compact glossary](#)**In this entry:** sum rule; constant multiple rule; product rule; quotient rule; chain rule; composite derivative.**Definition and key ideas**

Differentiation is linear: $(af + bg)' = af' + bg'$. The product rule is $(fg)' = f'g + fg'$ and the quotient rule is $(f/g)' = (f'g - fg')/g^2$ where $g \neq 0$. The chain rule says $(f \circ g)'(x) = f'(g(x))g'(x)$, provided g is differentiable at x and f at $g(x)$.

For nested expressions, work from the outside inward and include one derivative factor for each layer. In Leibniz notation $dy/dx = (dy/du)(du/dx)$, but this is a theorem, not unrestricted fraction cancellation. Evaluate at a convenient point before expanding large products when possible. The derivative of a product is generally not the product of derivatives.

Problem 040.1[See solution](#)

If $f(1) = 2$ and $f'(1) = -3$, what is $\left. \frac{d}{dx} \frac{f(x^2)}{xf(x^3)} \right|_{x=1}$?

- (A) $-5/2$ (B) -1 (C) 0 (D) $1/2$ (E) $5/2$

Problem 040.2[See solution](#)

What is the derivative at $x = 0$ of $F(x) = e^{\sin x} \log(2 + \cos x)$?

- (A) 0 (B) $1/3$ (C) $\log 2$ (D) $\log 3$ (E) $1 + \log 3$

Problem 040.3[See solution](#)

A differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + y) = f(x)f(y)$ for all real x, y , with $f(0) = 1$ and $f'(0) = 3$. Which identity follows?

- (A) $f'(x) = 3$ (B) $f'(x) = 3f(x)$
 (C) $f'(x) = f(x)^3$ (D) $f'(x) = 3xf(x)$
 (E) $f'(x) = f(x)/3$

Problem 040.4[See solution](#)

If $f(1) = 2$, $f'(1) = 3$, $g(2) = 1$, and $g'(2) = 4$, what is $(f \circ g \circ f)'(1)$?

- (A) 48 (B) 36 (C) 24 (D) 12 (E) 72

Problem 040.5[See solution](#)

Let $u(x) = |x|$ and $v(x) = x|x|$. Which statement about the product $u(x)v(x)$ at zero is correct?

- (A) It is not differentiable because u is not.
- (B) It is differentiable with derivative 1.
- (C) It is differentiable with derivative 0, even though the product rule cannot be invoked using $u'(0)$.
- (D) It is discontinuous.
- (E) Its derivative has unequal one-sided limits at zero.

2. Calculus I

The Derivative ▸ Inverse-function rule

041 Derivative of an inverse function

Book pp. 48. [Find in compact glossary](#)**In this entry:** inverse-function derivative; local inverse; reciprocal slope.**Definition and key ideas**

If f is continuously differentiable near a and $f'(a) \neq 0$, it has a differentiable local inverse g near $b = f(a)$, with $g'(b) = 1/f'(a)$. More generally, the same derivative formula holds for a continuous local inverse when f is differentiable at a with nonzero derivative. The input to g' is the output b of f , not a .

Differentiate $f(g(y)) = y$ to derive the formula. A second differentiation gives $g''(b) = -f''(a)/(f'(a))^3$ when the required second derivatives exist. A globally invertible function can have zero derivative, but its inverse need not have a finite derivative at the corresponding point; $f(x) = x^3$ illustrates this distinction.

Problem 041.1[See solution](#)

Let g be the inverse of $f(x) = x^3 + x$. What is $g'(2)$?

- (A) $1/13$ (B) $1/4$ (C) $1/2$ (D) 2 (E) 4

Problem 041.2[See solution](#)

Let g be the inverse of $f(x) = x + e^x$. What is $g''(1)$?

- (A) $-1/8$ (B) $-1/4$ (C) 0 (D) $1/4$ (E) $1/8$

Problem 041.3[See solution](#)

Let g be the inverse of $f(x) = x^3$. What is $\lim_{h \rightarrow 0} [g(h) - g(0)]/h$ in the extended real sense?

- (A) 0
(B) 1
(C) $-\infty$
(D) $+\infty$
(E) The two one-sided limits have opposite signs.

Problem 041.4[See solution](#)

If $f(2) = 7$, $f'(2) = 3$, and g is a differentiable local inverse, what is $[g(y)^2]'$ at $y = 7$?

- (A) $3/4$ (B) 4 (C) $2/3$ (D) 12 (E) $4/3$

Problem 041.5[See solution](#)

For $f(x) = x^3 - 3x$, let g_1, g_2 be local inverses near output zero with $g_1(0) = 0$ and $g_2(0) = \sqrt{3}$. What is $g_2'(0) - g_1'(0)$?

- (A) $-1/2$ (B) $-1/6$ (C) $1/6$ (D) $1/3$ (E) $1/2$

2. Calculus I

The Derivative ▸ Table of differentials

042 Elementary derivatives and differentials

Book pp. 48-49, 97. [Find in compact glossary](#)

In this entry: power rule; exponential derivative; logarithmic derivative; trigonometric derivatives; inverse trigonometric derivatives; differential.

Definition and key ideas

For differentiable u , $d(u^k) = ku^{k-1}du$, $d(e^u) = e^u du$, $d(a^u) = a^u \log(a)du$ for $a > 0$, and $d(\log|u|) = du/u$ where $u \neq 0$. The usual power formula treats k as constant. Also $d \sin u = \cos u du$, $d \cos u = -\sin u du$, $d \tan u = \sec^2 u du$, $d \cot u = -\csc^2 u du$, $d \sec u = \sec u \tan u du$, and $d \csc u = -\csc u \cot u du$.

Inverse derivatives are $d \arcsin u = du/\sqrt{1-u^2}$, $d \arccos u = -du/\sqrt{1-u^2}$, and $d \arctan u = du/(1+u^2)$ on their differentiability domains. The differential $df = f'(x)dx$ records the linear response to an increment. Keep real domains and radians in mind: inverse sine and cosine derivatives fail at arguments ± 1 , and logarithms require the stated domain.

For $a > 0$, $a \neq 1$, and $u > 0$, $d(\log_a u) = du/(u \log a)$. Changing the logarithm base contributes the constant factor $1/\log a$.

Problem 042.1

[See solution](#)

What is $\left. \frac{d}{dx} \arcsin(\log x) \right|_{x=\sqrt{e}}$?

- (A) $1/\sqrt{e}$ (B) $2/\sqrt{3e}$ (C) $1/\sqrt{3e}$ (D) $2/\sqrt{e}$ (E) $\sqrt{3}/(2\sqrt{e})$

Problem 042.2

[See solution](#)

On $(0, \pi/2)$ let $F(x) = \log(\sec x + \tan x) - \log(\csc x - \cot x)$. What is $F'(\pi/4)$?

- (A) $-2\sqrt{2}$ (B) $-\sqrt{2}$ (C) 0 (D) $\sqrt{2}$ (E) $2\sqrt{2}$

Problem 042.3

[See solution](#)

What is $\left. \frac{d}{dx} \arctan \sqrt{x} \right|_{x=3}$?

- (A) $1/8$ (B) $1/(8\sqrt{3})$ (C) $\sqrt{3}/8$ (D) $1/(4\sqrt{3})$ (E) $1/6$

Problem 042.4

[See solution](#)

What is $\left. \frac{d}{dx} \log(\sin(x^2)) \right|_{x=\sqrt{\pi}/2}$?

- (A) $2\sqrt{\pi}$ (B) π (C) $\sqrt{\pi}/2$ (D) $\sqrt{\pi}$ (E) 1

Problem 042.5[See solution](#)

For $x > 1$, the derivative of $\log(x + \sqrt{x^2 - 1})$ is

- (A) $x/\sqrt{x^2 - 1}$ (B) $1/x$ (C) $1/(x^2 - 1)$ (D) $\sqrt{x^2 - 1}/x$ (E) $1/\sqrt{x^2 - 1}$

Problem 042.6[See solution](#)

If $y = 2^{\log_3 x}$ for $x > 0$, what is the differential dy at $x = 27$?

- (A) $\frac{8 \log 2}{27 \log 3} dx$ (B) $\frac{8}{27} dx$ (C) $\frac{8 \log 3}{27 \log 2} dx$ (D) $8 \log 2 dx$ (E) $\frac{\log 2}{27} dx$

Problem 042.7[See solution](#)

What is the derivative of $\arccos x + \log |x|$ at $x = -1/2$?

- (A) $2 - 2/\sqrt{3}$ (B) $-2 - 2/\sqrt{3}$ (C) $-2 + 2/\sqrt{3}$ (D) $2 + 2/\sqrt{3}$ (E) It is undefined.

2. Calculus I

Linear Approximations Using Differentials ▸ Local approximation

043 Linearization, differentials, and approximation error

Book pp. 50. [Find in compact glossary](#)

In this entry: linear approximation; linearization; differential approximation; relative error; tangent approximation.

Definition and key ideas

If f is differentiable at a , its linearization is $L(x) = f(a) + f'(a)(x - a)$ and $f(a + h) = f(a) + f'(a)h + o(h)$. Thus $\Delta f \approx df = f'(a)\Delta x$ for small changes. A nonzero derivative is not required; a zero derivative simply gives a constant first-order approximation.

Relative error is approximately $\Delta f/f \approx (f'(a)/f(a))\Delta x$ when $f(a) \neq 0$. For powers $y = x^k$, the fractional output change is approximately k times the fractional input change. Concavity predicts the direction of error: a convex graph lies above its tangent, while a concave graph lies below. A local approximation is not an exact equality, and a quantified guarantee needs a remainder bound.

If $|f''(x)| \leq M$ on the segment between a and $a + h$, Taylor's theorem gives $|f(a + h) - f(a) - f'(a)h| \leq Mh^2/2$. This turns a tangent approximation into a numerical error guarantee.

Problem 043.1

[See solution](#)

A positive quantity x is measured 0.4% too high. Using differentials, by approximately what percentage is $y = x^{-3/2}$ in error?

- (A) 0.6% too high
- (B) 0.4% too high
- (C) 0.6% too low
- (D) 0.4% too low
- (E) 1.5% too low

Problem 043.2

[See solution](#)

Using the tangent line at $x = 16$ to approximate $\sqrt{16.8}$ gives which value and direction of error?

- (A) 4.1, an overestimate
- (B) 4.1, an underestimate
- (C) 4.2, an overestimate
- (D) 4.2, an underestimate
- (E) 4.05, an exact value

Problem 043.3[See solution](#)

The linearization of $f(x) = \cos x$ at zero predicts no change when x changes from 0 to h . What is $\lim_{h \rightarrow 0} [f(h) - f(0)]/h^2$?

- (A) -1 (B) $-1/2$ (C) 0 (D) $1/2$ (E) 1

Problem 043.4[See solution](#)

For $f(x) = \log x$, let L be the linearization at $x = 1$. Which bound holds for $|f(1.1) - L(1.1)|$?

- (A) At most $1/200$ (B) Greater than $1/20$
(C) At least $1/10$ (D) Exactly $1/100$
(E) Exactly zero

Problem 043.5[See solution](#)

The tangent approximation to x^{-1} at $x = 2$ is evaluated at $2 + h$. What is its error, actual value minus approximation?

- (A) $-h^2/[4(2 + h)]$ (B) 0 (C) $h^2/(2 + h)$ (D) $h/[2(2 + h)]$ (E) $h^2/[4(2 + h)]$

2. Calculus I

Implicit Differentiation ▸ Local implicit functions

044 Implicit differentiation and the implicit function theorem

Book pp. 51. [Find in compact glossary](#)

In this entry: implicit equation; implicit differentiation; implicit function theorem; dependent variable; independent variable.

Definition and key ideas

An equation $F(x, y) = c$ may determine y locally as a function of x without giving an explicit formula. Differentiate while treating $y = y(x)$; thus $d(y^k)/dx = ky^{k-1}y'$. If F has continuous first partial derivatives near (a, b) , $F(a, b) = c$, and $F_y(a, b) \neq 0$, the implicit function theorem gives a unique local continuously differentiable branch through (a, b) and $y' = -F_x/F_y$.

The conclusion is local; an entire curve may have several branches. A vanishing F_y means this sufficient theorem does not apply, not automatically that no function exists. If $F_x \neq 0$ instead, solve locally for x as a function of y . For second derivatives, differentiate the first derivative relation and include every product and chain-rule term.

Problem 044.1

[See solution](#)

On the branch of $x^2 + xy + y^2 = 3$ through $(1, 1)$, what is y'' at that point?

- (A) -2 (B) $-2/3$ (C) 0 (D) $2/3$ (E) 2

Problem 044.2

[See solution](#)

Near which point of $x^2 + y^2 = 1$ does the implicit function theorem directly give x as a differentiable function of y but not y as a differentiable function of x ?

- (A) $(0, 1)$ (B) $(0, -1)$ (C) $(1, 0)$ (D) $(1/\sqrt{2}, 1/\sqrt{2})$ (E) At no point

Problem 044.3

[See solution](#)

At $(0, 0)$, the equation $F(x, y) = y^3 - x^3 = 0$ has $F_x = F_y = 0$. Which conclusion is correct?

- (A) No continuous local function $y(x)$ exists.
 (B) Exactly three real local branches exist.
 (C) The equation has the unique real branch $y = x$, even though the usual nonzero-partial-derivative hypothesis fails.
 (D) The only real solution is $(0, 0)$.
 (E) The branch must have a vertical tangent.

Problem 044.4

[See solution](#)

For $e^{xy} + y = 2$ near $(0, 1)$, what is dy/dx ?

- (A) 1 (B) -1 (C) $1/2$ (D) 0 (E) -2

Problem 044.5[See solution](#)

How many nonzero points on $x^3 + y^3 = 6xy$ have a horizontal tangent with the curve locally expressible as a differentiable function $y(x)$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

2. Calculus I

Higher-Order Derivatives ▸ Repeated differentiation

045 Higher-order derivatives

Book pp. 51-52. [Find in compact glossary](#)**In this entry:** second derivative; higher-order derivative; nth derivative; Leibniz product formula.**Definition and key ideas**

The second derivative f'' is the derivative of f' ; repeated differentiation gives $f^{(n)}$, also denoted $d^n f/dx^n$. Existence at one order does not automatically imply existence at the next. Higher derivatives describe changes in slope, concavity, local polynomial coefficients, and approximation error.

For repeated products, the Leibniz formula is $(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$. For $(a - bx)^{-1}$, the pattern is $n!b^n(a - bx)^{-n-1}$. Exponentials reproduce themselves, trigonometric derivatives cycle every four differentiations, and a polynomial vanishes after more differentiations than its degree. Recognize these patterns rather than computing long derivative lists.

Problem 045.1[See solution](#)If $f(x) = 1/(2 - 3x)$ and $n \geq 1$, what is $f^{(n)}(0)$?

- (A) $n!3^n/2^n$ (B) $n!3^n/2^{n+1}$ (C) $n!(-3)^n/2^{n+1}$ (D) $3^n/2^{n+1}$ (E) $n!2^n/3^{n+1}$

Problem 045.2[See solution](#)For $f(x) = x^2e^x$, what is $f^{(101)}(0)$?

- (A) 101 (B) 202 (C) 5050 (D) 10100 (E) 101!

Problem 045.3[See solution](#)Let $f(x) = x|x|$. What is the highest order of derivative of f that exists at zero?

- (A) No first derivative exists.
 (B) First order only
 (C) Second order only
 (D) Third order only
 (E) Derivatives of all orders exist.

Problem 045.4[See solution](#)What is the 101st derivative of $\sin x$ at $x = \pi/3$?

- (A) $1/2$ (B) $\sqrt{3}/2$ (C) 0 (D) $-1/2$ (E) $-\sqrt{3}/2$

Problem 045.5[See solution](#)For $f(x) = \log(1 + x)$ and $n \geq 1$, what is $f^{(n)}(0)$?

- (A) $(-1)^n n!$ (B) $(-1)^{n-1}/n!$ (C) $(-1)^{n-1}(n-1)!$ (D) $(-1)^{n-1}n!$ (E) $1/n$

2. Calculus I

Curve Sketching ▸ Properties of the first derivative

046 Critical points, stationary points, and monotonicity

Book pp. 52-54. [Find in compact glossary](#)

In this entry: critical point; stationary point; increasing function; decreasing function; first-derivative test.

Definition and key ideas

An interior domain point is critical if its derivative is zero or does not exist; it is stationary when its derivative is zero. A positive derivative throughout an interval implies strict increase there, and a negative derivative implies strict decrease. These are interval conclusions, not claims about global behavior from a derivative value at a single point.

The first-derivative test classifies a critical point by a sign change: positive to negative gives a local maximum; negative to positive gives a local minimum; no sign change need give neither. Include nondifferentiable domain points, but do not count excluded points such as poles as critical points of the function. A repeated factor of even multiplicity in f' usually does not change its sign; odd multiplicity does.

Problem 046.1

[See solution](#)

A differentiable function satisfies $f'(x) = x^2(x - 1)^3(x + 2)$. At which points does f have local extrema?

- (A) $x = 0$ only
- (B) $x = -2, 0$ only
- (C) $x = 0, 1$ only
- (D) $x = -2, 1$ only
- (E) $x = -2, 0, 1$

Problem 046.2

[See solution](#)

For $f(x) = |x|^{2/3}(x - 2)$, which statement about $x = 0$ is correct?

- (A) It is a differentiable local minimum.
- (B) It is a differentiable local maximum.
- (C) It is a nondifferentiable local maximum.
- (D) It is a nondifferentiable local minimum.
- (E) It is not a critical point.

Problem 046.3[See solution](#)

A differentiable $f : \mathbb{R} \rightarrow \mathbb{R}$ has $f'(x) \geq 0$ everywhere, and $f'(x) = 0$ at only countably many points. Which must hold?

- (A) $f'(x) > 0$ everywhere.
- (B) f is strictly increasing.
- (C) f is constant on some interval.
- (D) f has a strict local maximum.
- (E) f is bounded.

Problem 046.4[See solution](#)

How many stationary points does $f(x) = x + 1/x$ have on its natural real domain?

- (A) 1 (B) 3 (C) 4 (D) 2 (E) 0

Problem 046.5[See solution](#)

For $f(x) = x^3 - 3ax$, with $a > 0$, what is the difference between its local maximum and local minimum values?

- (A) $2a^{3/2}$ (B) $6a^{3/2}$ (C) $4a^{3/2}$ (D) $4a$ (E) $2\sqrt{a}$

2. Calculus I

Curve Sketching ▸ Properties of the second derivative

047 Concavity, convexity, and inflection points

Book pp. 52-53. [Find in compact glossary](#)**In this entry:** concave up; convex function; concave down; inflection point; second-derivative sign.**Definition and key ideas**

For twice differentiable functions, $f'' \geq 0$ on an interval implies convexity, also called concavity up; slopes are nondecreasing and the graph lies above each tangent. The opposite inequalities describe concavity down. Strict positivity or negativity provides strict curvature, but a zero second derivative at one point alone supplies little information.

An inflection point is a point on the graph where concavity changes across the point. Possible candidates satisfy $f'' = 0$ or failure of the second derivative, but these conditions are not sufficient. Verify the change of concavity. An inflection point need not be stationary, and a stationary point need not be an extremum. These distinctions are frequent conceptual traps.

Without assuming differentiability, convexity means $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$ for $0 \leq t \leq 1$. Thus the graph lies below its chords. Secant slopes on successive intervals are nondecreasing, which gives both upper and lower bounds from known function values.

Problem 047.1[See solution](#)

What is the slope of the tangent at the nonzero inflection point of $f(x) = x^4 - 4x^3$?

- (A) -32 (B) -16 (C) 0 (D) 16 (E) 32

Problem 047.2[See solution](#)

Which function has an inflection point at zero although its second derivative is undefined there?

- (A) $|x|$ (B) x^4 (C) $|x|^{2/3}$ (D) $x^{1/3}$ (E) $-x^2$

Problem 047.3[See solution](#)

On which interval is $f(x) = xe^{-x}$ both increasing and concave down?

- (A) $(1, 2)$ (B) $(-\infty, 1)$ (C) $(-\infty, 2)$ (D) $(2, \infty)$ (E) $(0, 2)$

Problem 047.4[See solution](#)

A convex function $f : [-1, 1] \rightarrow \mathbb{R}$ satisfies $f(-1) = 2$, $f(0) = 0$, and $f(1) = 2$. What is the sharp interval of possible values of $f(1/2)$?

- (A) $[0, 1]$ (B) $[-1, 1]$ (C) $[-2, 2]$ (D) $[0, 2]$ (E) $[-1, 2]$

Problem 047.5[See solution](#)

A twice differentiable f satisfies $f'' > 0$ on (a, b) . For $a < x < y < b$, which must hold?

- (A) $f'(x) > f'(y)$
- (B) The secant slope equals $f'(x)$
- (C) $f'(x) < [f(y) - f(x)]/(y - x) < f'(y)$
- (D) $f(y) > f(x)$
- (E) $f(x) > 0$

2. Calculus I

Curve Sketching ▸ Classifying critical points

048 Second-derivative and higher-derivative tests

Book pp. 54. [Find in compact glossary](#)

In this entry: second-derivative test; higher-derivative test; degenerate critical point; local maximum; local minimum.

Definition and key ideas

At a stationary point a , $f''(a) > 0$ implies a strict local minimum and $f''(a) < 0$ a strict local maximum, assuming the usual local differentiability needed by the test. If the second derivative is zero, the test is inconclusive. The functions x^4 , $-x^4$, and x^3 illustrate minimum, maximum, and neither despite the same first two derivatives at zero.

More generally, suppose f is sufficiently differentiable near a , its derivatives of orders 1 through $m - 1$ vanish there, and $f^{(m)}(a) \neq 0$. The leading local change is $f^{(m)}(a)(x - a)^m/m!$. Even m gives a minimum for positive coefficient and maximum for negative coefficient; odd m gives neither. If every derivative vanishes, this finite-order test says nothing.

Problem 048.1

[See solution](#)

Suppose f is six times continuously differentiable near zero, $f^{(j)}(0) = 0$ for $1 \leq j \leq 5$, and $f^{(6)}(0) = -12$. Which conclusion follows?

- (A) Zero is a strict local minimum.
- (B) Zero is a strict local maximum.
- (C) Zero is an inflection point.
- (D) f is constant near zero.
- (E) No conclusion about extrema is possible.

Problem 048.2

[See solution](#)

Which function has neither a local maximum nor a local minimum at zero?

- (A) $e^{x^2} - 1 - x^2$ (B) $\cos x - 1 + x^2/2$ (C) $\sin x - x$ (D) $x^4 + x^6$ (E) $-x^6$

Problem 048.3

[See solution](#)

A smooth real function satisfies $f(0) = 0$ and $f^{(n)}(0) = 0$ for every positive integer n . What can be concluded about an extremum at zero?

- (A) It is a strict minimum.
- (B) It is a strict maximum.
- (C) It is neither a maximum nor a minimum.
- (D) The function is constant near zero.
- (E) The data do not determine any of the preceding conclusions.

Problem 048.4[See solution](#)

Let $f(0) = 0$ and $f(x) = x^4(2 + \sin(1/x))$ for $x \neq 0$. Which holds at zero?

- (A) A strict local maximum
- (B) A strict local minimum, with $f''(0) = 0$
- (C) Neither extremum, because the sine oscillates
- (D) A strict local minimum, with $f''(0) > 0$
- (E) No first derivative

Problem 048.5[See solution](#)

For which real a is zero a strict local minimum of $f(x) = ax^4 + x^6$?

- (A) $a = 0$ only (B) $a > 0$ (C) $a < 0$ (D) Every real a (E) $a \geq 0$

2. Calculus I

Theorems Concerning Differentiable Functions ▸ Rolle's theorem

049 Rolle's theorem and repeated zeros

Book pp. 54. [Find in compact glossary](#)**In this entry:** Rolle theorem; horizontal tangent; zero counting.**Definition and key ideas**

If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$ with $a < b$, then some $c \in (a, b)$ satisfies $f'(c) = 0$. Endpoint differentiability is unnecessary, but continuity at endpoints and differentiability throughout the open interval are essential. A corner can invalidate the theorem.

Apply Rolle's theorem between successive distinct zeros: m distinct zeros of f force at least $m - 1$ distinct zeros of f' . Repetition bounds root counts by higher derivatives. This is often more efficient than solving an equation. The theorem proves existence, not the exact number or location of stationary points.

Problem 049.1[See solution](#)

A function f is three times differentiable on $\mathbb{R}$ and $f^{(3)}(x) > 0$ for every x . What is the largest possible number of distinct real zeros of f ?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) No finite upper bound

Problem 049.2[See solution](#)

Which hypothesis of Rolle's theorem fails for $f(x) = |x|$ on $[-1, 1]$?

- (A) Continuity at -1
(B) Continuity at 1
(C) Equality of endpoint values
(D) Differentiability at every point of $(-1, 1)$
(E) All hypotheses hold.

Problem 049.3[See solution](#)

For $f(x) = x^{2/3}(1 - x)$ on $[0, 1]$, which point is guaranteed by Rolle's theorem and satisfies $f'(c) = 0$?

- (A) $c = 1/5$
(B) $c = 1/3$
(C) $c = 2/5$
(D) $c = 1/2$
(E) No point, since $f'(0)$ is undefined.

2. Calculus I

Theorems Concerning Differentiable Functions ▷ Mean value theorem

050 Mean value theorem and derivative bounds

Book pp. 55. [Find in compact glossary](#)

In this entry: mean value theorem; secant slope; constant derivative theorem; Lipschitz bound; monotonicity from derivative.

Definition and key ideas

If f is continuous on $[a, b]$ and differentiable on (a, b) , some $c \in (a, b)$ satisfies $f'(c) = [f(b) - f(a)]/(b - a)$. Thus an instantaneous rate equals the average rate on the interval. The theorem implies that a function with zero derivative on an interval is constant, and that positive derivative throughout an interval implies strict increase.

If $m \leq f' \leq M$, then $m(b - a) \leq f(b) - f(a) \leq M(b - a)$. In particular, $|f'| \leq K$ gives the Lipschitz estimate $|f(x) - f(y)| \leq K|x - y|$. Connectedness matters: zero derivative on two disjoint intervals need not give the same constant on both. The derivative need not be continuous for the mean value theorem itself.

Problem 050.1

[See solution](#)

If f is differentiable on $[0, 3]$, $f(0) = 2$, and $1 \leq f'(x) \leq 4$ for $0 < x < 3$, which interval must contain $f(3)$?

- (A) $[1, 4]$ (B) $[2, 12]$ (C) $[3, 12]$ (D) $[5, 14]$ (E) $[6, 24]$

Problem 050.2

[See solution](#)

Let f be differentiable on $\mathbb{R}$ with $|f'(x)| \leq 1/3$ for all x . How many fixed points can f have?

- (A) Exactly zero
(B) At most one, and one is not guaranteed
(C) Exactly one
(D) Exactly two
(E) Infinitely many are possible

Problem 050.3

[See solution](#)

If $f(0) = 0$ and $2 \leq f'(x) \leq 5$ on $(0, 1)$, which is the sharp guaranteed interval for $\int_0^1 f(x) dx$? Assume f is continuously differentiable.

- (A) $[1/2, 5]$ (B) $[0, 5]$ (C) $[2, 5/2]$ (D) $[2, 5]$ (E) $[1, 5/2]$

Problem 050.4[See solution](#)

A differentiable function on $\mathbb{R} \setminus \{0\}$ satisfies $f'(x) = 0$ everywhere in its domain and $f(-1) = 3$. Which are the possible values of $f(1)$?

- (A) 3 only
- (B) -3 only
- (C) 0 only
- (D) All positive real numbers, and no others
- (E) All real numbers

Problem 050.5[See solution](#)

Let f be continuous on $[0, 1]$ and differentiable on $(0, 1)$. Which equality must hold for some $c \in (0, 1)$?

- (A) $f'(c) = c[f(1) - f(0)]$
- (B) $f'(c) = 2c[f(1) - f(0)]$
- (C) $f'(c) = 2[f(1) - f(0)]$
- (D) $f'(c) = f(c)$
- (E) $f'(c) = f(1) + f(0)$

2. Calculus I

Max/Min Problems ▷ Finding extrema

051 Optimization and constrained extrema

Book pp. 55–57. [Find in compact glossary](#)

In this entry: optimization; max/min problem; local extremum; absolute extremum; endpoint test; constraint reduction.

Definition and key ideas

A local extremum compares nearby domain points, whereas an absolute extremum compares the entire domain. At an interior local extremum where the derivative exists, it must vanish. On a closed bounded interval, a continuous function attains extrema, so compare values at all interior critical points and both endpoints. A zero derivative is a candidate, not a conclusion.

Translate a word problem into an objective function and its actual feasible domain. Eliminate one variable using the constraint, differentiate, and check that the candidate is admissible. On open or unbounded domains, also investigate limiting behavior at excluded boundaries; a supremum may not be attained. A second-derivative test gives local classification but does not alone establish global optimality.

Problem 051.1

[See solution](#)

For nonnegative x, y with $x + 2y = 12$, what is the maximum of x^2y ?

- (A) 64 (B) 96 (C) 128 (D) 192 (E) 256

Problem 051.2

[See solution](#)

For $a, b > 0$, what is the maximum of $x^a(1 - x)^b$ on $[0, 1]$?

- (A) $a^ab^b/(a+b)^{a+b}$ (B) $(a+b)^{a+b}/(a^ab^b)$ (C) $a^a/(a+b)^a$ (D) $b^b/(a+b)^b$ (E) 1

Problem 051.3

[See solution](#)

A rectangle with sides parallel to the axes is inscribed in $x^2/9 + y^2/4 = 1$. What is its largest area?

- (A) 12 (B) 6 (C) 9 (D) 24 (E) 18

Problem 051.4

[See solution](#)

Among positive x, y satisfying $xy = 16$, what is the minimum of $x + 4y$?

- (A) 8 (B) 20 (C) 12 (D) 16 (E) 32

Problem 051.5

[See solution](#)

A right circular cylinder is inscribed in a sphere of radius 3. What height maximizes its volume?

- (A) $3\sqrt{2}$ (B) $\sqrt{3}$ (C) 3 (D) 6 (E) $2\sqrt{3}$

Problem 051.6[See solution](#)

At which points does $f(x) = x^3 - 3x$ attain its absolute minimum on $[-2, 3/2]$?

- (A) $x = -2$ only (B) $x = -1$ only
(C) $x = 1$ only (D) $x = -2$ and $x = 1$
(E) $x = -1$ and $x = 3/2$

2. Calculus I

Related Rates ▸ Differentiating a geometric constraint

052 Related rates and time differentiation

Book pp. 57-58. [Find in compact glossary](#)**In this entry:** related rates; instantaneous rate; time derivative; rate of change.

Definition and key ideas

Related-rates problems connect time-dependent quantities through a geometric or algebraic constraint. Differentiate the constraint with respect to time before inserting the particular instantaneous values. For example, $x^2 + y^2 = L^2$ with fixed L gives $x \, dx/dt + y \, dy/dt = 0$. If a supposedly fixed quantity actually varies, its derivative must also be included.

Track units and signs: a decreasing length has negative derivative, whereas its speed of decrease is the positive magnitude. Similar triangles may provide a relation among hidden variables, such as radius and height in a cone. Rates can become large near singular configurations, so a numerical answer should also be checked against the geometry.

Problem 052.1

[See solution](#)

Water fills a cone whose height is twice its radius at every water level. If the water volume increases at 3π cubic units per minute, how fast does the water height rise when it is 3?

- (A) $1/3$ (B) $2/3$ (C) 1 (D) $4/3$ (E) 3

Problem 052.2

[See solution](#)

A 13-unit ladder has its base moving horizontally away from a wall at 2 units per second. When the base is 5 units from the wall, what is the rate of change of the area of the triangle enclosed by the ladder, wall, and floor?

- (A) $-119/12$ (B) $-5/6$ (C) $5/6$ (D) $119/12$ (E) 12

Problem 052.3

[See solution](#)

A point moves on $y = x^2$ with $dx/dt = 3$. How fast is its distance from the origin changing when $x = 1$?

- (A) 9 (B) $6/\sqrt{2}$ (C) $3\sqrt{5}$ (D) $9/\sqrt{2}$ (E) $3/\sqrt{2}$

Problem 052.4

[See solution](#)

A sphere's volume grows at 12π units per second. At radius 2, how fast does its surface area grow?

- (A) 24π (B) 48π (C) 3π (D) 12π (E) 6π

Problem 052.5[See solution](#)

A searchlight is 30 meters from a straight wall. Its beam makes angle θ with the perpendicular to the wall and rotates at $d\theta/dt = 1/10$ radian per second. How fast does the spot move along the wall when it is 40 meters from the nearest point on the wall?

- (A) 3 m/s (B) 4 m/s (C) 5 m/s (D) $25/3$ m/s (E) $40/3$ m/s

Problem 052.6[See solution](#)

A gas obeys $PV = kT$ with fixed $k > 0$. At an instant, pressure increases at 3% per second and absolute temperature decreases at 1% per second. What is the instantaneous relative rate V'/V ?

- | | |
|-----------------------|-----------------------|
| (A) -4% per second | (B) -2% per second |
| (C) 2% per second | (D) 3% per second |
| (E) 4% per second | |

2. Calculus I

Indefinite Integration (Antidifferentiation) ▸ Definitions and elementary formulas

053 Antiderivatives and indefinite integration

Book pp. 59-60. [Find in compact glossary](#)**In this entry:** antiderivative; indefinite integral; integrand; constant of integration; variable of integration; power integration rule.**Definition and key ideas**

An antiderivative of f on an interval is a differentiable function F with $F' = f$. The notation $\int f(x) dx = F(x) + C$ describes all antiderivatives there; any two differ by a constant by the mean value theorem. The integrand is $f(x)$, and dx identifies the variable. On disconnected domains, separate components may have different constants.

Reverse known derivative formulas: $\int x^p dx = x^{p+1}/(p+1) + C$ for $p \neq -1$ on the appropriate real domain, while $\int dx/x = \log|x| + C$. Integrals are linear. Initial values determine constants, and differentiation checks candidate answers. The formula $\int a^x dx = a^x/\log a + C$ assumes $a > 0$ and $a \neq 1$; $a = 1$ is a separate constant-integrand case.

Riemann integrability alone does not guarantee an antiderivative on the entire interval. A jump in the integrand can prevent any differentiable primitive from existing across that point, even though a definite integral is well defined.

Problem 053.1[See solution](#)

On $\mathbb{R} \setminus \{0\}$, which is the most general antiderivative of $1/x$?

- (A) $\log x + C$
- (B) $\log|x| + C$ with one shared constant required
- (C) $\log|x| + C_+$ for $x > 0$ and $\log|x| + C_-$ for $x < 0$, with independent constants
- (D) $1/x^2 + C$
- (E) No antiderivative exists.

Problem 053.2[See solution](#)

If $F'(x) = 3x^2 - 2/x$ for $x > 0$ and $F(1) = 4$, what is $F(e)$?

- (A) $e^3 - 2$ (B) $e^3 + 1$ (C) $e^3 + 3$ (D) $e^3 + 5$ (E) $3e^2 - 2/e$

Problem 053.3[See solution](#)

If $F' = f$ on an interval and $G' = 2f + 3$, which expression is constant?

- (A) $G - F - 3x$ (B) $G - 2F - 3x$ (C) $G - 2F + 3x$ (D) $G - 2F$ (E) $2G - F$

Problem 053.4[See solution](#)

For $a > 0$, let F_a be the antiderivative of a^x with $F_a(0) = 0$. Which formula is valid for every $a > 0$?

- (A) $F_a(x) = a^x / \log a$
- (B) $F_a(x) = (a^x - 1) / \log a$ for $a \neq 1$, and $F_1(x) = x$
- (C) $F_a(x) = (a^x - 1) / a$
- (D) $F_a(x) = (a^x - 1) / \log a$ for $a \neq 1$, and $F_1(x) = 0$
- (E) $F_a(x) = xa^x$

Problem 053.5[See solution](#)

Let $f(x) = 0$ for $x < 0$ and $f(x) = 1$ for $x \geq 0$. Does f have an antiderivative on $(-1, 1)$?

- (A) Yes: $\max\{x, 0\}$.
- (B) Yes: $|x|$.
- (C) Yes, after choosing different integration constants on the two sides.
- (D) No, because differentiability at zero cannot be achieved.
- (E) No, because f is unbounded.

2. Calculus I

Techniques of Integration ▸ Integration by Substitution

054 Integration by substitution

Book pp. 59-61. [Find in compact glossary](#)**In this entry:** u-substitution; change of variable; reverse chain rule; transformed limits.**Definition and key ideas**

Substitution reverses the chain rule: $\int f(g(x))g'(x) dx = F(g(x)) + C$ when $F' = f$. Set $u = g(x)$ and $du = g'(x)dx$; rewrite the entire integrand in the new variable. For definite integrals, use corresponding limits $u = g(a)$ and $u = g(b)$, or first return to the original variable before applying the original limits.

Look for an inner expression whose derivative occurs elsewhere in the integrand. Completing a square, factoring, or a trigonometric identity may reveal the pattern. Do not leave a mixture of x and u without an explicit relation. A substitution that is not one-to-one can still implement the chain-rule identity when used correctly; geometric changes of variable may require splitting branches.

Problem 054.1[See solution](#)What is $\int_0^1 \frac{x^3}{1+x^2} dx$?

- (A) $\frac{1}{2} \log 2$ (B) $\frac{1}{2}(1 - \log 2)$ (C) $1 - \log 2$ (D) $\log 2 - \frac{1}{2}$ (E) $1/4$

Problem 054.2[See solution](#)What is $\int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin^4 x} dx$?

- (A) $\pi/16$ (B) $\pi/8$ (C) $\pi/4$ (D) $\log 2/2$ (E) $1/2$

Problem 054.3[See solution](#)What is $\int_{-1}^1 2x \cos(x^2) dx$?

- (A) 0
(B) $2 \sin 1$
(C) $\sin 1$
(D) $2(1 - \cos 1)$
(E) The substitution $u = x^2$ is invalid, so the integral does not exist.

Problem 054.4[See solution](#)A continuous f satisfies $\int_1^4 f(u) du = 6$. What is $\int_0^1 xf(1+3x^2) dx$?

- (A) 1 (B) 2 (C) 3 (D) 6 (E) 18

Problem 054.5[See solution](#)

What is the value of the convergent improper integral $\int_0^\infty \frac{\log x}{1+x^2} dx$?

- (A) $-\pi/2$ (B) $-\log 2$ (C) 0 (D) $\log 2$ (E) $\pi/2$

2. Calculus I

Techniques of Integration ▸ Integration by Parts

055 Integration by parts

Book pp. 61-62. [Find in compact glossary](#)**In this entry:** integration by parts; reverse product rule; reduction formula; cyclic integral.**Definition and key ideas**

The product rule rearranges to $\int u dv = uv - \int v du$. For definite integrals, $\int_a^b uv' dx = [uv]_a^b - \int_a^b u'v dx$. Choose u so differentiating it simplifies the remaining integral and choose dv so it has a manageable antiderivative. Logarithms and inverse trigonometric functions often make good choices for u .

Repeated integration by parts reduces polynomial factors against exponentials or trigonometric functions. Sometimes the original integral reappears, producing an algebraic equation to solve. Boundary terms matter, especially in definite and improper integrals; evaluate their limits rather than dropping them. Differentiation is a quick verification of an indefinite result.

Problem 055.1[See solution](#)What is $\int_0^1 x(\log x)^2 dx$?

- (A) $1/8$
 (B) $1/4$
 (C) $1/2$
 (D) 1
 (E) The integral diverges.

Problem 055.2[See solution](#)What is $\int_0^\pi e^x \sin x dx$?

- (A) $(e^\pi - 1)/2$ (B) $(e^\pi + 1)/2$ (C) $e^\pi + 1$ (D) $e^\pi - 1$ (E) 0

Problem 055.3[See solution](#)What is $\int_0^\pi x \sin(x) dx$?

- (A) $-\pi$ (B) $1 + \pi$ (C) 2π (D) $\frac{\pi}{2}$ (E) π

Problem 055.4[See solution](#)For integers $n \geq 1$, define $I_n = \int_0^1 x^n e^x dx$. Which recurrence is correct?

- (A) $I_n = e + nI_{n-1}$ (B) $I_n = e - nI_{n-1}$
 (C) $I_n = 1 - nI_{n-1}$ (D) $I_n = (e - I_{n-1})/n$
 (E) $I_n = nI_{n-1}$

Problem 055.5[See solution](#)

What is $\int_0^1 \arctan x \, dx$?

- (A) $\pi/4$ (B) $\pi/4 - \log 2$ (C) $\pi/4 - \frac{1}{2} \log 2$ (D) $\pi/4 + \frac{1}{2} \log 2$ (E) $\frac{1}{2} \log 2$

2. Calculus I

Techniques of Integration ▸ Trig Substitutions

056 Trigonometric substitutions and radical integrals

Book pp. 61-63. [Find in compact glossary](#)

In this entry: trigonometric substitution; Pythagorean substitution; radical integral; power-reduction identity.

Definition and key ideas

The substitutions $u = a \sin \theta$, $u = a \tan \theta$, and $u = a \sec \theta$ simplify radicals $\sqrt{a^2 - u^2}$, $\sqrt{a^2 + u^2}$, and $\sqrt{u^2 - a^2}$, respectively, with $a > 0$. Their differentials are $a \cos \theta d\theta$, $a \sec^2 \theta d\theta$, and $a \sec \theta \tan \theta d\theta$. Choose an angular range that resolves signs correctly.

The identities $1 - \sin^2 \theta = \cos^2 \theta$ and $1 + \tan^2 \theta = \sec^2 \theta$ remove square roots, but $\sqrt{\cos^2 \theta} = |\cos \theta|$ before a range is chosen. Power-reduction formulas, such as $\sin^2 \theta = (1 - \cos 2\theta)/2$, then finish many integrals. In definite integrals convert endpoints; in indefinite integrals convert back using the substitution or a triangle.

Problem 056.1

[See solution](#)

For $a > 0$, what is $\int_0^a \frac{x^2}{\sqrt{a^2 - x^2}} dx$?

- (A) $\pi a/4$ (B) $\pi a^2/8$ (C) $\pi a^2/4$ (D) $\pi a^3/4$ (E) $a^2/2$

Problem 056.2

[See solution](#)

What is $\int_0^{\sqrt{3}} \frac{dx}{(1+x^2)^{3/2}}$?

- (A) $1/2$ (B) $\sqrt{3}/2$ (C) $\pi/3$ (D) $\sqrt{3}$ (E) $2/\sqrt{3}$

Problem 056.3

[See solution](#)

What is $\int_2^4 \frac{\sqrt{x^2-4}}{x} dx$?

- (A) $2\sqrt{3} - 2\pi/3$ (B) $2\sqrt{3} - \pi/3$ (C) $4\sqrt{3} - 2\pi/3$ (D) $2\pi/3$ (E) $2\sqrt{3} + 2\pi/3$

Problem 056.4[See solution](#)

What is $\int_0^{\sqrt{5}} \frac{x^2}{\sqrt{x^2+4}} dx$?

- (A) $-2 \operatorname{asinh}\left(\frac{\sqrt{5}}{2}\right) + 1 + \frac{3\sqrt{5}}{2}$
(B) $-\operatorname{asinh}\left(\frac{\sqrt{5}}{2}\right) + \frac{3\sqrt{5}}{4}$
(C) $-4 \operatorname{asinh}\left(\frac{\sqrt{5}}{2}\right) + 3\sqrt{5}$
(D) $-\frac{3\sqrt{5}}{2} + 2 \operatorname{asinh}\left(\frac{\sqrt{5}}{2}\right)$
(E) $-2 \operatorname{asinh}\left(\frac{\sqrt{5}}{2}\right) + \frac{3\sqrt{5}}{2}$

Problem 056.5[See solution](#)

What is $\int_{-4}^{-2\sqrt{2}} \frac{dx}{x\sqrt{x^2-4}}$?

- (A) $-\pi/12$ (B) $-\pi/24$ (C) 0 (D) $\pi/24$ (E) $\pi/12$

2. Calculus I

Techniques of Integration ▸ The Method of Partial Fractions

057 Partial fractions and rational integration

Book pp. 63-64. [Find in compact glossary](#)**In this entry:** partial-fraction decomposition; proper rational function; polynomial division; repeated linear factor; irreducible quadratic factor.**Definition and key ideas**

For $P(x)/Q(x)$, first divide if $\deg P \geq \deg Q$, then factor the proper denominator over $\mathbb{R}$. A factor $(x - a)^m$ requires terms $A_1/(x - a) + \cdots + A_m/(x - a)^m$. A repeated irreducible quadratic $q(x)^m$ requires a linear numerator over each power: $(B_jx + C_j)/q(x)^j$. All lower powers are necessary.

Clear denominators and compare coefficients, or use convenient roots to isolate constants. Linear denominators give logarithms or powers; irreducible quadratics commonly give logarithms and arctangents after completing a square. Decomposition does not remove genuine poles, so splitting and improper-limit checks remain necessary when a definite interval crosses a denominator zero.

Problem 057.1[See solution](#)

In $\frac{1}{x(x-1)^2} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$, what is $A + B + C$?

- (A) -1 (B) 0 (C) 1 (D) 2 (E) 3

Problem 057.2[See solution](#)

What is $\int_1^2 \frac{dx}{x(x^2+1)}$?

- (A)
- $\log 2$
- (B)
- $\frac{1}{2} \log(8/5)$
- (C)
- $\frac{1}{2} \log(5/2)$
- (D)
- $\arctan 2 - \pi/4$
- (E)
- $\log(5/4)$

Problem 057.3[See solution](#)

What is $\int_0^1 \frac{1}{(x+1)(x+2)} dx$?

- (A)
- $\log\left(\frac{16}{9}\right)$
- (B)
- $-\frac{\log(3)}{2} + \log(2)$
- (C)
- $-\log\left(\frac{4}{3}\right)$
- (D)
- $\log\left(\frac{4}{3}\right) + 1$
- (E)
- $\log\left(\frac{4}{3}\right)$

Problem 057.4[See solution](#)

How many independent coefficient slots occur in the standard real partial-fraction template for $\frac{x^5+1}{(x-1)^3(x^2+x+1)^2}$ before the coefficients are determined?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Problem 057.5[See solution](#)

How does the ordinary improper integral $\int_{-1}^1 \frac{dx}{x(x+2)}$ behave?

- (A) It equals $-\frac{1}{2} \log 3$.
- (B) It equals 0.
- (C) It equals $\frac{1}{2} \log 3$.
- (D) It converges absolutely.
- (E) It diverges because the two integrals at the interior pole do not converge separately.

Problem 057.6[See solution](#)

What is $\int_0^1 \frac{x^3+x^2+x+2}{x^2+1} dx$?

- (A) $1/2 + \pi/4$ (B) $1 + \pi/4$ (C) $3/2 + \pi/4$ (D) $3/2 + \log 2$ (E) $2 + \pi/2$

2. Calculus I

Definite Integration ▷ Area as a limit of sums

058 Riemann sums and definite integrals

Book pp. 64-66. [Find in compact glossary](#)

In this entry: Riemann sum; partition; mesh; tagged partition; definite integral; Riemann integrability; sum of squares.

Definition and key ideas

For a partition $a = x_0 < \cdots < x_n = b$ and tags $\xi_i \in [x_{i-1}, x_i]$, a Riemann sum is $\sum_i f(\xi_i)(x_i - x_{i-1})$. Its limiting value as the mesh $\max_i(x_i - x_{i-1})$ tends to zero, independent of partitions and tags, is the definite integral when f is Riemann integrable. Continuous functions on closed bounded intervals are integrable, as are bounded functions with only finitely many jump discontinuities.

For equal widths, write $\Delta x = (b - a)/n$ and $x_i = a + i\Delta x$ to identify an integral hidden in a sum. Nonuniform partitions work too. Useful finite sums include $\sum_{k=1}^n k = n(n+1)/2$ and $\sum_{k=1}^n k^2 = n(n+1)(2n+1)/6$. Integrals represent signed accumulation, not necessarily positive geometric area.

Problem 058.1

[See solution](#)

What is $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2}$?

- (A) $1/2$ (B) $\log 2$ (C) $\pi/4$ (D) $\pi/2$ (E) 1

Problem 058.2

[See solution](#)

What is $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{2k-1}{n^2} \sin\left(\frac{k^2}{n^2}\right)$?

- (A) $\sin 1$ (B) $\cos 1$ (C) $1 - \cos 1$ (D) $\frac{1}{2} \sin 1$ (E) $1/2$

Problem 058.3

[See solution](#)

Let D equal 1 at rational points of $[0, 1]$ and 0 at irrational points. On equal partitions, all-rational tags give Riemann sums 1, while all-irrational tags give sums 0. Which conclusion follows?

- (A) The integral is $1/2$.
 (B) The integral is 0.
 (C) The integral is 1.
 (D) D is not Riemann integrable.
 (E) Only the rational-tag sums are permitted.

Problem 058.4[See solution](#)

Partition $[0, 1]$ at $0, 1/(2n), 2/(2n), \dots, 1/2, 1$. Let S_n be the left-endpoint sum for $f(x) = x^2$. What is $\lim S_n$?

- (A) $1/24$ (B) $1/8$ (C) $1/6$ (D) $1/4$ (E) $1/3$

Problem 058.5[See solution](#)

Let $S_n = \frac{1}{n} \sum_{k=1}^n (k/n)^2$. What is $\lim_{n \rightarrow \infty} n(S_n - 1/3)$?

- (A) 0 (B) $1/6$ (C) $1/3$ (D) $1/2$ (E) 1

2. Calculus I

The Fundamental Theorem of Calculus ▸ Integration and differentiation

059 Fundamental theorem of calculus and variable limits

Book pp. 66-68. [Find in compact glossary](#)

In this entry: fundamental theorem of calculus; accumulation function; evaluation theorem; dummy variable; variable limits; Leibniz rule for endpoints.

Definition and key ideas

For continuous f on $[a, b]$, the accumulation function $A(x) = \int_a^x f(t)dt$ satisfies $A'(x) = f(x)$ in the interior. If $F' = f$, then $\int_a^b f(x)dx = F(b) - F(a)$. These two statements link differentiation with integration. The integration variable is a dummy variable; using t avoids confusion with a changing endpoint x .

With differentiable endpoints $u(x), v(x)$ and continuous integrand, $\frac{d}{dx} \int_{u(x)}^{v(x)} f(t)dt = f(v(x))v'(x) - f(u(x))u'(x)$. Include the chain-rule factors and the minus sign at the lower endpoint. The formula stated here assumes that the integrand has no additional x dependence; an x -dependent integrand requires an additional differentiated-integrand term under appropriate hypotheses.

If $K(x, t)$ and its partial derivative K_x are continuous on the relevant region, the full Leibniz rule is $\frac{d}{dx} \int_{u(x)}^{v(x)} K(x, t)dt = K(x, v(x))v'(x) - K(x, u(x))u'(x) + \int_{u(x)}^{v(x)} K_x(x, t)dt$. The last term is essential when the integrand depends on the parameter.

Problem 059.1

[See solution](#)

If $F(x) = \int_{x^2}^{e^x} \sin(t^2) dt$, what is $F'(0)$?

- (A) 0 (B) $\sin 1$ (C) $\cos 1$ (D) 1 (E) $2 \sin 1$

Problem 059.2

[See solution](#)

If $F(x) = \int_0^{x^2} e^{t^2} dt$, what is $F''(1)$?

- (A) $2e$ (B) $4e$ (C) $6e$ (D) $8e$ (E) $10e$

Problem 059.3

[See solution](#)

Let $f(t) = -1$ for $t < 0$ and $f(t) = 1$ for $t \geq 0$, and set $A(x) = \int_0^x f(t) dt$. Which statement about A at zero holds?

- (A) $A'(0) = 1$
 (B) $A'(0) = -1$
 (C) $A'(0) = 0$
 (D) A is continuous but not differentiable at zero.
 (E) A is discontinuous at zero.

Problem 059.4[See solution](#)

Let $F(x) = \int_0^x (x-t)e^{t^2} dt$. What is $F''(1)$?

- (A) 0 (B) 1 (C) e (D) $2e$ (E) $\int_0^1 e^{t^2} dt$

Problem 059.5[See solution](#)

A continuously differentiable f on $[0, 1]$ satisfies $\int_0^x f(t) dt = xf(x) - x^2$ for every $x \in [0, 1]$, and $f(0) = 3$. What is $f(1)$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

2. Calculus I

The Fundamental Theorem of Calculus ▸ Properties of definite integrals

060 Integral properties, signed area, and piecewise integration

Book pp. 67, 99. [Find in compact glossary](#)

In this entry: linearity of the integral; reversal of limits; additivity over intervals; integral comparison; signed area; geometric area; piecewise integral; floor-function integral.

Definition and key ideas

Definite integrals are linear, reverse sign when limits are reversed, and add over adjacent intervals: $\int_a^c f = \int_a^b f + \int_b^c f$. If $f \leq g$ on $[a, b]$ with $a < b$, then $\int_a^b f \leq \int_a^b g$. Consequently $|\int f| \leq \int |f|$. Reversing the order of endpoints changes the sign and therefore changes inequality direction.

An integral counts area above the horizontal axis positively and below it negatively. Geometric area is $\int_a^b |f(x)| dx$, evaluated by splitting where the sign changes. Piecewise and floor functions similarly require splitting at formula changes. Changing finitely many values does not change a Riemann integral, though it may destroy continuity. A nonnegative continuous function with integral zero must vanish identically.

Problem 060.1

[See solution](#)

What is the geometric area between $y = x^2 - 1$ and the x -axis on $[-1, 2]$?

- (A) 0 (B) $4/3$ (C) 2 (D) $8/3$ (E) 4

Problem 060.2

[See solution](#)

What is $\int_{-3/2}^{3/2} \lfloor x \rfloor dx$?

- (A) -3 (B) $-3/2$ (C) -1 (D) 0 (E) $3/2$

Problem 060.3

[See solution](#)

Which assertion is necessarily true? I. If continuous f on $[0, 1]$ satisfies $\int_0^1 f(x)^2 dx = 0$, then $f = 0$ everywhere. II. The same conclusion follows from $\int_0^1 f(x) dx = 0$ for continuous f . III. It follows for any nonnegative Riemann-integrable f with integral zero.

- (A) I only (B) II only (C) I and II only (D) I and III only (E) I, II, and III

Problem 060.4

[See solution](#)

If f is odd, continuous on $[-3, 3]$, and $\int_0^3 f(x) dx = 5$, what is $\int_{-3}^3 (f(x) + x^2) dx$?

- (A) 23 (B) 18 (C) 28 (D) 10 (E) 5

Problem 060.5

[See solution](#)

What is $\int_0^2 \max\{x, x^2\} dx$?

- (A) 3 (B) $19/6$ (C) $8/3$ (D) $17/6$ (E) $7/2$

Problem 060.6[See solution](#)

If $\int_0^2 f(x)dx = 3$ and $\int_2^5 f(x)dx = -1$, what is $\int_5^0 [2f(x) - 1]dx$?

- (A) -9 (B) -1 (C) 1 (D) 4 (E) 9

2. Calculus I

The Average Value of a Function ▸ Integral averages

061 Average value and the integral mean value theorem

Book pp. 68-69. [Find in compact glossary](#)**In this entry:** average value of a function; mean value theorem for integrals; average rate.**Definition and key ideas**

The average value of an integrable function on $[a, b]$ with $a < b$ is $f_{\text{avg}} = (b - a)^{-1} \int_a^b f(x) dx$. For continuous f , the integral mean value theorem guarantees a point $c \in [a, b]$ at which $f(c) = f_{\text{avg}}$. Geometrically, a constant-height rectangle at the average value has the same signed area.

The average lies between the minimum and maximum when they exist. Continuity guarantees that the average is actually attained; integrability alone does not. Distinguish the average value of f from the average rate of change $[f(b) - f(a)]/(b - a)$, which is an average of the derivative when the fundamental theorem applies. In physical applications, specify whether averaging is over time, distance, or another variable.

For a nonnegative integrable weight w with positive integral, the weighted average is $\int_a^b wf / \int_a^b w$. If f is continuous, this value lies between its minimum and maximum and is attained. Average velocity is displacement divided by elapsed time, while average speed is total distance divided by elapsed time.

Problem 061.1[See solution](#)

At which point $c \in [1, e]$ does $\log c$ equal the average value of $\log x$ on that interval?

- (A) $e/2$ (B) $\sqrt{e}$ (C) $\exp(1/(e - 1))$ (D) $(1 + e)/2$ (E) e^{e-1}

Problem 061.2[See solution](#)

Let f be continuous on $[0, 2]$ with average value 3. What is the average value on $[0, 1]$ of $g(x) = xf(x^2 + 1)$ if $\int_0^1 f(t) dt = 2$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Problem 061.3[See solution](#)

The average of f on $[0, 2]$ is 3 and on $[2, 5]$ is 8. What is its average on $[0, 5]$?

- (A) 6 (B) 7 (C) $11/2$ (D) $13/2$ (E) 5

Problem 061.4[See solution](#)

Which function on $[0, 1]$ has an average value that it never attains?

- (A) $f(x) = x$
- (B) $f(x) = x^2$
- (C) $f(x) = |x - 1/2|$
- (D) $f(x) = 0$ for $x < 1/2$, and $f(x) = 1$ for $x \geq 1/2$
- (E) $f(x) = \sin(2\pi x)$

Problem 061.5[See solution](#)

If f is continuous on $[0, 1]$, which expression must equal $f(c)$ for some $c \in [0, 1]$?

- (A) $\int_0^1 3xf(x)dx$ (B) $f(0) + f(1)$ (C) $2 \int_0^1 xf(x)dx$ (D) $\int_0^1 (f(x) + 1)dx$ (E) $\int_0^1 xf(x)dx$

Problem 061.6[See solution](#)

A particle has position $s(t) = t^3 - 3t$ for $0 \leq t \leq 2$. What are its average velocity and average speed over this interval, respectively?

- (A) 1, 1 (B) 1, 3 (C) 3, 1 (D) 2, 3 (E) -1, 3

2. Calculus I

Finding the Area Between Two Curves ▸ Choosing the integration variable

062 Area between curves

Book pp. 69-71. [Find in compact glossary](#)

In this entry: area between curves; upper minus lower; right minus left; intersection bounds; horizontal slicing.

Definition and key ideas

If $f(x) \geq g(x)$ on $[a, b]$, the region between their graphs has area $\int_a^b [f(x) - g(x)] dx$. If their ordering changes, split at intersections or integrate $|f - g|$. Horizontal slices give $\int_c^d [x_{\text{right}}(y) - x_{\text{left}}(y)] dy$ and may avoid several vertical pieces.

Solve for actual intersections and inspect which boundary is active on each interval. An intersection equation can include points outside the requested quadrant or region. Auxiliary horizontal or vertical lines may truncate the region before the next curve intersection. The integration variable is chosen for a simple geometric description, not fixed by the original equations.

For a continuous increasing bijection $f : [0, a] \rightarrow [0, b]$ with $f(0) = 0$, the graph partitions a rectangle, yielding $\int_0^a f(x) dx + \int_0^b f^{-1}(y) dy = ab$. Switching between vertical and horizontal slices can determine one integral without an inverse formula.

Problem 062.1

[See solution](#)

What is the area enclosed by $y = x$ and $y = x^3$ over all their bounded regions?

- (A) $1/4$ (B) $1/2$ (C) $3/4$ (D) 1 (E) 2

Problem 062.2

[See solution](#)

What is the area of the region in the first quadrant lying below $y = 4$ and between $y = x^2$ and $y = 4x$?

- (A) $2/3$ (B) $4/3$ (C) $8/3$ (D) $10/3$ (E) 4

Problem 062.3

[See solution](#)

A continuous strictly increasing $f : [0, 2] \rightarrow [0, 4]$ satisfies $f(0) = 0$, $f(2) = 4$, and $\int_0^2 f(x) dx = 3$. What is $\int_0^4 f^{-1}(y) dy$?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 8

Problem 062.4

[See solution](#)

What is the area between $y = \sqrt{x}$ and $y = x^2$ for $0 \leq x \leq 1$?

- (A) $1/3$ (B) $1/2$ (C) 1 (D) $1/6$ (E) $2/3$

Problem 062.5

[See solution](#)

The line $y = a$ with $a > 0$ and the parabola $y = x^2$ enclose area $32/3$. What is a ?

- (A) 16 (B) 3 (C) 8 (D) 4 (E) 2

2. Calculus I

Polar Coordinates ▸ Coordinates and curves

063 Polar coordinates and standard polar curves

Book pp. 71-72. [Find in compact glossary](#)

In this entry: polar coordinates; Cartesian coordinates; negative radius; polar circle; cardioid; polar-to-Cartesian conversion.

Definition and key ideas

Polar coordinates satisfy $x = r \cos \theta$ and $y = r \sin \theta$. For nonnegative r , $r = \sqrt{x^2 + y^2}$; if signed radii are allowed, (r, θ) and $(-r, \theta + \pi)$ represent the same point. Angles are determined only modulo 2π , and $\tan \theta = y/x$ by itself does not identify the quadrant.

Curves $r = 2a \cos \theta$ and $r = 2a \sin \theta$ are circles centered at $(a, 0)$ and $(0, a)$, each of radius $|a|$. A cardioid has form $r = A(1 \pm \cos \theta)$ or its sine variant, with orientation determined by the sign and trigonometric function. Before integrating, identify how often an interval traces the curve; polar representations are not unique and naive angle ranges may double-count.

A cardioid is also the curve traced by a point on a circle rolling without slipping around the outside of a fixed circle of the same radius. This equal-radius rolling construction produces its single cusp.

Problem 063.1

[See solution](#)

Which Cartesian equation describes $r = 4 \cos \theta - 6 \sin \theta$?

- (A) $(x - 2)^2 + (y + 3)^2 = 13$ (B) $(x - 4)^2 + (y + 6)^2 = 52$
(C) $(x + 2)^2 + (y - 3)^2 = 13$ (D) $x^2 + y^2 = 4x + 6y$
(E) $x^2 + y^2 = 13$

Problem 063.2

[See solution](#)

Which pair of polar coordinates represents the Cartesian point $(-\sqrt{3}, 1)$?

- (A) $(-2, \pi/6)$ (B) $(2, \pi/6)$ (C) $(-2, -\pi/6)$ (D) $(2, 7\pi/6)$ (E) $(-2, 5\pi/6)$

Problem 063.3

[See solution](#)

A point on a circle rolling externally around an equal circle traces a cardioid. In suitable coordinates its equation is $r = 2(1 - \cos \theta)$. Which point is farthest from the origin?

- (A) $(4, 0)$ (B) $(-4, 0)$ (C) $(0, 4)$ (D) $(0, -4)$ (E) $(0, 0)$

Problem 063.4[See solution](#)

The curve $r \cos \theta = 2$ is best described as

- (A) A horizontal line
- (B) A parabola
- (C) A hyperbola
- (D) A circle through the origin
- (E) A vertical line

Problem 063.5[See solution](#)

What is the Cartesian equation of $r^2 \cos(2\theta) = 4$?

- (A) $xy = 2$ (B) $x^2 - y^2 = 16$ (C) $x^2 + y^2 = 4$ (D) $x^2 - y^2 = 4$ (E) $(x - y)^2 = 4$

Problem 063.6[See solution](#)

As θ runs from 0 to 2π , how many times does $r = 2 \cos \theta$, with signed radii allowed, trace its circle?

- (A) One half (B) Once (C) Twice (D) Three times (E) Four times

2. Calculus I

Polar Coordinates ▸ Area in polar coordinates

064 Polar area and overlap regions

Book pp. 72-73, 100. [Find in compact glossary](#)

In this entry: polar area formula; sector area; radial bounds; overlap of polar regions; rose curve; lemniscate.

Definition and key ideas

A polar sector swept once by $0 \leq r \leq R(\theta)$ over $\alpha \leq \theta \leq \beta$ has area $\frac{1}{2} \int_{\alpha}^{\beta} R(\theta)^2 d\theta$. With inner and outer radii, use $\frac{1}{2} \int (R_{\text{outer}}^2 - R_{\text{inner}}^2) d\theta$. Squaring a radius is required because sector area is quadratic in length.

For overlap, the active outer radius is often the smaller of two positive radial boundary functions, so split where they meet. Check angular restrictions, origin intersections, and repeated tracing before choosing bounds. Solving $r_1(\theta) = r_2(\theta)$ finds same-angle intersections, but the origin and points represented with different signs or angles may require separate attention.

A rose $r = A \sin(n\theta)$ or $A \cos(n\theta)$ is split into petals by successive relevant zeros of r ; integrate over one complete petal and account for distinct petals rather than repeated traces. For the lemniscate $r^2 = A^2 \cos 2\theta$, only angles with $\cos 2\theta \geq 0$ give real radii; its two loops can be integrated separately.

Problem 064.1

[See solution](#)

What is the area enclosed by $r = 3(1 + \cos \theta)$?

- (A) $9\pi/2$ (B) 9π (C) $27\pi/2$ (D) 18π (E) 27π

Problem 064.2

[See solution](#)

What is the common area of the two disks bounded by $r = 2 \cos \theta$ and $r = 2 \sin \theta$?

- (A) $\pi/2 - 1$ (B) $\pi - 2$ (C) $\pi - 1$ (D) $2\pi - 2$ (E) 1

Problem 064.3

[See solution](#)

What is the area of one petal of $r = 2 \sin(3\theta)$?

- (A) $\pi/3$ (B) $\pi/2$ (C) π (D) $\pi/6$ (E) $2\pi/3$

Problem 064.4

[See solution](#)

What is the total area enclosed by $r^2 = 4 \cos(2\theta)$?

- (A) 2 (B) π (C) 4 (D) 8 (E) 2π

Problem 064.5

[See solution](#)

What is the area inside $r = 2 \cos \theta$ but outside $r = 1$?

- (A) $\pi/2$ (B) $\pi - \sqrt{3}$ (C) $\pi/3 - \sqrt{3}/2$ (D) $\pi/3 + \sqrt{3}/2$ (E) $2\pi/3$

Problem 064.6[See solution](#)

How many distinct points are common to the cardioids $r = 1 + \cos \theta$ and $r = 1 - \cos \theta$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

2. Calculus I

Volumes of Solids of Revolution ▸ Disks and washers

065 Volumes of revolution: disks and washers

Book pp. 73-75. [Find in compact glossary](#)

In this entry: solid of revolution; disk method; washer method; outer radius; inner radius; cross-sectional area.

Definition and key ideas

Rotating a region about an axis creates a solid of revolution. A slice perpendicular to the axis yields a disk of area πR^2 or a washer of area $\pi(R^2 - r^2)$, where radii are distances from the axis. Integrate this cross-sectional area along the axis: $V = \int A(s)ds$. Rotating about a horizontal axis usually suggests vertical slices and dx ; a vertical axis suggests horizontal slices and dy .

For shifted axes subtract the axis coordinate before squaring. If the region crosses the axis, rotating separate signed sides can overlap, so identify the actual cross-section. A change in which boundary is farthest from the axis requires a split. Distinguish the square of the difference from the difference of squares: washers use the latter.

Problem 065.1

[See solution](#)

The region $0 \leq x \leq 1$, $x^2 \leq y \leq x$ is revolved about $y = 2$. What is its volume?

- (A) $\pi/30$ (B) $7\pi/30$ (C) $\pi/3$ (D) $8\pi/15$ (E) π

Problem 065.2

[See solution](#)

The region enclosed by $y = x^2$ and $y = 2 - x^2$ is revolved about the y -axis. What is its volume?

- (A) $\pi/2$ (B) π (C) $4\pi/3$ (D) 2π (E) $8\pi/3$

Problem 065.3

[See solution](#)

The region $0 \leq x \leq 2$, $-1 \leq y \leq x$ is revolved about the x -axis. What is its volume?

- (A) $8\pi/3$ (B) $10\pi/3$ (C) $14\pi/3$ (D) 4π (E) $16\pi/3$

Problem 065.4

[See solution](#)

The region under $y = \sqrt{x}$, above the x -axis, and between $x = 0$ and $x = 4$ is revolved about the y -axis. What is its volume?

- (A) $64\pi/5$ (B) $128\pi/5$ (C) 32π (D) 16π (E) $32\pi/5$

Problem 065.5

[See solution](#)

A region has vertical slices $g(x) \leq y \leq f(x)$ on $a \leq x \leq b$, where $0 \leq g(x) \leq f(x) \leq 1$. Its volumes when revolved about $y = 2$ and $y = 3$ are 14π and 22π , respectively. What is its area?

- (A) 2 (B) 4 (C) 7 (D) 8 (E) 11

2. Calculus I

Arc Length ▸ Length by integration

066 Arc length

Book pp. 75-76, 101. [Find in compact glossary](#)**In this entry:** arc length; smooth curve; line element; rectifiable curve; parametric arc length.**Definition and key ideas**

For a continuously differentiable graph $y = f(x)$ on $[a, b]$, its length is $L = \int_a^b \sqrt{1 + [f'(x)]^2} dx$. This follows from the line element $ds = \sqrt{dx^2 + dy^2}$. If $x = g(y)$ is simpler, integrate $\sqrt{1 + [g'(y)]^2}$ with respect to y instead. Singular endpoint slopes may still yield finite length via an improper integral.

For a continuously differentiable parametrization, the general formula, counting repeated traversals, is

$$L = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2} dt.$$

Algebra often makes $1 + (f')^2$ a perfect square; take its nonnegative square root. Arc length differs from area under a graph, and an interval of zero net displacement can still have positive length.

A curve is rectifiable when the supremum of the lengths of its inscribed polygonal paths is finite. Continuous differentiability on a compact parameter interval guarantees rectifiability, but continuity alone does not. In three dimensions the speed is $\sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}$; the same integral counts distance traveled, including repeated traversals.

Problem 066.1[See solution](#)

What is the length of $y = x^{3/2}$ for $0 \leq x \leq 4/3$?

- (A) $28/27$ (B) $56/27$ (C) $8/3$ (D) 4 (E) $7/3$

Problem 066.2[See solution](#)

What is the first-quadrant arc length of $x^{2/3} + y^{2/3} = 4$, from $(8, 0)$ to $(0, 8)$?

- (A) 8 (B) 12 (C) 16 (D) 24 (E) 4π

Problem 066.3[See solution](#)

What is the length of $y = \log(\cos x)$ for $0 \leq x \leq \pi/3$?

- (A) $\pi/3$ (B) $\sqrt{3}$ (C) $\log(2 + \sqrt{3})$ (D) $2\log(2 + \sqrt{3})$ (E) $\log 2$

Problem 066.4[See solution](#)

What is the length of the parametrized curve $(3 \cos t, 3 \sin t, 4t)$ for $0 \leq t \leq 2\pi$?

- (A) 8π (B) 25π (C) 6π (D) 10π (E) 14π

Problem 066.5[See solution](#)

A curve is parametrized by $(x(t), y(t)) = (t^2, 0)$ for $-1 \leq t \leq 1$. What are its parametrized arc length and the length of its geometric image, respectively?

- (A) 0, 1 (B) 1, 1 (C) 2, 1 (D) 2, 2 (E) 1, 2

Problem 066.6[See solution](#)

Which graph on $[0, 1]$ has infinite arc length, using the continuous extension at zero when needed?

- (A) $y = \sqrt{x}$ (B) $y = x^{1/3}$ (C) $y = x^2 \log x$ (D) $y = x \sin(1/x^2)$ (E) $y = \sqrt{1 - x^2}$

2. Calculus I

The Natural Exponential and Logarithm Functions ▸ Definitions and calculus

067 The natural exponential, logarithm, and the number e

Book pp. 76–78. [Find in compact glossary](#)

In this entry: natural exponential; natural logarithm; Euler number e ; exponential limit; integral definition of logarithm; transcendental number.

Definition and key ideas

The natural exponential e^x is the positive exponential whose derivative equals itself, normalized by $e^0 = 1$. Its inverse is $\log x = \ln x$ on $x > 0$, satisfying $\log x = \int_1^x dt/t$. Consequently $(\log x)' = 1/x$ and $a^x = e^{x \log a}$ for $a > 0$. The book uses $\log$ for the natural logarithm.

The constant e satisfies $e = \lim_{n \rightarrow \infty} (1 + 1/n)^n$. More generally, $\log(1 + t)/t \rightarrow 1$ as $t \rightarrow 0$, which controls many variable-power limits. The number e is transcendental: it is not a root of any nonzero polynomial with integer coefficients. This is stronger than irrationality and does not follow merely from its nonrepeating decimal expansion.

The numerical value is $e \approx 2.718281828$. The inequality $\log x \leq x - 1$ holds for all $x > 0$, with equality only at $x = 1$; it follows by comparing $1/t$ with 1 in the defining integral or by differentiating the difference.

Problem 067.1

[See solution](#)

For fixed real a, b , what is $\lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^{bn}$?

- (A) a^b (B) e^{a+b} (C) e^{ab} (D) $e^{a/b}$ (E) 1 for every a, b

Problem 067.2

[See solution](#)

If $a > 1$ and $\int_1^a \frac{dt}{t} = A$, what is $\int_{a^{-2}}^{a^3} \frac{dt}{t}$?

- (A) A (B) $2A$ (C) $3A$ (D) $5A$ (E) $6A$

Problem 067.3

[See solution](#)

A polynomial p with rational coefficients satisfies $p(e) = p(1)$. What must $p(2) - p(1)$ equal?

- (A) 0 (B) 1 (C) $e - 1$ (D) $e^2 - 1$ (E) It is not determined.

Problem 067.4

[See solution](#)

What is $\lim_{x \rightarrow 0} (e^{5x} - e^{2x})/x$?

- (A) 7 (B) 3 (C) 2 (D) 5 (E) 1

Problem 067.5[See solution](#)

For which real c does $\log x \leq c(x - 1)$ hold for every $x > 0$?

- (A) $c \geq 1$ (B) $c \leq 1$ (C) $c = 1$ only (D) $c > 0$ (E) No such c

2. Calculus I

The Natural Exponential and Logarithm Functions ▸ Variable exponents

068 Logarithmic differentiation and variable powers

Book pp. 79. [Find in compact glossary](#)

In this entry: logarithmic differentiation; variable exponent; logarithmic derivative; exponential rewriting.

Definition and key ideas

For positive $u(x)$ and differentiable $v(x)$, write $u(x)^{v(x)} = \exp(v(x) \log u(x))$. Differentiation gives $\frac{d}{dx}u^v = u^v[v' \log u + vu'/u]$. This formula includes both the change of exponent and the change of base; the ordinary power rule only covers constant exponents.

For a nonzero differentiable product or quotient, differentiating $\log |f|$ often turns multiplication into addition and division into subtraction. Recover f' by multiplying the logarithmic derivative by f . Real variable powers require positive bases unless a special domain has been explicitly justified. Logarithmic differentiation can also identify critical points of a positive function using the simpler derivative of its logarithm.

The same method applies to implicit equations with positive variable powers: take logarithms on both sides before differentiating, and include the derivative of every dependent variable. To classify an extremum of a positive function, the sign of its logarithmic derivative is the sign of its ordinary derivative.

Problem 068.1

[See solution](#)

What is $\left. \frac{d}{dx} x^{\sin x} \right|_{x=\pi/2}$?

- (A) 0 (B) 1 (C) $\pi/2$ (D) $\log(\pi/2)$ (E) $1 + \log(\pi/2)$

Problem 068.2

[See solution](#)

What is $\left. \frac{d}{dx} (x+1)^{x^2} \right|_{x=1}$?

- (A) $1 + \log 2$ (B) $1 + 2 \log 2$ (C) $1 + 4 \log 2$ (D) $2 + 4 \log 2$ (E) $4 \log 2$

Problem 068.3

[See solution](#)

For $x > 1$, let $f(x) = x^3(x-1)^2/(x+1)^4$. What is $f'(2)/f(2)$?

- (A) $1/6$ (B) $5/6$ (C) $7/6$ (D) $13/6$ (E) $29/6$

Problem 068.4[See solution](#)

At which $x > 0$ does $x^{1/x}$ attain its global maximum?

- (A) $1/e$
- (B) 1
- (C) $\sqrt{e}$
- (D) e
- (E) No maximum is attained.

Problem 068.5[See solution](#)

A differentiable positive branch $y = y(x)$ through $(2, 4)$ satisfies $y^x = x^y$. What is $y'(2)$?

- (A) 1 (B) 2 (C) $\frac{2 - 2 \log 2}{1/2 - \log 2}$ (D) $\frac{1/2 - \log 2}{2 - 2 \log 2}$ (E) $\frac{2 + 2 \log 2}{1/2 + \log 2}$

2. Calculus I

L'Hopital's Rule ▸ Limits by differentiation

069 L'Hopital's rule and indeterminate forms

Book pp. 79-81. [Find in compact glossary](#)

In this entry: L'Hopital rule; indeterminate form; zero over zero; infinity over infinity; indeterminate product; indeterminate power; growth hierarchy.

Definition and key ideas

For a quotient approaching $0/0$ or ∞/∞ , under differentiability on a punctured interval and a nonvanishing denominator derivative, L'Hopital's rule identifies the quotient limit with $\lim f'/g'$ when that latter limit exists, including suitable infinite limits. One-sided limits and limits at infinity have corresponding versions. Differentiate numerator and denominator separately, not by the quotient rule.

The rule is not valid for arbitrary quotients. Convert $0 \cdot \infty$ and $\infty - \infty$ to a quotient by algebra, and handle 1^∞ , 0^0 , or ∞^0 by first taking logarithms of positive expressions. Recheck hypotheses before each repeated application. Taylor expansions or elementary cancellation are often shorter. Powers dominate logarithms, and exponentials with base greater than one dominate fixed powers at infinity.

Problem 069.1

[See solution](#)

What is $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{1 - \cos x}$?

- (A) 0
- (B) $1/2$
- (C) 1
- (D) 2
- (E) The limit does not exist.

Problem 069.2

[See solution](#)

What is $\lim_{x \rightarrow 0} (1 + 3x)^{\cot x}$, with x sufficiently close to zero that the base is positive?

- (A) 1 (B) e (C) e^2 (D) e^3 (E) $3e$

Problem 069.3[See solution](#)

For $x > 0$, let $F(x) = (x + \sin x)/x$. Which statement correctly describes its limit at infinity and the derivative quotient in L'Hopital's rule?

- (A) $F \rightarrow 1$ and the derivative quotient tends to 1.
- (B) $F \rightarrow 1$ but the derivative quotient has no limit.
- (C) F has no limit because the derivative quotient has no limit.
- (D) $F \rightarrow 0$.
- (E) $F \rightarrow +\infty$.

Problem 069.4[See solution](#)

What is $\lim_{x \rightarrow 0^+} x^x$?

- (A) 0 (B) e^{-1} (C) 1 (D) e (E) ∞

Problem 069.5[See solution](#)

What is $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\log(1+x)} \right)$?

- (A) -1 (B) $-1/2$ (C) 0 (D) $1/2$ (E) 1

Problem 069.6[See solution](#)

What is $\lim_{x \rightarrow \infty} x^{1/\sqrt{x}}$?

- (A) 0 (B) 1 (C) $\sqrt{e}$ (D) e (E) $+\infty$

2. Calculus I

Improper Integrals ▸ Definition and evaluation

070 Improper integrals and independent limiting operations

Book pp. 81–84. [Find in compact glossary](#)

In this entry: improper integral; unbounded interval; endpoint singularity; interior singularity; convergent improper integral; divergent improper integral; Cauchy principal value; improper integral of the first kind; improper integral of the second kind; absolute convergence of an improper integral; conditional convergence of an improper integral.

Definition and key ideas

An integral with an infinite endpoint or unbounded integrand is defined by a limit of proper integrals. For example, $\int_a^\infty f = \lim_{b \rightarrow \infty} \int_a^b f$ when the limit is finite. An interior singularity at c requires both $\int_a^c f$ and $\int_c^b f$ to converge separately. Integrals over the whole real line similarly require two independent tails.

Symmetric cancellation is not ordinary convergence: a Cauchy principal value may exist when the improper integral diverges. Split at every singularity and verify each piece before adding. A substitution may simplify evaluation, but does not waive convergence checks. Finite interval length does not guarantee convergence when the integrand blows up, and an infinite interval does not force divergence if decay is sufficient.

An improper integral is absolutely convergent if the integral of the absolute value is finite. It is conditionally convergent if the original improper integral converges but the absolute-value integral does not. Oscillation can produce conditional convergence on one infinite tail, which is different from cancelling two independently divergent tails by a principal value.

An infinite limit of integration makes an integral improper of the first kind. An unbounded integrand at a finite endpoint or an interior point makes it improper of the second kind. Both causes can occur together and must each be handled by a separate limiting check.

Problem 070.1

[See solution](#)

Which statement is correct about $\int_{-1}^1 \frac{dx}{x}$?

- (A) It converges to 0.
- (B) It converges to $2 \log 2$.
- (C) It diverges, but its symmetric Cauchy principal value is 0.
- (D) It diverges and has no symmetric principal value.
- (E) It converges conditionally to 1.

Problem 070.2[See solution](#)

What is $\int_0^\infty \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx$?

- (A) $1/2$
- (B) 1
- (C) 2
- (D) $\sqrt{\pi}$
- (E) The integral diverges.

Problem 070.3[See solution](#)

How does $\int_1^\infty \sin x/x dx$ behave?

- (A) It converges absolutely.
- (B) It converges, but its absolute-value integral diverges.
- (C) It diverges to $+\infty$.
- (D) It diverges to $-\infty$.
- (E) It has no improper limit because $\sin x$ oscillates.

Problem 070.4[See solution](#)

For $0 < a < b$, what is $\int_0^\infty \frac{dx}{(x+a)(x+b)}$?

- (A) $\frac{\log(b/a)}{b-a}$
- (B) $\frac{\log(b/a)}{a+b}$
- (C) $\frac{\log(a/b)}{b-a}$
- (D) $1/(ab)$
- (E) It diverges because both partial-fraction terms have divergent integrals.

Problem 070.5[See solution](#)

What is $\int_0^1 \log x dx$?

- (A) -1 (B) It diverges (C) 0 (D) $-1/2$ (E) 1

2. Calculus I

Improper Integrals ▸ Convergence criteria

071 Improper p-integrals and comparison near singularities

Book pp. 81-84, 101-102. [Find in compact glossary](#)

In this entry: p-integral; comparison test for integrals; limit comparison for integrals; logarithmic singularity; endpoint convergence.

Definition and key ideas

The integral $\int_1^\infty x^{-p} dx$ converges exactly when $p > 1$, whereas $\int_0^1 x^{-p} dx$ converges exactly when $p < 1$. These opposite inequalities reflect behavior at infinity versus near zero. A logarithmic substitution gives $\int_e^\infty dx/[x(\log x)^p]$, which converges exactly when $p > 1$ and then equals $1/(p - 1)$.

For nonnegative integrands, a smaller function than a convergent majorant has convergent integral; a larger function than a divergent minorant has divergent integral. If $f/g \rightarrow L$ with $0 < L < \infty$ near the problematic endpoint, their integrals have the same convergence behavior. Check every endpoint separately: a function can be harmless at infinity but nonintegrable at zero.

Problem 071.1

[See solution](#)

For which real p does $\int_0^\infty \frac{dx}{x^p(1+x)}$ converge?

- (A) $p < 0$ (B) $p > 1$ (C) $0 < p < 1$ (D) $p = 1$ (E) All real p

Problem 071.2

[See solution](#)

For which real q does $\int_{e^e}^\infty \frac{dx}{x \log x (\log \log x)^q}$ converge?

- (A) $q < 0$ (B) $q < 1$ (C) $q \leq 1$ (D) $q > 1$ (E) Every $q > 0$

Problem 071.3

[See solution](#)

For which real p does $\int_0^1 x^{-p} |\log x| dx$ converge?

- (A) $p \leq 1$ (B) $p > 1$ (C) $p > 0$ (D) $p < 1$ (E) Every real p

Problem 071.4

[See solution](#)

Without evaluating it exactly, which bound follows for $I = \int_e^\infty \frac{2+\sin x}{x(\log x)^2} dx$?

- (A) $0 \leq I \leq 1$ (B) $1 \leq I \leq 3$ (C) $3 \leq I \leq 5$ (D) $I = 2$ (E) I diverges.

Problem 071.5[See solution](#)

Positive continuous functions f, g on $(0, 1]$ satisfy $f(x)/g(x) \rightarrow 0$ as $x \rightarrow 0^+$ and $\int_0^1 g(x)dx = +\infty$. What follows about $\int_0^1 f(x)dx$?

- (A) It must converge.
- (B) It must diverge.
- (C) It must equal zero.
- (D) It may converge or diverge.
- (E) The assumptions are impossible.

2. Calculus I

Infinite Series ▸ Definitions and exact sums

072 Infinite series, partial sums, geometric and telescoping sums

Book pp. 84–85. [Find in compact glossary](#)

In this entry: infinite series; partial sum; sum of a series; geometric series; common ratio; telescoping series.

Definition and key ideas

The series $\sum_{n=1}^{\infty} a_n$ converges to S when its partial sums $s_N = \sum_{n=1}^N a_n$ tend to the finite number S . The terms and the partial sums are different sequences. A geometric series with first term A and ratio r has sum $A/(1-r)$ for $|r| < 1$; its finite sum is $A(1-r^N)/(1-r)$ for $r \neq 1$.

A telescoping series simplifies because adjacent terms cancel in finite partial sums; retain both beginning and end terms before taking a limit. Removing finitely many terms preserves convergence but changes the sum. Convergent series can be added and multiplied by constants term by term. Do not manipulate divergent sums as ordinary numbers, and verify the starting index before applying a familiar formula.

For $|r| < 1$, $\sum_{n=1}^{\infty} nr^n = r/(1-r)^2$. This follows by subtracting r times the weighted series from itself, leaving $r + r^2 + \cdots = r/(1-r)$; it is also obtained by differentiating the geometric power series inside its convergence interval.

Problem 072.1

[See solution](#)

What is $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$?

- (A) $1/4$ (B) $1/2$ (C) $3/4$ (D) 1 (E) $3/2$

Problem 072.2

[See solution](#)

What is $\sum_{n=0}^{\infty} \frac{(-2)^n}{3^{n+1}}$?

- (A) -1 (B) $-1/5$ (C) $1/5$ (D) $1/3$ (E) 1

Problem 072.3

[See solution](#)

What is $\sum_{n=0}^{\infty} \frac{1+2+2^2+\cdots+2^n}{3^{n+1}}$?

- (A) $1/2$ (B) 1 (C) $3/2$ (D) 2 (E) 3

Problem 072.4

[See solution](#)

What is $\sum_{n=1}^{\infty} n/3^n$?

- (A) $9/4$ (B) $3/2$ (C) $1/2$ (D) $3/4$ (E) $2/3$

Problem 072.5[See solution](#)

The partial sums of a series satisfy $s_N = (3N + 1)/(N + 2)$ for every integer $N \geq 1$. What are its first term and its total sum, respectively?

- (A) $1/2, 3$ (B) $4/3, 3$ (C) $5/6, 3$ (D) $4/3, 1/2$ (E) $3, 3$

2. Calculus I

Infinite Series ▸ Basic convergence tests

073 The term test, harmonic series, and p-seriesBook pp. 85. [Find in compact glossary](#)**In this entry:** nth-term test; necessary condition for series convergence; harmonic series; p-series.**Definition and key ideas**

If $\sum a_n$ converges, then $a_n \rightarrow 0$, because $a_n = s_n - s_{n-1}$. Therefore nonzero limiting terms or failure of a term limit prove divergence. The converse is false: the harmonic series $\sum 1/n$ diverges although its terms decrease to zero.

The p -series $\sum_{n=1}^{\infty} n^{-p}$ converges exactly for $p > 1$. It is the standard benchmark for rational expressions in n and many radical expressions after rationalization. Compare the effective power at infinity, not only the largest exponent appearing somewhere in the expression. For signed series, a comparison of magnitudes may prove absolute convergence, but a divergent magnitude series does not by itself prove divergence of the signed series.

Problem 073.1[See solution](#)

For which real p does $\sum_{n=1}^{\infty} \frac{n}{1+n^p}$ converge?

- (A) $p > 0$ (B) $p > 1$ (C) $p \geq 2$ (D) $p > 2$ (E) No real p

Problem 073.2[See solution](#)

Which statement describes $\sum_{n=1}^{\infty} (\sqrt{n^2+1} - n)$?

- (A) It converges because its terms tend to zero.
 (B) It converges by geometric comparison.
 (C) It diverges because its terms do not tend to zero.
 (D) Its terms tend to zero, but the series diverges.
 (E) Its partial sums alternate in sign.

Problem 073.3[See solution](#)

Which series diverges because its terms fail to tend to zero?

- (A) $\sum_{n=1}^{\infty} 1/n^2$ (B) $\sum_{n=1}^{\infty} 1/[n(n+1)]$
 (C) $\sum_{n=1}^{\infty} 1/n$ (D) $\sum_{n=1}^{\infty} (-1)^n/n$
 (E) $\sum_{n=1}^{\infty} n/(n+1)$

Problem 073.4[See solution](#)

For which real p does $\sum_{n=1}^{\infty} 1/\lfloor \sqrt{n} \rfloor^p$ converge?

- (A) $p > 0$ (B) $p > 1$ (C) $p > 2$ (D) $p \geq 2$ (E) No real p

Problem 073.5[See solution](#)

If $a_n > 0$ and $\sum a_n$ converges, which must be true?

(A) $na_n \rightarrow 0$

(B) $\sum \sqrt{a_n}$ converges

(C) $a_n \rightarrow 0$

(D) $a_{n+1} < a_n$ eventually

(E) $a_n \leq 1/n^2$ eventually

2. Calculus I

Infinite Series ▸ The comparison test

074 Direct and limit comparison for series

Book pp. 86-87. [Find in compact glossary](#)

In this entry: comparison test; direct comparison; limit comparison; nonnegative series; eventual inequalities.

Definition and key ideas

For eventually nonnegative terms with $a_n \leq b_n$, convergence of $\sum b_n$ implies convergence of $\sum a_n$; divergence of $\sum a_n$ implies divergence of $\sum b_n$. The other two directions are invalid. Only eventual inequalities matter because finite initial changes do not affect convergence.

A useful consequence is limit comparison: if $a_n, b_n > 0$ eventually and $a_n/b_n \rightarrow L$ with $0 < L < \infty$, then the two series have the same convergence behavior. Bounded oscillatory numerators are often handled by direct comparison, while rational functions and radicals are often handled by dominant terms. For signed series apply these positive-term arguments to absolute values when appropriate.

Problem 074.1

[See solution](#)

Which of the following converge: I. $\sum_{n \geq 1} \sin^2 n/n^2$; II. $\sum_{n \geq 1} n^2/(n^3+1)$; III. $\sum_{n \geq 1} (\sqrt{n^2+1}-n)^2$?

(A) I only (B) II only (C) I and II only (D) I and III only (E) I, II, and III

Problem 074.2

[See solution](#)

Assume $a_n \geq 0$ and $\sum a_n < \infty$. Which series need not converge?

(A) $\sum a_n^2$ (B) $\sum a_n/(1+a_n)$ (C) $\sum a_n/(1+n)$ (D) $\sum \sqrt{a_n}$ (E) $\sum \min(a_n, 1)$

Problem 074.3

[See solution](#)

How does $\sum_{n=1}^{\infty} [1/n - \log(1+1/n)]$ behave?

- (A) It diverges because $\sum 1/n$ diverges.
- (B) It diverges to $-\infty$.
- (C) It converges to a finite positive number.
- (D) Its terms fail to tend to zero.
- (E) It equals zero.

Problem 074.4

[See solution](#)

If $0 \leq a_n \leq b_n$ and $\sum a_n$ diverges, which must hold?

- (A) b_n is increasing
- (B) $a_n/b_n \rightarrow 1$
- (C) $b_n \rightarrow \infty$
- (D) $\sum b_n$ diverges
- (E) $\sum (b_n - a_n)$ diverges

Problem 074.5[See solution](#)

The series $\sum_{n=1}^{\infty} \log(1 + 1/n^2)$

- (A) Diverges by comparison with $\sum 1/n$
- (B) Diverges because its terms approach 1
- (C) Has negative terms
- (D) Converges by limit comparison with $\sum 1/n^2$
- (E) Converges conditionally

2. Calculus I

Infinite Series ▸ The ratio test

075 Ratio test

Book pp. 86-87. [Find in compact glossary](#)**In this entry:** ratio test; factorial series; absolute ratio; inconclusive ratio.**Definition and key ideas**

For a series with eventually nonzero terms, set $L = \lim_{n \rightarrow \infty} |a_{n+1}/a_n|$ when it exists. If $L < 1$, the series converges absolutely; if $L > 1$ or $L = \infty$, its terms fail to tend to zero and the series diverges. If $L = 1$, the test is inconclusive. This formulation applies to signed terms by using absolute values.

Factorials and exponentials often simplify dramatically in adjacent ratios. Cancel exact factors before taking limits, and remember that $(1 + 1/n)^n \rightarrow e$. Both convergent $\sum 1/n^2$ and divergent $\sum 1/n$ have limiting ratio one, so an equality case always requires another argument. A nonexistent ratio limit does not imply divergence.

Problem 075.1[See solution](#)

For $a > 0$, the ratio test proves convergence of $\sum_{n=1}^{\infty} \frac{n!}{(an)^n}$ precisely under which strict inequality?

- (A) $a > 1$ (B) $a > e$ (C) $a > 1/e$ (D) $a < 1/e$ (E) $a < e$

Problem 075.2[See solution](#)

For $a_n = n^3 2^n / n!$, what is $\lim a_{n+1}/a_n$, and what does it imply about $\sum a_n$?

- (A) 0, convergence (B) 1, convergence
(C) 2, convergence (D) 2, divergence
(E) ∞ , divergence

Problem 075.3[See solution](#)

Let $a_n = n!e^n/n^n$. Its adjacent ratio tends to 1. Using $(1 + 1/n)^n < e$, which conclusion follows?

- (A) $\sum a_n$ converges by the ratio test.
(B) $a_n \rightarrow 0$, so the series converges.
(C) a_n is increasing and $a_n \geq e$, so the series diverges by the term test.
(D) The ratio-test limit alone proves divergence.
(E) The terms alternate, so the series converges.

Problem 075.4[See solution](#)

For which real x does the ratio test alone prove convergence of $\sum_{n=1}^{\infty} nx^n/4^n$?

- (A) $|x| < 4$ (B) $|x| < 1$ (C) $x < 4$ (D) $|x| \leq 4$ (E) Every real x

Problem 075.5[See solution](#)

A positive sequence has $a_{2n-1} = 8 \cdot 4^{-n}$ and $a_{2n} = 4^{-n}$ for $n \geq 1$. Which statement is correct?

- (A) The adjacent ratio tends to 2, so the series diverges.
- (B) The adjacent ratio tends to $1/8$, so the sum is $8/7$.
- (C) The adjacent ratio has no limit, but the series converges to 3.
- (D) Infinitely many adjacent ratios exceed 1, so the series diverges.
- (E) The series converges to 9.

2. Calculus I

Infinite Series ▷ The root test

076 Root test and limsup formulationBook pp. 86-87. [Find in compact glossary](#)**In this entry:** root test; n th root; limsup root test; absolute convergence test.**Definition and key ideas**

For $\sum a_n$, let $L = \lim |a_n|^{1/n}$ if the limit exists. If $L < 1$, the series converges absolutely; if $L > 1$, it diverges; if $L = 1$, the test is inconclusive. The stronger version uses $L = \limsup |a_n|^{1/n}$ with the same strict conclusions, even when the ordinary limit fails to exist.

This test is suited to expressions raised to the n th power. Fixed polynomial factors disappear under the root because $n^{k/n} \rightarrow 1$. The ratio and root tests can give different levels of convenience; an oscillating adjacent ratio may coexist with a decisive root estimate. As with the ratio test, the boundary case one requires another convergence argument.

Problem 076.1[See solution](#)

Let $a_n = 2^{-n}$ for even n and $a_n = 3^{-n}$ for odd n . Which is correct?

- (A) $\sum a_n$ diverges because the root limit does not exist.
- (B) $\sum a_n$ converges and $\limsup a_n^{1/n} = 1/2$.
- (C) $\sum a_n$ converges and $\limsup a_n^{1/n} = 1/3$.
- (D) $\sum a_n$ diverges by the ratio test.
- (E) a_n does not tend to zero.

Problem 076.2[See solution](#)

For $c > 0$, the root test guarantees convergence of $\sum_{n=1}^{\infty} \frac{(1 + 1/n)^{n^2}}{c^n}$ precisely when which inequality holds?

- (A) $c > 1$ (B) $c > 2$ (C) $c > e$ (D) $c > e^2$ (E) $c < e$

Problem 076.3[See solution](#)

Let $a_n = 1$ when n is a power of 2, and $a_n = 2^{-n}$ otherwise. What are $\limsup a_n^{1/n}$ and the behavior of $\sum a_n$?

- (A) $1/2$; convergence
- (B) 1 ; convergence
- (C) 1 ; divergence
- (D) 2 ; divergence
- (E) 0 ; convergence

Problem 076.4[See solution](#)

For which $c > 0$ does the root test prove convergence of $\sum_{n=1}^{\infty} n^3/c^n$?

- (A) Every $c > 0$ (B) $c > 1$ (C) $c \geq 1$ (D) $0 < c < 1$ (E) $c > 3$

Problem 076.5[See solution](#)

The root test applied to $a_n = 1/n^2$ yields which conclusion?

- (A) It is inconclusive
- (B) It proves convergence
- (C) It proves divergence
- (D) It proves conditional convergence
- (E) It shows the terms do not vanish

2. Calculus I

Infinite Series ▸ The integral test

077 Integral test and remainder bounds

Book pp. 86-88. [Find in compact glossary](#)**In this entry:** integral test; eventually decreasing terms; integral remainder estimate; logarithmic series.**Definition and key ideas**

If f is positive, continuous, and decreasing on $[N_0, \infty)$ and $a_n = f(n)$, then $\sum a_n$ and $\int_{N_0}^{\infty} f(x)dx$ either both converge or both diverge. Eventual decrease suffices; the finite beginning is irrelevant. Their values are usually different, so the test is about convergence rather than equality of sums and integrals.

For a convergent positive decreasing series, the tail after N satisfies $\int_{N+1}^{\infty} f(x)dx \leq \sum_{n=N+1}^{\infty} a_n \leq \int_N^{\infty} f(x)dx$. These bounds give guaranteed numerical accuracy. A logarithmic substitution is especially useful for series involving $1/[n(\log n)^p]$ or iterated logarithms. Check positivity and monotonicity on a sufficiently late interval.

Problem 077.1[See solution](#)

For which p does $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ converge?

- (A) $p > 0$ (B) $p \geq 1$ (C) $p > 1$ (D) $p < 1$ (E) All real p

Problem 077.2[See solution](#)

Let $R_{100} = \sum_{n=101}^{\infty} 1/n^2$. Which bound follows from the integral test?

- (A) $1/102 \leq R_{100} \leq 1/101$ (B) $1/101 \leq R_{100} \leq 1/100$
 (C) $1/100 \leq R_{100} \leq 1/99$ (D) $R_{100} = 1/100$
 (E) $R_{100} \leq 1/10000$

Problem 077.3[See solution](#)

On $[1, \infty)$, let $f(x) = e^{-x} + g(x)$, where g consists of disjoint triangular spikes centered at each integer $n \geq 2$, each of height 1 and base width 2^{-n} , and is zero elsewhere. Which is true?

- (A) Both $\int_1^{\infty} f$ and $\sum_{n=1}^{\infty} f(n)$ converge.
 (B) The integral converges but the series diverges.
 (C) The series converges but the integral diverges.
 (D) Both diverge.
 (E) f cannot be positive and continuous.

Problem 077.4[See solution](#)

What is $\sum_{n=1}^{\infty} e^{-n} - \int_1^{\infty} e^{-x} dx$?

- (A) 0 (B) $1/e$ (C) $1/(e-1)$ (D) $1/[e(e-1)]$ (E) $1/(e+1)$

Problem 077.5[See solution](#)

For which real p does $\sum_{n=3}^{\infty} 1/[n \log n (\log \log n)^p]$ converge?

- (A) $p \geq 1$ (B) No p (C) $p > 0$ (D) $p < 1$ (E) $p > 1$

2. Calculus I

Alternating Series > Alternating-series test

078 Alternating series and their error estimates

Book pp. 88. [Find in compact glossary](#)

In this entry: alternating series; Leibniz alternating-series test; alternating remainder bound; eventual monotonicity.

Definition and key ideas

If $b_n \geq 0$, b_n decreases eventually, and $b_n \rightarrow 0$, then $\sum(-1)^{n+1}b_n$ converges. All three ingredients matter: alternating signs alone do not imply convergence, and a zero term limit without suitable control does not suffice. A finite exceptional beginning does not affect convergence.

When the decrease holds from the start, the error after N terms has magnitude at most b_{N+1} and the sign of the first omitted term. Thus consecutive partial sums bracket the sum. This often gives a simpler approximation guarantee than Taylor's derivative remainder. If the term magnitudes fail to decrease, look for absolute convergence or another argument rather than applying this test automatically.

If monotonicity begins only later, apply the remainder theorem to the omitted tail once that tail is decreasing to zero. The finitely many earlier irregular magnitudes do not affect this tail estimate. Alternating signs and terms tending to zero alone are insufficient.

Problem 078.1[See solution](#)

For the alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}/n^3$, what is the smallest N for which the standard bound $|R_N| \leq 1/(N+1)^3$ is at most 10^{-6} ?

- (A) 98 (B) 99 (C) 100 (D) 101 (E) 1000

Problem 078.2[See solution](#)

Let $S = \sum_{n=1}^{\infty}(-1)^{n+1}/\sqrt{n}$ and s_N be its N th partial sum. Which must hold?

- (A) $s_{2N-1} < S < s_{2N}$ (B) $s_{2N} < S < s_{2N-1}$
 (C) $S = s_{2N}$ (D) $|S - s_N| \geq 1/\sqrt{N+1}$
 (E) The series diverges.

Problem 078.3[See solution](#)

How does $\sum_{n=1}^{\infty}[(-1)^{n-1}/\sqrt{n} + 1/n]$ behave?

- (A) It converges absolutely.
 (B) It converges conditionally.
 (C) It diverges to $+\infty$.
 (D) It diverges to $-\infty$.
 (E) Its partial sums remain bounded but fail to converge.

Problem 078.4[See solution](#)

For $b_n = n^2/2^n$, what is the smallest integer $N \geq 0$ such that the omitted magnitudes $b_{N+1}, b_{N+2}, \dots$ are nonincreasing and hence admit the standard alternating-tail bound?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 078.5[See solution](#)

Let $S = \log 2$ and $s_N = \sum_{n=1}^N (-1)^{n-1}/n$. Which is an exact expression for $S - s_N$?

- (A) $(-1)^N \int_0^1 \frac{t^N}{1+t} dt$
(B) $(-1)^{N+1} \int_0^1 \frac{t^N}{1+t} dt$
(C) $(-1)^N \int_0^1 \frac{t^{N+1}}{1+t} dt$
(D) $(-1)^N/(N+1)$
(E) $\int_0^1 \frac{t^N}{1-t} dt$

2. Calculus I

Alternating Series > Absolute versus conditional convergence

079 Absolute and conditional convergence

Book pp. 88-89. [Find in compact glossary](#)**In this entry:** absolute convergence; conditional convergence; absolute convergence implies convergence; signed series.**Definition and key ideas**

A series $\sum a_n$ is absolutely convergent if $\sum |a_n|$ converges. Absolute convergence implies convergence. A convergent series is conditionally convergent when its absolute-value series diverges; the alternating harmonic series is the standard example. A divergent series is neither absolutely nor conditionally convergent.

Absolute convergence makes arbitrary bounded changes of signs harmless and permits rearrangement without changing the sum. Conditional convergence is more delicate: arbitrary rearrangement is not generally legitimate. If magnitudes have a convergent positive majorant, no alternating pattern is needed. Conversely, failure of absolute convergence does not determine whether the original signed series converges; analyze its cancellations separately.

Problem 079.1[See solution](#)

For which real p is $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^p}$ conditionally convergent?

- (A) $p > 1$ (B) $p \geq 1$ (C) $0 < p \leq 1$ (D) $0 \leq p < 1$ (E) $p \leq 0$

Problem 079.2[See solution](#)

If $\sum a_n$ converges absolutely and $|b_n| \leq 7$ for every n , which conclusion must hold?

- (A) $\sum b_n$ converges.
 (B) $\sum a_n b_n$ converges absolutely.
 (C) $\sum a_n/b_n$ converges.
 (D) $b_n \rightarrow 0$.
 (E) $\sum \sqrt{|a_n|}$ converges.

Problem 079.3[See solution](#)

For which p is $\sum_{n=2}^{\infty} (-1)^n / [n(\log n)^p]$ conditionally convergent?

- (A) Every real p (B) $0 < p \leq 1$ (C) $p \leq 1$ (D) $p > 1$ (E) $p < 0$

Problem 079.4[See solution](#)

Let $S = 1 - 1/2 + 1/3 - 1/4 + \cdots$. Rearrange by taking two positive terms, then one negative term, repeatedly: $1 + 1/3 - 1/2 + 1/5 + 1/7 - 1/4 + \cdots$. What is the new sum?

- (A) $S/2$
- (B) S
- (C) $3S/2$
- (D) $2S$
- (E) The rearranged series diverges.

Problem 079.5[See solution](#)

If $\sum a_n$ converges conditionally, define $P_N = \sum_{n=1}^N \max(a_n, 0)$ and $Q_N = \sum_{n=1}^N \max(-a_n, 0)$. What must happen?

- (A) P_N and Q_N both have finite limits.
- (B) $P_N \rightarrow \infty$ and Q_N has a finite limit.
- (C) $Q_N \rightarrow \infty$ and P_N has a finite limit.
- (D) $P_N \rightarrow \infty$ and $Q_N \rightarrow \infty$.
- (E) At least one of P_N, Q_N oscillates.

2. Calculus I

Power Series ▷ Convergence sets

080 Power series, radius, and interval of convergence

Book pp. 89-90. [Find in compact glossary](#)

In this entry: power series; center of a power series; radius of convergence; interval of convergence; endpoint testing; Cauchy–Hadamard formula.

Definition and key ideas

A power series centered at a is $\sum_{n=0}^{\infty} c_n(x - a)^n$. There is a radius $R \in [0, \infty]$ such that it converges absolutely for $|x - a| < R$ and diverges for $|x - a| > R$. Each boundary point must be tested separately. The real interval of convergence includes precisely the interior and whichever endpoints converge.

When $L = \lim |c_{n+1}/c_n|$ exists, $R = 1/L$, with $1/0 = \infty$ and $1/\infty = 0$. More generally $1/R = \limsup |c_n|^{1/n}$. Factorials may produce radius zero or infinity. The ratio test alone does not decide the endpoints, where alternating, comparison, or term tests may be required. Shifting the center changes the interval location, not this basic structure.

Problem 080.1

[See solution](#)

What is the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n3^n}$?

- (A) $(-1, 5)$ (B) $[-1, 5)$ (C) $(-1, 5]$ (D) $[-1, 5]$ (E) $(1, 3)$

Problem 080.2

[See solution](#)

What is the radius of convergence of $\sum_{n=0}^{\infty} \frac{n!(2n)!}{(3n)!} x^n$?

- (A) $4/27$ (B) $3/2$ (C) $9/4$ (D) $27/4$ (E) ∞

Problem 080.3

[See solution](#)

A real power series centered at 1 converges at $x = 4$ and diverges at $x = -2$. What is its interval of convergence?

- (A) $(-2, 4)$
(B) $[-2, 4)$
(C) $(-2, 4]$
(D) $[-2, 4]$
(E) It cannot be determined.

Problem 080.4[See solution](#)

What is the interval of convergence of $\sum_{n=1}^{\infty} (x+2)^{2n}/(n9^n)$?

- (A) $(-5, 1)$ (B) $(-5, 1]$ (C) $(-11, 7)$ (D) $[-5, 1)$ (E) $[-5, 1]$

Problem 080.5[See solution](#)

Let $c_{2n} = 2^n$ and $c_{2n+1} = 0$ for $n \geq 0$. What is the radius of convergence of $\sum_{m=0}^{\infty} c_m x^m$?

- (A) 0 (B) $1/2$ (C) $1/\sqrt{2}$ (D) $\sqrt{2}$ (E) ∞

Problem 080.6[See solution](#)

What are the radii of $\sum_{n=0}^{\infty} n!x^n$ and $\sum_{n=0}^{\infty} x^n/(n!)^2$, respectively?

- (A) 0, 0 (B) 0, ∞ (C) 1, 1 (D) ∞ , 0 (E) ∞ , ∞

2. Calculus I

Power Series ▸ Functions Defined by Power Series

081 Termwise differentiation and integration of power series

Book pp. 90-91. [Find in compact glossary](#)

In this entry: termwise differentiation; termwise integration; unchanged radius; endpoint caution; power-series regularity.

Definition and key ideas

Inside its open radius of convergence, a power series defines a continuous function with derivatives of every order. It can be differentiated and integrated term by term: $(\sum c_n(x-a)^n)' = \sum_{n \geq 1} n c_n(x-a)^{n-1}$, and an antiderivative is $C + \sum c_n(x-a)^{n+1}/(n+1)$. Both new series have the same radius.

Endpoint behavior may change: differentiation can destroy convergence and integration can create it. The interior theorem does not automatically authorize differentiation at a convergent endpoint. Start with the geometric series and differentiate or integrate to sum expressions with factors n , n^2 , or $1/n$. Keep starting indices and constants of integration correct.

Problem 081.1

[See solution](#)

For $|x| < 1$, what is $\sum_{n=1}^{\infty} n^2 x^n$?

- (A) $x/(1-x)^2$ (B) $x(1+x)/(1-x)^3$ (C) $x^2/(1-x)^3$ (D) $2x/(1-x)^3$ (E) $x/(1-x^2)$

Problem 081.2

[See solution](#)

The series $\sum_{n=1}^{\infty} x^n/n^2$ converges at $x = 1$. What happens to its termwise derivative series at $x = 1$?

- (A) It converges absolutely.
 (B) It converges conditionally.
 (C) It diverges to $+\infty$.
 (D) It converges to zero.
 (E) Its radius becomes zero.

Problem 081.3

[See solution](#)

Integrating the geometric power series twice from zero produces $\sum_{n=0}^{\infty} x^{n+2}/[(n+1)(n+2)]$. What is the interval of convergence of this new series?

- (A) $(-1, 1)$ (B) $[-1, 1)$ (C) $(-1, 1]$ (D) $[-1, 1]$ (E) $\mathbb{R}$

Problem 081.4[See solution](#)

What is $\sum_{n=0}^{\infty} 1/[(2n+1)4^n]$?

- (A) $\log 4$ (B) $2\log 3$ (C) $\log 2$ (D) $\frac{1}{2}\log 3$ (E) $\log 3$

Problem 081.5[See solution](#)

A power series for f centered at zero has radius exactly 2. After termwise differentiation and substitution, the series for $g(x) = f'(3x+1)$ is centered at which point and has what radius?

- (A) $-1/3, 2/3$ (B) $1/3, 2/3$ (C) $-1, 2$ (D) $-1/3, 2$ (E) $-1/3, 6$

2. Calculus I

Power Series ▷ Operations on series

082 Power-series algebra, substitution, and coefficient extraction

Book pp. 91. [Find in compact glossary](#)

In this entry: power-series substitution; Cauchy product; coefficient extraction; power-series multiplication; power-series division; geometric-series transformation.

Definition and key ideas

Within a region of absolute convergence, power series may be added, multiplied, and composed under the appropriate convergence condition. Their product coefficients are finite convolutions: the coefficient of x^n in $(\sum a_k x^k)(\sum b_j x^j)$ is $\sum_{k=0}^n a_k b_{n-k}$. Division near the center is possible when the divisor has nonzero constant term.

Substitute $g(x)$ into a known series only where $g(x)$ lies in its convergence interval. For a requested coefficient, retain only terms that can contribute to that degree; this is usually faster than taking many derivatives. The series for $\arctan x$ follows by integrating $1/(1+x^2) = \sum (-1)^n x^{2n}$. Boundary validity needs its own argument.

For division, multiply the unknown coefficient series by the denominator and equate coefficients; the nonzero constant term determines each next coefficient. Cauchy products of absolutely convergent series converge to the product of the sums, but two conditionally convergent series can have a divergent Cauchy product.

Problem 082.1

[See solution](#)

What is the coefficient of x^4 in the Maclaurin series of e^{x+x^2} ?

- (A) $1/24$ (B) $13/24$ (C) $25/24$ (D) $7/6$ (E) $3/2$

Problem 082.2

[See solution](#)

What is the coefficient of x^5 in $\log(1+x)/(1-x)$ about zero?

- (A) $1/5$ (B) $47/60$ (C) $77/60$ (D) $137/60$ (E) 5

Problem 082.3

[See solution](#)

For which real x does $\sum_{n=0}^{\infty} (x^2 - 1)^n$ converge?

- (A) $(-\sqrt{2}, \sqrt{2})$ (B) $[-\sqrt{2}, \sqrt{2}]$ (C) $(-\sqrt{2}, 0) \cup (0, \sqrt{2})$ (D) $(-1, 1)$ (E) $\mathbb{R} \setminus \{-\sqrt{2}, \sqrt{2}\}$

Problem 082.4

[See solution](#)

What is the coefficient of x^6 in the power series about zero for $1/(1-x-x^2)$?

- (A) 5 (B) 8 (C) 13 (D) 21 (E) 32

Problem 082.5[See solution](#)

Let $a_n = (-1)^n / \sqrt{n+1}$ and $c_n = \sum_{k=0}^n a_k a_{n-k}$. Which describes $\sum_{n=0}^{\infty} c_n$?

- (A) It converges absolutely to $(\sum a_n)^2$.
- (B) It converges conditionally to $(\sum a_n)^2$.
- (C) It diverges because c_n does not tend to zero.
- (D) It diverges to $+\infty$ because every c_n is positive.
- (E) It is a geometric series with ratio $-1/2$.

2. Calculus I

Taylor Series ▷ Coefficients and centers

083 Taylor coefficients, Taylor series, and analyticity

Book pp. 92, 94. [Find in compact glossary](#)

In this entry: Taylor coefficient; Taylor series; Maclaurin series; expansion center; analytic function; smooth nonanalytic function.

Definition and key ideas

If f has a power-series representation about a , its coefficients are uniquely $f^{(n)}(a)/n!$, and the representation is its Taylor series $\sum_{n \geq 0} f^{(n)}(a)(x - a)^n/n!$. At $a = 0$ it is called the Maclaurin series. The coefficient of $(x - a)^n$ is the derivative divided by $n!$, not the derivative itself.

Existence of derivatives of every order does not guarantee equality with the Taylor series. A function is analytic near a when it equals a convergent power series there. For example, $f(x) = e^{-1/x^2}$ for $x \neq 0$ with $f(0) = 0$ is smooth and has every derivative zero at zero, yet is positive elsewhere. To justify a Taylor representation, show that the remainder tends to zero.

A nonzero analytic function has isolated zeros locally: after factoring out its first nonzero Taylor power, the remaining factor stays nonzero near the center. Consequently, zeros accumulating at an interior point force the analytic function to vanish in a neighborhood of that point. Smooth functions alone need not satisfy this conclusion.

Problem 083.1

[See solution](#)

What is the coefficient of $(x - 1)^3$ in the Taylor series of $\log x$ about $x = 1$?

- (A) $-1/3$ (B) $1/6$ (C) $1/3$ (D) $1/2$ (E) 2

Problem 083.2

[See solution](#)

Which assertion is true for every function f infinitely differentiable on a neighborhood of zero?

- (A) Its Taylor series has positive radius of convergence.
(B) Its Taylor series equals f near zero.
(C) If all derivatives at zero vanish, then f vanishes nearby.
(D) If f has a power-series representation near zero, its coefficients are uniquely $f^{(n)}(0)/n!$.
(E) Its Taylor series converges for every real x .

Problem 083.3[See solution](#)

A real function f is analytic on a neighborhood of zero and $f(1/n) = 0$ for every sufficiently large integer n . What must be true?

- (A) f is odd near zero.
- (B) f is identically zero on some neighborhood of zero.
- (C) f has a nonzero derivative of finite order at zero.
- (D) Its Taylor series has radius zero.
- (E) f is zero for every real argument.

Problem 083.4[See solution](#)

Suppose f is infinitely differentiable near zero and $f^{(n)}(0) = (n!)^2$ for every $n \geq 0$. What is the radius of convergence of its Maclaurin series?

- (A) 0 (B) $1/2$ (C) 1 (D) 2 (E) ∞

Problem 083.5[See solution](#)

If f is analytic near zero and all its odd derivatives at zero vanish, which holds sufficiently near zero?

- | | |
|---------------------|-------------------------|
| (A) $f(x) = 0$ | (B) $f(-x) = f(x)$ |
| (C) f is constant | (D) f is a polynomial |
| (E) $f(-x) = -f(x)$ | |

2. Calculus I

Taylor Series ▷ Standard expansions

084 Standard Maclaurin expansions and leading-order limits

Book pp. 92-93, 104. [Find in compact glossary](#)

In this entry: exponential series; sine series; cosine series; logarithm series; arctangent series; leading nonzero term; order of vanishing.

Definition and key ideas

Essential expansions are $e^x = \sum x^n/n!$, $\sin x = \sum (-1)^n x^{2n+1}/(2n+1)!$, and $\cos x = \sum (-1)^n x^{2n}/(2n)!$, all valid for every real x . Also $\log(1+x) = \sum_{n \geq 1} (-1)^{n+1} x^n/n$ for $-1 < x \leq 1$, and $\arctan x = \sum_{n \geq 0} (-1)^n x^{2n+1}/(2n+1)$ for $-1 \leq x \leq 1$. The geometric series $\sum x^n = 1/(1-x)$ requires $|x| < 1$.

To evaluate a complicated limit, identify the first nonzero term after cancellation. If $f(x) = cx^m + o(x^m)$ with $c \neq 0$, then $f(x)/x^m \rightarrow c$. Multiplying leading terms adds orders and multiplies coefficients. Keep enough terms to survive subtraction; $e^x \approx 1+x$ is inadequate when both of those terms cancel.

Problem 084.1

[See solution](#)

What is $\lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{x^3}$?

- (A) $-1/3$ (B) $-1/6$ (C) 0 (D) $1/6$ (E) $1/3$

Problem 084.2

[See solution](#)

For which integer k is $\lim_{x \rightarrow 0} \frac{(e^{x^2} - 1 - x^2)(\log(1+x) - x)}{x^k}$ finite and nonzero, and what is the limit?

- (A) $k = 4$, limit $-1/4$ (B) $k = 5$, limit $-1/2$
 (C) $k = 6$, limit $-1/4$ (D) $k = 6$, limit $1/4$
 (E) $k = 8$, limit $-1/4$

Problem 084.3

[See solution](#)

What is $\sum_{n=0}^{\infty} \frac{(-1)^n (1 - 3^{-(2n+1)})}{2n+1}$?

- (A) $\arctan(1/4)$ (B) $\arctan(1/2)$ (C) $\pi/6$ (D) $\pi/3$ (E) $\pi/4 + \arctan(1/3)$

Problem 084.4

[See solution](#)

For $h(x) = (e^x - e^{-x})/(e^x + e^{-x})$, what is $h^{(3)}(0)$?

- (A) -6 (B) -2 (C) $-1/3$ (D) $1/3$ (E) 2

Problem 084.5[See solution](#)

What is $\lim_{x \rightarrow 0} (e^{\sin x} - e^x)/x^3$?

- (A) 0 (B) $-1/6$ (C) $1/3$ (D) $1/6$ (E) $-1/3$

2. Calculus I

Taylor Polynomials ▸ Approximation and error

085 Taylor polynomials and rigorous remainder bounds

Book pp. 93-94, 104. [Find in compact glossary](#)

In this entry: Taylor polynomial; Taylor theorem; Lagrange remainder; truncation error; error bound; degree versus number of terms.

Definition and key ideas

The degree- n Taylor polynomial about a is $P_n(x) = \sum_{j=0}^n f^{(j)}(a)(x-a)^j/j!$. If f has the required $n+1$ derivatives on the intervening interval, Taylor's theorem gives $f(x) - P_n(x) = f^{(n+1)}(\xi)(x-a)^{n+1}/(n+1)!$ for some point ξ between a and x . A bound $|f^{(n+1)}| \leq M$ yields error at most $M|x-a|^{n+1}/(n+1)!$.

The sign of the remainder can show whether an approximation is high or low. Use a valid derivative bound over the entire interval, not just at the center. Degree and number of nonzero terms differ: a sine polynomial of degree five has three nonzero terms. A sufficient upper bound exceeding a tolerance does not prove that the true error exceeds it; minimum-term questions need a lower bound or a direct remainder argument as well.

The finite Taylor theorem does not require the infinite Taylor series to converge at the evaluation point: it needs the stated differentiability on the intervening interval. Combining Taylor expansions at symmetric points can cancel odd powers and yield derivative-approximation bounds.

Problem 085.1

[See solution](#)

For $P_3(x) = 1 + x + x^2/2 + x^3/6$, use $e^{1/2} < 2$ to bound $e^{1/2} - P_3(1/2)$. Which statement follows?

- (A) $-1/192 < R < 0$ (B) $0 < R < 1/384$
 (C) $0 < R < 1/192$ (D) $1/192 < R < 1/96$
 (E) $R = 1/384$

Problem 085.2

[See solution](#)

What is the smallest number of nonzero Maclaurin terms of $\sin x$ needed to approximate $\sin(1/2)$ with absolute error less than 10^{-4} ?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 085.3

[See solution](#)

Let $f \in C^4([-h, h])$, where $h > 0$, and suppose $|f^{(4)}(x)| \leq M$. Which bound follows for $|[f(h) - 2f(0) + f(-h)]/h^2 - f''(0)|$?

- (A) $Mh^4/24$ (B) $Mh^2/24$ (C) $Mh^2/12$ (D) $Mh/6$ (E) $M/2$

Problem 085.4[See solution](#)

The quadratic Taylor polynomial of $\log x$ about 1 is $P_2(x) = (x-1) - (x-1)^2/2$. Which statement about approximation at $x = 3$ is correct?

- (A) Taylor's theorem fails because the Taylor series diverges there.
- (B) The remainder is zero since $P_2(3) = 0$.
- (C) The remainder is $\log 3 = 8/(3\xi^3)$ for some $1 < \xi < 3$.
- (D) The remainder is negative because the quadratic coefficient is negative.
- (E) The remainder equals $8/3$, obtained by setting $\xi = 1$.

Problem 085.5[See solution](#)

A smooth function satisfies $f(0) = f'(0) = f''(0) = f^{(3)}(0) = 0$. What upper bound on $|f(1) - P_2(1)|$ follows from these data alone?

- (A) 0
- (B) $1/6$
- (C) $1/2$
- (D) 1
- (E) No finite upper bound follows.

3. Calculus II

Analytic Geometry of $\mathbb{R}^3$ ▷ Coordinates and Vectors**086 Euclidean space, vectors, norm, and distance**Book pp. 105-107. [Find in compact glossary](#)

In this entry: Euclidean space; Cartesian coordinates; octant; first octant; standard unit vectors; vector components; scalar multiplication; norm; unit vector; distance.

Definition and key ideas

A point of $\mathbb{R}^3$ is an ordered triple (x, y, z) . The coordinate planes divide space into eight octants; the first has $x, y, z > 0$. The standard unit vectors are $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, and $\mathbf{k} = (0, 0, 1)$. Vector addition and scalar multiplication act componentwise. The displacement from P to Q is $Q - P$, whose Euclidean norm is $\|Q - P\| = \sqrt{\sum_i (Q_i - P_i)^2}$.

A nonzero vector v determines the unit vector $v/\|v\|$. Points on a segment have the form $(1 - t)P + tQ$ with $0 \leq t \leq 1$; allowing all real t gives its containing line. Distance ratios along a line become ratios of parameter differences, but an external division point requires t outside $[0, 1]$.

Problem 086.1[See solution](#)

Let $P = (1, -2, 0)$ and $Q = (3, 1, 6)$. A point R on their line lies beyond Q and satisfies $\|R - P\| = 3\|R - Q\|$. What is $\|R\|^2$?

- (A) 57 (B) 81 (C) $\frac{413}{4}$ (D) 109 (E) $\frac{441}{4}$

Problem 086.2[See solution](#)

If $u, v \in \mathbb{R}^3$, $\|u\| = 3$, $\|v\| = 2$, and $\|u + v\| = 4$, what is $\|2u - v\|^2$?

- (A) 25 (B) 28 (C) 31 (D) 34 (E) 40

Problem 086.3[See solution](#)

A unit vector u lies in the first octant and its components are in the ratio $1 : 2 : 3$. What is $\|u - \mathbf{i}\|^2$?

- (A) $1 - 1/\sqrt{14}$ (B) $2 - 2/\sqrt{14}$ (C) $2 + 2/\sqrt{14}$ (D) $13/14$ (E) 1

Problem 086.4[See solution](#)

What is the squared distance from $(2, 3, 2)$ to the closed segment joining $(1, 0, 0)$ and $(1, 2, 0)$?

- (A) 1 (B) 5 (C) 6 (D) 9 (E) 14

Problem 086.5[See solution](#)

Which equation describes all points equidistant from $P = (1, 2, 0)$ and $Q = (3, 0, 2)$?

- (A) $x - y + z = 0$ (B) $x - y + z = 2$
 (C) $x + y + z = 2$ (D) $x - y + z = 4$
 (E) $x^2 + y^2 + z^2 = 4$

3. Calculus II

Analytic Geometry of R3 ▷ The Dot Product

087 Dot product, angle, and orthogonality

Book pp. 107-108. [Find in compact glossary](#)**In this entry:** dot product; scalar product; angle between vectors; orthogonal vectors; perpendicular vectors.**Definition and key ideas**

The dot product is $u \cdot v = \sum_i u_i v_i = \|u\| \|v\| \cos \theta$ for nonzero vectors. It is symmetric and bilinear: $u \cdot (av + bw) = a(u \cdot v) + b(u \cdot w)$. Two vectors are orthogonal when their dot product is zero. The zero vector is orthogonal to every vector, although its angle with another vector is undefined.

The identity $\|u \pm v\|^2 = \|u\|^2 + \|v\|^2 \pm 2u \cdot v$ often replaces coordinate calculations. Cauchy–Schwarz, $|u \cdot v| \leq \|u\| \|v\|$, bounds angles and linear expressions, with equality precisely for linearly dependent vectors. Dot products are distributive over addition; this is not an associativity law for a product of three vectors.

Problem 087.1[See solution](#)

Nonzero vectors u, v satisfy $\|u\| = 2\|v\|$ and $(u + v) \perp (u - 2v)$. What is the angle between u and v ?

- (A) 0 (B) $\pi/6$ (C) $\pi/3$ (D) $\pi/2$ (E) $2\pi/3$

Problem 087.2[See solution](#)

For unit vectors u, v, w with $u + v + w = 0$, what is $(u - v) \cdot (v - w)$?

- (A) -3 (B) $-\frac{3}{2}$ (C) $-\frac{1}{2}$ (D) $\frac{1}{2}$ (E) $\frac{3}{2}$

Problem 087.3[See solution](#)

What is the cosine of the angle between $u = (1, 2, -2)$ and $v = (2, -1, 2)$?

- (A) $\frac{4}{9}$ (B) $-\frac{2}{9}$ (C) $-\frac{8}{9}$ (D) $\frac{5}{9}$ (E) $-\frac{4}{9}$

Problem 087.4[See solution](#)

What is the maximum of $x + 2y + 2z$ subject to $x^2 + y^2 + z^2 = 1$ and $x = y$?

- (A) $\sqrt{13}/2$ (B) $\sqrt{17}/2$ (C) $\sqrt{34}/2$ (D) 3 (E) $\sqrt{17}$

Problem 087.5[See solution](#)

Vectors $u, v, w \in \mathbb{R}^3$ satisfy $u = v + w$, $u \cdot v = 0$, and $u \cdot w = 0$. Which conclusion must hold?

- (A) $v = w = 0$
- (B) $v \cdot w = 0$
- (C) $u = 0$ and $v = -w$
- (D) u, v, w are three nonzero mutually perpendicular vectors.
- (E) The angle between u and v is $\pi/2$.

3. Calculus II

Analytic Geometry of R3 ▷ The Dot Product

088 Scalar and vector projection

Book pp. 107-108. [Find in compact glossary](#)**In this entry:** scalar projection; vector projection; orthogonal decomposition.**Definition and key ideas**

For $a \neq 0$, the scalar projection of b onto the directed vector a is $(a \cdot b)/\|a\|$; it can be negative. The vector projection is $\text{proj}_a b = (a \cdot b)a/(a \cdot a)$. Thus b splits uniquely as a component parallel to a plus a component perpendicular to a .

Projection solves a nearest-point problem: the point ta closest to b has $t = (a \cdot b)/(a \cdot a)$. Consequently the squared distance to the line spanned by a is $\|b\|^2 - (a \cdot b)^2/\|a\|^2$. For an affine line, first subtract a point on that line. Do not confuse a signed scalar projection with the nonnegative length of a projected vector.

Reflection across a line through the origin preserves the parallel component and reverses the perpendicular component, so it maps b to $2\text{proj}_a b - b$. Reversing the chosen direction of the line changes a scalar projection's sign but leaves its vector projection unchanged.

Problem 088.1[See solution](#)What is the minimum value of $\|(2, -1, 3) - t(1, 2, 2)\|^2$ over real t ?

- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

Problem 088.2[See solution](#)

Vectors a, b have lengths 2, 3. The scalar projection of b onto a is -1 . What is the length of the component of a perpendicular to b ?

- (A)
- $\frac{2}{3}$
- (B)
- $\frac{2\sqrt{2}}{3}$
- (C)
- $\sqrt{3}$
- (D)
- $\frac{4\sqrt{2}}{3}$
- (E) 2

Problem 088.3[See solution](#)What is the point of the affine line $r = (1, 0, 1) + t(1, 1, 0)$ nearest to $(4, 1, 3)$?

- (A) (2, 1, 1) (B) (3, 2, 1) (C)
- $(7/2, 5/2, 1)$
- (D) (4, 3, 1) (E) (1, 0, 1)

Problem 088.4[See solution](#)

For $a \neq 0$, the scalar projection of b onto a is -2 , and $p = \text{proj}_a b$. What are the scalar and vector projections of b onto $-3a$, respectively?

- (A)
- $-6, 3p$
- (B)
- $6, -p$
- (C)
- $2, p$
- (D)
- $2, -p$
- (E)
- $-2, p$

Problem 088.5[See solution](#)

Let L be the line spanned by $(1, 2, 2)$. Reflection across L keeps the component parallel to L and negates the perpendicular component. What is the image of $(3, 0, 0)$?

(A) $(-7/3, 4/3, 4/3)$ **(B)** $(1/3, 2/3, 2/3)$ **(C)** $(-3, 0, 0)$ **(D)** $(7/3, -4/3, -4/3)$ **(E)** $(1, 2, 2)$

3. Calculus II

Analytic Geometry of $\mathbb{R}^3$ ▷ The Cross Product

089 Cross product, orientation, and area

Book pp. 109-110. [Find in compact glossary](#)

In this entry: cross product; vector product; right-hand rule; normal vector; parallelogram area; triangle area; vector triple-product identity.

Definition and key ideas

In oriented $\mathbb{R}^3$, $u \times v = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)$. It is perpendicular to both inputs, and its direction follows the right-hand rule. Its magnitude is $\|u\|\|v\|\sin\theta$. The product is bilinear and anticommutative, $v \times u = -u \times v$, and vanishes exactly when the inputs are dependent.

The parallelogram spanned by u, v has area $\|u \times v\|$; the triangle has half that area. A useful computational identity is $\|u \times v\|^2 = \|u\|^2\|v\|^2 - (u \cdot v)^2$. The cross product is not associative. A nonzero cross product of two directions in a plane supplies its normal.

The vector triple-product identity is $a \times (b \times c) = b(a \cdot c) - c(a \cdot b)$; the parentheses matter. For an orthogonal linear map Q , $(Qu) \times (Qv) = \det(Q)Q(u \times v)$. Rotations preserve orientation, while a reflection introduces the minus sign $\det(Q) = -1$.

Problem 089.1

[See solution](#)

If $\|u\| = 3$, $\|v\| = 4$, and $u \cdot v = 6$, what is the area of the triangle with vertices $0, u + v, u - v$?

- (A) $3\sqrt{3}$ (B) $6\sqrt{3}$ (C) $12\sqrt{3}$ (D) 18 (E) 24

Problem 089.2

[See solution](#)

Let $u = (1, 1, 0)$ and $v = (0, 1, 1)$. Which unit vector is normal to their plane and has positive second component?

- (A) $\frac{(1,1,1)}{\sqrt{3}}$ (B) $\frac{(1,-1,1)}{\sqrt{3}}$ (C) $\frac{(-1,1,-1)}{\sqrt{3}}$ (D) $\frac{(-1,1,1)}{\sqrt{3}}$ (E) $\frac{(1,0,-1)}{\sqrt{2}}$

Problem 089.3

[See solution](#)

Unit vectors u, v satisfy $u \cdot v = 1/2$. What is $v \cdot [u \times (u \times v)]$?

- (A) -1 (B) $-3/4$ (C) $-1/2$ (D) 0 (E) $3/4$

Problem 089.4

[See solution](#)

Among vectors v satisfying $(1, 1, 0) \times v = (1, -1, 2)$, which has minimum norm?

- (A) $(0, 2, 1)$ (B) $(-1, 1, 1)$ (C) $(1, 3, 1)$ (D) $(-1, 1, -1)$ (E) $(1, -1, 1)$

Problem 089.5[See solution](#)

Let $R(x, y, z) = (-x, y, z)$, reflection in the yz -plane. If $u \times v = (1, 2, 3)$, what is $(Ru) \times (Rv)$?

- (A) $(-1, 2, 3)$
- (B) $(1, 2, 3)$
- (C) $(1, -2, -3)$
- (D) $(-1, -2, -3)$
- (E) It cannot be determined.

3. Calculus II

Analytic Geometry of R3 ▷ The Triple Scalar Product

090 Scalar triple product and volume

Book pp. 110-111. [Find in compact glossary](#)

In this entry: scalar triple product; parallelepiped; coplanarity; oriented volume; tetrahedron volume; Gram determinant.

Definition and key ideas

The scalar triple product $(u \times v) \cdot w$ equals the determinant whose rows are u, v, w . Its absolute value is the volume of their parallelepiped, while its sign records orientation. Cyclically permuting the three inputs preserves the value; swapping two negates it.

The triple product vanishes exactly when the vectors are linearly dependent, equivalently when their directions lie in a common plane. A tetrahedron with vertices P, Q, R, S has volume $|((Q - P) \times (R - P)) \cdot (S - P)|/6$. Always use displacement vectors from one common vertex; raw position vectors work only when that vertex is the origin.

If G is the Gram matrix $G_{ij} = v_i \cdot v_j$ of three vectors, their squared scalar triple product is $\det G$: writing the vectors as rows of M gives $G = MM^T$, hence $\det G = (\det M)^2$. This computes volume from lengths and pairwise angles.

Problem 090.1

[See solution](#)

The scalar triple product $[u, v, w] = (u \times v) \cdot w$ equals 5. What is $[u + v, v + w, w + u]$?

- (A) 0 (B) 5 (C) 10 (D) 15 (E) 20

Problem 090.2

[See solution](#)

What is the volume of the tetrahedron with vertices $(1, 0, 0), (1, 2, 0), (1, 0, 3), (5, 1, 1)$?

- (A) 2 (B) 3 (C) 4 (D) 8 (E) 24

Problem 090.3

[See solution](#)

For how many real numbers t are $(1, 0, 0), (0, 1, 0), (0, 0, 1)$, and (t, t^2, t^3) coplanar?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinitely many

Problem 090.4

[See solution](#)

Vectors u, v have lengths 2, 3 and make angle $\pi/6$. Among vectors w of length 4, what is the largest possible parallelepiped volume $|(u \times v) \cdot w|$?

- (A) 6 (B) 12 (C) $12\sqrt{3}$ (D) 24 (E) $24\sqrt{3}$

Problem 090.5

[See solution](#)

Unit vectors u, v, w satisfy $u \cdot v = u \cdot w = v \cdot w = 1/2$. What is the volume of the tetrahedron with vertices $0, u, v, w$?

- (A) $1/12$ (B) $\sqrt{2}/12$ (C) $\sqrt{3}/12$ (D) $\sqrt{2}/6$ (E) $1/6$

3. Calculus II

Analytic Geometry of R3 ▷ Lines in 3-Space

091 Parametric and symmetric equations of lines

Book pp. 112-113. [Find in compact glossary](#)**In this entry:** line in space; direction vector; parametric equations; symmetric equations; skew lines.**Definition and key ideas**

A line through P with nonzero direction v is $r(t) = P + tv$, $t \in \mathbb{R}$. If every component of v is nonzero, eliminating t gives $(x - P_x)/v_x = (y - P_y)/v_y = (z - P_z)/v_z$. If $v_i = 0$, retain the equation $x_i = P_i$ instead of dividing by zero.

To intersect a line with a plane, substitute its parameterization into the plane equation. Two spatial lines can be parallel, intersecting, coincident, or skew: skew lines are nonparallel and do not intersect. When testing intersection use independent parameters for the two lines; equating their parameters without justification can miss intersections.

For nonparallel directions u, v , the distance between lines $P + tu$ and $Q + sv$ is $|(Q - P) \cdot (u \times v)| / \|u \times v\|$. The unit common normal measures the invariant perpendicular separation; components along either line can be removed by choosing t, s . A nonzero numerator detects skewness, while zero gives intersecting nonparallel lines.

Problem 091.1[See solution](#)

For what value of a do the lines $(x, y, z) = (t, 1 + t, 2 - t)$ and $(x, y, z) = (1 + s, 2 - s, a + 2s)$ intersect?

- (A) -1 (B) 0 (C) 1 (D) 2 (E) 3

Problem 091.2[See solution](#)

The line through $(2, -1, 3)$ parallel to $(0, 2, -1)$ meets $x + y + z = 8$ at Q . What is the distance from Q to $(2, -1, 3)$?

- (A) $2\sqrt{5}$ (B) $3\sqrt{5}$ (C) $4\sqrt{5}$ (D) $5\sqrt{5}$ (E) $6\sqrt{5}$

Problem 091.3[See solution](#)

What is the distance between the lines $r = (0, 0, 0) + t(1, 1, 0)$ and $r = (0, 1, 2) + s(0, 1, 1)$?

- (A) $\sqrt{3}/6$ (B) $\sqrt{3}/3$ (C) $2\sqrt{3}/3$ (D) $\sqrt{3}$ (E) $2\sqrt{3}$

Problem 091.4[See solution](#)

Which symmetric description gives the entire line $(x, y, z) = (2, 1, -3) + t(0, 3, -2)$?

- (A) $x = 2, \quad (y - 1)/3 = (z + 3)/(-2)$
- (B) $(y - 1)/3 = (z + 3)/(-2)$ with no restriction on x
- (C) $x = 2, \quad (y - 1)/2 = (z + 3)/(-3)$
- (D) $x = 2, \quad (y + 1)/3 = (z - 3)/(-2)$
- (E) $x = 0, \quad (y - 1)/3 = (z + 3)/(-2)$

Problem 091.5[See solution](#)

How are the lines $r = (1, 2, 3) + t(1, -1, 2)$ and $r = (3, 0, 7) + s(-2, 2, -4)$ related?

- (A) They are skew.
- (B) They are distinct and parallel.
- (C) They intersect at exactly one point.
- (D) They coincide.
- (E) They are perpendicular.

3. Calculus II

Analytic Geometry of $\mathbb{R}^3$ ▷ Planes in 3-Space

092 Planes and normal equations

Book pp. 113–114. [Find in compact glossary](#)**In this entry:** plane; normal equation; coordinate plane; parallel planes; angle between planes.**Definition and key ideas**

A plane through P with nonzero normal n is $n \cdot (x - P) = 0$, or $n_1x + n_2y + n_3z = d$. Its normal is determined by the coefficients of the coordinates. Three noncollinear points determine a unique plane; cross two independent displacement vectors to obtain its normal.

Parallel planes have proportional normals. The acute angle between two planes is the acute angle between their normals, so use an absolute value in the cosine formula. A line lies in a plane precisely when one of its points satisfies the plane equation and its direction is perpendicular to the normal. The direction of the intersection of two nonparallel planes is their normals' cross product.

Planes with proportional normals coincide only when their constants have the same proportionality factor; otherwise they are distinct parallel planes. For a whole family of plane equations, equate coefficients of the free parameter to find points belonging to every plane.

Problem 092.1[See solution](#)

A plane contains the line $r(t) = (1, 0, 1) + t(1, 2, 0)$ and the point $(0, 1, 2)$. Which is its equation?

- (A) $2x - y + 3z = 5$ (B) $2x - y - 3z = -1$
 (C) $x + 2y + z = 2$ (D) $x - 2y + 3z = 4$
 (E) $2x + y + 3z = 5$

Problem 092.2[See solution](#)

The planes $x + y + z = 1$ and $x - y + 2z = 3$ intersect in a line. Which vector is parallel to that line?

- (A) $(-3, 1, -2)$ (B) $(3, -1, -2)$ (C) $(3, 1, -2)$ (D) $(1, 3, -2)$ (E) $(1, -3, 2)$

Problem 092.3[See solution](#)

For which real k is the acute angle between the planes $x = 0$ and $x + ky + z = 1$ equal to $\pi/3$?

- (A) $k = 0$ (B) $k = \pm 1$ (C) $k = \pm\sqrt{2}$ (D) $k = \pm\sqrt{3}$ (E) No real k

Problem 092.4[See solution](#)

For which real a do the planes $2x - y + 2z = 3$ and $4x - 2y + az = 12$ have no common point?

- (A) $a = 0$ (B) $a = 2$ (C) $a = 4$ (D) $a = 6$ (E) No real a

Problem 092.5[See solution](#)

What is the intersection of all planes $\lambda x + (\lambda - 1)y + z = 2\lambda + 1$, where λ ranges over all real numbers?

- (A) The empty set
- (B) The single point $(2, 0, 1)$
- (C) The line $(2 - t, t, 1 + t)$, $t \in \mathbb{R}$
- (D) The plane $x + y + z = 3$
- (E) The line $(2 + t, t, 1 - t)$, $t \in \mathbb{R}$

3. Calculus II

Analytic Geometry of R3 ▷ Cylinders

093 Cylindrical surfaces and generators

Book pp. 115-116. [Find in compact glossary](#)

In this entry: cylinder; right circular cylinder; elliptic cylinder; parabolic cylinder; generator; element of a cylinder.

Definition and key ideas

A cylindrical surface is the union of lines through a base curve, all parallel to a fixed nonzero direction called its generator direction; these individual lines are its elements. The cylinder need not be circular. An equation independent of z , such as $x^2 + 4y^2 = 1$, describes a surface invariant under translation along the z -axis.

For an oblique cylinder, parameterize a base point $P(s)$ and write $P(s) + tv$. Eliminate s, t using linear combinations constant along v . This is more reliable than assuming a missing coordinate: only axis-parallel cylinders necessarily have a missing coordinate in the given Cartesian system.

A right circular cylinder has circular sections perpendicular to its axis. An elliptic cylinder has elliptical perpendicular sections; an axis-aligned example is $x^2/a^2 + y^2/b^2 = 1$ with z free. A parabolic cylinder, such as $z = x^2$ with y free, is obtained by translating a parabola. A tilted section of a circular cylinder can be an ellipse even though its perpendicular sections are circles.

Problem 093.1

[See solution](#)

The parabola $(s, s^2, 0)$ generates a cylinder in direction $(1, -1, 2)$. Which equation describes it?

- (A) $y + z = (x - z)^2$ (B) $y + \frac{z}{2} = (x - \frac{z}{2})^2$
 (C) $y - \frac{z}{2} = (x + \frac{z}{2})^2$ (D) $y + 2z = (x - 2z)^2$
 (E) $y = (x - z)^2$

Problem 093.2

[See solution](#)

The surface $(x - z)^2 + 4(y + 2z)^2 = 1$ is a cylinder. Which vector gives a generator direction?

- (A) $(1, 2, 1)$ (B) $(1, -2, 1)$ (C) $(1, -2, -1)$ (D) $(2, 1, 1)$ (E) $(0, 0, 1)$

Problem 093.3

[See solution](#)

The cylinder $x^2 + y^2 = 1$ is cut by the plane $z = x$. What is the area of the elliptical region enclosed by the intersection, measured in that plane?

- (A) $\pi/\sqrt{2}$ (B) π (C) $\sqrt{2}\pi$ (D) 2π (E) $\sqrt{3}\pi$

Problem 093.4

[See solution](#)

The curve $(s, 0, s^2)$ is translated parallel to the y -axis. Which surface results?

- (A) $z = y^2$ (B) $z = x^2$ (C) $x^2 + z^2 = 1$ (D) $z = x^2 + y^2$ (E) $y = x^2$

Problem 093.5[See solution](#)

Which surface is a circular cylinder with axis parallel to $(1, 0, 0)$?

(A) $x^2 + (z + 2)^2 = 9$

(B) $x^2 + y^2 = 9$

(C) $x^2 - y^2 - z^2 = 9$

(D) $(y - 1)^2 + (z + 2)^2 = 9$

(E) $x^2 + y^2 + z^2 = 9$

3. Calculus II

Analytic Geometry of R3 ▷ Surfaces of Revolution

094 Surfaces of revolution

Book pp. 117-118. [Find in compact glossary](#)**In this entry:** surface of revolution; axis of revolution; radial distance; ellipsoid of revolution.**Definition and key ideas**

A surface of revolution is obtained by rotating a plane curve about a line in its plane. Rotation about the z -axis preserves z and the distance $\sqrt{x^2 + y^2}$ from the axis; rotation about the x -axis preserves x and $\sqrt{y^2 + z^2}$.

If the curve $z = f(x)$ in the xz -plane rotates about the x -axis, the result is $y^2 + z^2 = f(x)^2$, with the original restrictions on x . More general implicit equations require considering both possible signs of the radial coordinate. Squaring can introduce unwanted branches, so retain restrictions such as nonnegative heights or a limited generating interval.

An ellipsoid of revolution is an ellipsoid with two equal semiaxes, obtained by rotating an ellipse about one of its axes. About the z -axis its equation is $(x^2 + y^2)/a^2 + z^2/c^2 = 1$. The equal semiaxes a are perpendicular to the rotation axis, while c is along it. For a translated axis, measure radial distance from that line rather than from the coordinate axis.

Problem 094.1[See solution](#)

The segment $z = 2 - x$, $0 \leq x \leq 2$, in the xz -plane is revolved about the z -axis. Which describes the resulting surface?

- (A) $z = 2 - x^2 - y^2$ (B) $x^2 + y^2 = (2 - z)^2$, $0 \leq z \leq 2$
 (C) $x^2 + y^2 = z^2$ (D) $x^2 + z^2 = (2 - y)^2$
 (E) $x^2 + y^2 + z^2 = 4$

Problem 094.2[See solution](#)

The parabola $z = y^2$ in the yz -plane is revolved about the y -axis. At $y = 2$, what is the area of the circular cross-section enclosed by the surface?

- (A) 4π (B) 8π (C) 12π (D) 16π (E) 32π

Problem 094.3[See solution](#)

The curve $y = x^2$, $x \geq 0$, in the xy -plane is revolved around the y -axis. Which equation describes the surface?

- (A) $y = x^2 + z^2$ (B) $y = x^2 - z^2$ (C) $y^2 = x^2 + z^2$ (D) $y^2 = x^2 - z^2$ (E) $x = y^2 + z^2$

Problem 094.4[See solution](#)

The parabola $(x, y, z) = (t^2, 0, t)$ is rotated about the line $x = 1, y = 0$. Which equation describes the resulting surface?

- (A) $x^2 + y^2 = (z^2 - 1)^2$
- (B) $(x - 1)^2 + y^2 = (z^2 - 1)^2$
- (C) $(x - 1)^2 + y^2 = z^4$
- (D) $(x - 1)^2 + y^2 = z^2 - 1$
- (E) $x^2 + (y - 1)^2 = (z - 1)^4$

Problem 094.5[See solution](#)

The ellipse $x^2/9 + z^2/4 = 1$ in the xz -plane is rotated about the z -axis. Which equation describes the surface?

- (A) $(x^2 + y^2)/9 + z^2/4 = 1$
- (B) $x^2/9 + (y^2 + z^2)/4 = 1$
- (C) $(x^2 + y^2)/4 + z^2/9 = 1$
- (D) $x^2 + y^2 + z^2 = 9$
- (E) $(x^2 + y^2)/9 - z^2/4 = 1$

3. Calculus II

Analytic Geometry of $\mathbb{R}^3$ ▸ Level Curves and Level Surfaces

095 Level sets, contours, and quadric surfaces

Book pp. 119-120. [Find in compact glossary](#)**In this entry:** level curve; level surface; contour curve; cross-section; circular paraboloid; elliptic paraboloid; hyperboloid of one sheet.**Definition and key ideas**

The level set of f at value c is $\{p : f(p) = c\}$. For $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ it is usually a curve, and for $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ usually a surface, but it may be empty, a point, or singular. A contour of the graph $z = f(x, y)$ at height c is its intersection with $z = c$; projecting this contour to the xy -plane gives the level curve.

Quadrics are recognized through cross-sections. The elliptic paraboloid $z = x^2/a^2 + y^2/b^2$ has elliptic horizontal sections. The one-sheet hyperboloid $x^2/a^2 + y^2/b^2 - z^2/c^2 = 1$ has nonempty elliptic sections at every height. A sphere satisfies $(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$. For comparison, $z^2 = x^2 + y^2$ is a circular double cone, and $z^2/c^2 - x^2/a^2 - y^2/b^2 = 1$ is a two-sheet hyperboloid, with sections only for $|z| \geq c$.

The elliptic paraboloid becomes a circular paraboloid when $a = b$. Its zero-height level is a point, positive-height levels are ellipses (circles when $a = b$), and negative-height levels are empty. A circular double cone has two nappes meeting at its vertex; unlike a two-sheet hyperboloid, it is connected through that vertex.

Problem 095.1[See solution](#)

The area enclosed by the horizontal section at height h of $x^2/4 + y^2 - z^2 = 1$ is twice the area of the section at height 0. What is $|h|$?

- (A) $\frac{1}{2}$ (B) 1 (C) $\sqrt{2}$ (D) 2 (E) $\sqrt{3}$

Problem 095.2[See solution](#)

For $f(x, y) = x^2 - y^2$, which description of the level set $f^{-1}(0)$ is correct?

- (A) A point
 (B) A circle
 (C) One straight line
 (D) Two intersecting straight lines
 (E) A hyperbola with two disjoint branches

Problem 095.3[See solution](#)

For $F(x, y, z) = x^2 + y^2 - z^2$, which lists the surfaces $F = -1$, $F = 0$, and $F = 1$, in that order?

- (A) Sphere; point; sphere
- (B) One-sheet hyperboloid; circular double cone; two-sheet hyperboloid
- (C) Two-sheet hyperboloid; circular double cone; one-sheet hyperboloid
- (D) Two-sheet hyperboloid; plane; one-sheet hyperboloid
- (E) Elliptic paraboloid; circular cylinder; one-sheet hyperboloid

Problem 095.4[See solution](#)

For the graph $z = x^2 + 4y^2$, which pair correctly gives the level curve of value 9 and its contour in three-dimensional space?

- (A) $x^2 + 4y^2 = 9$ in $\mathbb{R}^2$; the same ellipse at $z = 9$
- (B) $x^2 + 4y^2 = 9$ in $\mathbb{R}^2$; the same ellipse at $z = 0$
- (C) $x^2 + 4y^2 = 81$ in $\mathbb{R}^2$; the same ellipse at $z = 9$
- (D) $x^2 + y^2 = 9$ in $\mathbb{R}^2$; a circle at $z = 9$
- (E) The single point $(0, 0)$; the point $(0, 0, 9)$

Problem 095.5[See solution](#)

Which level set of $f(x, y) = x^2 + y^2 - 4x + 6y$ is a single point?

- (A) $f^{-1}(0)$ (B) $f^{-1}(13)$ (C) $f^{-1}(-14)$ (D) $f^{-1}(-13)$ (E) $f^{-1}(-12)$

Problem 095.6[See solution](#)

For which real c does the level surface $(x - 1)^2 + (y + 2)^2 + (z - 3)^2 = c$ meet the plane $z = 0$ in exactly one point?

- (A) $c = 0$ (B) $c = 3$ (C) $c = 9$ (D) $c > 9$ (E) $c \geq 9$

3. Calculus II

Analytic Geometry of R3 ▷ Cylindrical Coordinates

096 Cylindrical coordinates

Book pp. 121. [Find in compact glossary](#)**In this entry:** cylindrical coordinates; cylindrical radius; azimuthal angle; coordinate surface.**Definition and key ideas**

Cylindrical coordinates (r, θ, z) combine polar coordinates in the xy -plane with height: $x = r \cos \theta$, $y = r \sin \theta$, $r \geq 0$, and z is unchanged. Thus $r^2 = x^2 + y^2$. The angle is defined modulo 2π , and is arbitrary on the z -axis.

The surface $r = a > 0$ is a circular cylinder, while $z = c$ is a horizontal plane. With $r \geq 0$, a fixed value of θ gives a vertical half-plane bounded by the z -axis, not the whole plane. Cylindrical coordinates are especially efficient for rotational symmetry; angle-dependent expressions may simplify through $r^2 \cos 2\theta = x^2 - y^2$ and $r^2 \sin 2\theta = 2xy$.

Problem 096.1[See solution](#)

The surface $z = r^2 \sin(2\theta)$ is intersected by $y = x$. What is the resulting relation between z and x ?

- (A) $z = x^2$ (B) $z = 2x^2$ (C) $z = -2x^2$ (D) $z = 4x^2$ (E) $z = 0$

Problem 096.2[See solution](#)

A point has cylindrical coordinates $(2, 5\pi/6, -3)$. What is its squared distance from $(1, 0, 1)$?

- (A) 19 (B) 21 (C) $21 - 2\sqrt{3}$ (D) $21 + 2\sqrt{3}$ (E) 25

Problem 096.3[See solution](#)

Which cylindrical bounds describe $(x - 1)^2 + y^2 \leq 1$, $0 \leq z \leq 1$, using $r \geq 0$?

- (A) $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$, $0 \leq z \leq 1$ (B) $0 \leq r \leq 2 \cos \theta$, $-\pi/2 \leq \theta \leq \pi/2$, $0 \leq z \leq 1$
 (C) $0 \leq r \leq 2 \sin \theta$, $-\pi/2 \leq \theta \leq \pi/2$, $0 \leq z \leq 1$ (D) $1 \leq r \leq 2$, $0 \leq \theta \leq \pi$, $0 \leq z \leq 1$
 (E) $0 \leq r \leq \cos \theta$, $-\pi/2 \leq \theta \leq \pi/2$, $0 \leq z \leq 1$

Problem 096.4[See solution](#)

With $r \geq 0$ and unrestricted z , what Cartesian set is represented by $\theta = \pi/4$?

- (A) $x = y$ with no sign restriction
- (B) $x = y \geq 0$ with unrestricted z
- (C) $x = -y \geq 0$ with unrestricted z
- (D) $x = y = 0$
- (E) $x^2 + y^2 = 1$

Problem 096.5[See solution](#)

For $t > 0$, let a point have cylindrical coordinates $(r, \theta, z) = (t^2, 1/t, t)$. What happens as $t \rightarrow 0^+$?

- (A) The Cartesian point has no limit because the azimuth has no limit.
- (B) The Cartesian point tends to $(0, 0, 0)$ although its azimuth has no limit.
- (C) The Cartesian point tends to $(1, 0, 0)$.
- (D) The Cartesian point tends to $(0, 0, 1)$.
- (E) The distance from the origin tends to infinity.

3. Calculus II

Analytic Geometry of R3 ▷ Spherical Coordinates

097 Spherical coordinates

Book pp. 121-122. [Find in compact glossary](#)**In this entry:** spherical coordinates; radial distance; polar angle; azimuth; sphere; cone.**Definition and key ideas**

Using the book's order (ρ, ϕ, θ) , $\rho \geq 0$ is distance from the origin, $0 \leq \phi \leq \pi$ is the angle from the positive z -axis, and θ is azimuth. The conversion is $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$, so $x^2 + y^2 + z^2 = \rho^2$.

A constant positive ρ gives a sphere. A fixed ϕ usually gives one nappe of a cone, with exceptional cases $\phi = 0, \pi/2, \pi$ yielding an axis ray or plane. A fixed azimuth gives a half-plane. Conventions for the two angles vary between texts: identify the polar angle by its role in $z = \rho \cos \phi$. Translating a sphere generally creates a nonconstant radial bound.

At a pole, azimuth is arbitrary, and at the origin both angles are arbitrary. For a sphere whose interior does not contain the origin, a ray may have zero, one, or two intersections; substituting spherical coordinates gives a quadratic in ρ . Check both its discriminant and the requirement $\rho \geq 0$ rather than keeping every algebraic root.

Problem 097.1[See solution](#)

In spherical coordinates, the sphere $x^2 + y^2 + (z - 2)^2 = 4$ has a nonzero radial coordinate satisfying which equation?

- (A) $\rho = 2 \cos \phi$ (B) $\rho = 4 \cos \phi$ (C) $\rho = 4 \sin \phi$ (D) $\rho = 2 \sin \phi \cos \theta$ (E) $\rho = 4$

Problem 097.2[See solution](#)

A point lies on $x^2 + y^2 + z^2 = 25$, has polar angle $\pi/3$, and azimuth $3\pi/4$. What is $x + y + z$?

- (A) 0 (B) $\frac{5}{2}$ (C) 5 (D) $\frac{5\sqrt{3}}{2}$ (E) $\frac{5(1+\sqrt{6})}{2}$

Problem 097.3[See solution](#)

Using polar angle ϕ measured from the positive z -axis, which spherical equation describes $z = \sqrt{x^2 + y^2}$ away from the origin?

- (A) $\phi = 3\pi/4$ (B) $\rho = 1$ (C) $\phi = \pi/4$ (D) $\phi = \pi/2$ (E) $\theta = \pi/4$

Problem 097.4[See solution](#)

Using (ρ, ϕ, θ) with ϕ polar and $\rho \geq 0$, which lists the sets $\phi = 0$, $\phi = \pi/2$, and $\theta = 0$, respectively?

- (A) Positive z -axis ray; xy -plane; half-plane $y = 0, x \geq 0$
- (B) Entire z -axis; xy -plane; entire xz -plane
- (C) Positive x -axis ray; yz -plane; entire xy -plane
- (D) Cone of opening $\pi/2$; positive z -axis ray; sphere
- (E) Origin only; upper hemisphere; entire xz -plane

Problem 097.5[See solution](#)

A ray from the origin with polar angle $\phi \in [0, \pi]$ meets the sphere $x^2 + y^2 + (z - 3)^2 = 1$ in two distinct points precisely for which angles?

- (A) $0 \leq \phi < \arcsin(1/3)$
- (B) $0 \leq \phi \leq \arcsin(1/3)$
- (C) $0 \leq \phi < \arccos(1/3)$
- (D) $\arcsin(1/3) < \phi < \pi - \arcsin(1/3)$
- (E) $0 \leq \phi < \pi/2$

3. Calculus II

Partial Derivatives ▸ Definition and Geometric Interpretation

098 Partial derivatives and coordinate traces

Book pp. 122-124. [Find in compact glossary](#)**In this entry:** partial derivative; independent variable; coordinate trace; partial derivative as slope.**Definition and key ideas**

For $f(x, y)$, $f_x(a, b) = \lim_{h \rightarrow 0} [f(a + h, b) - f(a, b)]/h$ when this limit exists; f_y is defined by holding x fixed. Algebraically, treat all other independent variables as constants. Geometrically, f_x is the slope of the tangent to the graph's trace in the vertical plane $y = b$.

Existence of the partial derivatives at a point alone does not imply continuity or differentiability there. For piecewise formulas, compute derivatives at exceptional points from the definition instead of substituting into a formula valid only elsewhere. A tangent direction to the trace $y = b$ is $(1, 0, f_x)$, whereas the trace $x = a$ has direction $(0, 1, f_y)$.

If $f_x = 0$, the one-variable mean value theorem makes f constant in x on each connected horizontal interval of its domain; disconnected slices need not share that constant. Conversely, integrating a specified x -partial gives an arbitrary function of the other variables, not just one numerical constant. A derivative along a path changes every input specified by that path.

Problem 098.1[See solution](#)

The plane $x = 1$ cuts $z = \arctan(xy) + x^2y$ in a curve. Which vector is tangent to this curve at $(1, 1, 1 + \pi/4)$?

- (A) $(1, 0, 3/2)$ (B) $(0, 2, 3)$ (C) $(0, 3, 2)$ (D) $(2, 0, 3)$ (E) $(0, 1, 2)$

Problem 098.2[See solution](#)

Define $f(0, 0) = 0$ and $f(x, y) = xy/(x^2 + y^2)$ elsewhere. Which statement is true at the origin?

- (A) f is differentiable
 (B) f is continuous but not differentiable
 (C) $f_x = f_y = 0$ but f is discontinuous
 (D) f_x exists but f_y does not
 (E) Neither partial derivative exists

Problem 098.3[See solution](#)

Let $f(x, y) = \log(x/y)$ for $x, y > 0$, and $g(x) = f(x, x^2)$. What are $f_x(1, 1)$ and $g'(1)$, respectively?

- (A) $1, 1$ (B) $-1, -1$ (C) $1, -1$ (D) $-1, 1$ (E) $0, 0$

Problem 098.4[See solution](#)

Let $D = \{(x, y) : x \neq 0\}$. Which statement about a continuously differentiable function on D with $f_x = 0$ everywhere is correct?

- (A) f must be constant on all of D .
- (B) $f(-1, 0) = f(1, 0)$ must hold.
- (C) f cannot take both values 0 and 1.
- (D) f may equal 0 on $x < 0$ and 1 on $x > 0$.
- (E) No such function can be continuously differentiable.

Problem 098.5[See solution](#)

A continuously differentiable f satisfies $f_x(x, y) = 2xy + e^x$ and $f(0, y) = \sin y$. What is $f_y(1, 0)$?

- (A) 0 (B) 1 (C) 2 (D) e (E) $e + 1$

3. Calculus II

Partial Derivatives ▸ Higher-Order Partial Derivatives

099 Mixed partials and equality of differentiation orders

Book pp. 124-125. [Find in compact glossary](#)

In this entry: higher-order partial derivative; mixed partial derivative; Clairaut theorem; Schwarz theorem.

Definition and key ideas

Second partial derivatives differentiate first partial derivatives. Here f_{xy} means $(f_x)_y$; in fraction notation this is $\partial^2 f / (\partial y \partial x)$. For higher orders, read subscript notation left to right and operator fractions right to left.

If the second partial derivatives are continuous on a neighborhood of a point, the mixed partials agree there: $f_{xy} = f_{yx}$. Smoothness similarly allows reordering higher derivatives. Mere existence of the two mixed derivatives is insufficient; exceptional points in piecewise functions must be checked from difference quotients. Equal mixed partials are also the local compatibility condition behind potential functions.

For an analytic two-variable expansion, the coefficient of $x^m y^n$ equals $\partial^{m+n} f(0,0) / (\partial y^n \partial x^m)$ divided by $m!n!$. For smooth sums, choose the order of differentiation separately for each term; a derivative of order above a polynomial factor's degree annihilates that factor.

Problem 099.1

[See solution](#)

Let $f(x, y) = e^{xy}$. What is $f_{xxy}(1, 0)$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) e

Problem 099.2

[See solution](#)

Let $f(0, 0) = 0$ and $f(x, y) = xy(x^2 - 2y^2)/(x^2 + y^2)$ elsewhere. What is $f_{xy}(0, 0) - f_{yx}(0, 0)$?

- (A) -3 (B) -1 (C) 0 (D) 1 (E) 3

Problem 099.3

[See solution](#)

For $f(x, y) = e^{x^2 y}$, what is $\partial^9 f / (\partial y^3 \partial x^6)|_{(0,0)}$?

- (A) 0 (B) 120 (C) 360 (D) 720 (E) 4320

Problem 099.4

[See solution](#)

For which real k can a C^2 function on $\mathbb{R}^2$ satisfy $f_x = 2xy + 3y^2$ and $f_y = x^2 + kxy + \sin y$?

- (A) 0 (B) 2 (C) 3 (D) 6 (E) No real k

Problem 099.5[See solution](#)

Let $f(x, y) = x^2e^{y^2} + y^2 \log(1 + x)$ for $x > -1$. What is $\partial^7 f / (\partial x^3 \partial y^4)$?

- (A) 0 (B) $12e^{y^2}$ (C) $24/(1 + x)^3$ (D) $48y/(1 + x)^3$ (E) $120e^{y^2}$

3. Calculus II

Partial Derivatives ▸ The Tangent Plane to a Surface

100 Tangent planes and implicit partial differentiation

Book pp. 126-127. [Find in compact glossary](#)**In this entry:** tangent plane; implicit differentiation; implicit function condition; regular level surface.**Definition and key ideas**

If f is differentiable at (a, b) , the graph $z = f(x, y)$ has tangent plane $z - f(a, b) = f_x(a, b)(x - a) + f_y(a, b)(y - b)$. This includes the horizontal plane when both first partials vanish; no nonzero-gradient requirement is needed for a graph.

If a C^1 equation $F(x, y, z) = c$ satisfies $F_z \neq 0$ at a point, it locally defines $z = f(x, y)$, and $f_x = -F_x/F_z$, $f_y = -F_y/F_z$. Its tangent plane is $F_x(x - a) + F_y(y - b) + F_z(z - d) = 0$. More generally a nonzero full gradient suffices for a regular level surface, even when $F_z = 0$; one may then solve for a different coordinate.

A vanishing gradient of a particular defining function makes that gradient test inconclusive; it does not by itself prove that the geometric surface is singular. For instance, cubing a regular defining equation preserves its zero set while making its gradient vanish there.

Problem 100.1[See solution](#)

Near $(1, 1, 1)$, the equation $xz + yz^2 = 2$ defines $z = f(x, y)$. What is $f_x + 2f_y$ at $(1, 1)$?

- (A) -2 (B) -1 (C) $-\frac{1}{3}$ (D) $\frac{1}{3}$ (E) 1

Problem 100.2[See solution](#)

Which point lies in the tangent plane to $z = x^2 + xy + y^2$ at $(1, -1, 1)$?

- (A) $(0, 0, 1)$ (B) $(2, 1, 0)$ (C) $(1, 1, 0)$ (D) $(0, 1, -1)$ (E) $(2, 0, 2)$

Problem 100.3[See solution](#)

What is the tangent plane at the origin to the graph $z = |xy|$?

- (A) $z = 0$
 (B) $z = x + y$
 (C) $z = x - y$
 (D) $x = 0$
 (E) No tangent plane exists because the formula contains an absolute value.

Problem 100.4[See solution](#)

For the sphere $x^2 + y^2 + z^2 = 1$ at $(1, 0, 0)$, which statement is correct?

- (A) The surface is singular because $F_z = 0$.
- (B) It is locally a smooth graph $z = f(x, y)$ with horizontal tangent plane.
- (C) It is locally a smooth graph $x = g(y, z)$ with tangent plane $x = 1$.
- (D) The tangent plane is $z = 0$.
- (E) Every plane through $(1, 0, 0)$ is tangent.

Problem 100.5[See solution](#)

Let $F(x, y, z) = (z - x - y^2)^3$. Which is the tangent plane to $F = 0$ at the origin?

- (A) $z = 0$
- (B) $z = x$
- (C) $z = y$
- (D) $x + y + z = 0$
- (E) No tangent plane exists because $\nabla F(0) = 0$.

3. Calculus II

Partial Derivatives ▸ Linear Approximations

101 Total differential and linear approximation

Book pp. 127-128. [Find in compact glossary](#)**In this entry:** linear approximation; linearization; total differential; differentiability; first-order error.**Definition and key ideas**

Differentiability at a means $f(a+h) = f(a) + L(h) + o(\|h\|)$ for a linear map L ; for scalar-valued f , $L(h) = \nabla f(a) \cdot h$. The total differential is $df = \sum_i f_{x_i} dx_i$, and the linearization is $f(a) + df$. Continuous first partials near a guarantee differentiability.

The differential estimates a small change rather than the exact finite change. For products of powers, dividing by the original value yields useful relative-error formulas, such as $dV/V = 2dr/r + dh/h$ for $V = \pi r^2 h$. If the first-order term vanishes, the change may still be nonzero at second order; do not mistake a zero differential for constancy.

Problem 101.1[See solution](#)

For $f(x, y) = x^2 e^y$, what is the coefficient of t in its linear approximation along $(x, y) = (2 + 3t, -t)$ near $t = 0$?

- (A) -4 (B) 0 (C) 4 (D) 8 (E) 12

Problem 101.2[See solution](#)

The radius of a cylinder increases by 1% while its height decreases by 2%. What is the exact ratio of its new volume to its old volume?

- (A) 1 (B) 0.999698 (C) 1.0002 (D) 0.98 (E) 1.02

Problem 101.3[See solution](#)

Define $f(0, 0) = 0$ and $f(x, y) = x^2 y / (x^2 + y^2)$ elsewhere. Which statement holds at the origin?

- (A) f is discontinuous.
 (B) f is continuous, has both partials zero, but is not differentiable.
 (C) f is differentiable with zero differential.
 (D) f_x does not exist.
 (E) f_y does not exist.

Problem 101.4[See solution](#)

For positive x, y, z , let $Q = x^2 / (yz^3)$. If small relative input errors have magnitudes at most 1%, 2%, and 0.5%, respectively, what is the largest possible magnitude of the first-order relative error in Q ?

- (A) 1.5% (B) 2.5% (C) 3.5% (D) 5.5% (E) 7%

Problem 101.5[See solution](#)

A differentiable f has $f(0, 0) = 0$ and $\nabla f(0, 0) = (2, -1)$. If $\lim_{t \rightarrow 0} [f(t, t^2) - 2t]/t^2$ exists as a finite real number, which values can it have?

- (A) Only -1
- (B) Only 0
- (C) Only 1
- (D) Only nonnegative values
- (E) Any real value

3. Calculus II

Partial Derivatives ▸ The Chain Rule for Partial Derivatives

102 Multivariable chain rule and held-fixed variables

Book pp. 128-131. [Find in compact glossary](#)

In this entry: multivariable chain rule; dependent variable; intermediate variable; independent variable; total derivative; cyclic partial derivative identity.

Definition and key ideas

For $z = F(u(x, y), v(x, y), x)$, the total dependence on x gives $z_x = F_u u_x + F_v v_x + F_x$, with each derivative of F evaluated at the corresponding intermediate variables. A dependency diagram organizes the rule: multiply derivatives along every path, then add over paths. Along a curve $r(t)$, $d(f \circ r)/dt = \nabla f(r(t)) \cdot r'(t)$.

What is held fixed matters. For a regular relation among x, y, z , the cyclic identity is

$$(\partial x / \partial y)_z (\partial y / \partial z)_x (\partial z / \partial x)_y = -1,$$

provided the relevant derivatives and reciprocals exist. The minus sign comes from implicit differentiation, so these derivatives cannot be canceled as ordinary fractions.

For C^2 scalar f and a twice differentiable path r , $(f \circ r)'' = r'^T H_f r' + \nabla f \cdot r''$. The Hessian term measures change along the current velocity; the additional term accounts for acceleration of the path.

Problem 102.1

[See solution](#)

Let $z = u^2 + vx$, $u = x + y^2$, and $v = xy$. What is z_x at $(x, y) = (1, 2)$?

- (A) 12 (B) 14 (C) 16 (D) 18 (E) 20

Problem 102.2

[See solution](#)

Positive variables satisfy $x^2 + y^2 + z^2 = 1$. What is $(\partial x / \partial y)_z (\partial y / \partial z)_x (\partial z / \partial x)_y$?

- (A) -1 (B) 0 (C) 1 (D) xyz (E) $-xyz$

Problem 102.3

[See solution](#)

If $F(x, y) = x^2 + y^2$, $x = t^2$, and $y = t + 1$, what is dF/dt at $t = 1$?

- (A) 8 (B) -8 (C) 4 (D) 16 (E) 9

Problem 102.4

[See solution](#)

Let $z = F(u, v, y)$, $u = f(v, x)$, and $v = g(x, y)$. At the relevant point, $F_u = 2$, $F_v = 3$, $F_y = 5$, $f_v = 7$, and $g_y = 11$, with all other arguments held fixed in these derivatives. What is $\partial z / \partial y$ holding x fixed?

- (A) 38 (B) 82 (C) 159 (D) 187 (E) 192

Problem 102.5[See solution](#)

At $r(0)$, a C^2 function has gradient $(1, 2)$ and Hessian $\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$. If $r'(0) = (1, -1)$ and $r''(0) = (0, 3)$, what is $(f \circ r)''(0)$?

(A) 4 (B) 6 (C) 8 (D) 10 (E) 12

3. Calculus II

Directional Derivatives and the Gradient ▸ Rates of Change

103 Directional derivative and steepest ascent

Book pp. 132-133. [Find in compact glossary](#)

In this entry: directional derivative; gradient; del operator; steepest ascent; steepest descent; maximum rate of change.

Definition and key ideas

The gradient is $\nabla f = (f_{x_1}, \dots, f_{x_n})$. For a differentiable function and unit vector u , the directional derivative is $D_u f = \nabla f \cdot u$. If the direction is supplied as a nonunit vector v , normalize it first when the requested rate is per unit distance.

Cauchy–Schwarz gives $-\|\nabla f\| \leq D_u f \leq \|\nabla f\|$. If the gradient is nonzero, the maximizing direction is $\nabla f / \|\nabla f\|$ and the minimizing direction is its negative. If the gradient is zero all first-order directional rates vanish. Without differentiability, existing directional derivatives need not be given by a linear function of direction.

Problem 103.1[See solution](#)

At P , a differentiable function has directional derivatives 3 along $(1, 0)$ and 4 along $(3, 4)$. What is its maximum directional derivative at P ?

- (A) 4 (B) $\frac{\sqrt{193}}{4}$ (C) $\frac{\sqrt{265}}{4}$ (D) 5 (E) $\sqrt{34}$

Problem 103.2[See solution](#)

For $f(x, y, z) = xy + yz + zx$, what is the directional derivative at $(1, 2, -1)$ toward the origin?

- (A) $-\frac{2}{\sqrt{6}}$ (B) 0 (C) $\frac{1}{\sqrt{6}}$ (D) $\frac{2}{\sqrt{6}}$ (E) $\sqrt{6}$

Problem 103.3[See solution](#)

For $f(x, y) = 3x + 4y$, there are two unit directions u, v with directional derivative 3. What is $u \cdot v$?

- (A) -1 (B) $-7/25$ (C) 0 (D) $7/25$ (E) $3/5$

Problem 103.4[See solution](#)

For $f(x, y) = x^3 - 3xy^2$ at the origin, which statement is correct?

- (A) The maximum unit directional derivative is 3.
 (B) The maximum unit directional derivative is 1.
 (C) All directional derivatives vanish and the origin is a local minimum.
 (D) All directional derivatives vanish and the origin is a local maximum.
 (E) All directional derivatives vanish, but the origin is neither a local maximum nor a local minimum.

Problem 103.5[See solution](#)

Set $f(0, 0) = 0$ and $f(x, y) = x^3y/(x^6 + y^2)$ elsewhere. Which statement is true at the origin?

- (A) f is differentiable with zero gradient.
- (B) Every directional derivative is zero, but f is discontinuous.
- (C) Exactly one directional derivative exists.
- (D) f is continuous but has no directional derivatives.
- (E) Some directional derivative is infinite.

3. Calculus II

Directional Derivatives and the Gradient ▸ Normals to Level Sets

104 Gradient normals and tangent directions

Book pp. 134. [Find in compact glossary](#)**In this entry:** normal to a level set; tangent vector; normal line; intersection of level surfaces.**Definition and key ideas**

If f is C^1 and $\nabla f(P) \neq 0$, its gradient is normal to the regular level set through P . Every tangent vector v satisfies $\nabla f(P) \cdot v = 0$. The normal line is $P + t\nabla f(P)$, and the tangent plane to a level surface is $\nabla f(P) \cdot (x - P) = 0$.

For the intersection of two regular level surfaces $f = c$ and $g = d$, independent gradients give the tangent direction $\nabla f \times \nabla g$. At a critical point with vanishing gradient the normal equation becomes vacuous, so the surface must be analyzed separately. A tangent direction need not remain on the curved surface as a straight line; it describes only first-order motion.

Problem 104.1[See solution](#)

At $(1, 1, 1)$, which vector is tangent to the curve where $x^2 + y^2 + z^2 = 3$ and $x + 2y + 3z = 6$ intersect?

- (A) $(1, 1, -2)$ (B) $(1, -2, 1)$ (C) $(2, -1, -1)$ (D) $(1, 0, -1)$ (E) $(0, 3, -2)$

Problem 104.2[See solution](#)

At $P = (1, 1, 1)$, consider unit tangent directions to $x^2 + y^2 + z^2 = 3$. What is the maximum directional derivative of $f = x - y + 2z$ among these directions?

- (A) $\sqrt{2}$ (B) 2 (C) $\sqrt{14/3}$ (D) $\sqrt{6}$ (E) 3

Problem 104.3[See solution](#)

The normal line to $xy + z^2 = 5$ at $(1, 1, 2)$ meets $z = 0$ at $(a, b, 0)$. What is $a + b$?

- (A) -1 (B) 0 (C) 1 (D) 2 (E) 3

Problem 104.4[See solution](#)

Which describes $x^2 + y^2 - z^2 = 0$ at the origin?

- (A) A regular surface with tangent plane $z = 0$
 (B) A regular surface with tangent plane $x + y - z = 0$
 (C) A surface with a unique normal line parallel to the z -axis
 (D) A cone vertex with no single tangent plane in the usual smooth-surface sense
 (E) An isolated point of the surface

Problem 104.5[See solution](#)

A twice differentiable curve lies on $x^2 + y^2 + z^2 = 3$, with $r(0) = (1, 1, 1)$ and $r'(0) = (1, -2, 1)$. What is $(1, 1, 1) \cdot r''(0)$?

- (A) -6 (B) -3 (C) 0 (D) 3 (E) 6

3. Calculus II

Max/Min Problems ▷ Critical Points and the Second-Derivative Test

105 Critical points, Hessian, and saddle points

Book pp. 135–136. [Find in compact glossary](#)

In this entry: critical point; stationary point; local maximum; local minimum; saddle point; Hessian matrix; second-derivative test.

Definition and key ideas

An interior local extremum of a differentiable function must be stationary: $\nabla f = 0$. A critical point may also refer to a point where the derivative does not exist. A saddle point has arbitrarily nearby values both above and below its value. For C^2 functions of two variables, set $D = f_{xx}f_{yy} - f_{xy}^2$, the determinant of the Hessian matrix.

At a stationary point, $D > 0$ and $f_{xx} > 0$ imply a strict local minimum; $D > 0$ and $f_{xx} < 0$ imply a strict local maximum; $D < 0$ implies a saddle. When $D = 0$ the test is inconclusive. Positive behavior on coordinate axes does not exclude a saddle along oblique or curved paths. In higher dimensions the signs of the Hessian's eigenvalues play the analogous role.

Problem 105.1[See solution](#)

For what real values of a does $f(x, y) = x^2 + 2axy + y^2$ have a strict local minimum at the origin?

- (A) $a > 0$ (B) $a < 1$ (C) $|a| < 1$ (D) $|a| \leq 1$ (E) All real a

Problem 105.2[See solution](#)

The function $f(x, y) = x^4 + y^4 - 3x^2y^2$ has which behavior at the origin?

- (A) Strict local minimum
(B) Strict local maximum
(C) Saddle point
(D) No critical point
(E) Locally constant

Problem 105.3[See solution](#)

At the origin, $f(x, y) = x^2 + y^2 + 2x$ has a positive definite Hessian. Which conclusion is correct?

- (A) The origin is a strict local minimum.
(B) The origin is a strict local maximum.
(C) The origin is a saddle critical point.
(D) The origin is not a critical point and is not a local extremum.
(E) The Hessian determinant must be zero.

Problem 105.4[See solution](#)

For $f(x, y) = \sqrt{x^2 + y^2} - x/2$, which describes the origin?

- (A) A strict global minimum where differentiability fails
- (B) A differentiable strict minimum with positive definite Hessian
- (C) A saddle point
- (D) A strict local maximum
- (E) A point where the gradient is nonzero

Problem 105.5[See solution](#)

At the origin, the Hessian determinant is zero for both $f = x^4 + y^4$ and $g = (y - x^2)^2 - x^6$. How are the points classified?

- (A) Both strict local minima
- (B) Both saddle points
- (C) f : strict local minimum; g : saddle point
- (D) f : saddle point; g : strict local minimum
- (E) Neither function has a critical point there

Problem 105.6[See solution](#)

What are all stationary points of $f(x, y) = x^3 - 3x - y^2$, with their classifications?

- (A) $(1, 0)$ minimum; $(-1, 0)$ saddle
- (B) $(1, 0)$ saddle; $(-1, 0)$ strict local maximum
- (C) $(1, 0)$ strict local maximum; $(-1, 0)$ minimum
- (D) Both $(1, 0)$ and $(-1, 0)$ are saddles.
- (E) Only $(0, 0)$, a saddle

3. Calculus II

Max/Min Problems ▸ Absolute Extrema on a Closed Region

106 Absolute extrema and boundary analysis

Book pp. 136-137. [Find in compact glossary](#)

In this entry: absolute maximum; absolute minimum; closed and bounded region; compact region; extreme value theorem; boundary critical point.

Definition and key ideas

A continuous function on a nonempty compact subset of Euclidean space attains its absolute maximum and minimum. Closed and bounded subsets of $\mathbb{R}^n$ are compact. To locate extrema on a region, examine interior stationary or nondifferentiable points and all boundary pieces, including corners and endpoints.

A boundary restriction often reduces to a one-variable problem. Compare actual function values at all candidates; classifying each interior point is unnecessary for this comparison. Without compactness, extrema can fail to be attained even when a finite supremum or infimum exists. The condition $\nabla f = 0$ does not apply to every boundary extremum.

Problem 106.1[See solution](#)

What is the maximum of xy on $x \geq 0$, $y \geq 0$, $x + 2y \leq 6$?

- (A) 3 (B) 4 (C) $\frac{9}{2}$ (D) 6 (E) 9

Problem 106.2[See solution](#)

What is the minimum of $x^2 + y^2 - 2x - 4y$ on $0 \leq x, y \leq 1$?

- (A) -5 (B) -4 (C) -3 (D) -2 (E) 0

Problem 106.3[See solution](#)

What is the minimum of $(x - 3)^2 + (y - 4)^2$ on the closed disk $x^2 + y^2 \leq 4$?

- (A) 3 (B) 9 (C) 25 (D) 49 (E) 81

Problem 106.4[See solution](#)

On $D = \{(x, y) : 0 < x^2 + y^2 \leq 1\}$, what can be said about $f(x, y) = x^2 + y^2$?

- (A) It attains both minimum 0 and maximum 1.
(B) Its infimum is 0, not attained, and its maximum is 1.
(C) Its minimum is 1 and it has no maximum.
(D) Neither its infimum nor its supremum is finite.
(E) It attains a positive minimum smaller than 1.

Problem 106.5[See solution](#)

What is the absolute maximum of $|x| + y^2$ on the closed disk $x^2 + y^2 \leq 1$?

- (A) 1 (B) $5/4$ (C) $\sqrt{2}$ (D) $3/2$ (E) 2

3. Calculus II

Max/Min Problems ▸ Max/Min with a Constraint; Lagrange Multipliers

107 Constrained extrema and Lagrange multipliers

Book pp. 137-139. [Find in compact glossary](#)

In this entry: constraint; constrained extremum; Lagrange multiplier; regular constraint; elimination method.

Definition and key ideas

To optimize f subject to $g = c$, one can eliminate a variable, preserving all branches and domain restrictions, or solve $\nabla f = \lambda \nabla g$ together with $g = c$. The Lagrange condition is necessary at a constrained local extremum when f, g are C^1 and $\nabla g \neq 0$. Geometrically the relevant level sets share a tangent direction.

The equations produce candidates, not an automatic classification. Check singular constraint points where $\nabla g = 0$, endpoints when present, and every candidate's objective value. Dividing by a variable can discard valid cases. With multiple constraints the gradient of f is a linear combination of their gradients, assuming these gradients are independent.

Problem 107.1

[See solution](#)

What is the maximum of $x + 2y + 2z$ subject to $x^2 + y^2 + z^2 = 9$?

- (A) 3 (B) 6 (C) 9 (D) 12 (E) 18

Problem 107.2

[See solution](#)

What is the minimum of $x^2 + y^2$ subject to $2x^2 + 2xy + 2y^2 = 6$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Problem 107.3

[See solution](#)

Minimize $f(x, y) = x$ subject to $g(x, y) = x^2 + y^2 = 0$. Which statement is correct?

- (A) No minimum exists.
(B) The minimum is 0, and some λ satisfies $\nabla f = \lambda \nabla g$ there.
(C) The minimum is 0, but no such multiplier exists at the feasible point.
(D) The minimum is negative.
(E) There are infinitely many feasible points.

Problem 107.4[See solution](#)

For $f(x, y) = x^2y$ restricted to $x^2 + y^2 = 1$, how are $(0, 1)$ and $(0, -1)$ classified?

- (A) Both strict local minima
- (B) Both strict local maxima
- (C) $(0, 1)$ is a strict local minimum and $(0, -1)$ a strict local maximum, neither absolute.
- (D) $(0, 1)$ is an absolute minimum and $(0, -1)$ an absolute maximum.
- (E) Neither is a constrained local extremum.

Problem 107.5[See solution](#)

What is the maximum of xyz subject to $x + y + z = 0$ and $x^2 + y^2 + z^2 = 6$?

- (A) 1 (B) 2 (C) 3 (D) $2\sqrt{3}$ (E) 6

3. Calculus II

Line Integrals ▸ Line Integrals with Respect to Arc Length

108 Smooth curves, arc length, and scalar line integrals

Book pp. 139-143. [Find in compact glossary](#)

In this entry: parameterized curve; smooth curve; piecewise smooth curve; arc length element; scalar line integral; curve integral; contour integral; path integral; linear density; mass of a wire.

Definition and key ideas

A regular smooth parameterization $r : [a, b] \rightarrow \mathbb{R}^n$ has continuous derivative with $r'(t) \neq 0$. Piecewise smooth curves join finitely many smooth pieces. For increasing parameter, $ds = \|r'(t)\|dt$ and $\int_C f ds = \int_a^b f(r(t))\|r'(t)\|dt$. The case $f = 1$ gives arc length.

This scalar line integral is unchanged by reversing orientation. For a nonnegative linear density f , it is the mass of a wire; for a nonnegative height above a plane curve, it is the area of a vertical curtain. A parameterization that retraces a curve counts that portion repeatedly. Do not substitute dx for ds unless the geometry and orientation warrant it.

Problem 108.1

[See solution](#)

A wire follows $r(t) = (3 \cos t, 3 \sin t, 4t)$, $0 \leq t \leq \pi$, with density z . What is its mass?

- (A) $5\pi^2$ (B) $10\pi^2$ (C) $15\pi^2$ (D) $20\pi^2$ (E) $25\pi^2$

Problem 108.2

[See solution](#)

A plane curve has length $L > 0$ and is traversed once by a regular parametrization. If a positive continuous density ρ on the curve satisfies $\int_C \rho ds = M$, what is the sharp lower bound for $\int_C \rho^2 ds$ in terms of L and M ?

- (A) M/L
 (B) M^2/L
 (C) M^2L
 (D) M/L^2
 (E) 0, with no positive bound possible.

Problem 108.3

[See solution](#)

Let C be $r(t) = (t, t^2)$, $0 \leq t \leq 1$. What is $\int_C \sqrt{1 + 4x^2} ds$?

- (A) $\frac{7}{3}$ (B) $-\frac{7}{3}$ (C) $\frac{7}{6}$ (D) $\frac{10}{3}$ (E) $\frac{14}{3}$

Problem 108.4

[See solution](#)

A wire follows the entire boundary of the unit square $0 \leq x, y \leq 1$ and has linear density $|x - y|$. What is its mass?

- (A) $1/2$ (B) 1 (C) $3/2$ (D) 2 (E) 4

Problem 108.5[See solution](#)

For the parametrized path $r(t) = (\cos t, -\sin t)$, $0 \leq t \leq 3\pi$, what is $\int_C (2 + x) ds$ with the indicated traversal?

- (A) 0 (B) 2π (C) 3π (D) 4π (E) 6π

3. Calculus II

Line Integrals ▸ The Line Integral of a Vector Field

109 Vector fields, work, and oriented line integrals

Book pp. 143-147. [Find in compact glossary](#)

In this entry: vector field; vector-valued function; work integral; orientation; differential one-form; circulation.

Definition and key ideas

A vector field assigns a vector $F(p)$ to each point of a domain. Along an oriented piecewise smooth curve $r(t)$, its line integral is $\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt$. In the plane, $F = (M, N)$ gives the differential expression $M dx + N dy$; in space add the third component times dz .

This integral is the work of a force along the path; around a closed curve it is circulation. It depends in general on the entire path. Reversal changes its sign, unlike $\int f ds$. Splitting the path into successive pieces gives a sum. Parameterize both the field's coordinates and the differential consistently before integrating.

A differential one-form $M dx + N dy$ acts on a tangent displacement (u, v) by the linear pairing $Mu + Nv$. Writing $dr = T ds$ with oriented unit tangent T gives work $\int_C F \cdot T ds$ and the bound $|\int_C F \cdot dr| \leq \int_C \|F\| ds$. Continuity makes equality conditions pointwise along a regular traversal.

Problem 109.1

[See solution](#)

For $F = (y, x^2)$, what is the work along $y = x^2$ from $(0, 0)$ to $(1, 1)$ minus the work along the straight segment with the same endpoints?

- (A) $-\frac{1}{6}$ (B) 0 (C) $\frac{1}{6}$ (D) $\frac{1}{3}$ (E) $\frac{1}{2}$

Problem 109.2

[See solution](#)

For $F = (-y, x, z)$ and $r(t) = (\cos t, \sin t, t)$, $0 \leq t \leq 2\pi$, what is $\int_C F \cdot dr$?

- (A) 2π (B) $2\pi^2$ (C) $2\pi + 2\pi^2$ (D) $4\pi + 2\pi^2$ (E) 0

Problem 109.3

[See solution](#)

Let C go from $(0, 0)$ to $(1, 0)$ to $(1, 1)$ and then straight back to $(0, 0)$. What is $\oint_C x dy$?

- (A) -1 (B) $-1/2$ (C) 0 (D) $1/2$ (E) 1

Problem 109.4

[See solution](#)

Let $F = (-y, x)$, and let C traverse the unit circle once clockwise. What are $\int_C F \cdot dr$ and $\int_C \|F\| ds$, respectively?

- (A) $2\pi, 2\pi$ (B) $-2\pi, 2\pi$ (C) $-2\pi, -2\pi$ (D) $0, 2\pi$ (E) $2\pi, 0$

Problem 109.5[See solution](#)

A continuous force field has norm at most 3 along a regular smooth curve of length 5. Its work along the oriented curve is 15. What must hold at every point of the traversal?

- (A) The force is zero.
- (B) The force is perpendicular to the tangent.
- (C) The force has magnitude 3 and points along the unit tangent.
- (D) The force has magnitude 3 and points opposite to the unit tangent.
- (E) The force is conservative on its entire domain.

3. Calculus II

Line Integrals ▸ The Fundamental Theorem of Calculus for Line Integrals

110 Potential functions and conservative fields

Book pp. 147-150. [Find in compact glossary](#)**In this entry:** scalar field; gradient field; potential function; conservative field; path independence; fundamental theorem for line integrals; closed curve.**Definition and key ideas**

A potential for F is a scalar function ϕ with $F = \nabla\phi$. If ϕ is continuously differentiable near a piecewise smooth path from A to B , then $\int_C F \cdot dr = \phi(B) - \phi(A)$. Potentials differ by constants on a connected domain. A conservative field has path-independent integrals; on an open connected domain, this is equivalent to possessing a potential and to all closed-path integrals vanishing.

To construct a planar potential, integrate M with respect to x , append an unknown function of y , and use $\phi_y = N$ to determine it. For C^1 fields, $M_y = N_x$ is necessary. Its sufficiency needs a domain hypothesis, addressed later: the equality alone does not remove holes or singularities.

Problem 110.1[See solution](#)

For $F = (2xy + \cos x, x^2 + 3y^2)$, what is the integral along any piecewise smooth path from $(0, 1)$ to $(\pi, 2)$?

- (A) $2\pi^2 + 7$ (B) $\pi^2 + 7$ (C) $2\pi^2 + 9$ (D) $2\pi^2 + 6$ (E) $\pi^2 + 9$

Problem 110.2[See solution](#)

For which real a, b is $F = (ay^2 + 2x, 2bxy + \cos y)$ conservative on $\mathbb{R}^2$?

- (A) $a = b$ (B) $a = -b$ (C) $a = 2b$ (D) $a = b = 0$ (E) All a, b

Problem 110.3[See solution](#)

On $D = \{(x, y) : x \neq 0\}$, let $F = (1/x, 0)$, $\phi = \log|x|$, and $\psi = \log|x| + \operatorname{sgn}(x)$. Which statement is correct?

- (A) Only ϕ is a potential.
 (B) Only ψ is a potential.
 (C) Both are potentials, and their difference is one constant on all of D .
 (D) Both are potentials, but their difference has different constants on the two components.
 (E) Neither is a potential because D is disconnected.

Problem 110.4[See solution](#)

Let ϕ be C^1 and $F = \phi^2 \nabla\phi$. For a piecewise smooth path from A to B with $\phi(A) = -1$ and $\phi(B) = 2$, what is $\int_C F \cdot dr$?

- (A) 1 (B) $7/3$ (C) 3 (D) 7 (E) 9

Problem 110.5[See solution](#)

For $F = (-2x, -2y)/(x^2 + y^2)^2$ on the punctured plane, what is its integral around any closed piecewise smooth curve avoiding the origin?

- (A) 0
- (B) 2π
- (C) -2π
- (D) 2π times the number of turns around the origin
- (E) It is undefined for every curve surrounding the origin.

3. Calculus II

Double Integrals ▸ Iterated Integrals and Reversing Order

111 Double integrals, Fubini, and region bounds

Book pp. 150-154. [Find in compact glossary](#)

In this entry: double integral; area element; Riemann sum; signed volume; iterated integral; order of integration; Fubini theorem.

Definition and key ideas

The double integral $\iint_R f \, dA$ is the limit of sums $\sum f(P_i)\Delta A_i$ as the partition mesh tends to zero, when this limit exists independently of sample points. It gives signed volume; physical volume between surfaces is $\iint_R (z_{\text{top}} - z_{\text{bottom}}) \, dA$. The integral of 1 gives area.

For continuous functions on compact suitable regions, Fubini's theorem permits evaluation by iterated integrals. A vertically simple region $a \leq x \leq b$, $g(x) \leq y \leq h(x)$ gives $\int_a^b \int_{g(x)}^{h(x)} f \, dy \, dx$. Reversing order requires redescribing the region, sometimes in pieces. The outer bounds are constant when computing a fixed numerical double integral. For improper integrals, changing order requires additional justification such as absolute integrability.

Problem 111.1[See solution](#)

What is $\int_0^1 \int_x^1 e^{y^2} \, dy \, dx$?

- (A) $e - 1$
- (B) $\frac{e-1}{2}$
- (C) $\frac{e}{2}$
- (D) $\frac{e+1}{2}$
- (E) No elementary value exists

Problem 111.2[See solution](#)

What is the volume between $z = 4 - x - y$ and $z = 0$ above the triangle $x, y \geq 0$, $x + y \leq 2$?

- (A) $\frac{8}{3}$ (B) 4 (C) $\frac{16}{3}$ (D) 6 (E) 8

Problem 111.3[See solution](#)

What is $\int_0^1 \int_0^{1-x} xy \, dy \, dx$?

- (A) $\frac{25}{24}$ (B) $\frac{1}{12}$ (C) $\frac{1}{48}$ (D) $\frac{1}{24}$ (E) $-\frac{1}{24}$

Problem 111.4[See solution](#)

For continuous f , which expression equals $\int_0^1 \int_y^{2-y} f(x, y) dx dy$?

- (A) $\int_0^2 \int_0^x f(x, y) dy dx$
- (B) $\int_0^1 \int_0^x f(x, y) dy dx + \int_1^2 \int_0^{2-x} f(x, y) dy dx$
- (C) $\int_0^2 \int_x^{2-x} f(x, y) dy dx$
- (D) $\int_0^1 \int_x^{2-x} f(x, y) dy dx$
- (E) $\int_0^1 \int_0^{2-x} f(x, y) dy dx + \int_1^2 \int_0^x f(x, y) dy dx$

Problem 111.5[See solution](#)

Over $R = [0, 1]^2$, what are the signed integral of $y - x^2$ and the physical volume between $z = y - x^2$ and $z = 0$, respectively?

- (A) $1/6, 1/6$ (B) $1/6, 11/30$ (C) $11/30, 11/30$ (D) $-1/6, 11/30$ (E) $1/6, 1/3$

Problem 111.6[See solution](#)

For $f(x, y) = (x^2 - y^2)/(x^2 + y^2)^2$, let $I = \lim_{\epsilon \downarrow 0} \int_{\epsilon}^1 \int_0^1 f(x, y) dy dx$ and $J = \lim_{\epsilon \downarrow 0} \int_{\epsilon}^1 \int_0^1 f(x, y) dx dy$. What are I, J ?

- (A) $0, 0$ (B) $\pi/4, \pi/4$ (C) $\pi/4, -\pi/4$ (D) $-\pi/4, \pi/4$ (E) Both limits diverge.

3. Calculus II

Double Integrals ▸ Double Integrals in Polar Coordinates

112 Polar area element and radial integration

Book pp. 154-155. [Find in compact glossary](#)**In this entry:** polar double integral; polar area element; radial bound; polar substitution.**Definition and key ideas**

Under $x = r \cos \theta$, $y = r \sin \theta$ with $r \geq 0$, the area element is $dA = r \, dr \, d\theta$. Therefore transform the integrand, the region, and the area element together. For $\alpha \leq \theta \leq \beta$ and $r_1(\theta) \leq r \leq r_2(\theta)$, integrate $f(r \cos \theta, r \sin \theta)r$ over those bounds.

The extra factor r accounts for the increasing width of angular sectors. Shifted circles often have nonconstant bounds: $x^2 + y^2 = 2ax$ becomes $r = 2a \cos \theta$, with angles restricted so that the bound is nonnegative. Avoid counting a region twice when a polar curve retraces itself. Symmetry can shorten angular intervals only when the transformed integrand shares the symmetry.

Problem 112.1[See solution](#)Let $D = \{(x, y) : x^2 + y^2 \leq 2x\}$. What is $\iint_D (x^2 + y^2) \, dA$?

- (A)
- π
- (B)
- $\frac{3\pi}{2}$
- (C)
- 2π
- (D)
- 3π
- (E)
- 4π

Problem 112.2[See solution](#)What is $\iint_D \frac{1}{x^2 + y^2} \, dA$ for $D = \{1 \leq x^2 + y^2 \leq 9, y \geq 0\}$?

- (A)
- $\pi \log 2$
- (B)
- $\pi \log 3$
- (C)
- $2\pi \log 3$
- (D)
- 3π
- (E)
- 4π

Problem 112.3[See solution](#)For $D = \{(x, y) : 0 \leq x^2 + y^2 \leq 1\}$, what is $\iint_D e^{-x^2 - y^2} \, dA$?

- (A)
- $-\frac{\pi}{2e} + \frac{\pi}{2}$
- (B)
- $-\frac{2\pi}{e} + 2\pi$
- (C)
- $-\frac{\pi}{e} + \pi$
- (D)
- $-\frac{\pi}{e} + 1 + \pi$
- (E)
- $-\pi + \frac{\pi}{e}$

Problem 112.4[See solution](#)Let D be the sector $0 \leq r \leq 2$, $0 \leq \theta \leq \pi/4$. What is $\iint_D (x^2 - y^2) \, dA$?

- (A) 0 (B) 1 (C) 2 (D) 4 (E)
- 2π

Problem 112.5[See solution](#)On $D = \{0 < x^2 + y^2 < e^{-2}\}$, put $r = \sqrt{x^2 + y^2}$. For which real p does $\iint_D [r^2(\log(1/r))^p]^{-1} \, dA$ converge, and what is its value?

- (A) $p > 0$; $2\pi/p$
 (B) $p > 1$; $2\pi/(p-1)$
 (C) $p < 1$; $2\pi/(1-p)$
 (D) $p > 2$; $2\pi/(p-2)$
 (E) It diverges for every p .

Problem 112.6[See solution](#)

What is the total area enclosed by the three petals of the polar rose $r = \cos(3\theta)$?

- (A) $\pi/12$ (B) $\pi/6$ (C) $\pi/4$ (D) $\pi/2$ (E) π

3. Calculus II

Green's Theorem ▷ Circulation and Area

113 Green's theorem, positive orientation, and area

Book pp. 156-158. [Find in compact glossary](#)

In this entry: simple closed curve; positive orientation; boundary orientation; Green theorem; area from a line integral.

Definition and key ideas

A simple closed curve returns to its starting point and has no other self-intersections. For a bounded planar region with piecewise smooth simple boundary C oriented counterclockwise, and M, N continuously differentiable on an open neighborhood of the region and boundary, Green's theorem says $\oint_C M dx + N dy = \iint_R (N_x - M_y) dA$. Positive boundary orientation keeps the region on the left.

Consequently $\text{area}(R) = \oint_C x dy = -\oint_C y dx = \frac{1}{2} \oint_C (x dy - y dx)$. For a region with holes, the outer boundary is counterclockwise and inner boundaries clockwise. Smoothness only on the boundary is insufficient: a singularity in the enclosed region can invalidate the theorem. An awkward integrand may simplify dramatically after taking $N_x - M_y$.

Problem 113.1

[See solution](#)

Let C be the ellipse $x^2/4 + y^2 = 1$, traversed clockwise. What is $\oint_C (e^x - 3y) dx + (5x + \sin y) dy$?

- (A) -16π (B) -8π (C) 0 (D) 8π (E) 16π

Problem 113.2

[See solution](#)

Let C be the positively oriented entire boundary of the annulus $1 \leq x^2 + y^2 \leq 4$. What is $\oint_C (-y^3 dx + x^3 dy)$?

- (A) 9π (B) $\frac{45\pi}{2}$ (C) 24π (D) 30π (E) 45π

Problem 113.3

[See solution](#)

The positively oriented simple closed curve $x = 2 \cos t - \cos 2t$, $y = 2 \sin t - \sin 2t$, $0 \leq t \leq 2\pi$, encloses what area?

- (A) 3π (B) 4π (C) 6π (D) 8π (E) 12π

Problem 113.4

[See solution](#)

A C^1 planar field has scalar curl 2 on a neighborhood of the closed annulus $1 \leq x^2 + y^2 \leq 4$. Its counterclockwise circulation around the outer circle is 10π . What is its counterclockwise circulation around the inner circle?

- (A) -4π (B) 4π (C) 6π (D) 10π (E) 16π

Problem 113.5[See solution](#)

A closed walk traverses a radius-1 circle counterclockwise, follows a connector to a disjoint radius-2 circle, traverses it clockwise, then returns along the same connector. What is $\frac{1}{2} \oint_C (x \, dy - y \, dx)$?

- (A) -5π (B) -3π (C) 0 (D) 3π (E) 5π

3. Calculus II

Green's Theorem ▸ Path Independence and Gradient Fields

114 Convexity, simple connectivity, and the hole obstruction

Book pp. 158-159. [Find in compact glossary](#)

In this entry: convex region; simply connected region; curl-free planar field; closed-path criterion; hole obstruction.

Definition and key ideas

A set is convex if it contains the line segment joining every pair of its points. An open connected region is simply connected if every loop can be continuously contracted to a point within it. For planar domains this expresses the absence of holes. Convex domains are simply connected, but the converse fails.

On a simply connected open planar domain, a C^1 field (M, N) with $M_y = N_x$ has a potential. On a punctured plane this implication can fail: $(-y, x)/(x^2 + y^2)$ has equal cross-partial derivatives but integrates to 2π around the positively oriented unit circle. Path independence, existence of a potential, and vanishing of all closed-path integrals remain equivalent on open connected domains; simple connectivity is a sufficient global hypothesis for the derivative test, not a hypothesis required for those three equivalences themselves.

Problem 114.1

[See solution](#)

For $F = (-y, x)/(x^2 + y^2)$, what is the integral around $r(t) = (2 \cos t, 2 \sin t)$, $0 \leq t \leq 4\pi$?

- (A) 0 (B) π (C) 2π (D) 4π (E) 8π

Problem 114.2

[See solution](#)

A C^1 vector field (M, N) on the annulus $1 < x^2 + y^2 < 4$ satisfies $M_y = N_x$. Which additional condition guarantees a potential on this annulus?

- (A) It is bounded
 (B) Its integral around every closed piecewise smooth path is zero
 (C) It has no zero vector values
 (D) Its integral along one radial segment is zero
 (E) It is tangent to both boundary circles

Problem 114.3

[See solution](#)

What is the counterclockwise integral of $(-y dx + x dy)/(x^2 + y^2)$ around $(x - 3)^2 + y^2 = 1$?

- (A) It is undefined (B) 0 (C) π (D) 2π (E) -2π

Problem 114.4[See solution](#)

Let $D = \mathbb{R}^2 \setminus \{(x, 0) : x \leq 0\}$. Which statement about D and $F = (-y, x)/(x^2 + y^2)$ on D is correct?

- (A) D is convex and F has no potential.
- (B) D is not convex but is simply connected, and F has a potential.
- (C) D is not simply connected and F has no potential.
- (D) D is convex and every potential must be constant.
- (E) F fails the equal-cross-partial condition on D .

Problem 114.5[See solution](#)

A continuously differentiable curl-free planar field is defined on a convex open set. Which must hold?

- (A) Its divergence vanishes
- (B) Its potential is unique without a constant
- (C) Its integral around every closed piecewise smooth curve is zero
- (D) The field is identically zero
- (E) The domain is bounded

3. Calculus II

Chapter 3 Review Questions > Vector Identities

115 Vector triple product and cross-product equations

Book pp. 160. [Find in compact glossary](#)

In this entry: vector triple product; BAC-CAB identity; cross-product equation.

Definition and key ideas

The vector triple product identity is $a \times (b \times c) = b(a \cdot c) - c(a \cdot b)$. It shows explicitly that cross products are not associative and reduces nested cross products to dot products and scalar multiplication. The related expression $(a \times b) \times c = b(a \cdot c) - a(b \cdot c)$ has a different arrangement.

To solve $a \times v = b$ with $a \neq 0$, a necessary condition is $a \cdot b = 0$. When this holds, all solutions are $v = -(a \times b)/\|a\|^2 + ta$. The free component parallel to a is invisible to crossing with a . If one also requires $v \perp a$, the solution is unique.

Problem 115.1

[See solution](#)

Nonzero perpendicular vectors $a, b \in \mathbb{R}^3$ are given. There exists a vector x satisfying both $a \times x = b$ and $x \times b = a$ exactly when

- (A) $\|a\| = 1$. (B) $\|b\| = 1$. (C) $\|a\| = \|b\|$. (D) $\|a\|\|b\| = 1$. (E) $a = -b$.

Problem 115.2

[See solution](#)

Unit vectors a, b satisfy $a \cdot b = 1/2$. What is $a \times (b \times a)$?

- (A) $a - \frac{1}{2}b$ (B) $b - \frac{1}{2}a$ (C) $\frac{1}{2}a - b$ (D) $a + b$ (E) $\frac{1}{2}(a + b)$

Problem 115.3

[See solution](#)

For fixed $a \neq 0$, define the linear map $T(v) = a \times v$. Which identity holds for every v ?

- (A) $T^3(v) = 0$
 (B) $T^3(v) = \|a\|^2 T(v)$
 (C) $T^3(v) = -\|a\|^2 T(v)$
 (D) $T^2(v) = -\|a\|^2 v$ for all v
 (E) $T^2(v) = \|a\|^2 v$ for all v

Problem 115.4

[See solution](#)

If $a \neq 0$, the equation $a \times x = b$ is solvable exactly when

- (A) $\|a\| = \|b\|$ (B) $b = 0$ (C) $a \cdot b = 0$ (D) $a \cdot b = 1$ (E) $a \times b = 0$

Problem 115.5[See solution](#)

If $(a \times b) \cdot c = 3$, what is $(a \times b) \times (a \times c)$?

- (A) $3a$ (B) $-3a$ (C) $3b$ (D) $3c$ (E) 0

3. Calculus II

Chapter 3 Review Questions > Distance to a Plane

116 Perpendicular distance to a plane

Book pp. 161. [Find in compact glossary](#)**In this entry:** point-to-plane distance; orthogonal foot; distance between parallel planes.**Definition and key ideas**

For a plane $n \cdot x = d$ with $n \neq 0$, the distance from P is $|n \cdot P - d|/\|n\|$. The signed numerator measures normal displacement in the scale set by n , and dividing by its norm converts it to Euclidean length. The orthogonal foot is $Q = P - [(n \cdot P - d)/\|n\|^2]n$.

For parallel planes written with the same normal, the separation is $|d_1 - d_2|/\|n\|$. If the normals are proportional but unequal, rescale the equations first. Squared distance to a plane is often convenient in optimization and avoids absolute values.

Reflection across a plane sends P to $2Q - P$, where Q is its orthogonal foot. Equating distances to two planes generally produces two angle-bisector planes. For a sphere of radius R whose center is distance d from a plane, the minimum and maximum point-to-plane distances are $\max\{d - R, 0\}$ and $d + R$.

Problem 116.1[See solution](#)

What is the distance between $2x - y + 2z = 3$ and $4x - 2y + 4z = 18$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 116.2[See solution](#)

The point $P = (2, 1, 0)$ is reflected across $x + 2y + 2z = 1$. What is the reflected point?

- (A)
- $(\frac{4}{3}, -\frac{1}{3}, -\frac{4}{3})$
- (B)
- $(\frac{5}{3}, \frac{1}{3}, -\frac{2}{3})$
- (C)
- $(0, -3, -4)$
- (D)
- $(\frac{8}{3}, \frac{7}{3}, \frac{4}{3})$
- (E)
- $(1, -1, -2)$

Problem 116.3[See solution](#)

Let Q be the perpendicular projection of $P = (1, 2, 3)$ onto the plane $2x - y + 2z = 3$. What is $\|Q\|^2$?

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 14

Problem 116.4[See solution](#)

Which equation describes the points equidistant from the planes $x = 0$ and $x + y = 0$?

- (A)
- $y = 0$
- (B)
- $y = x$
- (C)
- $(x + y)^2 = x^2$
- (D)
- $(x + y)^2 = 2x^2$
- (E)
- $x^2 + y^2 = 1$

Problem 116.5[See solution](#)

On the sphere centered at $(1, 1, 1)$ with radius 3, what are the minimum and maximum distances to the plane $x + 2y + 2z = 11$?

- (A) 1, 5 (B) 0, 5 (C) 0, 3 (D) 1, 3 (E) 2, 5

3. Calculus II

Chapter 3 Review Questions > Continuity at the Origin

117 Multivariable limits and continuity

Book pp. 162. [Find in compact glossary](#)

In this entry: multivariable limit; path test; continuity in several variables; polar estimate; squeeze theorem.

Definition and key ideas

A limit $f(x, y) \rightarrow L$ at (a, b) must hold along every approach within the domain, not just coordinate axes or straight lines. Continuity means this limit equals the function value. Two paths with different limits disprove a limit, but checking any finite collection of paths cannot prove it.

A reliable positive argument uses a uniform bound, often after $x = r \cos \theta$, $y = r \sin \theta$: if $|f - L| \leq Cr^\alpha$ for all angles with $\alpha > 0$, then the limit is L . Angular formulas must be bounded independently of angle. Curved paths such as $y = x^2$ are especially effective when denominators contain unequal powers.

Problem 117.1[See solution](#)

Define $f(0, 0) = 0$ and $f(x, y) = x^2y/(x^4 + y^2)$ elsewhere. Which statement is true?

- (A) The limit at the origin is zero
- (B) The limit at the origin is $1/2$
- (C) All straight-line limits are zero, but the full limit fails
- (D) Only the coordinate-axis limits are zero
- (E) The function is unbounded near the origin

Problem 117.2[See solution](#)

For which real p is $f(0, 0) = 0$, $f(x, y) = xy/(x^2 + y^2)^p$ continuous at the origin?

- (A) $p < 0$ (B) $p \leq 0$ (C) $p < 1$ (D) $p \leq 1$ (E) All real p

Problem 117.3[See solution](#)

What is $\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{1+x^2+y^2} - \cos \sqrt{x^2+y^2}}{x^2+y^2}$?

- (A) 0
- (B) $1/2$
- (C) 1
- (D) 2
- (E) The limit does not exist.

Problem 117.4[See solution](#)

What is $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \sin(1/(x^2 + y^2))$?

- (A) ∞
- (B) -1
- (C) 1
- (D) The limit does not exist
- (E) 0

Problem 117.5[See solution](#)

Let $D = \{(x, y) : y \neq 0, |x| \leq y^2\}$. What is the limit of $x^2/(x^2 + y^2)$ as $(x, y) \rightarrow (0, 0)$ within D ?

- (A) 0
- (B) $1/2$
- (C) 1
- (D) It depends on the sign of y .
- (E) The limit does not exist.

4. Differential Equations

Introduction ▸ Differential equations and initial conditions

118 Differential equations, order, solutions, integral curves, and IVPs

Book pp. 167-169. [Find in compact glossary](#)

In this entry: differential equation; ordinary differential equation; partial differential equation; order; solution; general solution; particular solution; integral curve; initial condition; initial value problem.

Definition and key ideas

A differential equation relates an unknown function to its derivatives. An ordinary differential equation (ODE) uses one independent variable; a partial differential equation (PDE) uses partial derivatives in several variables. Its order is the highest derivative present, even when that derivative occurs nonlinearly. An apparently nonlinear equation may simplify algebraically, so factor or complete a square in its derivative before choosing a solution method. A solution is a sufficiently differentiable function satisfying the equation throughout an interval or region.

A general solution describes a family; a particular solution selects a member. An initial value problem (IVP) adds values of the function and, for higher order equations, derivatives at one point. Integral curves are solution graphs, sometimes described implicitly by $F(x, y) = C$. Verify them by differentiation. For $y' = f(x)$, all solutions are antiderivatives plus a constant. Existence or uniqueness is not automatic: standard first-order uniqueness holds locally when the right side is continuous and locally Lipschitz in the dependent variable.

Problem 118.1

[See solution](#)

If $y'' = 6x$, $y(1) = 2$, and $y'(1) = 5$, what is $y(2)$?

- (A) 7 (B) 9 (C) 10 (D) 11 (E) 13

Problem 118.2

[See solution](#)

The integral curve $y^3 + xy = 2$ locally defines y as a function of x near $(1, 1)$. What is $y''(1)$?

- (A) $-1/8$ (B) $-1/32$ (C) $1/32$ (D) $1/8$ (E) $3/32$

Problem 118.3

[See solution](#)

For an unknown function $u(x, y)$, the equation $(\partial^3 u / \partial x^2 \partial y)^2 + \partial^4 u / \partial y^4 = 0$ is which type?

- (A) A second-order nonlinear ODE
(B) A third-order linear PDE
(C) A fourth-order nonlinear PDE
(D) A sixth-order nonlinear PDE
(E) A fourth-order linear ODE

Problem 118.4[See solution](#)

How many continuously differentiable solutions on $\mathbb{R}$ satisfy $y' = 3|y|^{2/3}$ and $y(0) = 0$?

- (A) None (B) Exactly one (C) Exactly two (D) Exactly three (E) Infinitely many

Problem 118.5[See solution](#)

Which function satisfies $(y')^2 - 2xy' + x^2 = 0$ and $y(0) = 1$ on $\mathbb{R}$?

- (A) $1 + x$ (B) $1 - x^2/2$ (C) $1 + x^2/2$ (D) $1 + x^3/3$ (E) $e^{x^2/2}$

4. Differential Equations

Separable Equations ▸ Separation of variables

119 Separable ODEs, equilibrium solutions, and maximal intervals

Book pp. 169. [Find in compact glossary](#)

In this entry: separable equation; separation of variables; equilibrium solution; constant solution; maximal interval; finite-time blow-up.

Definition and key ideas

An equation $y' = f(x)g(y)$ is separable: where $g(y) \neq 0$, write $dy/g(y) = f(x) dx$ and integrate both sides. A solution can remain implicit. If $g(c) = 0$, the constant function $y = c$ is an equilibrium solution, which division by $g(y)$ would remove. Check these solutions separately.

Antiderivatives involving logarithms require absolute values, such as $\int dy/y = \log|y| + C$. After solving, choose the branch and the maximal interval containing the initial point on which the original equation and solution are defined. A finite denominator zero can cause blow-up. Conversely, if integration yields $e^y = H(x)$, then $H(x) > 0$ is essential; $y = \log|H(x)|$ would incorrectly add a branch.

Problem 119.1

[See solution](#)

The solution of $y' = y^2$, $y(0) = 2$, has which maximal interval of existence?

- (A) $\mathbb{R}$ (B) $(-\infty, 1/2)$ (C) $(1/2, \infty)$ (D) $(-1/2, 1/2)$ (E) $(0, 1/2)$

Problem 119.2

[See solution](#)

If $y' = y(1 - y)$ and $y(0) = 1/3$, what is $y(\log 4)$?

- (A) $1/2$ (B) $3/5$ (C) $2/3$ (D) $3/4$ (E) $4/5$

Problem 119.3

[See solution](#)

For $y' = -e^{-y}$ and $y(0) = 0$, which is the solution on its maximal interval containing zero?

- (A) $y = \log(1 - x)$ on $(-\infty, 1)$ (B) $y = \log|1 - x|$ on $\mathbb{R} \setminus \{1\}$
 (C) $y = -\log(1 - x)$ on $(-\infty, 1)$ (D) $y = \log(1 + x)$ on $(-1, \infty)$
 (E) $y = 1 - x$ on $\mathbb{R}$

Problem 119.4[See solution](#)

The IVP $y' = e^{-y^2}$, $y(0) = 0$, has which behavior?

- (A) It is constant.
- (B) It is increasing on $\mathbb{R}$ and tends to $+\infty$ as $x \rightarrow +\infty$.
- (C) It blows up at a finite positive x .
- (D) It increases to the finite limit 1.
- (E) It is not separable.

Problem 119.5[See solution](#)

What are all constant solutions of $y' = y^2(y - 2)(y + 1)$?

- | | |
|---------------------|----------------------|
| (A) Every constant | (B) $y = -1, 2$ only |
| (C) $y = 0$ only | (D) $y = -1, 0, 2$ |
| (E) $y = 0, 2$ only | |

4. Differential Equations

Homogeneous Equations ▸ Scaling and the substitution $y=xv$

120 Homogeneous functions and first-order homogeneous equations

Book pp. 170. [Find in compact glossary](#)

In this entry: homogeneous function; homogeneous degree; homogeneous first-order equation; substitution $y=xv$; scale invariance; Euler identity for homogeneous functions.

Definition and key ideas

A function H is homogeneous of degree n if $H(tx, ty) = t^n H(x, y)$ whenever the scaling and expressions are defined. A first-order equation $M dx + N dy = 0$ is called homogeneous in this sense when M and N have the same degree. Where division is valid, its slope depends only on the ratio y/x .

Set $y = xv$, so $y' = v + xv'$, to reduce $y' = F(y/x)$ to $xv' = F(v) - v$, a separable equation. Ratios v with $F(v) = v$ give straight-line solutions and can disappear if one divides by $F(v) - v$. The substitution works on intervals avoiding $x = 0$. This meaning of homogeneous is distinct from a linear equation having zero forcing term.

For differentiable H homogeneous of degree n , differentiating $H(tx, ty) = t^n H(x, y)$ with respect to t at $t = 1$ gives Euler's identity $xH_x + yH_y = nH$. This is the identity used to recover one partial derivative from the other and the function value.

Problem 120.1

[See solution](#)

For $x > 0$, the IVP $y' = y/x + (y/x)^2$, $y(1) = 1$, has which solution near $x = 1$?

- (A) $y = x/(1 - \log x)$ (B) $y = x/(1 + \log x)$
 (C) $y = 1/(1 - x)$ (D) $y = x \log x$
 (E) $y = x^2$

Problem 120.2

[See solution](#)

Which family solves $y' = (x^2 + y^2)/(2xy)$ on regions where $xy \neq 0$?

- (A) $x^2 + y^2 = C$ (B) $y^2 = x^2 + Cx$ (C) $y^2 = x^2 + Cx^2$ (D) $xy = C$ (E) $y = x \log |x| + C$

Problem 120.3

[See solution](#)

A differentiable function $F(x, y)$ is homogeneous of degree 3. If $F(1, 2) = 5$ and $F_x(1, 2) = 1$, what is $F_y(1, 2)$?

- (A) 5 (B) 2 (C) 7 (D) 14 (E) 15

Problem 120.4[See solution](#)

On $x > 0$, which are all straight-line solutions $y = kx$ of $y' = (y/x)^2 + 2y/x - 2$?

- (A) $y = x$ and $y = -2x$ (B) $y = 2x$ and $y = -x$
(C) $y = x$ and $y = 2x$ (D) $y = -x$ and $y = -2x$
(E) $y = 0$ only

Problem 120.5[See solution](#)

On $x, y > 0$, which function is homogeneous of degree 2?

- (A) $x^2 \log y$ (B) $x^2 + y^3$ (C) $x^2 \log(y/x) + y^2$ (D) $x^2 + \log(y/x)$ (E) $\log(x^2 + y^2)$

4. Differential Equations

Exact Equations ▸ Total differentials and potentials

121 Exact equations, total differentials, and potential functions

Book pp. 170-171. [Find in compact glossary](#)

In this entry: total differential; exact differential; exact equation; potential function; mixed partial derivatives; level curve; simply connected domain.

Definition and key ideas

The total differential of F is $dF = F_x dx + F_y dy$. An equation $M dx + N dy = 0$ is exact if $M = F_x$ and $N = F_y$ for some potential F ; its solution curves lie in levels $F = C$. For continuously differentiable M, N on a simply connected open region, the condition $M_y = N_x$ guarantees exactness; without a domain condition it guarantees only local exactness.

To find F , integrate M with respect to x , including an unknown function $h(y)$, and determine h by matching $F_y = N$. The implicit function theorem supplies a local graph $y(x)$ when $N \neq 0$. An entire level set can have several components, so a point on the same level need not lie on the same connected solution curve; this distinction matters when interpreting an IVP.

Problem 121.1

[See solution](#)

The solution through $(1, 1)$ of $(2xy + 1) dx + (x^2 + 3y^2) dy = 0$ lies on which level set?

- (A) $x^2y + x + y^3 = 2$ (B) $x^2y + x + y^3 = 3$
 (C) $xy^2 + x + y^3 = 3$ (D) $x^2y + y + x^3 = 3$
 (E) $x^2 + y^2 = 2$

Problem 121.2

[See solution](#)

On $\mathbb{R}^2$, $(\cos(xy) - xy \sin(xy)) dx + N(x, y) dy$ is exact, and $N(0, y) = y^2$. What is $N(x, y)$?

- (A) $-x^2 \sin(xy) + y^2$ (B) $x^2 \sin(xy) + y^2$
 (C) $-x \sin(xy) + y^2$ (D) $x \cos(xy) + y^2$
 (E) $-x^2 \cos(xy) + y^2$

Problem 121.3

[See solution](#)

Which is the maximal graph solution through $(0, 1)$ of $-2x dx + 2y dy = 0$?

- (A) $y = \sqrt{1 + x^2}$ for all real x
 (B) $y = -\sqrt{1 + x^2}$ for all real x
 (C) The full two-branch set $y^2 - x^2 = 1$ as one graph
 (D) $y = 1 + x$
 (E) $y = \sqrt{1 - x^2}$ for $|x| < 1$

Problem 121.4[See solution](#)

For the exact relation $2x \, dx + 2y \, dy = 0$ through $(1, 0)$, which statement is correct?

- (A) There is no smooth integral curve.
- (B) The circle $x^2 + y^2 = 1$ is an integral curve, but there is no C^1 graph $y(x)$ through the point on an open interval about $x = 1$.
- (C) The unique local graph is $y = 0$.
- (D) The relation is not exact.
- (E) The circle has a horizontal tangent there.

Problem 121.5[See solution](#)

How many C^1 graph solutions on an open interval about zero satisfy $y y' = x$ and $y(0) = 0$?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) Infinitely many distinct local branches

Problem 121.6[See solution](#)

On the punctured plane, consider $[(x - y) \, dx + (x + y) \, dy]/(x^2 + y^2) = 0$. Which describes its nonconstant integral curves locally and its exactness globally?

- (A) Circles; globally exact
- (B) Radial rays; globally exact
- (C) Logarithmic spirals $\log r + \theta = C$; locally exact but not globally exact
- (D) Logarithmic spirals $\log r - \theta = C$; globally exact
- (E) No local potential exists anywhere.

4. Differential Equations

Nonexact Equations and Integrating Factors ▷ Factors depending on one variable

122 Integrating factors for nonexact equations

Book pp. 172-173. [Find in compact glossary](#)**In this entry:** nonexact equation; integrating factor; one-variable integrating factor; equivalent differential equation.**Definition and key ideas**

A nonvanishing function μ is an integrating factor when $\mu M dx + \mu N dy$ is exact. Multiplication preserves the equation only where μ is finite and nonzero. Thus singularities and zeros of a proposed factor must be treated separately; an integrating factor need not exist globally on an arbitrary region.

If $(M_y - N_x)/N$ depends only on x , an integrating factor is $\mu(x) = \exp(\int (M_y - N_x)/N dx)$. If $(N_x - M_y)/M$ depends only on y , use $\mu(y) = \exp(\int (N_x - M_y)/M dy)$. These formulas follow by expanding $(\mu M)_y = (\mu N)_x$. Be especially careful about the sign in the y -dependent formula.

Problem 122.1[See solution](#)

For $x > 0$, an integrating factor depending only on x for $(3y + x) dx + x dy = 0$ is

- (A) x^{-2} (B) x^{-1} (C) x (D) x^2 (E) e^x

Problem 122.2[See solution](#)

For $y > 0$, which expression is constant along solutions of $y dx + (2x + y) dy = 0$?

- (A) $xy + y^2/2$ (B) $xy^2 + y^3/3$ (C) $x^2y + y^2$ (D) $xy^2 + y^3$ (E) $xy + y^3/3$

Problem 122.3[See solution](#)

Multiplying $y' - 1 = 0$ by y gives $y(y' - 1) = 0$. Which function solves the multiplied equation but not the original one?

- (A) $y = x$ (B) $y = x + 1$ (C) $y = x - 1$ (D) $y = 0$ (E) $y = 2x$

Problem 122.4[See solution](#)

On $x, y > 0$, which is an integrating factor for $\frac{y}{x+y} dx + \frac{x}{x+y} dy = 0$?

- (A) $x + y$ (B) $1/(x + y)$ (C) xy (D) $(x + y)^2$ (E) $1/(xy)$

Problem 122.5[See solution](#)

Which integrating factor makes $y' + 3y = x$ directly integrable by the product rule?

- (A) e^{-3x} (B) e^{3x} (C) $3x$ (D) x^3 (E) $1/x$

4. Differential Equations

First-Order Linear Equations ▸ Standard form and integrating factor

123 First-order linear ODEs and the product-derivative method

Book pp. 174-175. [Find in compact glossary](#)

In this entry: first-order linear equation; standard form; linear integrating factor; inhomogeneous term; variation of constants.

Definition and key ideas

A first-order linear equation has standard form $y' + P(x)y = Q(x)$. On an interval where P, Q are continuous, define $\mu(x) = e^{\int P(x) dx}$. Multiplying gives $(\mu y)' = \mu Q$, so $y = \mu^{-1} (C + \int \mu Q dx)$. An initial value fixes C and yields a unique solution on that interval.

First divide by the coefficient of y' where it is nonzero; forgetting this step changes the integrating factor. A nonzero constant multiple of μ works equally well. For $P(x) = a/x$, work on one side of 0 and use $|x|^a$ as needed. Equations linear in x as a function of y can sometimes be solved by reversing dependent and independent variables.

Variation of constants writes $y = v(x)e^{-\int P(x) dx}$, replacing the constant in a homogeneous solution by a function. Substitution yields $v' = e^{\int P(x) dx} Q(x)$, giving the same formula as the integrating-factor method. If solving for x as a function of y , verify that the resulting curve can be inverted on the requested portion.

Problem 123.1

[See solution](#)

For $x > 0$, $y' + 2y/x = x^2$ and $y(1) = 1$. What is $y(2)$?

- (A) $7/5$ (B) $8/5$ (C) $9/5$ (D) 2 (E) $11/5$

Problem 123.2

[See solution](#)

For $x > 0$, which solution of $y' + y/x = \cos x$ has a finite limit as $x \rightarrow 0^+$?

- (A) $\sin x + \cos x/x$ (B) $\sin x + (\cos x - 1)/x$
 (C) $\cos x + (\sin x - 1)/x$ (D) $\sin x + 1/x$
 (E) $\cos x$

Problem 123.3

[See solution](#)

On $x > -1$, multiplying the entire equation $(1+x)y' + 2y = \sin x$ by which function makes its left side an exact product derivative?

- (A) $1+x$ (B) $(1+x)^2$ (C) $1/(1+x)$ (D) e^{2x} (E) e^{x^2}

Problem 123.4[See solution](#)

The integral curve of $dy/dx = 1/(y + 2x)$ through $(0, 1)$ reaches $y = 2$ at which x-coordinate?

- (A) $(e^2 - 1)/2$ (B) $(3e^2 - 5)/4$ (C) $(3e^2 - 3)/4$ (D) $(5e^2 - 3)/4$ (E) $(3e^{-2} - 5)/4$

Problem 123.5[See solution](#)

What initial value $y(0)$ makes the solution of $y' + 2y = \sin x$ periodic with period 2π ?

- (A) $-2/5$ (B) $-1/5$ (C) 0 (D) $1/5$ (E) $2/5$

4. Differential Equations

Higher-Order Linear Equations with Constant Coefficients ▸ Linear structure and general solutions

124 Linear differential operators, superposition, and solution spaces

Book pp. 175-177. [Find in compact glossary](#)

In this entry: linear differential equation; constant coefficient; homogeneous linear equation; nonhomogeneous linear equation; superposition; complementary solution; particular solution; linear independence; fundamental set.

Definition and key ideas

A linear ODE is $a_n(x)y^{(n)} + \cdots + a_0(x)y = g(x)$; the unknown and its derivatives occur only to first power and are not multiplied together. It is homogeneous when $g = 0$. Its homogeneous solutions form a vector space, since a linear differential operator L satisfies $L(au + bv) = aL(u) + bL(v)$.

If a_n is nonzero and the normalized coefficients are continuous on an interval, the homogeneous solution space has dimension n . A fundamental set consists of n linearly independent solutions, meaning no nontrivial constant linear combination vanishes identically. For $L[y] = g$, every solution is $y_h + y_p$, where y_p is any one particular solution and y_h is an arbitrary homogeneous solution. The nonhomogeneous solution set is an affine translate, not generally a vector space.

Problem 124.1

[See solution](#)

Functions u, v satisfy $L[u] = f$ and $L[v] = 2f$, where L is a linear differential operator and $f \not\equiv 0$. Which is necessarily a homogeneous solution?

- (A) $u + v$ (B) $2u - v$ (C) uv (D) $u^2 - v$ (E) $u - v$

Problem 124.2

[See solution](#)

The functions e^x and e^{-x} form a fundamental set for a second-order homogeneous equation. If x^2 is a particular solution of the corresponding nonhomogeneous equation, which function is also a solution of that same nonhomogeneous equation?

- (A) $2x^2$ (B) $x^2 + \cosh x$ (C) $x^2 + xe^x$ (D) $x^2 + 1$ (E) $x^2 + \sin x$

Problem 124.3

[See solution](#)

For $y'' - y = 0$, let $u = e^x + ae^{-x}$ and $v = ae^x + e^{-x}$. For which real a do u, v form a fundamental set?

- (A) $a \neq 0$ (B) $a \neq 1$ (C) $a \neq -1$ (D) $a \notin \{-1, 1\}$ (E) Every real a

Problem 124.4[See solution](#)

Which is a linear homogeneous ODE with nonconstant coefficients?

(A) $\frac{d}{dx}(x^2y) + e^xy'' = 0$

(B) $\frac{d}{dx}(y^2) + y'' = 0$

(C) $y'' + xy' = 1$

(D) $y'' + y = 0$

(E) $(y')^2 + y = 0$

Problem 124.5[See solution](#)

What is the dimension of the real vector space of C^1 solutions on $\mathbb{R}$ of $xy' - 2y = 0$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinite

4. Differential Equations

Higher-Order Linear Equations with Constant Coefficients ▷ Auxiliary equation: distinct real roots

125 Characteristic polynomial and distinct exponential modes

Book pp. 175–177. [Find in compact glossary](#)

In this entry: auxiliary equation; characteristic equation; characteristic polynomial; exponential mode; distinct characteristic roots.

Definition and key ideas

For $L[y] = a_n y^{(n)} + \cdots + a_0 y = 0$ with constant coefficients, substituting $y = e^{rx}$ gives $L[e^{rx}] = p(r)e^{rx}$, where $p(r) = a_n r^n + \cdots + a_0$. Thus every root of the characteristic or auxiliary polynomial produces a solution mode. The method extends to arbitrary finite order, not only second order.

If all n roots are distinct real numbers $r_1, \dots, r_n$, the general real solution is $\sum C_j e^{r_j x}$. A zero root contributes the constant mode 1. Initial conditions determine the coefficients by a linear system. Long-time behavior can often be read directly: positive roots give growing modes, negative roots decaying modes, and boundedness can force coefficients to vanish.

Problem 125.1

[See solution](#)

A solution of $y'' - y' - 6y = 0$ satisfies $y(0) = 4$ and is bounded as $x \rightarrow \infty$. What is $y(\log 2)$?

- (A) 1 (B) 2 (C) 4 (D) 8 (E) 16

Problem 125.2

[See solution](#)

How many linearly independent solutions of $y''' - 4y' = 0$ are bounded on all of $\mathbb{R}$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 125.3

[See solution](#)

If $y'' + y' - 2y = 0$, $y(0) = 2$, and $y'(0) = 1$, what is $y(1)$?

- (A) $(e + 5e^{-2})/3$ (B) $(5e - e^{-2})/3$ (C) $(5e + e^{-2})/3$ (D) $2e$ (E) $e + e^{-2}$

Problem 125.4

[See solution](#)

Both e^x and e^{2x} solve $y''' + ay'' + by' - 6y = 0$, where a, b are constants. What are a, b ?

- (A) $-3, 2$ (B) $-6, 11$ (C) $6, 11$ (D) $-6, -11$ (E) $-3, 11$

Problem 125.5

[See solution](#)

What is the largest possible number of distinct real zeros of a nonzero solution of $y'' - 3y' + 2y = 0$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinitely many

4. Differential Equations

Higher-Order Linear Equations with Constant Coefficients ▷ Auxiliary equation: repeated roots

126 Repeated characteristic roots and polynomial factors

Book pp. 176-177; 179. [Find in compact glossary](#)

In this entry: repeated characteristic root; root multiplicity; polynomial-exponential solution; zero characteristic root.

Definition and key ideas

A real characteristic root r of multiplicity m contributes the m solutions $e^{rx}, xe^{rx}, \dots, x^{m-1}e^{rx}$. Merely repeating e^{rx} does not create independent solutions. In particular, a zero root of multiplicity m contributes $1, x, \dots, x^{m-1}$.

The substitution $y = e^{rx}v$ explains this rule: $(D - r)^m(e^{rx}v) = e^{rx}v^{(m)}$, where $D = d/dx$. For a repeated quadratic root, the solution is $(C_1 + C_2x)e^{rx}$. Combine the contributions from all characteristic roots, counting multiplicity, to obtain exactly as many independent modes as the differential equation's order.

More generally, for a polynomial p , $p(D)(e^{rx}v) = e^{rx}p(D + r)v$. This shift identity follows repeatedly from $D(e^{rx}v) = e^{rx}(D + r)v$. It determines both which polynomial-exponential modes are annihilated and the first one that survives.

Problem 126.1

[See solution](#)

If $y'' - 4y' + 4y = 0$, $y(0) = 1$, and $y'(0) = 0$, what is $y(1)$?

- (A) $-e^2$ (B) 0 (C) e^2 (D) $2e^2$ (E) $3e^2$

Problem 126.2

[See solution](#)

What is the dimension of the space of polynomial solutions of $y^{(5)} - 2y^{(4)} + y''' = 0$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 126.3

[See solution](#)

What is the least order of a nonzero homogeneous linear ODE with constant coefficients having $x^3e^{2x} + xe^{-x}$ as a solution?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 8

Problem 126.4

[See solution](#)

For $(D + 1)^3y = 0$, where $D = d/dx$, what are the dimensions of the subspaces bounded on $[0, \infty)$ and on all of $\mathbb{R}$, respectively?

- (A) 0, 0 (B) 1, 0 (C) 3, 0 (D) 3, 1 (E) 3, 3

Problem 126.5[See solution](#)

For $y = x^2e^x$, what is $y''' - 3y' + 2y$?

- (A) 0 (B) $2e^x$ (C) $6e^x$ (D) $6xe^x$ (E) $6x^2e^x$

4. Differential Equations

Higher-Order Linear Equations with Constant Coefficients ▷ Auxiliary equation: nonreal roots

127 Complex characteristic roots, oscillation, and damping

Book pp. 176-177. [Find in compact glossary](#)

In this entry: complex characteristic root; oscillatory solution; angular frequency; damped oscillation; complex exponential; real solution basis.

Definition and key ideas

For real coefficients, nonreal characteristic roots occur in pairs $\alpha \pm i\beta$, $\beta > 0$. Euler's identity converts the exponential modes into the real basis $e^{\alpha x} \cos \beta x$, $e^{\alpha x} \sin \beta x$. The factor $e^{\alpha x}$ is the amplitude envelope; β is the angular frequency, and the undamped oscillation period is $2\pi/\beta$.

If the pair has multiplicity m , multiply both real modes by $1, x, \dots, x^{m-1}$. A negative α yields decay, a positive one growth, and zero gives pure oscillation. A damped oscillation need not itself be periodic even though its sine and cosine factors repeat. Initial conditions require differentiating the exponential envelope as well as the trigonometric factors.

Problem 127.1

[See solution](#)

For $y'' + 4y' + 13y = 0$, $y(0) = 0$, and $y'(0) = 3$, what is the first positive zero of y ?

- (A) $\pi/6$ (B) $\pi/3$ (C) $\pi/2$ (D) $2\pi/3$ (E) π

Problem 127.2

[See solution](#)

Which is the general real solution of $y^{(4)} + 2y'' + y = 0$?

- (A) $A \cos x + B \sin x$ (B) $Ae^x + Be^{-x} + Cx + D$
 (C) $(A + Bx) \cos x + (C + Dx) \sin x$ (D) $A \cos x + B \sin x + C \cos 2x + D \sin 2x$
 (E) $(A + Bx)e^x + (C + Dx)e^{-x}$

Problem 127.3

[See solution](#)

The solution of $y'' + 2y' + 5y = 0$ has $y(0) = 0$ and $y'(0) = 2$. Where is its first positive local maximum?

- (A) $\frac{1}{2} \arctan(1/2)$ (B) $\frac{1}{2} \arctan 2$ (C) $\pi/4$ (D) $\pi/2$ (E) $\arctan 2$

Problem 127.4[See solution](#)

Which statement holds for every nonzero real solution of $y'' + 2y' + 5y = 0$?

- (A) It is periodic with period π .
- (B) It has only finitely many zeros.
- (C) It has infinitely many zeros but is not periodic.
- (D) Its amplitude grows exponentially.
- (E) It tends to a nonzero constant.

Problem 127.5[See solution](#)

A decaying oscillatory solution of $y'' + by' + cy = 0$ has consecutive zeros separated by $\pi/2$. The absolute values at consecutive local extrema have ratio $e^{-\pi}$. What are b, c ?

- (A) 2, 5 (B) 4, 8 (C) 4, 13 (D) 2, 8 (E) 8, 4

4. Differential Equations

Higher-Order Linear Equations with Constant Coefficients ▷ Particular solutions of nonhomogeneous equations

128 Particular-solution guessing, undetermined coefficients, and resonance

Book pp. 177; 179. [Find in compact glossary](#)

In this entry: undetermined coefficients; forcing term; resonance; particular-solution ansatz; verification by substitution.

Definition and key ideas

For a constant-coefficient equation $L[y] = g$, a forcing term built from polynomials, exponentials, sine, and cosine suggests a particular solution of the same form with unknown coefficients. Substitute the trial expression and match coefficients. For polynomial forcing, include all lower powers through its degree, not just the leading term. A sum of forcing terms can be handled term by term using linearity.

If a trial function belongs to the homogeneous solution space, multiply the trial family by enough powers of x to escape it. For forcing $e^{\lambda x}$, multiply by x^m when λ is a characteristic root of multiplicity m ; analogous rules apply to sinusoidal forcing. This is resonance. A quick direct substitution into answer choices can be more efficient than constructing the full general solution.

Problem 128.1

[See solution](#)

Which is a particular solution of $y'' - 3y' + 2y = e^x$?

- (A) e^x (B) xe^x (C) $-xe^x$ (D) x^2e^x (E) $\frac{1}{2}e^x$

Problem 128.2

[See solution](#)

What is the solution of $y'' + y = \cos x$, $y(0) = 0$, $y'(0) = 0$?

- (A) $\sin x$ (B) $x \sin x$ (C) $\frac{1}{2}x \sin x$ (D) $\frac{1}{2}x \cos x$ (E) $1 - \cos x$

Problem 128.3

[See solution](#)

Which is a particular solution of $y'' - y = x^2$?

- (A) $-x^2$ (B) $-x^2 - 2$ (C) $x^2 + 2$ (D) $-x^2 + 2$ (E) $x^2 - 2$

Problem 128.4

[See solution](#)

Which is a particular solution of $y'' - 4y' + 4y = e^{2x}$?

- (A) e^{2x} (B) xe^{2x} (C) $\frac{1}{2}x^2e^{2x}$ (D) x^2e^{2x} (E) $\frac{1}{6}x^3e^{2x}$

Problem 128.5[See solution](#)

How many solutions of $y'' + y = \sin x$ are bounded on $[0, \infty)$?

- (A) None
- (B) Exactly one
- (C) Exactly two
- (D) A one-parameter family
- (E) Every solution

Problem 128.6[See solution](#)

Which is a particular solution of $y'' + y = x + \sin 2x$?

- (A) $x + \sin 2x$ (B) $x - \frac{1}{3} \sin 2x$ (C) $-x + \frac{1}{3} \sin 2x$ (D) $x + \frac{1}{2}x \sin 2x$ (E) $\frac{1}{2}x^2 - \frac{1}{3} \sin 2x$

4. Differential Equations

Chapter 4 Review Questions ▸ Growth proportional to population

129 Exponential growth, decay, doubling time, and half-life

Book pp. 178; 376. [Find in compact glossary](#)

In this entry: exponential growth; exponential decay; proportional rate; growth constant; doubling time; half-life.

Definition and key ideas

A quantity whose instantaneous rate is proportional to its amount satisfies $N' = kN$. The solution is $N(t) = N(0)e^{kt}$. Positive k gives growth and negative k decay. Equal time increments multiply the amount by the same factor, independently of the starting time.

If $N(t + T) = aN(t)$ with $a > 0$, then $k = \log a/T$. Doubling time is $\log 2/k$ for $k > 0$, and half-life is $\log 2/|k|$ for $k < 0$. Distinguish a multiplicative increase by a factor a from an additive increase by a units or a percentage increase of $a\%$. In decay or growth applications, the model and proportionality assumption determine the exponential law.

A sum of quantities with different decay constants is generally not one exponential, so its half-time cannot be obtained by averaging individual half-lives. Also a multiplier observed at one fixed time step alone does not imply the differential law $N' = kN$: a positive periodic factor can preserve that step multiplier while changing the instantaneous rate.

Problem 129.1

[See solution](#)

A population follows $N' = kN$ and increases by 50% in three hours. In how many hours, measured from the same initial time, does it become four times its initial size?

- (A) 6 (B) $3 \log 4 / \log(3/2)$ (C) $4 \log 3 / \log 2$ (D) $3 \log(3/2) / \log 4$ (E) 12

Problem 129.2

[See solution](#)

A radioactive sample has 80% of its initial amount after two days and follows exponential decay. What fraction remains after five days?

- (A) $(4/5)^2$ (B) $(4/5)^{5/2}$ (C) $1/2$ (D) $(4/5)^5$ (E) $2/5$

Problem 129.3

[See solution](#)

Two samples initially have equal amounts and decay exponentially with half-lives 2 and 3 years. After how many years is the first sample one-fourth the second?

- (A) 4 (B) 6 (C) 8 (D) 12 (E) 18

4. Differential Equations

Chapter 4 Review Questions ▸ Area conditions and differential modeling

130 Converting integral or geometric conditions into ODEs

Book pp. 178; 376-377. [Find in compact glossary](#)

In this entry: integral equation; differential modeling; area condition; Fundamental Theorem of Calculus; boundary consistency.

Definition and key ideas

A condition involving an area or accumulated quantity can often be converted into an ODE by the Fundamental Theorem of Calculus. If $A(x) = \int_0^x f(t) dt$, then $A'(x) = f(x)$ for continuous f . A geometric relation between $A(x)$ and $xf(x)$ becomes a first-order equation after applying the product rule.

Differentiation may discard an additive constant, so solutions of the resulting ODE must be checked in the original integral relation and against endpoint conditions. For example, $\int_0^x f(t) dt = \alpha x f(x)$ on $x > 0$ leads to $\alpha x f'(x) = (1 - \alpha)f(x)$. Its power-law solutions must have the correct integrability and behavior at 0.

The tangent to $y = f(x)$ at a point with $f'(x) \neq 0$ meets the x-axis at $x - f(x)/f'(x)$. A condition on that intercept can therefore be translated into an ODE, then checked in the original geometry just as an area condition is checked in its original integral relation.

Problem 130.1

[See solution](#)

A positive differentiable function on $(0, \infty)$, continuous at 0, satisfies $\int_0^x f(t) dt = \frac{1}{4}xf(x)$ and $f(1) = 2$. What is $f(2)$?

- (A) 4 (B) 8 (C) 12 (D) 16 (E) 32

Problem 130.2

[See solution](#)

A differentiable function satisfies $f(x) = 2 + \int_0^x (x - t)f(t) dt$. What is $f(x)$?

- (A) $2e^x$ (B) $2 \cos x$ (C) $2 \cosh x$ (D) $2 \sinh x$ (E) $2 + x^2$

Problem 130.3

[See solution](#)

A differentiable function satisfies $f(x) = 1 + \int_0^x 2f(t)dt$. What is $f(1)$?

- (A) e (B) 1 (C) $2e$ (D) 3 (E) e^2

Problem 130.4

[See solution](#)

A positive differentiable curve $y = f(x)$ for $x > 0$ passes through $(1, 2)$. Its tangent at every point $(x, f(x))$ meets the x-axis at $(2x, 0)$. What is $f(x)$?

- (A) $2x$ (B) $2x^2$ (C) $2/x$ (D) $2/\sqrt{x}$ (E) $2e^{1-x}$

Problem 130.5[See solution](#)

Which differentiable functions on $(0, \infty)$ satisfy $\int_1^x f(t) dt = xf(x)$ for every $x > 0$?

- (A) None
- (B) Only $f = 0$
- (C) Every constant function
- (D) Every $f(x) = Ce^x$
- (E) Every $f(x) = Cx$

4. Differential Equations

Chapter 4 Review Questions ▸ Power-product integrating factors

131 Monomial integrating factors and coefficient matching

Book pp. 179; 380. [Find in compact glossary](#)

In this entry: monomial integrating factor; power-product integrating factor; coefficient matching; singular axis.

Definition and key ideas

When a factor is suggested in the form $\mu(x, y) = x^m y^n$, impose exactness after multiplication: $\partial_y(x^m y^n M) = \partial_x(x^m y^n N)$. Matching coefficients of distinct monomials often gives linear equations for the unknown exponents m, n . Then integrate the resulting exact differential to find the potential.

Work on a region where the powers are real and the factor is nonzero, for example $x, y > 0$ for arbitrary real exponents. If m or n is negative, an axis is excluded; if positive, the factor vanishes there. In either case, study any possible axis solutions using the original equation. Multiplication alone does not establish equivalence on those exceptional sets.

Problem 131.1[See solution](#)

On $x, y > 0$, the form $(2y^2 + 3y/x) dx + (2xy + 4) dy$ has an integrating factor $x^m y^n$. What is $m + n$?

- (A) -1 (B) 0 (C) 1 (D) 2 (E) 3

Problem 131.2[See solution](#)

On $x, y > 0$, multiplying $(2y + 1/x) dx + x dy = 0$ by x makes it exact. Which level family describes its solutions?

- (A) $x^2 y + x = C$ (B) $xy + \log x = C$
(C) $x^2 y + x^2 = C$ (D) $xy^2 + x = C$
(E) $x^2 + y^2 = C$

Problem 131.3[See solution](#)

On $x, y > 0$, for which real pairs (m, n) is $x^m y^n$ an integrating factor for $y dx + x dy$?

- (A) Only $(0, 0)$
(B) Exactly those with $m + n = 0$
(C) Exactly those with $m = n$
(D) Exactly those with $m + n = -1$
(E) No real pair

Problem 131.4[See solution](#)

On $x, y > 0$, which pair (m, n) makes $x^m y^n$ an integrating factor for $(y + 1) dx + (x^2 + 1) dy$?

- (A) $(0, 0)$ (B) $(-2, 0)$ (C) $(0, -1)$ (D) $(-2, -1)$ (E) No such pair exists.

Problem 131.5[See solution](#)

Multiplying $y dx - x dy = 0$ by the monomial factor y^{-2} gives $d(x/y) = 0$ on $y \neq 0$. Which solution curve of the original relation is omitted by this transformation?

- (A) $y = 0$ (B) $y = x$ (C) $y = 2x$ (D) $y = -x$ (E) $y = 3x$

4. Differential Equations

Chapter 4 Review Questions ▸ Orthogonal trajectories

132 Orthogonal trajectories and eliminating a family parameter

Book pp. 180; 382. [Find in compact glossary](#)

In this entry: orthogonal trajectory; orthogonal family; negative reciprocal slope; parameter elimination; tangent direction.

Definition and key ideas

Two curve families are orthogonal trajectories if their tangent lines are perpendicular at every regular intersection. If one family has slope field $y' = F(x, y)$, its orthogonal field is $y' = -1/F(x, y)$ where both finite slopes are defined. Horizontal and vertical tangents must be handled by tangent directions rather than a reciprocal formula with division by zero.

Starting from an implicit family $G(x, y, c) = 0$, differentiate with respect to x and eliminate the parameter c using the original equation. Only then take the negative reciprocal and solve the resulting ODE. The parameter must not remain in a slope field intended to describe the entire family. Singular intersections may require separate treatment.

For level families $F = C$ and $G = K$ with nonzero gradients, checking $\nabla F \cdot \nabla G = 0$ directly verifies orthogonality. A proposed family may have two different tangents at one point from different members; then no curve through that point can be perpendicular to both unless those tangents are parallel. Parameter elimination need not produce a single-valued slope field.

Problem 132.1

[See solution](#)

Which family gives orthogonal trajectories to $y = cx^2$ away from the coordinate axes?

- (A) $x^2 + y^2 = C$ (B) $x^2 + 2y^2 = C$ (C) $2x^2 + y^2 = C$ (D) $xy = C$ (E) $y = Cx^{-2}$

Problem 132.2

[See solution](#)

The circles $x^2 + y^2 = 2cy$ are orthogonal to which family?

- (A) $x^2 + y^2 = 2kx$ (B) $x^2 + y^2 = k$ (C) $y = kx$ (D) $x^2 - y^2 = k$ (E) $xy = k$

Problem 132.3

[See solution](#)

Which family is orthogonal to $x^2 - y^2 = C$ at every regular intersection away from the origin?

- (A) $x^2 + y^2 = K$ (B) $xy = K$ (C) $x - y = K$ (D) $x^2 + 2y^2 = K$ (E) $y/x = K$

Problem 132.4[See solution](#)

Which is the complete family of orthogonal trajectories to $x^2 + y^2 = C > 0$ in the punctured plane?

- (A) All circles centered at the origin
- (B) Only the nonvertical lines $y = kx$
- (C) All radial lines, including the vertical line, with components separated at the omitted origin
- (D) All horizontal lines
- (E) All parabolas $y = kx^2$

Problem 132.5[See solution](#)

Two members of $y = cx + c^2$ pass through $(0, 1)$. Which tangent direction at that point is perpendicular to both members?

- (A) Slope 1
- (B) Slope -1
- (C) Slope 0
- (D) Vertical
- (E) No nonzero tangent direction has this property.

4. Differential Equations

Chapter 4 Review Questions ▸ First-order partial differential equations

133 Linear first-order PDEs, characteristic directions, and level curves

Book pp. 180; 383. [Find in compact glossary](#)

In this entry: first-order PDE; transport equation; characteristic curve; directional derivative; gradient; level curve.

Definition and key ideas

For constants a, b , not both zero, the PDE $au_x + bu_y = 0$ says the directional derivative of u along (a, b) vanishes. Thus u is constant on lines in that direction, the characteristic curves. Locally the general differentiable solution is $u(x, y) = F(bx - ay)$, since $bx - ay$ is unchanged along each characteristic.

Initial values along a curve can determine F if the curve crosses the characteristics rather than following one. The chain rule verifies a proposed solution quickly. More generally, $au_x + bu_y = c$ means u changes linearly at rate c along a parameterized characteristic. Domain geometry can affect the global form, so local characteristic descriptions need compatible connected regions for global claims.

For variable coefficients, characteristic curves solve $x' = a(x, y)$ and $y' = b(x, y)$, with $u' = c(x, y)$ along them. They need not be straight lines. A global description by one invariant also requires consistency across characteristic components: a connected domain can intersect one characteristic line in separate pieces on which the constants differ.

Problem 133.1

[See solution](#)

A differentiable function satisfies $3u_x - 2u_y = 0$ and $u(0, y) = e^y$ for all real y . What is $u(3, 1)$?

- (A) e^{-1} (B) 1 (C) e (D) e^2 (E) e^3

Problem 133.2

[See solution](#)

A differentiable function satisfies $u_x + u_y = 2$ on $\mathbb{R}^2$ and $u(x, 0) = x^2$. What is $u(1, 2)$?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Problem 133.3

[See solution](#)

For $u_x + u_y = 0$ on $\mathbb{R}^2$, compare the data (I) $u(t, t) = t$ and (II) $u(t, t) = 1$ for all real t . How many C^1 solutions does each set of data admit?

- (A) I: none; II: exactly one
(B) I: none; II: infinitely many
(C) I: exactly one; II: exactly one
(D) I: infinitely many; II: infinitely many
(E) I: exactly one; II: none

Problem 133.4[See solution](#)

On $x, y > 0$, which family, with arbitrary C^1 function F , gives all C^1 solutions of $xu_x - yu_y = 0$?

- (A) $u = F(xy)$
- (B) $u = F(x/y)$
- (C) $u = F(x + y)$
- (D) $u = F(x^2 - y^2)$
- (E) $u = F(\arctan(y/x))$

Problem 133.5[See solution](#)

Let $D = \mathbb{R}^2 \setminus \{(0, y) : y \leq 0\}$. Can a smooth u on this connected domain satisfy $u_x = 0$, $u(-1, -1) = 0$, and $u(1, -1) = 1$?

- (A) No, because connectedness forces u to be constant.
- (B) No, because $u_x = 0$ forces the same value at any two points of equal height.
- (C) Yes; horizontal characteristic slices can be disconnected.
- (D) Yes, but only if u is discontinuous.
- (E) Only if the PDE is replaced by $u_y = 0$.

5. Linear Algebra

Solutions of Linear Systems ▸ Consistency and Number of Solutions

134 Linear equations and consistent systems

Book pp. 181-182. [Find in compact glossary](#)

In this entry: linear equation; linear system; coefficient; unknown; solution; consistent system; inconsistent system.

Definition and key ideas

A linear equation has the form $a_1x_1 + \cdots + a_nx_n = b$ with fixed coefficients. A linear system asks for a single vector satisfying every equation. It is consistent if it has at least one solution and inconsistent otherwise. Over $\mathbb{R}$ or $\mathbb{C}$, a linear system has zero, one, or infinitely many solutions.

If u, v are two distinct solutions, every $(1 - t)u + tv$ is also a solution, giving infinitely many. A system with fewer equations than unknowns need not be consistent, but if it is consistent it cannot have a unique solution. The zero-one-infinity statement depends on the infinite scalar field: over a finite field, a system can have a finite number greater than one.

Problem 134.1

[See solution](#)

For which (a, b) does the system $x + y + z = 1$, $2x + 2y + 2z = a$, $x - y = b$ have infinitely many real solutions?

- (A) $a = 2$, any b (B) $a = 2, b = 0$ (C) $a = 1$, any b (D) Any $a, b = 0$ (E) No such pair

Problem 134.2

[See solution](#)

A real system $Ax = b$ has two distinct solutions u, v . Which vector must also be a solution?

- (A) $u + v$ (B) $2u$ (C) $3u - 2v$ (D) $u - v$ (E) 0

Problem 134.3

[See solution](#)

All arithmetic is performed modulo 3. How many vectors (x_1, x_2, x_3, x_4) satisfy $x_1 + 2x_2 + x_3 + x_4 = 1$?

- (A) 3 (B) 9 (C) 27 (D) 54 (E) 81

Problem 134.4

[See solution](#)

For a system of two linear equations in four real unknowns, which list gives all possible numbers of solutions as the coefficients and constants vary?

- (A) 0 or 1 only
(B) 1 or infinitely many only
(C) Infinitely many only
(D) 0 or infinitely many only
(E) 0, 1, or infinitely many

Problem 134.5[See solution](#)

If $Ax = b$ is solvable for every $b \in \mathbb{R}^m$ and A is $m \times n$, which must hold?

- (A) $\text{rank } A = n \leq m$
- (B) $A^T A = I_n$
- (C) $m = n$
- (D) Every solution is unique
- (E) $\text{rank } A = m \leq n$

5. Linear Algebra

Matrices and Matrix Algebra ▸ Dimensions and Matrix Operations

135 Matrices, entries, and triangular form

Book pp. 183–184. [Find in compact glossary](#)

In this entry: matrix; matrix dimensions; matrix entry; row; column; diagonal entry; main diagonal; square matrix; upper triangular matrix; lower triangular matrix; zero matrix; matrix addition; scalar multiple; diagonal matrix.

Definition and key ideas

An $m \times n$ matrix has m rows and n columns; a_{ij} is its entry in row i , column j . A square matrix has $m = n$, and its main diagonal consists of a_{ii} . Upper triangular means $a_{ij} = 0$ for $i > j$, and lower triangular means $a_{ij} = 0$ for $i < j$. A diagonal matrix has zero entries away from its main diagonal.

Matrices of the same size add entrywise, and scalar multiplication acts on every entry. The zero matrix is the additive identity. Counting the independently selectable entries gives dimensions of structured matrix spaces: an upper triangular $n \times n$ matrix has $n(n+1)/2$ free entries, while a diagonal one has n . Constraints on entries can reduce these counts.

A matrix is strictly upper triangular if its entries on and below the diagonal vanish. Every square matrix has a unique decomposition into a strictly upper triangular matrix and a lower triangular matrix, obtained by assigning each entry to its permitted part. Extra symmetries must be combined with forced zeros before counting free entries.

Problem 135.1[See solution](#)

How many entries of a real 5×5 upper triangular matrix with all diagonal entries fixed equal to 1 may be chosen independently?

- (A) 5 (B) 10 (C) 15 (D) 20 (E) 25

Problem 135.2[See solution](#)

If a matrix A is both upper and lower triangular and satisfies $A^2 = I_4$, how many real 4×4 matrices A are possible?

- (A) 4 (B) 8 (C) 16 (D) 24 (E) Infinitely many

Problem 135.3[See solution](#)

Real matrices A, B, C, D all have size 2×3 and satisfy $2A + B = C$ and $A - B = D$. Which expression equals $A + 2B$?

- (A) $C + D$ (B) $C - D$ (C) $2C - D$ (D) $(C + 2D)/3$ (E) $(2C - D)/3$

Problem 135.4[See solution](#)

For an arbitrary real $n \times n$ matrix M , how many decompositions $M = U + L$ exist when U is upper triangular with every diagonal entry zero and L is lower triangular?

- (A) None unless M is diagonal
- (B) Exactly one
- (C) Exactly n
- (D) Infinitely many for every M
- (E) One if M is symmetric, and none otherwise

Problem 135.5[See solution](#)

An upper triangular real 9×9 matrix $A = (a_{ij})$ also satisfies $a_{ij} = a_{10-i, 10-j}$ for all indices. How many entries can be chosen independently?

- (A) 1 (B) 4 (C) 5 (D) 9 (E) 45

5. Linear Algebra

Matrices and Matrix Algebra ▸ The Transpose

136 Transpose and row-column duality

Book pp. 184. [Find in compact glossary](#)**In this entry:** transpose; row vector; column vector; transpose of a product.**Definition and key ideas**

The transpose A^T of an $m \times n$ matrix is the $n \times m$ matrix with $(A^T)_{ij} = a_{ji}$. It interchanges rows and columns, so a row vector becomes a column vector. Transposition is linear and involutive: $(A + B)^T = A^T + B^T$, $(cA)^T = cA^T$, and $(A^T)^T = A$.

For products, $(AB)^T = B^T A^T$: the order reverses. Over real scalars, $x^T y$ is the dot product of column vectors. Thus $x^T A^T A x = \|Ax\|^2$, a useful identity showing that $A^T A$ and A have the same nullspace. For complex vectors use conjugate transpose, not ordinary transpose, to obtain the corresponding nonnegative norm identity.

The conjugate transpose is $A^* = \overline{A}^T$, where the bar conjugates each entry. The trace $\text{tr } B$ is the sum of diagonal entries of a square matrix; for real A , $\text{tr}(A^T A) = \sum_{i,j} a_{ij}^2$. A matrix with $A^T = -A$ is skew-symmetric, and $x^T A x = 0$ for every real x .

Problem 136.1[See solution](#)

If a real 3×5 matrix A has rank 2, what is the nullity of $A^T A$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 136.2[See solution](#)

For compatible matrices A, B, C , which is always equal to $(ABC)^T$?

- (A) $A^T B^T C^T$ (B) $C^T B^T A^T$ (C) $B^T C^T A^T$ (D) CBA (E) $A^T C^T B^T$

Problem 136.3[See solution](#)

For $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \end{bmatrix}$, what is $\text{tr}(A^T A)$?

- (A) 8 (B) 32 (C) 16 (D) -16 (E) 17

Problem 136.4[See solution](#)

Let $A = \begin{pmatrix} 1 \\ i \end{pmatrix}$, where $i^2 = -1$, and let $A^* = \overline{A}^T$. Which pair is $(A^T A, A^* A)$?

- (A) (2, 2) (B) (0, 0) (C) (0, 2) (D) (2, 0) (E) (-2, 2)

Problem 136.5[See solution](#)

A real square matrix satisfies $A^T = -A$. For any real column vector x of compatible size, what is $x^T Ax$?

- (A) $\|x\|^2$
- (B) $-\|x\|^2$
- (C) $\det A$
- (D) 0
- (E) It can be any real number.

5. Linear Algebra

Matrices and Matrix Algebra ▷ Matrix Multiplication

137 Matrix multiplication and noncommutativity

Book pp. 184-186. [Find in compact glossary](#)

In this entry: matrix product; compatible dimensions; commuting matrices; associativity; distributivity; noncommutativity.

Definition and key ideas

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$ with $(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$. Each entry is a row-column dot product. The product may exist even when BA does not, or the two products may have different dimensions.

Matrix multiplication is associative and distributive but usually noncommutative. A pair commutes when $AB = BA$. Scalar-looking factorizations can fail: $(A + B)^2 = A^2 + AB + BA + B^2$, and this becomes $A^2 + 2AB + B^2$ only when the two matrices commute. Cancellation by a singular matrix is invalid. Nonzero matrices can have a zero product.

Problem 137.1[See solution](#)

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. What is $\text{tr}((A + B)^2 - A^2 - B^2)$?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 10

Problem 137.2[See solution](#)

Which real 2×2 matrix commutes with $D = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$?

- (A) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
(B) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
(C) $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$
(D) $\begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$
(E) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

Problem 137.3[See solution](#)

If $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$, what is $\text{tr}((AB - BA)^2)$?

- (A) 32 (B) 16 (C) 64 (D) 33 (E) -32

Problem 137.4[See solution](#)

Real 3×3 matrices A, B satisfy $\text{rank } A = 2$ and $AB = 0$. What is the greatest possible rank of B ?

- (A) 0
(B) 1
(C) 2
(D) 3
(E) The conditions are impossible.

Problem 137.5[See solution](#)

Nonzero real column vectors u, v satisfy $v^T u = -2$. Set $M = uv^T$. Which expression is M^5 ?

- (A) $-32M$ (B) $16M$ (C) $-16M$ (D) $32M$ (E) $16I$

5. Linear Algebra

Matrices and Matrix Algebra ▷ Identity Matrices and Inverses

138 Identity matrices and one-sided inverses

Book pp. 186–188. [Find in compact glossary](#)

In this entry: identity matrix; Kronecker delta; left inverse; right inverse; inverse matrix; invertible matrix; singular matrix; nonsingular matrix.

Definition and key ideas

The identity I_n has entries δ_{ij} , where the Kronecker delta is 1 for $i = j$ and 0 otherwise. Compatible multiplication by an identity changes nothing. A left inverse of A satisfies $LA = I$; a right inverse satisfies $AR = I$. For square matrices either condition implies both, and the inverse is unique.

An invertible matrix is also called nonsingular; a noninvertible square matrix is singular. Products satisfy $(AB)^{-1} = B^{-1}A^{-1}$, and $(A^T)^{-1} = (A^{-1})^T$. Rectangular one-sided inverses need not be two-sided: a matrix with more columns than rows may have a right inverse but cannot have a left inverse. Always multiply equations on the side that cancels the intended factor.

Problem 138.1[See solution](#)

Let A be 2×3 and B be 3×2 , with $AB = I_2$. Which statement must be true?

- (A) $BA = I_3$
- (B) BA is singular and $(BA)^2 = BA$
- (C) A has a two-sided inverse
- (D) B has nontrivial nullspace
- (E) $BA = 0$

Problem 138.2[See solution](#)

Invertible square matrices satisfy $AXB = C$. Which expression equals X ?

- (A) $A^{-1}B^{-1}C$
- (B) $CA^{-1}B^{-1}$
- (C) $B^{-1}CA^{-1}$
- (D) $A^{-1}CB^{-1}$
- (E) $B^{-1}A^{-1}C$

Problem 138.3[See solution](#)

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$. Among all real 3×2 matrices B satisfying $AB = I_2$, what is the minimum of the sum of squares of the entries of B ?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) No minimum exists.

Problem 138.4[See solution](#)

If A, B, C are invertible real matrices of the same size, what is $(A^T B^{-1} C)^{-1}$?

- (A) $C^{-1}B(A^{-1})^T$
- (B) $(A^{-1})^T B C^{-1}$
- (C) $C^{-1}B^{-1}(A^{-1})^T$
- (D) $CB(A^{-1})^T$
- (E) $C^{-1}BA^T$

Problem 138.5[See solution](#)

An $m \times n$ real matrix A has matrices L and R satisfying $LA = I_n$ and $AR = I_m$. Which must hold?

- (A) $m < n$ and $L \neq R$
- (B) $m > n$ and $L \neq R$
- (C) $m = n$ and $L = R$
- (D) $L = R$ but m and n may differ
- (E) $m = n$ but L and R may differ

5. Linear Algebra

Matrices and Matrix Algebra ▷ Matrix Powers and Polynomial Identities

139 Matrix powers and polynomial identities

Book pp. 187-188. [Find in compact glossary](#)

In this entry: matrix power; matrix polynomial; polynomial identity; inverse from a polynomial identity.

Definition and key ideas

For square A , define $A^0 = I$, A^k as repeated multiplication, and $p(A) = c_0I + c_1A + \cdots + c_dA^d$. Scalar constants become multiples of the identity. Powers of one matrix commute with each other, even though unrelated matrices need not commute.

A relation $A^2 - aA + bI = 0$ with $b \neq 0$ immediately gives $A^{-1} = (aI - A)/b$. Polynomial identities also reduce high powers recursively. Do not conclude that a product of two matrix polynomials being zero forces either factor to be zero: matrices have zero divisors. Later, Cayley–Hamilton supplies such an identity from every characteristic polynomial.

If $N^r = 0$, then $(I - N)^{-1} = I + N + \cdots + N^{r-1}$. Every polynomial in A commutes with A , and every matrix commuting with A commutes with its polynomials. The converse assertion that every commuting matrix is a polynomial in A is false in general.

Problem 139.1[See solution](#)

If $A^2 - 3A + 2I = 0$, what is A^4 ?

- (A) $7A - 6I$ (B) $8A - 7I$ (C) $15A - 14I$ (D) $16A - 15I$ (E) $9A - 8I$

Problem 139.2[See solution](#)

A square matrix satisfies $A^3 + 2A^2 - A + 3I = 0$. Which expression is its inverse?

- (A) $\frac{1}{3}(A^2 + 2A - I)$
 (B) $\frac{1}{3}(-A^2 - 2A + I)$
 (C) $A^2 + 2A - I$
 (D) $-A^2 + 2A + I$
 (E) It need not be invertible

Problem 139.3[See solution](#)

A real 2×2 matrix satisfies $(A - I)(A - 2I) = 0$. Which statement must hold?

- (A) $A = I$
 (B) $A = 2I$
 (C) $A - I$ is invertible
 (D) $(2I - A)^2 = 2I - A$
 (E) A has two equal diagonal entries

Problem 139.4[See solution](#)

A square matrix N satisfies $N^3 = 0$ but $N^2 \neq 0$. Which expression is $(I - N)^{-1}$?

- (A) $I + N$ (B) $I - N + N^2$ (C) $I + N + N^2$ (D) N^{-1} (E) $I + N + 2N^2$

Problem 139.5[See solution](#)

For real square matrices A, B of the same size and real polynomials p, q , which statement is NOT always true?

- (A) A commutes with A^3 .
(B) $p(A)$ commutes with $q(A)$.
(C) If $AB = BA$, then B commutes with $p(A)$.
(D) If $AB = BA$, then $B = p(A)$ for some polynomial p .
(E) $p(A)^T = p(A^T)$.

5. Linear Algebra

Gaussian Elimination ▸ Row Operations and Echelon Form

140 Gaussian elimination and echelon form

Book pp. 188-190. [Find in compact glossary](#)

In this entry: coefficient matrix; augmented matrix; Gaussian elimination; elementary row operation; row equivalence; echelon form; pivot; back-substitution; reduced row echelon form.

Definition and key ideas

The augmented matrix $[A \mid b]$ records a system $Ax = b$. Elementary row operations swap rows, multiply a row by a nonzero scalar, or add a multiple of one row to another. Each preserves the solution set and has an inverse operation. Gaussian elimination creates echelon form: nonzero rows precede zero rows, and each leading nonzero entry lies to the right of the one above it.

These leading entries are pivots. Back-substitution solves from the bottom upward; columns without pivots in the coefficient portion correspond to free variables. A row $[0 \cdots 0 \mid c]$ with $c \neq 0$ proves inconsistency. Reduced row echelon form additionally makes pivots 1 and the only nonzero entries in their columns; it is unique although echelon forms need not be.

Row equivalent matrices have the same row space and the same reduced row echelon form; having only the same rank is insufficient. A simultaneous set of row combinations preserves every system precisely when its transformation is invertible. One must distinguish simultaneous use of old rows from sequential elementary operations.

Problem 140.1[See solution](#)

For what value of a is $x + y + z = 1$, $x + 2y + 3z = 2$, $2x + 3y + az = 4$ inconsistent?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) All real a

Problem 140.2[See solution](#)

A system has augmented echelon matrix $\begin{pmatrix} 1 & 2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$, with four unknowns. What is its general solution?

- (A) $(3 - 2s + t, s, 2 - 4t, t)$ (B) $(3 - 2s - t, s, 2 + 4t, t)$
 (C) $(3 + 2s + t, s, 2 - 4t, t)$ (D) $(3 - 2s + t, s, 2 + 4t, t)$
 (E) $(3, s, 2, t)$

Problem 140.3[See solution](#)

An augmented matrix has a row $(0, 0, 0 \mid 2)$ after elimination. What follows?

- (A) There is one free variable
- (B) The homogeneous system is inconsistent
- (C) There are two solutions
- (D) The coefficient matrix is invertible
- (E) The system is inconsistent

Problem 140.4[See solution](#)

Which matrix is row equivalent to $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$?

- (A) $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}$
- (B) $\begin{pmatrix} 2 & 4 & 1 \\ 1 & 2 & 1 \end{pmatrix}$
- (C) $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \end{pmatrix}$
- (D) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}$
- (E) $\begin{pmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Problem 140.5[See solution](#)

The two rows of any augmented matrix are simultaneously replaced by $R_1 + R_2$ and $R_1 + tR_2$, using the old rows in both expressions. For which real t does this transformation preserve the solution set for every two-equation system?

- (A) Every t (B) Only $t = 1$ (C) Every $t \neq 0$ (D) Every $t \neq 1$ (E) Only $t = -1$

5. Linear Algebra

Gaussian Elimination ▸ Free Variables and Homogeneous Systems

141 Homogeneous systems and affine solution spaces

Book pp. 191-193. [Find in compact glossary](#)

In this entry: free variable; parameter; homogeneous system; nonhomogeneous system; trivial solution; particular solution; general solution; affine solution space.

Definition and key ideas

A homogeneous system is $Ax = 0$ and always has the trivial solution $x = 0$. Its solutions form a vector space. If a system $Ax = b$ is consistent and x_p is one particular solution, every solution is $x = x_p + x_h$, where $Ax_h = 0$.

With n unknowns and coefficient rank r , a consistent system has $n - r$ independent free parameters. A nonhomogeneous solution set is a translate of a nullspace and is usually an affine space rather than a vector subspace. Differences of solutions are homogeneous solutions; sums of two solutions generally solve $Ax = 2b$, not $Ax = b$.

When $b \neq 0$, the linear span of the nonempty affine solution set has dimension $\dim \ker A + 1$, because a particular solution lies outside the kernel. The solution of least Euclidean norm is characterized by being perpendicular to every homogeneous direction; adding a perpendicular direction would increase its squared norm.

Problem 141.1

[See solution](#)

A real 4×7 matrix has rank 3, and $Ax = b$ is consistent. What is the dimension of its affine solution set?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 7

Problem 141.2

[See solution](#)

The solutions of a consistent real system are $x = (1, 2, 0) + t(1, -1, 1)$, $t \in \mathbb{R}$. Which value of t gives the solution of least Euclidean norm?

- (A) -1 (B) $-1/3$ (C) 0 (D) $1/3$ (E) 1

Problem 141.3

[See solution](#)

If $Ax = b$ has one solution x_0 and $\dim \ker A = 2$, what is its full solution set?

- (A) $x_0 + \ker A$, an affine plane
(B) The image of A
(C) A line through x_0
(D) A two-dimensional vector space in every case
(E) Exactly two points

Problem 141.4[See solution](#)

A real matrix with three columns has rank one. Each of $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ solves $Ax = b$, where $b \neq 0$. What is the full solution set?

- (A) $x_1 + x_2 + x_3 = 0$
- (B) $x_1 + x_2 + x_3 = 1$
- (C) $x_1 = x_2 = x_3$
- (D) Only the three specified points
- (E) All of $\mathbb{R}^3$

Problem 141.5[See solution](#)

A real 3×5 matrix A has rank three, and $b \neq 0$. What is the dimension of the linear span of all solutions of $Ax = b$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

5. Linear Algebra

Solving Matrix Equations Using an Inverse ▸ The Two-by-Two Formula

142 Inverse solution formula and the two-by-two inverse

Book pp. 193-194. [Find in compact glossary](#)**In this entry:** matrix equation; inverse solution formula; two-by-two inverse; unique solvability.**Definition and key ideas**

If A is square and invertible, $Ax = b$ has the unique solution $A^{-1}b$ for every b . For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, invertibility is equivalent to $ad - bc \neq 0$, and $A^{-1} = (ad - bc)^{-1} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

A matrix equation $AX = B$ solves one system for each column of B , yielding $X = A^{-1}B$. For $XA = B$ the inverse must instead multiply on the right. Explicit inversion can be slower than elimination, but the formula is useful for parameter dependence and conceptual questions. If a square matrix is singular, no right-hand side has a unique solution.

For two-by-two matrices, $\text{tr}(A^{-1}) = \text{tr } A / \det A$, where trace means the sum of diagonal entries. A small nonzero determinant can make the inverse large; as matrices approach a singular limit, a small right-hand side need not give a small solution. Inverse action can also be computed from known images by linearity.

Problem 142.1[See solution](#)

For $A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$, what is the sum of the entries of $A^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$?

- (A) -3 (B) -1 (C) 0 (D) 1 (E) 3

Problem 142.2[See solution](#)

The equations $tx + y = 1$, $x + ty = 1$ have a unique real solution for which t ?

- (A) $t \neq 0$ (B) $t \neq 1$ (C) $t \neq -1$ (D) $t \neq \pm 1$ (E) Every real t

Problem 142.3[See solution](#)

For $\varepsilon > 0$, let $A_\varepsilon = \begin{pmatrix} 1 & 1 \\ 1 & 1 + \varepsilon \end{pmatrix}$. If $A_\varepsilon x = (0, \varepsilon)^T$, what is $\lim_{\varepsilon \rightarrow 0^+} x$?

- (A) $(0, 0)^T$
 (B) $(1, 1)^T$
 (C) $(-1, 1)^T$
 (D) $(1, -1)^T$
 (E) The norm of x tends to infinity.

Problem 142.4[See solution](#)

An invertible real 2×2 matrix satisfies $A(1, 1)^T = (2, 0)^T$ and $A(1, -1)^T = (0, 4)^T$. What is $A^{-1}(1, 1)^T$?

- (A) $(1/2, 1/4)^T$ (B) $(3/4, 1/4)^T$ (C) $(1/4, 3/4)^T$ (D) $(3, 1)^T$ (E) $(1/2, -1/4)^T$

Problem 142.5[See solution](#)

An invertible real 2×2 matrix A has $\text{tr } A = \text{tr}(A^{-1}) = 3$. What is $\det A$?

- (A) -3 (B) -1 (C) 1 (D) 3 (E) 9

5. Linear Algebra

Vector Spaces ▸ Vector Spaces and Subspaces

143 Vector space axioms and the subspace test

Book pp. 195. [Find in compact glossary](#)

In this entry: real vector space; vector space; scalar field; closure; subspace; zero vector; trivial subspace; proper subspace.

Definition and key ideas

A vector space over a field has vector addition and scalar multiplication satisfying the usual associativity, commutativity of addition, zero and additive inverse laws, distributive laws, and compatibility with scalar multiplication, including $1v = v$. The elements can be functions or matrices rather than geometric arrows.

A subset of an existing vector space is a subspace exactly when it is nonempty and closed under linear combinations $au + bv$. It must contain zero, but that alone does not suffice. Homogeneous linear constraints define subspaces; a nonzero constant constraint usually defines an affine translate. Intersections of subspaces are subspaces; their union is a subspace only when one contains the other.

The trivial subspace is $\{0\}$; a proper subspace is not the whole vector space.

For finite-dimensional subspaces, $\dim(U + V) = \dim U + \dim V - \dim(U \cap V)$, where $U + V = \{u + v : u \in U, v \in V\}$. The field and the chosen operations are part of a vector-space structure: a real subspace need not be a complex one, and with altered operations the zero vector need not be the usual numerical zero. A bijection carrying both operations to those of a known vector space verifies all axioms.

Problem 143.1[See solution](#)

Which subset of $\mathbb{R}^3$ is a vector subspace?

- (A) $x + y + z = 1$ (B) $x^2 + y^2 + z^2 = 1$
 (C) $xy = 0$ (D) $x + y = 0, y - z = 0$
 (E) $x, y, z \geq 0$

Problem 143.2[See solution](#)

If U, V are subspaces of a real vector space and $U \cup V$ is a subspace, which statement must hold?

- (A) $U = V$
 (B) $U \cap V = \{0\}$
 (C) $U \subseteq V$ or $V \subseteq U$
 (D) $U + V = \{0\}$
 (E) Both have dimension one

Problem 143.3[See solution](#)

On the set of positive real numbers define $u \oplus v = uv$ and $c \odot u = u^c$ for every real c . Which statement is correct?

- (A) This is not a real vector space because it omits the number 0.
- (B) It is a real vector space with zero vector 1 and additive inverse of u equal to $1/u$.
- (C) It is a real vector space with zero vector 1 and additive inverse $-u$.
- (D) It is a real vector space with zero vector 0.
- (E) It fails closure under negative scalar multiplication.

Problem 143.4[See solution](#)

If U, V are subspaces with dimensions 4, 5 and $\dim(U \cap V) = 2$, what is $\dim(U + V)$?

- (A) 5 (B) 2 (C) 11 (D) 7 (E) 9

Problem 143.5[See solution](#)

For subspaces U, V , which is always a subspace?

- (A) $u_0 + U$ for arbitrary u_0
- (B) $U \cup V$
- (C) $U \setminus V$
- (D) $\{u \in U : u \neq 0\}$
- (E) $U \cap V$

Problem 143.6[See solution](#)

The set $W = i\mathbb{R}$ is considered inside $\mathbb{C}$ with ordinary addition. Which statement describes its subspace status?

- (A) It is a subspace over both $\mathbb{R}$ and $\mathbb{C}$.
- (B) It is a subspace over $\mathbb{R}$ but not over $\mathbb{C}$.
- (C) It is a subspace over $\mathbb{C}$ but not over $\mathbb{R}$.
- (D) It is a subspace over neither field.
- (E) Its subspace status depends on whether 0 is included.

5. Linear Algebra

Vector Spaces ▸ The Nullspace

144 Nullspace and homogeneous directions

Book pp. 196. [Find in compact glossary](#)**In this entry:** nullspace; null space; kernel of a matrix; nullity of a matrix.**Definition and key ideas**

For an $m \times n$ matrix A , its nullspace is $\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}$. It is a subspace of the input space $\mathbb{R}^n$, not generally of $\mathbb{R}^m$. Solve the homogeneous system by elimination and express the answer as a linear combination of vectors multiplying independent free parameters.

Those parameter vectors form a nullspace basis; their number is the nullity. Row operations preserve the nullspace because they preserve solutions. A square matrix is invertible exactly when its nullspace is $\{0\}$, but a rectangular matrix can also have trivial nullspace when its columns are independent.

For a square matrix, $\ker A \subseteq \ker A^2 \subseteq \cdots$, since another application of A still sends zero to zero. These inclusions can be strict; a vector may map into the kernel without itself being in the kernel.

Problem 144.1[See solution](#)

What is a basis for the nullspace of $\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \end{pmatrix}$?

- (A) $\{(1, -1, 1)\}$ (B) $\{(-1, 1, 1)\}$
 (C) $\{(1, 1, -1)\}$ (D) $\{(1, 0, -1), (0, 1, -2)\}$
 (E) $\{(0, 0, 0)\}$

Problem 144.2[See solution](#)

A real $m \times n$ matrix A satisfies $Ax = 0$ only for $x = 0$. Which statement necessarily follows?

- (A) $m < n$
 (B) $m = n$
 (C) $m \geq n$
 (D) A is square and invertible
 (E) A^T has trivial nullspace

Problem 144.3[See solution](#)

A linear map $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ satisfies $Ae_1 = 0$, $Ae_2 = e_1$, and $Ae_3 = e_2$ for the standard basis. What is $\ker A^2$?

- (A) $\{0\}$ (B) $\text{span}\{e_1\}$ (C) $\text{span}\{e_1, e_2\}$ (D) $\text{span}\{e_2, e_3\}$ (E) $\mathbb{R}^3$

Problem 144.4[See solution](#)

What is the nullity of the 4×4 matrix whose every entry is 1?

- (A) 1 (B) 0 (C) 4 (D) 3 (E) 2

Problem 144.5[See solution](#)

If $\ker A \cap \ker B = \{0\}$ for maps $A, B : \mathbb{R}^n \rightarrow \mathbb{R}^m$, which map is injective?

- (A) $A - B$ (B) $x \mapsto (Ax, Bx)$ (C) B (D) $A + B$ (E) A

5. Linear Algebra

Vector Spaces ▸ Linear Combinations

145 Linear combinations and span

Book pp. 196-197. [Find in compact glossary](#)**In this entry:** linear combination; span; spanning set; trivial linear combination.**Definition and key ideas**

A linear combination of $v_1, \dots, v_k$ is $a_1v_1 + \dots + a_kv_k$. Their span is the set of all such combinations, the smallest subspace containing them. Membership in a span becomes a linear system with the spanning vectors as columns and the target as right-hand side.

Adding a vector already in the span changes neither the span nor its dimension. Replacing one vector by itself plus a multiple of another preserves the span; multiplying a vector by a nonzero scalar does too. These facts underlie elimination. The span of the empty set is conventionally $\{0\}$. A spanning set can contain substantial redundancy.

The trivial linear combination is the one with every coefficient zero. A span uses arbitrary scalars and finite sums, so even nonnegative generators can span functions of either sign. One always has $\text{span}(S \cap T) \subseteq \text{span } S \cap \text{span } T$, but equality can fail.

Problem 145.1[See solution](#)

For what a is $(1, a, 3)$ in the span of $(1, 1, 0)$ and $(0, 1, 1)$?

- (A) -2 (B) 0 (C) 2 (D) 3 (E) 4

Problem 145.2[See solution](#)

Linearly independent vectors u, v are given. For which real a does $\text{span}\{u + av, au + v\} = \text{span}\{u, v\}$?

- (A) $a \neq 0$ (B) $a \neq 1$ (C) $a \neq -1$ (D) $a \neq \pm 1$ (E) All real a

Problem 145.3[See solution](#)

In $\mathbb{R}^2$ let $S = \{(1, 0), (0, 1)\}$ and $T = \{(1, 1), (1, -1)\}$. What is the pair $(\text{span}(S \cap T), \text{span } S \cap \text{span } T)$?

- (A) $(\{0\}, \{0\})$ (B) $(\{0\}, \mathbb{R}^2)$ (C) $(\mathbb{R}^2, \{0\})$ (D) $(\mathbb{R}^2, \mathbb{R}^2)$ (E) $(\emptyset, \mathbb{R}^2)$

Problem 145.4[See solution](#)

Let $U = \text{span}\{(1, 1, 0), (0, 1, 1)\}$ and $W = \text{span}\{(1, 0, 1), (0, 1, 0)\}$. Which vector spans $U \cap W$?

- (A) $(1, 1, 1)$ (B) $(1, 2, 1)$ (C) $(1, 0, -1)$ (D) $(0, 1, 0)$ (E) $(1, -2, 1)$

Problem 145.5[See solution](#)

In the real polynomial space P_2 , what is the span of $\{(x - a)^2 : a \in \mathbb{R}\}$?

- (A) Only multiples of x^2
- (B) Only even polynomials in P_2
- (C) Only polynomials of degree exactly two
- (D) All of P_2
- (E) Only polynomials nonnegative on $\mathbb{R}$

5. Linear Algebra

Vector Spaces ▸ Linear Independence

146 Linear independence and dependence

Book pp. 196–197. [Find in compact glossary](#)**In this entry:** linearly independent; linearly dependent; dependence relation; redundant vector.**Definition and key ideas**

Vectors $v_1, \dots, v_k$ are linearly independent if $\sum_i a_i v_i = 0$ forces every $a_i = 0$. Otherwise they are dependent. Dependence means at least one vector is a linear combination of the others; it does not mean every vector can always be discarded. A set containing zero is dependent, and more than n vectors in an n -dimensional space are dependent.

To test independence, solve the homogeneous system having these vectors as columns. For exactly n vectors in $\mathbb{R}^n$, a nonzero determinant is equivalent to independence. Pairwise nonparallel vectors need not be independent when there are three or more. Nonzero pairwise orthogonal vectors are independent.

Problem 146.1[See solution](#)

As functions on $\mathbb{R}$, the three functions $1, \sin^2 x, \cos^2 x$ are

- (A) linearly independent
- (B) linearly dependent although no two are scalar multiples
- (C) all scalar multiples of one function
- (D) independent on $(0, \pi/2)$ but dependent on $\mathbb{R}$
- (E) dependent only if complex scalars are allowed

Problem 146.2[See solution](#)

If u, v, w are linearly independent, which list is necessarily linearly independent?

- (A) $u + v, v + w, u + 2v + w$
- (B) $u + v, u - v, w$
- (C) $u, 2u, v$
- (D) $u + v, 2u + 2v, w$
- (E) $u - v, v - u, w$

Problem 146.3[See solution](#)

For which values of t are the columns of $\begin{bmatrix} t & 1 & 0 \\ 2 & t & 1 \\ 0 & 1 & 1 \end{bmatrix}$ dependent? Give the product of all such real values.

- (A) -3 (B) 2 (C) -1 (D) -4 (E) -2

Problem 146.4[See solution](#)

Nonzero vectors $v_1, \dots, v_k$ in a real inner-product space are pairwise orthogonal. Which argument proves their linear independence?

- (A) Every pair spans the whole space.
- (B) The sum of their norms is nonzero.
- (C) Taking the inner product of $\sum a_i v_i = 0$ with v_j gives $a_j \|v_j\|^2 = 0$.
- (D) No two have equal lengths.
- (E) Every orthogonal vector must be the zero vector.

Problem 146.5[See solution](#)

Four vectors have exactly one independent dependence relation, $2v_1 - v_2 + v_3 = 0$. How many of their four three-vector sublists are bases for the span of all four?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

5. Linear Algebra

Vector Spaces ▸ Bases and Dimension

147 Basis, dimension, and minimal spanning sets

Book pp. 197-198. [Find in compact glossary](#)

In this entry: basis; minimal spanning set; dimension; finite-dimensional space; infinite-dimensional space; basis extension.

Definition and key ideas

A basis is a linearly independent spanning set. Every vector has a unique expression as a finite linear combination of basis vectors. All bases of a finite-dimensional vector space have the same size, its dimension. If no finite spanning set exists, the space is infinite dimensional.

In an n -dimensional space, any n independent vectors form a basis, and any n spanning vectors form a basis. An independent list can be extended to a basis; a finite spanning list can be reduced to one. A proper subspace of a finite-dimensional space has smaller dimension. Counting listed generators alone does not determine dimension until redundancy is removed.

Problem 147.1[See solution](#)

Let $V = \{(x_1, x_2, x_3, x_4) : x_1 + x_2 + x_3 + x_4 = 0\}$. Which list is a basis of V ?

- (A) $(1, -1, 0, 0), (0, 1, -1, 0), (0, 0, 1, -1)$
- (B) $(1, -1, 0, 0), (2, -2, 0, 0), (0, 0, 1, -1)$
- (C) $(1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0)$
- (D) $(1, -1, 0, 0), (0, 0, 1, -1)$
- (E) $(1, 1, -1, -1), (2, 2, -2, -2), (0, 1, -1, 0)$

Problem 147.2[See solution](#)

Six vectors span a four-dimensional real vector space. What is the least number that must be removed to leave a basis, with suitable choices of vectors to remove?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 147.3[See solution](#)

Let W be the vector space of all real polynomials that vanish at 0, with no degree bound. Which is a basis of W ?

- (A) $\{1, x, x^2, \dots\}$
- (B) $\{x, x^2, x^3, \dots\}$
- (C) $\{x, x^2, \dots, x^{100}\}$
- (D) $\{x^2, x^3, x^4, \dots\}$
- (E) $\{x, 2x, 3x, \dots\}$

Problem 147.4[See solution](#)

How many two-element subsets of $\{1, x, x^2, x^3\}$ can be added to $\{1 + x, x + x^2\}$ to form a basis of P_3 ?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Problem 147.5[See solution](#)

Which assertion about a five-dimensional space is always true?

- (A) Every list of four vectors is independent
- (B) Every list of six vectors is dependent
- (C) Every independent list spans
- (D) Every spanning list has exactly five vectors
- (E) Every list of five vectors is a basis

5. Linear Algebra

Vector Spaces ▷ The Rank, Column Space, and Row Space

148 Rank, column space, and row space

Book pp. 198–200. [Find in compact glossary](#)**In this entry:** rank; row rank; column rank; row space; column space; pivot column; full rank.**Definition and key ideas**

For an $m \times n$ matrix, the column space lies in $\mathbb{R}^m$ and the row space in $\mathbb{R}^n$. Their dimensions agree; the common number is the rank, at most $\min(m, n)$. The equation $Ax = b$ is consistent exactly when b lies in the column space.

Row reduction preserves row space and rank, so the nonzero rows of an echelon matrix form a row-space basis. It generally changes column space: use the pivot columns of the original matrix to obtain a column-space basis. Equivalently row-reduce A^T to obtain column-space information. Full rank means rank $\min(m, n)$, not necessarily invertibility for a rectangular matrix.

The identities $\text{Row}(A) = \text{Col}(A^T)$ and $\text{Col}(A) = \text{Row}(A^T)$ also give membership tests. If $A^T y = 0$, every vector in the column space is perpendicular to y ; hence $y^T b \neq 0$ proves that $Ax = b$ is inconsistent.

Problem 148.1[See solution](#)

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$. For which b is $(1, b, 0)^T$ in its column space?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Every real b

Problem 148.2[See solution](#)

A 3×4 matrix has reduced row echelon form $\begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$. Which original columns necessarily form a basis for its column space?

- (A) Columns 1, 2 (B) Columns 1, 2, 3
 (C) Columns 1, 2, 4 (D) Columns 2, 3
 (E) All four columns

Problem 148.3[See solution](#)

A real 4×6 matrix A has rank three, and $b \in \mathbb{R}^6$. Which condition is equivalent to b belonging to the row space of A ?

- (A) $Ax = b$ has a solution.
- (B) $A^T x = b$ has a solution with $x \in \mathbb{R}^4$.
- (C) $Ab = 0$.
- (D) b belongs to a four-dimensional row space.
- (E) b is a column of A .

Problem 148.4[See solution](#)

Which operation necessarily preserves the column space of A ?

- (A) Replacing a column by zero
- (B) Right multiplication by an invertible matrix
- (C) Left multiplication by every invertible matrix
- (D) Deleting a pivot column
- (E) Transposing A

Problem 148.5[See solution](#)

For a real matrix A and compatible vector b , suppose there is a vector y with $A^T y = 0$ and $y^T b = 1$. Which conclusion follows?

- (A) $Ax = b$ has a unique solution.
- (B) $Ax = b$ has infinitely many solutions.
- (C) $Ax = b$ has no solution.
- (D) A has full row rank.
- (E) b lies in the row space of A .

5. Linear Algebra

Vector Spaces ▸ Other Vector Spaces

149 Polynomial, function, and matrix spaces

Book pp. 200. [Find in compact glossary](#)

In this entry: polynomial space; function space; bounded functions; continuous functions; differentiable functions; matrix vector space; matrix units.

Definition and key ideas

The polynomial space P_n consists of polynomials of degree at most n , including zero. Its standard basis $1, x, \dots, x^n$ gives dimension $n + 1$. Polynomials of degree exactly n do not form a vector space. The space $M_{m \times n}(\mathbb{R})$ of real matrices has dimension mn , with basis matrices having one entry 1 and all others zero.

Spaces of all continuous, differentiable, or bounded functions on a fixed nondegenerate interval are generally infinite dimensional. Linear conditions on values, derivatives, or integrals define subspaces and can be counted as independent constraints. Differentiation is linear and sends P_n into P_{n-1} ; multiplication of functions is not one of the required vector-space operations.

The matrix unit E_{ij} has a single 1 in position (i, j) and zeros elsewhere; every matrix is uniquely $\sum a_{ij}E_{ij}$. A bounded function satisfies $|f(x)| \leq M$ for some finite constant on its entire domain. Continuous functions have no discontinuities, and differentiable functions have derivatives at every interior point; each of these classes, with a fixed common domain, is closed under finite linear combinations. Infinite dimension can be proved by finding independent lists of arbitrarily large finite length.

Problem 149.1

[See solution](#)

What is the dimension of $\{p \in P_4 : p(0) = p(1) = 0\}$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 149.2

[See solution](#)

What is the dimension of the real 3×3 matrices whose three row sums are all zero?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Problem 149.3

[See solution](#)

What is the dimension of the real 4×4 matrices whose row sums and column sums are all zero?

- (A) 9 (B) 8 (C) 10 (D) 12 (E) 16

Problem 149.4[See solution](#)

What is the dimension of the real vector space of bounded continuous functions on $\mathbb{R}$?

- (A) 1
- (B) 2
- (C) Countably many functions, so it is not a vector space
- (D) Infinite
- (E) It is not a vector space because sums need not be bounded.

Problem 149.5[See solution](#)

What is the dimension of the even real polynomials of degree at most 10 satisfying $p(1) = 0$?

- (A) 3 (B) 4 (C) 6 (D) 10 (E) 5

Problem 149.6[See solution](#)

Let E_{ij} be the standard matrix units in $M_{2 \times 3}(\mathbb{R})$. Which set is a basis for the matrices satisfying $a_{11} + a_{23} = 0$?

- (A) All six E_{ij}
- (B) $E_{11} - E_{23}, E_{12}, E_{13}, E_{21}, E_{22}$
- (C) $E_{11} + E_{23}, E_{12}, E_{13}, E_{21}, E_{22}$
- (D) $E_{12}, E_{13}, E_{21}, E_{22}$
- (E) $E_{11}, E_{23}, E_{12}, E_{13}, E_{21}$

5. Linear Algebra

Determinants ▸ Permutations and the Defining Formula

150 Permutation signs and determinant definition

Book pp. 200-202. [Find in compact glossary](#)

In this entry: determinant; permutation; transposition; even permutation; odd permutation; sign of a permutation; Leibniz determinant formula.

Definition and key ideas

A permutation of $\{1, \dots, n\}$ is a bijection of that set. Every permutation is a product of transpositions, and the parity of their number is well defined. Its sign is $+1$ for even parity and -1 for odd parity. The determinant is $\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}$.

Each term chooses one entry from every row and column. For 2×2 matrices this is $ad - bc$; for 3×3 there are six signed terms. The usual three-by-three diagonal shortcut does not extend to higher sizes. A permutation matrix has determinant equal to the permutation sign, and a cycle of length k has sign $(-1)^{k-1}$.

A transposition swaps exactly two elements and fixes the others.

An inversion is a pair $i < j$ with $\sigma(i) > \sigma(j)$; the sign is $(-1)^{\text{number of inversions}}$. Signs multiply under composition. A determinant monomial using any row or column twice is absent, and zeros can eliminate nearly all permutation terms in a sparse matrix.

Problem 150.1[See solution](#)

A permutation matrix acts by a five-cycle on the first five coordinates and exchanges coordinates 6 and 7. What is its determinant?

- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

Problem 150.2[See solution](#)

In the determinant of a 4×4 matrix, what is the coefficient of $a_{13}a_{21}a_{34}a_{42}$?

- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

Problem 150.3[See solution](#)

In the polynomial expansion of the determinant of a general 4×4 matrix, what is the coefficient of $a_{11}a_{23}a_{33}a_{44}$?

- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

Problem 150.4[See solution](#)

If σ and τ are any permutations of a finite set, what is the sign of $\sigma\tau\sigma^{-1}\tau^{-1}$?

- (A) Always $+1$
- (B) Always -1
- (C) The sign of σ
- (D) The sign of τ
- (E) It is undefined unless σ and τ commute.

Problem 150.5[See solution](#)

What is the determinant of $\begin{pmatrix} 1 & t & 0 & 0 \\ 0 & 2 & t & 0 \\ 0 & 0 & 3 & t \\ t & 0 & 0 & 4 \end{pmatrix}$?

- (A) $24 + t^4$ (B) $24 - t^4$ (C) $24 - 4t$ (D) $24t - t^4$ (E) 24

5. Linear Algebra

Determinants ▸ Properties and Elimination

151 Determinant laws and row operations

Book pp. 202-204. [Find in compact glossary](#)

In this entry: determinant multilinearity; alternating determinant; determinant row operations; triangular determinant; determinant of a product; determinant scaling; determinant of a transpose.

Definition and key ideas

The determinant is linear in each row separately, changes sign under a row interchange, and equals 1 at the identity. Multiplying a single row by c multiplies the determinant by c ; adding a multiple of a different row leaves it unchanged. A triangular determinant is the product of diagonal entries.

For square matrices of the same size, $\det(AB) = \det A \det B$, $\det A^T = \det A$, and $\det(cA) = c^n \det A$. An invertible matrix satisfies $\det A^{-1} = 1/\det A$. A zero row, repeated rows, or any dependent row collection gives determinant zero. The determinant is not linear in the whole matrix, so $\det(A + B)$ is not generally $\det A + \det B$.

For block upper triangular matrices, $\det \begin{pmatrix} B & C \\ 0 & D \end{pmatrix} = \det B \det D$ when the diagonal blocks are square. Exchanging blocks of r and s rows introduces sign $(-1)^{rs}$, since every row of one block passes every row of the other. Simultaneous row combinations can instead be recorded by a left multiplier and its determinant.

Problem 151.1[See solution](#)

If A is 3×3 with $\det A = -2$, what is $\det(2A^{-1}A^T)$?

- (A) -8 (B) -4 (C) 1 (D) 4 (E) 8

Problem 151.2[See solution](#)

A matrix B is obtained from A by swapping two rows, multiplying one row by -3 , and adding five times a row to a different row. If $\det B = 12$, what is $\det A$?

- (A) -36 (B) -4 (C) 4 (D) 12 (E) 36

Problem 151.3[See solution](#)

A 3×3 matrix A has rows r_1, r_2, r_3 and determinant 5. A matrix B has rows $r_1 + r_2, r_2 + r_3, r_3 + r_1$. What is $\det B$?

- (A) 0 (B) 5 (C) 10 (D) 15 (E) 40

Problem 151.4[See solution](#)

A real 5×5 matrix satisfies $A^T = -A$. What must its determinant be?

- (A) -1
- (B) 0
- (C) 1
- (D) A positive number
- (E) Any real number is possible.

Problem 151.5[See solution](#)

For 3×3 matrices B, C with $\det B = 2$ and $\det C = -3$, what is $\det \begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix}$?

- (A) -36 (B) -6 (C) 0 (D) 6 (E) 36

5. Linear Algebra

Determinants ▸ Laplace Expansions

152 Minors, cofactors, and Laplace expansion

Book pp. 204-205; 217. [Find in compact glossary](#)**In this entry:** minor; cofactor; Laplace expansion; cofactor expansion; determinant test for collinearity.**Definition and key ideas**

Deleting row i and column j from A gives a submatrix whose determinant is the (i, j) minor. Its cofactor is $C_{ij} = (-1)^{i+j} \det A_{ij}$. Expanding along any row gives $\det A = \sum_j a_{ij} C_{ij}$, and expanding along any column gives $\sum_i a_{ij} C_{ij}$.

Choose a row or column with many zeros to shorten the calculation. Signs alternate in a checkerboard beginning with $+$ at $(1, 1)$. The cofactor already includes this sign; do not attach it a second time. If all entries except a_{ij} are fixed, the determinant depends linearly on a_{ij} with coefficient C_{ij} , because that minor does not involve the varied entry.

For different row indices i, k , $\sum_j a_{ij} C_{kj} = 0$, by expansion of a matrix with repeated rows. Laplace expansion can also produce recurrences for structured determinants. Although determinants are not linear in an arbitrary matrix change, $\det(A + tuv^T)$ is affine in t : any term choosing two updated rows vanishes because those rows are proportional.

Three planar points $(x, y), (a, b), (c, d)$ are collinear exactly when $\det \begin{pmatrix} x & y & 1 \\ a & b & 1 \\ c & d & 1 \end{pmatrix} = 0$. With two

distinct fixed points this is the equation of their line. If the two fixed points coincide, two rows are identical and the condition holds for the entire plane.

Problem 152.1[See solution](#)

What is the determinant of $\begin{pmatrix} 1 & 0 & 2 & 0 \\ 3 & 4 & 0 & 0 \\ 0 & 0 & 5 & 6 \\ 0 & 0 & 7 & 8 \end{pmatrix}$?

(A) -16 (B) -8 (C) 8 (D) 16 (E) 40 **Problem 152.2**[See solution](#)

For $A(t) = \begin{pmatrix} 1 & 2 & 3 \\ 0 & t & 4 \\ 2 & 1 & 0 \end{pmatrix}$, what is $d(\det A(t))/dt$?

(A) -6 (B) -4 (C) 0 (D) 4 (E) 6

Problem 152.3[See solution](#)

Let C_{ij} denote cofactors of a 4×4 matrix A . What is $\sum_{j=1}^4 a_{1j}C_{2j}$?

- (A) $\det A$
- (B) $-\det A$
- (C) 0
- (D) $\operatorname{tr} A$
- (E) It depends on the entries of A .

Problem 152.4[See solution](#)

Let D_n be the determinant of the $n \times n$ matrix having 2 on the main diagonal, -1 immediately above and below it, and zeros elsewhere. What is D_5 ?

- (A) 1 (B) 5 (C) 6 (D) 16 (E) 32

Problem 152.5[See solution](#)

For a fixed real $n \times n$ matrix A and column vectors u, v , consider $D(t) = \det(A + tuv^T)$. Which statement always holds?

- (A) D is constant.
- (B) D has degree exactly one.
- (C) D is of the form $a + bt$, possibly identically zero.
- (D) D has degree exactly n .
- (E) D contains only even powers of t .

Problem 152.6[See solution](#)

For fixed real a, b, c, d , let L be the set of (x, y) for which $\det \begin{pmatrix} x & y & 1 \\ a & b & 1 \\ c & d & 1 \end{pmatrix} = 0$. Which description is always correct?

- (A) L is a line for every choice of the constants.
- (B) L is a line if $(a, b) \neq (c, d)$ and is all of $\mathbb{R}^2$ otherwise.
- (C) L is empty if $(a, b) = (c, d)$.
- (D) L consists only of the two fixed points.
- (E) L is a circle unless $b = d$.

5. Linear Algebra

Determinants ▸ The Adjugate Matrix

153 Adjugate and the invertibility criterion

Book pp. 205-206. [Find in compact glossary](#)

In this entry: adjugate; classical adjoint; cofactor matrix; adjugate inverse formula; determinant criterion for invertibility.

Definition and key ideas

The adjugate is the transpose of the cofactor matrix: $(\text{adj } A)_{ij} = C_{ji}$. It satisfies $A \text{adj } A = \text{adj } A A = (\det A)I$, even for singular matrices. Consequently A is invertible exactly when $\det A \neq 0$, with $A^{-1} = \text{adj } A / \det A$.

For invertible $n \times n$ matrices, $\det(\text{adj } A) = (\det A)^{n-1}$. The identity extends to singular matrices for $n \geq 2$. If rank is at most $n - 2$, every $(n - 1)$ -minor vanishes and the adjugate is zero. If rank is $n - 1$, the adjugate is nonzero but its columns lie in the one-dimensional kernel, so it has rank one. Classical adjoint here is distinct from conjugate transpose.

Problem 153.1

[See solution](#)

If A is invertible 4×4 with $\det A = 3$, what is $\det(\text{adj}(2A))$?

- (A) 27 (B) 432 (C) 1728 (D) 110592 (E) 331776

Problem 153.2

[See solution](#)

A 4×4 real matrix has rank 2. What is $\text{adj } A$?

- (A) I_4 (B) A (C) 0 (D) A matrix of rank one (E) A matrix of rank two

Problem 153.3

[See solution](#)

If an $n \times n$ matrix, $n \geq 2$, has rank $n - 1$, what is the rank of its adjugate?

- (A) It cannot be determined
(B) 1
(C) $n - 1$
(D) n
(E) 0

Problem 153.4

[See solution](#)

For $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$, what is the $(1, 3)$ entry of $\text{adj } A$?

- (A) 0 (B) -6 (C) 6 (D) -3 (E) 2

Problem 153.5[See solution](#)

If A is invertible and $\operatorname{adj} A = 2A^{-1}$, what is $\det A$?

- (A) -1
- (B) 2
- (C) It depends on matrix size
- (D) 1
- (E) 4

5. Linear Algebra

Determinants ▸ Cramer's Rule

154 Cramer's rule and a selected unknown

Book pp. 206-207. [Find in compact glossary](#)**In this entry:** Cramer rule; column replacement determinant.**Definition and key ideas**

For an invertible $n \times n$ matrix A and system $Ax = b$, replace column j of A by b to obtain A_j . Cramer's rule states $x_j = \det A_j / \det A$. It follows from determinant multilinearity by writing b as the combination of columns with coefficients x_i .

This can be efficient when only one unknown is requested, especially if the replaced column contains many zeros. The nonzero-denominator hypothesis is essential: when $\det A = 0$, this formula cannot distinguish no solutions from infinitely many. Replace a column corresponding to the unknown, not a row, under the usual column-vector convention.

For a row-vector system $x^T A = b^T$, the corresponding rule replaces row j by b^T . If the same b is added to every column of an invertible A with $Ax = b$, multilinearity gives determinant $(1 + \sum_j x_j) \det A$, because terms using b twice vanish.

Problem 154.1[See solution](#)

The system $x + 2y + z = 0$, $2x + y - z = 3$, $x - y + 2z = 1$ has a unique solution. What is y ?

- (A) $-\frac{1}{2}$ (B) $-\frac{1}{3}$ (C) 0 (D) $\frac{1}{3}$ (E) $\frac{2}{3}$

Problem 154.2[See solution](#)

An invertible matrix A has columns c_1, c_2, c_3 . A new matrix has columns $c_1, 2c_1 - 3c_2 + 4c_3, c_3$. What is the ratio of its determinant to $\det A$?

- (A) -4 (B) -3 (C) 2 (D) 3 (E) 4

Problem 154.3[See solution](#)

Let $A = \text{diag}(1, 0, 0)$ and $b = (0, 1, 0)^T$. If A_j replaces column j of A by b , which statement is correct?

- (A) All three $\det A_j$ vanish, and $Ax = b$ is inconsistent.
(B) All three $\det A_j$ vanish, and $Ax = b$ has infinitely many solutions.
(C) Exactly one $\det A_j$ is nonzero, and the system is inconsistent.
(D) Exactly one $\det A_j$ is nonzero, and the system has a unique solution.
(E) All three $\det A_j$ are nonzero.

Problem 154.4[See solution](#)

An invertible matrix A and a row vector b^T satisfy $x^T A = b^T$. Which determinant ratio gives x_j ?

- (A) Replace row j of A by b^T , then divide its determinant by $\det A$.
- (B) Replace column j of A by b , then divide by $\det A$.
- (C) Replace row j by b^T , then multiply by $\det A$.
- (D) Divide $\det A$ by the determinant after replacing row j .
- (E) Replace both row and column j , then divide by $\det A$.

Problem 154.5[See solution](#)

An invertible 3×3 matrix A has columns c_1, c_2, c_3 , and $Ax = b$. The matrix with columns $c_1 + b, c_2 + b, c_3 + b$ has determinant $7 \det A$. What is $x_1 + x_2 + x_3$?

- (A) 3 (B) 4 (C) 6 (D) 7 (E) 8

5. Linear Algebra

Linear Transformations ▸ Linearity and Component Vectors

155 Linear transformations and coordinates in a basis

Book pp. 207-208. [Find in compact glossary](#)

In this entry: linear transformation; linear map; linear operator; component vector; coordinate vector; image of a basis.

Definition and key ideas

A map $T : V \rightarrow W$ is linear if $T(u + v) = T(u) + T(v)$ and $T(cu) = cT(u)$. If $V = W$ it is a linear operator. Every linear map sends zero to zero, though this alone does not prove linearity. Given an ordered basis $B = (b_1, \dots, b_n)$, the coordinate vector $[x]_B$ records the unique coefficients in $x = \sum_i k_i b_i$.

The values $T(b_i)$ determine T everywhere, since $T(x) = \sum_i k_i T(b_i)$. Conversely arbitrary choices for the images of a basis define a unique linear map. Coordinates depend on the ordered basis; a coordinate vector is generally different from the original vector in standard coordinates.

Prescribing images on a dependent list defines a linear map only if every dependence relation among inputs is respected by the outputs. After prescribing compatible images on r independent vectors in an n -dimensional domain, an m -dimensional codomain leaves $m(n - r)$ free scalar parameters, one arbitrary output vector for each basis vector still unspecified.

Problem 155.1[See solution](#)

A linear map satisfies $T(1, 2) = (3, 1)$ and $T(2, -1) = (1, 4)$. What is $T(5, 0)$?

- (A) $(-1, 7)$ (B) $(5, 9)$ (C) $(7, 6)$ (D) $(9, 5)$ (E) $(15, 0)$

Problem 155.2[See solution](#)

Let $B = (1 + x, 1 - x)$ be an ordered basis of P_1 . If $T(1 + x) = 1$ and $T(1 - x) = x$, what is $T(3 + x)$?

- (A) $1 + 2x$ (B) $2 + x$ (C) $3 + x$ (D) $2 - x$ (E) $1 - 2x$

Problem 155.3[See solution](#)

For which real a can a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ satisfy $T(1, 0) = (1, 2)$, $T(0, 1) = (2, -1)$, and $T(1, 1) = (a, 1)$?

- (A) No real a (B) $a = 1$ only (C) $a = 2$ only (D) $a = 3$ only (E) Every real a

Problem 155.4[See solution](#)

Which map from the real polynomial space P_2 to $\mathbb{R}$ is linear?

- (A) $p \mapsto p(0)^2$ (B) $p \mapsto p(0) + 1$
(C) $p \mapsto p(1)p(-1)$ (D) $p \mapsto p(1) - 2p'(0)$
(E) $p \mapsto |p(0)|$

Problem 155.5[See solution](#)

Linear maps $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ satisfy $T(1, 1, 0) = 0$, $T(0, 1, 1) = (1, 0)$, and $T(1, 0, -1) = (-1, 0)$. How many independent real parameters describe all such maps?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

5. Linear Algebra

Linear Transformations ▸ Standard Matrix Representative

156 Matrix representation of a linear map

Book pp. 208. [Find in compact glossary](#)**In this entry:** standard basis; standard matrix; matrix representative; column image rule.**Definition and key ideas**

For $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, its standard matrix is $A = [T(e_1) \cdots T(e_n)]$, where e_i are the standard coordinate vectors. Thus $T(x) = Ax$. The input dimension determines the number of columns; the output dimension determines the number of rows.

With arbitrary ordered bases of the input and output, column j records the output coordinates of $T(b_j)$, not necessarily its standard coordinates. Linear differential and integral operators on polynomial spaces can be represented the same way. A transformation involving a constant nonzero translation cannot be represented solely by multiplication by a matrix of the stated dimensions.

Problem 156.1[See solution](#)

On P_2 with ordered basis $(1, x, x^2)$, let $T(p) = p + (1 + x)p'$. What is the matrix of T ?

(A) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{pmatrix}$

(B) $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 2 & 3 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$

(D) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{pmatrix}$

(E) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Problem 156.2[See solution](#)

A linear map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ sends (x, y, z) to $(x - y, 2y + z)$. What is $T(1, 1, 1) + T(2, -1, 0)$?

- (A) $(2, 1)$ (B) $(3, 1)$ (C) $(3, 3)$ (D) $(4, 1)$ (E) $(4, 3)$

Problem 156.3[See solution](#)

On P_3 , let $T(p) = xp' + p$. What is the trace of the matrix of T^2 in the monomial basis?

- (A) 10 (B) 14 (C) 16 (D) 30 (E) 36

Problem 156.4[See solution](#)

Let $T : P_2 \rightarrow P_3$ be $T(p)(x) = \int_0^x p(t) dt$, represented in increasing monomial bases. What are the matrix dimensions and its $(3, 2)$ entry?

- (A) 3×4 and $1/2$ (B) 4×3 and $1/2$ (C) 4×3 and 2 (D) 3×4 and 2 (E) 4×3 and 0

Problem 156.5[See solution](#)

Differentiation maps P_2 to P_1 . Using input basis $(1, x, x^2)$ and output basis $(1 + x, 1 - x)$, which is its matrix?

- (A) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$
(B) $\begin{pmatrix} 0 & 1/2 & 1 \\ 0 & 1/2 & -1 \end{pmatrix}$
(C) $\begin{pmatrix} 0 & 1/2 & -1 \\ 0 & 1/2 & 1 \end{pmatrix}$
(D) $\begin{pmatrix} 0 & 0 \\ 1/2 & 1/2 \\ 1 & -1 \end{pmatrix}$
(E) $\begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & -1 \end{pmatrix}$

Problem 156.6[See solution](#)

For $F(x, y) = (2x - y + 1, x + y)$, which statement is correct?

- (A) There is a unique 2×2 matrix A with $F(v) = Av$ for every v .
(B) No such 2×2 matrix exists, because $F(0) \neq 0$.
(C) No such matrix exists, because the two components are different.
(D) Every 2×2 matrix with determinant three represents F .
(E) F is linear after merely reordering the ordinary basis vectors.

5. Linear Algebra

Linear Transformations ▸ The Rank Plus Nullity Theorem

157 Kernel, image, and rank-nullity

Book pp. 208-209. [Find in compact glossary](#)

In this entry: kernel; range; image; nullity; rank-nullity theorem; injective linear map; surjective linear map; isomorphism.

Definition and key ideas

The kernel of $T : V \rightarrow W$ is $\{v : T(v) = 0\}$, and its image or range is $\{T(v) : v \in V\}$. Both are subspaces, respectively of the domain and codomain. Their dimensions are the nullity and rank. For finite-dimensional V , $\dim \ker T + \dim \operatorname{im} T = \dim V$.

A linear map is injective exactly when its kernel is zero and surjective exactly when its image equals the codomain. With equal finite domain and codomain dimensions, injectivity and surjectivity are equivalent. A bijective linear map is an isomorphism. These conclusions need the correct dimension: subtract nullity from the domain dimension, not from the codomain dimension.

For a subspace $U \subseteq V$, the restricted map has kernel $U \cap \ker T$, so $\dim T(U) = \dim U - \dim(U \cap \ker T)$ when U is finite dimensional. Infinite-dimensional endomorphisms can be surjective without being injective, or conversely; differentiation and multiplication by x on polynomial spaces provide the two examples.

Problem 157.1[See solution](#)

A linear map $T : \mathbb{R}^7 \rightarrow \mathbb{R}^5$ has a four-dimensional kernel. Which is necessarily true?

- (A) It is onto
- (B) It is one-to-one
- (C) Its image has dimension 3
- (D) Its image has dimension 4
- (E) Its image has dimension 5

Problem 157.2[See solution](#)

Let $T : P_4 \rightarrow P_4$ be $T(p) = xp' - 2p$. What is its rank?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 157.3[See solution](#)

On the vector space $\mathbb{R}[x]$ of all real polynomials, differentiation D is

- (A) injective and surjective
- (B) injective but not surjective
- (C) surjective but not injective
- (D) neither injective nor surjective
- (E) not a linear map

Problem 157.4[See solution](#)

A linear map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^3$ has rank three. As U ranges over four-dimensional subspaces of $\mathbb{R}^5$, which values can $\dim T(U)$ take?

- (A) Only 1 (B) Only 2 (C) Only 3 (D) 1 or 2 (E) 2 or 3

Problem 157.5[See solution](#)

Define $E : P_2 \rightarrow \mathbb{R}^3$ by $E(p) = (p(0), p(1), p(2))$. Which statement is correct?

- (A) E is an isomorphism.
- (B) E is onto but has a one-dimensional kernel.
- (C) E is one-to-one but not onto.
- (D) E has rank two.
- (E) E is not linear because evaluation changes the argument.

5. Linear Algebra

Linear Transformations ▸ A Note on Inverses and Compositions

158 Composition and inverse transformations

Book pp. 209. [Find in compact glossary](#)**In this entry:** composition of linear maps; inverse linear transformation; matrix order in composition.**Definition and key ideas**

The composition $T \circ S$ applies S first and T second. If their matrices are B and A with compatible intermediate bases, the composition matrix is AB . Composition remains linear and is associative, but reversing the order generally changes the map or may make it undefined.

A bijective linear map has a linear inverse; the inverse matrix is the matrix inverse. For invertible maps $(T \circ S)^{-1} = S^{-1} \circ T^{-1}$. Kernels can also be tracked through composition: $\ker(T \circ S) = S^{-1}(\ker T)$, where this denotes a preimage set and does not require that S be invertible.

A useful composition identity is: if $I_V - TS$ is invertible for $S : V \rightarrow W$ and $T : W \rightarrow V$, then $(I_W - ST)^{-1} = I_W + S(I_V - TS)^{-1}T$. Checking both multiplication orders proves the result without assuming that S or T has an inverse.

Problem 158.1[See solution](#)

On $\mathbb{R}^2$ define $S(x, y) = (x + y, y)$ and $T(x, y) = (x, y - x)$. What is $(T \circ S)^{-1}(a, b)$?

- (A) $(a - b, b)$ (B) $(-b, a + b)$ (C) $(b, a - b)$ (D) $(a, b - a)$ (E) $(a + b, a)$

Problem 158.2[See solution](#)

Linear maps $S : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ and $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ satisfy $T \circ S = I_{\mathbb{R}^2}$. What is $\dim \ker T$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Cannot be determined

Problem 158.3[See solution](#)

Let $S(x, y, z) = (x + y, y + z, 0)$ and $T(a, b, c) = (a - b, c)$. What is $\ker(T \circ S)$?

- (A) $\{(s, -s, s) : s \in \mathbb{R}\}$ (B) $\{(s, t, s) : s, t \in \mathbb{R}\}$
 (C) $\{(s, s, 0) : s \in \mathbb{R}\}$ (D) $\{(0, t, 0) : t \in \mathbb{R}\}$
 (E) $\{0\}$

Problem 158.4[See solution](#)

On $\mathbb{R}[x]$, let $D(p) = p'$ and $M(p) = xp$. Which operator equals $D \circ M - M \circ D$?

- (A) The zero operator (B) The identity operator (C) D (D) M (E) $-I$

Problem 158.5[See solution](#)

Let $S : V \rightarrow W$ and $T : W \rightarrow V$ be linear maps. If $I_V - TS$ is invertible, which is the inverse of $I_W - ST$?

(A) $I_W - S(I_V - TS)^{-1}T$

(B) $I_W + S(I_V - TS)^{-1}T$

(C) $S(I_V - TS)^{-1}T$

(D) $I_W + ST$

(E) $I_W - ST$

5. Linear Algebra

Eigenvalues and Eigenvectors ▸ Characteristic Equation

159 Eigenvalues, eigenvectors, and characteristic polynomial

Book pp. 209-210. [Find in compact glossary](#)

In this entry: eigenvalue; characteristic value; eigenvector; characteristic vector; characteristic polynomial; characteristic equation.

Definition and key ideas

For a square matrix A , a nonzero vector v satisfying $Av = \lambda v$ is an eigenvector with eigenvalue λ . Eigenvalues solve $\det(A - \lambda I) = 0$. The monic convention $\chi_A(t) = \det(tI - A)$ differs only by $(-1)^n$ from the book convention and has the same roots.

Eigenvectors are never zero, although zero may be an eigenvalue; this occurs exactly when A is singular. Eigenvalues may depend on the scalar field: a real rotation can have no real eigenvectors but have complex ones. For an eigenpair, $p(A)v = p(\lambda)v$; if A is invertible, $A^{-1}v = \lambda^{-1}v$. Only scalar multiples of the same eigenvector, not coordinatewise reciprocals, are relevant.

A real matrix of odd size has a real eigenvalue because its real characteristic polynomial has odd degree. The statement about eigenvectors of $p(A)$ is one-way: different eigenvalues of A can merge under p , creating new eigenvectors of $p(A)$. If every nonzero vector is an eigenvector of A , comparing basis vectors and their sums forces A to be a scalar multiple of the identity.

Problem 159.1

[See solution](#)

An invertible matrix has eigenvector v with eigenvalue 2. What is the eigenvalue of $A^3 - 3A + 4A^{-1}$ on v ?

- (A) 2 (B) 3 (C) 4 (D) 6 (E) 8

Problem 159.2

[See solution](#)

For $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$, which statement is true?

- (A) Its real eigenvalues are 2, -2
(B) It has eigenvalue 0
(C) Its complex eigenvalues are $2i, -2i$
(D) Every real vector is an eigenvector
(E) Its characteristic polynomial is $t^2 - 4$

Problem 159.3[See solution](#)

Which statement holds for every real 5×5 matrix?

- (A) All five eigenvalues are real.
- (B) At least one eigenvalue is real.
- (C) Zero is an eigenvalue.
- (D) Every eigenvalue is positive.
- (E) No eigenvalue can be repeated.

Problem 159.4[See solution](#)

Let $A = \text{diag}(1, -1)$ and $v = (1, 1)^T$. Which statement is true?

- (A) v is an eigenvector of A with eigenvalue 1.
- (B) v is an eigenvector of A with eigenvalue -1 .
- (C) v is an eigenvector of A^2 but is not an eigenvector of A .
- (D) v is not an eigenvector of A^2 .
- (E) $Av = 0$.

Problem 159.5[See solution](#)

Every nonzero vector in $\mathbb{R}^3$ is an eigenvector of a real matrix A , and $\text{tr } A = 9$. What is A ?

- (A) I
- (B) $3I$
- (C) $9I$
- (D) Any diagonal matrix with trace nine
- (E) The data are impossible.

5. Linear Algebra

Eigenvalues and Eigenvectors ▸ Trace and Determinant

160 Trace and eigenvalue sums and products

Book pp. 211. [Find in compact glossary](#)

In this entry: trace; sum of eigenvalues; product of eigenvalues; algebraic multiplicity; two-by-two characteristic polynomial.

Definition and key ideas

The trace is $\text{tr } A = \sum_i a_{ii}$. For an $n \times n$ matrix, its complex eigenvalues counted with algebraic multiplicity sum to the trace and multiply to the determinant. For 2×2 matrices, $\chi_A(t) = t^2 - (\text{tr } A)t + \det A$.

Algebraic multiplicity means root multiplicity in the characteristic polynomial, not the number of independent eigenvectors. Repeated eigenvalues must be counted repeatedly in these identities. Also $\text{tr}(AB) = \text{tr}(BA)$ for compatible rectangular factors, even though the products can differ. Trace is linear, whereas determinant is not linear in the matrix.

For any positive integer k , $\text{tr}(A^k) = \sum_i \lambda_i^k$, with algebraic multiplicities, even if A is not diagonalizable. In size three the coefficient of t in the monic characteristic polynomial is $[(\text{tr } A)^2 - \text{tr}(A^2)]/2$. For an idempotent $A^2 = A$, the decomposition into image and kernel gives $\text{tr } A = \text{rank } A$.

Problem 160.1

[See solution](#)

A real 3×3 matrix has trace 7, determinant 12, and eigenvalue 3. What is the sum of the squares of its three eigenvalues, counting multiplicity?

- (A) 13 (B) 15 (C) 17 (D) 19 (E) 21

Problem 160.2

[See solution](#)

A 2×2 matrix A has trace 1 and determinant -2 . What is $\det(A + 2I)$?

- (A) 0 (B) 2 (C) 4 (D) 6 (E) 8

Problem 160.3

[See solution](#)

For real $n \times n$ matrices with $n \geq 1$, can $AB - BA = I_n$ hold?

- (A) Yes, for every n
(B) Yes, exactly when n is even
(C) Yes, exactly when A is singular
(D) No, because the traces of the two sides differ
(E) No, because all square matrices commute

Problem 160.4[See solution](#)

A real 8×8 matrix satisfies $A^2 = A$ and $\text{tr } A = 3$. What is $\dim \ker A$?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 8

Problem 160.5[See solution](#)

A real 3×3 matrix has $\text{tr } A = 3$, $\text{tr}(A^2) = 5$, and $\det A = 0$. What is its monic characteristic polynomial?

- (A) $t^3 - 3t^2 + 2t$ (B) $t^3 - 3t^2 + 5t$ (C) $t^3 + 3t^2 + 2t$ (D) $t^3 - 5t^2 + 3t$ (E) $t^3 - 3t^2 - 2t$

5. Linear Algebra

Eigenvalues and Eigenvectors ▸ Eigenspaces

161 Eigenspaces and geometric multiplicity

Book pp. 212. [Find in compact glossary](#)

In this entry: eigenspace; geometric multiplicity; zero eigenspace; independent eigenvectors.

Definition and key ideas

The eigenspace for λ is $E_\lambda = \ker(A - \lambda I)$, consisting of all corresponding eigenvectors together with zero. Its dimension is the geometric multiplicity. For an eigenvalue, this dimension is at least one and no greater than its algebraic multiplicity.

Find an eigenspace by row-reducing $A - \lambda I$. Eigenvectors for distinct eigenvalues are linearly independent, but repeated roots do not guarantee enough independent eigenvectors to form a basis. The zero eigenspace is exactly the nullspace. An eigenvalue can have algebraic multiplicity three and geometric multiplicity only one.

Eigenspaces for distinct eigenvalues have zero intersection and their sum is direct. If $v = u + w$ with $Au = \lambda u$, $Aw = \mu w$, and $\lambda \neq \mu$, then $u = (Av - \mu v)/(\lambda - \mu)$. This isolates an eigenspace component without finding all coordinates of an eigenbasis.

Problem 161.1

[See solution](#)

Let $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. What is the dimension of its eigenspace for 2?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) It has no eigenvalue 2

Problem 161.2

[See solution](#)

A 4×4 matrix has distinct eigenvalues 1, 2, 3, with eigenspace dimensions 2, 1, 1 respectively. What is its determinant?

- (A) 6 (B) 12 (C) 18 (D) 24 (E) Cannot be determined

Problem 161.3[See solution](#)

A 4×4 matrix has exactly one distinct eigenvalue, namely 2, and a two-dimensional eigenspace. Is it diagonalizable?

- (A) Yes
- (B) The data are insufficient
- (C) Only if its trace is zero
- (D) No
- (E) Only if its determinant is nonzero

Problem 161.4[See solution](#)

Suppose $v = u + w$, where $Au = 2u$ and $Aw = 5w$. Which expression recovers u from v and Av ?

- (A) $(Av - 2v)/3$ (B) $(5v - Av)/3$ (C) $(Av - 5v)/3$ (D) $(2v + Av)/7$ (E) $(5Av - v)/3$

Problem 161.5[See solution](#)

An eigenvalue has algebraic multiplicity three. Which is impossible for its geometric multiplicity?

- (A) All of 1, 2, 3 are impossible
- (B) 3
- (C) 4
- (D) 2
- (E) 1

5. Linear Algebra

The Cayley-Hamilton Theorem ▸ Reducing Matrix Powers

162 Cayley–Hamilton theorem

Book pp. 213. [Find in compact glossary](#)**In this entry:** Cayley-Hamilton theorem; characteristic identity; reduction of matrix powers.**Definition and key ideas**

Cayley–Hamilton states that every square matrix satisfies its own characteristic polynomial: $\chi_A(A) = 0$. If $\chi_A(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_0$, substitute A for t and c_0I for the constant term. Consequently every nonnegative power of A can be expressed as a polynomial of degree less than n in A .

For a two-by-two matrix this gives $A^2 - (\text{tr } A)A + (\det A)I = 0$, so all powers reduce to combinations of I, A . For larger matrices the reduction is not necessarily linear in A . Divide the desired scalar polynomial by χ_A and evaluate the remainder at A ; this often avoids direct multiplication.

An annihilating polynomial p satisfies $p(A) = 0$ but need not have the same degree or multiplicities as χ_A . A repeated root requires care: from $\chi_A(t) = (t - \lambda)^2$ one gets $(A - \lambda I)^2 = 0$, which allows terminating binomial expansions even without diagonalizability. Polynomial identities may also invert $A - cI$ after factoring $p(t) - p(c)$.

Problem 162.1[See solution](#)

A real 2×2 matrix has characteristic polynomial $(t - 1)^2$. Without assuming diagonalizability, which formula is guaranteed?

- (A) $A^{100} = I$ (B) $A^{100} = 100A$
 (C) $A^{100} = 100A - 99I$ (D) $A^{100} = 99A - 100I$
 (E) $A^{100} = 2^{99}A$

Problem 162.2[See solution](#)

For $A = \text{diag}(-1, 0, 1)$, which assertion is true?

- (A) $A^2 = aA + bI$ for some real a, b .
 (B) $A^3 = A$, but A^2 is not a linear combination of I and A .
 (C) $A^2 = I$.
 (D) A is invertible.
 (E) A does not satisfy its characteristic polynomial.

Problem 162.3[See solution](#)

A real 3×3 matrix satisfies $A^2 - 3A + 2I = 0$. Which polynomial could be its characteristic polynomial?

- (A) $t^2 - 3t + 2$ (B) $(t - 1)(t - 2)(t - 3)$
(C) $(t - 1)^2(t - 2)$ (D) $t^3 - 1$
(E) $(t - 1)^2(t + 2)$

Problem 162.4[See solution](#)

A matrix A has characteristic polynomial $t^3 - t - 1$. Which expression is $(A - I)^{-1}$?

- (A) $A^2 - I$ (B) $A^2 + A$ (C) $A^2 - A$ (D) $I - A^2$ (E) $A + I$

Problem 162.5[See solution](#)

A real 3×3 matrix has characteristic polynomial t^3 . Which assertion must hold?

- (A) $A = 0$ (B) $A^2 = 0$ (C) $A^3 = 0$ (D) $\text{rank } A = 1$ (E) $\dim \ker A = 2$

5. Linear Algebra

Chapter 5 Review Questions > Counterexamples to Scalar Rules

163 Nilpotent and involutory matrices

Book pp. 214. [Find in compact glossary](#)

In this entry: nilpotent matrix; nilpotency index; involutory matrix; matrix cancellation failure; zero divisor.

Definition and key ideas

A matrix is nilpotent if $N^k = 0$ for some positive integer k ; the smallest such k is its nilpotency index. A nonzero nilpotent matrix shows why $A^2 = 0$ need not imply $A = 0$. A matrix is involutory if $A^2 = I$, equivalently $A = A^{-1}$; this does not force $A = I$ or $A = -I$.

Singular matrices defeat cancellation: $AB = AC$ only implies $A(B - C) = 0$. For nilpotent N with $N^k = 0$, the finite geometric series gives $(I - N)^{-1} = I + N + \cdots + N^{k-1}$. Also $(I + N)^m$ can be computed by the binomial theorem because I commutes with N .

For $N^2 = 0$, $\text{im } N \subseteq \ker N$, so $\text{rank } N \leq \lfloor n/2 \rfloor$. A nilpotent on an n -dimensional space has index at most n , and if its kernel is one-dimensional the index is exactly n . For a real involution, $v = (v + Av)/2 + (v - Av)/2$ gives its unique decomposition into the $+1$ and -1 eigenspaces.

Problem 163.1[See solution](#)

Real square matrices N, M of the same size satisfy $N^2 = M^2 = 0$ and $NM = MN$. Which conclusion must hold?

- (A) $N + M = 0$. (B) $(N + M)^2 = 0$.
 (C) $(N + M)^3 = 0$. (D) $NM = 0$.
 (E) $N + M$ is invertible.

Problem 163.2[See solution](#)

A real 2×2 matrix satisfies $A^2 = I$ and $\text{tr } A = 0$. What is $\det A$?

- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

Problem 163.3[See solution](#)

A nilpotent linear operator on $\mathbb{R}^5$ has rank four. What is its nilpotency index?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 163.4[See solution](#)

Let $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $M = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Which statement is correct?

- (A) $N + M$ is nilpotent. (B) $N + M$ is involutory.
(C) $NM = MN$. (D) $NM = 0$.
(E) N is invertible.

Problem 163.5[See solution](#)

A real matrix satisfies $A^2 = I$. For any vector v , set $v_+ = (v + Av)/2$ and $v_- = (v - Av)/2$. Which statement must hold?

- (A) $Av_+ = -v_+$ and $Av_- = v_-$
(B) $Av_+ = v_+$ and $Av_- = -v_-$
(C) $v_+ = v_-$
(D) $v_- = 0$ for every v
(E) $v_+ = 0$ for every v

Problem 163.6[See solution](#)

A real 7×7 matrix satisfies $N^2 = 0$. What is the greatest possible rank of N ?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 6

5. Linear Algebra

Chapter 5 Review Questions > Symmetric Matrix Spaces

164 Symmetric and skew-symmetric matrices

Book pp. 216. [Find in compact glossary](#)

In this entry: symmetric matrix; skew-symmetric matrix; symmetric matrix space; dimension of symmetric matrices.

Definition and key ideas

A real square matrix is symmetric if $A^T = A$ and skew-symmetric if $A^T = -A$. A symmetric $n \times n$ matrix is determined by its n diagonal entries and $n(n-1)/2$ entries above the diagonal, so the symmetric space has dimension $n(n+1)/2$.

A real skew-symmetric matrix has zero diagonal and dimension $n(n-1)/2$. Every real square matrix splits uniquely as $A = (A + A^T)/2 + (A - A^T)/2$, its symmetric and skew-symmetric parts. The intersection of these two spaces is zero. These are linear subspaces; requiring positive definiteness instead is not a subspace condition.

In the entrywise squared norm, the symmetric part $(A + A^T)/2$ is the closest symmetric matrix to A . Symmetric and skew-symmetric matrices are perpendicular for the inner product $\text{tr}(B^T C)$. Also $x^T A^2 x = -\|Ax\|^2$ for skew-symmetric A . A real symmetric matrix is positive definite when $x^T A x > 0$ for every nonzero real x ; these matrices form a class closed under addition and positive scaling, but not a vector subspace.

Problem 164.1[See solution](#)

What is the dimension of the real symmetric 4×4 matrices with trace zero?

- (A) 5 (B) 6 (C) 8 (D) 9 (E) 10

Problem 164.2[See solution](#)

For $A = \begin{pmatrix} 1 & 4 \\ -2 & 3 \end{pmatrix}$, what is the minimum of $\sum_{i,j} (a_{ij} - s_{ij})^2$ over real symmetric matrices S ?

- (A) 0 (B) 6 (C) 9 (D) 18 (E) 36

Problem 164.3[See solution](#)

What is the dimension of the real symmetric $n \times n$ matrices whose diagonal entries are all zero?

- (A) $n(n-1)/2$ (B) n^2 (C) $n^2 - n$ (D) $n(n+1)/2$ (E) $n-1$

Problem 164.4[See solution](#)

For a nonzero real column vector x and a real skew-symmetric matrix A , which inequality always holds?

- (A) $x^T A^2 x > 0$ (B) $x^T A^2 x = \|Ax\|^2$
(C) $x^T A^2 x = -\|Ax\|^2 \leq 0$ (D) $x^T A^2 x \geq \|x\|^2$
(E) $x^T A^2 x = -\|x\|^2$

Problem 164.5[See solution](#)

If real symmetric matrices A, B have symmetric product AB , what must hold?

- (A) A is invertible (B) $AB = 0$ (C) $AB = BA$ (D) $A = B$ (E) B is diagonal

Problem 164.6[See solution](#)

Let $\mathcal{P}$ be the positive definite real symmetric $n \times n$ matrices, with $n \geq 1$. Which claim is true?

- (A) $\mathcal{P}$ is a vector subspace.
(B) If $A, B \in \mathcal{P}$, then $A - B \in \mathcal{P}$.
(C) If $A, B \in \mathcal{P}$ and $0 < t < 1$, then $tA + (1 - t)B \in \mathcal{P}$.
(D) If $A \in \mathcal{P}$, then $-A \in \mathcal{P}$.
(E) The zero matrix belongs to $\mathcal{P}$.

5. Linear Algebra

Chapter 5 Review Questions > Rotations and Reflections

165 Rotation and reflection operators

Book pp. 217. [Find in compact glossary](#)

In this entry: rotation matrix; reflection matrix; orthogonal transformation; orientation-preserving map; orientation-reversing map.

Definition and key ideas

Counterclockwise planar rotation by θ has matrix $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. Reflection across the x -axis has matrix $S = \text{diag}(1, -1)$; reflection across a line at angle α is $R_\alpha S R_{-\alpha}$. Both preserve Euclidean lengths and angles.

Such matrices are orthogonal: $Q^T Q = I$, so $Q^{-1} = Q^T$. Rotations have determinant 1, reflections determinant -1 . In the plane, $S R_\theta = R_{-\theta} S$, making order essential. The composition of reflections in two lines is a rotation through twice the directed angle from the first line to the second.

An invertible real map preserves orientation when its determinant is positive and reverses orientation when it is negative; its absolute determinant is the volume scale. This condition alone does not imply orthogonality. Every planar orthogonal map is a rotation or a reflection: a reflection has eigenvalues 1, -1 and trace zero, and its fixed eigenspace is the reflecting line.

Problem 165.1[See solution](#)

First reflect in the line making angle $\pi/6$ with the positive x -axis, then reflect in the line making angle $\pi/3$. What is the resulting map?

- (A) Rotation by $-\pi/3$
- (B) Rotation by $\pi/6$
- (C) Rotation by $\pi/3$
- (D) Reflection in angle $\pi/4$
- (E) The identity

Problem 165.2[See solution](#)

Let R rotate the plane by $\pi/3$ and S reflect it across the x -axis. What is $(SR)^2$?

- (A) I (B) $-I$ (C) R (D) R^2 (E) S

Problem 165.3[See solution](#)

The matrix $Q = \frac{1}{5} \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$ represents reflection in which line?

- (A) $y = 2x$ (B) $y = x/2$ (C) $y = -x/2$ (D) $y = -2x$ (E) $y = x$

Problem 165.4[See solution](#)

A real orthogonal 2×2 matrix has trace one. Which description is forced?

- (A) Reflection in a line through the origin
- (B) Rotation by $\pi/3$ or $-\pi/3$, modulo 2π
- (C) Rotation by $\pi/3$ only
- (D) Rotation by $2\pi/3$ or $-2\pi/3$
- (E) A length-changing map of determinant one

Problem 165.5[See solution](#)

The map $S(x, y) = (x + y, y)$ has determinant one. Which statement correctly describes it?

- (A) It is a rotation.
- (B) It is a reflection.
- (C) It preserves orientation and area but not every length.
- (D) It reverses orientation and preserves every length.
- (E) It is orthogonal because its determinant is one.

5. Linear Algebra

Chapter 5 Review Questions > The Cross Product as a Linear Map

166 Cross-product linear operators

Book pp. 218. [Find in compact glossary](#)**In this entry:** cross-product matrix; cross-product linear map; skew-symmetric operator.**Definition and key ideas**

For fixed $v = (a, b, c)$, the map $T(x) = v \times x$ is linear, with standard matrix $[v]_{\times} = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}$. Its columns are $v \times e_i$. This matrix is skew-symmetric.

If $v \neq 0$, the kernel is $\text{span}\{v\}$ and the image is the plane perpendicular to v , so rank is 2. The vector triple product gives $T^2(x) = v(v \cdot x) - \|v\|^2 x$, or $[v]_{\times}^2 = vv^T - \|v\|^2 I$. Thus its complex eigenvalues are $0, \pm i\|v\|$, while its only real eigenvalue is zero.

For these operators, $T_u T_v - T_v T_u = T_{u \times v}$. Also $I + T_v$ is always invertible, with inverse $(I - T_v + vv^T)/(1 + \|v\|^2)$. These identities follow from the triple-product formula and the fact that crossing with v kills the component parallel to v .

Problem 166.1[See solution](#)

For $v = (1, 2, 2)$ and $T(x) = v \times x$, what is $T^2(x)$ whenever $x \perp v$?

- (A) $-9x$ (B) $-3x$ (C) 0 (D) $3x$ (E) $9x$

Problem 166.2[See solution](#)

If $v \neq 0$ and $T(x) = v \times x$ on $\mathbb{R}^3$, what is the rank of T^2 ?

- (A) 0
(B) 1
(C) 2
(D) 3
(E) It depends on the coordinates of v

Problem 166.3[See solution](#)

For $v = (1, 2, -1)$, define $T(x) = v \times x$ on $\mathbb{R}^3$. What is $\det(I + T)$?

- (A) -7 (B) 8 (C) 7 (D) 14 (E) $\frac{7}{2}$

Problem 166.4[See solution](#)

For vectors $u, v \in \mathbb{R}^3$, write $T_u(x) = u \times x$ and $T_v(x) = v \times x$. Which operator equals $T_u T_v - T_v T_u$?

- (A) The zero operator (B) $T_{u \times v}$ (C) $-T_{u \times v}$ (D) T_{u+v} (E) $(u \cdot v)I$

Problem 166.5[See solution](#)

For fixed $v \in \mathbb{R}^3$ let $T(x) = v \times x$ and $r^2 = \|v\|^2$. Which expression is $(I + T)^{-1}$, where vv^T denotes the outer-product operator?

(A) $(I + T)/(1 + r^2)$

(B) $(I - T)/(1 + r^2)$

(C) $(I - T + vv^T)/(1 + r^2)$

(D) $I - T + T^2$

(E) $(I + T + vv^T)/(1 + r^2)$

Problem 166.6[See solution](#)

The linear map $T(x) = v \times x$ has standard matrix $\begin{pmatrix} 0 & -3 & -2 \\ 3 & 0 & -1 \\ 2 & 1 & 0 \end{pmatrix}$. What is v ?

(A) $(1, 2, 3)$ **(B)** $(-1, 2, -3)$ **(C)** $(1, -2, 3)$ **(D)** $(3, -2, 1)$ **(E)** $(-3, 2, -1)$

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ Divisibility

167 Divisibility, parity, primes, and coprimality

Book pp. 220–221. [Find in compact glossary](#)

In this entry: divides; divisor; factor; multiple; even; odd; prime; composite; relatively prime; pairwise relatively prime; divisibility tests; square-free integer.

Definition and key ideas

For integers $a \neq 0$ and b , $a \mid b$ means $b = ac$ for some $c \in \mathbb{Z}$. An integer is even if divisible by 2, and odd otherwise. A prime is an integer $p > 1$ whose positive divisors are 1, p ; a composite integer is an integer greater than 1 that is not prime. Integers are relatively prime if their greatest common divisor is 1. Pairwise relatively prime means every distinct pair has this property.

Divisibility by both a and b implies divisibility by ab when $\gcd(a, b) = 1$. Digit tests follow by reducing powers of 10: digit sums work modulo 3, 9, alternating digit sums modulo 11, and the last k digits suffice modulo 2^k or 5^k . The numbers 0, 1, and negative integers are not primes under the standard definition.

An integer is square-free if no prime square divides it; the positive integer 1 is included. Euclid's lemma says that a prime dividing a product divides at least one factor. Prime-exponent parities characterize perfect squares. A gcd of one for three or more integers does not imply pairwise coprimality, as 6, 10, 15 demonstrate.

Problem 167.1

[See solution](#)

What is the greatest positive integer that divides $n^5 - n$ for every integer n ?

- (A) 10 (B) 20 (C) 30 (D) 60 (E) 120

Problem 167.2

[See solution](#)

An integer is divisible by 12 and by 18. Which is the largest integer that must divide it?

- (A) 24 (B) 30 (C) 36 (D) 72 (E) 216

Problem 167.3

[See solution](#)

How many choices of decimal digits a, b make the four-digit integer $4a2b$ divisible by 72?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 167.4[See solution](#)

Positive integers a, b, c have greatest common divisor one, and ab, bc, ca are all perfect squares. Which statement must hold?

- (A) $a = b = c = 1$
- (B) a, b, c are pairwise relatively prime.
- (C) a, b, c are all perfect squares.
- (D) Exactly one of a, b, c is even.
- (E) abc is prime.

Problem 167.5[See solution](#)

For which positive integers m does $m \mid a^2$ imply $m \mid a$ for every integer a ?

- (A) Exactly the primes
- (B) Exactly the perfect squares
- (C) Exactly those with no squared prime factor
- (D) Exactly the odd positive integers
- (E) Exactly the even positive integers

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ The Division Algorithm

168 Quotient and remainder

Book pp. 221. [Find in compact glossary](#)**In this entry:** division algorithm; quotient; remainder; floor function.**Definition and key ideas**

For every integer b and positive integer a , unique integers q, r satisfy $b = qa + r$ with $0 \leq r < a$. The quotient is $q = \lfloor b/a \rfloor$; the remainder is the nonnegative representative of the congruence class of b modulo a . The dividend may be negative.

In a quotient-remainder problem, the inequality on r is part of the answer. Truncation toward zero gives the wrong quotient for a negative nonintegral ratio. Repeated division yields base expansions and the Euclidean algorithm; comparing two representations proves uniqueness because a multiple of a with absolute value less than a must vanish.

In base $b \geq 2$, each digit is a remainder in $\{0, \dots, b-1\}$; repeated quotient division recovers the preceding digits. If two remainders modulo m sum to at least m , their sum contributes one extra quotient unit (a carry); otherwise it contributes none. When solving backwards for a divisor, retain the strict remainder bound.

Problem 168.1[See solution](#)

If $-137 = 9q + r$ with $0 \leq r < 9$, what is $q + r$?

- (A) -18 (B) -16 (C) -9 (D) -7 (E) -5

Problem 168.2[See solution](#)

An even integer N leaves remainder 17 on division by 35. What is the remainder of N^2 on division by 70?

- (A) 9 (B) 17 (C) 29 (D) 44 (E) 59

Problem 168.3[See solution](#)

An integer N has base-seven expansion $(356)_7$. What is $\lfloor N/7 \rfloor + \lfloor N/49 \rfloor$?

- (A) 8 (B) 26 (C) 29 (D) 35 (E) 41

Problem 168.4[See solution](#)

For an integer $m \geq 2$, how many ordered pairs (r, s) with $0 \leq r, s < m$ satisfy $\lfloor (r+s)/m \rfloor = 1$?

- (A) $m(m-1)/2$ (B) $m(m+1)/2$ (C) m^2 (D) $(m-1)^2$ (E) $m-1$

Problem 168.5[See solution](#)

For how many positive integers d does dividing 100 by d give equal quotient and remainder?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 8

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ Primes

169 Prime factorization and the distribution of primes

Book pp. 221. [Find in compact glossary](#)

In this entry: fundamental theorem of arithmetic; prime factorization; prime-counting function; prime number theorem; Bertrand postulate; prime gaps; prime-adic valuation; factorial trailing zeros.

Definition and key ideas

Every integer $n > 1$ has a factorization $n = \prod p^{v_p(n)}$ unique up to the order of the prime factors; $v_p(n)$ is the exponent of p . Consequently the number of positive divisors is $\prod_p (v_p(n) + 1)$. Factorial valuations are $v_p(n!) = \sum_{j \geq 1} \lfloor n/p^j \rfloor$, which count multiples of successive powers of p .

There are infinitely many primes and arbitrarily long strings of composite integers. Bertrand's postulate gives a prime $n < p < 2n$ for integer $n > 1$. The prime-counting function $\pi(x)$ satisfies $\pi(x) \sim x/\log x$, meaning the ratio tends to 1, not that the approximation is an exact formula. The reciprocal-prime series diverges. These distribution results are separate from unique factorization.

The prime-counting function $\pi(x)$ is the number of primes at most x . Prime gaps of any prescribed finite length follow from the consecutive numbers $(k+1)! + 2, \dots, (k+1)! + k + 1$. This is compatible with Bertrand's relative interval bound. Although primes have density zero by the prime number theorem, their reciprocal sum diverges; their squared-reciprocal sum converges by comparison with $\sum 1/n^2$.

Problem 169.1[See solution](#)

How many trailing zeros occur in the base-ten expansion of $250!$?

- (A) 50 (B) 60 (C) 61 (D) 62 (E) 63

Problem 169.2[See solution](#)

What is the smallest positive integer having exactly 18 positive divisors?

- (A) 108 (B) 144 (C) 180 (D) 216 (E) 288

Problem 169.3[See solution](#)

How many positive divisors of 10800 are perfect squares?

- (A) 8 (B) 12 (C) 16 (D) 24 (E) 60

Problem 169.4[See solution](#)

For an integer $k \geq 2$, which list is guaranteed to consist of k consecutive composite positive integers?

- (A) $k! + 1, \dots, k! + k$ (B) $(k + 1)! + 2, \dots, (k + 1)! + k + 1$
(C) $k + 1, \dots, 2k$ (D) $2k + 1, \dots, 3k$
(E) $k^2, \dots, k^2 + k - 1$

Problem 169.5[See solution](#)

Using the prime number theorem, what is $\lim_{x \rightarrow \infty} \frac{\pi(x^2)}{x\pi(x)}$?

- (A) 0 (B) $1/2$ (C) 1 (D) 2 (E) $+\infty$

Problem 169.6[See solution](#)

For every integer $n > 1$, which statement is true of $\binom{2n}{n}$?

- (A) It is a perfect square.
(B) It is not a perfect square.
(C) It is prime.
(D) It is divisible by every prime at most n .
(E) It is odd.

Problem 169.7[See solution](#)

If $p_1 < p_2 < \dots$ are all primes, which statement is correct?

- (A) Both $\sum 1/p_j$ and $\sum 1/p_j^2$ converge.
(B) Both series diverge.
(C) $\sum 1/p_j$ diverges and $\sum 1/p_j^2$ converges.
(D) $\sum 1/p_j$ converges and $\sum 1/p_j^2$ diverges.
(E) Neither convergence statement is known.

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ The Greatest Common Divisor and the Least Common Multiple; The Euclidean Algorithm

170 GCD, LCM, Euclid, and Bezout

Book pp. 221–223. [Find in compact glossary](#)

In this entry: greatest common divisor; least common multiple; Euclidean algorithm; Bezout identity; extended Euclidean algorithm.

Definition and key ideas

For nonzero integers, the greatest common divisor is the largest positive common divisor; the least common multiple is the least positive common multiple. In prime factorizations, take the minimum exponent for the gcd and the maximum for the lcm. For positive a, b , $\gcd(a, b) \operatorname{lcm}(a, b) = ab$.

Euclid repeatedly uses $\gcd(a, b) = \gcd(b, a - qb)$ until the remainder is zero. The last nonzero remainder is the gcd. Reversing the substitutions gives Bezout's identity $ax + by = \gcd(a, b)$ with integers x, y . This proves that the set of all integer linear combinations of a, b is exactly the set of multiples of their gcd; it also computes modular inverses without factorization.

Problem 170.1

[See solution](#)

Positive integers a, b have gcd 12 and lcm 420. How many ordered pairs (a, b) are possible?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 12

Problem 170.2

[See solution](#)

What is the multiplicative inverse of 37 modulo 101?

- (A) 11 (B) 30 (C) 41 (D) 63 (E) 71

Problem 170.3

[See solution](#)

For a nonnegative integer n , which formula gives $\gcd(2n + 1, n^2 + n + 1)$?

- (A) 1 for every n
(B) 3 for every n
(C) 3 if $n \equiv 1 \pmod{3}$, and 1 otherwise
(D) $2n + 1$ for every n
(E) 3 if $n \equiv 0 \pmod{3}$, and 1 otherwise

Problem 170.4

[See solution](#)

Positive integers a, b and integers x, y satisfy $ax + by = \gcd(a, b)$. Which quantity must equal one?

- (A) $\gcd(a, b)$ (B) $\gcd(a, x)$ (C) $\gcd(b, y)$ (D) $\gcd(x, y)$ (E) $\gcd(a + b, x + y)$

Problem 170.5[See solution](#)

For positive integers a, b, c , which expression always equals $\gcd(\text{lcm}(a, b), \text{lcm}(a, c))$?

- (A) $\text{lcm}(a, \gcd(b, c))$ (B) $\gcd(a, \text{lcm}(b, c))$ (C) $a \gcd(b, c)$ (D) $\text{lcm}(a, b, c)$ (E) $\gcd(a, b, c)$

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ The Diophantine Equation $ax + by = c$

171 Linear Diophantine equations

Book pp. 223-224. [Find in compact glossary](#)**In this entry:** Diophantine equation; integer solution; linear Diophantine equation; Fermat last theorem.**Definition and key ideas**

A Diophantine equation seeks solutions in integers, sometimes with additional positivity restrictions. For $ab \neq 0$ and $d = \gcd(a, b)$, $ax + by = c$ is solvable in integers exactly when $d \mid c$. If (x_0, y_0) is one solution, all solutions are $x = x_0 + (b/d)t$, $y = y_0 - (a/d)t$, with $t \in \mathbb{Z}$.

Obtain one solution from Bezout's identity multiplied by c/d , then impose bounds on the integer parameter. The same parametrization applies when $c = 0$; one may use $(0, 0)$ as the particular solution. Higher-degree Diophantine equations can behave very differently: Fermat's last theorem says $x^n + y^n = z^n$ has no nonzero integer solution for integer $n > 2$.

The homogeneous equation uses the primitive direction $(b/d, -a/d)$, not $(b, -a)$ unless $d = 1$. Bounds or an objective such as $x^2 + y^2$ are imposed after parametrizing; a quadratic objective reduces to checking integers nearest its real minimizer. Clearing a common denominator transfers a rational power equation to an integer one, allowing Fermat's theorem to rule out nonzero rational solutions as well.

Problem 171.1[See solution](#)How many pairs of nonnegative integers (x, y) satisfy $18x + 30y = 252$?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 171.2[See solution](#)What is the least positive integer x for which there exists an integer y satisfying $26x + 39y = 65$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 171.3[See solution](#)Which parametrization gives all integer solutions of $12x + 18y = 0$, with t ranging over all integers?

- (A) $(x, y) = (18t, -12t)$ (B) $(x, y) = (3t, -2t)$
 (C) $(x, y) = (2t, -3t)$ (D) $(x, y) = (3t, 2t)$
 (E) $(x, y) = (t, -t)$

Problem 171.4[See solution](#)How many ordered pairs of rational numbers (u, v) satisfy $u^3 + v^3 = 1$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinitely many

Problem 171.5[See solution](#)

Among integer solutions of $7x + 11y = 1$, what is the minimum of $x^2 + y^2$?

- (A) 5 (B) 10 (C) 13 (D) 25 (E) 34

Problem 171.6[See solution](#)

For how many integers k with $1 \leq k \leq 100$ does $28x + 42y = k^2 + 3$ have an integer solution?

- (A) 7 (B) 12 (C) 14 (D) 16 (E) 28

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ Congruences

172 Congruence arithmetic and valid cancellation

Book pp. 225. [Find in compact glossary](#)

In this entry: congruence; modulus; residue class; modular arithmetic; cancellation modulo n ; Chinese remainder theorem.

Definition and key ideas

For $n > 0$, $a \equiv b \pmod{n}$ means $n \mid (a - b)$. Congruences may be added, subtracted, multiplied, and raised to nonnegative integer powers. Cancellation must track the modulus: $ac \equiv bc \pmod{n}$ implies $a \equiv b \pmod{n/\gcd(c, n)}$, and preserves modulus n when c is a unit modulo n .

A useful extension is the Chinese remainder theorem: congruences with pairwise coprime moduli specify exactly one residue class modulo their product. With moduli m, n that are not coprime, $x \equiv a \pmod{m}$, $x \equiv b \pmod{n}$ is consistent exactly when $a \equiv b \pmod{\gcd(m, n)}$, and then has one class modulo $\text{lcm}(m, n)$.

A residue class a modulo m splits into exactly k classes modulo km : $a, a + m, \dots, a + (k - 1)m$. CRT also combines independent root choices of polynomial congruences over coprime factors. For a composite modulus, a product can be zero while neither factor is zero modulo that whole modulus.

Problem 172.1[See solution](#)

What is the least positive integer x satisfying $x \equiv 2 \pmod{7}$ and $x \equiv 5 \pmod{11}$?

- (A) 16 (B) 23 (C) 38 (D) 49 (E) 60

Problem 172.2[See solution](#)

For integers a, b , $18a \equiv 18b \pmod{84}$ is equivalent to which condition?

- (A) $a \equiv b \pmod{84}$ (B) $a \equiv b \pmod{42}$
(C) $a \equiv b \pmod{28}$ (D) $a \equiv b \pmod{14}$
(E) $a \equiv b \pmod{7}$

Problem 172.3[See solution](#)

Which value of b makes the system $x \equiv 5 \pmod{12}$, $x \equiv b \pmod{18}$ consistent?

- (A) 1 (B) 3 (C) 8 (D) 11 (E) 14

Problem 172.4[See solution](#)

How many residue classes modulo 54 satisfy both $x \equiv 2 \pmod{6}$ and $x \equiv 5 \pmod{9}$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 6

Problem 172.5[See solution](#)

How many residue classes modulo 15 solve $x^2 \equiv 1 \pmod{15}$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 8

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ Congruences

173 Fermat, Euler, and modular powers

Book pp. 225. [Find in compact glossary](#)**In this entry:** Fermat little theorem; Euler totient function; Euler theorem; modular exponentiation.**Definition and key ideas**

For prime p , Fermat's little theorem gives $a^p \equiv a \pmod{p}$ for every integer a , and $a^{p-1} \equiv 1 \pmod{p}$ when $p \nmid a$. The Euler totient $\varphi(n)$ counts residue classes coprime to n and equals $n \prod_{p|n} (1 - 1/p)$. Euler's extension says $a^{\varphi(n)} \equiv 1 \pmod{n}$ when $\gcd(a, n) = 1$.

Reduce an exponent modulo a known period only after checking that the base is invertible. Repeated squaring computes powers even when no theorem applies. A number satisfying a Fermat congruence need not be prime, and the order of an individual unit can be much smaller than $\varphi(n)$.

Problem 173.1[See solution](#)What is the remainder of 7^{222} on division by 13?

- (A) 1 (B) 3 (C) 5 (D) 8 (E) 12

Problem 173.2[See solution](#)What is the remainder of 3^{1000} on division by 40?

- (A) 1 (B) 3 (C) 9 (D) 21 (E) 27

Problem 173.3[See solution](#)How many residue classes a modulo 360 satisfy $a^{96} \equiv 1 \pmod{360}$?

- (A) 48 (B) 72 (C) 96 (D) 120 (E) 360

Problem 173.4[See solution](#)What is the remainder of 2^{120} on division by 72?

- (A) 0 (B) 1 (C) 8 (D) 16 (E) 64

Problem 173.5[See solution](#)Since $2^{10} = 1024 = 3 \cdot 341 + 1$, which statement is correct?

- (A) 341 must be prime.
 (B) $2^{340} \not\equiv 1 \pmod{341}$.
 (C) $2^{340} \equiv 1 \pmod{341}$ although 341 is composite.
 (D) The order of 2 modulo 341 is 340.
 (E) Fermat's little theorem is false.

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ The Congruence Equation $ax \equiv b \pmod{n}$

174 Linear congruences and solution counts

Book pp. 226. [Find in compact glossary](#)**In this entry:** linear congruence; modular inverse; number of congruence solutions.**Definition and key ideas**

Let $d = \gcd(a, n)$ with $n > 0$. The congruence $ax \equiv b \pmod{n}$ has a solution exactly when $d \mid b$. When solvable, divide all three quantities by d and invert a/d modulo n/d ; this gives one class modulo n/d .

In the original modulus n , that class consists of exactly d distinct solutions, separated by n/d . Thus uniqueness modulo n/d must not be confused with uniqueness modulo n . Bounds such as $L < x < U$ reduce the final step to counting members of an arithmetic progression.

Problem 174.1[See solution](#)How many residues x modulo 126 satisfy $42x \equiv 84 \pmod{126}$?

- (A) 1 (B) 3 (C) 21 (D) 42 (E) 63

Problem 174.2[See solution](#)Which of the following integers x with $100 < x < 150$ satisfies $35x \equiv 10 \pmod{55}$?

- (A) 103 (B) 112 (C) 115 (D) 127 (E) 143

Problem 174.3[See solution](#)For how many residue classes b modulo 60 does $24x \equiv b \pmod{60}$ have exactly twelve solution classes modulo 60?

- (A) 1 (B) 5 (C) 12 (D) 24 (E) 60

Problem 174.4[See solution](#)For an integer $n > 1$, how many ordered pairs of residues (a, b) modulo n make $x \mapsto ax + b$ a permutation of all residue classes modulo n ?

- (A)
- $\varphi(n)$
- (B)
- n
- (C)
- $n\varphi(n)$
- (D)
- n^2
- (E)
- $n - \varphi(n)$

Problem 174.5[See solution](#)How many solution classes of $12x \equiv 12 \pmod{60}$ are represented by integers relatively prime to 60?

- (A) 3 (B) 4 (C) 5 (D) 8 (E) 12

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Binary Structures and the Definition of a Group

175 Binary operations, semigroups, and monoids

Book pp. 227-228. [Find in compact glossary](#)**In this entry:** binary operation; closure; associativity; binary structure; semigroup; monoid; identity.**Definition and key ideas**

A binary operation on S is a function $S \times S \rightarrow S$; closure and being defined on every ordered pair are built into the definition. An associative binary operation makes S a semigroup. A two-sided identity e satisfies $ea = ae = a$ for all a ; a semigroup with identity is a monoid. A two-sided identity, if present, is unique.

These properties must be checked for the specified set and operation. Subtraction is closed on $\mathbb{Z}$ but not associative; addition on positive integers is associative but has no identity in that set. A table in which each symbol occurs once in each row and column need not be associative, so that condition alone does not prove a group.

A nonempty set together with a specified binary operation is a binary structure. A right identity satisfies $x * e = x$ and a left identity satisfies $e * x = x$; either property alone need not give uniqueness. Cancellation means $ax = ay \Rightarrow x = y$ and $xa = ya \Rightarrow x = y$. Every finite cancellative monoid is a group, by repetition of powers, but finiteness is essential.

Problem 175.1[See solution](#)

On $\mathbb{R}$, define $x * y = x + y + xy$. Which description is correct?

- (A) Not closed
- (B) Closed but not associative
- (C) An associative operation with no identity
- (D) A monoid that is not a group
- (E) A group

Problem 175.2[See solution](#)

A finite monoid satisfies both cancellation laws. Which conclusion must hold?

- (A) It is a group.
- (B) It is commutative.
- (C) Its order is prime.
- (D) Every element is its own inverse.
- (E) It has at most two elements.

Problem 175.3[See solution](#)

On a set S with at least two elements define $x * y = x$. Which description is correct?

- (A) The operation is not associative.
- (B) Every element is a two-sided identity.
- (C) Every element is a right identity, but no left identity exists.
- (D) Every element is a left identity, but no right identity exists.
- (E) It is a group.

Problem 175.4[See solution](#)

On $\{0, 1, 2\}$ define $x * y \equiv -x - y \pmod{3}$. Each row and column of its operation table contains all three elements. Which statement is true?

- (A) The structure must be a group.
- (B) The operation is associative but has no identity.
- (C) The operation is not associative.
- (D) The operation is not closed.
- (E) The identity is zero.

Problem 175.5[See solution](#)

A binary operation on a nonempty set has an element l with $l * x = x$ for all x , and an element r with $x * r = x$ for all x . Associativity is not assumed. Which conclusion follows?

- (A) $l = r$, and this element is a two-sided identity.
- (B) The operation is associative.
- (C) The operation is commutative.
- (D) Every element has an inverse.
- (E) l and r must be distinct.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Binary Structures and the Definition of a Group

176 Groups, inverses, and commutativity

Book pp. 228-229. [Find in compact glossary](#)**In this entry:** group; inverse; Abelian group; commutative operation; finite group; infinite group; group order; cancellation law.**Definition and key ideas**

A group is a monoid in which every element has a two-sided inverse. It is Abelian if $ab = ba$ for every pair of elements. Its order $|G|$ is its cardinality, finite or infinite. In a group, $ax = b$ and $ya = b$ have the unique solutions $x = a^{-1}b$ and $y = ba^{-1}$.

Inverses reverse order: $(ab)^{-1} = b^{-1}a^{-1}$. Cancellation is valid, but factors cannot be commuted without a hypothesis. For example, $(ab)^2 = a^2b^2$ for every a, b holds exactly when the group is Abelian. The identity has order one even though the symbol used for it may be 0, 1, or an identity matrix.

Standard infinite Abelian groups include $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ under addition; nonzero $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ under multiplication; and fixed-size matrices with entries in any of those four sets under addition. The set restriction is essential: omitting zero breaks additive closure, while including zero prevents a nontrivial multiplicative group.

Problem 176.1[See solution](#)In a group, $a^{-1}xba = c$. What is x ?

- (A) $aca^{-1}b^{-1}$ (B) $acb^{-1}a^{-1}$ (C) $a^{-1}cb^{-1}a$ (D) $baca^{-1}$ (E) $ca^{-1}b^{-1}a$

Problem 176.2[See solution](#)Every element of a group G satisfies $g^2 = e$. Which statement must hold?

- (A) $|G| = 2$
 (B) G is cyclic
 (C) G is finite
 (D) G is Abelian
 (E) G has no proper nontrivial subgroup

Problem 176.3[See solution](#)

Elements a, b of a group have orders two and three, respectively. How many group elements x can satisfy $ax = xb$?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) It depends on the group.

Problem 176.4[See solution](#)

If a group satisfies $(ab)^2 = a^2b^2$ for all a, b , which follows?

- (A) The group is trivial
- (B) The group is finite
- (C) The group is abelian
- (D) The group is cyclic
- (E) Every element has order two

Problem 176.5[See solution](#)

Which is the inverse of aba^{-1} in an arbitrary group?

- (A) $a^{-2}b^{-1}$ (B) $ab^{-1}a$ (C) $ab^{-1}a^{-1}$ (D) $a^{-1}ba$ (E) $a^{-1}b^{-1}a$

Problem 176.6[See solution](#)

Which of the following, with the indicated ordinary operation, is an infinite Abelian group?

- (A) Nonzero rational numbers under addition
- (B) Positive integers under multiplication
- (C) All real 2×3 matrices under addition
- (D) All real 2×2 matrices under multiplication
- (E) Positive real numbers under addition

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Examples of Groups

177 General and special linear groups

Book pp. 229. [Find in compact glossary](#)

In this entry: general linear group; special linear group; matrix group; $GL(n, F)$; $SL(n, F)$; unimodular matrix.

Definition and key ideas

For a field F , $GL(n, F)$ is the group of invertible $n \times n$ matrices under multiplication. Its subgroup $SL(n, F)$ consists of determinant-one matrices. The determinant is multiplicative, so determinant restrictions often identify subgroups; a restriction such as determinant 2 does not.

Standard notation $GL(n, \mathbb{Z})$ means matrices invertible over $\mathbb{Z}$, equivalently integer matrices of determinant ± 1 . All nonsingular integer matrices form only a monoid. Over $\mathbf{F}_q$, choose independent columns to get $|GL(n, \mathbf{F}_q)| = \prod_{j=0}^{n-1} (q^n - q^j)$. The determinant maps onto $\mathbf{F}_q^*$, so the special linear group has $1/(q-1)$ as many elements.

An integer matrix of determinant ± 1 is called unimodular here; its integer inverse shows that it preserves the greatest common divisor of vector coordinates. In particular it preserves primitive integer vectors, whose coordinate gcd is one. Positive determinant matrices form a subgroup over $\mathbb{R}$, whereas a fixed determinant other than one need not define a subgroup.

Problem 177.1[See solution](#)

How many elements does $SL(2, \mathbf{F}_3)$ contain?

- (A) 12 (B) 18 (C) 24 (D) 36 (E) 48

Problem 177.2[See solution](#)

For how many integers k is $\begin{pmatrix} 2 & 3 \\ k & 2 \end{pmatrix}$ invertible over $\mathbb{Z}$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) Infinitely many

Problem 177.3[See solution](#)

Which subset of $GL(2, \mathbb{R})$ is a subgroup under multiplication?

- (A) $\{A : \det A > 0\}$
 (B) $\{A : \det A < 0\}$
 (C) $\{A : \det A = 2\}$
 (D) $\{A : \operatorname{tr} A = 0\}$
 (E) $\{A : \text{every entry of } A \text{ is positive}\}$

Problem 177.4[See solution](#)

Let $A \in \text{GL}(2, \mathbb{Z})$ and let $v \in \mathbb{Z}^2$ have relatively prime coordinates. Which must hold for Av ?

- (A) Its coordinates are positive.
- (B) Its Euclidean norm equals that of v .
- (C) Its coordinates are relatively prime.
- (D) At least one coordinate equals one.
- (E) It equals v or $-v$.

Problem 177.5[See solution](#)

If $A \in \text{SL}(2, \mathbb{R})$ and $A^2 = I$, which conclusion is forced?

- (A) $A = I$ (B) $A = -I$ (C) $A = I$ or $A = -I$ (D) $\text{tr } A = 0$ (E) A is singular

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Examples of Groups

178 Roots of unity as a finite group

Book pp. 229. [Find in compact glossary](#)**In this entry:** root of unity; primitive root of unity; multiplicative roots-of-unity group.**Definition and key ideas**

The complex n th roots of unity are $U_n = \{e^{2\pi ik/n} : 0 \leq k < n\}$. They form a multiplicative group of order n . A primitive n th root has order exactly n , so $e^{2\pi ik/n}$ is primitive precisely when $\gcd(k, n) = 1$. There are $\varphi(n)$ primitive roots.

Multiplying roots adds their exponent indices modulo n , connecting this geometric group to $\mathbb{Z}/n\mathbb{Z}$. Intersections satisfy $U_m \cap U_n = U_{\gcd(m, n)}$. For a fixed root of unity, its order can be computed arithmetically instead of by repeated complex multiplication.

For $\zeta = e^{2\pi i/n}$, the finite geometric sum $\sum_{k=0}^{n-1} \zeta^{mk}$ equals n when $n \mid m$ and zero otherwise. This roots-of-unity filter can extract selected exponents in a polynomial expansion. A power equation in U_n becomes a congruence on exponent indices; primitiveness need not be preserved when two primitive roots are multiplied.

Problem 178.1[See solution](#)

How many complex numbers satisfy both $z^{84} = 1$ and $z^{126} = 1$?

- (A) 14 (B) 21 (C) 28 (D) 42 (E) 84

Problem 178.2[See solution](#)

If $\zeta = e^{2\pi i/45}$, what is the order of ζ^{12} ?

- (A) 3 (B) 5 (C) 9 (D) 15 (E) 45

Problem 178.3[See solution](#)

Let $\zeta = e^{2\pi i/5}$. What is $\sum_{k=0}^4 (1 + \zeta^k)^6$?

- (A) 5 (B) 10 (C) 30 (D) 35 (E) 64

Problem 178.4[See solution](#)

How many elements $z \in U_{60}$ satisfy $z^{14} = e^{2\pi i/30}$?

- (A) 0 (B) 1 (C) 2 (D) 14 (E) 30

Problem 178.5[See solution](#)

If u and v are each primitive eighth roots of unity, which list gives all possible orders of uv ?

- (A) 8 only (B) 4 only (C) 2 or 4 only (D) 1, 2, or 4 (E) 1, 2, 4, or 8

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Examples of Groups

179 Permutations and cycle decompositions

Book pp. 229; 327. [Find in compact glossary](#)

In this entry: permutation; symmetric group; cycle notation; disjoint cycles; transposition; permutation order.

Definition and key ideas

The symmetric group S_n consists of all bijections of an n -element set, composed as functions; $|S_n| = n!$. Read a product of permutations right to left unless another convention is stated. Every permutation decomposes into disjoint cycles, and its order is the least common multiple of the cycle lengths.

A transposition swaps two elements, and every cycle can be expressed as a product of transpositions. There are $(k - 1)!$ distinct cycles on a fixed set of k symbols, because cyclic rotations of the written list give the same permutation. S_n is non-Abelian for $n \geq 3$. Conjugation relabels a cycle and preserves cycle type; cycle structure can distinguish elements and count permutations with specified orders.

Conjugation obeys $\tau(a_1 \cdots a_k)\tau^{-1} = (\tau(a_1) \cdots \tau(a_k))$. Squaring a cycle of odd length keeps that length; squaring a $2k$ -cycle splits it into two k -cycles. Thus a permutation is a square precisely when each even cycle length occurs an even number of times: pair those cycles to build roots, and take roots within odd cycles.

Problem 179.1[See solution](#)

How many elements of S_7 have order exactly 6 and consist of a 3-cycle, a 2-cycle, and two fixed points?

- (A) 210 (B) 280 (C) 420 (D) 630 (E) 840

Problem 179.2[See solution](#)

Let $\sigma = (1\,2\,3\,4)(5\,6\,7)$ in S_7 . What is the order of σ^2 ?

- (A) 3 (B) 4 (C) 6 (D) 8 (E) 12

Problem 179.3[See solution](#)

With composition read right to left, what is $(1\,2\,3)(2\,4)$ in S_4 ?

- (A) $(1\,2\,4\,3)$ (B) $(1\,4\,2\,3)$ (C) $(1\,3\,4\,2)$ (D) $(1\,2)(3\,4)$ (E) The identity

Problem 179.4[See solution](#)

For $\sigma = (1\,2\,3\,4\,5)$, how many $\tau \in S_5$ satisfy $\tau\sigma\tau^{-1} = \sigma$?

- (A) 1 (B) 4 (C) 5 (D) 20 (E) 24

Problem 179.5[See solution](#)

Which permutation in S_{10} is the square of another permutation?

- (A) $(1\ 2)(3\ 4\ 5\ 6)$
- (B) $(1\ 2)(3\ 4)$
- (C) $(1\ 2)(3\ 4\ 5\ 6\ 7\ 8)$
- (D) $(1\ 2\ 3\ 4)(5\ 6\ 7\ 8\ 9\ 10)$
- (E) $(1\ 2)(3\ 4\ 5\ 6\ 7\ 8\ 9\ 10)$

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Examples of Groups

180 Dihedral groups and geometric symmetry

Book pp. 229. [Find in compact glossary](#)**In this entry:** dihedral group; rotation; reflection; symmetry group.**Definition and key ideas**

The dihedral group D_n is the symmetry group of a regular n -gon and has $2n$ elements. It is generated by a rotation r through $2\pi/n$ and a reflection s , with $r^n = s^2 = e$ and $sr s = r^{-1}$. Every element has the form r^k or $r^k s$. Some authors write this same group as D_{2n} , so check the convention.

The relation $sr^k = r^{-k}s$ reduces words efficiently. Every reflection has order 2, whereas rotation orders are $n/\gcd(n, k)$. For $n \geq 3$ the group is non-Abelian. When n is odd, its center is trivial; when n is even, its center consists of the identity and the half-turn.

The center consists of elements commuting with every group element; commuting with both generators r, s is sufficient here. Two reflections $s, r^k s$ generate $\langle r^k, s \rangle$ of order $2n/\gcd(n, k)$. Conjugating a reflection by r^j changes its exponent by $2j$, so for even n the reflections split into two classes according to exponent parity; for odd n there is one class.

Problem 180.1[See solution](#)

In $D_9 = \langle r, s : r^9 = s^2 = e, sr s = r^{-1} \rangle$, what is $(r^4 s)(r^7 s)$?

- (A) r^2 (B) r^3 (C) r^6 (D) $r^2 s$ (E) $r^6 s$

Problem 180.2[See solution](#)

How many elements of order exactly 2 are in the symmetry group of a regular decagon?

- (A) 5 (B) 9 (C) 10 (D) 11 (E) 20

Problem 180.3[See solution](#)

In the symmetry group D_{12} of a regular twelve-gon, how many elements commute with the basic rotation r but do not commute with a fixed reflection s ?

- (A) 2 (B) 6 (C) 10 (D) 12 (E) 22

Problem 180.4[See solution](#)

In D_{15} , what is the order of the subgroup generated by s and $r^6 s$?

- (A) 5 (B) 6 (C) 10 (D) 15 (E) 30

Problem 180.5[See solution](#)

In D_{10} , how many distinct conjugates of a fixed reflection s occur?

- (A) 2 (B) 5 (C) 10 (D) 19 (E) 20

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Examples of Groups

181 Residue groups and multiplication tables

Book pp. 230-231. [Find in compact glossary](#)**In this entry:** additive group modulo n ; unit group modulo n ; multiplication table; group table; Cayley table.**Definition and key ideas**

The additive group $\mathbb{Z}/n\mathbb{Z}$ consists of residue classes with addition modulo n , has order n , and is generated by 1. Multiplication modulo n on all classes is a monoid. Its invertible classes form $(\mathbb{Z}/n\mathbb{Z})^*$, whose order is $\varphi(n)$; a class is invertible exactly when it is coprime to n .

For prime p , every nonzero class is a unit, giving a group of order $p - 1$. In a group table each row and each column is a permutation of the elements, by cancellation. Such a table can reveal identities and inverses quickly, but the Latin-square property alone does not guarantee associativity.

To read a Cayley table, use the first factor's row and the second factor's column. The identity row and column reproduce the headers; the location of the identity in an element's row gives its right inverse. A symmetric group table indicates commutativity. For composite n , merely removing zero from all residues need not give a group: nonzero zero divisors can destroy closure.

Problem 181.1[See solution](#)What is the order of the unit group $(\mathbb{Z}/84\mathbb{Z})^*$?

- (A) 12 (B) 18 (C) 24 (D) 36 (E) 42

Problem 181.2[See solution](#)Which group is isomorphic to $(\mathbb{Z}/8\mathbb{Z})^*$?

- (A)
- $\mathbb{Z}/2\mathbb{Z}$
- (B)
- $\mathbb{Z}/4\mathbb{Z}$
- (C)
- $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$
- (D)
- $\mathbb{Z}/8\mathbb{Z}$
- (E)
- S_3

Problem 181.3[See solution](#)How many elements of the unit group $(\mathbb{Z}/40\mathbb{Z})^*$ have order exactly two?

- (A) 1 (B) 3 (C) 4 (D) 7 (E) 8

Problem 181.4[See solution](#)

A group has the table below, with the left factor selecting the row. Which statement is correct?

$*$	a	b	c	d
a	d	a	b	c
b	a	b	c	d
c	b	c	d	a
d	c	d	a	b

- (A) The identity is a .
- (B) The identity is b , the inverse of a is c , and the group is cyclic.
- (C) The identity is b , and every element is its own inverse.
- (D) The identity is c , and $a^{-1} = b$.
- (E) The group is non-Abelian.

Problem 181.5[See solution](#)

For which integers $n \geq 2$ do all nonzero residue classes modulo n form a group under multiplication modulo n ?

- (A) All n
- (B) Exactly the odd n
- (C) Exactly the primes
- (D) Exactly the prime powers
- (E) Exactly the square-free n

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Cyclic Groups

182 Cyclic groups and generators

Book pp. 231-232. [Find in compact glossary](#)**In this entry:** cyclic group; generator; infinite cyclic group; Klein four-group.**Definition and key ideas**

A group is cyclic if $G = \langle a \rangle = \{a^k : k \in \mathbb{Z}\}$ for some a . Integer exponents, including negative ones, are essential for infinite cyclic groups. Every infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$; every finite cyclic group of order n is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

If a has finite order n , then a^k generates G exactly when $\gcd(k, n) = 1$, so G has $\varphi(n)$ generators. Infinite cyclic groups have exactly two generators, a and a^{-1} . Cyclic implies Abelian, but the Klein four-group $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ shows the converse fails.

Every nonzero rational has a rational half, which prevents $(\mathbb{Q}, +)$ from being cyclic: a generator's half cannot be its integer multiple. In V_4 , any two distinct nonidentity elements generate the whole group, although no single element does. For an infinite-order element, restricting to nonnegative powers yields a monoid whose only subgroup is the identity subgroup.

Problem 182.1[See solution](#)

A cyclic group has order 90. How many of its elements generate the entire group?

- (A) 12 (B) 18 (C) 24 (D) 30 (E) 36

Problem 182.2[See solution](#)

Let $G = \langle a \rangle$ be infinite cyclic. Which set generates G ?

- (A) $\{a^6, a^{15}\}$ (B) $\{a^{10}, a^{25}\}$ (C) $\{a^{14}, a^{21}\}$ (D) $\{a^{15}, a^{22}\}$ (E) $\{a^{18}, a^{24}\}$

Problem 182.3[See solution](#)

Let $G = \langle a \rangle$ be infinite cyclic and $P = \{a^k : k \geq 0\}$. Which subgroup of G is the largest one contained in P ?

- (A) G
 (B) P
 (C) $\langle a^2 \rangle$
 (D) $\{e\}$
 (E) No subgroup is contained in P .

Problem 182.4[See solution](#)

How many ordered pairs of elements generate the Klein four-group V_4 ?

- (A) 3 (B) 4 (C) 6 (D) 8 (E) 12

Problem 182.5[See solution](#)

How many elements individually generate the additive group $(\mathbb{Q}, +)$?

- (A) 0
- (B) 1
- (C) 2
- (D) Countably infinitely many
- (E) Uncountably infinitely many

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Subgroups

183 Subgroups and the center

Book pp. 232-233. [Find in compact glossary](#)**In this entry:** subgroup; proper subgroup; trivial subgroup; subgroup test; center of a group.**Definition and key ideas**

A subgroup $H \leq G$ is a subset that is a group under the same operation. The efficient subgroup test is: H is nonempty and $ab^{-1} \in H$ for all $a, b \in H$. A proper subgroup is different from G , and the trivial subgroup is $\{e\}$. For finite nonempty subsets, closure under multiplication alone suffices.

The center $Z(G) = \{z : zg = gz \text{ for all } g \in G\}$ is an Abelian subgroup, and in fact normal. Intersections of subgroups are subgroups, but unions need not be: $H \cup K$ is a subgroup exactly when one of H, K contains the other. The trivial group has only one subgroup, so the usual two examples need not be distinct.

For finite subgroups, the product set $HK = \{hk : h \in H, k \in K\}$ has cardinality $|H||K|/|H \cap K|$, even when HK is not itself a subgroup. This gives intersection bounds. A union of three proper subgroups can be a whole group, as in V_4 , despite the restriction on unions of two. In the quotient notation used below, $G/Z(G)$ identifies elements differing by a central factor.

Problem 183.1[See solution](#)

Which group is the union of three proper subgroups?

- (A) $\mathbb{Z}_4$
- (B) $\mathbb{Z}_6$
- (C) $\mathbb{Z}_8$
- (D) The Klein four-group V_4
- (E) $\mathbb{Z}_9$

Problem 183.2[See solution](#)For $n \geq 2$, which matrices comprise the center of $\text{GL}(n, \mathbb{R})$?

- (A) All diagonal invertible matrices
- (B) All determinant-one matrices
- (C) All nonzero scalar matrices
- (D) Only the identity matrix
- (E) All symmetric invertible matrices

Problem 183.3[See solution](#)

If H, K are subgroups of orders 12, 18 in a group of order 36, what is the least possible order of $H \cap K$?

- (A) 6 (B) 3 (C) 2 (D) 1 (E) 12

Problem 183.4[See solution](#)

If $G/Z(G)$ is cyclic, which must be true?

- (A) G is finite (B) G is abelian (C) $|G|$ is prime (D) $Z(G) = \{e\}$ (E) G is cyclic

Problem 183.5[See solution](#)

For a subgroup $H \leq G$, which condition is sufficient for H to be normal?

- (A) G is finite (B) H is abelian (C) $|H| = 2$ (D) $[G : H] = 2$ (E) H is cyclic

Problem 183.6[See solution](#)

Which subsets of the multiplicative group $\mathbb{R}^*$ must be subgroups? (I) $\{x : x^2 \in \mathbb{Q}\}$; (II) $\{x : |x| \geq 1\}$; (III) any finite nonempty subset closed under multiplication.

- (A) I only (B) II only (C) I and III only (D) II and III only (E) I, II, and III

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Cyclic Subgroups

184 Element order and generated subgroups

Book pp. 233-234. [Find in compact glossary](#)

In this entry: order of an element; cyclic subgroup; generated subgroup; finitely generated group; group exponent.

Definition and key ideas

The order of $a \in G$ is the least positive m with $a^m = e$, or infinity if none exists; it equals $|\langle a \rangle|$. If a has finite order n , then $a^k = e$ exactly when $n \mid k$, and $\text{ord}(a^k) = n / \gcd(n, k)$.

The subgroup generated by a set consists of all finite products of generators and their inverses. A group is finitely generated if a finite set suffices. In Abelian groups, the order of a product divides the least common multiple of the factor orders, with equality when those orders are coprime. Without commutativity, this product rule can fail. The exponent of a finite group is the lcm of all element orders.

The exponent of a finite group need not equal any one element order, as S_4 shows. Without commutativity even a product of two involutions can have infinite order. A finite generating list of positive rational numbers can involve only finitely many primes in all numerators and denominators, so it cannot generate the entire multiplicative group.

Problem 184.1[See solution](#)

An element g has finite order, g^6 has order 10, and g^{10} has order 6. What is the order of g ?

(A) 10 (B) 12 (C) 30 (D) 60 (E) 120

Problem 184.2[See solution](#)

In an Abelian group, a has order 8 and b has order 15. What is the order of ab ?

(A) 15 (B) 23 (C) 40 (D) 60 (E) 120

Problem 184.3[See solution](#)

What is the exponent of the symmetric group S_4 , defined as the least common multiple of all element orders?

(A) 4 (B) 6 (C) 8 (D) 12 (E) 24

Problem 184.4[See solution](#)

Consider the group of bijections of $\mathbb{R}$ under composition. Let $s(x) = -x$ and $t(x) = 2 - x$, both of order two. What is the order of $s \circ t$?

(A) 1 (B) 2 (C) 3 (D) 4 (E) Infinite

Problem 184.5[See solution](#)

Which statement describes the multiplicative group of positive rational numbers?

- (A) It is generated by 2 and 3.
- (B) It is cyclic.
- (C) It is finitely generated but not cyclic.
- (D) It is not finitely generated.
- (E) It is not Abelian.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Generators and Relations

185 Group presentations and word reduction

Book pp. 233-234. [Find in compact glossary](#)**In this entry:** group presentation; generators and relations; defining relation; word in generators.**Definition and key ideas**

A presentation $\langle S \mid R \rangle$ describes the group generated by the symbols S , subject exactly to the relations R and their consequences. Relations permit replacement and simplification of words. For example, $\langle a, b \mid a^2 = b^2 = e, ab = ba \rangle$ presents the Klein four-group.

A relation $a^n = e$ by itself only implies that the order of a specified element divides n . In the one-generator presentation with that sole relation, the resulting group has order exactly n . To prove a presented group has a proposed size, word reduction gives an upper bound; a concrete group realizing the relations often gives the matching lower bound.

Conjugation respects powers: $ab^ma^{-1} = (aba^{-1})^m$. Iterating a conjugation relation can impose unexpected additional power relations. In contrast, $\langle a, b \mid ab = ba \rangle$ has no torsion relation and is the infinite group $\mathbb{Z}^2$; commuting words alone do not bound the group order.

Problem 185.1[See solution](#)

In the presentation $\langle a, b \mid aba^{-1} = b^2 \rangle$, which word equals $a^3b^2a^{-3}$?

- (A) b^6 (B) b^8 (C) b^{12} (D) b^{16} (E) b^{32}

Problem 185.2[See solution](#)

What is the order of the group $\langle a, b \mid a^4 = e, b^2 = e, ab = ba, b = a^2 \rangle$?

- (A) 1 (B) 2 (C) 4 (D) 8 (E) 16

Problem 185.3[See solution](#)

What is the order of $\langle a, b \mid a^4 = e, b^2 = e, bab = a^{-1} \rangle$?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) Infinite.

Problem 185.4[See solution](#)

Which group is presented by $\langle a, b \mid ab = ba \rangle$, with no other defining relations?

- (A) $\mathbb{Z}_2 \times \mathbb{Z}_2$ (B) $\mathbb{Z}$ (C) $\mathbb{Z} \times \mathbb{Z}$ (D) The trivial group (E) S_3

Problem 185.5[See solution](#)

What is the order of $\langle a, b \mid a^3 = e, b^5 = e, aba^{-1} = b^2 \rangle$?

- (A) 1 (B) 3 (C) 5 (D) 15 (E) Infinite

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Some Theorems Concerning Subgroups

186 Lagrange, Cauchy, and Sylow existence

Book pp. 234-235. [Find in compact glossary](#)

In this entry: Lagrange theorem; Cauchy theorem; Sylow first theorem; Sylow subgroup; subgroup order; index of a subgroup.

Definition and key ideas

Lagrange's theorem states that $|H|$ divides $|G|$ for a subgroup H of a finite group G ; the quotient $[G : H] = |G|/|H|$ is its index. Consequently every element order divides $|G|$, and every group of prime order is cyclic. The converse fails for arbitrary finite groups: a divisor need not occur as a subgroup order.

Cauchy's theorem guarantees an element, hence subgroup, of order p for each prime $p \mid |G|$. If p^k is the largest power of p dividing $|G|$, Sylow's first theorem guarantees a subgroup of order p^k ; subgroups of order p^j exist for every $0 \leq j \leq k$. Finite Abelian groups have subgroups of every divisor order. Finite cyclic groups have exactly one subgroup of each divisor order, and every subgroup of a cyclic group is cyclic.

The alternating group A_4 has order twelve but no subgroup of order six, a standard failure of the converse of Lagrange. In an Abelian group, subgroups exist for every divisor order, but they need not be cyclic. Cauchy plus commutativity does guarantee a cyclic subgroup whose order is the product of any selected distinct prime divisors, by multiplying elements of those coprime orders.

Problem 186.1[See solution](#)

A group has order 180. Which order is guaranteed for one of its subgroups by Sylow existence?

- (A) 10 (B) 12 (C) 15 (D) 20 (E) 9

Problem 186.2[See solution](#)

How many subgroups does a cyclic group of order 360 have?

- (A) 12 (B) 18 (C) 24 (D) 30 (E) 36

Problem 186.3[See solution](#)

A finite group contains an element of order seven but no element of order three or five. Which could be its order?

- (A) 21 (B) 28 (C) 35 (D) 42 (E) 70

Problem 186.4[See solution](#)

How many subgroups of order six does A_4 , the group of even permutations of four symbols, have?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 186.5[See solution](#)

Every Abelian group of order 72 must contain a cyclic subgroup of which order?

- (A) 4 (B) 6 (C) 8 (D) 9 (E) 12

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ The Concept of Isomorphism

187 Isomorphism and structural invariants

Book pp. 235-237. [Find in compact glossary](#)**In this entry:** isomorphism; isomorphic groups; structural invariant; nonisomorphic groups.**Definition and key ideas**

A group isomorphism is a bijection $f : G \rightarrow H$ satisfying $f(ab) = f(a)f(b)$. It transfers the entire group structure, including identity, inverses, element orders, subgroup structure, commutativity, and cyclicity. Names of elements and symbols for operations are irrelevant.

To disprove isomorphism, find one differing invariant; equal order alone is far from sufficient. To prove isomorphism, give a well-defined bijective operation-preserving map, or apply a classification theorem. For example $x \mapsto e^x$ is an isomorphism $(\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \cdot)$, while $(\mathbb{R}, +)$ is not isomorphic to $(\mathbb{R}^*, \cdot)$ because the latter has an element of order two.

Problem 187.1[See solution](#)Which group is isomorphic to the multiplicative group $\{1, -1, i, -i\}$?

- (A) $\mathbb{Z}_2 \times \mathbb{Z}_2$ (B) $\mathbb{Z}_4$ (C) S_3 (D) $(\mathbb{Z}, +)$ (E) $(\mathbb{Z}/8\mathbb{Z})^*$

Problem 187.2[See solution](#)Which property distinguishes $(\mathbb{Q}, +)$ from $(\mathbb{Q}_{>0}, \cdot)$ and therefore proves that they are not isomorphic?

- (A) Only the first is countable
 (B) Only the first is Abelian
 (C) Only the first admits a group square root of every element
 (D) Only the first has an identity
 (E) Only the first is infinite

Problem 187.3[See solution](#)Which group is isomorphic to $\mathbb{Z}_{12}$?

- (A) $\mathbb{Z}_3 \times \mathbb{Z}_4$
 (B) $\mathbb{Z}_2 \times \mathbb{Z}_6$
 (C) $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$
 (D) $S_3 \times \mathbb{Z}_2$
 (E) The alternating group A_4

Problem 187.4[See solution](#)

Which invariant distinguishes $\mathbb{Z}_8 \times \mathbb{Z}_2$ from $\mathbb{Z}_4 \times \mathbb{Z}_4$?

- (A) The number of identity elements
- (B) Commutativity
- (C) Whether the group is finite
- (D) The largest element order
- (E) Group order

Problem 187.5[See solution](#)

Which group is isomorphic to the additive group $\mathbb{Q}$?

- (A) The multiplicative group $\mathbb{Q}_{>0}$
- (B) The multiplicative group $\{2^q : q \in \mathbb{Q}\}$
- (C) $\mathbb{Q} \times \mathbb{Z}_2$
- (D) The additive group $\mathbb{Z}$
- (E) The circle group

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ The Classification of Finite Abelian Groups

188 Direct products, direct sums, and element orders

Book pp. 237-240. [Find in compact glossary](#)**In this entry:** direct product of groups; finite direct sum; coordinatewise operation; order in a direct product; cyclic product criterion.**Definition and key ideas**

The direct product $G \times H$ has pairs as elements and coordinatewise operations. Its order is $|G||H|$ when both factors are finite, and an element (g, h) of finite component orders has order $\text{lcm}(\text{ord } g, \text{ord } h)$. For finitely many Abelian factors, direct product and direct sum agree.

A product $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic exactly when $\gcd(m, n) = 1$, and then is isomorphic to $\mathbb{Z}_{mn}$. For more finite cyclic factors, the orders must be pairwise coprime. Infinite direct products and direct sums differ: the sum allows only finitely many nonidentity coordinates, whereas the product does not impose that restriction.

The support of a sequence is the set of coordinates where it is not the identity. A direct sum permits only finite support. Thus the countable sum of copies of $\mathbb{Z}_2$ is countable, while their full product is uncountable. Subgroups can also link coordinates: the diagonal $\{(g, g)\}$ is a subgroup of $G \times G$ isomorphic to G .

Problem 188.1[See solution](#)What is the order of $(6, 10, 9)$ in $\mathbb{Z}_{18} \times \mathbb{Z}_{25} \times \mathbb{Z}_{21}$?

- (A) 15 (B) 21 (C) 35 (D) 105 (E) 315

Problem 188.2[See solution](#)How many elements of $\mathbb{Z}_4 \times \mathbb{Z}_{12}$ have order exactly 4?

- (A) 4 (B) 6 (C) 8 (D) 12 (E) 16

Problem 188.3[See solution](#)Let $S = \bigoplus_{j=1}^{\infty} \mathbb{Z}_2$ and $P = \prod_{j=1}^{\infty} \mathbb{Z}_2$. Which cardinality statement is correct?

- (A) Both are finite.
 (B) Both are countably infinite.
 (C) S is countably infinite and P is uncountable.
 (D) S is uncountable and P is countably infinite.
 (E) Both have exactly two elements.

Problem 188.4[See solution](#)

Which element belongs to the cyclic subgroup generated by $(1, 1)$ in $\mathbb{Z} \times \mathbb{Z}_2$?

- (A) $(0, 1)$ (B) $(1, 0)$ (C) $(2, 0)$ (D) $(3, 0)$ (E) $(4, 1)$

Problem 188.5[See solution](#)

Let $D = \{(g, g) : g \in S_3\} \subseteq S_3 \times S_3$. Which statement is true?

- (A) D is not closed under multiplication.
(B) D is a subgroup of order six.
(C) D is a cyclic subgroup of order six.
(D) D has order thirty-six.
(E) D is an Abelian subgroup.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ The Classification of Finite Abelian Groups

189 Elementary divisors and invariant factors

Book pp. 238-240. [Find in compact glossary](#)**In this entry:** fundamental theorem of finite Abelian groups; elementary divisors; invariant factors; integer partitions; finite Abelian group classification.**Definition and key ideas**

Every finite Abelian group is a product of cyclic groups of prime-power order. The multiset of prime powers, called elementary divisors, is unique. Equivalently, it is $\mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_r}$ with $1 < d_1 \mid d_2 \mid \cdots \mid d_r$; the d_i are the unique invariant factors.

To convert elementary divisors to invariant factors, sort the powers of each prime in increasing order, right-align the lists, and multiply columns. The number of isomorphism classes of Abelian groups of order $\prod p_i^{a_i}$ is $\prod P(a_i)$, where $P(a)$ is the number of integer partitions of a . Invariant factors also show that the largest element order is d_r .

An integer partition writes a positive integer as a sum of positive parts with order ignored. For example the partitions of four are 4, 3 + 1, 2 + 2, 2 + 1 + 1, and 1 + 1 + 1 + 1. A group with r nontrivial cyclic p -power factors has p^r elements killed by p , including the identity. Its number of index- p subgroups is $(p^r - 1)/(p - 1)$, the number of nonzero linear conditions on the reduced prime coordinates up to nonzero scaling. The least number of generators of a nontrivial finite Abelian group is the number of its invariant factors.

Problem 189.1[See solution](#)How many isomorphism classes of Abelian groups have order $2^4 3^3$?

- (A) 8 (B) 10 (C) 12 (D) 15 (E) 20

Problem 189.2[See solution](#)What are the invariant factors of $\mathbb{Z}_4 \times \mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_3$?

- (A) 12, 72 (B) 24, 36 (C) 4, 8, 27 (D) 8, 108 (E) 3, 288

Problem 189.3[See solution](#)

A finite Abelian group has order sixteen, exactly three elements of order two, and no element of order greater than four. Which group is it isomorphic to?

- (A)
- $\mathbb{Z}_{16}$
- (B)
- $\mathbb{Z}_8 \times \mathbb{Z}_2$
- (C)
- $\mathbb{Z}_4 \times \mathbb{Z}_4$
- (D)
- $\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2$
- (E)
- $\mathbb{Z}_2^4$

Problem 189.4[See solution](#)What is the smallest number of generators of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 9

Problem 189.5[See solution](#)

How many subgroups of index three are in $\mathbb{Z}_9 \times \mathbb{Z}_3 \times \mathbb{Z}_5$?

- (A) 1 (B) 2 (C) 4 (D) 6 (E) 8

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Group Homomorphisms

190 Homomorphisms and maps from cyclic groups

Book pp. 240-242. [Find in compact glossary](#)

In this entry: group homomorphism; monomorphism; epimorphism; endomorphism; automorphism; homomorphic image.

Definition and key ideas

A homomorphism preserves multiplication, hence the identity and inverses. A monomorphism is injective, an epimorphism is surjective for groups, an endomorphism has the same domain and codomain, and an automorphism is a bijective endomorphism. Images and inverse images of subgroups under a homomorphism are subgroups.

The order of $f(g)$ divides the finite order of g ; equality is guaranteed when f is injective, not for arbitrary homomorphisms. A map from $\mathbb{Z}_m$ is determined by the image of 1, which must satisfy $mf(1) = 0$. Thus there are $\gcd(m, n)$ homomorphisms $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$. Automorphisms of $\mathbb{Z}_n$ correspond to units modulo n .

A proposed formula on residue classes must give the same output for every representative; respecting the generator relation is essential. The trivial homomorphism sends every element to the identity. Between finite groups of coprime orders it is the only homomorphism, since element orders can only decrease by divisibility. Inversion $g \mapsto g^{-1}$ is always bijective but is a homomorphism exactly for Abelian groups.

Problem 190.1

[See solution](#)

How many group homomorphisms are there from $\mathbb{Z}_{18}$ to $\mathbb{Z}_{30}$?

- (A) 3 (B) 5 (C) 6 (D) 9 (E) 15

Problem 190.2

[See solution](#)

How many surjective homomorphisms $\mathbb{Z}_{24} \rightarrow \mathbb{Z}_8$ are there?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 12

Problem 190.3

[See solution](#)

Which formula defines a group homomorphism $f : \mathbb{Z}_6 \rightarrow \mathbb{Z}_4$?

- | | |
|----------------------------|--------------------------|
| (A) $f([x]_6) = [x]_4$ | (B) $f([x]_6) = [2x]_4$ |
| (C) $f([x]_6) = [x + 1]_4$ | (D) $f([x]_6) = [x^2]_4$ |
| (E) $f([x]_6) = [3x]_4$ | |

Problem 190.4[See solution](#)

Compare the endomorphisms $f : \mathbb{Z} \rightarrow \mathbb{Z}$ and $g : \mathbb{Q} \rightarrow \mathbb{Q}$, both given by multiplication by two. Which statement is correct?

- (A) Both are automorphisms.
- (B) f is a monomorphism but not an epimorphism, while g is an automorphism.
- (C) f is an epimorphism but not a monomorphism, while g is an automorphism.
- (D) Neither is injective.
- (E) Neither is a homomorphism.

Problem 190.5[See solution](#)

Finite groups G, H have orders twenty-seven and twenty, respectively. How many group homomorphisms $G \rightarrow H$ exist?

- (A) 0 (B) 1 (C) 3 (D) 9 (E) 20

Problem 190.6[See solution](#)

Let $f : \mathbb{Z}^2 \rightarrow \mathbb{Z}_{12}$ be $f(a, b) = 3a + 4b \pmod{12}$, and let $H = \{0, 6\}$. What is $f^{-1}(H)$?

- (A) $2\mathbb{Z} \times 3\mathbb{Z}$ (B) $3\mathbb{Z} \times 2\mathbb{Z}$ (C) $6\mathbb{Z} \times \mathbb{Z}$ (D) $\mathbb{Z} \times 6\mathbb{Z}$ (E) $\{(a, b) : a + b \equiv 0 \pmod{6}\}$

Problem 190.7[See solution](#)

For a group G , the inversion map $j(g) = g^{-1}$ is a bijection. When is it a group endomorphism?

- (A) For every group
- (B) Exactly when G is finite
- (C) Exactly when G is cyclic
- (D) Exactly when G is Abelian
- (E) Only for the trivial group

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Group Homomorphisms

191 Kernels and normal subgroups

Book pp. 241-243. [Find in compact glossary](#)

In this entry: kernel of a group homomorphism; normal subgroup; conjugation; trivial kernel; fiber of a homomorphism.

Definition and key ideas

The kernel of $f : G \rightarrow H$ is $\ker f = \{g : f(g) = e_H\}$. It is a subgroup, and f is injective exactly when its kernel is trivial. For finite G , $|G| = |\ker f| |f(G)|$: every nonempty fiber has the same size as the kernel.

A subgroup N is normal when $gNg^{-1} = N$ for every $g \in G$. Kernels are normal because $f(gng^{-1}) = f(g)e_Hf(g)^{-1} = e_H$. Every subgroup of an Abelian group is normal, but a subgroup of a non-Abelian group need not be. Normality is the condition needed to put a well-defined group operation on cosets.

A fiber over $h \in H$ is $f^{-1}(\{h\}) = \{g : f(g) = h\}$; a nonempty fiber is a coset of the kernel. The determinant map $GL_n(\mathbb{R}) \rightarrow \mathbb{R}^\times$ has kernel $SL_n(\mathbb{R})$. A nontrivial homomorphism from a prime-order group is injective, since its kernel cannot be the whole group. Infinite groups may be isomorphic to proper subgroups: $k \mapsto mk$ is an isomorphism $\mathbb{Z} \rightarrow m\mathbb{Z}$ for $m > 1$; finite groups cannot have this property.

Problem 191.1[See solution](#)

A homomorphism from a group of order 72 has an image of order 12. How many elements map to a fixed element of its image?

- (A) 3 (B) 6 (C) 12 (D) 24 (E) 60

Problem 191.2[See solution](#)

Let $f : \mathbb{Z}_{36} \rightarrow \mathbb{Z}_{24}$ be defined by $f(x) = 10x$. What is $|\ker f|$?

- (A) 2 (B) 3 (C) 6 (D) 9 (E) 12

Problem 191.3[See solution](#)

Let $n \geq 2$ and let $f : GL_n(\mathbb{R}) \rightarrow \mathbb{R}^\times$ be $f(A) = \det A$. Which statement about the fiber $f^{-1}(-2)$ is correct?

- (A) It is the subgroup $SL_n(\mathbb{R})$.
 (B) It is a coset of $SL_n(\mathbb{R})$ but is not a subgroup.
 (C) It is empty when n is even.
 (D) It is a normal subgroup of index two.
 (E) It contains exactly two matrices.

Problem 191.4[See solution](#)

Let G have prime order p , and let $f : G \rightarrow H$ be a homomorphism whose image contains an element other than the identity. Which must be true?

- (A) f is onto H .
- (B) H has order p .
- (C) f is injective.
- (D) H is Abelian.
- (E) $f(g) = g$ for every $g \in G$.

Problem 191.5[See solution](#)

Which subgroup of S_3 cannot be the kernel of a homomorphism from S_3 to any group?

- (A) $\{e\}$
- (B) A_3
- (C) S_3
- (D) $\{e, (12)\}$
- (E) Each listed subgroup can be such a kernel.

Problem 191.6[See solution](#)

For an integer $m > 1$, define $f : (\mathbb{Z}, +) \rightarrow (m\mathbb{Z}, +)$ by $f(k) = mk$. Which statement is correct?

- (A) f is not a homomorphism because $m \neq 1$.
- (B) f is injective but not surjective.
- (C) f is surjective but not injective.
- (D) f is an isomorphism onto a proper subgroup of $\mathbb{Z}$.
- (E) f is an automorphism of $\mathbb{Z}$.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Group Homomorphisms

192 Inner automorphisms and automorphism groups

Book pp. 243–244. [Find in compact glossary](#)

In this entry: inner automorphism; automorphism group; inner automorphism group; conjugation map.

Definition and key ideas

For $a \in G$, the map $c_a(g) = aga^{-1}$ is an automorphism, with inverse $c_{a^{-1}}$. The set $\text{Aut}(G)$ is a group under composition, and inner automorphisms form a subgroup $\text{Inn}(G)$ because $c_a c_b = c_{ab}$.

The map $a \mapsto c_a$ has kernel $Z(G)$, so a finite group satisfies $|\text{Inn}(G)| = |G|/|Z(G)|$. All inner automorphisms are the identity exactly when G is Abelian. An Abelian group can nonetheless have many nonidentity automorphisms; inner and arbitrary automorphism are distinct notions.

Composition is the group operation in $\text{Aut}(G)$. More precisely, $\alpha c_a \alpha^{-1} = c_{\alpha(a)}$, so $\text{Inn}(G)$ is normal in $\text{Aut}(G)$. An automorphism may permute central elements; being central does not mean being fixed by every automorphism.

Problem 192.1

[See solution](#)

A finite group has order 60 and center of order 5. How many distinct inner automorphisms does it have?

- (A) 5 (B) 10 (C) 12 (D) 15 (E) 60

Problem 192.2

[See solution](#)

What is the order of the automorphism $f(x) = 5x$ of the additive group $\mathbb{Z}_{24}$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 8

Problem 192.3

[See solution](#)

Let $V = \mathbb{Z}_2 \times \mathbb{Z}_2$. What is the ordered pair $(|\text{Aut}(V)|, |\text{Inn}(V)|)$?

- (A) (1, 1) (B) (3, 1) (C) (6, 1) (D) (6, 3) (E) (6, 6)

Problem 192.4

[See solution](#)

For $a \in G$, write $c_a(g) = aga^{-1}$. If $\alpha \in \text{Aut}(G)$, which identity always holds, with composition read from right to left?

- | | |
|---|--|
| (A) $\alpha c_a \alpha^{-1} = c_a$ | (B) $\alpha c_a \alpha^{-1} = c_{\alpha(a)}$ |
| (C) $\alpha c_a \alpha^{-1} = c_{a^{-1}}$ | (D) $\alpha c_a \alpha^{-1} = \alpha$ |
| (E) $\alpha c_a \alpha^{-1} = c_{\alpha^{-1}(a)}$ | |

Problem 192.5[See solution](#)

A group G has no automorphism other than the identity. Which statement must be true?

- (A) G has exactly one element.
- (B) G is infinite.
- (C) G is non-Abelian.
- (D) G is Abelian and every $g \in G$ satisfies $g^2 = e$.
- (E) G contains an element of order four.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Rings

193 Rings, unity, and subrings

Book pp. 244–245. [Find in compact glossary](#)

In this entry: ring; commutative ring; ring with unity; subring; distributive laws; additive identity; multiplicative identity.

Definition and key ideas

A ring has an Abelian additive group, an associative multiplication, and both distributive laws. A commutative ring also has commutative multiplication. A ring with unity has a multiplicative identity 1; when discussing domains and fields, require $1 \neq 0$. The book allows rings without unity and subrings that need not share the ambient identity.

With that convention, a nonempty subset is a subring when closed under subtraction and multiplication. Some modern texts instead require rings and subrings to preserve 1, so conventions must be stated. The additive identity always absorbs multiplication: $0r = r0 = 0$. Matrix rings of size at least two are standard noncommutative examples, while $n\mathbb{Z}$ is a ring without unity for $n > 1$.

The two distributive laws are $r(s + t) = rs + rt$ and $(r + s)t = rt + st$. For example, $r0 = r(0 + 0) = r0 + r0$ gives $r0 = 0$ by additive cancellation; similarly $0r = 0$. The two distributive laws must both be checked when a proposed multiplication is not commutative.

Problem 193.1[See solution](#)

Under the book convention allowing a subring to have a different identity, what is the multiplicative identity of $S = \{0, 3, 6, 9, 12\} \subseteq \mathbb{Z}_{15}$?

- (A) 0 (B) 3 (C) 6 (D) 9 (E) 12

Problem 193.2[See solution](#)

Which subset of $M_2(\mathbb{R})$ is a subring under ordinary operations?

- (A) All matrices of trace zero
(B) All matrices of determinant one
(C) All symmetric matrices
(D) All upper triangular matrices
(E) All invertible matrices

Problem 193.3[See solution](#)

On the additive group $\mathbb{Z}_6$, define a new multiplication by $a * b = 0$ for all a, b . With ordinary addition, this structure is which of the following?

- (A) Not a ring because multiplication has no cancellation
- (B) Not a ring because $1 * 1 \neq 1$
- (C) A commutative ring without a multiplicative identity
- (D) A field of characteristic six
- (E) A noncommutative ring with identity zero

Problem 193.4[See solution](#)

Let R be a unital ring and $e \in R$ satisfy $e^2 = e$. Set $S = \{ere : r \in R\}$. Which statement always holds?

- (A) S is a subring whose identity is e .
- (B) S is a subring whose identity must equal 1_R .
- (C) S is commutative.
- (D) S has no nonzero zero divisors.
- (E) S is an additive subgroup but need not be closed under multiplication.

Problem 193.5[See solution](#)

On $\mathbb{Z}$, retain ordinary addition and define $a * b = a$. Which of the following laws hold?

I. $(a * b) * c = a * (b * c)$.

II. $a * (b + c) = a * b + a * c$.

III. $(a + b) * c = a * c + b * c$.

- (A) I only (B) II only (C) I and II only (D) I and III only (E) All three

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Rings

194 Ring characteristic

Book pp. 245; 247. [Find in compact glossary](#)**In this entry:** characteristic of a ring; prime characteristic; characteristic zero; additive order of unity.**Definition and key ideas**

The characteristic of a ring is the least positive integer n such that $nr = 0$ for every r , or zero if there is no such integer. In a unital ring it suffices to find the additive order of 1, because $(n1)r = nr$. Thus $\text{char}(\mathbb{Z}_n) = n$, and $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ have characteristic zero.

An integral domain or field has characteristic zero or a prime: if $n = ab$ were a least positive composite characteristic, $(a1)(b1) = 0$ would contradict the lack of zero divisors. General rings can have composite characteristic. The characteristic of a finite direct product is the lcm of the positive characteristics of its factors.

An infinite ring can have positive characteristic, as $\mathbf{F}_p[x]$ shows. Conversely, characteristic zero does not force every nonzero element to have infinite additive order: $(0, 1)$ has order two in $\mathbb{Z} \times \mathbb{Z}_2$. Prime characteristic alone does not imply a domain; matrix rings over $\mathbf{F}_p$ have zero divisors.

Problem 194.1[See solution](#)What is the characteristic of $M_3(\mathbb{Z}_{12}) \times \mathbb{Z}_{18}$?

- (A) 6 (B) 12 (C) 18 (D) 36 (E) 216

Problem 194.2[See solution](#)

A field has exactly 125 elements. What is its characteristic?

- (A) 0 (B) 3 (C) 5 (D) 25 (E) 125

Problem 194.3[See solution](#)Let p be prime. Which statement about the polynomial ring $\mathbf{F}_p[x]$ is correct?

- (A) It has p elements and characteristic p .
 (B) It is infinite and has characteristic zero.
 (C) It is infinite and has characteristic p .
 (D) It is a field of characteristic p .
 (E) It has p^2 elements and characteristic p^2 .

Problem 194.4[See solution](#)In the ring $R = \mathbb{Z} \times \mathbb{Z}_2$, let $a = (0, 1)$. What are $\text{char}(R)$ and the additive order of a , respectively?

- (A) 0 and 2 (B) 2 and 2 (C) 0 and infinite (D) 2 and infinite (E) 1 and 2

Problem 194.5[See solution](#)

Let $R = M_2(\mathbf{F}_2)$. Which pair of statements about R is correct?

- (A) Its characteristic is zero, and it is a domain.
- (B) Its characteristic is two, and it is a field.
- (C) Its characteristic is four, and it has zero divisors.
- (D) Its characteristic is two, and it has nonzero zero divisors.
- (E) Its characteristic is two, and every nonzero element is a unit.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Rings

195 Polynomial, Gaussian-integer, and function rings

Book pp. 245–246. [Find in compact glossary](#)

In this entry: polynomial ring; indeterminate; Gaussian integer; quadratic integer ring; pointwise addition; pointwise multiplication; function ring.

Definition and key ideas

The polynomial ring $R[x]$ consists of finite formal sums $a_0 + a_1x + \cdots + a_nx^n$, with coefficientwise addition and convolution multiplication. Over a domain, the degree of a nonzero product is the sum of degrees. The Gaussian integers $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ form a subring of $\mathbb{C}$, with multiplicative norm $N(a + bi) = a^2 + b^2$.

Real-valued functions on a set form a ring under pointwise addition and multiplication, with constant functions 0, 1 as identities. Pointwise multiplication is different from composition. Function rings usually have zero divisors: two nonzero functions supported on disjoint subsets multiply to zero. A quadratic set such as $\mathbb{Z}[\sqrt{d}]$ is closed under multiplication because $(\sqrt{d})^2 = d$, whereas a proposed span must always be checked for closure.

An indeterminate is a formal symbol: polynomial equality means equality of all coefficients, not merely agreement after evaluating on a specified set. Distinct polynomials can induce the same function over a finite field. Pointwise operations mean $(f + g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$, not $f(g(x))$. Over a commutative unital ring, a polynomial is a unit exactly when its constant coefficient is a unit and all higher coefficients are nilpotent; nonconstant units are therefore possible when the coefficient ring has nonzero nilpotents.

Problem 195.1[See solution](#)

How many units are in the Gaussian integer ring $\mathbb{Z}[i]$?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) Infinitely many

Problem 195.2[See solution](#)

In the ring of all real-valued functions on $\mathbb{R}$, which condition is equivalent to f being a unit?

- (A) f is continuous
(B) f is bounded
(C) $f(x) \neq 0$ for every x
(D) f is injective
(E) f is constant and nonzero

Problem 195.3[See solution](#)

Let $S = \{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$ and $T = \{a + b\sqrt[3]{2} : a, b \in \mathbb{Z}\}$. Under the usual operations in $\mathbb{R}$, which are subrings?

- (A) S only
- (B) T only
- (C) Both S and T
- (D) Neither S nor T
- (E) Both, but neither has a unity

Problem 195.4[See solution](#)

In $\mathbb{Z}_4[x]$, which polynomial is a unit?

- (A) $1 + x$ (B) 2 (C) $2 + x$ (D) $1 + 2x$ (E) x

Problem 195.5[See solution](#)

In the ring $C([-1, 1], \mathbb{R})$ under pointwise operations, let $f(x) = \max(x, 0)$ and $g(x) = \max(-x, 0)$. Which statement is correct?

- (A) f and g are both units.
- (B) $f + g$ is a unit.
- (C) f and g are nonzero elements with $fg = 0$.
- (D) $f \circ g$ equals fg as functions.
- (E) fg is the constant function one.

Problem 195.6[See solution](#)

Let p be prime and let $q(x) = x^p - x \in \mathbf{F}_p[x]$. Which statement is correct?

- (A) q is the zero polynomial.
- (B) q is a nonzero polynomial that induces the zero function on $\mathbf{F}_p$.
- (C) q induces a bijection of $\mathbf{F}_p$.
- (D) q has exactly one root in $\mathbf{F}_p$.
- (E) q is a unit of $\mathbf{F}_p[x]$.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Rings

196 Nilpotent elements

Book pp. 246-247. [Find in compact glossary](#)**In this entry:** nilpotent element; nilpotency index; sum of commuting nilpotents.**Definition and key ideas**

An element a is nilpotent if $a^m = 0$ for some positive integer m ; the least such m is its nilpotency index. A nonzero nilpotent is a zero divisor and cannot be a unit in a nonzero unital ring. If $a^m = 0$, then $1 - a$ is a unit with inverse $1 + a + \cdots + a^{m-1}$.

If a, b commute and satisfy $a^m = b^n = 0$, then $(a + b)^{m+n-1} = 0$: each binomial term has an exponent large enough to vanish. Commutativity matters; two nilpotent matrices can have an invertible sum. In $\mathbb{Z}_N$, an element is nilpotent exactly when every prime divisor of N divides that element.

If a unit u commutes with a nilpotent a , then $u + a = u(1 + u^{-1}a)$ is a unit, because $u^{-1}a$ is nilpotent. Without commutation this can fail. An idempotent satisfies $e^2 = e$; an element that is both idempotent and nilpotent must be zero. For $N = \prod p^{\alpha_p}$ and a nilpotent residue a , its index is the least k with $k v_p(a) \geq \alpha_p$ for every prime factor p of N , where $v_p(a)$ counts factors of p .

Problem 196.1[See solution](#)How many nilpotent elements are in $\mathbb{Z}_{72}$?

- (A) 6 (B) 8 (C) 12 (D) 18 (E) 24

Problem 196.2[See solution](#)Let $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $N = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$ in $M_2(\mathbb{R})$. Which conclusion follows?

- (A) A unit plus a nilpotent element is always a unit.
 (B) A unit plus a nilpotent element can fail to be a unit when they do not commute.
 (C) The sum of a unit and a nilpotent element must be zero.
 (D) N has nilpotency index three.
 (E) U is nilpotent.

Problem 196.3[See solution](#)What is the nilpotency index of 12 in $\mathbb{Z}_{288}$?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 196.4[See solution](#)

In an associative algebra over $\mathbb{R}$, commuting elements a, b satisfy $a^2 = 0$, $b^3 = 0$, and $ab^2 \neq 0$. What is the nilpotency index of $a + b$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) It need not be nilpotent.

Problem 196.5[See solution](#)

An element a of a ring satisfies both $a^m = 0$ for some positive integer m and $a^2 = a$. Which conclusion is forced?

- (A) $a = 0$
- (B) $a = 1$
- (C) $m = 2$
- (D) a is a nonzero zero divisor.
- (E) The ring has characteristic two.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Ring Homomorphisms

197 Ring homomorphisms and evaluation

Book pp. 247-249. [Find in compact glossary](#)

In this entry: ring homomorphism; ring endomorphism; evaluation homomorphism; substitution homomorphism; kernel of a ring homomorphism; unital homomorphism.

Definition and key ideas

A ring homomorphism preserves addition and multiplication, hence zero and additive inverses. The book does not require preservation of unity; a unital homomorphism additionally sends 1 to 1. Under the nonunital convention, both the zero map and identity map are endomorphisms of $\mathbb{Z}$. Kernels and images are additive subgroups and subrings.

For a commutative ring R , evaluation $R[x] \rightarrow R, f \mapsto f(a)$, is a homomorphism. For a non-commutative coefficient ring this requires a to commute with coefficients; it is not automatic. A homomorphism $\mathbb{Z} \rightarrow R$ is determined by the idempotent image of 1. Preserving one operation alone, as with the additive map $n \mapsto 2n$, does not make a ring homomorphism.

For $m, n > 1$, a unital map $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ exists exactly when $n \mid m$; it is then reduction of residue classes. The determinant on a full matrix ring is multiplicative but generally not additive. Complex conjugation preserves both operations and is a ring automorphism. Evaluation at a matrix fixes all coefficients, so its substituted matrix must commute with each coefficient.

Problem 197.1[See solution](#)

Under the convention that ring homomorphisms need not preserve 1, how many ring homomorphisms $\mathbb{Z} \rightarrow \mathbb{Z}_6$ are there?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 6

Problem 197.2[See solution](#)

Let $E : \mathbb{Q}[x] \rightarrow \mathbb{Q}$ evaluate polynomials at $x = 3$. Which polynomial generates its kernel as an ideal?

- (A) x (B) $x - 3$ (C) $x^2 - 9$ (D) $x + 3$ (E) $3x$

Problem 197.3[See solution](#)

Which condition on positive integers $m, n > 1$ is necessary and sufficient for a unital ring homomorphism $\mathbb{Z}_m \rightarrow \mathbb{Z}_n$ to exist?

- (A) $\gcd(m, n) = 1$
 (B) $m \mid n$
 (C) $n \mid m$
 (D) $m = n$
 (E) m and n are both prime

Problem 197.4[See solution](#)

In $M_2(\mathbb{Q})[x]$, the indeterminate x commutes with every coefficient. For which matrices A does evaluation $\sum B_k x^k \mapsto \sum B_k A^k$ define a ring homomorphism into $M_2(\mathbb{Q})$?

- (A) All invertible A
- (B) All diagonal A
- (C) All symmetric A
- (D) Exactly the scalar matrices $A = cI$
- (E) Exactly the nilpotent matrices

Problem 197.5[See solution](#)

Which of the following maps preserve both addition and multiplication?

I. $\mathbb{Z} \rightarrow \mathbb{Z}, n \mapsto 2n$.

II. $M_2(\mathbb{R}) \rightarrow \mathbb{R}, A \mapsto \det A$.

III. $\mathbb{C} \rightarrow \mathbb{C}, z \mapsto \bar{z}$.

- (A) I only (B) II only (C) III only (D) I and III only (E) II and III only

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Ring Homomorphisms

198 Frobenius endomorphism

Book pp. 248-249. [Find in compact glossary](#)**In this entry:** Frobenius endomorphism; freshman dream in prime characteristic.**Definition and key ideas**

In a commutative ring of prime characteristic p , $F(a) = a^p$ is a ring endomorphism. The binomial coefficients $\binom{p}{k}$ vanish in the ring for $0 < k < p$, yielding $(a + b)^p = a^p + b^p$; commutativity also gives $(ab)^p = a^p b^p$.

Frobenius need not be injective when nilpotents exist and need not be surjective in every infinite field. It is injective in an integral domain and bijective in a finite field. Iteration gives $F^r(a) = a^{p^r}$. One must not apply the binomial simplification in characteristic zero or to arbitrary noncommuting matrices.

In fact, Frobenius is injective exactly when there are no nonzero nilpotents. For a finite field of size p^n , every z satisfies $z^{p^n} = z$, so the inverse of $z \mapsto z^p$ is $z \mapsto z^{p^{n-1}}$. In $\mathbf{F}_p(t)$ the element t has no p th root: degrees in a putative equation $a(t)^p = t b(t)^p$ give a contradiction modulo p .

Problem 198.1[See solution](#)In $\mathbf{F}_5[x]$, what is $(x + 2)^{25}$?

- (A) $x^{25} + 2$ (B) $x^{25} + 4$ (C) $x^{25} + 2x^5 + 2$ (D) $x^{25} + x + 2$ (E) $x^{25} + 1$

Problem 198.2[See solution](#)

Let R be a commutative ring of prime characteristic p . Its Frobenius endomorphism is injective if and only if which condition holds?

- (A) R is finite.
 (B) R is a field.
 (C) R has no nonzero nilpotent element.
 (D) R has no nonzero zero divisor.
 (E) Every element satisfies $a^p = a$.

Problem 198.3[See solution](#)

In the field with 16 elements, what is the inverse function of the Frobenius automorphism $F(z) = z^2$?

- (A) $z \mapsto z^2$ (B) $z \mapsto z^4$ (C) $z \mapsto z^8$ (D) $z \mapsto z^{16}$ (E) $z \mapsto z^{15}$

Problem 198.4[See solution](#)

Let $R = \{a + b\varepsilon : a, b \in \mathbf{F}_3\}$, where $\varepsilon^2 = 0$ and expressions are unique. How many elements map to 1 under $F(u) = u^3$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 9

Problem 198.5[See solution](#)

On which of the following rings is $u \mapsto u^2$ a ring endomorphism?

- I. $\mathbb{Z}_4$.
II. $M_2(\mathbf{F}_2)$.
III. A field with four elements.
(A) I only (B) II only (C) III only (D) I and III only (E) II and III only

Problem 198.6[See solution](#)

In the rational-function field $K = \mathbf{F}_p(t)$, where p is prime, which statement about $F(u) = u^p$ is correct?

- (A) It is neither injective nor surjective.
(B) It is injective but not surjective.
(C) It is surjective but not injective.
(D) It is an automorphism.
(E) It is not a ring homomorphism.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Ring Homomorphisms

199 Ideals and absorption

Book pp. 249-250. [Find in compact glossary](#)**In this entry:** ideal; two-sided ideal; left ideal; right ideal; principal ideal; ideal generated by elements.**Definition and key ideas**

A two-sided ideal I in a ring R is an additive subgroup satisfying $rI \subseteq I$ and $Ir \subseteq I$ for every $r \in R$. One-sided versions impose only one absorption condition. In a commutative unital ring, the principal ideal (a) consists of multiples ra ; (a, b) consists of sums $ra + sb$.

The kernel of a ring homomorphism is an ideal in the domain ring. An ideal in a unital ring that contains 1, or any unit, equals the entire ring. A subring need not be an ideal, since closure under multiplication by its own members is weaker than absorption by all ambient elements. In $\mathbb{Z}$, every ideal is $n\mathbb{Z}$, and $(a, b) = \gcd(a, b)\mathbb{Z}$.

A left ideal requires $rI \subseteq I$ for every ambient r ; a right ideal requires $Ir \subseteq I$. In $\mathbb{Z}_n$, every ideal is generated by a divisor of n , and (a, b) is the set of multiples of $\gcd(n, a, b)$. In a full matrix ring over a field, the only two-sided ideals are zero and the whole ring; multiplication on both sides by matrix units proves this.

Problem 199.1[See solution](#)What is the ideal of $\mathbb{Z}$ generated by 84 and 126?

- (A)
- $6\mathbb{Z}$
- (B)
- $14\mathbb{Z}$
- (C)
- $21\mathbb{Z}$
- (D)
- $42\mathbb{Z}$
- (E)
- $252\mathbb{Z}$

Problem 199.2[See solution](#)How many ideals does $\mathbb{Z}_{30}$ have?

- (A) 3 (B) 4 (C) 6 (D) 8 (E) 10

Problem 199.3[See solution](#)Let $L \subset M_2(\mathbb{R})$ be the set of matrices whose second column is zero. Which statement is correct?

- (A) L is a left ideal but not a right ideal.
 (B) L is a right ideal but not a left ideal.
 (C) L is a two-sided ideal.
 (D) L is not an additive subgroup.
 (E) L contains the identity matrix.

Problem 199.4[See solution](#)

Let D be the diagonal matrices in $M_2(\mathbb{R})$. Which gives a valid proof that D is not a two-sided ideal?

- (A) D is not closed under addition.
- (B) D is not closed under multiplication.
- (C) D contains the identity but is a proper subset of the matrix ring.
- (D) D contains the zero matrix.
- (E) D is commutative.

Problem 199.5[See solution](#)

Let F be a field and I a two-sided ideal of $M_n(F)$ containing a matrix A with nonzero entry a_{ij} . Which conclusion must hold?

- (A) I consists only of diagonal matrices.
- (B) I consists only of scalar matrices.
- (C) I has dimension one over F .
- (D) $I = M_n(F)$.
- (E) Every matrix in I is invertible.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Integral Domains

200 Zero divisors, cancellation, and integral domains

Book pp. 250–251. [Find in compact glossary](#)**In this entry:** zero divisor; left zero divisor; right zero divisor; integral domain; cancellation in a ring.

Definition and key ideas

A nonzero element is a zero divisor if it multiplies some nonzero element to zero; in a noncommutative ring one distinguishes sides. An integral domain is a commutative unital ring with $1 \neq 0$ and no nonzero zero divisors. Thus $ab = 0$ forces $a = 0$ or $b = 0$, and multiplication by a nonzero element can be canceled.

In $\mathbb{Z}_n$, a nonzero class is a zero divisor exactly when it is not coprime to n ; $\mathbb{Z}_n$ is a domain exactly when n is prime. A product of two nonzero rings is not a domain: for nonzero a, b in its factors, $(a, 0)(0, b) = 0$. Matrix rings and unrestricted function rings supply many zero divisors, even when their scalar coefficients lie in a field.

Here the side names follow the source: if $ab = 0$ with $a, b \neq 0$, the left factor a is a left zero divisor and the right factor b is a right zero divisor. Some texts reverse this terminology, so the equation is decisive. Nonzero elements that are not zero divisors need not be units in an infinite ring; multiplication by such an element can be injective without being surjective.

Problem 200.1

[See solution](#)How many nonzero zero divisors are in $\mathbb{Z}_{45}$?

- (A) 12 (B) 20 (C) 21 (D) 24 (E) 30

Problem 200.2

[See solution](#)

Which ring is an integral domain?

- (A) $\mathbb{Z}_{21}$
(B) $M_2(\mathbb{Q})$
(C) $\mathbb{Q} \times \mathbb{Q}$
(D) $\mathbb{Z}[\sqrt{3}]$
(E) All real-valued functions on $\mathbb{R}$

Problem 200.3

[See solution](#)

In an integral domain R , nonzero elements a, b satisfy $a^2 = b^2$. Which conclusion always follows?

- (A) $a = b$ (B) $a = -b$ (C) $a = b$ or $a = -b$ (D) $a + b = 1$ (E) a and b are units

Problem 200.4[See solution](#)

On a vector space with basis $e_0, e_1, \dots$, define $Re_j = e_{j+1}$ and $Le_0 = 0$, $Le_{j+1} = e_j$. In its endomorphism ring, call A a left zero divisor if $AB = 0$ for some $B \neq 0$, and a right zero divisor if $BA = 0$ for some $B \neq 0$. Which statement about L is correct?

- (A) It is neither kind.
- (B) It is a left zero divisor only.
- (C) It is a right zero divisor only.
- (D) It is both kinds.
- (E) It is a two-sided unit.

Problem 200.5[See solution](#)

In $C([0, 1], \mathbb{R})$ under pointwise operations, let $f(x) = x$. Which statement is correct?

- (A) f is a unit.
- (B) f is a nonzero zero divisor.
- (C) f is neither a unit nor a zero divisor.
- (D) f is nilpotent.
- (E) f is idempotent.

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▸ Fields

201 Units, division rings, fields, and finite domains

Book pp. 251-253. [Find in compact glossary](#)

In this entry: unit; division ring; field; subfield; finite field; quadratic field; quaternion division ring; strictly skew field.

Definition and key ideas

A unit has a two-sided multiplicative inverse. A division ring is a nonzero unital ring in which every nonzero element is a unit; a field is a commutative division ring. Every field is a domain, but $\mathbb{Z}$ is a domain that is not a field. Every finite domain is a field because multiplication by a nonzero element is injective and therefore surjective on a finite set.

For nonsquare rational d , the quadratic field $\mathbb{Q}(\sqrt{d})$ consists of $a + b\sqrt{d}$ with rational coefficients; invert by multiplying by the conjugate. Matrix representations can be fields when restricted to a suitable commuting subring. Real quaternions have the form $a + bi + cj + dk$ with real coefficients and $i^2 = j^2 = k^2 = ijk = -1$. The inverse of a nonzero quaternion is its conjugate divided by $a^2 + b^2 + c^2 + d^2$. They form a noncommutative division ring, illustrating why commutativity is substantive.

A subfield is a subset that is itself a field under the ambient operations, with the same 1. Every subfield of $\mathbb{C}$ contains $\mathbb{Q}$. In a product of unital rings, a tuple is a unit exactly when every coordinate is a unit. Quaternion multiplication satisfies $ij = k$, $jk = i$, $ki = j$, and reversing any of these products changes its sign. A matrix model for $\mathbb{C}$ is $a + bi \mapsto \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$, whose determinant is $a^2 + b^2$. The quaternions have the matrix model $\left\{ \begin{pmatrix} z & w \\ -\bar{w} & \bar{z} \end{pmatrix} : z, w \in \mathbb{C} \right\}$, with determinant $|z|^2 + |w|^2$; they are also called a strictly skew field.

Problem 201.1

[See solution](#)

What is $(2 + \sqrt{3})^{-1}$ in $\mathbb{Q}(\sqrt{3})$?

- (A) $2 - \sqrt{3}$ (B) $-2 + \sqrt{3}$ (C) $(2 - \sqrt{3})/7$ (D) $(2 + \sqrt{3})/7$ (E) $(2 - \sqrt{3})/3$

Problem 201.2

[See solution](#)

Let R be a finite commutative ring with identity and no nonzero zero divisors. If R has 49 elements, how many units does it contain?

- (A) 6 (B) 7 (C) 42 (D) 48 (E) 49

Problem 201.3[See solution](#)

In the real quaternions, let q, r satisfy $(1 + i)q = j$ and $r(1 + i) = j$. Which pair (q, r) is correct?

- (A) $((j - k)/2, (j + k)/2)$ (B) $((j + k)/2, (j - k)/2)$
 (C) $((j - k)/2, (j - k)/2)$ (D) $(j - k, j + k)$
 (E) $((j + k)/2, (j + k)/2)$

Problem 201.4[See solution](#)

For real t , set $K_t = \left\{ \begin{pmatrix} a & tb \\ b & a \end{pmatrix} : a, b \in \mathbb{R} \right\}$. For which t is K_t a field under ordinary matrix operations?

- (A) $t > 0$ (B) $t \geq 0$ (C) $t < 0$ (D) $t \neq 0$ (E) All real t

Problem 201.5[See solution](#)

What is $\mathbb{Q}(\sqrt{2}) \cap \mathbb{Q}(\sqrt{3})$ as subfields of $\mathbb{R}$?

- (A) $\mathbb{Z}$ (B) $\mathbb{Q}$ (C) $\mathbb{Q}(\sqrt{6})$ (D) $\mathbb{Q}(\sqrt{2})$ (E) $\mathbb{Q}(\sqrt{3})$

Problem 201.6[See solution](#)

How many units does $\mathbb{Z} \times \mathbb{Z}_{12} \times \mathbf{F}_4$ have?

- (A) 12 (B) 16 (C) 24 (D) 32 (E) 48

Problem 201.7[See solution](#)

Let $H(z, w) = \begin{pmatrix} z & w \\ -\bar{w} & \bar{z} \end{pmatrix}$ for $z, w \in \mathbb{C}$. Which formula gives $H(z, w)^{-1}$ when $(z, w) \neq (0, 0)$?

- (A) $H(z, w)/(|z|^2 + |w|^2)$
 (B) $H(\bar{z}, -w)/(|z|^2 + |w|^2)$
 (C) $H(\bar{z}, \bar{w})/(|z|^2 - |w|^2)$
 (D) $H(-z, w)/(|z|^2 + |w|^2)$
 (E) $H(w, z)/(|z|^2 + |w|^2)$

6. Number Theory and Abstract Algebra

Part B: Abstract Algebra ▷ Fields

202 Idempotents and Boolean rings

Book pp. 253; 256. [Find in compact glossary](#)**In this entry:** idempotent element; Boolean ring; orthogonal idempotents.**Definition and key ideas**

An element e is idempotent if $e^2 = e$. In a domain or division ring, $e(e - 1) = 0$ forces $e = 0$ or 1 . A commutative ring can have additional idempotents; commuting idempotents have idempotent product. The term orthogonal describes idempotents e, f with $ef = fe = 0$.

A Boolean ring is one in which every element is idempotent. Expanding $(x + x)^2 = x + x$ gives $2x = 0$, and expanding $(x + y)^2 = x + y$ then gives $xy = yx$. Thus every Boolean ring is commutative. In $\mathbb{Z}_n$, choosing residue 0 or 1 independently modulo each distinct prime-power factor yields exactly $2^{\omega(n)}$ idempotents, where $\omega(n)$ counts distinct prime divisors.

Problem 202.1[See solution](#)How many idempotents does $\mathbb{Z}_{180}$ contain?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 12

Problem 202.2[See solution](#)

In a nonzero Boolean ring with identity, which statement must hold?

- (A) Every element is a unit
(B) The only unit is 1
(C) The ring is an integral domain
(D) The ring is infinite
(E) The characteristic is zero

Problem 202.3[See solution](#)In a unital ring, idempotents e, f satisfy $ef = fe = 0$. Which statement must hold?

- (A) $e + f$ is idempotent.
(B) $e + f = 1$
(C) $e = f$
(D) $e - f$ is nilpotent.
(E) e and f are both units.

Problem 202.4[See solution](#)

A ring R , not assumed to have unity, satisfies $x^2 = x$ for every $x \in R$. Which conclusion must hold?

- (A) R is a field.
- (B) R has exactly two elements.
- (C) $xy = 0$ for all x, y .
- (D) $2x = 0$ for every x , and R is commutative.
- (E) R has characteristic zero.

Problem 202.5[See solution](#)

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$. Which statement is correct?

- (A) Neither A nor B is idempotent.
- (B) A is idempotent but B is not.
- (C) Both are idempotent, and so is AB .
- (D) Both are idempotent, but AB is not.
- (E) $AB = BA$

7. Additional Topics

Logic ▸ Sentential Calculus

203 Propositions and logical connectives

Book pp. 257-258. [Find in compact glossary](#)

In this entry: logic; sentential calculus; propositional calculus; sentence; proposition; negation; conjunction; disjunction; conditional; biconditional; if and only if.

Definition and key ideas

A proposition is a statement with a truth value. Negation $\neg P$ reverses that value; conjunction $P \wedge Q$ requires both statements; inclusive disjunction $P \vee Q$ requires at least one. The implication $P \Rightarrow Q$ is false only when P is true and Q is false, so it equals $\neg P \vee Q$. A biconditional $P \Leftrightarrow Q$ says that their truth values agree.

The antecedent is a sufficient condition for the consequent, and the consequent is a necessary condition for the antecedent. A false antecedent makes an implication true; this is vacuous truth. Logical disjunction is inclusive unless exclusive disjunction is expressly specified. Parentheses are especially important in formulas containing more than one implication.

Logic, or sentential/propositional calculus in this section, studies truth-functional combinations of propositions. In $P \Rightarrow Q$, P is the antecedent and Q the consequent. Exclusive disjunction means exactly one statement is true, expressed by $(P \vee Q) \wedge \neg(P \wedge Q)$.

Problem 203.1[See solution](#)

For how many of the eight truth assignments to P, Q, R is $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ true?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 203.2[See solution](#)

Which formula is equivalent to the negation of $(P \vee Q) \Rightarrow R$?

- (A) $(\neg P \wedge \neg Q) \vee R$ (B) $(P \vee Q) \wedge \neg R$
 (C) $(\neg P \vee \neg Q) \wedge R$ (D) $P \vee (Q \wedge \neg R)$
 (E) $(P \wedge Q) \wedge \neg R$

Problem 203.3[See solution](#)

A statement says: “For x to belong to A , it is necessary that x belong to B , and sufficient that x belong to C . Which inclusion expresses exactly this statement?

- (A) $A \subseteq B \subseteq C$ (B) $B \subseteq A \subseteq C$ (C) $C \subseteq A \subseteq B$ (D) $A = B = C$ (E) $B \subseteq C \subseteq A$

Problem 203.4[See solution](#)

Suppose $(P \vee Q) \wedge \neg(P \wedge Q) \wedge R$ is true. Which expression must be false?

- (A) P (B) Q (C) $P \Leftrightarrow Q$ (D) R (E) $P \vee Q$

Problem 203.5[See solution](#)

Over the integers, consider the following statements.

I. $\forall n (n^2 = 2 \Rightarrow n^3 = 8)$.

II. $\exists n (n^2 = 2 \Rightarrow n^3 = 8)$.

III. $\exists n (n^2 = 2 \wedge n^3 = 8)$.

Which are true?

(A) I only (B) II only (C) I and II only (D) II and III only (E) All three

7. Additional Topics

Logic ▸ Logical Equivalences and Truth Tables

204 Truth tables, tautologies, and contraposition

Book pp. 258; 322-323; 327. [Find in compact glossary](#)

In this entry: truth table; logical equivalence; tautology; contradiction; contrapositive; converse; inverse of an implication; De Morgan laws; distributive laws; excluded middle; quantifier negation.

Definition and key ideas

Two formulas are logically equivalent if they agree on every truth assignment. A tautology is always true and a contradiction always false. The law of excluded middle is the tautology $P \vee \neg P$. De Morgan's laws are $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ and $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$. Distributivity holds in both directions between conjunction and disjunction.

The implication $P \Rightarrow Q$ is equivalent to its contrapositive $\neg Q \Rightarrow \neg P$, but not generally to its converse $Q \Rightarrow P$ or inverse $\neg P \Rightarrow \neg Q$. The source labels contraposition as contradiction; these are different notions. A useful extension for mathematical statements is quantifier negation: $\neg \forall x P(x)$ means $\exists x \neg P(x)$, and $\neg \exists x P(x)$ means $\forall x \neg P(x)$.

A truth table lists every assignment of truth values to the variables and evaluates the formula on each row. Double negation gives $\neg \neg P \equiv P$. Conjunction and disjunction are each commutative and associative. The distributive identities are $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$ and $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$. These identities justify algebraic simplification without checking every row separately.

Problem 204.1

[See solution](#)

Assume $P \Rightarrow Q$, $Q \Rightarrow R$, and $\neg P \Rightarrow \neg R$. Which must hold?

- (A) P, Q, R are all true
- (B) P, Q, R are all false
- (C) P, Q, R have the same truth value
- (D) P and R have opposite truth values
- (E) Q is true

Problem 204.2

[See solution](#)

What is the negation of $\forall x \in \mathbb{R} \exists y \in \mathbb{R} (y > x \text{ and } f(y) > 0)$?

- (A) $\forall x \forall y (y \leq x \text{ or } f(y) \leq 0)$
- (B) $\exists x \forall y (y \leq x \text{ or } f(y) \leq 0)$
- (C) $\exists x \exists y (y \leq x \text{ and } f(y) \leq 0)$
- (D) $\forall x \exists y (y \leq x \text{ and } f(y) \leq 0)$
- (E) $\exists x \forall y (y > x \text{ and } f(y) \leq 0)$

Problem 204.3[See solution](#)

The implication $P \Rightarrow Q$ is true, but its converse and inverse are both false. Which truth assignment is possible?

- (A) $P = T, Q = T$
- (B) $P = T, Q = F$
- (C) $P = F, Q = T$
- (D) $P = F, Q = F$
- (E) No such assignment exists.

Problem 204.4[See solution](#)

Which is logically equivalent to $(P \vee Q) \Rightarrow R$?

- (A) $R \Rightarrow (P \vee Q)$
- (B) $P \wedge Q \wedge R$
- (C) $(P \wedge Q) \Rightarrow R$
- (D) $(P \Rightarrow R) \vee (Q \Rightarrow R)$
- (E) $(P \Rightarrow R) \wedge (Q \Rightarrow R)$

Problem 204.5[See solution](#)

Classify $F = (P \Rightarrow Q) \wedge P \wedge \neg Q$ and $G = P \vee \neg P$.

- (A) Both are tautologies.
- (B) Both are contradictions.
- (C) F is a tautology and G a contradiction.
- (D) F is a contradiction and G a tautology.
- (E) Both are true for some assignments and false for others.

Problem 204.6[See solution](#)

Which formula is equivalent to $\neg\neg P \wedge (Q \vee (R \vee Q))$?

- (A) $(P \wedge Q) \vee (P \wedge R)$
- (B) $(P \vee Q) \wedge (P \vee R)$
- (C) $P \vee (Q \wedge R)$
- (D) $(P \wedge Q) \vee R$
- (E) $P \wedge Q \wedge R$

7. Additional Topics

Set Theory ▸ Subsets and Complements

205 Sets, subsets, and complements

Book pp. 259. [Find in compact glossary](#)

In this entry: set; element; membership; set-builder notation; family of sets; empty set; subset; proper subset; superset; universal set; complement; relative complement; set difference.

Definition and key ideas

A set is determined by its elements, with $x \in A$ denoting membership. Set-builder notation $\{x : P(x)\}$ describes elements satisfying a property. The empty set has no elements. The assertion $A \subseteq B$ means every element of A belongs to B ; a proper subset additionally differs from B . Equality can be proved by the two inclusions.

Relative to a fixed universe U , the complement is $A^c = U \setminus A$; the difference $A \setminus B$ consists of elements in A but not B . Membership and inclusion are different: $\emptyset \subseteq A$ always, but $\emptyset \in A$ only when the empty set is itself a listed element. A family of sets is simply a set whose members are sets.

The notation $B \supseteq A$ says that B is a superset of A , equivalently $A \subseteq B$. We write $A \subsetneq B$ for strict inclusion. A universal set fixes which objects are under discussion; changing it can change the complement even when A stays the same. A set ignores repetition and the order in which its members are listed.

Problem 205.1[See solution](#)

Let $A = \{\emptyset, \{\emptyset\}\}$. How many members of A are also subsets of A ?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 205.2[See solution](#)

For arbitrary sets A, B, C with $A \subseteq B$, which statement must be true?

- (A) $C \setminus A \subseteq C \setminus B$ (B) $C \setminus B \subseteq C \setminus A$ (C) $A \setminus C = B \setminus C$ (D) $B \setminus A = \emptyset$ (E) $A \in B$

Problem 205.3[See solution](#)

A set A has five elements and $B \subset A$ has two elements. How many subsets of A contain B ?

- (A) 10 (B) 32 (C) 16 (D) 4 (E) 8

Problem 205.4[See solution](#)

If $A \subseteq B$ and $B \subseteq C$, which identity must hold?

- (A) $C \subseteq B$ (B) $A = C$ (C) $A \cap C = A$ (D) $A \cup C = A$ (E) $B \setminus A = \emptyset$

Problem 205.5[See solution](#)

Let $S = \{n \in \mathbb{Z} : n^2 < 9\}$ and $T = \{n^2 : n \in S\}$. What is $(|S \setminus T|, |T \setminus S|)$?

- (A) (2, 2) (B) (3, 1) (C) (1, 3) (D) (3, 2) (E) (5, 3)

7. Additional Topics

Set Theory ▸ Union and Intersection; Venn Diagrams

206 Unions, intersections, and symmetric difference

Book pp. 260; 263. [Find in compact glossary](#)

In this entry: union; intersection; disjoint sets; symmetric difference; indexed family; indexed union; indexed intersection; Venn diagram; set De Morgan laws.

Definition and key ideas

The union $A \cup B$ uses membership in at least one set, whereas $A \cap B$ uses membership in both. Disjoint sets have empty intersection. Symmetric difference $A \triangle B = (A \setminus B) \cup (B \setminus A)$ means membership in exactly one. Indexed unions use an existential quantifier; indexed intersections use a universal one.

De Morgan's laws exchange unions and intersections when taking complements, including infinite families. Venn diagrams encode the possible membership patterns and help expose false identities. Symmetric difference is associative and satisfies $A \triangle A = \emptyset$; its behavior matches addition modulo two of membership indicators. Set difference, in contrast, is not associative.

For an indexed family $\{A_i : i \in I\}$, $\bigcup_{i \in I} A_i = \{x : \exists i \in I, x \in A_i\}$ and $\bigcap_{i \in I} A_i = \{x : \forall i \in I, x \in A_i\}$. A Venn diagram represents membership regions by overlapping shapes; shading selects a set expression. The symmetric difference of three sets selects elements in one or three of them, not elements in exactly one only.

Problem 206.1

[See solution](#)

For arbitrary sets A, B, C , which expression equals $A \setminus (B \setminus C)$?

- (A) $(A \setminus B) \setminus C$ (B) $(A \setminus B) \cup (A \cap C)$
 (C) $(A \cap B) \setminus C$ (D) $(A \cup B) \cap C$
 (E) $A \setminus (B \cup C)$

Problem 206.2

[See solution](#)

If $A \triangle B = A \triangle C$, what must hold?

- (A) $A = \emptyset$ (B) $B = C$ (C) $B \cap C = \emptyset$ (D) $A = B = C$ (E) $B = C^c$

Problem 206.3

[See solution](#)

Finite sets A, B, C have sizes $(10, 12, 14)$, pairwise intersection sizes $|A \cap B| = 4$, $|A \cap C| = 5$, $|B \cap C| = 6$, and triple intersection size 2. What is $|A \triangle B \triangle C|$?

- (A) 10 (B) 12 (C) 14 (D) 16 (E) 18

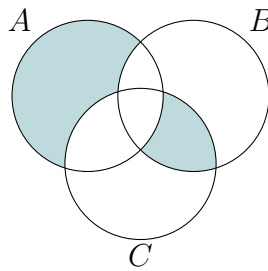
Problem 206.4[See solution](#)

Let U be a nonempty set and define $A_{n,m} = U$ if $m \geq n$, and $A_{n,m} = \emptyset$ otherwise, for positive integers n, m . What is the ordered pair $(\bigcap_n \bigcup_m A_{n,m}, \bigcup_m \bigcap_n A_{n,m})$?

- (A) (U, U)
- (B) $(\emptyset, \emptyset)$
- (C) $(U, \emptyset)$
- (D) $(\emptyset, U)$
- (E) It depends on the cardinality of U .

Problem 206.5[See solution](#)

Which expression describes the shaded region in this Venn diagram?



- | | |
|--|---------------------------|
| (A) $(A \setminus (B \cup C)) \cup ((B \cap C) \setminus A)$ | (B) $A \Delta B \Delta C$ |
| (C) $(A \cup B) \setminus C$ | (D) $(A \cap B) \cup C$ |
| (E) $A \setminus (B \cap C)$ | |

7. Additional Topics

Set Theory ▸ Cartesian Products; Intervals of the Real Line

207 Cartesian products and real intervals

Book pp. 260-262. [Find in compact glossary](#)

In this entry: Cartesian product; ordered pair; open interval; closed interval; half-open interval; unbounded interval; extended real numbers.

Definition and key ideas

The Cartesian product $A \times B = \{(a, b) : a \in A, b \in B\}$ records ordered pairs. For finite sets, $|A \times B| = |A||B|$; if either factor is empty, the product is empty. An interval in $\mathbb{R}$ contains every real number between any two of its points. Parentheses exclude finite endpoints and brackets include them.

The extended real line adjoins $-\infty, +\infty$ for bounds and limits; in ordinary real intervals these are not included endpoints. Coordinate conditions describe products exactly, so boundary membership requires checking each coordinate. Infinite unions and intersections of intervals demand attention to whether an endpoint ever occurs or survives every set.

Explicitly, $(a, b) = \{x : a < x < b\}$, $[a, b] = \{x : a \leq x \leq b\}$, and $[a, b)$ or $(a, b]$ includes exactly the indicated endpoint. In $\mathbb{R}$, $[a, \infty)$ is unbounded but has only real elements. In the extended real line, an interval such as $[a, +\infty]$ can instead include the adjoined element $+\infty$; the ambient space must be stated.

Problem 207.1[See solution](#)

What is $\bigcap_{n=1}^{\infty} (-1/n, 1 + 1/n)$?

- (A) $(0, 1)$ (B) $[0, 1]$ (C) $[0, 1)$ (D) $(0, 1]$ (E) $\emptyset$

Problem 207.2[See solution](#)

Let $A = (0, 2]$, $B = [1, 3)$, and $C = [2, 4]$. What is $(A \times B) \cap (B \times C)$?

- (A) $(0, 3) \times [1, 4]$ (B) $[1, 2] \times [2, 3)$ (C) $(1, 2) \times (2, 3)$ (D) $[1, 2] \times [2, 3]$ (E) $\emptyset$

Problem 207.3[See solution](#)

What is $\bigcup_{n=1}^{\infty} [1/n, 1 - 1/n]$, interpreting reversed-endpoint intervals as empty?

- (A) $[0, 1)$ (B) $[0, 1]$ (C) $(0, 1]$ (D) $(0, 1)$ (E) $\{1/2\}$

Problem 207.4[See solution](#)

Let $I = \bigcap_{n=1}^{\infty} [n, \infty)$ as subsets of $\mathbb{R}$, and $J = \bigcap_{n=1}^{\infty} [n, +\infty]$ as subsets of the extended real line. Which pair (I, J) is correct?

- (A) $(\emptyset, \emptyset)$ (B) $(\{+\infty\}, \{+\infty\})$ (C) $(\emptyset, \{+\infty\})$ (D) $(\mathbb{R}, \{+\infty\})$ (E) $(\emptyset, [-\infty, +\infty])$

Problem 207.5[See solution](#)

For arbitrary sets A, B , which condition is equivalent to $A \times B = B \times A$?

- (A) $A = B$
- (B) $A \subseteq B$
- (C) $A \cap B \neq \emptyset$
- (D) $A = B$ or at least one of A, B is empty
- (E) $|A| = |B|$

7. Additional Topics

Set Theory ▸ Cardinality

208 Cardinality, countability, and algebraic numbers

Book pp. 263–264. [Find in compact glossary](#)

In this entry: cardinality; equinumerous sets; bijection; finite set; infinite set; countably infinite; countable set; aleph-null; algebraic number; countable union.

Definition and key ideas

Two sets have the same cardinality when a bijection exists between them. A set is countably infinite if it is in bijection with $\mathbb{N}$, and countable if finite or countably infinite. Integers, rationals, finite tuples of integers, and algebraic numbers are countable. An algebraic number is a root of a nonzero polynomial with integer coefficients.

A finite product and a countable union of countable sets are countable, under the usual set-theoretic framework. For algebraic numbers, count polynomials by their finite coefficient lists and use that each nonzero polynomial has finitely many roots. Infinite countable products are different: all binary sequences already form an uncountable set. A countably infinite set can be bijective with a proper subset, unlike a finite set.

The symbol $\aleph_0$ (aleph-null) denotes $|\mathbb{N}|$. Cardinal comparisons can be proved by injections: $|A| \leq |B|$ means that an injection $A \rightarrow B$ exists. For example, selecting distinct rationals from disjoint nonempty real intervals shows that such a family is at most countable, even though each interval is uncountable.

Problem 208.1

[See solution](#)

Which set is uncountable?

- (A) All finite subsets of $\mathbb{Z}$
- (B) All polynomials with rational coefficients
- (C) All real algebraic numbers
- (D) All functions $\mathbb{N} \rightarrow \{0, 1\}$
- (E) All ordered pairs of rational numbers

Problem 208.2[See solution](#)

How many real numbers can be roots of nonzero polynomials whose coefficients all belong to $\{0, 1\}$?

- (A) None
- (B) Finitely many but nonzero
- (C) Countably infinitely many
- (D) Continuum many
- (E) More than continuum many

Problem 208.3[See solution](#)

A family of pairwise disjoint, nonempty open intervals in $\mathbb{R}$ is given. Which conclusion must hold?

- (A) The family is finite.
- (B) The family is at most countable.
- (C) Every interval has rational endpoints.
- (D) The union is countable.
- (E) All intervals have the same length.

Problem 208.4[See solution](#)

Write $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Which formula defines a bijection from $\mathbb{N}_0 \times \mathbb{N}_0$ onto the positive integers?

- (A) $f(m, n) = m + n + 1$
- (B) $f(m, n) = 2m + n + 1$
- (C) $f(m, n) = 2^m 3^n$
- (D) $f(m, n) = 2^m(2n + 1)$
- (E) $f(m, n) = (m + 1)(n + 1)$

Problem 208.5[See solution](#)

If A is countably infinite and $B \subseteq A$, which must hold?

- (A) B is countably infinite
- (B) B is finite
- (C) $\mathcal{P}(B)$ is countable
- (D) $A \setminus B$ is finite
- (E) B is finite or countably infinite

7. Additional Topics

Set Theory ▸ Cardinality

209 Power sets, uncountability, and the continuum

Book pp. 264-265. [Find in compact glossary](#)

In this entry: power set; Cantor theorem; uncountable set; continuum; transcendental number; cardinal inequality; diagonal argument.

Definition and key ideas

The power set $\mathcal{P}(A)$ is the set of all subsets of A . If $|A| = n$, it has 2^n elements. Cantor's theorem states $|\mathcal{P}(A)| > |A|$ for every set A : a purported surjection $f : A \rightarrow \mathcal{P}(A)$ misses the diagonal set $\{a : a \notin f(a)\}$.

The continuum $\mathfrak{c} = |\mathbb{R}|$ equals $2^{\aleph_0}$ and is strictly larger than $\aleph_0$. Every nondegenerate real interval has cardinality $\mathfrak{c}$; so do irrationals and transcendental real numbers. A transcendental number is not algebraic. The displayed chain $\aleph_0 < \mathfrak{c} < 2^{\mathfrak{c}}$ does not claim these are consecutive infinite cardinalities.

Algebraic numbers are closed under field operations; consequently a nonconstant rational fractional-linear transformation $(at + b)/(ct + d)$ with rational coefficients and $ad - bc \neq 0$ sends a transcendental t to a transcendental value whenever defined. Explicit interval-to-line bijections, such as $x \mapsto x/(1 - x^2)$ on $(-1, 1)$, prove equality of cardinalities without treating length as cardinality.

Problem 209.1

[See solution](#)

If $|A| = 3$ and $|B| = 4$, what is $|\mathcal{P}(\mathcal{P}(A) \times B)|$?

- (A) 2^{12} (B) 2^{16} (C) 2^{24} (D) 2^{32} (E) 2^{64}

Problem 209.2

[See solution](#)

Which set has cardinality strictly larger than $\mathbb{R}$?

- (A) $\mathbb{R} \times \mathbb{R}$
 (B) $\mathbb{R} \setminus \mathbb{Q}$
 (C) $\mathcal{P}(\mathbb{N})$
 (D) All finite sequences of real numbers
 (E) $\mathcal{P}(\mathbb{R})$

Problem 209.3

[See solution](#)

If $|A| = 3$, how many functions $A \rightarrow \mathcal{P}(A)$ are there?

- (A) 729 (B) 512 (C) 256 (D) 24 (E) 64

Problem 209.4[See solution](#)

Let t be transcendental over $\mathbb{Q}$. Which of the following must also be transcendental?

- (A) $t - t$ (B) t/t (C) $(2t + 1)/(t + 3)$ (D) $\sqrt{2}$ (E) $\sqrt{3} + \sqrt{5}$

Problem 209.5[See solution](#)

Which function is a bijection from $(-1, 1)$ onto $\mathbb{R}$?

- (A) $x \mapsto x/(1 + x^2)$ (B) $x \mapsto x/(1 - x^2)$
(C) $x \mapsto x^2/(1 - x^2)$ (D) $x \mapsto \arcsin x$
(E) $x \mapsto \tan(\pi x)$

Problem 209.6[See solution](#)

For a function $f : A \rightarrow \mathcal{P}(A)$, set $D = \{a \in A : a \notin f(a)\}$. Which statement is always true?

- (A) $D = \emptyset$
(B) $D = A$
(C) $D \notin \mathcal{P}(A)$
(D) $D \neq f(a)$ for every $a \in A$
(E) $D = f(a)$ for exactly one $a \in A$

7. Additional Topics

Graph Theory ▸ Vertices, Edges, and Degree

210 Simple graphs and vertex degrees

Book pp. 265–266. [Find in compact glossary](#)

In this entry: graph; vertex; edge; simple graph; undirected graph; adjacent vertices; degree; odd vertex; even vertex; complete graph; graph order; graph size; handshaking lemma; loop; graph complement; regular graph; graphical degree sequence.

Definition and key ideas

A finite simple undirected graph consists of vertices and unordered pairs of distinct vertices called edges, with no loops or repeated edges. Its order is the number n of vertices, and its size is the number m of edges. The degree counts incident edges, or equivalently neighbors in a simple graph. A complete graph K_n has every possible edge and size $\binom{n}{2}$.

The handshaking identity $\sum_v \deg(v) = 2m$ implies an even number of odd-degree vertices. Degree sequences can disprove proposed graphs. Some review diagrams go beyond simple graphs by allowing loops; a loop contributes two to degree in the standard convention. Always distinguish this from merely counting distinct neighboring vertices.

The complement of a simple graph on a fixed vertex set has exactly the missing edges; its degrees are $n - 1 - \deg(v)$. A regular graph has the same degree at every vertex. A degree sequence is graphical if some simple graph realizes it. Loops and parallel edges violate simplicity, and in their presence degree counts incidences, not just distinct neighbors.

Problem 210.1

[See solution](#)

A simple graph has 12 vertices. Seven have degree 5 and the other five all have degree d . Which value of d is impossible?

- (A) 1 (B) 3 (C) 4 (D) 5 (E) 7

Problem 210.2

[See solution](#)

A simple graph on eight vertices has six vertices of degree 5 and two of degree 4. How many edges are missing from the complete graph?

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Problem 210.3

[See solution](#)

Every finite simple graph with at least two vertices must have which property?

- (A) A vertex of degree zero
(B) A vertex adjacent to all others
(C) Two vertices of equal degree
(D) An even number of vertices
(E) A connected complement

Problem 210.4[See solution](#)

Which sequence cannot be the degrees of the vertices of a simple graph?

- (A) (3, 3, 3, 3, 3, 3) (B) (2, 2, 2, 2, 2, 2)
(C) (4, 4, 4, 4, 4, 4) (D) (5, 5, 3, 1, 1, 1)
(E) (5, 1, 1, 1, 1, 1)

Problem 210.5[See solution](#)

A graph allows loops and parallel edges. At a vertex v there are two loops, three edges to w , and one edge to u , where u, v, w are distinct. What are the degree of v and the number of distinct other vertices adjacent to v , respectively?

- (A) (6, 2) (B) (8, 2) (C) (8, 3) (D) (4, 2) (E) (6, 3)

7. Additional Topics

Graph Theory ▸ Graph Isomorphism; Review Matrices

211 Graph isomorphisms and incidence matrices

Book pp. 266; 323; 326. [Find in compact glossary](#)

In this entry: graph isomorphism; adjacency preservation; adjacency matrix; incidence matrix; unoriented incidence matrix; graph invariant.

Definition and key ideas

A graph isomorphism is a vertex bijection preserving adjacency in both directions. The drawing's shape and line crossings are irrelevant unless a crossing is designated as a vertex. Vertex count, edge count, degree multiset, connectedness, and cycle counts are invariants, although agreeing on a few invariants does not prove isomorphism.

For a simple graph, the adjacency matrix has entry 1 when two vertices are adjacent and 0 otherwise. The unoriented incidence matrix has rows for vertices and columns for edges, with 1 at the two endpoints of each edge. Each column sums to two, row sums are degrees, and $BB^T = D + A$, where D is the degree matrix and A the adjacency matrix. These conventions require adjustment if loops are allowed.

For an adjacency matrix A , the entry $(A^k)_{ij}$ counts walks of length k from i to j . Thus $\text{tr}(A^2) = 2|E|$ and $\text{tr}(A^3) = 6$ times the number of triangles in a simple undirected graph. Equal invariants are necessary for an isomorphism; a proposed vertex bijection must still preserve adjacency in both directions.

Problem 211.1

[See solution](#)

Let B be the unoriented incidence matrix of a simple graph. If $\text{tr}(BB^T) = 26$, how many edges does the graph have?

- (A) 6 (B) 13 (C) 26 (D) 39 (E) 52

Problem 211.2

[See solution](#)

Which invariant proves that a cycle on six vertices is not isomorphic to the disjoint union of two triangles?

- (A) Number of vertices
(B) Number of edges
(C) Degree sequence
(D) Number of connected components
(E) Number of vertices of odd degree

Problem 211.3[See solution](#)

For the adjacency matrix A of a simple graph, $\text{tr}(A^3) = 42$. How many triangles does the graph contain?

- (A) 6 (B) 7 (C) 14 (D) 21 (E) 42

Problem 211.4[See solution](#)

Let B be the unoriented vertex-edge incidence matrix of a simple 3-regular graph on eight vertices. What is $\text{tr}((BB^T)^2)$?

- (A) 24 (B) 48 (C) 72 (D) 96 (E) 120

Problem 211.5[See solution](#)

Two graphs have vertex set $\mathbb{Z}_5$. In G , distinct vertices are adjacent when their difference is ± 1 ; in H , when their difference is ± 2 . Which map, computed modulo five, is an isomorphism $G \rightarrow H$?

- (A) $v \mapsto v$ (B) $v \mapsto 2v$ (C) $v \mapsto v^2$ (D) $v \mapsto v^3$ (E) $v \mapsto 0$

7. Additional Topics

Graph Theory ▸ Paths, Trees, and Subgraphs

212 Walks, circuits, trees, and spanning subgraphs

Book pp. 266; 319; 321; 327. [Find in compact glossary](#)

In this entry: walk; trail; path; closed walk; circuit; cycle; connected graph; connected component; forest; tree; subgraph; spanning subgraph; spanning tree.

Definition and key ideas

A walk follows adjacent vertices and may repeat vertices or edges. A trail repeats no edge; a path repeats no vertex. A closed trail is a circuit, and a cycle is a closed path with only the start/end vertex repeated. Terminology varies in the book, so inspect the intended repetition rule. A graph is connected when every two vertices are joined by a path.

A connected component is a maximal connected subgraph. A subgraph uses subsets of the original vertices and edges, with every retained edge keeping its endpoints. A forest has no cycles; a connected forest is a tree. A forest with n vertices and c components has $n - c$ edges. A spanning subgraph retains all vertices, and a spanning tree is a spanning subgraph that is a tree. Every finite connected graph has a spanning tree with $n - 1$ edges; a connected graph of size m can therefore lose at most $m - n + 1$ edges while retaining connectedness.

Problem 212.1

[See solution](#)

A connected graph has 17 vertices and 29 edges. What is the maximum number of edges that can be deleted while keeping all vertices connected?

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 16

Problem 212.2

[See solution](#)

A tree has four vertices of degree 3, two vertices of degree 4, and all remaining vertices have degree 1. How many degree-one vertices are there?

- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

Problem 212.3

[See solution](#)

A forest has c connected components. A vertex of degree d and all of its incident edges are deleted. How many connected components does the remaining forest have?

- (A) $c + d$ (B) $c + d - 1$ (C) $c - d$ (D) $c - d + 1$ (E) c

Problem 212.4[See solution](#)

In a simple graph, a, b, c, d, e are distinct vertices and the sequence a, b, c, a, d, e, a follows six distinct edges. Under the terminology in this guide, the sequence is which of the following?

- (A) A path but not a trail
- (B) A cycle
- (C) A circuit but not a cycle
- (D) A walk but not a trail
- (E) Neither a walk nor a circuit

Problem 212.5[See solution](#)

A graph has vertices 1, 2, 3, 4 and edges 12, 23, 34, 41, 13. How many spanning trees does it have?

- (A) 4 (B) 5 (C) 6 (D) 8 (E) 10

7. Additional Topics

Algorithms ▸ Tracing and Termination

213 Algorithms, loops, and invariants

Book pp. 267; 321; 329. [Find in compact glossary](#)

In this entry: algorithm; input; output; assignment; conditional branch; while loop; iteration; termination; deterministic algorithm; loop invariant; Collatz iteration.

Definition and key ideas

An algorithm specifies unambiguous operations on inputs and, for a terminating algorithm, produces its output after finitely many steps. Assignment changes a variable, whereas an equality tests a relation. A while loop tests its condition before each iteration; distinguish the initial input from values explicitly printed after updates. A deterministic algorithm chooses its next step uniquely from its current state.

A loop invariant is a property preserved by each iteration, useful for proving correctness. Termination needs a separate argument, often a nonnegative integer that decreases. For Euclid, the gcd is invariant and the positive remainder decreases. Collatz iteration replaces a positive even integer n by $n/2$ and an odd integer by $3n + 1$. A finite trace verifies only that input; the book's $3x + 1$ example is not a proof of termination for all positive integers.

An input is the initial supplied data; an output is a value returned or printed at a specified point. The symbol $x := E$ assigns the current value of the expression E to x . A conditional branch selects instructions according to a truth test. A loop invariant remains true through updates; a decreasing nonnegative integer (often called a termination measure) proves that infinitely many updates are impossible.

Problem 213.1

[See solution](#)

Initialize $a = 5$, $b = 1$. While $a > 0$, first replace b by $2b + a$, then replace a by $a - 2$. What is the final value of b ?

- (A) 17 (B) 21 (C) 31 (D) 35 (E) 37

Problem 213.2

[See solution](#)

Euclid repeatedly replaces (a, b) by $(b, a \bmod b)$ while $b > 0$. Starting at $(221, 85)$, how many replacements occur?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 213.3[See solution](#)

For positive integer inputs a, b , repeat the following until $a = b$: subtract the smaller value from the larger. Which pair consists of a preserved quantity and a strictly decreasing nonnegative quantity, respectively?

- (A) $(a + b, \gcd(a, b))$ (B) $(\gcd(a, b), a + b)$
(C) $(ab, |a - b|)$ (D) $(a - b, ab)$
(E) $(\min(a, b), \max(a, b))$

Problem 213.4[See solution](#)

Start the Collatz loop at $n = 6$. While $n > 1$, replace n by $n/2$ if even and by $3n + 1$ if odd, and then print the new value. How many values are printed?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

Problem 213.5[See solution](#)

An integer x is input. While $x \neq 0$, replace x by $1 - x$. For exactly which inputs does this procedure terminate?

- (A) All integers
(B) All nonnegative integers
(C) Only 0 and 1
(D) Only even integers
(E) Only integers at most one

7. Additional Topics

Combinatorics ▸ The Multiplication Principle

214 Sum and product counting principles

Book pp. 267–268. [Find in compact glossary](#)

In this entry: combinatorics; addition principle; multiplication principle; fundamental counting principle; complementary counting.

Definition and key ideas

The addition principle counts disjoint alternatives by adding their sizes. The multiplication principle counts sequences of choices when every permitted partial choice has the prescribed number of continuations: multiply the numbers at each stage. Probabilistic independence is not required; choosing without replacement is a standard product count.

Restrictions often suggest the best order of decisions. Choose the most constrained positions first, even if they occur later in the final object. Count a complement when the prohibited class is simpler than the desired class. Double counting occurs when a description generates the same object more than once, so specify whether objects, positions, and containers are labeled.

Combinatorics studies discrete structures and their enumeration. A useful product-counting technique assigns each underlying object a state, such as membership in the first set only, the second only, or both. If an ordered construction describes each desired unordered result exactly r times, divide by r ; the multiplicity must be constant.

Problem 214.1

[See solution](#)

How many four-digit positive integers have distinct digits, are even, and use only the digits 0, 1, 2, 3, 4, 5?

- (A) 120 (B) 132 (C) 144 (D) 156 (E) 180

Problem 214.2

[See solution](#)

How many strings of length five over $\{A, B, C, D\}$ contain at least one A and at least one B ?

- (A) 390 (B) 480 (C) 570 (D) 600 (E) 625

Problem 214.3

[See solution](#)

How many functions $f : \{1, \dots, 8\} \rightarrow \{1, \dots, 5\}$ have $f(1), f(2), f(3)$ all distinct, with no restriction on the other images?

- (A) 5^8 (B) $5 \cdot 4 \cdot 3 \cdot 5^5$ (C) $\binom{5}{3} 5^5$ (D) $5! 5^3$ (E) $\binom{8}{3} 5 \cdot 4 \cdot 3$

Problem 214.4

[See solution](#)

How many ways can two indistinguishable rooks be placed on a 4×5 board so that they share neither a row nor a column?

- (A) 60 (B) 80 (C) 100 (D) 120 (E) 240

Problem 214.5[See solution](#)

Let A have six elements. How many ordered pairs of subsets (S, T) satisfy $S \cup T = A$ and $S \cap T \neq \emptyset$?

- (A) 2^6 (B) $3^6 - 2^6$ (C) $3^6 - 1$ (D) $4^6 - 3^6$ (E) $2^6(2^6 - 1)$

7. Additional Topics

Combinatorics ▷ Permutations and Combinations

215 Permutations, repeated symbols, and circular orders

Book pp. 268-269; 325; 328. [Find in compact glossary](#)

In this entry: permutation count; factorial; ordered selection; arrangement; permutation with repeated symbols; circular permutation.

Definition and key ideas

An ordered selection of k distinct objects from n has $P(n, k) = n!/(n - k)!$ possibilities. For an integer $n \geq 1$, the factorial is $n! = 1 \cdot 2 \cdots n$, with $0! = 1$. Arrangements of a multiset with n total symbols and multiplicities $m_1, \dots, m_r$ number $n!/(m_1! \cdots m_r!)$, because exchanging identical symbols creates no new arrangement.

If n distinct people sit on a circle and rotations count as the same, fixing one person leaves $(n - 1)!$ orders. Reflections remain distinct unless stated otherwise. Adjacency restrictions can be handled by treating a required adjacent pair as a block; prohibited adjacency is often counted by subtraction. Whether positions are labeled changes the count.

When reflections as well as rotations are identified for distinct people, the circular count is halved for $n \geq 3$, because no reflected seating fixes all distinct labels. Relative-order constraints can instead be handled by symmetry: each ordering of a specified group of k distinct objects occurs in exactly $1/k!$ of all permutations.

Problem 215.1

[See solution](#)

How many distinct arrangements of the letters in BANANA have no two A letters adjacent?

- (A) 6 (B) 10 (C) 12 (D) 18 (E) 24

Problem 215.2

[See solution](#)

Six distinct people sit around a circular table, with rotations identified but reflections distinct. Two specified people must not sit next to each other. How many seatings are possible?

- (A) 48 (B) 60 (C) 72 (D) 96 (E) 120

Problem 215.3

[See solution](#)

How many permutations of $1, 2, \dots, 7$ place 1 before 2 before 3, and also place 4 before 5?

- (A) 210 (B) 420 (C) 630 (D) 840 (E) 2520

Problem 215.4

[See solution](#)

How many different strings of length five can be made using at most three A symbols, two B symbols, and one C symbol?

- (A) 30 (B) 40 (C) 50 (D) 60 (E) 90

Problem 215.5[See solution](#)

Eight distinct people sit around a circular table. Rotations and reflections count as the same seating. Two specified people must sit opposite each other. How many seatings are there?

- (A) 180 (B) 360 (C) 720 (D) 1440 (E) 2520

7. Additional Topics

Combinatorics ▷ Permutations and Combinations

216 Combinations, binomial coefficients, and partitions into teams

Book pp. 269; 322; 326. [Find in compact glossary](#)

In this entry: combination; binomial coefficient; binomial theorem; Pascal identity; multinomial coefficient; labeled groups; unlabeled groups; multinomial theorem; hockey-stick identity.

Definition and key ideas

An unordered k -element selection from n distinct objects has $\binom{n}{k} = n!/[k!(n-k)!]$ possibilities. The binomial theorem is $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$, and Pascal's identity $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ follows by separating selections according to one distinguished object.

Assigning n distinct objects to labeled groups of fixed sizes $n_1, \dots, n_r$ gives the multinomial coefficient $n!/(n_1! \cdots n_r!)$. If equal-sized groups are unlabeled, divide additionally by the factorial of the number of interchangeable groups. Do not divide for labels such as distinct tasks or offices. This distinction resolves ambiguous committee wording in some review questions.

The multinomial theorem extends the binomial theorem: the coefficient of $x_1^{n_1} \cdots x_r^{n_r}$ in $(x_1 + \cdots + x_r)^n$ is $n!/(n_1! \cdots n_r!)$ when $\sum n_i = n$. Summing Pascal's identity gives $\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$, often called the hockey-stick identity.

Problem 216.1

[See solution](#)

In how many ways can eight distinct people be separated into four unlabeled pairs?

- (A) 70 (B) 105 (C) 210 (D) 420 (E) 2520

Problem 216.2

[See solution](#)

What is the coefficient of x^7 in $(1+x)^{10}(1-x)^4$?

- (A) -72 (B) -24 (C) 0 (D) 24 (E) 72

Problem 216.3

[See solution](#)

Eight distinct people are assigned to labeled teams A, B, C of sizes 3, 3, 2, respectively. Two specified people must be together on one of the three-person teams. How many assignments are possible?

- (A) 60 (B) 90 (C) 120 (D) 240 (E) 560

Problem 216.4

[See solution](#)

What is $\sum_{k=3}^8 \binom{k}{3}$?

- (A) 56 (B) 70 (C) 84 (D) 126 (E) 210

Problem 216.5[See solution](#)

What is the coefficient of $x^2y^3z^2$ in $(x - 2y + z)^7$?

(A) -1680 (B) -840 (C) -210 (D) 840 (E) 1680

7. Additional Topics

Combinatorics ▸ With Repetitions Allowed

217 Repetition and stars and bars

Book pp. 270. [Find in compact glossary](#)

In this entry: ordered selection with repetition; combination with repetition; multiset selection; stars and bars; weak composition; positive composition.

Definition and key ideas

Selecting an ordered list of k objects from n types with repetition allowed gives n^k lists. An unordered selection is a multiset, equivalently a tuple of nonnegative multiplicities with sum k . Stars and bars gives $\binom{n+k-1}{k}$ such tuples.

Positive solutions of $x_1 + \cdots + x_r = N$ number $\binom{N-1}{r-1}$ after replacing x_i by $1 + y_i$. Lower bounds are handled by subtracting the bounds. Upper bounds require excluding violations, often by inclusion-exclusion. The formula counts labeled coordinates or boxes; unordered integer partitions are a different problem.

A weak composition of N into r parts is an ordered r -tuple of nonnegative integers summing to N ; a positive composition requires every part to be at least one. If the number of positive parts is unrestricted and $N \geq 1$, there are 2^{N-1} compositions, by choosing separator positions between N ones. Nondecreasing lists correspond bijectively to multisets through their multiplicities.

Problem 217.1

[See solution](#)

How many integer solutions satisfy $x_1 + x_2 + x_3 + x_4 = 12$, $x_1 \geq 2$, $x_2 \geq 1$, $x_3, x_4 \geq 0$?

(A) 165 (B) 180 (C) 210 (D) 220 (E) 286

Problem 217.2

[See solution](#)

How many nonnegative integer triples satisfy $x + y + z = 10$ with $x, y, z \leq 5$?

(A) 15 (B) 18 (C) 21 (D) 24 (E) 30

Problem 217.3

[See solution](#)

A six-character string uses an alphabet of four symbols. Repetition is allowed, positions two and five must agree, and positions one and six must differ. How many strings are possible?

(A) 256 (B) 384 (C) 768 (D) 1024 (E) 3072

Problem 217.4

[See solution](#)

How many ordered representations of 8 as a sum of one or more positive integers are there, allowing any number of summands?

(A) 64 (B) 128 (C) 256 (D) 512 (E) 22

Problem 217.5[See solution](#)

How many integer sequences satisfy $1 \leq a_1 \leq a_2 \leq a_3 \leq a_4 \leq a_5 \leq 5$ and $a_3 = 3$?

- (A) 18 (B) 24 (C) 30 (D) 36 (E) 60

7. Additional Topics

Combinatorics ▸ The Pigeonhole Principle

218 Pigeonhole and averaging arguments

Book pp. 271; 328. [Find in compact glossary](#)

In this entry: pigeonhole principle; generalized pigeonhole principle; residue-class pigeonholes; averaging principle.

Definition and key ideas

If N objects occupy k boxes, some box contains at least $\lceil N/k \rceil$ objects. Equivalently, more than $k(r-1)$ objects force some box to contain at least r . The proof is a counting contradiction: if all boxes contained at most $r-1$, the total could not exceed $k(r-1)$.

The skill is choosing boxes that encode a useful relation, such as residue classes, intervals, or common odd parts of integers. Averaging gives complementary bounds: some box has at most $\lfloor N/k \rfloor$ objects. To show a guarantee is sharp, exhibit a distribution or subset immediately below the threshold that avoids the desired property.

Problem 218.1

[See solution](#)

What is the smallest m such that every m -element subset of $\{1, 2, \dots, 20\}$ contains two numbers, one dividing the other?

- (A) 6 (B) 9 (C) 10 (D) 11 (E) 12

Problem 218.2

[See solution](#)

What is the smallest number of integers that guarantees two have difference divisible by 17?

- (A) 16 (B) 17 (C) 18 (D) 33 (E) 34

Problem 218.3

[See solution](#)

Five points lie in the closed unit square. What is the smallest number d such that every such configuration has two distinct points at distance at most d ?

- (A) $1/4$ (B) $1/2$ (C) $\sqrt{2}/2$ (D) 1 (E) $\sqrt{2}$

Problem 218.4

[See solution](#)

Objects have one of four colors. What is the smallest number of objects that guarantees three disjoint groups of four objects each, with every group monochromatic? The groups may have the same color.

- (A) 16 (B) 17 (C) 20 (D) 21 (E) 24

Problem 218.5

[See solution](#)

Seven real numbers have sum 28. Among all such seven-tuples, what is the largest possible value of the smallest sum of three distinct entries?

- (A) 8 (B) 10 (C) 12 (D) 14 (E) 16

7. Additional Topics

Probability and Statistics ▸ Elementary Probability

219 Equally likely outcomes and geometric probability

Book pp. 271-273. [Find in compact glossary](#)

In this entry: probability; favorable outcomes; equally likely outcomes; sampling without replacement; geometric probability; uniform distribution.

Definition and key ideas

For a finite sample space with equally likely elementary outcomes, $P(A) = |A|/|S|$. The equal-likelihood hypothesis matters: unordered dice sums, for example, are not equally likely. Sampling without replacement changes the available population after each draw and naturally leads to combinations or conditional products.

For a point chosen uniformly from a region, probability is the measure of the favorable subregion divided by the total measure. Two independent uniform coordinates in $[0, 1]$ give uniform area on the unit square. Curved constraints become integration problems; independence and uniformity must be specified rather than inferred from the phrase at random.

Problem 219.1

[See solution](#)

Three numbers are selected uniformly without replacement from $\{1, \dots, 10\}$. What is the probability that their sum is even?

- (A) $1/3$ (B) $5/12$ (C) $1/2$ (D) $7/12$ (E) $2/3$

Problem 219.2

[See solution](#)

Independent random variables X, Y are uniform on $[0, 1]$. What is $P(XY \leq 1/4)$?

- (A) $1/4$ (B) $(1 + \log 2)/4$ (C) $(1 + 2 \log 2)/4$ (D) $(1 + \log 2)/2$ (E) $3/4$

Problem 219.3

[See solution](#)

Independent X, Y are uniform on $[0, 1]$. What is $P(X + Y > 3/2)$?

- (A) $1/16$ (B) $1/8$ (C) $1/4$ (D) $3/8$ (E) $1/2$

Problem 219.4

[See solution](#)

Independent X, Y are uniform on $[0, 1]$. What is $P(|X - Y| < 1/3)$?

- (A) $1/9$ (B) $2/9$ (C) $4/9$ (D) $5/9$ (E) $2/3$

Problem 219.5

[See solution](#)

First choose K uniformly from $\{1, 2, 3\}$. Given $K = k$, choose X uniformly from $\{1, \dots, k\}$. What is $P(X = 1)$?

- (A) $1/3$ (B) $1/2$ (C) $5/9$ (D) $11/18$ (E) $2/3$

7. Additional Topics

Probability and Statistics ▸ Probability Spaces

220 Sample spaces, events, and probability measures

Book pp. 273–274. [Find in compact glossary](#)

In this entry: sample space; outcome; event; algebra of sets; Boolean algebra of sets; sigma-algebra; probability measure; probability space; countable additivity.

Definition and key ideas

A probability space is $(\Omega, \mathcal{F}, P)$, with outcomes $\omega \in \Omega$, an event sigma-algebra $\mathcal{F}$, and a probability measure. A sigma-algebra contains Ω and is closed under complements and countable unions. A Boolean algebra of sets requires only finite unions. The standard measure axioms are nonnegativity, $P(\Omega) = 1$, and countable additivity on disjoint events.

The book states a finite-additivity version; for general infinite spaces the standard countable form is needed. It yields $P(\emptyset) = 0$, $P(A^c) = 1 - P(A)$, monotonicity, and continuity under increasing unions or decreasing intersections. A probability-zero event need not be empty, and a probability-one event need not contain every outcome.

A finite event algebra can be described by its atoms: nonempty events having no smaller nonempty event in the algebra. Every event is a union of these disjoint atoms. In an infinite space, closure under finite operations does not imply closure under countable unions. Countable additivity means $P(\bigcup_n A_n) = \sum_n P(A_n)$ for pairwise disjoint measurable events, including when each singleton has probability zero.

Problem 220.1

[See solution](#)

Events $A_1 \supseteq A_2 \supseteq \cdots$ satisfy $P(A_n) = 1/3 + 1/(n+3)$. What is $P(\bigcap_{n=1}^{\infty} A_n)$?

- (A) 0 (B) 1/4 (C) 1/3 (D) 7/12 (E) 1

Problem 220.2

[See solution](#)

A finite sample space is partitioned into four nonempty blocks. How many events belong to the sigma-algebra consisting of all unions of these blocks?

- (A) 4 (B) 8 (C) 12 (D) 16 (E) 24

Problem 220.3[See solution](#)

On $\mathbb{N}$, let $\mathcal{A}$ consist of finite sets and sets with finite complement. Assign $P(A) = 0$ to finite sets and $P(A) = 1$ to sets with finite complement. Which statement is correct?

- (A) $\mathcal{A}$ is a sigma-algebra and P is countably additive.
- (B) $\mathcal{A}$ is an algebra but not a sigma-algebra; P is finitely additive but fails countable additivity.
- (C) $\mathcal{A}$ is not closed under complements.
- (D) $P(\mathbb{N}) = 0$
- (E) P fails finite additivity on disjoint members of $\mathcal{A}$.

Problem 220.4[See solution](#)

A point is chosen uniformly from $[0, 1]$. For $A = \mathbb{Q} \cap [0, 1]$ and $B = [0, 1] \setminus \mathbb{Q}$, which statement is correct?

- (A) $P(A) = P(B) = 1/2$
- (B) $P(A) = 0$ and $A = \emptyset$
- (C) $P(A) = 0$, $P(B) = 1$, and both events are nonempty
- (D) $P(A) > 0$ because A is infinite
- (E) $P(B) = 1$ implies $B = [0, 1]$

Problem 220.5[See solution](#)

A probability measure on the positive integers assigns mass $c/[n(n+1)]$ to outcome n , where c is a constant. What is $P(\{4, 5, 6, \dots\})$?

- (A) $1/5$ (B) $1/4$ (C) $1/3$ (D) $3/4$ (E) $4/5$

7. Additional Topics

Probability and Statistics ▷ Probability Spaces

221 Independence, exclusion, and inclusion-exclusion

Book pp. 274-275; 322; 327. [Find in compact glossary](#)

In this entry: independent events; mutually independent events; pairwise independence; mutually exclusive events; inclusion-exclusion; conditional probability; Bayes theorem; law of total probability.

Definition and key ideas

Events A, B are independent when $P(A \cap B) = P(A)P(B)$. Mutually exclusive events have empty intersection; positive-probability mutually exclusive events are not independent. Always $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Events are mutually independent when every finite subcollection satisfies $P(A_{i_1} \cap \cdots \cap A_{i_k}) = \prod_{j=1}^k P(A_{i_j})$. For three or more events, pairwise independence need not imply mutual independence.

Conditional probability, a useful extension, is $P(A | B) = P(A \cap B)/P(B)$ when $P(B) > 0$. Bayes' formula is $P(A | B) = P(B | A)P(A)/P(B)$ when both conditioning probabilities are positive; it follows by expressing the intersection in either order. Independence is preserved when either event is complemented. If no independence or joint-distribution information is supplied, marginal probabilities alone generally do not determine a union probability.

For a finite partition $H_1, \dots, H_r$ of the sample space, $P(B) = \sum_i P(B | H_i)P(H_i)$, omitting zero-probability partition members. This gives the denominator in Bayes computations. Without independence, $\max(0, P(A) + P(B) - 1) \leq P(A \cap B) \leq \min(P(A), P(B))$; these are the sharp bounds obtained from the four disjoint membership regions.

Problem 221.1

[See solution](#)

Events A, B are independent with $P(A) = 2/5$ and $P(B) = 3/4$. What is $P(A | A \cup B)$?

- (A) $4/17$ (B) $6/17$ (C) $8/17$ (D) $1/2$ (E) $3/5$

Problem 221.2

[See solution](#)

Events A, B, C are pairwise independent, each of probability $1/2$, and $P(A \cap B \cap C) = 0$. What is $P(A \cup B \cup C)$?

- (A) $1/2$ (B) $5/8$ (C) $3/4$ (D) $7/8$ (E) 1

Problem 221.3

[See solution](#)

Independent events A, B have probabilities $\frac{1}{3}$ and $\frac{3}{5}$. What is the probability that exactly one occurs?

- (A) $2/15$ (B) $4/15$ (C) $8/15$ (D) $3/5$ (E) $14/15$

Problem 221.4[See solution](#)

Bag A contains four red balls and one blue ball; bag B contains two red and three blue. Select bag A with probability $1/3$ and bag B with probability $2/3$, then draw two balls without replacement from that bag. Given that both are red, what is the probability that bag A was chosen?

- (A) $1/3$ (B) $1/2$ (C) $3/5$ (D) $2/3$ (E) $3/4$

Problem 221.5[See solution](#)

Events A, B are independent and satisfy $A \cap B = \emptyset$. Which conclusion must hold?

- (A) $A = \emptyset$
(B) $B = \emptyset$
(C) At least one of A, B is empty.
(D) $P(A) = 0$ or $P(B) = 0$
(E) $P(A) + P(B) = 1$

Problem 221.6[See solution](#)

Only $P(A) = 3/5$ and $P(B) = 3/4$ are known. What is the full possible range of $P(A | B)$?

- (A) $[0, 1]$ (B) $[7/15, 4/5]$ (C) $[3/5, 3/4]$ (D) $[1/5, 4/5]$ (E) $[7/20, 3/5]$

7. Additional Topics

Probability and Statistics ▸ Bernoulli Trials

222 Bernoulli trials and the binomial distribution

Book pp. 276-277; 282. [Find in compact glossary](#)

In this entry: Bernoulli trial; success probability; failure probability; binomial distribution; binomial probability mass function; binomial mode.

Definition and key ideas

A Bernoulli random variable takes values 1 and 0 with probabilities p and $1 - p$. The sum of n independent Bernoulli variables with common p is binomial: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$. Its mean is np and variance $np(1 - p)$. Independence and a fixed success probability are essential hypotheses.

For $0 < p < 1$, comparing successive masses gives $P(X = k + 1)/P(X = k) = [(n - k)/(k + 1)][p/(1 - p)]$. Thus the unique mode is $\lfloor (n + 1)p \rfloor$ unless $(n + 1)p$ is an integer, in which case that integer and the preceding integer tie. Sampling without replacement is generally hypergeometric rather than binomial.

The probability of an even binomial count is $P(X \text{ even}) = [1 + (1 - 2p)^n]/2$, obtained by adding the binomial expansions with signs $+p$ and $-p$. A sum of independent Bernoulli variables with unequal success probabilities generally does not have a binomial distribution. At the boundary cases $p = 0$ and $p = 1$, the binomial law is concentrated at zero and n , respectively.

Problem 222.1

[See solution](#)

For $X \sim \text{Bin}(8, p)$ with $0 < p < 1$, suppose $P(X = 3) = P(X = 4)$. What is p ?

- (A) $1/3$ (B) $2/5$ (C) $4/9$ (D) $1/2$ (E) $5/9$

Problem 222.2

[See solution](#)

A fair coin is tossed eight times. Given that exactly three tosses are heads, what is the probability that no two heads are consecutive?

- (A) $1/4$ (B) $5/14$ (C) $3/7$ (D) $1/2$ (E) $5/7$

Problem 222.3

[See solution](#)

If $X \sim \text{Bin}(4, \frac{1}{3})$, what is the probability that X is even?

- (A) $40/81$ (B) $1/2$ (C) $41/81$ (D) $2/3$ (E) $16/27$

Problem 222.4

[See solution](#)

Independent variables X, Y are Bernoulli with success probabilities $p, q \in (0, 1)$, respectively. If $X + Y$ has a $\text{Bin}(2, r)$ distribution, which condition is necessary?

- (A) $p + q = 1$ (B) $p = q = r$ (C) $r = pq$ (D) $r = p + q$ (E) $p = 1 - q = r$

Problem 222.5[See solution](#)

A binomial random variable X has mean 4 and variance 3. What is $P(X = 1)$?

- (A) $(3/4)^{16}$ (B) $4(3/4)^{15}$ (C) $3(1/4)^{15}$ (D) $16(3/4)^{15}$ (E) $4(1/4)^{15}$

7. Additional Topics

Probability and Statistics ▸ Random Variables

223 Random variables and cumulative distribution functions

Book pp. 277-278. [Find in compact glossary](#)

In this entry: random variable; measurability; distribution function; cumulative distribution function; CDF; discrete random variable; probability mass; atom; right continuity.

Definition and key ideas

A real random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$; measurability ensures that $\{X \leq t\}$ is an event. Its cumulative distribution function is $F_X(t) = P(X \leq t)$. Every CDF is nondecreasing, right-continuous, and tends to 0 and 1 at the two infinite endpoints.

For $a < b$, $P(a < X \leq b) = F(b) - F(a)$. The mass at a is the jump $P(X = a) = F(a) - F(a^-)$, so $P(a \leq X \leq b) = F(b) - F(a^-)$. A discrete random variable is supported on a finite or countable set of values, whose probabilities sum to one. A continuous CDF has no atoms; this does not by itself imply that a density exists. Discrete, continuous, and mixed distributions demand careful endpoint treatment.

Here $F(a^-) = \lim_{t \uparrow a} F(t) = P(X < a)$. An atom is a point with positive probability. A mixed law can have both atoms and an absolutely continuous part. Measurability means that inverse images of open subsets of $\mathbb{R}$ are events; equivalently, it is enough to check all sets $\{X \leq t\}$. A continuous CDF is atomless, but absolute continuity is the stronger condition required for a density.

Problem 223.1

[See solution](#)

A random variable has $F(t) = 0$ for $t < 0$, $F(t) = 1/4$ for $0 \leq t < 1$, $F(t) = 3/4$ for $1 \leq t < 2$, and $F(t) = 1$ for $t \geq 2$. What is $P(0 < X < 2 \mid X \geq 1)$?

- (A) $1/4$ (B) $1/2$ (C) $2/3$ (D) $3/4$ (E) 1

Problem 223.2

[See solution](#)

If X has CDF F and $P(X = 2) = 1/5$, while $F(2) = 7/10$ and $F(5) = 9/10$, what is $P(2 \leq X \leq 5)$?

- (A) $1/5$ (B) $3/10$ (C) $2/5$ (D) $1/2$ (E) $7/10$

Problem 223.3

[See solution](#)

A function F is zero for $t < 0$, equals c at $t = 0$, equals $(1+t)/2$ for $0 < t < 1$, and equals one for $t \geq 1$. For which c is F a cumulative distribution function?

- (A) 0 only (B) $1/2$ only (C) 1 only (D) Every $c \in [0, 1/2]$ (E) Every $c \in [0, 1]$

Problem 223.4[See solution](#)

If X has CDF F and $Y = -X$, which formula is the CDF of Y , valid even when X has atoms?

- (A) $F(-t)$ (B) $1 - F(-t)$ (C) $1 - F((-t)^-)$ (D) $F(t^-)$ (E) $1 - F(t)$

Problem 223.5[See solution](#)

Let $\Omega = \{1, 2, 3, 4\}$ and $\mathcal{F} = \{\emptyset, \{1, 2\}, \{3, 4\}, \Omega\}$. Which tuple $(X(1), X(2), X(3), X(4))$ defines a measurable real random variable on this event space?

- (A) $(0, 1, 0, 1)$ (B) $(0, 0, 1, 1)$ (C) $(0, 1, 1, 0)$ (D) $(1, 2, 3, 4)$ (E) $(0, 0, 0, 1)$

Problem 223.6[See solution](#)

A real random variable has a CDF that is continuous everywhere. Which conclusion must hold?

- (A) It has a density with respect to length.
(B) It has a finite expectation.
(C) $P(X = a) = 0$ for every real a .
(D) Its CDF is strictly increasing.
(E) Its support is an interval.

7. Additional Topics

Probability and Statistics ▸ Random Variables

224 Densities and transformations of continuous variables

Book pp. 278. [Find in compact glossary](#)

In this entry: probability density function; absolutely continuous distribution; density normalization; exponential distribution; survival probability; transformed random variable.

Definition and key ideas

A density f is nonnegative and integrates to one, with $P(a < X \leq b) = \int_a^b f(x) dx$. A distribution admitting such an f is absolutely continuous, and its CDF is $F(t) = \int_{-\infty}^t f(x) dx$. The density can exceed one; it is interval integrals, not pointwise heights, that are probabilities.

To transform a random variable, often compute the new CDF directly: $P(g(X) \leq y)$. For a differentiable monotone transformation, the density gains the absolute derivative of the inverse transformation. Exponential density $\lambda e^{-\lambda x}$ on $x \geq 0$ has survival probability $P(X > t) = e^{-\lambda t}$ and the memoryless property $P(X > s + t | X > s) = e^{-\lambda t}$.

For a differentiable one-to-one transformation $Y = g(X)$, $f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$ on the transformed support. For finitely many smooth inverse branches, add their contributions. Exponential formulas assume $\lambda > 0$ and nonnegative times. Independent exponential waiting times have an exponential minimum whose rate is the sum of their rates.

Problem 224.1

[See solution](#)

Let X have density $f(x) = 2x$ on $(0, 1)$ and zero elsewhere. What is the density of $Y = X^2$ on $(0, 1)$?

- (A) 1 (B) $2y$ (C) $1/(2\sqrt{y})$ (D) $2\sqrt{y}$ (E) y^{-1}

Problem 224.2

[See solution](#)

An exponential random variable X satisfies $P(X > 3) = 1/8$. What is $P(X > 5 | X > 2)$?

- (A) $1/32$ (B) $1/16$ (C) $1/8$ (D) $1/4$ (E) $1/2$

Problem 224.3

[See solution](#)

A density is $f(x) = c(1 - 2|x|)$ for $|x| < 1/2$ and zero elsewhere. What are c and $P(|X| \leq 1/4)$, respectively?

- (A) $(1, 3/8)$ (B) $(2, 1/2)$ (C) $(2, 3/4)$ (D) $(4, 3/4)$ (E) $(1, 3/4)$

Problem 224.4[See solution](#)

Let X be uniform on $(-2, 1)$ and $Y = X^2$. What is the density of Y , away from its endpoints?

- (A) $1/(6\sqrt{y})$ for $0 < y < 4$, and zero elsewhere
- (B) $1/(3\sqrt{y})$ for $0 < y < 4$, and zero elsewhere
- (C) $1/(3\sqrt{y})$ for $0 < y < 1$, $1/(6\sqrt{y})$ for $1 < y < 4$, and zero elsewhere
- (D) $1/(6\sqrt{y})$ for $0 < y < 1$, $1/(3\sqrt{y})$ for $1 < y < 4$, and zero elsewhere
- (E) $1/4$ for $0 < y < 4$, and zero elsewhere

Problem 224.5[See solution](#)

Independent exponential random variables X, Y have rates 2 and 3. What is the density of $T = \min(X, Y)$ for $t > 0$?

- (A) e^{-5t} (B) $5e^{-5t}$ (C) $2e^{-2t} + 3e^{-3t}$ (D) $6e^{-6t}$ (E) $6(e^{-2t} - e^{-3t})$

7. Additional Topics

Probability and Statistics ▸ Expectation, Variance, and Standard Deviation

225 Expectation, indicators, and moments

Book pp. 279. [Find in compact glossary](#)

In this entry: expectation; mean; expected value; linearity of expectation; indicator random variable; moment; law of the unconscious statistician.

Definition and key ideas

For an integrable random variable, its mean or expectation is $E[X] = \int x f(x) dx$ when a density exists, or $\sum_x x P(X = x)$ in the discrete case. More generally $E[g(X)]$ uses $g(x)$ in that integral or sum. The k th raw moment is $E[X^k]$, when it exists. Integrability means $E|X| < \infty$; an arbitrary random variable need not have a finite mean.

Linearity $E[aX + bY] = aE[X] + bE[Y]$ does not require independence. Indicators satisfy $E[1_A] = P(A)$, so writing a count as a sum of indicators is often the fastest method. An expected value need not be one of the variable's possible values. Nonlinear transformations do not commute with expectation in general.

For $g(X)$, the law of the unconscious statistician uses the distribution of X directly: $E[g(X)] = \sum_x g(x)P(X = x)$ or $\int g(x)f_X(x)dx$, when the expectation exists. A signed expectation is undefined if its positive and negative parts both have infinite means; canceling divergent tails by symmetry is not legitimate. For a binomial count, falling products such as $X(X - 1)$ count ordered distinct successes, yielding $E[X(X - 1)] = n(n - 1)p^2$.

Problem 225.1

[See solution](#)

A uniformly random permutation of $\{1, \dots, 20\}$ is selected. What is the expected number of fixed points?

- (A) $1/20$ (B) $1/2$ (C) 1 (D) 2 (E) 10

Problem 225.2

[See solution](#)

Let X have density $3x^2$ on $(0, 1)$. What is $E[1/X]$?

- (A) 1 (B) $4/3$ (C) $3/2$ (D) 2 (E) Infinite

Problem 225.3

[See solution](#)

Independently place 5 labeled balls into 3 boxes, each box equally likely for each ball. What is the expected number of empty boxes?

- (A) $32/243$ (B) $32/81$ (C) $1/81$ (D) $211/81$ (E) $160/243$

Problem 225.4[See solution](#)

Let X have density $f(x) = 1/[\pi(1+x^2)]$ on $\mathbb{R}$. Which statement about its expectation is correct?

- (A) $E[X] = 0$ by symmetry.
- (B) $E[X] = 1/\pi$
- (C) $E[X] = +\infty$
- (D) $E[X] = -\infty$
- (E) $E[X]$ is undefined because its positive and negative parts both have infinite expectation.

Problem 225.5[See solution](#)

Let X count successes in three independent Bernoulli trials, each with success probability $1/3$. What is the third raw moment $E[X^3]$?

- (A) 1 (B) $5/3$ (C) $7/3$ (D) $29/9$ (E) 9

7. Additional Topics

Probability and Statistics ▸ Expectation, Variance, and Standard Deviation

226 Variance, standard deviation, covariance, and correlation

Book pp. 279-280. [Find in compact glossary](#)

In this entry: variance; standard deviation; second moment; covariance; correlation; variance under scaling.

Definition and key ideas

For a variable with finite second moment, $\text{Var}(X) = E[(X - EX)^2] = E[X^2] - (EX)^2$. Standard deviation is its nonnegative square root. Translation leaves variance unchanged and scaling gives $\text{Var}(aX + b) = a^2 \text{Var}(X)$, so standard deviation scales by $|a|$.

A useful extension is covariance $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$. Then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$. Independence implies zero covariance when moments exist, but the converse need not hold. Correlation divides covariance by the two positive standard deviations and lies between -1 and 1 .

For finite, strictly positive variances, the correlation coefficient is $\rho_{X,Y} = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$. It measures the direction and strength of linear association, is unchanged by translations, and satisfies $\rho_{aX+b, cY+d} = \text{sgn}(ac)\rho_{X,Y}$ for nonzero a, c . It is undefined when either standard deviation is zero. The Cauchy–Schwarz inequality gives $|\rho| \leq 1$. Equality holds precisely when $Y = \alpha X + \beta$ almost surely for a nonzero α ; the sign of ρ is the sign of α . The phrase almost surely means with probability one. Zero correlation does not exclude a nonlinear dependence.

For constants a, b , $\text{Var}(aX + bY) = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\text{Cov}(X, Y)$. Completing the square shows that $\text{Var}(X - tY)$ is minimized at $t = \text{Cov}(X, Y)/\sigma_Y^2$ when $\sigma_Y^2 > 0$, with minimum $\sigma_X^2 - \text{Cov}(X, Y)^2/\sigma_Y^2$. When $\sigma_X^2 > 0$ as well, this equals $\sigma_X^2(1 - \rho^2)$. This links optimal linear prediction to correlation.

Problem 226.1

[See solution](#)

Random variables satisfy $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, and $\text{Cov}(X, Y) = 2$. What is $\text{Var}(3X - 2Y)$?

- (A) 24 (B) 36 (C) 48 (D) 60 (E) 96

Problem 226.2

[See solution](#)

Let X have density $2x$ on $(0, 1)$. What is its variance?

- (A) $1/36$ (B) $1/18$ (C) $1/12$ (D) $1/9$ (E) $1/6$

Problem 226.3

[See solution](#)

Random variables have $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, and $\text{Cov}(X, Y) = -3$. What is the correlation of $U = 5 - 2X$ and $V = 1 + 3Y$?

- (A) $-1/2$ (B) $-1/6$ (C) $1/6$ (D) $1/2$ (E) 1

Problem 226.4[See solution](#)

Let X take values $-1, 0, 1$ with equal probabilities, and let $Y = X^2$. Which statement is correct?

- (A) X, Y are independent and have correlation 0.
- (B) X, Y are dependent and have correlation 0.
- (C) X, Y have correlation 1.
- (D) X, Y have correlation -1 .
- (E) The correlation is undefined because Y is a function of X .

Problem 226.5[See solution](#)

If $\text{Var}(X) = 9$, $\text{Var}(Y) = 4$, and $\text{Cov}(X, Y) = 3$, which pair gives the minimizing t and minimum value of $\text{Var}(X - tY)$?

- (A) $(3/4, 27/4)$ (B) $(3/2, 9)$ (C) $(3/4, 9/4)$ (D) $(-3/4, 27/4)$ (E) $(1/3, 7)$

Problem 226.6[See solution](#)

Suppose $E[X] = 1$, $\sigma_X = 2$, $E[Y] = -3$, $\sigma_Y = 6$, and $\rho_{X,Y} = -1$. Which statement correctly combines the resulting relation with the definition for a constant variable C ?

- (A) $Y = -3X$ almost surely; $\rho_{C,Y} = 0$.
- (B) $Y = -3X$ almost surely; $\rho_{C,Y}$ is undefined.
- (C) $Y = -3X - 6$ almost surely; $\rho_{C,Y}$ is undefined.
- (D) $Y = 3X - 6$ almost surely; $\rho_{C,Y} = 0$.
- (E) No almost-sure relation follows; $\rho_{C,Y}$ is undefined.

7. Additional Topics

Probability and Statistics ▸ The Normal Distribution

227 Normal distributions and standardization

Book pp. 280-281. [Find in compact glossary](#)**In this entry:** normal distribution; Gaussian distribution; normal density; standard normal distribution; z-score; standardization; standard normal CDF.**Definition and key ideas**

A normal variable $X \sim N(\mu, \sigma^2)$ with $\sigma > 0$ has density $[\sigma\sqrt{2\pi}]^{-1} \exp[-(x - \mu)^2/(2\sigma^2)]$. Its mean is μ and variance σ^2 . Standardizing gives $Z = (X - \mu)/\sigma \sim N(0, 1)$, with CDF Φ .

Interval probabilities become $\Phi((b - \mu)/\sigma) - \Phi((a - \mu)/\sigma)$, and symmetry gives $\Phi(-z) = 1 - \Phi(z)$. The second parameter in $N(\mu, \sigma^2)$ is variance, not standard deviation. Linear combinations of independent normal variables are normal, with means combining linearly and variances adding after the appropriate squared coefficients.

The standard normal CDF is $\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt$. Useful values are $\Phi(0) = 1/2$, $\Phi(1) \approx 0.8413$, $\Phi(2) \approx 0.9772$, and $\Phi(3) \approx 0.99865$. Because the distribution has a density, single points have probability zero, so strict versus nonstrict endpoints do not change its probabilities. The density is symmetric about μ and decreases with distance from μ ; among intervals of a fixed length, the interval centered at μ has largest probability.

Problem 227.1[See solution](#)

Independent variables satisfy $X \sim N(2, 9)$ and $Y \sim N(-1, 4)$. In terms of the standard normal CDF, what is $P(X + 2Y > 5)$?

- (A) $1 - \Phi(1)$ (B) $1 - \Phi(5/7)$ (C) $1 - \Phi(5/\sqrt{17})$ (D) $\Phi(1)$ (E) $1 - \Phi(5/13)$

Problem 227.2[See solution](#)

For $X \sim N(\mu, \sigma^2)$ with $\sigma > 0$, suppose $P(X \leq 4) = \Phi(-1)$ and $P(X \leq 10) = \Phi(2)$. What is $\mu + \sigma$?

- (A) 6 (B) 7 (C) 8 (D) 9 (E) 10

Problem 227.3[See solution](#)

A random variable has density $f(x) = Ce^{-2x^2+8x}$ on $\mathbb{R}$, where $C > 0$. Which pair gives C and $E[X^2]$?

- (A) $(e^{-8}\sqrt{2/\pi}, 17/4)$ (B) $(e^8\sqrt{2/\pi}, 17/4)$
 (C) $(e^{-8}/\sqrt{2\pi}, 5)$ (D) $(e^{-8}\sqrt{2/\pi}, 9/2)$
 (E) $(e^{-8}\sqrt{2/\pi}, 4)$

Problem 227.4[See solution](#)

For a standard normal variable Z , what is $P(|Z| > 2 \mid |Z| > 1)$?

- (A) $1 - \Phi(2)$ (B) $2[1 - \Phi(2)]$ (C) $\frac{1 - \Phi(2)}{1 - \Phi(1)}$ (D) $\frac{\Phi(2) - \Phi(1)}{1 - \Phi(1)}$ (E) $\frac{1 - \Phi(1)}{1 - \Phi(2)}$

Problem 227.5[See solution](#)

Let $X \sim N(3, 4)$, with the second parameter denoting variance. Which pair gives the real a maximizing $P(a \leq X \leq a + 4)$ and that maximum probability?

- (A) $(1, 2\Phi(1) - 1)$ (B) $(3, \Phi(2) - 1/2)$
(C) $(1, 2\Phi(2) - 1)$ (D) $(-1, 2\Phi(1) - 1)$
(E) $(3, 2\Phi(1) - 1)$

7. Additional Topics

Probability and Statistics ▷ The Normal Approximation to the Binomial Distribution

228 Normal approximation and continuity correction

Book pp. 282. [Find in compact glossary](#)

In this entry: normal approximation to binomial; continuity correction; binomial standard deviation; approximation regime.

Definition and key ideas

For $X \sim \text{Bin}(n, p)$, a normal approximation uses mean np and variance $np(1 - p)$. It is useful when both expected successes and failures are reasonably large; extreme tails or very small p may require greater care or a different approximation. The result is approximate, not an exact equality.

Continuity correction matches each integer mass to a unit interval. Thus $P(a \leq X \leq b)$ is approximated by $\Phi((b + 1/2 - np)/\sigma) - \Phi((a - 1/2 - np)/\sigma)$ with $\sigma = \sqrt{np(1 - p)}$. For $P(X \geq k)$ the corrected boundary is $k - 1/2$, while $P(X > k)$ uses $k + 1/2$.

For a sequence of binomial models, a normal limit for the standardized count holds when both np and $n(1 - p)$ tend to infinity. For fixed $0 < p < 1$, the sample proportion $\hat{p} = X/n$ has standard deviation $\sqrt{p(1 - p)/n}$. A normal-design margin of error d at a two-sided standard-normal cutoff z uses $n \geq z^2 p(1 - p)/d^2$; when p is unknown, the bound $p(1 - p) \leq 1/4$ gives a conservative design rule. This is an approximation-based calculation, not an exact finite-sample probability guarantee.

Problem 228.1

[See solution](#)

For $X \sim \text{Bin}(100, 1/2)$, which is the continuity-corrected normal approximation to $P(X \geq 60)$?

- (A) $1 - \Phi(2.1)$ (B) $1 - \Phi(2)$ (C) $1 - \Phi(1.9)$ (D) $\Phi(1.9)$ (E) $\Phi(2.1) - \Phi(1.9)$

Problem 228.2

[See solution](#)

For $X \sim \text{Bin}(400, 1/4)$, which is the continuity-corrected normal approximation to $P(90 \leq X < 110)$?

- (A) $\Phi(10/\sqrt{75}) - \Phi(-10/\sqrt{75})$
 (B) $\Phi(9.5/\sqrt{75}) - \Phi(-10.5/\sqrt{75})$
 (C) $\Phi(10.5/\sqrt{75}) - \Phi(-9.5/\sqrt{75})$
 (D) $\Phi(9.5/75) - \Phi(-10.5/75)$
 (E) $\Phi(10.5/\sqrt{75}) - \Phi(-10.5/\sqrt{75})$

Problem 228.3[See solution](#)

For fixed $\alpha > 0$, let $X_n \sim \text{Bin}(n, n^{-\alpha})$. For which α do both the expected number of successes and the expected number of failures tend to infinity, supporting a standardized normal limit?

- (A) $0 < \alpha < 1$ (B) $\alpha = 1$ only (C) $\alpha > 1$ (D) $0 < \alpha \leq 1$ (E) Every $\alpha > 0$

Problem 228.4[See solution](#)

If $X_n \sim \text{Bin}(n, 1/4)$, what is $\lim_{n \rightarrow \infty} P(|X_n/n - 1/4| \leq \sqrt{3}/(2\sqrt{n}))$?

- (A) $\Phi(2)$ (B) $2\Phi(1) - 1$ (C) $2\Phi(2) - 1$ (D) $1 - \Phi(2)$ (E) 1

Problem 228.5[See solution](#)

Using a normal approximation without continuity correction, a proportion estimate is to have two-sided margin 0.01 at cutoff $z = 1.96$. With the true proportion unknown, use the conservative variance bound $p(1 - p) \leq 1/4$. What minimum integer sample size does this design rule prescribe?

- (A) 2401 (B) 2500 (C) 4802 (D) 9604 (E) 38416

7. Additional Topics

Point-Set Topology ▷ Introduction

229 Topology, open and closed sets, and neighborhoods

Book pp. 283-284. [Find in compact glossary](#)**In this entry:** topology; topological space; open set; closed set; neighborhood; standard topology.**Definition and key ideas**

A topology $\mathcal{T}$ on X contains $\emptyset$ and X , is closed under arbitrary unions, and is closed under finite intersections. Its members are open; a set is closed when its complement is open. A neighborhood of x contains an open set containing x and need not itself be open. In the standard topology on $\mathbb{R}$, an open set contains a small interval about each of its points.

Arbitrary intersections of closed sets and finite unions of closed sets are closed. A set can be both open and closed, or neither. To find the smallest topology containing specified sets, repeatedly add their finite intersections and arbitrary unions; on a finite underlying set this process terminates.

Problem 229.1[See solution](#)

Let $X = \{a, b, c, d\}$. How many sets belong to the smallest topology on X containing $\{a, b\}$ and $\{b, c\}$?

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

Problem 229.2[See solution](#)

For two topologies $\mathcal{T}_1, \mathcal{T}_2$ on the same set, which must be true?

- (A) $\mathcal{T}_1 \cap \mathcal{T}_2$ and $\mathcal{T}_1 \cup \mathcal{T}_2$ are topologies.
(B) Only their intersection is necessarily a topology.
(C) Only their union is necessarily a topology.
(D) Neither is necessarily a topology.
(E) Their union is a topology only if they are equal.

Problem 229.3[See solution](#)

Let $X = \{a, b, c\}$ with topology $\{\emptyset, \{a\}, \{a, b\}, X\}$. What is the closure of $\{b\}$?

- (A) $\{b\}$ (B) X (C) $\{b, c\}$ (D) $\{c\}$ (E) $\{a, b\}$

Problem 229.4[See solution](#)

In the usual topology on $\mathbb{R}$, let $N = [-1, 1] \cup \{2\}$. For exactly which real x is N a neighborhood of x ?

- (A) $x \in N$ (B) $x \in [-1, 1]$ (C) $x \in (-1, 1)$ (D) $x \in \{-1, 1, 2\}$ (E) No real x

Problem 229.5[See solution](#)

Let F_n be closed subsets and U_n open subsets of $\mathbb{R}$, for $n \geq 1$. Which assertions always hold?

I. $\bigcap_n F_n$ is closed.

II. $\bigcup_n F_n$ is closed.

III. $\bigcap_n U_n$ is open.

(A) I only (B) II only (C) I and II only (D) I and III only (E) All three

7. Additional Topics

Point-Set Topology ▷ Introduction

230 Hausdorff separation

Book pp. 283. [Find in compact glossary](#)

In this entry: Hausdorff space; separation of points; uniqueness of limits; T_1 separation; cofinite topology; cocountable topology.

Definition and key ideas

A space is Hausdorff if distinct points have disjoint open neighborhoods. Every metric space is Hausdorff. Hausdorff spaces have closed singletons and unique limits of convergent sequences, though those latter properties alone should not be substituted for the definition in arbitrary spaces.

Every compact subset of a Hausdorff space is closed. A finite Hausdorff space is discrete: for each point, intersect finitely many open neighborhoods that exclude each other point. The Hausdorff hypothesis is essential in many compactness and homeomorphism statements.

A T_1 space is one in which every singleton is closed. The cofinite topology consists of the empty set and sets with finite complement; the cocountable topology replaces finite by countable.

On an infinite cofinite space, disjoint nonempty open sets do not exist. On an uncountable cocountable space the same obstruction holds, although every convergent sequence is eventually constant.

Problem 230.1

[See solution](#)

A Hausdorff topology is placed on a set of five points. How many open sets does it have?

- (A) 5 (B) 6 (C) 10 (D) 16 (E) 32

Problem 230.2

[See solution](#)

Let X be Hausdorff, let $K \subseteq X$ be compact, and suppose (x_n) in K converges in X to x . Which must hold?

- (A) x is isolated in K . (B) K is finite. (C) $x \in K$. (D) K is open. (E) X is compact.

Problem 230.3

[See solution](#)

An infinite set has the cofinite topology. Which is true?

- (A) It is discrete
(B) Every singleton is open
(C) It is not compact
(D) It is Hausdorff but not T_1
(E) It is T_1 but not Hausdorff

Problem 230.4[See solution](#)

A convergent sequence in a Hausdorff space has

- (A) An open range
- (B) At most one limit
- (C) An eventually constant tail
- (D) Only finitely many distinct terms
- (E) Exactly one isolated point in its range

Problem 230.5[See solution](#)

An uncountable set X has the topology consisting of $\emptyset$ and sets with countable complement. Which statement is correct?

- (A) X is Hausdorff.
- (B) Every sequence converges to every point.
- (C) X is not Hausdorff, but every convergent sequence has a unique limit.
- (D) Every singleton is open.
- (E) Some sequence has two distinct limits.

7. Additional Topics

Point-Set Topology ▷ Introduction

231 Discrete, indiscrete, finer, and coarser topologies

Book pp. 283. [Find in compact glossary](#)**In this entry:** discrete topology; indiscrete topology; trivial topology; finer topology; coarser topology.**Definition and key ideas**

The discrete topology is the power set $\mathcal{P}(X)$; the indiscrete topology is $\{\emptyset, X\}$. A topology $\mathcal{T}_f$ is finer than $\mathcal{T}_c$ when $\mathcal{T}_c \subseteq \mathcal{T}_f$. Finer means more open sets and therefore more conditions for convergence.

The identity from the finer topology to the coarser one is continuous. The reverse identity need not be. Every function from a discrete space is continuous, and every function into an indiscrete space is continuous. A discrete space is compact exactly when it is finite; every indiscrete space is compact.

Problem 231.1[See solution](#)

An infinite set X has the discrete topology and the same set Y has the indiscrete topology. Which statement about the identity bijection $f : X \rightarrow Y$ is correct?

- (A) It is continuous and open.
- (B) It is continuous but not open.
- (C) It is open but not continuous.
- (D) It is neither continuous nor open.
- (E) It is a homeomorphism.

Problem 231.2[See solution](#)

A set of four points has the indiscrete topology. How many functions from this space to $\mathbb{R}$ with its usual topology are continuous?

- (A) 0
- (B) 1
- (C) 4
- (D) Exactly one for each real number.
- (E) Every function from the set to $\mathbb{R}$.

Problem 231.3[See solution](#)

Let $X = \{a, b\}$ with $a \neq b$, and let x_n alternate between a and b . What are its sets of limits in the discrete and indiscrete topologies, respectively?

- (A) $(\emptyset, X)$ (B) $(X, \emptyset)$ (C) $(\{a\}, \{b\})$ (D) $(\emptyset, \emptyset)$ (E) (X, X)

Problem 231.4[See solution](#)

If $\mathcal{T}_1 \subseteq \mathcal{T}_2$, which identity map is necessarily continuous?

- (A) Continuity requires equality of the topologies
- (B) Both identities are homeomorphisms
- (C) $(X, \mathcal{T}_2) \rightarrow (X, \mathcal{T}_1)$
- (D) $(X, \mathcal{T}_1) \rightarrow (X, \mathcal{T}_2)$
- (E) Neither identity is continuous

Problem 231.5[See solution](#)

An infinite set X is given once with the discrete topology and once with the indiscrete topology. Which statement about compact subsets, with their relative topologies, is correct?

- (A) Only finite subsets are compact in either topology.
- (B) Every subset is compact in either topology.
- (C) Only finite subsets are compact in the discrete topology; every subset is compact in the indiscrete topology.
- (D) Every subset is compact in the discrete topology; only finite subsets are compact in the indiscrete topology.
- (E) No nonempty subset is compact in the indiscrete topology.

7. Additional Topics

Point-Set Topology ▸ The Subspace Topology

232 Subspace or relative topology

Book pp. 284. [Find in compact glossary](#)**In this entry:** subspace topology; relative topology; relatively open; relatively closed.**Definition and key ideas**

For $S \subseteq X$, the subspace topology is $\{S \cap U : U \text{ open in } X\}$. A relatively closed subset similarly equals $S \cap F$ for a closed subset F of X . Relative openness or closedness always depends on the stated ambient space.

If S itself is open in X , every relatively open subset of S is open in X ; the analogous closed statement also holds. In subspaces of $\mathbb{R}$, isolated points become open singletons even though singletons are never open in the ambient real line.

For $A \subseteq S \subseteq X$, relative closure satisfies $\overline{A}^S = S \cap \overline{A}^X$. Interior has no such general intersection formula; a thin set may have empty ambient interior while being open in itself. Every relatively open subset of S is ambient open exactly when S itself is ambient open.

Problem 232.1[See solution](#)

Let $S = [0, 1] \cup \{2\}$ have the subspace topology from $\mathbb{R}$. Which subset is both open and closed in S ?

- (A) $[0, 1/2)$ (B) $[0, 1/2]$ (C) $\{0\}$ (D) $[0, 1]$ (E) $(0, 1)$

Problem 232.2[See solution](#)

Let $S = \{0\} \cup \{1/n : n \in \mathbb{N}, n \geq 1\}$ in $\mathbb{R}$. Which subset is open in S but not closed in S ?

- (A) $\{0\}$ (B) $\{1\}$ (C) $S \setminus \{0\}$ (D) $\{0, 1\}$ (E) $\emptyset$

Problem 232.3[See solution](#)

For $S \subseteq \mathbb{R}$, which condition is equivalent to every relatively open subset of S being open in $\mathbb{R}$?

- (A) S is closed in $\mathbb{R}$.
(B) S is open in $\mathbb{R}$.
(C) S is finite.
(D) S is dense in $\mathbb{R}$.
(E) S has the discrete relative topology.

Problem 232.4[See solution](#)

In the subspace $\mathbb{Q} \subset \mathbb{R}$, which set is both open and closed?

- (A) $\{q \in \mathbb{Q} : q \leq 1\}$ (B) $\{1\}$
(C) $\{q \in \mathbb{Q} : 0 < q < 1\}$ (D) $\{q \in \mathbb{Q} : q < 1\}$
(E) $\{q \in \mathbb{Q} : q < \sqrt{2}\}$

Problem 232.5[See solution](#)

Let $A \subseteq S \subseteq X$, with S carrying the subspace topology. Which identity always holds?

- (A) $\overline{A}^S = \overline{A}^X$ (B) $\text{int}_S A = \text{int}_X A$
(C) $\overline{A}^S = S \cap \overline{A}^X$ (D) $\text{int}_S A = S \cap \text{int}_X A$
(E) $\overline{A}^X \subseteq S$

7. Additional Topics

Point-Set Topology ▷ The Interior, Exterior, Boundary, Limit Points, and Closure of a Set

233 Interior, exterior, and boundary

Book pp. 284-286. [Find in compact glossary](#)**In this entry:** interior; exterior; boundary; clopen set.**Definition and key ideas**

The interior $\text{int } A$ is the largest open subset of A ; the exterior is $\text{int}(X \setminus A)$. A point lies in the boundary ∂A when every neighborhood meets both A and its complement. Thus $\partial A = \overline{A} \cap \overline{X \setminus A} = \overline{A} \setminus \text{int } A$.

Interior, boundary, and exterior partition X . The boundary is closed and is unchanged on replacing A by its complement. A set is clopen when it is both open and closed; equivalently, it has empty boundary. Dense sets can have empty interior and a very large boundary: both $\mathbb{Q}$ and its complement have boundary $\mathbb{R}$.

Boundaries obey $\partial(A \cup B) \subseteq \partial A \cup \partial B$, with possible strict inclusion. If A is open or closed, its boundary has empty interior, and therefore $\partial(\partial A) = \partial A$. This last identity need not hold for an arbitrary set, such as the rationals.

Problem 233.1[See solution](#)

In $\mathbb{R}$, let $A = (\mathbb{Q} \cap [0, 1]) \cup (2, 3)$. What is ∂A ?

- (A) $\{0, 1, 2, 3\}$ (B) $[0, 1] \cup \{2, 3\}$ (C) $[0, 1] \cup [2, 3]$ (D) $\mathbb{Q} \cap [0, 1]$ (E) $\mathbb{R}$

Problem 233.2[See solution](#)

Let $A = \{1/n : n \geq 1\} \cup [2, 4]$ in $\mathbb{R}$. What is the exterior of A ?

- (A) $(-\infty, 0) \cup (1, 2) \cup (4, \infty)$ (B) $\mathbb{R} \setminus A$
 (C) $\mathbb{R} \setminus (\{0\} \cup \{1/n : n \geq 1\} \cup [2, 4])$ (D) $(-\infty, 0) \cup (4, \infty)$
 (E) $\mathbb{R} \setminus [0, 4]$

Problem 233.3[See solution](#)

For arbitrary subsets A, B of a topological space, which inclusion always holds?

- (A) $\partial(A \cup B) \subseteq \partial A \cup \partial B$ (B) $\partial A \cup \partial B \subseteq \partial(A \cup B)$
 (C) $\partial(A \cup B) \subseteq \partial A \cap \partial B$ (D) $\partial A \subseteq A$
 (E) $\partial A \cap \partial B = \emptyset$

Problem 233.4[See solution](#)

What is the boundary of $\mathbb{Z}$ in $\mathbb{R}$?

- (A) $\mathbb{R} \setminus \mathbb{Z}$ (B) $\mathbb{R}$ (C) $\emptyset$ (D) $\{0\}$ (E) $\mathbb{Z}$

Problem 233.5[See solution](#)

If A is a closed subset of $\mathbb{R}$, which statement must be true?

- (A) $\partial A = \emptyset$ (B) ∂A is open. (C) $\partial(\partial A) = \partial A$ (D) $\partial A = \text{int } A$ (E) $\partial A = A$

7. Additional Topics

Point-Set Topology ▷ The Interior, Exterior, Boundary, Limit Points, and Closure of a Set

234 Limit points, derived set, closure, and density

Book pp. 285-286. [Find in compact glossary](#)

In this entry: limit point; accumulation point; cluster point; derived set; closure; isolated point; dense subset.

Definition and key ideas

A limit point of A is a point whose every neighborhood meets $A \setminus \{x\}$; the derived set A' consists of all such points. The closure $\overline{A}$ consists of points whose every neighborhood meets A , so $\overline{A} = A \cup A'$. It is the smallest closed set containing A .

A point $x \in A$ is isolated if some neighborhood N satisfies $N \cap A = \{x\}$. Such a point belongs to the closure but is not a limit point. A subset is dense in X if its closure equals X . In metric spaces, points of the closure are limits of sequences in A ; this sequential characterization should not be assumed for arbitrary topologies.

Problem 234.1

[See solution](#)

For $A = \{1/n : n \geq 1\} \cup \{2 + 1/n : n \geq 1\}$ in $\mathbb{R}$, what is the derived set of the derived set, $(A')'$?
(A) $\emptyset$ **(B)** $\{0\}$ **(C)** $\{2\}$ **(D)** $\{0, 2\}$ **(E)** $A \cup \{0, 2\}$

Problem 234.2

[See solution](#)

Let $A = \mathbb{Q} \cap (0, 1)$ and $B = (\mathbb{R} \setminus \mathbb{Q}) \cap (0, 1)$. Which equality is correct?

- (A)** $\overline{A \cap B} = \overline{A} \cap \overline{B}$ **(B)** $\overline{A} \cap \overline{B} = [0, 1]$
(C) $\overline{A \cup B} = (0, 1)$ **(D)** $A' = \emptyset$
(E) $\text{int}(A \cup B) = \emptyset$

Problem 234.3

[See solution](#)

Let $A = \{1/n + 1/m : n, m \in \mathbb{N}\}$. What is the set of limit points of its derived set?

- (A)** $\mathbb{Q} \cap [0, 2]$ **(B)** $\{1/n : n \in \mathbb{N}\}$ **(C)** $\emptyset$ **(D)** $\{0\}$ **(E)** $[0, 2]$

Problem 234.4

[See solution](#)

For any subset A of a metric space, which equality always holds?

- (A)** $\overline{A} = A$ **(B)** $\overline{A}$ is open **(C)** $\overline{\overline{A}} = \overline{A}$ **(D)** $\text{int } \overline{A} = \text{int } A$ **(E)** $\overline{A \cap B} = \overline{A} \cap \overline{B}$ for every B

Problem 234.5

[See solution](#)

If A and B are dense open subsets of $\mathbb{R}$, which must be true?

- (A)** $A \cap B = \mathbb{R}$ **(B)** $A \setminus B$ is empty **(C)** $A \cap B$ is dense **(D)** $A \cap B$ is closed **(E)** $A = B$

7. Additional Topics

Point-Set Topology ▷ Basis for a Topology

235 Topological bases and the lower-limit topology

Book pp. 286-287. [Find in compact glossary](#)

In this entry: basis; basis element; generated topology; lower-limit topology; Sorgenfrey line; second countable space.

Definition and key ideas

A basis covers the space and satisfies a local intersection condition: if $x \in B_1 \cap B_2$, a basis element B_3 contains x and lies in that intersection. All unions of basis elements form the generated topology. To compare two bases, test whether every point of each coarse basis element has a fine basis element inside it.

Intervals (a, b) generate the usual topology on $\mathbb{R}$, whereas $[a, b)$ generate the lower-limit topology. The latter is strictly finer. Its intervals $[a, b)$ are both open and closed, and no subset containing two distinct points is connected. Neighborhoods there control approach from the right.

The usual real topology has a countable basis of rational-endpoint open intervals. The lower-limit topology has no countable basis: refining $[x, x + 1)$ at each real x requires distinct basis elements with minimum x . A space with a countable basis is called second countable. In the lower-limit line, any two points are separated within their subspace by complementary open rays, so connected subsets contain at most one point.

Problem 235.1

[See solution](#)

In the lower-limit topology on $\mathbb{R}$, which sequence converges to 0?

- (A) $x_n = -1/n$ (B) $x_n = (-1)^n/n$ (C) $x_n = 1/n$ (D) $x_n = -1$ (E) $x_n = (-1)^n$

Problem 235.2

[See solution](#)

In the lower-limit topology, let $A = (0, 1)$. Which pair is $(\overline{A}, \partial A)$?

- (A) $([0, 1], \{0, 1\})$ (B) $([0, 1), \{0\})$ (C) $((0, 1], \{1\})$ (D) $((0, 1), \emptyset)$ (E) $([0, 1), \{0, 1\})$

Problem 235.3

[See solution](#)

Which families are bases for the usual topology on $\mathbb{R}$?

I. Open intervals with rational endpoints.

II. All open intervals of length one.

III. All closed intervals $[a, b]$ with $a < b$.

- (A) I only (B) II only (C) I and II only (D) I and III only (E) All three

Problem 235.4[See solution](#)

In the lower-limit topology on $\mathbb{R}$, which description of $[0, 1)$ is correct?

- (A) Open, closed, and connected
- (B) Open, closed, and disconnected
- (C) Open but not closed, and connected
- (D) Closed but not open, and disconnected
- (E) Neither open nor closed

Problem 235.5[See solution](#)

What can be said about the size of a basis for the lower-limit topology on $\mathbb{R}$?

- (A) A finite basis exists.
- (B) A countably infinite basis exists.
- (C) Every basis is uncountable.
- (D) Every basis consists only of the intervals $[a, b)$.
- (E) No basis exists.

7. Additional Topics

Point-Set Topology ▸ The Product Topology

236 Product topology and open rectangles

Book pp. 287. [Find in compact glossary](#)**In this entry:** product topology; open rectangle; coordinate projection.**Definition and key ideas**

The product topology on $X \times Y$ has basis $U \times V$, with U, V open in their respective factors. For finitely many Euclidean factors this agrees with the usual Euclidean topology. Coordinate projections are continuous and open. A map into a product is continuous exactly when its coordinate maps are continuous.

For nonempty factors, closure satisfies $\overline{A \times B} = \overline{A} \times \overline{B}$. A thin subset such as a line in $\mathbb{R}^2$ may be closed but has empty interior. Openness in the plane requires a full small rectangle or disk, not just openness along a curve.

Open coordinate projections of a subset do not by themselves make that subset open. Although projection need not send closed sets to closed sets in general, projection $X \times K \rightarrow X$ is a closed map when K is compact: a finite cover of the omitted fiber supplies one neighborhood missing the projection. The topology on each factor also matters when checking continuity of coordinate maps.

Problem 236.1[See solution](#)

Let $A = (\mathbb{Q} \cap [0, 1]) \times (0, 1)$ in $\mathbb{R}^2$. What is its boundary?

- (A) The four edges of $[0, 1]^2$.
- (B) $[0, 1]^2$
- (C) $\{0, 1\} \times [0, 1]$
- (D) A
- (E) $\emptyset$

Problem 236.2[See solution](#)

Let $\pi : \mathbb{R}^2 \rightarrow \mathbb{R}$ be $\pi(x, y) = x$. Which statement is false?

- (A) π is continuous.
- (B) π maps open sets to open sets.
- (C) π maps compact sets to compact sets.
- (D) π maps every closed set to a closed set.
- (E) π maps connected sets to connected sets.

Problem 236.3[See solution](#)

The codomain Y is $\mathbb{R} \times \mathbb{R}$ with the usual topology in the first factor and the discrete topology in the second. Which maps from the usual real line into Y are continuous?

I. $t \mapsto (t, t)$.

II. $t \mapsto (t, 0)$.

III. $t \mapsto (0, \lfloor t \rfloor)$.

(A) I only (B) II only (C) I and II only (D) II and III only (E) All three

Problem 236.4[See solution](#)

Let $S = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$. Which description is correct?

(A) S is open because both coordinate projections are open subsets of $\mathbb{R}$.

(B) S is closed with empty interior, although both coordinate projections equal $\mathbb{R}$.

(C) S has nonempty interior only at the origin.

(D) S is neither open nor closed.

(E) S is compact.

Problem 236.5[See solution](#)

Let C be a closed subset of $\mathbb{R} \times [0, 1]$ with the product topology. Which statement about its first-coordinate projection $\pi(C)$ must hold?

(A) It is compact.

(B) It is open.

(C) It is closed, but need not be bounded.

(D) It is an interval.

(E) It is all of $\mathbb{R}$ unless C is empty.

7. Additional Topics

Point-Set Topology ▷ Connectedness

237 Connectedness, separations, and connected components

Book pp. 288. [Find in compact glossary](#)**In this entry:** connected space; disconnected space; separation; connected component.**Definition and key ideas**

A separation is a representation of a space as two disjoint nonempty relatively open subsets. A space is connected if no separation exists; equivalently its only clopen subsets are $\emptyset$ and the whole space. Connected subsets of $\mathbb{R}$ are exactly intervals, with singletons and the empty set included.

Unions of connected sets with a common point are connected. If A is connected and $A \subseteq B \subseteq \overline{A}$, then B is connected; products of connected spaces are connected. A connected component is a maximal connected subset. Use a continuous map to a discrete set, or a visible clopen partition, to disprove connectedness.

Problem 237.1[See solution](#)

How many connected components does $\mathbb{R} \setminus (\{0\} \cup \{1/n : n \geq 1\})$ have?

- (A) 1
- (B) 2
- (C) 3
- (D) Countably infinitely many.
- (E) Uncountably many.

Problem 237.2[See solution](#)

Let $A = \{(x, \sin(1/x)) : 0 < x \leq 1\}$ and $B = A \cup (\{0\} \times [-1, 1])$. Which conclusion follows from connectedness and closure alone?

- (A) B is disconnected.
- (B) B is connected.
- (C) B is open in $\mathbb{R}^2$.
- (D) A is compact.
- (E) B has exactly two connected components.

Problem 237.3[See solution](#)

How many connected components does $\mathbb{R} \setminus \mathbb{Q}$ have?

- (A) One
- (B) Uncountably many
- (C) Countably infinitely many
- (D) Zero
- (E) Two

Problem 237.4[See solution](#)

A continuous map from a nonempty connected space to the discrete set $\{0, 1, 2\}$ must have image of size

- (A) Any of 1, 2, 3 (B) 0 (C) 2 (D) 3 (E) 1

Problem 237.5[See solution](#)

If connected subsets A, B have nonempty intersection, which must be connected?

- (A) The complement of $A \cup B$ (B) $A \setminus B$ (C) ∂A (D) $A \cup B$ (E) $A \cap B$

Problem 237.6[See solution](#)

How many connected components does $(\mathbb{R} \setminus \{0\})^3$ have in the product topology?

- (A) 2 (B) 3 (C) 6 (D) 8 (E) Infinitely many

7. Additional Topics

Point-Set Topology ▷ Connectedness

238 Paths, path connectedness, and convexity

Book pp. 288. [Find in compact glossary](#)**In this entry:** path; path-connected space; convex set; star-shaped set.**Definition and key ideas**

A path from x to y is a continuous map $p : [0, 1] \rightarrow X$ with endpoints x, y . A space is path connected if every pair of points can be joined by a path. Path connectedness implies connectedness because $[0, 1]$ is connected; the converse fails.

In Euclidean space, a convex subset contains the segment $(1 - t)x + ty$ between any two of its points, and is therefore path connected. A punctured plane remains path connected because a polygonal path can avoid the missing point, whereas removing a point from the real line disconnects it. These provide useful dimensional distinctions.

A set is star-shaped if it has a point c such that the segment from c to every point of the set lies in the set; unlike convexity, it need not contain segments between arbitrary pairs. Two segments through c give a path. A nonempty closed convex subset of finite-dimensional Euclidean space has a unique nearest point to any given point: compact restriction gives existence, and the strict midpoint distance inequality gives uniqueness.

Problem 238.1[See solution](#)

Which space is connected but not path connected?

- (A) $\mathbb{R} \setminus \{0\}$
- (B) $\mathbb{Q}$
- (C) $\{(x, \sin(1/x)) : 0 < x \leq 1\} \cup (\{0\} \times [-1, 1])$
- (D) $\mathbb{R}^2 \setminus \{(0, 0)\}$
- (E) $[0, 1]^2$

Problem 238.2[See solution](#)Let $A = \{(x, y) \in \mathbb{R}^2 : xy > 0\}$ and $B = \{(x, y) \in \mathbb{R}^2 : xy \geq 0\}$. Which statement is correct?

- (A) Both are connected.
- (B) Neither is connected.
- (C) A is connected and B is not.
- (D) A is disconnected and B is path connected.
- (E) A is path connected and B is disconnected.

Problem 238.3[See solution](#)

The set $\mathbb{R}^2 \setminus \{(0, 0)\}$ is

- (A) Convex
- (B) Compact
- (C) Connected but not path connected
- (D) Path connected but not convex
- (E) Disconnected

Problem 238.4[See solution](#)

Every nonempty star-shaped subset of $\mathbb{R}^n$ is

- (A) Open (B) Path connected (C) Bounded (D) Convex (E) Closed

Problem 238.5[See solution](#)

Let C be a nonempty closed convex subset of $\mathbb{R}^n$, and let $p \notin C$. Which statement must hold?

- (A) No point of C minimizes distance to p .
- (B) Exactly one point of C minimizes distance to p .
- (C) At least two points of C minimize distance to p .
- (D) C is bounded.
- (E) C is open.

7. Additional Topics

Point-Set Topology ▷ Compactness

239 Open covers and compactness

Book pp. 288-289. [Find in compact glossary](#)

In this entry: covering; open cover; finite subcover; compact space; finite intersection property; sequential compactness.

Definition and key ideas

A space is compact if every open cover has a finite subcover. To disprove compactness, produce one open cover for which no finite selection suffices. Finite spaces and indiscrete spaces are compact; an infinite discrete space fails because the cover by singletons has no finite subcover.

Closed subsets of compact spaces are compact, and products of finitely many compact spaces are compact. In metric spaces compactness is equivalent to sequential compactness, meaning every sequence has a convergent subsequence with its limit in the space. This equivalence is not a definition valid for all topological spaces.

An open cover is a family of open sets whose union contains the space or subset being covered; a finite subcover selects finitely many members of that same family. A family has the finite intersection property when every finite subfamily has nonempty intersection. For closed subsets of a compact space this property guarantees a nonempty full intersection; the nested-set question is a special case.

Problem 239.1

[See solution](#)

For $n \geq 2$, let $U_n = (1/n, 2) \cap (0, 1] = (1/n, 1]$, open in the subspace $(0, 1]$. Which statement is correct?

- (A) U_2 alone covers $(0, 1]$.
- (B) Two members cover $(0, 1]$.
- (C) Every finite subcollection covers $(0, 1]$.
- (D) This is an open cover with no finite subcover.
- (E) The collection does not cover $(0, 1]$.

Problem 239.2

[See solution](#)

Let K be compact and $F_1 \supseteq F_2 \supseteq \cdots$ be nonempty closed subsets of K . Which must be true?

- (A) Some F_n is finite.
- (B) $\bigcap_{n=1}^{\infty} F_n \neq \emptyset$.
- (C) The intersection is a singleton.
- (D) Some F_n equals the intersection.
- (E) Every F_n is open.

Problem 239.3[See solution](#)

An infinite subset of a compact metric space must have

- (A) A finite number of points
- (B) A limit point in the space
- (C) A limit point belonging to the subset
- (D) Nonempty interior
- (E) Positive diameter bounded below by one

Problem 239.4[See solution](#)

Let X, Y be compact spaces and $f : X \rightarrow \mathbb{R}$, $g : Y \rightarrow \mathbb{R}$ continuous. Which property must the set $C = \{(x, y) \in X \times Y : f(x) = g(y)\}$ have?

- (A) It is nonempty.
- (B) It is open in $X \times Y$.
- (C) It is connected.
- (D) It is finite.
- (E) It is compact.

Problem 239.5[See solution](#)

In an arbitrary topological space, a sequence x_n converges to x . Which property must $K = \{x\} \cup \{x_n : n \geq 1\}$ have, with its subspace topology?

- (A) It is finite.
- (B) It is compact.
- (C) It is Hausdorff.
- (D) Every subset of K is closed.
- (E) It is connected.

7. Additional Topics

Point-Set Topology ▷ Compactness

240 Boundedness, norm, and the Heine–Borel theorem

Book pp. 289. [Find in compact glossary](#)**In this entry:** Euclidean norm; bounded set; Heine–Borel theorem; closed and bounded.**Definition and key ideas**

The Euclidean norm is $\|x\|_2 = (\sum x_j^2)^{1/2}$, and a subset of $\mathbb{R}^n$ is bounded if it fits inside a finite-radius ball. The Heine–Borel theorem states that in finite-dimensional Euclidean space, compactness is equivalent to being closed and bounded.

Both conditions matter: closed unbounded sets and bounded nonclosed sets may fail compactness. Closed and bounded is not a universal characterization for arbitrary metric spaces. For subsets defined by equations and weak inequalities of continuous functions, closedness often follows by taking inverse images of closed sets.

The finite-dimensional hypothesis is essential. In ℓ^2 , the square-summable sequence space with norm $(\sum x_j^2)^{1/2}$, the standard unit vectors form a closed bounded set with pairwise distance $\sqrt{2}$ and no convergent subsequence. Thus even a closed bounded subset of a familiar normed space need not be compact in infinite dimension.

Problem 240.1[See solution](#)Which subset of $\mathbb{R}^2$ is compact?

- (A) $\{(x, y) : xy = 1\}$
- (B) $\{(x, y) : x^2 + y^2 < 1\}$
- (C) $\{(x, y) : x^2 + y^2 \leq 1, x + y \geq 1/2\}$
- (D) $\{(1/n, 0) : n \geq 1\}$
- (E) $\{(x, y) : 0 < x \leq 1, y = 1/x\}$

Problem 240.2[See solution](#)For a real parameter a , let $K_a = \{(x, y) : x^2 + ay^2 \leq 1\}$. For which a is K_a compact?

- (A) $a < 0$ (B) $a \leq 0$ (C) $a = 0$ (D) $a \geq 0$ (E) $a > 0$

Problem 240.3[See solution](#)

In the metric space ℓ^2 of square-summable real sequences with distance $d(x, y) = (\sum_{j \geq 1} (x_j - y_j)^2)^{1/2}$, let $E = \{e_n : n \geq 1\}$, where e_n has a single 1 in position n . Which description is correct?

- (A) E is compact.
- (B) E is bounded but not closed.
- (C) E is closed and bounded but not compact.
- (D) E is closed and unbounded.
- (E) E is neither closed nor bounded.

Problem 240.4[See solution](#)

Which set is bounded but not compact in $\mathbb{R}^2$?

- (A) $\{(x, y) : 0 < x^2 + y^2 \leq 1\}$
- (B) $[0, 1]^2$
- (C) $\{(x, y) : x^2 + y^2 \leq 1\}$
- (D) $\{(0, 0)\}$
- (E) $\{(x, y) : x^2 + y^2 = 1\}$

Problem 240.5[See solution](#)

A subset of $\mathbb{R}^n$ is compact if and only if it is

- (A) Bounded
- (B) Connected and bounded
- (C) Closed
- (D) Complete
- (E) Closed and bounded

7. Additional Topics

Point-Set Topology ▸ Metric Spaces

241 Metrics, open balls, and equivalent metric topologies

Book pp. 289-290. [Find in compact glossary](#)

In this entry: metric; metric space; triangle inequality; open ball; metric topology; Euclidean metric; square metric; sup metric; sup norm; bounded function; sup metric on bounded functions.

Definition and key ideas

A metric satisfies $d(x, y) \geq 0$, equality exactly for $x = y$, symmetry, and the triangle inequality $d(x, z) \leq d(x, y) + d(y, z)$. Open balls $B_d(x, r) = \{y : d(x, y) < r\}$ form a basis. Every metric topology is Hausdorff.

For $v \in \mathbb{R}^n$, the sup norm is $\|v\|_\infty = \max_{1 \leq j \leq n} |v_j|$, and the square or sup metric is $d_\infty(x, y) = \|x - y\|_\infty$. It and the Euclidean metric induce the same topology because $\|v\|_\infty \leq \|v\|_2 \leq \sqrt{n}\|v\|_\infty$. Equivalent topologies need not give identical notions of Cauchy sequence or boundedness unless stronger comparison bounds hold. Squaring an ordinary distance can destroy the triangle inequality. On the space of real-valued functions on a nonempty set E with $\sup_{x \in E} |f(x)| < \infty$, the analogous sup metric is $d_\infty(f, g) = \sup_{x \in E} |f(x) - g(x)|$; boundedness ensures this distance is finite.

To compute a sup distance, maximize the absolute difference over the entire function domain, not only at fixed sample points. Another equivalent metric on $\mathbb{R}$ is $d(x, y) = |x - y|/(1 + |x - y|)$: balls of radius $r < 1$ are ordinary intervals of radius $r/(1 - r)$, and the whole space is bounded in d . Thus equivalent topologies can have different bounded sets.

Problem 241.1

[See solution](#)

For which $p > 0$ does $d_p(x, y) = |x - y|^p$ define a metric on $\mathbb{R}$?

(A) Only $p = 1$. (B) $0 < p \leq 1$ (C) $p \geq 1$ (D) $0 < p < 2$ (E) Every $p > 0$.

Problem 241.2

[See solution](#)

In $\mathbb{R}^3$, what is the smallest C such that $\|x - y\|_2 \leq C\|x - y\|_\infty$ for all x, y ?

(A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 3 (E) $\sqrt{6}$

Problem 241.3

[See solution](#)

Which function is a metric on $\mathbb{R}$?

- (A) $d(x, y) = 0$ for every x, y
- (B) $d(x, y) = (x - y)^2$
- (C) $d(x, y) = |x + y|$
- (D) $d(x, y) = |x - y| - 1$
- (E) $d(x, y) = \min\{1, |x - y|\}$

Problem 241.4[See solution](#)

On the bounded real functions on $[0, 1]$, use the sup metric. For $f_n(x) = x^n$ and $n \geq 1$, what is $d_\infty(f_n, f_{2n})$?

- (A) 0 (B) $1/(2n)$ (C) $1/4$ (D) $1/2$ (E) 1

Problem 241.5[See solution](#)

For the metric $d(x, y) = |x - y|/(1 + |x - y|)$ on $\mathbb{R}$, what is $B_d(0, 1/2)$, and is the whole space bounded in this metric?

- (A) $(-1, 1)$; yes (B) $[-1, 1]$; yes (C) $(-1, 1)$; no (D) $(-1/2, 1/2)$; yes (E) $\mathbb{R}$; yes

7. Additional Topics

Point-Set Topology ▷ Continuous Functions

242 Topological and metric continuity

Book pp. 290-292. [Find in compact glossary](#)

In this entry: continuous map; inverse image criterion; continuity at a point; epsilon-delta continuity; uniform continuity; sequential criterion for continuity.

Definition and key ideas

A map $f : X \rightarrow Y$ is continuous when inverse images of open sets are open, equivalently inverse images of closed sets are closed. It suffices to check a basis for the target topology. Continuity at x means every neighborhood of $f(x)$ contains the image of some neighborhood of x .

In metric spaces this becomes: for every $\varepsilon > 0$ there is $\delta > 0$ such that $d_X(x, x_0) < \delta$ implies $d_Y(f(x), f(x_0)) < \varepsilon$. At a single point the preimage of each target neighborhood need only be a neighborhood of that point, not an open set everywhere; the stronger global assertion characterizes continuity throughout the domain.

A map between metric spaces is uniformly continuous if for every $\varepsilon > 0$ one $\delta > 0$ works for all pairs x, y : $d_X(x, y) < \delta$ implies $d_Y(f(x), f(y)) < \varepsilon$. Ordinary pointwise continuity allows δ to depend on the base point. Sequential continuity is equivalent to continuity for metric spaces: $x_n \rightarrow x$ must imply $f(x_n) \rightarrow f(x)$.

Problem 242.1[See solution](#)

Let $X = \{a, b, c\}$ with topology $\{\emptyset, \{a\}, \{a, b\}, X\}$. How many continuous maps from X to the discrete space $\{0, 1\}$ exist?

(A) 2 (B) 3 (C) 4 (D) 6 (E) 8

Problem 242.2[See solution](#)

Let $\mathcal{T}$ be the usual topology on $\mathbb{R}$ and $\mathcal{L}$ the lower-limit topology. Which maps are continuous: $f : (\mathbb{R}, \mathcal{L}) \rightarrow (\mathbb{R}, \mathcal{T})$, $f(x) = x$, and $g : (\mathbb{R}, \mathcal{L}) \rightarrow (\mathbb{R}, \mathcal{L})$, $g(x) = -x$?

- (A) Both.
- (B) f only.
- (C) g only.
- (D) Neither.
- (E) Both only on bounded intervals.

Problem 242.3[See solution](#)

Define $f(x) = x$ for rational x and $f(x) = -x$ for irrational x . Let $U = (-1, 2)$. Which statement is correct?

- (A) f is discontinuous at zero because $f^{-1}(U)$ is not open.
- (B) f is continuous at zero, although $f^{-1}(U)$ is not open in $\mathbb{R}$.
- (C) f is continuous everywhere.
- (D) $f^{-1}(U)$ does not contain a neighborhood of zero.
- (E) $f^{-1}(U) = (-1, 2)$

Problem 242.4[See solution](#)

A function between metric spaces is continuous if and only if it

- (A) Maps bounded sets to bounded sets
- (B) Is uniformly continuous
- (C) Maps open sets to open sets
- (D) Preserves limits of convergent sequences
- (E) Maps closed sets to closed sets

Problem 242.5[See solution](#)

Which function on $(0, 1)$ is continuous but not uniformly continuous?

- (A) $f(x) = 1/x$ (B) $f(x) = 3$ (C) $f(x) = x$ (D) $f(x) = \sin x$ (E) $f(x) = x^2$

7. Additional Topics

Point-Set Topology ▷ Continuous Functions

243 Continuous images and the extreme value theorem

Book pp. 292. [Find in compact glossary](#)

In this entry: continuous image of compact set; continuous image of connected set; extreme value theorem; absolute maximum; absolute minimum.

Definition and key ideas

Continuous maps preserve compactness and connectedness under direct images. Inverse images preserve openness and closedness, but generally not compactness or connectedness. The image of a nonempty compact connected space in $\mathbb{R}$ is therefore a closed bounded interval.

A continuous real-valued function on a nonempty compact space attains both an absolute minimum and an absolute maximum. On a compact metric space it is also uniformly continuous. Compactness must refer to the domain, and extrema can fail on open or unbounded domains.

Problem 243.1

[See solution](#)

Let X be nonempty, compact, and connected, and $f : X \rightarrow \mathbb{R}$ continuous with minimum -2 and maximum 5 . What is $f(X)$?

- (A) $\{-2, 5\}$
- (B) $(-2, 5)$
- (C) $[-2, 5]$
- (D) Any closed subset of $[-2, 5]$ containing both endpoints.
- (E) Any bounded subset containing both endpoints.

Problem 243.2

[See solution](#)

Let $K \subset \mathbb{R}^n$ be nonempty and compact, and $f : K \rightarrow (0, \infty)$ continuous. Which must be true?

- (A) $f(x) \geq 1$ for all x .
- (B) There is $c > 0$ such that $f(x) \geq c$ for all $x \in K$.
- (C) f is constant.
- (D) $1/f$ is unbounded.
- (E) $f(K)$ is open in $\mathbb{R}$.

Problem 243.3[See solution](#)

If K is compact and $f : K \rightarrow \mathbb{R}$ is continuous, which is impossible?

- (A) $f(K) = (0, 1)$
- (B) $f(K) = [0, 1]$
- (C) $f(K) = \{0, 1\}$
- (D) $f(K) = [-2, 3]$
- (E) $f(K) = \{0\}$

Problem 243.4[See solution](#)

For the continuous map $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \sin x$, let $K = \{0\}$. Which description of $f^{-1}(K)$ is correct?

- (A) Compact and connected
- (B) Compact but disconnected
- (C) Closed and connected but not compact
- (D) Closed, neither compact nor connected
- (E) Not closed

Problem 243.5[See solution](#)

Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous, and for $n \geq 2$ set $M_n = \max_{0 \leq x \leq 1-1/n} |f(x + 1/n) - f(x)|$. Which conclusion must hold?

- (A) $M_n = 0$ for every n .
- (B) $M_n \rightarrow 0$
- (C) $nM_n \rightarrow 0$
- (D) f is differentiable.
- (E) M_n is eventually strictly decreasing.

7. Additional Topics

Point-Set Topology ▷ Open Maps and Homeomorphisms

244 Open maps, homeomorphisms, and topological invariants

Book pp. 292-293. [Find in compact glossary](#)

In this entry: open map; homeomorphism; topological invariant; compact-to-Hausdorff theorem.

Definition and key ideas

An open map takes open subsets of its domain to open subsets of its codomain. A homeomorphism is a bijection whose map and inverse are both continuous; equivalently it is a continuous open bijection. Homeomorphisms preserve properties defined by topology, including compactness, connectedness, and the number of components after deleting specified points.

A continuous bijection from a compact space to a Hausdorff space is automatically a homeomorphism: closed domain subsets are compact, their images are compact and hence closed, so the inverse is continuous. Without these hypotheses a continuous bijection need not be a homeomorphism.

Problem 244.1

[See solution](#)

Which pair of spaces is homeomorphic?

- (A) $[0, 1]$ and $(0, 1)$
- (B) $\mathbb{R}$ and $(0, 1)$
- (C) $\mathbb{R}$ and the unit circle
- (D) $\mathbb{R}$ and $\mathbb{R}^2$
- (E) $[0, 1)$ and $[0, 1]$

Problem 244.2

[See solution](#)

A continuous bijection $f : X \rightarrow Y$ is given, where X is compact and Y is Hausdorff. Which must be true?

- (A) X is finite.
- (B) Y is connected.
- (C) f^{-1} is continuous.
- (D) Y is discrete.
- (E) Every subset of X is compact.

Problem 244.3[See solution](#)

Give $\mathbb{R}$ its usual topology and $\mathbb{N}_0$ the discrete topology. Define $f : \mathbb{R} \rightarrow \mathbb{N}_0$ by $f(x) = 0$ for $x \leq 0$ and $f(x) = 1$ for $x > 0$. Which statement is correct?

- (A) f is continuous and open.
- (B) f is continuous but not open.
- (C) f is open but not continuous.
- (D) f is neither open nor continuous.
- (E) f is a homeomorphism.

Problem 244.4[See solution](#)

The map $f : [0, 2\pi) \rightarrow S^1$ is $f(t) = (\cos t, \sin t)$, with both spaces carrying their usual subspace topologies. At which points of S^1 is the inverse function discontinuous?

- (A) No points
- (B) Only $(-1, 0)$
- (C) Only $(1, 0)$
- (D) Every point with second coordinate zero
- (E) Every point

Problem 244.5[See solution](#)

Which property is invariant under homeomorphism?

- (A) Being bounded in an ambient Euclidean space
- (B) Compactness
- (C) Euclidean diameter
- (D) Euclidean distance
- (E) The numerical value of area

7. Additional Topics

Real Analysis ▸ The Completeness of the Real Numbers

245 Bounds, supremum, infimum, and order completeness

Book pp. 293–294. [Find in compact glossary](#)

In this entry: upper bound; lower bound; bounded above; bounded below; least upper bound; supremum; greatest lower bound; infimum; least upper bound axiom.

Definition and key ideas

An upper bound u for $A \subseteq \mathbb{R}$ satisfies $a \leq u$ for every $a \in A$. The supremum is the least upper bound; the infimum is the greatest lower bound. A maximum or minimum must belong to the set, whereas a supremum or infimum need not.

Every nonempty real set bounded above has a finite real supremum, and the dual statement holds below. The nonempty hypothesis matters. If $s = \sup A$, then every $\varepsilon > 0$ admits $a \in A$ with $s - \varepsilon < a \leq s$. Rational numbers lack this completeness: rational sets can have irrational real suprema.

A lower bound ℓ satisfies $\ell \leq a$ for every $a \in A$; bounded below means that at least one such real ℓ exists. The empty set is bounded above and below vacuously but has no real supremum or infimum, since every real number is a bound and none is least or greatest as appropriate. An unattained finite supremum can always be approached by a strictly increasing sequence from the set.

Problem 245.1

[See solution](#)

Let $A = \{(-1)^n + 1/n : n = 1, 2, \dots\}$. What is $(\inf A, \sup A)$?

- (A) $(-1, 3/2)$ (B) $(-1, 2)$ (C) $(0, 3/2)$ (D) $(-2, 3/2)$ (E) $(-1, 1)$

Problem 245.2

[See solution](#)

Nonempty bounded subsets $A, B \subset \mathbb{R}$ satisfy $\sup A = 3$, $\inf A = -1$, $\sup B = 4$, and $\inf B = 2$. What is $\sup\{a - b : a \in A, b \in B\}$?

- (A) -5 (B) -1 (C) 1 (D) 2 (E) 7

Problem 245.3

[See solution](#)

Let $A = \{q \in \mathbb{Q} : 0 < q, q^2 < 2\}$. Which statement is correct?

- (A) A is not bounded above in $\mathbb{Q}$.
(B) A has rational supremum $3/2$.
(C) A has no supremum in $\mathbb{Q}$, but has real supremum $\sqrt{2}$.
(D) A has a maximum.
(E) The least rational upper bound is two.

Problem 245.4[See solution](#)

Let $A \subset \mathbb{R}$ be nonempty and bounded above, with $s = \sup A \notin A$. Which conclusion must hold?

- (A) A is countable.
- (B) A is an interval.
- (C) Every sequence in A converges to s .
- (D) There is a strictly increasing sequence of points of A converging to s .
- (E) A is closed.

Problem 245.5[See solution](#)

If $A \subset (0, \infty)$ has $\inf A = 2$ and $\sup A = 5$, what is $\sup\{1/a : a \in A\}$?

- (A) 5
- (B) 2
- (C) $1/5$
- (D) It depends on whether $2 \in A$
- (E) $1/2$

7. Additional Topics

Real Analysis ▸ The Completeness of the Real Numbers

246 Cauchy sequences and complete metric spaces

Book pp. 294. [Find in compact glossary](#)

In this entry: Cauchy sequence; complete space; metric completeness; closed subspace; total boundedness.

Definition and key ideas

A sequence is Cauchy if for every $\varepsilon > 0$ there is N such that $d(x_m, x_n) < \varepsilon$ whenever $m, n \geq N$. Every convergent sequence is Cauchy. A metric space is complete when every Cauchy sequence converges to a point of that space; $\mathbb{R}$ and $\mathbb{R}^n$ with usual metrics are complete.

A subspace of a complete metric space is complete exactly when it is closed. The condition $d(x_{n+1}, x_n) \rightarrow 0$ is much weaker than being Cauchy. Completeness depends on the metric, not just the topology: distinct metrics producing the same open sets can have different Cauchy sequences.

A metric space is totally bounded if for every $\varepsilon > 0$ finitely many balls of radius ε cover it. This is stronger than boundedness. A metric space is compact exactly when it is both complete and totally bounded: finite covers yield a Cauchy subsequence, and completeness supplies its limit. A Cauchy sequence with a convergent subsequence converges to the same limit, by the triangle inequality.

Problem 246.1

[See solution](#)

On $\mathbb{R}$, define $d(x, y) = |\arctan x - \arctan y|$. Which statement about $x_n = n$ is correct?

- (A) It is not Cauchy.
- (B) It is Cauchy and converges to 0.
- (C) It is Cauchy and has no limit in this metric space.
- (D) It converges to every real number.
- (E) The function d is not a metric.

Problem 246.2

[See solution](#)

Which subset of $\mathbb{R}$ is complete with the usual metric?

- (A) $\mathbb{Q}$ (B) $(0, 1]$ (C) $\{1/n : n \geq 1\}$ (D) $\{0\} \cup \{1/n : n \geq 1\}$ (E) $\mathbb{R} \setminus \mathbb{Z}$

Problem 246.3[See solution](#)

In the usual metric on $\mathbb{R}$, let $x_n = \log n$. Which statement is correct?

- (A) The sequence is Cauchy because $|x_{n+1} - x_n| \rightarrow 0$.
- (B) The sequence converges to one.
- (C) The consecutive distances tend to zero, but the sequence is not Cauchy.
- (D) The consecutive distances are bounded below by a positive constant.
- (E) The sequence is bounded but divergent.

Problem 246.4[See solution](#)

Every Cauchy sequence in a metric space is

- (A) Convergent
- (B) Contained in a finite set
- (C) Monotone
- (D) Bounded
- (E) Eventually constant

Problem 246.5[See solution](#)

A complete, totally bounded metric space is

- (A) Necessarily a Euclidean interval
- (B) Discrete
- (C) Compact
- (D) Necessarily finite
- (E) Necessarily connected

Problem 246.6[See solution](#)

A Cauchy sequence (x_n) in a metric space has a subsequence converging to x in that space. What must follow?

- (A) $x_n \rightarrow x$, even if the ambient space is incomplete.
- (B) The ambient space is complete.
- (C) The sequence is eventually constant.
- (D) Every point of the space is a subsequential limit.
- (E) The whole sequence may diverge if the space is incomplete.

7. Additional Topics

Real Analysis ▸ Lebesgue Measure

247 Outer measure and measurable sets

Book pp. 294-295. [Find in compact glossary](#)

In this entry: Lebesgue outer measure; Lebesgue measurable set; Caratheodory criterion; sigma-algebra.

Definition and key ideas

Lebesgue outer measure $m^*(A)$ is the infimum of the sums of lengths of countable open-interval covers of A . It is defined for every subset, is monotone and countably subadditive, and may equal infinity. A set E is measurable if $m^*(A) = m^*(A \cap E) + m^*(A \setminus E)$ for every $A \subseteq \mathbb{R}$.

Measurable sets form a sigma-algebra: they contain $\emptyset$ and are closed under complements and countable unions, hence countable intersections. Open and closed sets are measurable, and every subset of a set of outer measure zero is measurable. Arbitrary unions need not preserve measurability, and not every real subset is Lebesgue measurable.

Translation leaves outer measure unchanged, and $m^*(rA + b) = |r|m^*(A)$ for real $r \neq 0$; the proof transforms interval covers and then applies the inverse transformation. This formula needs no measurability assumption. For an arbitrary cut, countable subadditivity gives $m^*(A) \leq m^*(A \cap E) + m^*(A \setminus E)$; measurability requires equality for every test set A .

Problem 247.1

[See solution](#)

Enumerate $\mathbb{Q}$ as $(q_n)_{n \geq 1}$ and fix $\varepsilon > 0$. Set $U = \bigcup_{n \geq 1} (q_n - \varepsilon/2^{n+2}, q_n + \varepsilon/2^{n+2})$. Which must hold?

- (A) U is closed.
- (B) $m(U) = \infty$.
- (C) U is open and dense, with $m(U) \leq \varepsilon/2$.
- (D) $U = \mathbb{R}$.
- (E) $m(U) = 0$.

Problem 247.2

[See solution](#)

Which operation is guaranteed to preserve Lebesgue measurability for every family of measurable real sets?

- (A) An arbitrary union, even over an uncountable index set.
- (B) Taking every subset of a positive-measure set.
- (C) A countable intersection.
- (D) Taking an arbitrary image under any function.
- (E) Taking the inverse image under any function.

Problem 247.3[See solution](#)

A set $A \subseteq \mathbb{R}$, not assumed measurable, has outer measure $m^*(A) = 2/5$. What is $m^*(\{7 - 3a : a \in A\})$?

- (A) $2/15$
- (B) $2/5$
- (C) $6/5$
- (D) $33/5$
- (E) It is undefined unless A is measurable.

Problem 247.4[See solution](#)

A set $E \subseteq [0, 1]$ satisfies $m^*(E) = 3/4$ and $m^*([0, 1] \setminus E) = 3/4$. Which conclusion follows?

- (A) E is measurable with measure $3/4$.
- (B) E is not Lebesgue measurable.
- (C) $m^*([0, 1]) = 3/2$
- (D) E is countable.
- (E) No outer measure can be assigned to E .

Problem 247.5[See solution](#)

If E has outer measure zero, every subset of E is

- (A) Countable
- (B) Closed
- (C) Lebesgue measurable with measure zero
- (D) Open
- (E) Empty

7. Additional Topics

Real Analysis ▸ Lebesgue Measure

248 Measure, null sets, and countable additivity

Book pp. 295. [Find in compact glossary](#)

In this entry: Lebesgue measure; measure zero; null set; countable additivity; almost everywhere; monotonicity of measure; Cantor set; continuity of measure.

Definition and key ideas

Lebesgue measure assigns each interval its length, regardless of endpoint inclusion. For pairwise disjoint measurable sets E_n , $m(\bigcup E_n) = \sum m(E_n)$. It is monotone, and a countable set has measure zero. Removing or adding a null set does not change measure.

A property holds almost everywhere if the exceptional set has measure zero. Every positive-measure set is uncountable, but uncountable null sets exist, such as the middle-thirds Cantor set. Density, cardinality, and measure are different notions: $\mathbb{Q}$ is dense but null, while its complement has full measure on every interval.

The middle-thirds Cantor set is obtained by repeatedly deleting open middle thirds from $[0, 1]$. It is closed and uncountable but has measure zero. For increasing measurable sets, measure is continuous from below. For decreasing sets, $m(\bigcap E_n) = \lim m(E_n)$ requires $m(E_1) < \infty$ (or a finite-measure later member); without that condition, infinite masses can shrink to an empty intersection.

Problem 248.1

[See solution](#)

Let $E = \mathbb{Q} \cup \bigcup_{n \geq 1} [2^{-n}, 2^{-n} + 4^{-n}]$. What is $m(E)$?

- (A) 0 (B) $1/4$ (C) $1/3$ (D) $1/2$ (E) 1

Problem 248.2

[See solution](#)

A family of pairwise disjoint Lebesgue-measurable subsets of $[0, 1]$ has strictly positive measure for every member. What is the largest possible cardinality of this family?

- (A) One.
(B) Any finite cardinality, but never infinite.
(C) Countably infinite.
(D) The cardinality of $\mathbb{R}$.
(E) Greater than the cardinality of $\mathbb{R}$.

Problem 248.3[See solution](#)

Start with $[0, 1]$, remove the open middle third of each remaining interval, and repeat indefinitely. Let C be the intersection of all the remaining sets. Which description is correct?

- (A) C is countable and has measure zero.
- (B) C is uncountable and has measure zero.
- (C) C is uncountable and has measure $1/3$.
- (D) C is finite and nonempty.
- (E) C is not measurable.

Problem 248.4[See solution](#)

Let $E_n = [n, \infty)$ in $\mathbb{R}$. What is $(\lim_{n \rightarrow \infty} m(E_n), m(\bigcap_{n=1}^{\infty} E_n))$?

- (A) $(0, 0)$
- (B) (∞, ∞)
- (C) $(\infty, 0)$
- (D) $(1, 0)$
- (E) The second measure is undefined.

Problem 248.5[See solution](#)

If A, B are measurable, $m(A) = m(B) = 1$, and $m(A \triangle B) = 0$, what is $m(A \cap B)$?

- (A) 0
- (B) It cannot be determined
- (C) 2
- (D) $1/2$
- (E) 1

7. Additional Topics

Real Analysis ▸ Lebesgue Measurable Functions

249 Measurable functions and characteristic functions

Book pp. 295–296. [Find in compact glossary](#)

In this entry: measurable function; characteristic function; indicator function; pointwise limit; almost-everywhere limit.

Definition and key ideas

A real function is Lebesgue measurable when the inverse image of every open real set is measurable. Continuous functions are measurable, as are sums, differences, products, and absolute values of measurable functions. Pointwise limits of measurable functions are measurable; for Lebesgue measure this remains true for almost-everywhere limits and arbitrary changes on null sets.

The characteristic or indicator function χ_E equals 1 on E and 0 elsewhere. It is measurable exactly when E is measurable. Its discontinuities are exactly the boundary of E , which explains why a measurable function can be discontinuous everywhere.

Pointwise convergence requires $f_n(x) \rightarrow f(x)$ at every point. Almost-everywhere convergence permits a measurable exceptional set of measure zero. Because Lebesgue measure includes every subset of a null set, changing a measurable function arbitrarily on such a set preserves measurability; it need not preserve continuity or boundedness.

Problem 249.1

[See solution](#)

On $[0, 1]$, set $f(x) = x\chi_{\mathbb{Q}}(x)$. At which points is f continuous?

- (A) Every point.
- (B) Every irrational point.
- (C) Every rational point.
- (D) 0 only.
- (E) No point.

Problem 249.2

[See solution](#)

Let $E \subset \mathbb{R}$ be measurable and have both E and $\mathbb{R} \setminus E$ dense in $\mathbb{R}$. Which is necessarily true of χ_E ?

- (A) It is nonmeasurable.
- (B) It is continuous exactly on E .
- (C) It is continuous exactly on the complement.
- (D) It is measurable and nowhere continuous.
- (E) It is constant almost everywhere.

Problem 249.3[See solution](#)

Let $N \subset \mathbb{R}$ have Lebesgue measure zero, and let $h : N \rightarrow \mathbb{R}$ be arbitrary. Define $f = h$ on N and $f = 0$ outside N . Which statement must hold?

- (A) f is continuous.
- (B) f is bounded.
- (C) f vanishes at every point.
- (D) f is nonmeasurable unless h is continuous.
- (E) f is Lebesgue measurable.

Problem 249.4[See solution](#)

For a subset $E \subset \mathbb{R}$, its indicator χ_E is Lebesgue measurable exactly when

- (A) E is countable (B) E is open (C) $m(E) = 0$ (D) E is measurable (E) E is closed

Problem 249.5[See solution](#)

If measurable real functions f_n converge pointwise to a finite function f , which follows?

- (A) f is bounded
- (B) f is continuous
- (C) The convergence is uniform
- (D) f is integrable
- (E) f is measurable

7. Additional Topics

Real Analysis ▸ Lebesgue Measurable Functions

250 Simple functions and approximation from below

Book pp. 295–297. [Find in compact glossary](#)**In this entry:** simple function; step function; finite-valued function; simple-function approximation.**Definition and key ideas**

A measurable simple function takes finitely many values and can be written $s = \sum a_j \chi_{E_j}$ with measurable level sets E_j . The canonical level sets are disjoint. In elementary calculus a step function often means constant on finitely many intervals; measure theory allows arbitrary measurable level sets.

Every nonnegative measurable function is the increasing pointwise limit of nonnegative simple functions, using truncation and dyadic rounding. For the canonical disjoint representation of an integrable simple function, $\int s \, dm = \sum a_j m(E_j)$. Overlapping measurable sets are also permitted when each displayed summand is integrable; otherwise a displayed sum may contain undefined infinite cancellations. Determine absolute integrability from the actual level values and disjoint level sets.

A real measurable function admits uniform approximation by finite-valued simple functions exactly when it is bounded. A bounded range allows a finite grid of values with arbitrarily small error; an unbounded function cannot be uniformly close to even one finite-valued function.

Problem 250.1[See solution](#)Let $s = 2\chi_{[0,2]} - 3\chi_{[1,3]}$. What is $\int_{\mathbb{R}} |s| \, dm$?

- (A) 2 (B) 4 (C) 5 (D) 6 (E) 10

Problem 250.2[See solution](#)For $x \in [0, 1)$ define $s_n(x) = 2^{-n} \lfloor 2^n x \rfloor$. What is $\int_0^1 s_n(x) \, dx$?

- (A)
- $1/2$
- (B)
- $(1 - 2^{-n})/2$
- (C)
- $1 - 2^{-n}$
- (D)
- 2^{-n}
- (E)
- $(1 + 2^{-n})/2$

Problem 250.3[See solution](#)

Let f be a real measurable function on a nonempty set with a sigma algebra. Which condition is equivalent to the existence of finite-valued measurable simple functions converging uniformly to f ?

- (A) f is bounded.
 (B) f is nonnegative.
 (C) f has countable range.
 (D) f is integrable.
 (E) f vanishes outside a set of finite measure.

Problem 250.4[See solution](#)

For a nonnegative measurable function f , define $s_n = 2^{-n} \lfloor 2^n \min(f, n) \rfloor$. Which statement must hold?

- (A) Each s_n has at most n distinct values
- (B) $s_n \rightarrow f$ uniformly on every measure space
- (C) $s_n \downarrow f$ pointwise
- (D) $s_n = f$ for all sufficiently large n
- (E) $s_n \uparrow f$ pointwise

Problem 250.5[See solution](#)

On $\mathbb{R}$, set $s = \chi_{[0, \infty)} - \chi_{[1, \infty)}$. Which statement is correct?

- (A) s is not a simple function.
- (B) $\int s$ is undefined because the displayed terms each have infinite integral.
- (C) s is integrable and $\int s = 1$.
- (D) $\int |s| = \infty$.
- (E) $\int s = 0$.

7. Additional Topics

Real Analysis ▸ Lebesgue Integrable Functions

251 Lebesgue integral and positive and negative parts

Book pp. 296-297. [Find in compact glossary](#)

In this entry: Lebesgue integral; Lebesgue integrable function; positive part; negative part; absolute integrability.

Definition and key ideas

For nonnegative measurable f , define $\int f \, dm$ as the supremum of $\int s \, dm$ over nonnegative measurable simple $s \leq f$; the value may be infinity. Integrability requires this supremum to be finite, not merely each individual simple integral to be finite.

Write $f = f^+ - f^-$ where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$; both are nonnegative. A real measurable function is Lebesgue integrable when $\int |f| < \infty$, equivalently both parts have finite integrals. Then $\int f = \int f^+ - \int f^-$. The expression $\infty - \infty$ is never a valid integral.

For a nonnegative measurable function, $\int f = 0$ holds exactly when $f = 0$ almost everywhere. Indeed each set $\{f \geq 1/k\}$ must be null; their countable union contains every point where $f > 0$. This conclusion does not require pointwise vanishing.

Problem 251.1

[See solution](#)

For real p , define $f_p(x) = x^{-p}$ on $(0, 1)$ and 0 elsewhere. For which p is f_p Lebesgue integrable?

(A) $p < 0$ (B) $p \leq 0$ (C) $p < 1$ (D) $p \leq 1$ (E) Every real p .

Problem 251.2

[See solution](#)

An integrable real function satisfies $\int f \, dm = 3$ and $\int |f| \, dm = 11$. What is $\int f^- \, dm$?

(A) 3 (B) 4 (C) 7 (D) 8 (E) 11

Problem 251.3

[See solution](#)

For which real p is $f(x) = 1/[x(\log x)^p]$ Lebesgue integrable on (e, ∞) ?

(A) $p > 1$ (B) $p < 1$ (C) $p \geq 1$ (D) All real p (E) $p > 0$

Problem 251.4[See solution](#)

A nonnegative measurable function f on $[0, 1]$ satisfies $\int f \, dm = 0$. Which conclusion must hold?

- (A) $f(x) = 0$ for every x .
- (B) f is bounded.
- (C) $\{x : f(x) > 0\}$ has measure zero.
- (D) f is continuous.
- (E) $\{x : f(x) > 0\}$ is countable.

Problem 251.5[See solution](#)

Let $f(x) = 1/x$ on $(0, 1)$. For every nonnegative measurable simple function $s \leq f$, the integral of s is finite. What follows about f ?

- (A) f is integrable because all these simple integrals are finite.
- (B) $\int_0^1 f = \infty$; the simple integrals are not bounded by a common finite number.
- (C) f is not measurable.
- (D) $\int_0^1 f = 1$.
- (E) The integral of f is undefined even as a nonnegative extended integral.

7. Additional Topics

Real Analysis ▸ Lebesgue Integrable Functions

252 Riemann versus Lebesgue integration

Book pp. 297-298. [Find in compact glossary](#)

In this entry: Riemann integrability criterion; Lebesgue criterion; Dirichlet function; conditional improper integral.

Definition and key ideas

A bounded function on a closed bounded interval is Riemann integrable if and only if its discontinuity set has Lebesgue measure zero. Such a function is Lebesgue integrable with the same integral. Countably many discontinuities suffice but are not necessary: an uncountable null discontinuity set is allowed.

The Dirichlet function $\chi_{\mathbb{Q}}$ has Lebesgue integral zero but is nowhere continuous and is not Riemann integrable on a nondegenerate interval. Also distinguish a conditionally convergent improper Riemann integral from Lebesgue integrability, which requires absolute integrability. For example $\int_1^\infty \sin x/x \, dx$ converges improperly, but $\int_1^\infty |\sin x|/x \, dx$ diverges.

Problem 252.1

[See solution](#)

Let $C \subseteq [0, 1]$ be the middle-thirds Cantor set. Its measure is zero. Which statement about χ_C on $[0, 1]$ is true?

- (A) It is not Riemann integrable because C is uncountable.
- (B) It is Riemann integrable with integral 0.
- (C) It is Riemann integrable with integral 1.
- (D) It is not Lebesgue measurable.
- (E) Its Lebesgue integral is positive.

Problem 252.2

[See solution](#)

For $x \in [k, k+1)$, $k = 1, 2, \dots$, define $f(x) = (-1)^{k+1}/k$. Which description is correct?

- (A) f is Lebesgue integrable and $\int_1^\infty f = \log 2$.
- (B) The improper integral diverges because f has infinitely many jumps.
- (C) The improper integral converges, but $\int f^+ = \int f^- = \infty$.
- (D) The improper integral equals zero.
- (E) $\int_1^\infty |f| < \infty$ but the improper integral diverges.

Problem 252.3

[See solution](#)

For the Dirichlet function $f = \chi_{\mathbb{Q}}$ on $[0, 1]$, let P be any finite partition into nondegenerate subintervals. What are its lower Darboux sum, upper Darboux sum, and Lebesgue integral, respectively?

- (A) $(0, 0, 0)$ (B) $(0, 1, 0)$ (C) $(1, 1, 1)$ (D) $(0, 1, 1)$ (E) $(0, 1, \text{undefined})$

Problem 252.4[See solution](#)

On $[0, 1]$ define $f(0) = 0$ and $f(x) = x^{-1/2}$ for $x > 0$. Which statement is correct?

- (A) f is properly Riemann integrable with integral two.
- (B) f is not Lebesgue integrable because it is unbounded.
- (C) f has neither a Lebesgue nor an improper Riemann integral.
- (D) f is Lebesgue integrable and improperly Riemann integrable, but not properly Riemann integrable.
- (E) Changing the value at zero makes f properly Riemann integrable.

Problem 252.5[See solution](#)

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable. Which statement must hold?

- (A) Its Lebesgue integral exists and equals its Riemann integral.
- (B) It is continuous except at a countable set.
- (C) Every function equal to f almost everywhere is Riemann integrable.
- (D) It has only finitely many discontinuities.
- (E) It is uniformly continuous.

7. Additional Topics

Real Analysis ▸ Lebesgue Integrable Functions

253 Monotone and dominated convergence

Book pp. 298. [Find in compact glossary](#)

In this entry: monotone convergence theorem; Levi theorem; dominated convergence theorem; interchange of limit and integral.

Definition and key ideas

The monotone convergence theorem states that if $0 \leq f_n \uparrow f$ almost everywhere with all functions measurable, then $\int f_n \rightarrow \int f$, with infinite values allowed. Nonnegativity and monotonicity are the relevant hypotheses.

The dominated convergence theorem states that if every f_n is measurable, $f_n \rightarrow f$ almost everywhere and $|f_n| \leq g$ for a single integrable function g , then f is integrable and $\int f_n \rightarrow \int f$; in fact $\int |f_n - f| \rightarrow 0$. A uniform bound on the numbers $\int |f_n|$ is insufficient. Concentrating mass on shrinking sets is the standard counterexample.

The monotone convergence theorem is also called Levi's theorem. Its conclusion extends to increasing signed functions bounded below by $-g$ for one nonnegative integrable g : apply the nonnegative theorem to $f_n + g$ and subtract the finite integral of g .

Problem 253.1

[See solution](#)

On $(0, 1)$ let $f_n(x) = n\chi_{(0,1/n)}(x)$. Which pair equals $(\lim_n \int_0^1 f_n, \int_0^1 \lim_n f_n)$?

- (A) $(0, 0)$ (B) $(0, 1)$ (C) $(1, 0)$ (D) $(1, 1)$ (E) $(\infty, 0)$

Problem 253.2

[See solution](#)

Let $f_n(x) = \min\{n, x^{-1/2}\}$ on $(0, 1)$. What is $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx$?

- (A) 1 (B) $3/2$ (C) 2 (D) π (E) ∞

Problem 253.3

[See solution](#)

On a measure space, measurable real functions satisfy $f_n \uparrow f$ almost everywhere and $f_n \geq -g$ for one nonnegative integrable function g . Which conclusion is valid?

- (A) The integrals need not approach $\int f$ because some f_n may be negative.
(B) $\int f_n \rightarrow \int f$, with $+\infty$ allowed.
(C) $f_n \rightarrow f$ uniformly.
(D) $\int f$ must be finite.
(E) $\int |f_n - f| \rightarrow 0$ even when $\int f = \infty$.

Problem 253.4

[See solution](#)

What is $\lim_{n \rightarrow \infty} \int_0^1 \frac{nx}{1 + n^2 x^2} dx$?

- (A) 0 (B) $\log 2$ (C) $+\infty$ (D) $1/2$ (E) 1

Problem 253.5[See solution](#)

What is $\lim_{n \rightarrow \infty} \int_0^2 (1 - x/n)^n dx$?

- (A) $e^2 - 1$ (B) $1 - e^{-2}$ (C) $2e^{-2}$ (D) 1 (E) 2

7. Additional Topics

Real Analysis ▸ Lebesgue Integrable Functions

254 Lebesgue spaces and completeness

Book pp. 298. [Find in compact glossary](#)**In this entry:** L^p space; L^p norm; almost-everywhere equivalence; Banach space; Holder inequality; conjugate exponents.**Definition and key ideas**

For $1 \leq p < \infty$, L^p consists of measurable functions with $\int |f|^p < \infty$, identifying functions equal almost everywhere. Its norm is $\|f\|_p = (\int |f|^p)^{1/p}$. Identification is needed because a nonzero function supported on a null set has norm zero.

Each L^p is complete in this norm, so is a Banach space. On a finite-measure domain, $L^q \subseteq L^p$ for $q > p$, but the inclusion can fail on infinite-measure domains. Norm convergence does not generally imply pointwise convergence of the full sequence, although an almost-everywhere convergent subsequence can be extracted.

A Banach space is a normed vector space in which every Cauchy sequence for the norm converges in that norm to an element of the space. In L^p , the limit is unique as an almost-everywhere equivalence class, not as a pointwise-defined representative. Holder's inequality states that if $1 < p, q < \infty$ and $1/p + 1/q = 1$, then $\int |fg| \leq \|f\|_p \|g\|_q$. Such p, q are conjugate exponents. Applying this to f and the constant function one gives the finite-measure inclusions; $p = q = 2$ is Cauchy-Schwarz.

Problem 254.1[See solution](#)

On $(0, 1)$, for which $p \geq 1$ is $f(x) = x^{-1/3}$ in L^p ?

(A) Every $p \geq 1$. (B) $1 \leq p < 3$ (C) $1 \leq p \leq 3$ (D) $p > 3$ (E) Only $p = 3$.

Problem 254.2[See solution](#)

Which function on $(1, \infty)$ belongs to L^2 but not L^1 ?

(A) x^{-2} (B) e^{-x} (C) $x^{-3/4}$ (D) $x^{-1/4}$ (E) 1

Problem 254.3[See solution](#)

For $f(x) = x^{-1/4}[\log(1/x)]^{-1/2}$ on $(0, e^{-1})$, for which $p \geq 1$ is $f \in L^p$?

(A) $p = 4$ only (B) Every $p \geq 1$ (C) $1 \leq p < 4$ (D) $1 \leq p \leq 4$ (E) $p > 4$

Problem 254.4[See solution](#)

On a measure space with total measure 3, suppose $\|f\|_2 = 2$. What is the smallest universal upper bound for $\|f\|_1$?

(A) 2 (B) 4 (C) 3 (D) 6 (E) $2\sqrt{3}$

Problem 254.5[See solution](#)

Let $f \in L^3$ and $g \in L^{3/2}$ on the same measure space. Which quantity is necessarily finite?

- (A) $\int |f|^4$ (B) $\int |g|$ (C) $\int |g|^3$ (D) $\int |f|$ (E) $\int |fg|$

Problem 254.6[See solution](#)

For $1 \leq p < \infty$, a sequence (f_n) is Cauchy in $L^p([0, 1])$. Which conclusion must hold?

- (A) There is a unique pointwise representative f to which every $f_n(x)$ converges.
(B) The sequence converges uniformly to a continuous function.
(C) There is $f \in L^p$ with $\|f_n - f\|_p \rightarrow 0$, and any other such limit equals f almost everywhere.
(D) The sequence is eventually constant almost everywhere.
(E) The sequence converges pointwise everywhere, although its limit may not belong to L^p .

Problem 254.7[See solution](#)

For $n = 2^k + j$ with $k \geq 0$ and $0 \leq j < 2^k$, let $f_n = \chi_{[j2^{-k}, (j+1)2^{-k})}$ on $[0, 1)$. For fixed $1 \leq p < \infty$, which statement is correct?

- (A) $\|f_n\|_p \not\rightarrow 0$.
(B) f_n converges pointwise to zero everywhere.
(C) f_n converges uniformly to zero.
(D) $\|f_n\|_p \rightarrow 0$, but $f_n(x)$ fails to converge at every $x \in [0, 1)$.
(E) $f_n(x) \rightarrow 1$ for every x .

7. Additional Topics

Complex Variables ▸ Introduction

255 Complex arithmetic and conjugation

Book pp. 298-299. [Find in compact glossary](#)

In this entry: complex number; complex field; imaginary unit; real part; imaginary part; complex conjugate; complex plane; real axis; imaginary axis; pure imaginary number.

Definition and key ideas

A complex number is $z = x + iy$ with real x, y and $i^2 = -1$. Its real and imaginary parts are x, y , and its conjugate is $\bar{z} = x - iy$. Conjugation preserves sums and products, and $z\bar{z} = x^2 + y^2$. Division by a nonzero complex number is simplified by multiplying numerator and denominator by its conjugate.

Complex numbers form a field but cannot be ordered compatibly with the ordered-field axioms: a square would have to be nonnegative, contradicting $i^2 = -1$. For equations involving z and $\bar{z}$, either write real and imaginary components or use polar form when products and powers dominate.

The complex plane identifies $x + iy$ with the point (x, y) . The real axis consists of numbers with $y = 0$; the imaginary axis consists of numbers with $x = 0$. Numbers of the form iy are purely imaginary (some conventions reserve that phrase for $y \neq 0$). Conjugation reflects the plane in the real axis.

Problem 255.1

[See solution](#)

For real a, b , consider $z + \bar{z} = a + ib$. Which statement gives the number of complex solutions?

- (A) Exactly one for every a, b .
- (B) Exactly two when $b = 0$, and none otherwise.
- (C) Infinitely many when $b = 0$, and none otherwise.
- (D) No solutions unless $a = b = 0$, in which case there is one.
- (E) Infinitely many for every a, b .

Problem 255.2

[See solution](#)

How many distinct complex numbers satisfy $z^2 = \bar{z}$?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Problem 255.3[See solution](#)

Why can the field $\mathbb{C}$ not be given an order satisfying the ordered-field axioms?

- (A) Its addition is not commutative.
- (B) It has nonzero zero divisors.
- (C) Every ordered field must be finite.
- (D) In an ordered field every nonzero square is positive, but $i^2 = -1$.
- (E) The conjugate of a nonreal number is not unique.

Problem 255.4[See solution](#)

If $|z| = 1$ and $z + \bar{z} = 1$, what is $z^4 + \bar{z}^4$?

- (A) 1 (B) 0 (C) -2 (D) -1 (E) 2

Problem 255.5[See solution](#)

For real t , which point is omitted from the set $\{(t - i)/(t + i) : t \in \mathbb{R}\}$ on the unit circle?

- (A) i (B) 1 (C) $-i$ (D) -1 (E) No point is omitted.

7. Additional Topics

Complex Variables ▸ The Polar Form

256 Modulus, arguments, and polar geometry

Book pp. 299-300. [Find in compact glossary](#)

In this entry: modulus; absolute value; argument; principal argument; polar form; triangle inequality in $\mathbb{C}$.

Definition and key ideas

For $z = x + iy$, its modulus is $|z| = \sqrt{x^2 + y^2}$. For $z \neq 0$, write $z = r(\cos \theta + i \sin \theta)$ with $r = |z| > 0$. Arguments differ by multiples of 2π ; the principal argument convention here is $-\pi < \text{Arg } z \leq \pi$. Zero has modulus zero but has no argument. Products multiply moduli and add arguments, with adjustment back to the principal interval when needed.

The identity $|z - w|$ equals Euclidean distance makes complex equations into geometric loci. In particular, equal distances give a perpendicular bisector, and a nonunit ratio of distances gives an Apollonius circle. The inequalities $||z| - |w|| \leq |z + w| \leq |z| + |w|$ are useful without computing coordinates.

Problem 256.1[See solution](#)

Which set is described by $|z| + |z - 2| = 2$?

- (A) The circle $|z - 1| = 1$.
- (B) The closed real segment $[0, 2]$.
- (C) The two points 0 and 2 only.
- (D) The whole real axis.
- (E) The disk $|z - 1| \leq 1$.

Problem 256.2[See solution](#)

What is the center of the locus $|z - 1| = 2|z + 1|$?

- (A) $5/3$ (B) $-5/3$ (C) -1 (D) $1/3$ (E) 0

Problem 256.3[See solution](#)

Among points satisfying $|z - 1| = |z - i|$, what is the minimum value of $|z - 2|$?

- (A) $1/\sqrt{2}$ (B) 1 (C) $\sqrt{2}$ (D) 2 (E) $\sqrt{5}$

Problem 256.4[See solution](#)

If $|z| = 2$ and $\text{Re } z \geq 0$, what is the largest possible value of $|z - 3|$?

- (A) 3 (B) 5 (C) $\sqrt{13}$ (D) 2 (E) 1

Problem 256.5[See solution](#)

For $z \neq 0$, suppose $\arg z = \pi/5$ modulo 2π . What is the principal argument of $-\bar{z}^3$, in $(-\pi, \pi]$?

- (A) $4\pi/5$ (B) $-3\pi/5$ (C) $2\pi/5$ (D) $3\pi/5$ (E) $-2\pi/5$

7. Additional Topics

Complex Variables ▸ The Exponential Form

257 Euler formula, complex exponential, and de Moivre formula

Book pp. 300. [Find in compact glossary](#)

In this entry: Euler formula; exponential form; complex exponential; de Moivre formula; exponential periodicity.

Definition and key ideas

Euler's identity $e^{i\theta} = \cos \theta + i \sin \theta$ gives $z = re^{i\theta}$. For $z = x + iy$, $e^z = e^x(\cos y + i \sin y)$, so $|e^z| = e^x$ and e^z is never zero. Its periods are $2\pi ik$ for integers k , rather than real multiples of 2π .

For an integer n , de Moivre's formula is $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$, with nonzero base required for negative n . Equality $e^u = e^v$ means $u - v \in 2\pi i\mathbb{Z}$; it does not force $u = v$. This accounts for the infinitely many solutions of exponential equations.

Problem 257.1

[See solution](#)What is $(1 + i)^{12}$?

- (A) 64 (B) -64 (C) $64i$ (D) $-64i$ (E) $-32 + 32i$

Problem 257.2

[See solution](#)What is the full solution set of $e^{(1+i)z} = 1$?

- (A) $\{2\pi k : k \in \mathbb{Z}\}$ (B) $\{\pi k(1 + i) : k \in \mathbb{Z}\}$
 (C) $\{\pi k(1 - i) : k \in \mathbb{Z}\}$ (D) $\{2\pi ik : k \in \mathbb{Z}\}$
 (E) $\{0\}$

Problem 257.3

[See solution](#)What is $\sum_{k=0}^8 \binom{8}{k} \cos(k\pi/2)$?

- (A) 0 (B) 8 (C) 16 (D) 256 (E) -16

Problem 257.4

[See solution](#)Among all solutions of $e^z = -e$, what is the smallest value of $|z|$?

- (A) $\sqrt{1 + 4\pi^2}$ (B) $\sqrt{1 + \pi^2}$ (C) π (D) 1 (E) e

Problem 257.5[See solution](#)

What is the image of the strip $\{z : 0 < \operatorname{Im} z < 2\pi\}$ under $z \mapsto e^z$, and is the map one-to-one on that strip?

- (A) $\mathbb{C} \setminus \{0\}$; yes.
- (B) $\mathbb{C} \setminus [0, \infty)$; yes.
- (C) $\mathbb{C} \setminus (-\infty, 0]$; yes.
- (D) $\mathbb{C} \setminus [0, \infty)$; no.
- (E) The open unit disk; no.

7. Additional Topics

Complex Variables ▸ Complex Roots

258 Complex roots and roots of unity

Book pp. 300-301. [Find in compact glossary](#)**In this entry:** n th root; principal n th root; roots of unity; primitive root of unity.**Definition and key ideas**

For $w = Re^{i\theta} \neq 0$, its n distinct n th roots are $R^{1/n}e^{i(\theta+2\pi k)/n}$ for $k = 0, \dots, n-1$. They form equally spaced points on a circle. The principal root uses $\theta = \text{Arg } w$ and $k = 0$. Zero has only one distinct n th root, not n distinct roots.

The n th roots of unity are powers of $\omega = e^{2\pi i/n}$; ω^k is primitive exactly when $\gcd(k, n) = 1$. Their geometric sums vanish unless the exponent is divisible by n . Factoring $z^n - 1$ often evaluates products or sums over roots much faster than trigonometric arithmetic.

Problem 258.1[See solution](#)How many roots of $z^6 = -64$ have strictly positive real part?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 258.2[See solution](#)Let $\omega = e^{2\pi i/7}$. What is $\prod_{k=1}^6 (1 - \omega^k)$?

- (A) 1 (B) 6 (C) 7 (D)
- -7
- (E) 49

Problem 258.3[See solution](#)Let $\omega = e^{2\pi i/9}$. What is $\sum_{k=1}^8 \frac{1}{1 - \omega^k}$?

- (A) 8 (B) 4 (C) 1 (D) 0 (E)
- $9/2$

Problem 258.4[See solution](#)

Which is the monic polynomial of least degree whose roots are exactly the primitive twelfth roots of unity?

- (A)
- $z^4 + z^2 + 1$
- (B)
- $z^4 - z^2 + 1$
- (C)
- $z^6 + 1$
- (D)
- $z^4 + 1$
- (E)
- $z^2 - z + 1$

Problem 258.5[See solution](#)Let r_1, r_2, r_3 be the three roots of $z^3 = -8$. Taking each square root to be its principal value with argument in $(-\pi/2, \pi/2]$, what is $\sqrt{r_1} + \sqrt{r_2} + \sqrt{r_3}$?

- (A) 0 (B)
- $\sqrt{6}$
- (C)
- $i\sqrt{2}$
- (D)
- $\sqrt{6} + i\sqrt{2}$
- (E)
- $\sqrt{6} - i\sqrt{2}$

7. Additional Topics

Complex Variables ▸ Complex Logarithms

259 Multivalued logarithm and principal logarithm

Book pp. 302. [Find in compact glossary](#)**In this entry:** complex logarithm; principal logarithm; logarithm branch; branch cut.

Definition and key ideas

Every logarithm of $z \neq 0$ has form $\log |z| + i(\text{Arg } z + 2\pi k)$, $k \in \mathbb{Z}$. The principal value is $\text{Log } z = \log |z| + i \text{Arg } z$. It is the imaginary part of this value, not its own argument, that lies in $(-\pi, \pi]$.

A holomorphic logarithm requires a consistent branch on a suitable domain. The usual principal analytic branch lives on $\mathbb{C} \setminus (-\infty, 0]$ and has derivative $1/z$. Principal logarithms need not satisfy $\text{Log}(zw) = \text{Log } z + \text{Log } w$; the discrepancy can be an integer multiple of $2\pi i$.

A branch is a single-valued choice that varies continuously (and, for an analytic branch, holomorphically) on its chosen domain. A branch cut removes a curve or ray so that such a consistent choice becomes possible. The principal value assigns argument π to negative real numbers, but jumps across that ray; the principal analytic branch therefore excludes the ray itself as well as zero.

Problem 259.1

[See solution](#)

Let $z = w = e^{3\pi i/4}$. What is $\text{Log } z + \text{Log } w - \text{Log}(zw)$?

- (A) 0 (B) πi (C) $-\pi i$ (D) $2\pi i$ (E) $-2\pi i$

Problem 259.2

[See solution](#)

For $z \neq 0$, write $\theta = \text{Arg } z \in (-\pi, \pi]$. For which θ does $\text{Log}(z^2) = 2 \text{Log } z$ hold?

- (A) $-\pi < \theta \leq \pi$ (B) $-\pi/2 < \theta \leq \pi/2$
 (C) $-\pi/2 \leq \theta < \pi/2$ (D) $0 \leq \theta \leq \pi$
 (E) $\theta = 0$ only.

Problem 259.3

[See solution](#)

Using $\text{Arg } z \in (-\pi, \pi]$, what is $\text{Log}(e^{2+7\pi i/3})$?

- (A) $2 - 5\pi i/3$ (B) $e^2 + \pi i/3$ (C) $2 + 7\pi i/3$ (D) $2 - \pi i/3$ (E) $2 + \pi i/3$

Problem 259.4[See solution](#)

Can there be a continuous function L on the entire unit circle such that $e^{L(z)} = z$ at every point of the circle?

- (A) Yes, with $L(e^{it}) = it$ for every real t .
- (B) Yes, and such L must be constant.
- (C) Yes, if only real values of L are used.
- (D) No, because the exponential has zeros.
- (E) No, because a full circuit would change a continuous logarithm by $2\pi i$ at the same point.

Problem 259.5[See solution](#)

On the slit plane $\mathbb{C} \setminus (-\infty, 0]$, let L be analytic and satisfy $e^{L(z)} = z$ and $L(1) = 4\pi i$. What is $L(i)$?

- (A) $7\pi i/2$ (B) $4\pi i$ (C) $5\pi i/2$ (D) $\pi i/2$ (E) $9\pi i/2$

7. Additional Topics

Complex Variables ▸ Complex Powers

260 Complex powers and branch dependence

Book pp. 302-303. [Find in compact glossary](#)**In this entry:** complex power; principal value; multivalued power; failure of radical identities.**Definition and key ideas**

For $z \neq 0$, values of z^w are $\exp(w(\operatorname{Log} z + 2\pi ik))$; the principal value uses $k = 0$. These values are not always infinitely many: integer exponents are single-valued, rational real exponents have finitely many values, and other exponents can have infinitely many.

Familiar radical identities require care with branches. For example principal square roots give $\sqrt{-1}\sqrt{-1} = -1$ but $\sqrt{(-1)(-1)} = 1$. A chosen power branch is analytic where its logarithm branch is analytic, and differentiation there gives $(z^w)' = wz^{w-1}$ with consistent choices.

Problem 260.1[See solution](#)What are all possible values of i^i ?

- (A) $\{e^{-\pi/2}\}$ (B) $\{e^{-\pi/2-2\pi k} : k \in \mathbb{Z}\}$
 (C) $\{e^{i(\pi/2+2\pi k)} : k \in \mathbb{Z}\}$ (D) $\{\pm e^{-\pi/2}\}$
 (E) $\{e^{\pi/2+2\pi ik} : k \in \mathbb{Z}\}$

Problem 260.2[See solution](#)

For fixed $z \neq 0$, how many distinct values does $z^{3/8}$ have when all complex logarithms are allowed?

- (A) 1 (B) 3 (C) 4 (D) 8 (E) Infinitely many.

Problem 260.3[See solution](#)

On the principal slit plane let $F(z) = \exp(i \operatorname{Log} z)$. What is $F'''(1)$?

- (A) $-1 - i$ (B) $-1 + i$ (C) i (D) $1 + i$ (E) -1

Problem 260.4[See solution](#)

For fixed $z \neq 0$, let every logarithm of z be allowed in defining its powers. How many distinct values do z^2 , $z^{\sqrt{2}}$, and z^{1+i} have, respectively?

- (A) One, one, one.
 (B) One, infinitely many, infinitely many.
 (C) Two, infinitely many, infinitely many.
 (D) One, infinitely many, one.
 (E) Infinitely many, infinitely many, infinitely many.

Problem 260.5[See solution](#)

All powers and square roots below use principal complex values. Which identities are true? I. $\sqrt{-1}\sqrt{-1} = \sqrt{(-1)(-1)}$; II. $((-1)^2)^{1/2} = -1$; III. $(-1)^{2/3} = ((-1)^{1/3})^2$.

(A) I only (B) II only (C) III only (D) I and III only (E) I, II, and III

7. Additional Topics

Complex Variables ▸ The Trigonometric Functions

261 Complex trigonometric functions and equations

Book pp. 303-304. [Find in compact glossary](#)

In this entry: complex sine; complex cosine; complex tangent; complex cotangent; complex secant; complex cosecant; trigonometric identities over $\mathbb{C}$.

Definition and key ideas

Define $\cos z = (e^{iz} + e^{-iz})/2$ and $\sin z = (e^{iz} - e^{-iz})/(2i)$. Quotients define tangent, cotangent, secant, and cosecant where their denominators are nonzero. Addition formulas, $\sin^2 z + \cos^2 z = 1$, and period 2π remain valid over $\mathbb{C}$.

Unlike their restrictions to the real axis, complex sine and cosine are unbounded. To solve equations such as $\cos z = a$, put $w = e^{iz}$, solve a quadratic for nonzero w , then use all logarithms of w . Keeping all periods is essential to obtain the full solution set.

Explicitly, $\tan z = \sin z / \cos z$, $\cot z = \cos z / \sin z$, $\sec z = 1 / \cos z$, and $\csc z = 1 / \sin z$, wherever the indicated denominator is nonzero. Thus $\sec^2 z - \tan^2 z = 1$ and $\csc^2 z - \cot^2 z = 1$ on their respective domains. Each identity may extend across points where its separate terms are undefined, if their difference has a removable singularity.

Problem 261.1

[See solution](#)

What is the full solution set of $\cos z = 2$?

- (A) $z = 2\pi k \pm i \log(2 + \sqrt{3})$, $k \in \mathbb{Z}$ (B) $z = \pi k \pm \log(2 + \sqrt{3})$, $k \in \mathbb{Z}$
 (C) $z = 2\pi k \pm \log(2 + \sqrt{3})$, $k \in \mathbb{Z}$ (D) $z = \pi/2 + 2\pi k \pm i \log(2 + \sqrt{3})$, $k \in \mathbb{Z}$
 (E) No solutions.

Problem 261.2

[See solution](#)

Which are all zeros of $\sin z$ in $\mathbb{C}$?

- (A) $\{k\pi : k \in \mathbb{Z}\}$ (B) $\{ik\pi : k \in \mathbb{Z}\}$
 (C) $\{2k\pi : k \in \mathbb{Z}\}$ (D) $\{\pi/2 + k\pi : k \in \mathbb{Z}\}$
 (E) $\{k\pi + i\ell\pi : k, \ell \in \mathbb{Z}\}$

Problem 261.3[See solution](#)

Put $L = \log(3 + \sqrt{10})$. Which set gives all solutions of $\sin z = 3i$?

- (A) $\{k\pi + iL : k \in \mathbb{Z}\}$
- (B) $\{2k\pi \pm iL : k \in \mathbb{Z}\}$
- (C) $\{2k\pi + iL, (2k + 1)\pi - iL : k \in \mathbb{Z}\}$
- (D) There are no solutions.
- (E) $\{\pi/2 + k\pi + iL : k \in \mathbb{Z}\}$

Problem 261.4[See solution](#)

How many complex solutions does $\tan z = i$ have?

- (A) None.
- (B) Exactly one.
- (C) Exactly two.
- (D) Countably infinitely many.
- (E) Uncountably infinitely many.

Problem 261.5[See solution](#)

Where is the complex function $\tan z$ singular?

- (A) At $z = k\pi i$, with simple poles
- (B) Nowhere in $\mathbb{C}$
- (C) At $z = \pi/2 + k\pi$, with simple poles
- (D) At $z = k\pi$, with simple poles
- (E) At $z = \pi/2 + k\pi$, with removable singularities

Problem 261.6[See solution](#)

Initially define $F(z) = \sec^2 z - \tan^2 z$ where $\cos z \neq 0$ and $G(z) = \csc^2 z - \cot^2 z$ where $\sin z \neq 0$. Which statement is correct?

- (A) Both functions extend to the constant function one on all of $\mathbb{C}$.
- (B) Only F has any removable singularities.
- (C) Only G extends to an entire function.
- (D) Both have nonremovable poles at their excluded points.
- (E) Both extend to the constant function zero.

7. Additional Topics

Complex Variables ▸ The Hyperbolic Functions

262 Hyperbolic functions and complex trigonometric modulus

Book pp. 304-305. [Find in compact glossary](#)

In this entry: hyperbolic sine; hyperbolic cosine; $\sinh$; $\cosh$; hyperbolic identity; hyperbolic tangent; hyperbolic cotangent.

Definition and key ideas

The definitions $\cosh z = (e^z + e^{-z})/2$ and $\sinh z = (e^z - e^{-z})/2$ give $\cosh^2 z - \sinh^2 z = 1$. Also $\cos(iz) = \cosh z$, $\sin(iz) = i \sinh z$, $\cosh(iz) = \cos z$, and $\sinh(iz) = i \sin z$. For real inputs, $\cosh x \geq 1$ while $\sinh$ ranges over all real numbers.

Writing $z = x + iy$ gives $\sin z = \sin x \cosh y + i \cos x \sinh y$ and $\cos z = \cos x \cosh y - i \sin x \sinh y$. Consequently $|\sin z|^2 = \sin^2 x + \sinh^2 y$ and $|\cos z|^2 = \cos^2 x + \sinh^2 y$. These identities reduce modulus bounds to real inequalities.

The hyperbolic tangent is $\tanh z = \sinh z / \cosh z$, and the hyperbolic cotangent is $\coth z = \cosh z / \sinh z$, each defined when its denominator is nonzero. The exponential definitions also show that $\cosh$ is even and $\sinh$ is odd.

Problem 262.1

[See solution](#)

What is the minimum of $|\sin(x + i \log 2)|$ over real x ?

- (A) 0 (B) $1/2$ (C) $3/4$ (D) 1 (E) $5/4$

Problem 262.2

[See solution](#)

For $z = x + iy$, which expression equals $|\sin z|^2 + |\cos z|^2$?

- (A) 1 (B) $\cos(2x)$ (C) $\cosh(2y)$ (D) $\sinh(2y)$ (E) e^{2x}

Problem 262.3

[See solution](#)

What are all complex solutions of $\coth z = 2$, where $\coth z = \cosh z / \sinh z$?

- (A) $\frac{1}{2} \log 3 + 2\pi ik, k \in \mathbb{Z}$ (B) $\log 3 + \pi ik, k \in \mathbb{Z}$
 (C) $\frac{1}{2} \log 3 + \pi ik, k \in \mathbb{Z}$ (D) $-\frac{1}{2} \log 3 + \pi ik, k \in \mathbb{Z}$
 (E) No solutions.

Problem 262.4

[See solution](#)

Fix a real $y > 0$, and let $w = \sin(x + iy) = u + iv$ as real x varies. What curve is traced by w ?

- (A) $u^2 + v^2 = 1$ (B) $u^2 / \cosh^2 y + v^2 / \sinh^2 y = 1$
 (C) $u^2 / \sinh^2 y + v^2 / \cosh^2 y = 1$ (D) $u^2 - v^2 = 1$
 (E) $u^2 / \cosh^2 y - v^2 / \sinh^2 y = 1$

Problem 262.5[See solution](#)

Which identity holds for every complex z at which both sides are defined?

(A) $\tan(iz) = \tanh z$

(B) $\tan(iz) = i \tanh z$

(C) $\tan(iz) = i \coth z$

(D) $\tan(iz) = -i \tanh z$

(E) $\tan(iz) = -\tanh z$

7. Additional Topics

Complex Variables ▸ The Derivative of a Function of a Complex Variable

263 Complex differentiability

Book pp. 305-306. [Find in compact glossary](#)**In this entry:** complex derivative; difference quotient; direction-independent limit.**Definition and key ideas**

The complex derivative is $f'(z_0) = \lim_{h \rightarrow 0} (f(z_0 + h) - f(z_0))/h$, where h approaches zero through every complex direction. Agreement along the real and imaginary axes is necessary but by itself not sufficient. Differentiability implies continuity.

Complex differentiability at one point is weaker than analyticity on a neighborhood. For example $|z|^2$ is complex differentiable at 0 only, while $\bar{z}$ is nowhere complex differentiable. Polynomial, product, quotient, and chain rules apply, and elementary derivative formulas apply on domains where the functions and branches are defined.

Problem 263.1[See solution](#)

At which points is $f(z) = |z|^2$ complex differentiable?

(A) Every point. (B) The real axis. (C) The imaginary axis. (D) 0 only. (E) No point.

Problem 263.2[See solution](#)

Suppose f is complex differentiable at $\bar{a}$ and define $F(z) = \overline{f(\bar{z})}$ near a . Which formula is valid?

- (A) $F'(a) = f'(a)$
 (B) $F'(a) = -\overline{f'(\bar{a})}$
 (C) $F'(a) = \overline{f'(\bar{a})}$
 (D) F cannot be complex differentiable unless f is constant.
 (E) $F'(a) = \overline{f'(a)}$

Problem 263.3[See solution](#)

Let f be holomorphic near $1+i$, and define $g(t) = f((1+i)e^{it})$ for real t near zero. If $g'(0) = 3+i$, what is $f'(1+i)$?

(A) $1+2i$ (B) $-1-2i$ (C) $2-i$ (D) $-2+i$ (E) $3+i$

Problem 263.4[See solution](#)

Define $f(0) = 0$ and $f(z) = \bar{z}^2/z$ for $z \neq 0$. Which statement about f at zero is correct?

- (A) It has no real partial derivatives.
- (B) It is complex differentiable with derivative -1 .
- (C) It is discontinuous.
- (D) It is continuous and has both real partial derivatives, but is not complex differentiable.
- (E) It is complex differentiable with derivative 1 .

Problem 263.5[See solution](#)

If $f(z) = z^2 + \bar{z}$, what is the set of points at which f is complex differentiable?

- (A) The real axis (B) The empty set (C) The imaginary axis (D) $\{0\}$ (E) All of $\mathbb{C}$

7. Additional Topics

Complex Variables ▸ The Cauchy–Riemann Equations

264 Cauchy–Riemann equations

Book pp. 306–307. [Find in compact glossary](#)

In this entry: Cauchy–Riemann equations; necessary condition for complex differentiability; sufficient C1 criterion.

Definition and key ideas

For $f = u + iv$, complex differentiability implies $u_x = v_y$ and $u_y = -v_x$ at the point, and $f' = u_x + iv_x = v_y - iv_y$. If the four partial derivatives exist and are continuous on a neighborhood and the equations hold at the point, the derivative exists there.

Bare existence of partial derivatives and satisfaction of the equations at a point are not sufficient without a differentiability or regularity hypothesis. For polynomial real components the partials are automatically continuous, so solving the two equations gives exactly the differentiability set. Analyticity further requires an open neighborhood of such points.

The real derivative matrix of a holomorphic function is $J = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ when $f' = a + ib$. Consequently $J^T J = |f'|^2 I$ and $\det J = |f'|^2$. At a point with nonzero derivative, the linear approximation is a rotation followed by a positive scaling, rather than a reflection.

Problem 264.1

[See solution](#)

For which real a, b is $f(x + iy) = x^2 + ay^2 + ibxy$ entire?

- (A) $(a, b) = (1, 2)$ (B) $(a, b) = (-1, 2)$
(C) $(a, b) = (-1, -2)$ (D) $(a, b) = (1, -2)$
(E) $(a, b) = (0, 0)$

Problem 264.2

[See solution](#)

Let $f(x + iy) = xy + i(x^2 - y^2)$. At which points is it complex differentiable?

- (A) All points with $x = 0$. (B) All points with $y = 0$.
(C) All points with $x = y$. (D) 0 only.
(E) Every point.

Problem 264.3[See solution](#)

For real a, b , when is $f(x + iy) = e^{ax}(\cos by + i \sin by)$ entire?

- (A) Exactly when $a^2 + b^2 = 1$
- (B) Exactly when $a = b$
- (C) Exactly when $a = -b$
- (D) For all a, b
- (E) Exactly when $a = b = 0$

Problem 264.4[See solution](#)

At a point where a holomorphic function has nonzero derivative, let J be its real 2×2 derivative matrix. Which statement must hold?

- (A) $\det J < 0$
- (B) J is symmetric.
- (C) $\det J = 0$
- (D) $J^T J = (\det J)I$ and $\det J > 0$.
- (E) $\operatorname{tr} J = 0$

Problem 264.5[See solution](#)

If $f = u + iv$ is analytic and $u_x(1, 2) = 3$, $u_y(1, 2) = -4$, what is $f'(1 + 2i)$?

- (A) $3 + 4i$ (B) $-4 + 3i$ (C) $4 + 3i$ (D) $3 - 4i$ (E) 5

7. Additional Topics

Complex Variables ▸ The Cauchy–Riemann Equations

265 Harmonic functions and harmonic conjugates

Book pp. 307; 324; 328. [Find in compact glossary](#)**In this entry:** Laplace equation; harmonic function; harmonic conjugate.**Definition and key ideas**

A twice continuously differentiable real function is harmonic if $u_{xx} + u_{yy} = 0$. The real and imaginary parts of an analytic function are harmonic. A harmonic conjugate v of u makes $u + iv$ analytic and must satisfy $v_y = u_x$, $v_x = -u_y$.

On a simply connected plane domain every harmonic function has a global harmonic conjugate, unique up to an additive real constant; on domains with holes this can fail. To construct one, integrate a Cauchy–Riemann equation, retaining an unknown function of the other variable, then determine it from the second equation.

Harmonic functions are closed under real linear combinations, but generally not under products or nonlinear compositions. For instance, $\Delta(u^2) = 2|\nabla u|^2$ when u is harmonic, so u^2 is harmonic on a connected domain only when u is constant.

Problem 265.1[See solution](#)

Let $f = u + iv$ be entire with $u(x, y) = x^3 - 3xy^2 + 2x$ and $v(0, 0) = 5$. What is $v(1, 2)$?

- (A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Problem 265.2[See solution](#)

Which harmonic function on $\mathbb{C} \setminus \{0\}$ has no single-valued harmonic conjugate on that entire domain?

- (A) x (B) y (C) $x^2 - y^2$ (D) $\log \sqrt{x^2 + y^2}$ (E) $e^x \cos y$

Problem 265.3[See solution](#)

Let u be a real harmonic function on a connected open plane domain. If u^2 is also harmonic, which conclusion follows?

- (A) u must be identically zero.
(B) u must be constant, but need not be zero.
(C) u can be any affine function.
(D) u can be any harmonic polynomial.
(E) u must have no zeros.

Problem 265.4[See solution](#)

For real a, b, c , which condition is necessary and sufficient for $u(x, y) = ax^2 + bxy + cy^2$ to have a harmonic conjugate on all of $\mathbb{R}^2$?

- (A) $a = c$ (B) $b = 0$ (C) $a + c = 0$ (D) $a + b + c = 0$ (E) $b^2 = 4ac$

Problem 265.5[See solution](#)

On $\mathbb{C} \setminus \{0\}$, a harmonic conjugate of $u(x, y) = x/(x^2 + y^2)$ is which function?

- (A) $x/(x^2 + y^2)$
(B) $\log(x^2 + y^2)$
(C) $y/(x^2 + y^2)$
(D) $-y/(x^2 + y^2)$
(E) No global harmonic conjugate exists.

7. Additional Topics

Complex Variables ▸ Analytic Functions

266 Analytic, holomorphic, and entire functions

Book pp. 308. [Find in compact glossary](#)

In this entry: analytic function; holomorphic function; entire function; analytic rational function; higher complex derivatives; identity theorem; isolated zeros.

Definition and key ideas

A function is holomorphic on an open set when it is complex differentiable at every point of that set. In complex analysis this is equivalent to being analytic, meaning locally represented by a convergent power series. An entire function is holomorphic on all of $\mathbb{C}$. Polynomials, e^z , $\sin z$, and $\cos z$ are entire.

Rational functions are holomorphic away from denominator zeros, with cancellations possibly producing removable singularities. Holomorphic functions have derivatives of every order, and those derivatives are holomorphic. Differentiability on a line or at infinitely many isolated points does not imply analyticity on any open set.

The identity theorem states that two holomorphic functions on a connected open domain are identical if they agree at a set with an accumulation point inside that domain. Equivalently, a nonzero holomorphic function has isolated zeros. An accumulation point only on the boundary does not satisfy this hypothesis.

Problem 266.1

[See solution](#)

If f is entire and $\operatorname{Re} f(z) = 0$ for every z , which must be true?

- (A) $f(z) = 0$ for every z .
- (B) f is a constant purely imaginary function.
- (C) $f(z) = iz$.
- (D) f is unbounded.
- (E) f is not differentiable.

Problem 266.2

[See solution](#)

For $f(x + iy) = 2xy + i(x^2 + y^2)$, which statement is correct?

- (A) f is entire.
- (B) f is differentiable only at zero.
- (C) f is differentiable at every point of the imaginary axis and is analytic nowhere.
- (D) f is analytic precisely on the imaginary axis.
- (E) f is nowhere differentiable.

Problem 266.3[See solution](#)

An entire function satisfies $f(1/n) = 1/n^2$ for every positive integer n . What is $f(i)$?

- (A) The value is not determined.
- (B) -1
- (C) 1
- (D) 0
- (E) i

Problem 266.4[See solution](#)

Let F be the entire extension of $(z^5 - 5z + 4)/(z - 1)^2$ across $z = 1$. What is $F''(1)$?

- (A) 0 (B) 5 (C) 10 (D) 20 (E) 120

Problem 266.5[See solution](#)

An entire function satisfies $f(z)f(-z) = 0$ for every z . Which conclusion follows?

- (A) f is even and nonzero.
- (B) f is identically zero.
- (C) f is odd and nonzero.
- (D) f has exactly one zero.
- (E) f may be any polynomial with zero constant term.

7. Additional Topics

Complex Variables ▸ Complex Line Integrals

267 Complex line integrals and primitives

Book pp. 309. [Find in compact glossary](#)

In this entry: complex line integral; contour integral; parametrized path; primitive; antiderivative; path independence; orientation.

Definition and key ideas

For a piecewise smooth parametrized path $z(t)$, $a \leq t \leq b$, define $\int_C f(z) dz = \int_a^b f(z(t))z'(t) dt$. Reversing orientation changes the sign. The differential is complex dz , not the real arc-length element $|dz|$.

If $F' = f$ throughout a domain containing the path, then $\int_C f dz = F(z(b)) - F(z(a))$, so the integral depends only on endpoints. A holomorphic function on a domain with a hole need not possess a global primitive. Nonholomorphic integrands such as $\bar{z}$ generally require parametrization.

Problem 267.1

[See solution](#)

Let C be the unit circle traversed counterclockwise once. What is $\int_C \bar{z} dz$?

- (A) 0 (B) 2π (C) -2π (D) $2\pi i$ (E) $-2\pi i$

Problem 267.2

[See solution](#)

A piecewise smooth path C goes from 0 to $1 + i$. What is $\int_C z^2 dz$?

- (A) $(-2 + 2i)/3$
(B) $(2 + 2i)/3$
(C) $(-1 + i)/3$
(D) $2i$
(E) It depends on the path.

Problem 267.3

[See solution](#)

Let C traverse the triangle with vertices 0, 2, $2i$ counterclockwise. What is $\oint_C \bar{z} dz$?

- (A) 4 (B) 0 (C) $2i$ (D) $-4i$ (E) $4i$

Problem 267.4

[See solution](#)

Let C_1 be the straight segment from 0 to $1 + i$, and let C_2 run from 0 to 1 and then to $1 + i$, each segment straight. What is $\int_{C_2} \operatorname{Re} z dz - \int_{C_1} \operatorname{Re} z dz$?

- (A) 0 (B) $1/2$ (C) $i/2$ (D) $-i/2$ (E) $1 + i$

Problem 267.5[See solution](#)

Let C be the positively oriented circle $|z| = 2$. What is $\oint_C 2z/(z^2 + 1) dz$?

- (A) $-4\pi i$ (B) $2\pi i$ (C) $4\pi i$ (D) 0 (E) $8\pi i$

7. Additional Topics

Complex Variables ▸ Theorems Concerning Analytic Functions

268 Cauchy theorem and simply connected domains

Book pp. 310. [Find in compact glossary](#)

In this entry: Cauchy theorem; simply connected domain; closed contour; primitive existence; contour deformation.

Definition and key ideas

A domain is a connected open subset of $\mathbb{C}$. It is simply connected if every closed loop can be continuously contracted to a point inside the domain. If f is holomorphic on a simply connected domain, its integral around every piecewise smooth closed path is zero, and it has a primitive.

The absence of singularities on the contour alone is insufficient: what lies inside, or the topology of the domain, matters. The function $1/z$ is holomorphic on the punctured plane but integrates to $2\pi i$ around a counterclockwise circle enclosing the origin. Contours can be deformed through regions of holomorphy without changing the integral.

Problem 268.1

[See solution](#)

Which function has a single-valued holomorphic primitive on $\mathbb{C} \setminus \{0\}$?

- (A) $1/z$ (B) $1/z + 1/z^2$ (C) $1/z^2$ (D) e^z/z (E) $\cos z/z$

Problem 268.2

[See solution](#)

Let f be holomorphic on $1 < |z| < 4$. If its counterclockwise integral over $|z| = 2$ is $6\pi i$, what is its clockwise integral over $|z| = 3$?

- (A) 0 (B) $3\pi i$ (C) $6\pi i$ (D) $-6\pi i$ (E) $9\pi i$

Problem 268.3

[See solution](#)

Let $D = \mathbb{C} \setminus \{-it : t \geq 0\}$. For any piecewise smooth path C in D from -1 to 1 , what is $\int_C dz/z$?

- (A) 0
(B) πi
(C) $-\pi i$
(D) $2\pi i$
(E) It can have different values for different paths in D .

Problem 268.4[See solution](#)

On $D = \mathbb{C} \setminus \{-1, 1\}$ let $f(z) = 1/(z - 1) - 1/(z + 1)$. Its integral around $|z| = 2$ is zero. Does it have a single-valued primitive on D ?

- (A) Yes, because one closed contour integral vanishes.
- (B) Yes, because both poles lie inside the same circle.
- (C) Yes, with primitive $\log |z^2 - 1|$.
- (D) No, because a small counterclockwise circle about 1 has integral $2\pi i$.
- (E) No, because f is not holomorphic on D .

Problem 268.5[See solution](#)

Which assertion holds for every analytic function f on a simply connected open subset D of $\mathbb{C}$?

- (A) The function f is a polynomial.
- (B) The integral of f is zero around every closed piecewise smooth curve in D .
- (C) The function $1/f$ is analytic on D .
- (D) The function f has no zeros.
- (E) The function f is bounded.

7. Additional Topics

Complex Variables ▸ Theorems Concerning Analytic Functions

269 Morera theorem

Book pp. 310. [Find in compact glossary](#)

In this entry: Morera theorem; vanishing triangle integrals; analyticity from integration; local uniform convergence.

Definition and key ideas

Morera's theorem states that a continuous function on an open set is holomorphic if its integral around every triangle whose closed interior lies in the set is zero. Requiring every closed contour to have zero integral is a stronger sufficient assumption. Continuity is part of the theorem.

This converts integral information into differentiability. One application is that a function continuous across a line and holomorphic on both sides is holomorphic across the line: divide triangles along the line, apply Cauchy's theorem on each side, and pass to the boundary using continuity. Vanishing on a single contour is insufficient.

Uniform convergence on every compact subset is called local uniform convergence. Such a limit of holomorphic functions is holomorphic: it is continuous, and uniform convergence on each fixed triangle boundary permits passing the limit through the contour integral. Morera then supplies the derivative property without assuming convergence of derivatives in advance.

Problem 269.1

[See solution](#)

A continuous function f on the plane satisfies $\oint_{|z|=1} f(z) dz = 0$. Which example shows that this condition alone does not imply that f is entire?

- (A) $f(z) = z$ (B) $f(z) = e^z$ (C) $f(z) = |z|^2$ (D) $f(z) = 1$ (E) $f(z) = \sin z$

Problem 269.2

[See solution](#)

A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is continuous everywhere and holomorphic wherever $\operatorname{Im} z \neq 0$. Which must be true?

- (A) f is constant.
(B) f is entire.
(C) f is a polynomial.
(D) f is bounded.
(E) f vanishes on the real axis.

Problem 269.3[See solution](#)

A continuous function f on an open disk has zero integral around the boundary of every triangle contained in the disk. Which conclusion follows?

- (A) f is analytic on the disk.
- (B) f is constant.
- (C) f is real valued.
- (D) f extends to an entire function.
- (E) f is a polynomial.

Problem 269.4[See solution](#)

Let f_n be analytic on a domain D and converge uniformly on every compact subset to f . Which argument correctly proves that f is analytic?

- (A) Every f_n is bounded on all of D .
- (B) The limit is continuous, and uniform convergence on each triangle permits passing the limit through its boundary integral.
- (C) The limit must be a polynomial.
- (D) Uniform convergence forces f'_n to be constant.
- (E) Pointwise limits of analytic functions are always analytic.

Problem 269.5[See solution](#)

Define $f(0) = 1$ and $f(z) = 0$ for $z \neq 0$. Its integral around every triangle is zero. Why does Morera's theorem not imply that f is analytic?

- (A) Morera's theorem applies only to polynomials.
- (B) Triangle integrals must equal one.
- (C) The domain is not simply connected.
- (D) The required continuity hypothesis fails.
- (E) The function is unbounded.

7. Additional Topics

Complex Variables ▸ Theorems Concerning Analytic Functions

270 Cauchy integral and derivative formulas

Book pp. 310. [Find in compact glossary](#)**In this entry:** Cauchy integral formula; Cauchy derivative formula; simple closed contour; positive orientation.**Definition and key ideas**

If f is holomorphic on an open neighborhood of a simple closed contour and its interior, and a is inside, then for positive orientation $\int_C f(z)/(z-a)^{n+1} dz = 2\pi i f^{(n)}(a)/n!$, for $n \geq 0$. The $n = 0$ case is Cauchy's integral formula.

Identify the analytic numerator and the denominator power before differentiating: a power $n + 1$ requires the n th derivative. If a is outside the contour and no other singularities occur, the integral is zero. Clockwise orientation changes the sign, and a singularity on the contour invalidates the ordinary formula.

A simple closed contour is a closed curve with no self-intersections apart from the identified starting and ending point. Positive orientation is counterclockwise around its interior (the interior stays on the left as the curve is followed). These hypotheses give one positive winding around each interior point in the stated formula.

Problem 270.1[See solution](#)

For the counterclockwise circle $|z| = 2$, what is $\int_C z^2 e^z / (z-1)^3 dz$?

- (A) $2\pi i e$ (B) $5\pi i e$ (C) $7\pi i e$ (D) $14\pi i e$ (E) 0

Problem 270.2[See solution](#)

For the clockwise unit circle, what is $\int_C e^{2z} / z^5 dz$?

- (A) $4\pi i/3$ (B) $-4\pi i/3$ (C) $-8\pi i/3$ (D) $-16\pi i$ (E) 0

Problem 270.3[See solution](#)

For a off the circle $|z| = 2$, define $H(a) = \oint_{|z|=2} z^2 / (z-a) dz$ with positive orientation. Which formula is correct?

- (A) $H(a) = 2\pi i a^2$ for every such a .
 (B) $H(a) = 0$ for every such a .
 (C) $H(a) = 2\pi i a^2$ if $|a| < 2$, and $H(a) = 0$ if $|a| > 2$.
 (D) $H(a) = 0$ if $|a| < 2$, and $H(a) = 2\pi i a^2$ if $|a| > 2$.
 (E) $H(a) = 2\pi i a$ if $|a| < 2$, and zero otherwise.

Problem 270.4[See solution](#)

What is $\oint_{|z|=1} e^{z^2}/z^5 dz$, with counterclockwise orientation?

- (A) 0 (B) $4\pi i$ (C) $2\pi i$ (D) $12\pi i$ (E) πi

Problem 270.5[See solution](#)

Can the ordinary contour integral $\oint_{|z|=1} dz/(z-1)$ be evaluated directly by Cauchy's integral formula?

- (A) Yes; it is $2\pi i$.
(B) Yes; it is πi because the pole lies on the contour.
(C) Yes; it is zero because the pole is not strictly inside.
(D) No; the integrand is singular on the contour and the ordinary integral is undefined.
(E) Yes; it is $-2\pi i$.

7. Additional Topics

Complex Variables ▸ Theorems Concerning Analytic Functions

271 Cauchy derivative estimates and growth bounds

Book pp. 310. [Find in compact glossary](#)**In this entry:** Cauchy derivative estimate; coefficient estimate; polynomial growth.**Definition and key ideas**

If f is holomorphic on a neighborhood of the closed disk $|z - a| \leq R$ and $|f| \leq M$ on its boundary, then $|f^{(n)}(a)| \leq n!M/R^n$. Its Taylor coefficient c_n therefore satisfies $|c_n| \leq M/R^n$.

For entire functions one may let $R \rightarrow \infty$. If $|f(z)| \leq C(1 + |z|^\alpha)$ for constants $C > 0$ and $\alpha \geq 0$, all Taylor coefficients with integer index $n > \alpha$ vanish. Thus the entire function is a polynomial of degree at most $\lfloor \alpha \rfloor$. These estimates turn global growth restrictions into exact algebraic conclusions.

Coefficient estimates also control Taylor tails. If $|f| \leq M$ on $|z - a| = R$, then for $|z - a| \leq r < R$, the remainder after degree N is at most $M(r/R)^{N+1}/(1 - r/R)$. This follows by applying the coefficient bound term by term and summing a geometric series. When a bound depends on the chosen radius, optimizing that radius may improve the result.

Problem 271.1[See solution](#)

A function is holomorphic near $|z| \leq 3$ and satisfies $|f(z)| \leq 9$ on $|z| = 3$. What is the sharpest universal bound among the choices for $|f''(0)|$?

- (A) 1 (B) 2 (C) 3 (D) 6 (E) 9

Problem 271.2[See solution](#)

An entire function satisfies $|f(z)| \leq 10(1 + |z|^{3/2})$ for all z . What is the greatest possible degree of f ?

- (A) 0
(B) 1
(C) 2
(D) 3
(E) It need not be a polynomial.

Problem 271.3[See solution](#)

An entire function satisfies $|f(z)| \leq e^{|z|}$. What is the smallest bound for $|f''(0)|$ obtained by applying Cauchy's estimate on $|z| = R$ and optimizing over $R > 0$?

- (A) 2 (B) $e/2$ (C) $e^2/4$ (D) $e^2/2$ (E) $2e^2$

Problem 271.4[See solution](#)

A function is holomorphic near the closed unit disk, has $|f(z)| \leq 1$ for $|z| = 1$, and satisfies $f''(0) = 2$. Which function must it be on the disk?

- (A) 1
- (B) z
- (C) z^2
- (D) $-z^2$
- (E) The information does not determine f .

Problem 271.5[See solution](#)

Write $f(z) = \sum_{n \geq 0} a_n z^n$, where f is holomorphic near $|z| \leq 2$ and $|f| \leq 4$ on $|z| = 2$. For $|z| \leq 1$, what bound results from summing the individual Cauchy bounds for the tail $|f(z) - a_0 - a_1 z - a_2 z^2|$?

- (A) $1/2$ (B) 1 (C) 2 (D) 4 (E) 8

7. Additional Topics

Complex Variables ▸ Theorems Concerning Analytic Functions

272 Liouville theorem

Book pp. 310. [Find in compact glossary](#)**In this entry:** Liouville theorem; bounded entire function; fundamental theorem of algebra via analysis.

Definition and key ideas

Liouville's theorem says that every bounded entire function is constant. It follows from Cauchy's estimate for the first derivative by sending the radius to infinity. Entire is essential: nonconstant bounded holomorphic functions exist on the unit disk.

A useful strategy is to transform an unbounded-looking entire function into a bounded one. For example an entire function bounded away from zero has bounded entire reciprocal and is constant. Liouville also proves the fundamental theorem of algebra: a nonconstant polynomial with no zero would have a bounded entire reciprocal.

A bounded holomorphic function on a punctured disk extends across its center by the removable-singularity theorem. Thus a globally bounded holomorphic function on the punctured plane is also constant, after extension. On a disk, by contrast, $f(z) = z$ is bounded and nonconstant.

Problem 272.1

[See solution](#)

An entire function satisfies $|f(z)| \geq 2$ for every $z \in \mathbb{C}$. Which must be true?

- (A) f is constant.
- (B) $f(z) = 2$ everywhere.
- (C) f is linear.
- (D) f has a zero.
- (E) f is unbounded.

Problem 272.2

[See solution](#)

An entire function satisfies $\operatorname{Im} f(z) > 0$ everywhere. Which must follow?

- (A) f is constant.
- (B) $f(z) = i$ everywhere.
- (C) f is a nonconstant polynomial.
- (D) f has an essential singularity in $\mathbb{C}$.
- (E) f maps onto the upper half-plane.

Problem 272.3[See solution](#)

In a proof of the fundamental theorem of algebra, assume a nonconstant complex polynomial P has no zero. Why is $1/P$ bounded on the entire plane?

- (A) Every entire function is bounded.
- (B) Because $1/P$ tends to zero at infinity and is continuous on each fixed closed disk.
- (C) Because every polynomial has bounded modulus.
- (D) Because $|P(z)| \geq 1$ for every z .
- (E) Because every nonzero complex number has modulus at least one.

Problem 272.4[See solution](#)

For which of the following domains must every bounded holomorphic function be constant? I. $\mathbb{C}$; II. $\mathbb{C} \setminus \{0\}$; III. $\{z : |z| < 1\}$.

- (A) I only (B) II only (C) I and II only (D) I and III only (E) I, II, and III

Problem 272.5[See solution](#)

Which pair of entire functions shows that $f(z)g(z) = 1$ does not force either function to be constant?

- (A) $f(z) = z$, $g(z) = 1/z$
- (B) $f(z) = \sin z$, $g(z) = 1/\sin z$
- (C) $f(z) = e^z$, $g(z) = e^{-z}$
- (D) $f(z) = z + 1$, $g(z) = 1/(z + 1)$
- (E) $f(z) = \cos z$, $g(z) = 1/\cos z$

7. Additional Topics

Complex Variables ▸ Theorems Concerning Analytic Functions

273 Maximum modulus principle

Book pp. 310. [Find in compact glossary](#)

In this entry: maximum modulus principle; maximum principle; boundary maximum; minimum modulus principle.

Definition and key ideas

A nonconstant holomorphic function on a connected open set has no interior local maximum of its modulus. If it is continuous on the closure of a bounded domain, compactness ensures a maximum exists and the maximum is attained on the boundary. Continuity on the closure is essential for this boundary assertion.

If a holomorphic function has no zeros, apply the maximum principle to its reciprocal to obtain a minimum modulus principle. A function may have an interior minimum of modulus at a zero. The principle constrains modulus, rather than the complex value itself, because complex numbers have no compatible ordering.

Problem 273.1

[See solution](#)

What is $\max_{|z| \leq 1} |z^2 + 2z|$?

- (A) 1 (B) 2 (C) $\sqrt{5}$ (D) 3 (E) 4

Problem 273.2

[See solution](#)

A function f is holomorphic near the closed unit disk, has no zeros there, and satisfies $|f(z)| = 1$ for $|z| = 1$. Which must hold?

- (A) $f(z) = z$.
(B) f is constant of modulus 1 on the unit disk.
(C) $|f(0)| < 1$.
(D) f has a pole outside the disk.
(E) $f'(0) \neq 0$.

Problem 273.3

[See solution](#)

For $|z| < 1$, let $f(z) = \exp(-(1+z)/(1-z))$. Which statement is correct?

- (A) f is unbounded in the disk.
(B) f is bounded in the disk and extends continuously to $z = 1$ with value zero.
(C) f is bounded in the disk and extends continuously to $z = 1$ with value one.
(D) f is bounded in the disk but has no continuous extension to $z = 1$.
(E) f is identically zero.

Problem 273.4[See solution](#)

A function is analytic on a neighborhood of the closed unit disk, has $|f| \leq 1$ on its boundary, and satisfies $f(1/2) = 1$. What follows?

- (A) f has exactly one zero.
- (B) $f(z) = z$ on the disk.
- (C) f may be any finite Blaschke product.
- (D) The hypotheses are inconsistent.
- (E) $f \equiv 1$ on the disk.

Problem 273.5[See solution](#)

A zero-free function f is analytic on a neighborhood of the closed unit disk and satisfies $|f(z)| \geq 2$ on the unit circle. Which conclusion follows?

- (A) f is constant.
- (B) $|f(z)| \leq 2$ throughout the disk.
- (C) $f(0) = 2$
- (D) $f'(0) = 0$
- (E) $|f(z)| \geq 2$ throughout the disk.

7. Additional Topics

Complex Variables ▸ Taylor Series for Functions of a Complex Variable

274 Complex Taylor series and disks of convergence

Book pp. 311. [Find in compact glossary](#)

In this entry: complex Taylor series; power series; disk of convergence; radius of convergence; Cauchy–Hadamard formula.

Definition and key ideas

A function holomorphic near a has expansion $f(z) = \sum_{n \geq 0} f^{(n)}(a)(z - a)^n/n!$ throughout every disk centered at a on which it is holomorphic. For a general power series, $R^{-1} = \limsup_{n \rightarrow \infty} |c_n|^{1/n}$, with the usual conventions for zero and infinity. A plain limit may be used only when it exists.

Convergence is absolute for $|z - a| < R$ and fails for $|z - a| > R$; boundary behavior must be examined separately. For a rational function, the nearest nonremovable singularity determines the Taylor radius. Complex singularities matter even when the real function is smooth on the entire real axis.

Problem 274.1

[See solution](#)

What is the radius of convergence of the Taylor series of $1/(1 + z^2)$ about $z = 2$?

- (A) 1 (B) 2 (C) $\sqrt{5}$ (D) 3 (E) ∞

Problem 274.2

[See solution](#)

What is the radius of convergence of $\sum_{n=0}^{\infty} 4^n z^{2n}$?

- (A) $1/4$ (B) $1/2$ (C) 1 (D) 2 (E) 4

Problem 274.3

[See solution](#)

At which points on $|z| = 1$ does $\sum_{n=1}^{\infty} z^n/n$ converge?

- (A) At every point, absolutely.
(B) At no point.
(C) Only at $z = -1$.
(D) At every point except $z = 1$, and nowhere on the circle absolutely.
(E) At every point except $z = -1$.

Problem 274.4

[See solution](#)

After removing the singularity at $z = 1$, what is the radius of convergence about zero of $(z - 1)/[(z - 1)(z - 3)]$?

- (A) 4 (B) 3 (C) 2 (D) 1 (E) ∞

Problem 274.5[See solution](#)

What is the coefficient of z^7 in the Taylor series at zero of z^2e^{3z} ?

- (A) $3^5/5!$ (B) $3^7/7!$ (C) 0 (D) $3^7/5!$ (E) $3^5/7!$

7. Additional Topics

Complex Variables ▸ Singularities, Poles, and Laurent Series

275 Isolated singularities and their classification

Book pp. 311-314. [Find in compact glossary](#)

In this entry: singularity; isolated singularity; punctured disk; removable singularity; pole; order of a pole; simple pole; double pole; essential singularity.

Definition and key ideas

An isolated singularity at a means the function is holomorphic on some punctured disk $0 < |z - a| < R$. It is removable if the function extends holomorphically across a , a pole of order m if $(z - a)^m f(z)$ extends with a nonzero value at a , and essential if neither case holds.

A removable singularity has no negative Laurent powers; a pole has finitely many, ending in $(z - a)^{-m}$; an essential singularity has infinitely many negative powers in the local punctured-disk expansion. This classification must use an expansion centered at the singularity and valid near it, not an exterior annulus centered elsewhere.

A simple pole has order one and a double pole has order two. A bounded holomorphic function near an isolated missing point has a removable singularity. If other singularities accumulate at the point, no punctured disk of holomorphy exists, and the isolated-singularity trichotomy cannot be used.

Problem 275.1

[See solution](#)

What type of singularity at 0 has $f(z) = (\sin z - z)/z^5$?

(A) Removable. (B) Simple pole. (C) Double pole. (D) Pole of order 4. (E) Essential.

Problem 275.2

[See solution](#)

Classify the singularities at 0 of $f(z) = (1 - \cos z)/z^2$ and $g(z) = e^{1/z}$, respectively.

- (A) Simple pole; essential.
- (B) Removable; essential.
- (C) Double pole; simple pole.
- (D) Removable; removable.
- (E) Essential; essential.

Problem 275.3[See solution](#)

Consider $f(z) = 1/\sin(1/z)$ at points where it is defined near zero. Which description of $z = 0$ is correct?

- (A) A removable isolated singularity.
- (B) A simple pole.
- (C) An essential isolated singularity.
- (D) A pole of infinite order.
- (E) A nonisolated singular point, because poles accumulate at zero.

Problem 275.4[See solution](#)

A function is holomorphic on $0 < |z| < 1$ and satisfies $|f(z)| \leq |z|^{3/2}$. What must be true of its extension F at zero?

- (A) No holomorphic extension exists.
- (B) $F(0) = 0$ but $F'(0)$ may be nonzero.
- (C) $F(0) = F'(0) = 0$.
- (D) F is identically zero.
- (E) $F''(0) = 0$.

Problem 275.5[See solution](#)

At $z = 1/2$, what type of singularity does $\cos(\pi z)/(z - 1/2)^2$ have?

- | | |
|---------------------------|-------------------|
| (A) Essential | (B) Removable |
| (C) A pole of order three | (D) A simple pole |
| (E) A pole of order two | |

7. Additional Topics

Complex Variables ▸ Singularities, Poles, and Laurent Series

276 Laurent series, annuli, and principal parts

Book pp. 312-314. [Find in compact glossary](#)**In this entry:** Laurent series; Laurent coefficient; annulus; principal part; singular part; analytic part.**Definition and key ideas**

On an annulus $r < |z - a| < R$, a holomorphic function has a unique Laurent expansion $\sum_{n \in \mathbb{Z}} c_n(z - a)^n$. Negative powers form the principal part and nonnegative powers the analytic part. For a counterclockwise circle in the annulus, $c_n = (2\pi i)^{-1} \int f(z)/(z - a)^{n+1} dz$.

Different annuli about the same center can give different expansions. Rewrite rational terms as geometric series according to whether $|z - a|$ is smaller or larger than each pole distance. Nonnegative coefficients are not generally $f^{(n)}(a)/n!$ when the original function is singular or undefined at a ; derivatives of the analytic part give those coefficients.

Problem 276.1[See solution](#)

What is the coefficient of z^{-1} in the Laurent expansion of $1/((z - 1)(z - 3))$ valid for $1 < |z| < 3$?

- (A) -1 (B) $-1/2$ (C) 0 (D) $1/2$ (E) 1

Problem 276.2[See solution](#)

For $|z - 2| > 1$, which expansion equals $1/(z - 1)$?

- (A) $\sum_{n \geq 0} (z - 2)^n$ (B) $\sum_{n \geq 0} (-1)^n (z - 2)^n$
 (C) $\sum_{n \geq 0} (-1)^n (z - 2)^{-n-1}$ (D) $-\sum_{n \geq 0} (z - 2)^{-n-1}$
 (E) $\sum_{n \geq 0} (z - 2)^{-n}$

Problem 276.3[See solution](#)

In the Laurent expansion of $f(z) = e^{1/z} + z^2$ on $0 < |z| < \infty$, what is the analytic part?

- (A) z^2
 (B) $1 + z^2$
 (C) $e^{1/z}$
 (D) $1 + z + z^2$
 (E) There is no analytic part because f is undefined at zero.

Problem 276.4[See solution](#)

The expansion $1/(1 - z) = -\sum_{n=1}^{\infty} z^{-n}$ is valid for $|z| > 1$. What does it imply about the original rational function at zero and one?

- (A) Both points are essential singularities.
- (B) Zero is essential and one is removable.
- (C) Zero is ordinary and one is a simple pole.
- (D) Zero is a pole of infinite order and one is essential.
- (E) Both points are removable singularities.

Problem 276.5[See solution](#)

How many distinct maximal annuli centered at zero support Laurent expansions of $1/[(z - 1)(z + 2)(z - 4)]$?

- (A) 1 (B) 3 (C) 2 (D) 5 (E) 4

7. Additional Topics

Complex Variables ▸ The Residue Theorem

277 Residues and residue computation

Book pp. 314-315. [Find in compact glossary](#)**In this entry:** residue; coefficient of reciprocal term; simple-pole residue; higher-order-pole residue; logarithmic derivative.**Definition and key ideas**

The residue at an isolated singularity a is the coefficient of $(z - a)^{-1}$ in the local Laurent series. At a simple pole, $\text{Res}(f, a) = \lim_{z \rightarrow a} (z - a)f(z)$; for $f = p/q$ with p, q holomorphic near a , $q(a) = 0$ and $q'(a) \neq 0$, it equals $p(a)/q'(a)$.

At a pole of order m , the residue is $1/(m-1)!$ times the $(m-1)$ st derivative at a of $(z - a)^m f(z)$. Series multiplication is often faster for essential singularities or removable-looking quotients. A residue can vanish even at a genuine pole, such as $1/z^2$.

The logarithmic derivative F'/F has residue m at a zero of order m and residue $-m$ at a pole of order m : factor $F = (z - a)^{\pm m} h$ with $h(a) \neq 0$. Also, the residue of the derivative of any holomorphic function on a punctured disk is zero, as termwise differentiation of its Laurent series shows.

Problem 277.1[See solution](#)What is the residue at 0 of e^{2z}/z^3 ?

- (A) 1 (B) 2 (C) 4 (D) 8 (E) 0

Problem 277.2[See solution](#)What is the residue at 0 of $z^2 e^{1/z}$?

- (A) 0 (B) 1 (C)
- $1/2$
- (D)
- $1/6$
- (E)
- $1/24$

Problem 277.3[See solution](#)If f is holomorphic on a punctured disk about a , what must be the residue of f' at a ?

- (A) $\text{Res}(f, a)$
 (B) $-\text{Res}(f, a)$
 (C) 0
 (D) 1
 (E) It is undefined when f has an essential singularity.

Problem 277.4[See solution](#)Near a , let $F(z) = (z - a)^4 h(z)$, where h is holomorphic and $h(a) \neq 0$. What is the residue at a of $F'(z)/F(z)$?

- (A) 0 (B) 1 (C) 3 (D) 4 (E)
- $h'(a)/h(a)$

Problem 277.5[See solution](#)

What is the residue of $\cot z$ at $z = \pi$?

- (A) $1/\pi$ (B) 1 (C) 0 (D) -1 (E) π

7. Additional Topics

Complex Variables ▸ The Residue Theorem

278 Residue theorem and contour evaluation

Book pp. 315–317. [Find in compact glossary](#)

In this entry: residue theorem; sum of residues; enclosed singularity; winding number; argument principle.

Definition and key ideas

If f is holomorphic on and inside a positively oriented simple closed contour except for finitely many isolated singularities a_j inside, then $\int_C f(z) dz = 2\pi i \sum_j \text{Res}(f, a_j)$. There must be no singularity on the contour. Negative orientation negates the result.

For more general closed contours the contribution of each singularity is multiplied by its winding number, the signed number of turns around it. First locate the enclosed singularities, then compute only the needed residues. Poles outside the contour can influence analytic numerator values but do not directly enter the sum.

For a meromorphic function F with no zeros or poles on a simple positively oriented contour, $\frac{1}{2\pi i} \oint F'/F$ equals the number of enclosed zeros minus the number of enclosed poles, with multiplicities. This argument principle follows from the logarithmic-derivative residues. A trigonometric integral over a full period can also become a contour integral by $z = e^{i\theta}$, $\cos \theta = (z + z^{-1})/2$, and $d\theta = dz/(iz)$.

A meromorphic function is holomorphic except at isolated poles. In the argument principle it must have no zero or pole on the contour; zeros and poles inside are counted with their multiplicities.

Problem 278.1

[See solution](#)

For the counterclockwise circle $|z| = 2$, what is $\int_C dz/[z(z-1)(z-3)]$?

- (A) $-\pi i/3$ (B) $\pi i/3$ (C) $-\pi i$ (D) $2\pi i/3$ (E) 0

Problem 278.2

[See solution](#)

A closed contour avoids 0 and 2 and has winding numbers 2 about 0 and -1 about 2. What is $\int_C (3/z + 5/(z-2)) dz$?

- (A) $-2\pi i$ (B) $2\pi i$ (C) $8\pi i$ (D) $10\pi i$ (E) $22\pi i$

Problem 278.3

[See solution](#)

What is $\int_0^{2\pi} \frac{d\theta}{5 - 4\cos \theta}$?

- (A) $\pi/3$ (B) $2\pi/5$ (C) $2\pi/3$ (D) π (E) 2π

Problem 278.4[See solution](#)

What is $\oint_{|z|=1} e^{1/z} dz$ with counterclockwise orientation?

- (A) The integral does not exist.
- (B) $2\pi i$
- (C) $2\pi ie$
- (D) 0
- (E) πi

Problem 278.5[See solution](#)

Let $F(z) = (z - 1)^3(z - 4)/(z + 1)^2$. What is $\oint_{|z|=2} F'(z)/F(z) dz$ with positive orientation?

- (A) $-2\pi i$ (B) 0 (C) $2\pi i$ (D) $6\pi i$ (E) $10\pi i$

7. Additional Topics

Numerical Analysis ▸ Iterative Methods

279 Iterations, error, tolerance, and convergence rate

Book pp. 317-318. [Find in compact glossary](#)

In this entry: iterative method; initial guess; approximation error; error tolerance; rate of convergence; linear convergence; quadratic convergence; relative error; superlinear convergence.

Definition and key ideas

An iterative method produces approximations x_n to a target α . The error is $e_n = x_n - \alpha$; a tolerance specifies a desired bound on $|e_n|$ or a justified surrogate. Small successive changes alone do not guarantee proximity to the true answer unless an error estimate is known.

Convergence has order p if $|e_{n+1}|/|e_n|^p \rightarrow C$ with $0 < C < \infty$; linear convergence has $p = 1$ and $C < 1$, while quadratic convergence has $p = 2$. At quadratic convergence the number of correct digits approximately doubles per step. A local convergence theorem requires hypotheses near the particular root.

Absolute error is $|x_n - \alpha|$. Relative error, when $\alpha \neq 0$, is $|x_n - \alpha|/|\alpha|$; specify the exact-value denominator rather than substituting the approximation. Superlinear convergence means $|e_{n+1}|/|e_n| \rightarrow 0$, which does not by itself imply quadratic order. If future changes contract by a known ratio $q < 1$, a geometric tail yields $|x_n - \alpha| \leq q|x_n - x_{n-1}|/(1 - q)$.

Problem 279.1

[See solution](#)

If errors satisfy $e_{n+1} = 3e_n^2$ and $e_0 = 1/10$, what is e_2 ?

- (A) $3/100$ (B) $9/1000$ (C) $27/10000$ (D) $81/10000$ (E) $1/10000$

Problem 279.2

[See solution](#)

Suppose $x_n \rightarrow \alpha$ and $|x_{n+1} - \alpha| = \frac{1}{4}|x_n - \alpha|$ with nonzero errors. How many iterations guarantee reducing the initial error by at least a factor of 1000?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 10

Problem 279.3

[See solution](#)

An approximation sequence has positive errors $e_n = 2^{-n^2}$. Which statement is correct?

- (A) It converges linearly with asymptotic ratio $1/2$.
 (B) It converges quadratically with asymptotic constant one.
 (C) It does not converge.
 (D) $e_{n+1}/e_n \rightarrow 0$, but $e_{n+1}/e_n^2 \rightarrow \infty$.
 (E) $e_{n+1}/e_n \rightarrow 1$.

Problem 279.4[See solution](#)

A sequence converges to α , and from index n onward its successive changes satisfy $|x_{k+1} - x_k| \leq \frac{1}{4}|x_k - x_{k-1}|$. If $|x_n - x_{n-1}| = 1/1000$, what is the sharp bound on $|\alpha - x_n|$ obtained from these inequalities?

- (A) $1/4000$
- (B) $1/3000$
- (C) $1/1000$
- (D) $1/750$
- (E) No finite bound follows.

Problem 279.5[See solution](#)

A numerical approximation $a = 1.40$ to $\sqrt{2}$ has absolute error $\sqrt{2} - 1.40$. What is its relative error, measured against the exact value?

- (A) $1 - 7/10$ (B) $5\sqrt{2}/7 - 1$ (C) $7\sqrt{2}/10$ (D) $1 - 7\sqrt{2}/10$ (E) $\sqrt{2} - 7/5$

7. Additional Topics

Numerical Analysis ▸ Bisection Method

280 Bisection and certified root brackets

Book pp. 317-318. [Find in compact glossary](#)

In this entry: bisection method; sign-changing bracket; intermediate value theorem; bisection error bound.

Definition and key ideas

For continuous f on $[a, b]$ with $f(a)f(b) < 0$, bisection tests the midpoint and retains a half-interval with opposite endpoint signs, unless it finds an exact zero. Continuity guarantees a root in every retained bracket. A sign change is sufficient for a root, but not necessary: an even-multiplicity root need not change signs.

After n halvings the bracket length is $(b - a)/2^n$; its midpoint is within $(b - a)/2^{n+1}$ of a root in the bracket. State the iteration-count convention to avoid off-by-one errors. Bisection converges reliably under its hypotheses but reduces interval width only by a fixed factor at each step.

Problem 280.1

[See solution](#)

After n bisection halvings of an initial interval of length 3, its midpoint is used as the approximation. What is the least n guaranteeing error at most 10^{-3} ?

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Problem 280.2

[See solution](#)

Apply three bisection halvings to $f(x) = x^3 - 2$ starting with $[1, 2]$. What is the midpoint of the resulting bracket?

- (A) $9/8$ (B) $19/16$ (C) $5/4$ (D) $21/16$ (E) $11/8$

Problem 280.3

[See solution](#)

On $[1, 2]$, define $f(x) = -1$ for $x < \sqrt{2}$ and $f(x) = 1$ for $x \geq \sqrt{2}$. Suppose the bisection sign rule is applied repeatedly. What happens?

- (A) The brackets converge to $\sqrt{2}$, although f has no zero.
(B) The method reaches a zero after finitely many steps.
(C) The bracket lengths fail to tend to zero.
(D) The method must converge to one.
(E) Opposite endpoint signs prove that $f(\sqrt{2}) = 0$.

Problem 280.4[See solution](#)

Apply bisection to $f(x) = (x - 1)(x - 2)(x - 3)$ starting with $[0, 5]$. To which root do the retained brackets converge?

- (A) 1
- (B) 2
- (C) 3
- (D) The method must alternate among all three roots.
- (E) The limit depends on a free choice at every step.

Problem 280.5[See solution](#)

Which hypothesis guarantees that bisection can maintain a root-containing bracket for a function on $[a, b]$?

- (A) Differentiability on (a, b) alone
- (B) Opposite endpoint signs without continuity
- (C) Continuity on $[a, b]$ and $f(a)f(b) < 0$
- (D) Equal endpoint values
- (E) Monotonicity alone

Problem 280.6[See solution](#)

A continuous nonnegative function on $[0, 1]$ has value one at each of finitely many distinct sample points $x_1, \dots, x_N$. What can those data establish about zeros between sample points?

- (A) The function has no zero.
- (B) Every zero must be at a sampled point.
- (C) The function must have a sign-changing bracket.
- (D) An unsampled zero is still possible, despite every sampled value being positive.
- (E) There must be exactly one zero.

7. Additional Topics

Numerical Analysis ▸ Newton Method

281 Newton iteration and multiple roots

Book pp. 318; 320-321. [Find in compact glossary](#)**In this entry:** Newton method; Newton–Raphson method; tangent-line iteration; simple root; multiple-root convergence; Newton failure; modified Newton method.**Definition and key ideas**

Newton's iteration is $x_{n+1} = x_n - f(x_n)/f'(x_n)$ whenever the denominator is nonzero. It uses the tangent line's root as the next approximation. For a sufficiently smooth function with a simple root α and a sufficiently close initial guess, convergence is at least quadratic.

At a root of multiplicity $m > 1$, ordinary Newton iteration is generally linear with asymptotic error factor $(m - 1)/m$. A poor initial point can diverge, cycle, or reach a zero derivative. Thus quadratic convergence is not guaranteed merely by having a differentiable function or a plausible starting point.

A simple root satisfies $f(\alpha) = 0$ and $f'(\alpha) \neq 0$. A root of multiplicity m has local factorization $f(x) = (x - \alpha)^m g(x)$ with $g(\alpha) \neq 0$. When the multiplicity is known, the modified update $x_{n+1} = x_n - mf(x_n)/f'(x_n)$ removes the ordinary linear slowdown and, under the usual local smoothness conditions, restores at least quadratic convergence.

Problem 281.1[See solution](#)

Use Newton's method twice for $f(x) = x^2 - 3$, starting at $x_0 = 2$. What is x_2 ?

- (A) $7/4$ (B) $12/7$ (C) $97/56$ (D) $99/56$ (E) $26/15$

Problem 281.2[See solution](#)

For $f(x) = (x - 2)^4$, Newton iteration starts at $x_0 = 3$. What is $x_n - 2$ before reaching the root?

- (A) 4^{-n} (B) 2^{-n} (C) $(3/4)^n$ (D) $(1/4)^{2^n}$ (E) 3^n

Problem 281.3[See solution](#)

For Newton's method applied to $f(x) = \log x$, what is x_1 if $x_0 = e$?

- (A) $e/2$ (B) $1/e$ (C) 0 (D) $e - 1$ (E) 1

Problem 281.4[See solution](#)

Newton's method for $f(x) = x^2 - a$, $a > 0$, uses a positive iterate x_n . Which exact identity holds?

- (A) $x_{n+1} - \sqrt{a} = (x_n - \sqrt{a})^2/(2\sqrt{a})$
 (B) $x_{n+1} - \sqrt{a} = (x_n - \sqrt{a})^2/(2x_n)$
 (C) $x_{n+1} = a/x_n$
 (D) $x_{n+1} - \sqrt{a} = (x_n - \sqrt{a})/2$
 (E) $x_{n+1} - \sqrt{a} = -(x_n - \sqrt{a})^2/(2x_n)$

Problem 281.5[See solution](#)

For $f(x) = (x - 1)^3$, the modified Newton update $x_{n+1} = x_n - 3f(x_n)/f'(x_n)$ has what effect when $x_n \neq 1$?

- (A) It halves the error.
- (B) It multiplies the error by $2/3$.
- (C) It is undefined for every x_n .
- (D) It squares the error.
- (E) It reaches the root in one step.

Problem 281.6[See solution](#)

Newton iteration starts at zero for each of $f(x) = x^3 - 2x + 2$ and $g(x) = x^3 - 1$. Which pair of behaviors occurs, respectively?

- (A) Both converge quadratically to real roots.
- (B) f cycles between zero and one; the first update for g is undefined.
- (C) The first update for f is undefined; g cycles.
- (D) Both first updates are undefined.
- (E) f reaches a root in one step; g converges linearly.

8. Solutions to the Chapter Review Questions

Chapter 5 Review Questions ▸ Question 20: complex matrices

282 Hermitian matrices, conjugate transpose, and the spectral theorem

Book pp. 387. [Find in compact glossary](#)

In this entry: conjugate transpose; Hermitian matrix; unitary matrix; spectral theorem; Rayleigh quotient.

Definition and key ideas

The conjugate transpose of a complex matrix is $A^* = \overline{A}^T$. A square matrix is Hermitian if $A = A^*$; thus its diagonal entries are real and $a_{ji} = \overline{a_{ij}}$. A real Hermitian matrix is exactly a real symmetric matrix. A matrix U is unitary if $U^*U = I$, so its columns are an orthonormal basis for the complex inner product $\langle x, y \rangle = x^*y$.

The spectral theorem says that every Hermitian matrix has an orthonormal basis of eigenvectors and real eigenvalues: $A = UDU^*$ with U unitary and D real diagonal. Consequently $x^*Ax/(x^*x)$ lies between the smallest and largest eigenvalues. The trace is the sum of eigenvalues, the determinant is their product, and $\text{tr}(A^2)$ is the sum of their squares. Hermitian matrices form a real vector space of dimension n^2 , not a complex vector space: multiplying a nonzero Hermitian matrix by i generally destroys the condition.

The scalar $\mathcal{R}_H(x) = x^*Hx/(x^*x)$ for $x \neq 0$ is the Rayleigh quotient. For Hermitian H it is real; in an orthonormal eigenbasis it is a weighted average of the eigenvalues, with weights proportional to the squared absolute coordinates. Its minimum and maximum are the extreme eigenvalues. Conjugate transpose obeys $(AB)^* = B^*A^*$ and $(cA)^* = \overline{c}A^*$.

Problem 282.1

[See solution](#)

A Hermitian 3×3 matrix H satisfies $\text{tr } H = 6$, $\text{tr}(H^2) = 14$, and $\det H = 6$. What is $\text{tr}((H + I)^{-1})$?

- (A) $5/6$ (B) 1 (C) $13/12$ (D) $7/6$ (E) $3/2$

Problem 282.2

[See solution](#)

Let V be the real vector space of Hermitian 3×3 matrices H such that $H(1, 1, 1)^T = 0$ and $\text{tr } H = 0$. What is $\dim_{\mathbb{R}} V$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 282.3

[See solution](#)

Let $D = \text{diag}(1, 1, -1, -1)$. What is the real dimension of the space of Hermitian 4×4 matrices H satisfying $DH + HD = 0$?

- (A) 6 (B) 8 (C) 16 (D) 4 (E) 10

Problem 282.4[See solution](#)

For $H = \begin{pmatrix} 1 & i \\ -i & 3 \end{pmatrix}$, what is $|\det(H + iI)|^2$?

- (A) 9 (B) 25 (C) 17 (D) 5 (E) 13

Problem 282.5[See solution](#)

For arbitrary Hermitian matrices H, K of the same size, which matrices must be Hermitian? I. HK ; II. $HK + KH$; III. $i(HK - KH)$.

- (A) I only (B) II only (C) I and II only (D) II and III only (E) I, II, and III

Problem 282.6[See solution](#)

For $H = \begin{pmatrix} 2 & i \\ -i & 3 \end{pmatrix}$, what is the largest possible value of $x^*Hx/(x^*x)$ over nonzero $x \in \mathbb{C}^2$?

- (A) 3 (B) 5 (C) $(5 - \sqrt{5})/2$ (D) $(5 + \sqrt{5})/2$ (E) $\sqrt{7}$

8. Solutions to the Chapter Review Questions

Chapter 6 Review Questions > Question 8: a subgroup-order counterexample

283 Alternating groups and permutation parity

Book pp. 389. [Find in compact glossary](#)

In this entry: alternating group; even permutation; odd permutation; sign homomorphism; double transposition.

Definition and key ideas

The sign of a permutation is $+1$ if it can be written as an even number of transpositions and -1 otherwise; the parity is independent of the chosen expression. An r -cycle has sign $(-1)^{r-1}$, and signs multiply under composition. The alternating group A_n is the kernel of the sign homomorphism $S_n \rightarrow \{1, -1\}$. For $n \geq 2$ it is a normal subgroup of index 2 and has order $n!/2$.

Lagrange's theorem does not say that every divisor of a group's order occurs as a subgroup order. The group A_4 has order 12 but no subgroup of order 6. Its identity and three double transpositions form a normal Klein four subgroup V , and $A_4/V \cong \mathbb{Z}/3\mathbb{Z}$. For counting permutations with prescribed cycle lengths, divide $n!$ by the product of the cycle lengths and by the factorials of the multiplicities of repeated lengths; this corrects cyclic rotation within each cycle and exchange of equal-length cycles.

A double transposition is a product of two disjoint transpositions, such as $(12)(34)$; it is even and has order two. Conjugation merely relabels the four symbols, which explains why the identity and the three double transpositions in A_4 form a normal subgroup.

Problem 283.1

[See solution](#)

How many elements of A_6 have no fixed points and have exactly two cycles in their disjoint-cycle decomposition?

- (A) 40 (B) 90 (C) 120 (D) 130 (E) 180

Problem 283.2

[See solution](#)

How many surjective group homomorphisms are there from A_4 to the additive group $\mathbb{Z}/3\mathbb{Z}$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 283.3

[See solution](#)

If A_4 had a subgroup H of order six, which consequence would give a contradiction?

- (A) H would have index three.
(B) Every element of A_4 would have order six.
(C) The quotient of order two would force all eight 3-cycles of A_4 into H .
(D) A_4 would be cyclic because six divides twelve.
(E) H would have to contain an odd permutation.

Problem 283.4[See solution](#)

Let $a = (123)$ and $b = (124)$, with the rightmost permutation acting first. What is the commutator $aba^{-1}b^{-1}$?

- (A) The identity (B) $(12)(34)$ (C) $(13)(24)$ (D) $(14)(23)$ (E) (1234)

Problem 283.5[See solution](#)

What is the largest possible order of an element of A_7 ?

- (A) 7 (B) 6 (C) 10 (D) 12 (E) 5

9. Practice Test

Practice Test ▸ Question 26: surfaces of revolution

284 Torus and circular cross sections

Book pp. 410. [Find in compact glossary](#)**In this entry:** torus; major radius; minor radius; solid torus; annular cross section.**Definition and key ideas**

A ring torus is formed by revolving a circle of radius a about an axis in its plane whose distance from the center is $R > a$. Here R is the major radius and a the minor radius. With the axis chosen as the z -axis, its surface is $(\sqrt{x^2 + y^2} - R)^2 + z^2 = a^2$, equivalently $(x^2 + y^2 + z^2 + R^2 - a^2)^2 = 4R^2(x^2 + y^2)$. The solid torus uses the corresponding inequality in the first expression.

At height $|z| < a$, the solid's horizontal cross section is an annulus with radii $R \pm \sqrt{a^2 - z^2}$, so its area is $4\pi R\sqrt{a^2 - z^2}$. Integrating gives the volume $2\pi^2 Ra^2$. The surface area is $4\pi^2 Ra$. The condition $R > a$ is essential for this ordinary ring-torus description; it keeps the generating circle away from the axis and avoids overlapping radial regions.

A useful surface parametrization is $((R + a \cos v) \cos u, (R + a \cos v) \sin u, a \sin v)$ with both angles running through a full period. Its area element is $a(R + a \cos v) du dv$ for $R > a$. It explains the total area formula and permits partial-surface calculations that cannot be found by taking a simple fraction of the total.

Problem 284.1[See solution](#)

The surface obtained by revolving $(y - 2)^2 + z^2 = 1$ in the yz -plane about the z -axis intersects the sphere $x^2 + y^2 + z^2 = 5$. What is the total length of the intersection?

- (A) 2π (B) 4π (C) $4\pi\sqrt{2}$ (D) 8π (E) $4\pi\sqrt{5}$

Problem 284.2[See solution](#)

A solid torus has major radius 3, minor radius 2, and symmetry axis the z -axis. What is its volume between the planes $z = -1$ and $z = 1$?

- (A) $4\pi^2 + 6\pi\sqrt{3}$ (B) $8\pi^2 + 12\pi\sqrt{3}$ (C) $12\pi^2 + 8\pi\sqrt{3}$ (D) $16\pi^2$ (E) $24\pi^2$

Problem 284.3[See solution](#)

A solid torus has major radius 5 and minor radius 2, with axis the z -axis. What is the area of its cross section by $z = 1$?

- (A) $40\pi\sqrt{3}$ (B) 12π (C) $10\pi\sqrt{3}$ (D) 20π (E) $20\pi\sqrt{3}$

Problem 284.4[See solution](#)

A circle of radius 2 whose center is distance 5 from an axis in its plane is revolved about that axis. What is the area of the resulting torus surface?

- (A) $80\pi^2$ (B) $20\pi^2$ (C) $40\pi^2$ (D) $100\pi^2$ (E) 40π

Problem 284.5[See solution](#)

A ring torus has major radius three and minor radius one, with axis the z -axis. What is the area of the portion of its surface satisfying $\sqrt{x^2 + y^2} \geq 3$?

- (A) $6\pi^2$ (B) $6\pi^2 + 4\pi$ (C) $6\pi^2 - 4\pi$ (D) $12\pi^2$ (E) $4\pi^2 + 6\pi$

9. Practice Test

Practice Test ▸ Question 38: a quantitative continuity condition

285 Lipschitz and Holder conditions; constancy from a strong difference bound

Book pp. 413. [Find in compact glossary](#)

In this entry: Lipschitz continuity; Holder continuity; Holder exponent; uniform continuity; difference bound.

Definition and key ideas

A function on a subset of $\mathbb{R}$ is Lipschitz if $|f(x) - f(y)| \leq L|x - y|$ for one finite constant L and every pair in its domain. More generally it is Holder continuous with exponent $0 < \alpha \leq 1$ if $|f(x) - f(y)| \leq C|x - y|^\alpha$. Either bound implies uniform continuity. A bounded derivative on an interval gives a Lipschitz bound by the mean-value theorem; the converse does not imply differentiability everywhere, as $|x|$ illustrates.

The practice question uses the stronger inequality with exponent 2. If an exponent $\alpha > 1$ holds on an interval, the function is constant. Indeed divide $[x, y]$ into n equal pieces and add the bounds to get $|f(y) - f(x)| \leq C|y - x|^\alpha n^{1-\alpha} \rightarrow 0$. This conclusion depends on being able to subdivide within the domain. On a disconnected domain, the function can have different constants on different interval components, subject to the cross-component bounds. The terminology here names the condition used in the question.

Uniform continuity means that for every $\varepsilon > 0$ one $\delta > 0$ works for every pair of domain points: $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$. A Holder bound supplies δ from C, α, ε independently of the points. Even a Lipschitz function can fail to be differentiable at a point, so these estimates should not be used to assert existence of derivatives.

Problem 285.1

[See solution](#)

Let $E = [0, 1] \cup [2, 3]$. A function $f : E \rightarrow \mathbb{R}$ satisfies $f(0) = 0$ and $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in E$. What is the largest possible value of $f(3)$?

- (A) 0 (B) 1 (C) 2 (D) 4 (E) 9

Problem 285.2

[See solution](#)

What is the largest positive real number α for which there exists a finite constant C such that $|\sqrt{x} - \sqrt{y}| \leq C|x - y|^\alpha$ for all $x, y \in [0, 4]$?

- (A) $1/4$ (B) $1/2$ (C) 1 (D) 2 (E) No positive α works.

Problem 285.3[See solution](#)

A function on $[0, 1]$ satisfies $|f(x) - f(y)| \leq |x - y|/\log(e/|x - y|)$ whenever $x \neq y$. What must be true?

- (A) f is a nonconstant affine function.
- (B) f must equal zero.
- (C) f is strictly increasing.
- (D) f is constant.
- (E) f can be discontinuous.

Problem 285.4[See solution](#)

What is the optimal Lipschitz constant for $f(x) = x^{3/2}$ on $[0, 4]$?

- (A) 3 (B) 2 (C) 4 (D) $3/2$ (E) 6

Problem 285.5[See solution](#)

Suppose f is 2-Lipschitz and g is 3-Lipschitz, with the range of g contained in the domain of f . What is the smallest universal Lipschitz bound for $f \circ g$?

- (A) 3 (B) 9 (C) 2 (D) 5 (E) 6

Problem 285.6[See solution](#)

Let $f(0) = 0$ and $f(x) = x \sin(\log x)$ for $0 < x \leq 1$. Which statement is correct?

- (A) f is discontinuous at zero.
- (B) f is Lipschitz on $[0, 1]$ but has no right derivative at zero.
- (C) f has right derivative zero at zero but is not uniformly continuous.
- (D) f is uniformly continuous but cannot be Lipschitz.
- (E) f is differentiable at zero with right derivative one.

9. Practice Test

Practice Test ▸ Question 57: Cauchy mean-value formula

286 Cauchy mean-value theorem and simultaneous derivative ratios

Book pp. 418. [Find in compact glossary](#)

In this entry: Cauchy mean-value theorem; generalized mean-value theorem; common intermediate point.

Definition and key ideas

If f and g are continuous on $[a, b]$ and differentiable on (a, b) , where $a < b$, there is $c \in (a, b)$ such that $g'(c)(f(b) - f(a)) = f'(c)(g(b) - g(a))$. Apply Rolle's theorem to $(f(b) - f(a))(g(x) - g(a)) - (g(b) - g(a))(f(x) - f(a))$ to prove it. The cross-multiplied formulation is valid even if derivatives vanish.

If g' never vanishes on (a, b) , Rolle's theorem also ensures $g(b) \neq g(a)$, and the conclusion can be written $(f(b) - f(a))/(g(b) - g(a)) = f'(c)/g'(c)$. Applying the ordinary mean-value theorem separately to f and g does not supply a common point. Dividing by $f'(c)$ or $g'(c)$ without knowing it is nonzero is another invalid step. This theorem is a standard route to derivative-ratio estimates and to L'Hopital's rule.

With a common zero endpoint and $g' > 0$, Cauchy's theorem can compare f/g with the derivative ratio at a later point. If f'/g' is strictly increasing, the quotient rule then shows that f/g is strictly increasing as well. This uses the common intermediate point rather than separate applications of the ordinary theorem.

Problem 286.1

[See solution](#)

For $f(x) = x^2$ and $g(x) = x^3$ on $[0, 1]$, let c_f, c_g be their respective ordinary mean-value points, and let $c_C \in (0, 1)$ be the Cauchy mean-value point for the pair (f, g) . What is (c_f, c_g, c_C) ?

- (A) $(1/2, 1/\sqrt{3}, 2/3)$ (B) $(1/2, 1/\sqrt{3}, \sqrt{2/3})$
 (C) $(1/2, 2/3, 1/\sqrt{3})$ (D) $(1/\sqrt{3}, 1/2, 2/3)$
 (E) $(2/3, 2/3, 2/3)$

Problem 286.2

[See solution](#)

Apply the Cauchy mean-value theorem to $f(x) = \log x$ and $g(x) = x^{-1}$ on $[1, 4]$. Which number is the resulting point $c \in (1, 4)$?

- (A) $\frac{3}{4 \log 4}$ (B) $\frac{3 \log 4}{4}$ (C) $\frac{4 \log 4}{3}$ (D) $\frac{4}{3 \log 4}$ (E) $\sqrt{\frac{4 \log 4}{3}}$

Problem 286.3[See solution](#)

Apply the cross-multiplied Cauchy mean-value theorem to $f(x) = \cos x$ and $g(x) = \cos(2x)$ on $[0, \pi]$. Which statement is correct?

- (A) The theorem is inapplicable because $g(0) = g(\pi)$.
- (B) The unique interior point is $\pi/2$, but the derivative-ratio formulation is undefined there.
- (C) The unique interior point is $\pi/4$.
- (D) Every interior point works.
- (E) The unique point is $\pi/2$, and the derivative ratio equals zero.

Problem 286.4[See solution](#)

Let f, g be continuously differentiable on $[0, 1]$, with $f(0) = g(0) = 0$ and $g' > 0$. Suppose f'/g' is strictly increasing. What must hold on $(0, 1)$?

- (A) f/g is strictly increasing.
- (B) f/g is strictly decreasing.
- (C) f is everywhere positive.
- (D) f is constant.
- (E) f/g is constant.

Problem 286.5[See solution](#)

Suppose f, g are continuous on $[a, b]$, differentiable on (a, b) , $g' > 0$, and $2 \leq f'/g' \leq 5$. Which must hold?

- (A) $2 \leq [f(b) - f(a)]/[g(b) - g(a)] \leq 5$
- (B) $g(b) - g(a) = 1$
- (C) $f(b) - f(a) \geq 2$
- (D) The secant ratio can be any real number.
- (E) $f(b) - f(a) \leq 5$

9. Practice Test

Practice Test ▷ Question 58: a topology with finite complements

287 Cofinite topology and nonunique limits

Book pp. 419. [Find in compact glossary](#)**In this entry:** cofinite topology; finite-complement topology; T_1 separation; nonunique sequence limit.

Definition and key ideas

The cofinite topology on a set X consists of $\emptyset$ together with all subsets whose complements are finite. Its closed sets are the finite sets and X itself. On a finite set it is discrete. On an infinite set every two nonempty open sets intersect, so it is connected and is not Hausdorff; nevertheless every singleton is closed, which is the T_1 separation property.

Every subset is compact with its subspace topology: in a cover choose one nonempty member, which leaves only finitely many points uncovered, and cover those points using finitely many more members. A sequence converges to x precisely when it eventually avoids each finite subset not containing x . Thus a sequence of distinct points converges to every point. These examples explain why compact subsets need not be closed without a Hausdorff hypothesis, and why uniqueness of limits requires a separation assumption.

Problem 287.1

[See solution](#)

Give $\mathbb{R}$ its cofinite topology. How many limits does the sequence $x_n = n$ have in this space?

- (A) 0
- (B) 1
- (C) 2
- (D) Countably infinitely many.
- (E) Uncountably infinitely many.

Problem 287.2

[See solution](#)

Both the domain and codomain $\mathbb{R}$ carry the cofinite topology. Which functions are continuous?

I. $f(x) = x^2$. II. $g(x) = e^x$. III. $h(x) = \sin x$.

- (A) I only. (B) II only. (C) I and II only. (D) II and III only. (E) I, II, and III.

Problem 287.3

[See solution](#)

In the cofinite topology on $\mathbb{R}$, let $x_{2n} = 0$ and $x_{2n-1} = n$. What is the set of limits of this sequence?

- (A) $\{1, 2, 3, \dots\}$ (B) $\mathbb{R}$ (C) $\mathbb{R} \setminus \{0\}$ (D) The empty set (E) $\{0\}$

Problem 287.4[See solution](#)

An infinite set X has the cofinite topology. Which subsets of X are compact in their subspace topology?

- (A) Only finite subsets and X
- (B) Only finite subsets
- (C) Only X itself
- (D) Only countable subsets
- (E) Every subset

Problem 287.5[See solution](#)

A continuous map from an infinite cofinite space to a Hausdorff space must have which property?

- (A) Its image is infinite.
- (B) It is injective.
- (C) It is surjective.
- (D) Its image has exactly two points.
- (E) It is constant.

10. Practice Test Answers and Explanations

Practice Test Answers and Explanations > Question 46: Pascal triangle

288 Pascal triangle and binomial sum identitiesBook pp. 436. [Find in compact glossary](#)**In this entry:** Pascal triangle; Pascal identity; hockey-stick identity; Vandermonde identity; weighted binomial sum.**Definition and key ideas**

Number the rows and positions of Pascal's triangle starting at zero. Entry k in row n is $\binom{n}{k}$. The endpoints are 1 and each interior entry is the sum of its two parents: $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. This identity also follows by separating k -element subsets according to whether they contain a particular element. It gives the diagonal or hockey-stick sum $\sum_{j=r}^n \binom{j}{r} = \binom{n+1}{r+1}$.

Products of binomial expansions give Vandermonde's identity $\sum_k \binom{r}{k} \binom{s}{m-k} = \binom{r+s}{m}$, with out-of-range binomial coefficients interpreted as zero. Differentiating $(1+x)^n$ yields weighted row sums: $\sum_k k \binom{n}{k} = n2^{n-1}$ and $\sum_k k(k-1) \binom{n}{k} = n(n-1)2^{n-2}$. Combine these to obtain the sum weighted by k^2 . These are applications of earlier binomial coefficients; the triangle is first explicitly constructed in this solution.

Integration of the binomial expansion produces reciprocal weights, such as $\sum_{k=0}^n \binom{n}{k} / (k+1) = (2^{n+1} - 1) / (n+1)$. Modulo two, repeated squaring gives $(1+x)^{2^j} = 1+x^{2^j}$; multiplying the factors for the one-bits in the binary expansion of n counts the odd entries in row n . If that expansion has s one-bits, there are 2^s odd entries.

Problem 288.1[See solution](#)What is $\sum_{k=0}^6 \binom{6}{k}^2$?

- (A) 462 (B) 720 (C) 792 (D) 924 (E) 1024

Problem 288.2[See solution](#)What is $\sum_{k=0}^8 k^2 \binom{8}{k}$?

- (A) 2048 (B) 4096 (C) 4608 (D) 8192 (E) 9216

Problem 288.3[See solution](#)What is $\sum_{k=0}^8 (-1)^k \binom{8}{k}^2$?

- (A) 128 (B) 0 (C) -70 (D) 70 (E) 256

Problem 288.4[See solution](#)

How many entries in row ten of Pascal's triangle are odd, numbering the first row as row zero?

- (A) 2 (B) 4 (C) 5 (D) 6 (E) 10

Problem 288.5[See solution](#)

What is $\sum_{k=0}^6 \frac{1}{k+1} \binom{6}{k}$?

- (A) 9 (B) $64/7$ (C) $127/7$ (D) $128/7$ (E) 127

Problem 288.6[See solution](#)

What is $\sum_{k=3}^9 (10-k) \binom{k}{3}$?

- (A) 210 (B) 252 (C) 330 (D) 462 (E) 924

Supplement. Breadth beyond the book

Multivariable calculus

289 Jacobians and multivariable change of variables

Breadth supplement. [Find in compact glossary](#)

In this entry: Jacobian matrix; Jacobian determinant; change of variables; inverse function theorem.

Definition and key ideas

For a differentiable map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, its Jacobian matrix is $DT = (\partial T_i / \partial u_j)$. The determinant measures signed local volume scaling; integration uses its absolute value. If T is a one-to-one C^1 change of coordinates on suitable regions and its determinant is nonzero, then $\int_{T(U)} f(x) dx = \int_U f(T(u)) |\det DT(u)| du$. Boundary sets of measure zero do not change the integral. Count multiplicity if the parametrization covers a region more than once.

For T of class C^1 on an open neighborhood, the inverse function theorem gives a differentiable local inverse near a point where $\det DT \neq 0$, with derivative $(DT)^{-1}$. Local invertibility is not global injectivity. The columns of the Jacobian are the images of coordinate directions, so its determinant supplies the area/volume factor that substitution must include.

Problem 289.1

[See solution](#)

Let $R = \{(x, y) : 0 \leq x + y \leq 2, 0 \leq x - y \leq 1\}$. What is $\iint_R (x^2 - y^2) dx dy$?

- (A) $1/4$ (B) $1/2$ (C) 1 (D) 2 (E) 4

Problem 289.2

[See solution](#)

The map $T(u, v) = (u^2 - v^2, 2uv)$ maps the half disk $u^2 + v^2 < 1, v > 0$, onto the unit disk apart from a slit and its center. What is $\iint_{u^2+v^2<1, v>0} |\det DT| du dv$?

- (A) $\pi/4$ (B) $\pi/2$ (C) π (D) 2π (E) 4π

Problem 289.3

[See solution](#)

In the first quadrant, let R be defined by $1 \leq xy \leq 4$ and $1 \leq y/x \leq 9$. What is its area?

- (A) $6 \log 3$ (B) $3 \log 3/2$ (C) $3 \log 3$ (D) $4 \log 3$ (E) 9

Problem 289.4

[See solution](#)

What is $\iint_{x^2/4 + y^2/9 \leq 1} (x^2/4 + y^2/9) dx dy$?

- (A) 9π (B) $3\pi/2$ (C) π (D) 3π (E) 6π

Problem 289.5[See solution](#)

For $T(u, v) = (e^u \cos v, e^u \sin v)$ on $\mathbb{R}^2$, which statement is correct?

- (A) The Jacobian vanishes everywhere.
- (B) T has a differentiable local inverse at every point but is not globally one-to-one.
- (C) T is a bijection from $\mathbb{R}^2$ onto $\mathbb{R}^2$.
- (D) The Jacobian changes sign when v changes sign.
- (E) T has no local inverse at any point.

Supplement. Breadth beyond the book

Multivariable calculus

290 Triple integrals and solid coordinate systems

Breadth supplement. [Find in compact glossary](#)**In this entry:** triple integral; volume element; spherical volume element; cylindrical volume element.**Definition and key ideas**

A triple integral sums a density over a solid. Fubini's theorem permits iterated integration for continuous functions on compact regular solids, and more generally for integrable functions; Tonelli's theorem permits nonnegative integrands even before finiteness is known. In cylindrical coordinates $dV = r \, dr \, d\theta \, dz$. With polar angle ϕ measured from the positive z -axis, spherical coordinates give $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.

Derive bounds from the solid before manipulating the integrand. Sphere-cone intersections are usually spherical; vertical cylinders are usually cylindrical. The coordinate system changes the volume element, not merely the formula for the function. Symmetry makes odd integrands integrate to zero over regions symmetric in the relevant coordinate.

Problem 290.1[See solution](#)What is $\iiint_{x^2+y^2+z^2 \leq 1, z \geq \sqrt{x^2+y^2}} (x^2 + y^2 + z^2) \, dV$?

- (A) $\frac{\pi}{5}(2 - \sqrt{2})$ (B) $\frac{\pi}{3}(2 - \sqrt{2})$ (C) $\frac{2\pi}{5}$ (D) $\frac{\pi}{5}\sqrt{2}$ (E) $\frac{4\pi}{5}$

Problem 290.2[See solution](#)The region S is defined by $x, y, z \geq 0$ and $x + y + z \leq 1$. What is $\iiint_S xyz \, dV$?

- (A) $1/120$ (B) $1/360$ (C) $1/720$ (D) $1/1440$ (E) $1/5040$

Problem 290.3[See solution](#)What is $\iiint_{x^2+y^2+z^2 \leq 4} z^2 \, dV$?

- (A) $128\pi/5$ (B) $256\pi/15$ (C) $128\pi/15$ (D) $64\pi/15$ (E) $32\pi/15$

Problem 290.4[See solution](#)The solid S lies above $z = x^2 + y^2$ and below $z = 4$. What is its volume?

- (A) 4π (B) 32π (C) 16π (D) 8π (E) $16\pi/3$

Problem 290.5[See solution](#)What is the volume of $\{x, y, z \geq 0 : x/2 + y/3 + z/4 \leq 1\}$?

- (A) 2 (B) 12 (C) 4 (D) 6 (E) 24

Supplement. Breadth beyond the book

Multivariable calculus

291 Surface area and flux integrals

Breadth supplement. [Find in compact glossary](#)**In this entry:** regular surface parametrization; surface element; surface integral; flux; orientation.

Definition and key ideas

For a regular parametrized surface $\mathbf{r}(u, v)$, the vector $\mathbf{r}_u \times \mathbf{r}_v$ is normal to the surface, and $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv$. A scalar surface integral is $\iint_S f dS$. An oriented flux integral is $\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint \mathbf{F}(\mathbf{r}) \cdot (\mathbf{r}_u \times \mathbf{r}_v) du dv$. Reversing orientation changes flux's sign but not surface area.

For a graph $z = g(x, y)$, the upward vector area element is $(-g_x, -g_y, 1) dx dy$. Its length is $\sqrt{1 + g_x^2 + g_y^2}$. Distinguish the unit normal from the vector area element: normalizing and then forgetting the area factor is a frequent mistake.

Regular means that the parameter derivatives are linearly independent, equivalently $\mathbf{r}_u \times \mathbf{r}_v \neq 0$. A unit normal has length one; the vector area element combines that direction with the scalar area factor. A parametrization used for integration should cover the requested surface once, apart from harmless boundary identifications.

Problem 291.1

[See solution](#)

Let S be the graph $z = x^2 + y^2$ over $x^2 + y^2 \leq 1$, oriented upward. What is the flux of $\mathbf{F} = (x, y, z)$ through S ?

- (A) $-\pi$ (B) $-\pi/2$ (C) 0 (D) $\pi/2$ (E) π

Problem 291.2

[See solution](#)

What is the surface area of $z = x^2 + y^2$ above the unit disk?

- (A) π (B) $\frac{\pi}{3}(5\sqrt{5} - 1)$ (C) $\frac{\pi}{6}(5\sqrt{5} - 1)$ (D) $\frac{\pi}{6}(5\sqrt{5} + 1)$ (E) $5\pi/2$

Problem 291.3

[See solution](#)

What is the area of the surface parametrized by $\mathbf{r}(u, v) = (u \cos v, u \sin v, v)$ for $0 \leq u \leq 1$ and $0 \leq v \leq 2\pi$?

- (A) 2π (B) $\pi\sqrt{2}$
 (C) $\pi(\sqrt{2} + \log(1 + \sqrt{2}))$ (D) $2\pi(\sqrt{2} + \log(1 + \sqrt{2}))$
 (E) $\pi(\sqrt{2} - \log(1 + \sqrt{2}))$

Problem 291.4

[See solution](#)

Let S be the spherical cap $x^2 + y^2 + z^2 = 4$, $z \geq 1$, oriented outward. What are its surface area and the flux of $(0, 0, 1)$ through it, respectively?

- (A) $(3\pi, 3\pi)$ (B) $(4\pi, 4\pi)$ (C) $(4\pi, 3\pi)$ (D) $(8\pi, 4\pi)$ (E) $(4\pi, -3\pi)$

Problem 291.5[See solution](#)

A planar patch in the plane $x + y + z = 0$ has area one. Under the linear map $(x, y, z) \mapsto (2x, 3y, 4z)$, what is the area of its image?

- (A) 24 (B) 6 (C) $2\sqrt{61/3}$ (D) $\sqrt{29}$ (E) $2\sqrt{3}$

Supplement. Breadth beyond the book

Multivariable calculus

292 Divergence, curl, and differential identities

Breadth supplement. [Find in compact glossary](#)**In this entry:** divergence; curl; solenoidal field; irrotational field; Laplacian; vector Laplacian.**Definition and key ideas**

For $\mathbf{F} = (P, Q, R)$, divergence is $\nabla \cdot \mathbf{F} = P_x + Q_y + R_z$, a scalar measuring local outward flux. Curl is $\nabla \times \mathbf{F} = (R_y - Q_z, P_z - R_x, Q_x - P_y)$, measuring local circulation. A divergence-free field is solenoidal; a curl-free field is irrotational. The Laplacian of f is $\Delta f = \nabla \cdot \nabla f$.

When second derivatives are continuous, $\nabla \times \nabla f = 0$ and $\nabla \cdot (\nabla \times \mathbf{F}) = 0$. These are necessary tests for potential representations. A curl-free field on an open simply connected region is conservative, but holes can obstruct a global potential. Divergence-free alone does not imply curl-free or conservativity.

For a vector field, the vector Laplacian applies the scalar Laplacian to each component. With continuous second partial derivatives, $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \Delta \mathbf{F}$. The product rule is $\nabla \times (\phi \mathbf{F}) = \nabla \phi \times \mathbf{F} + \phi(\nabla \times \mathbf{F})$.

Problem 292.1[See solution](#)

For $\mathbf{F} = (axy, y^2 + bz, xz)$, which pair (a, b) makes $\nabla \cdot \mathbf{F}$ identically equal to x ?

- (A) $(0, 0)$ (B) $(2, 0)$ (C) $(-2, 0)$ only (D) $a = -2$ with any real b (E) No pair

Problem 292.2[See solution](#)

For every twice continuously differentiable vector field $\mathbf{F}$ on an open subset of $\mathbb{R}^3$, which identity holds, with the Laplacian applied componentwise?

- (A) $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \Delta \mathbf{F}$ (B) $\nabla \times (\nabla \times \mathbf{F}) = \Delta \mathbf{F}$
 (C) $\nabla \times (\nabla \times \mathbf{F}) = \mathbf{0}$ (D) $\nabla(\nabla \cdot \mathbf{F}) = \mathbf{0}$
 (E) $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) + \Delta \mathbf{F}$

Problem 292.3[See solution](#)

For $\mathbf{F} = (yz, xz, xy)$, which pair is $(\nabla \cdot \mathbf{F}, \nabla \times \mathbf{F})$?

- (A) $(0, (x, y, z))$ (B) $(0, \mathbf{0})$ (C) $(xyz, \mathbf{0})$ (D) $(0, (1, 1, 1))$ (E) $(x + y + z, \mathbf{0})$

Problem 292.4[See solution](#)

For a twice continuously differentiable scalar function f , which vector field is necessarily divergence free?

- (A) $\nabla \times (f, 0, 0)$
- (B) $f(x, y, z)$, interpreted as a radial multiple (fx, fy, fz)
- (C) ∇f
- (D) (f, f, f)
- (E) $(f_x, 0, 0)$

Problem 292.5[See solution](#)

If $\mathbf{F} = (-y, x, 0)$ and $\phi = z$, what is $\nabla \times (\phi\mathbf{F})$?

- (A) $(-x, -y, 0)$ (B) $(-x, -y, 2z)$ (C) $(x, y, 2z)$ (D) $(0, 0, z)$ (E) $(0, 0, 2z)$

Supplement. Breadth beyond the book

Multivariable calculus

293 The divergence theorem

Breadth supplement. [Find in compact glossary](#)

In this entry: divergence theorem; Gauss theorem; outward flux; closed surface.

Definition and key ideas

If a bounded solid V has a piecewise smooth boundary and $\mathbf{F}$ is continuously differentiable on a neighborhood of V , then $\iint_{\partial V} \mathbf{F} \cdot \mathbf{n} \, dS = \iiint_V \nabla \cdot \mathbf{F} \, dV$, where $\mathbf{n}$ points out of the solid. Internal flux cancels, leaving only the boundary. For a shell, the normal on an inner boundary points into the cavity.

Use the theorem to replace a difficult surface integral by a simpler volume integral. If the requested surface is open, add the missing caps and then subtract their flux. A field singular inside the region invalidates the direct theorem; remove a small neighborhood of the singularity and account for its boundary.

Problem 293.1

[See solution](#)

Let S be the boundary of the unit ball, oriented outward, and let $\mathbf{F} = (x^3, y^3, z^3)$. What is $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$?

- (A) $4\pi/5$ (B) $8\pi/5$ (C) $12\pi/5$ (D) 4π (E) 12π

Problem 293.2

[See solution](#)

Let S be the upper unit hemisphere, with outward normal, excluding its flat base. For $\mathbf{F} = (x, y, z + 1)$, the flux through S is

- (A) π (B) 2π (C) 3π (D) 4π (E) 0

Problem 293.3

[See solution](#)

Let $\mathbf{F} = (x, y, z)/(x^2 + y^2 + z^2)^{3/2}$. Its divergence is zero away from the origin. What is its outward flux through the unit sphere?

- (A) 0
(B) 2π
(C) 4π
(D) -4π
(E) The surface integral is undefined.

Problem 293.4[See solution](#)

A continuously differentiable vector field has zero outward flux through the boundary of every sufficiently small ball contained in its domain. What must hold throughout the domain?

- (A) The field is constant.
- (B) The field is zero.
- (C) Its curl is zero.
- (D) Its divergence is zero.
- (E) The field has a global scalar potential.

Problem 293.5[See solution](#)

Let S bound the region $1 \leq x^2 + y^2 + z^2 \leq 4$, oriented outward from that region. What is the flux of $\mathbf{F} = (x, y, z)/(x^2 + y^2 + z^2)^{3/2}$ through S ?

- (A) 12π (B) 4π (C) 8π (D) 0 (E) -4π

Supplement. Breadth beyond the book

Multivariable calculus

294 Stokes's theorem

Breadth supplement. [Find in compact glossary](#)**In this entry:** Stokes theorem; boundary orientation; circulation.**Definition and key ideas**

For an oriented piecewise smooth surface S with boundary C and a C^1 field on a neighborhood of S , Stokes's theorem states $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$. The orientation of C is induced by the surface normal through the right-hand rule. For an upward planar disk, positive boundary orientation is counterclockwise from above.

The key shortcut is that any admissible spanning surface with the same oriented boundary gives the same curl flux. A complicated cap can therefore be replaced by a disk. Green's circulation theorem is the planar instance. Check that the replacement surface does not cross a singularity of the field.

Problem 294.1[See solution](#)

Let C be the intersection of $x^2 + y^2 = 1$ and $z = x + y$, oriented counterclockwise as viewed from positive z . For $\mathbf{F} = (-y, z, x)$, what is $\oint_C \mathbf{F} \cdot d\mathbf{r}$?

- (A) -3π (B) $-\pi$ (C) 0 (D) π (E) 3π

Problem 294.2[See solution](#)

S is the upper unit hemisphere oriented outward. What is $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} dS$ for $\mathbf{F} = (y^3, z^3, x^3)$?

- (A) $-3\pi/4$ (B) $-3\pi/2$ (C) 0 (D) $3\pi/4$ (E) $3\pi/2$

Problem 294.3[See solution](#)

A smooth field $\mathbf{G}$ on $\mathbb{R}^3 \setminus \{0\}$ is divergence free and has outward flux one through the unit sphere. Can a smooth field $\mathbf{A}$ on the same whole domain satisfy $\nabla \times \mathbf{A} = \mathbf{G}$?

- (A) Yes, because every divergence-free field is a curl globally.
 (B) Yes, because the sphere has no boundary.
 (C) No, because Stokes would make the flux of this curl through the closed sphere zero.
 (D) No, because a divergence-free field cannot have nonzero flux through any sphere in its domain.
 (E) Only if $\mathbf{A}$ is constant.

Problem 294.4[See solution](#)

Let C be the circle $x^2 + y^2 = 4$, $z = 3$, oriented clockwise as viewed from above. For $\mathbf{F} = (-y, x, z^2)$, what is $\oint_C \mathbf{F} \cdot d\mathbf{r}$?

- (A) -4π (B) -8π (C) 8π (D) 0 (E) 4π

Problem 294.5[See solution](#)

A smooth vector field is defined on all of $\mathbb{R}^3$. What is the flux of its curl through any smooth closed oriented surface?

- (A) It need not have a fixed value.
- (B) The surface area
- (C) The enclosed volume
- (D) 0
- (E) The circulation around a great circle

Supplement. Breadth beyond the book

Differential equations

295 Existence, uniqueness, and maximal solution intervals

Breadth supplement. [Find in compact glossary](#)

In this entry: initial-value problem; local existence; local uniqueness; Picard-Lindelof theorem; maximal interval of existence.

Definition and key ideas

For $y' = F(t, y)$ with $y(t_0) = y_0$, continuity of F near (t_0, y_0) guarantees a local solution. A local Lipschitz condition in y gives uniqueness; continuity of $\partial F/\partial y$ near the initial point is a convenient sufficient condition. These are local conclusions, not promises of a solution for all real t . Even polynomial right sides can produce finite-time blowup.

A maximal interval is the largest interval containing the initial time on which that solution extends. When a uniqueness hypothesis fails, uniqueness may still hold in a particular problem, so failure of the theorem is not itself a proof of nonuniqueness. To demonstrate nonuniqueness, construct distinct solutions with the same initial condition and check them at the joining time.

The Picard–Lindelof theorem gives local existence and uniqueness for a continuous explicit right side that is locally Lipschitz in the dependent variable: on a small rectangle, $|F(t, y_1) - F(t, y_2)| \leq L|y_1 - y_2|$ for one finite L . An implicit equation may admit different explicit branches, each individually unique. The maximal interval must also stay inside the domain on which the differential equation itself is defined.

Problem 295.1

[See solution](#)

How many continuously differentiable solutions near $t = 0$ satisfy $(y')^2 = 1 + y^2$ and $y(0) = 0$?

- (A) None.
- (B) Exactly one.
- (C) Exactly two.
- (D) Countably infinitely many.
- (E) Uncountably infinitely many.

Problem 295.2

[See solution](#)

The maximal interval containing 0 for the solution of $y' = y^3$, $y(0) = 1$, is

- (A) $(-\infty, \infty)$ (B) $(-\infty, 1/2)$ (C) $(-1/2, 1/2)$ (D) $(1/2, \infty)$ (E) $(-\infty, 2)$

Problem 295.3

[See solution](#)

For the differential equation $y' = 2ty/(t^2 - 1)$, defined only when $t \neq \pm 1$, the initial condition is $y(0) = 0$. What is the maximal solution interval containing zero?

- (A) $\mathbb{R}$ (B) $(-\infty, 1)$ (C) $(-1, \infty)$ (D) $(-1, 1)$ (E) $\{0\}$

Problem 295.4[See solution](#)

Which initial-value problem is guaranteed by continuity of f and local continuity of $\partial f/\partial y$ to have a unique local solution near $(0, 0)$?

(A) $y' = |y|^{1/3}, y(0) = 0$

(B) $y' = \sqrt{|y|}, y(0) = 0$

(C) $y' = \sin(t + y^2), y(0) = 0$

(D) $y' = 1/y, y(0) = 0$

(E) $y' = 1/t, y(0) = 0$

Problem 295.5[See solution](#)

The right side of $y' = 1 + \sqrt{|y|}$ is continuous but not locally Lipschitz at zero. With $y(0) = 0$, which conclusion is correct?

(A) No local solution exists.

(B) Exactly two local solutions exist.

(C) Infinitely many waiting-time solutions exist.

(D) There is a unique local solution despite the failure of this sufficient Lipschitz condition.

(E) Every solution is constant.

Supplement. Breadth beyond the book

Differential equations

296 Autonomous equations and phase-line stability

Breadth supplement. [Find in compact glossary](#)

In this entry: autonomous differential equation; equilibrium solution; stable equilibrium; asymptotic stability; unstable equilibrium; phase line.

Definition and key ideas

An autonomous scalar equation has form $y' = f(y)$. Its equilibria are roots of f . A phase line marks the sign of f between successive roots: positive means solutions move upward and negative means downward. An equilibrium is stable if solutions starting sufficiently close stay close for future time; it is asymptotically stable if they also tend to the equilibrium.

For f continuously differentiable, $f'(a) < 0$ implies asymptotic stability of an equilibrium a , and $f'(a) > 0$ implies instability. If $f'(a) = 0$, inspect the sign directly; the derivative test is inconclusive. Under uniqueness, distinct solution curves do not cross, so a solution cannot pass through an equilibrium. A one-sided attracting point need not be stable from both sides.

Problem 296.1

[See solution](#)

For $y' = y(y - 1)(y - 2)$, which equilibria are asymptotically stable?

- (A) 0 only (B) 1 only (C) 2 only (D) 0 and 2 only (E) All three

Problem 296.2

[See solution](#)

The solution of $y' = y^2(1 - y)$ with $y(0) = -1$ has what limit as $t \rightarrow \infty$?

- (A) $-\infty$
(B) -1
(C) 0
(D) 1
(E) It reaches 0 in finite time and then ceases to exist.

Problem 296.3

[See solution](#)

For the two scalar equations $y' = -y^3$ and $z' = z^3$, how is the equilibrium zero classified, respectively?

- (A) Unstable; unstable.
(B) Asymptotically stable; unstable.
(C) Stable but not asymptotically stable; asymptotically stable.
(D) Asymptotically stable; asymptotically stable.
(E) The vanishing derivatives at zero prevent any conclusion.

Problem 296.4[See solution](#)

An autonomous equation has a continuously differentiable right side satisfying $f(y) = 0$ for every $y \in [-1, 1]$. Which statement about the equilibrium zero is forced?

- (A) It is unstable.
- (B) It is asymptotically stable.
- (C) It is stable but not asymptotically stable.
- (D) Every solution starting outside $[-1, 1]$ converges to zero.
- (E) Every solution starting near zero leaves $[-1, 1]$.

Problem 296.5[See solution](#)

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuously differentiable, which statement must hold for every periodic solution of $y' = f(y)$?

- (A) Its period is at least one.
- (B) Its mean over a period is zero.
- (C) It crosses an equilibrium twice in each period.
- (D) It is constant.
- (E) Its derivative never vanishes.

Supplement. Breadth beyond the book

Differential equations

297 Euler-Cauchy differential equations

Breadth supplement. [Find in compact glossary](#)**In this entry:** Euler-Cauchy equation; equidimensional equation; indicial equation.**Definition and key ideas**

An Euler-Cauchy equation, with $a \neq 0$, $ax^2y'' + bxy' + cy = 0$ is adapted to powers $y = x^m$ on $x > 0$. Substitution yields the indicial equation $am(m-1) + bm + c = 0$. Distinct real roots give independent powers; a repeated root m gives x^m and $x^m \log x$. Complex roots $\alpha \pm i\beta$ yield $x^\alpha \cos(\beta \log x)$ and $x^\alpha \sin(\beta \log x)$.

The change $t = \log x$ converts the equation to constant coefficients: $xy' = Y'$ and $x^2y'' = Y'' - Y'$ for $Y(t) = y(e^t)$. This also treats nonhomogeneous equations and explains logarithmic resonance. The point $x = 0$ is singular, so solve on intervals on one side of it and justify any proposed extension separately.

Problem 297.1[See solution](#)

For $x > 0$, the solution of $x^2y'' - 3xy' + 4y = 0$ with $y(1) = 0$ and $y'(1) = 1$ is

- (A) x^2 (B) $x^2 \log x$ (C) $x \log x$ (D) $\log x$ (E) $x^2(1 + \log x)$

Problem 297.2[See solution](#)

Which is a particular solution on $x > 0$ of $x^2y'' + xy' + y = x$?

- (A) x (B) $x/2$ (C) $x \log x$ (D) $\log x$ (E) $x^2/2$

Problem 297.3[See solution](#)

How many twice continuously differentiable functions on $(-1, 1)$ satisfy $x^2y'' - 2y = 0$ at every point and $y(0) = y'(0) = 0$?

- (A) None. (B) Exactly one. (C) Exactly two. (D) Exactly three. (E) Infinitely many.

Problem 297.4[See solution](#)

What is the solution of $x^2y'' + xy' + 4y = 0$, $y(1) = 1$, $y'(1) = 0$, on $x > 0$?

- (A) x^2 (B) x^{-2} (C) $\sin(2 \log x)$ (D) $\cos(2x)$ (E) $\cos(2 \log x)$

Problem 297.5[See solution](#)

Which is a particular solution of $x^2y'' - 2xy' + 2y = x^2$ for $x > 0$?

- (A) $x \log x$ (B) $x^2 \log x$ (C) $x^2(\log x)^2$ (D) $-x^2 \log x$ (E) $x^2/2$

Supplement. Breadth beyond the book

Differential equations

298 Linear differential systems and matrix exponentials

Breadth supplement. [Find in compact glossary](#)

In this entry: linear differential system; matrix exponential; fundamental matrix; Jordan-block solution.

Definition and key ideas

For a constant matrix A , the system $v' = Av$ with $v(0) = v_0$ has solution $v(t) = e^{tA}v_0$, where $e^{tA} = \sum_{k \geq 0} t^k A^k / k!$. If $A = PDP^{-1}$ is diagonalizable, then $e^{tA} = Pe^{tD}P^{-1}$. A repeated eigenvalue with a nontrivial Jordan block introduces powers of t multiplying the exponential.

For a block $A = \lambda I + N$ with $N^r = 0$, the commuting factors give $e^{tA} = e^{\lambda t}(I + tN + \cdots + t^{r-1}N^{r-1}/(r-1)!)$. A fundamental matrix has independent solution columns and is invertible. For a real constant system, all eigenvalues having negative real part implies asymptotic stability of the zero solution; eigenvalues on the imaginary axis require closer analysis.

For an arbitrary fundamental matrix Φ , the solution with initial vector v_0 at zero is $\Phi(t)\Phi(0)^{-1}v_0$. A nontrivial Jordan block for an eigenvalue of real part zero creates polynomial growth and prevents stability, whereas a diagonalizable system with purely imaginary eigenvalues has bounded oscillatory solutions.

Problem 298.1

[See solution](#)

Let $x' = 2x + y$, $y' = 2y$, with $x(0) = 0$, $y(0) = 1$. What is $x(t)$?

- (A) e^{2t} (B) te^{2t} (C) t^2e^{2t} (D) $(e^{2t} - 1)/2$ (E) $e^{2t} - e^t$

Problem 298.2

[See solution](#)

A real 2×2 matrix A has trace -2 and determinant 5 . Which statement must hold for every solution of $v' = Av$?

- (A) $v(t)$ is periodic.
 (B) $v(t) \rightarrow 0$ as $t \rightarrow \infty$.
 (C) Every nonzero solution is unbounded for $t > 0$.
 (D) $v(t)$ is a polynomial in t .
 (E) $v(t)$ is constant.

Problem 298.3

[See solution](#)

For $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$, what is the first component of $e^{tA}(1, 0)^T$?

- (A) $\cosh(2t)$ (B) $\cos(2t)$ (C) $\sin(2t)$ (D) e^{-2t} (E) e^{2t}

Problem 298.4[See solution](#)

A fundamental matrix for $v' = Av$ is $\Phi(t) = \begin{pmatrix} e^t & 0 \\ 0 & 2e^{-t} \end{pmatrix}$. What is the solution with $v(0) = (1, 2)^T$?

- (A) $(e^t, 4e^{-t})^T$ (B) $(e^t, e^{-t})^T$ (C) $(e^t, 2e^{-t})^T$ (D) $(2e^t, e^{-t})^T$ (E) $(e^{-t}, 2e^t)^T$

Problem 298.5[See solution](#)

A real 2×2 matrix has trace 0 and determinant -4 . What is the dimension of the real subspace of initial vectors whose solutions of $v' = Av$ tend to zero as $t \rightarrow \infty$?

- (A) 3
(B) 0
(C) 1
(D) 2
(E) It depends on the matrix.

Problem 298.6[See solution](#)

For $v' = Nv$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, which statement about the zero solution is correct?

- (A) It is asymptotically stable because both eigenvalues are zero.
(B) It is stable because neither eigenvalue has positive real part.
(C) It is unstable because arbitrarily small initial values can produce unbounded linear growth.
(D) Every solution is periodic.
(E) Every solution tends to a finite limit.

Supplement. Breadth beyond the book

Linear algebra

299 Inner products, orthogonality, and Gram-Schmidt

Breadth supplement. [Find in compact glossary](#)

In this entry: inner product; norm induced by an inner product; orthonormal basis; Gram-Schmidt process; Cauchy-Schwarz inequality.

Definition and key ideas

An inner product is a positive-definite sesquilinear form: over $\mathbb{C}$, it is linear in one argument, conjugate-linear in the other, and satisfies conjugate symmetry. Here $\langle x, y \rangle = x^*y$ is linear in the second argument. It induces $\|x\| = \sqrt{\langle x, x \rangle}$. Orthogonal vectors have zero inner product; an orthonormal list consists of unit, pairwise orthogonal vectors. Cauchy-Schwarz gives $|\langle x, y \rangle| \leq \|x\|\|y\|$.

Gram-Schmidt converts a linearly independent list $v_1, \dots, v_k$ to an orthonormal list spanning the same successive subspaces. Subtract $\sum_{j < k} q_j \langle q_j, v_k \rangle$ and normalize the nonzero residual. A zero residual signals dependence. Inner products of functions, such as $\int_a^b f(x)g(x) dx$, follow the same geometry.

Positive definite means that $\langle v, v \rangle$ is real and strictly positive for every nonzero v , and equals zero for $v = 0$. Equality in Cauchy-Schwarz occurs exactly when the two vectors are linearly dependent. For a dependent input list, discard each zero Gram-Schmidt residual and continue with the remaining vectors.

Problem 299.1

[See solution](#)

On $P_2(\mathbb{R})$ use $\langle f, g \rangle = \int_{-1}^1 fg$. The monic quadratic orthogonal to both 1 and x is

- (A) x^2 (B) $x^2 - 1/2$ (C) $x^2 - 1/3$ (D) $x^2 - x - 1/3$ (E) $x^2 - 2/3$

Problem 299.2

[See solution](#)

Gram-Schmidt is applied in order to $v_1 = (1, 1, 0)$, $v_2 = (1, 0, 1)$ and $v_3 = (0, 1, 1)$. The squared norm of the third residual, before normalization, is

- (A) $1/3$ (B) $2/3$ (C) 1 (D) $4/3$ (E) 2

Problem 299.3

[See solution](#)

A real continuous function f on $[0, 1]$ satisfies $\int_0^1 f = 0$ and $\int_0^1 f^2 = 1$. What is the largest possible value of $\int_0^1 xf(x) dx$?

- (A) $1/12$ (B) $1/6$ (C) $1/(2\sqrt{3})$ (D) $1/2$ (E) $1/\sqrt{3}$

Problem 299.4[See solution](#)

In $\mathbb{C}^2$ with $\langle v, w \rangle = v^*w$, what is the orthogonal projection of $w = (1, 1)$ onto the complex span of $v = (1, i)$?

- (A) $\frac{1}{2}(1 + i, 1 - i)$ (B) $\frac{1}{2}(1 - i, 1 + i)$ (C) $(1, 1)$ (D) $(0, 0)$ (E) $\frac{1}{2}(1 - i, -1 - i)$

Problem 299.5[See solution](#)

For real t , when does $\langle x, y \rangle = x^T \begin{pmatrix} 1 & t \\ t & 4 \end{pmatrix} y$ define an inner product on $\mathbb{R}^2$?

- (A) $|t| < 4$ (B) $|t| < 2$ (C) $|t| \leq 2$ (D) $t > 0$ (E) Every real t

Problem 299.6[See solution](#)

Gram–Schmidt is applied to $(1, 0, 1), (0, 1, 1), (1, 1, 2), (1, 1, 0)$ in that order, discarding a zero residual and continuing. How many nonzero orthonormal vectors result?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Supplement. Breadth beyond the book

Linear algebra

300 Orthogonal projections and least squares

Breadth supplement. [Find in compact glossary](#)

In this entry: orthogonal projection onto a subspace; least squares; normal equations; projection matrix.

Definition and key ideas

For a subspace W of a finite-dimensional inner product space, every vector has a unique decomposition $v = w + w^\perp$, with $w \in W$ and $w^\perp \perp W$. The projection $P_W v = w$ minimizes distance to W . If the columns of Q form an orthonormal basis of W , then $P_W = QQ^*$. If a real matrix A has independent columns, $P = A(A^T A)^{-1} A^T$.

A least-squares solution minimizes $\|Ax - b\|^2$ and satisfies the normal equations $A^T A x = A^T b$. The fitted vector Ax is unique even when the coefficient vector is not. A real matrix is an orthogonal projection exactly when $P^2 = P$ and $P^T = P$. Idempotence alone allows oblique projections.

Problem 300.1

[See solution](#)

Find the least-squares slope b of the line $y = a + bx$ fitted to $(0, 1), (1, 0), (2, 2)$.

- (A) 0 (B) $1/3$ (C) $1/2$ (D) $2/3$ (E) 1

Problem 300.2

[See solution](#)

P is the orthogonal projection in $\mathbb{R}^4$ onto the solutions of $x_1 + x_2 + x_3 + x_4 = 0$. Which pair equals $(\text{tr } P, \|Pe_1\|^2)$?

- (A) $(1, 1/4)$ (B) $(3, 1/4)$ (C) $(3, 3/4)$ (D) $(4, 3/4)$ (E) $(1, 3/4)$

Problem 300.3

[See solution](#)

Let $A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 0 \end{pmatrix}$ and $b = (1, 0, 1)^T$. Which describes the vectors x minimizing $\|Ax - b\|^2$?

- (A) There is a unique minimizer, with minimum zero.
 (B) There are infinitely many minimizers, all satisfying $x_1 + x_2 = 1/5$, with minimum $9/5$.
 (C) There are infinitely many minimizers, all satisfying $x_1 + x_2 = 1$, with minimum one.
 (D) There is no minimizer because $A^T A$ is singular.
 (E) Every x is a minimizer.

Problem 300.4[See solution](#)

A real matrix P satisfies $P^2 = P$. Which additional condition is equivalent to P being an orthogonal projection?

- (A) $\det P = 0$.
- (B) $\operatorname{tr} P$ is an integer.
- (C) All eigenvalues of P are real.
- (D) $\|Px\| \leq \|x\|$ for every real vector x , in the Euclidean norm.
- (E) P is diagonalizable over $\mathbb{R}$.

Problem 300.5[See solution](#)

A real matrix A has linearly independent columns. Which vector minimizes $\|Ax - b\|^2$?

- (A) $(A^T A)^{-1}b$ (B) $(AA^T)^{-1}Ab$ (C) $A^T b$ (D) $(A^T A)^{-1}A^T b$ (E) $A(A^T A)^{-1}b$

Supplement. Breadth beyond the book

Linear algebra

301 Change of basis, similarity, and diagonalization

Breadth supplement. [Find in compact glossary](#)

In this entry: coordinate vector; change-of-basis matrix; similar matrices; diagonalizable operator; algebraic multiplicity; geometric multiplicity.

Definition and key ideas

If columns of P are a new basis written in old coordinates, then $[v]_{\text{old}} = P[v]_{\text{new}}$ and the new matrix of an operator is $P^{-1}AP$. Similar matrices represent the same operator and share determinant, trace, characteristic polynomial, and eigenvalue multiplicities. They need not have the same individual entries or eigenvectors in coordinates.

A matrix is diagonalizable over its field exactly when it has a basis of eigenvectors. An eigenvalue's geometric multiplicity is its eigenspace dimension; its algebraic multiplicity is its multiplicity in the characteristic polynomial. Their equality for every eigenvalue, together with splitting of that polynomial, characterizes diagonalizability. Distinct eigenvalues suffice, but are not necessary. A repeated root by itself does not prove failure.

The minimal polynomial is the monic polynomial of least degree satisfying $m(A) = 0$. Since $p(P^{-1}AP) = P^{-1}p(A)P$, similarity preserves this polynomial as well as the characteristic polynomial. Geometric multiplicity can be computed as $\dim \ker(A - \lambda I) = n - \text{rank}(A - \lambda I)$.

Problem 301.1

[See solution](#)

$T(x, y) = (2x + y, 3y)$. Relative to the ordered basis $((1, 0), (1, 1))$, the matrix of T is

(A) $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$

(B) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

(C) $\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$

(D) $\begin{pmatrix} 2 & 2 \\ 0 & 3 \end{pmatrix}$

(E) $\begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$

Problem 301.2[See solution](#)

A real 4×4 matrix A satisfies $(A - I)(A - 2I) = 0$ and $\text{tr } A = 6$. What is $\det A$?

- (A) 2 (B) 4 (C) 6 (D) 8 (E) 16

Problem 301.3[See solution](#)

Let $T(x, y) = (2x, 3y)$. What is its matrix relative to the ordered basis $((1, 1), (1, -1))$?

- (A) $\frac{1}{2} \begin{pmatrix} 5 & 1 \\ 1 & 5 \end{pmatrix}$
(B) $\begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix}$
(C) $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$
(D) $\frac{1}{2} \begin{pmatrix} 5 & -1 \\ -1 & 5 \end{pmatrix}$
(E) $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$

Problem 301.4[See solution](#)

A real 5×5 matrix A has characteristic polynomial $(t - 1)^3(t - 2)^2$ and $\text{rank}(A - I) = 3$. Which conclusion must hold?

- (A) A is diagonalizable because it has real eigenvalues.
(B) The eigenspace for eigenvalue one has dimension three.
(C) A is not diagonalizable because the eigenspace for eigenvalue one has dimension two.
(D) A has a nonreal eigenvalue.
(E) A is singular.

Problem 301.5[See solution](#)

Which pair of real matrices cannot be similar?

- (A) Two matrices with different coordinate representations of the same linear map
(B) Two matrices with different orthonormal eigenbases
(C) Two matrices with different entries
(D) Two matrices with different eigenvectors
(E) Two matrices with different minimal polynomials

Supplement. Breadth beyond the book

Linear algebra

302 Rank inequalities and dimensions of intersections

Breadth supplement. [Find in compact glossary](#)

In this entry: Sylvester rank inequality; dimension formula; rank of a composition.

Definition and key ideas

For finite-dimensional subspaces, $\dim(U + V) = \dim U + \dim V - \dim(U \cap V)$. For compatible matrices $A : \mathbb{F}^n \rightarrow \mathbb{F}^m$ and $B : \mathbb{F}^p \rightarrow \mathbb{F}^n$, one has $\text{rank } A + \text{rank } B - n \leq \text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B)$. More precisely, $\text{rank}(AB) = \text{rank } B - \dim(\text{im } B \cap \ker A)$.

The precise formula explains both bounds: composition loses exactly the directions of the image of B that land in the kernel of A . For endomorphisms on an n -dimensional space, $\max(\text{nullity } A, \text{nullity } B) \leq \text{nullity}(AB) \leq \min(n, \text{nullity } A + \text{nullity } B)$, where $\text{nullity } T = \dim \ker T$.

Problem 302.1

[See solution](#)

A and B are linear operators on $\mathbb{R}^{10}$ with kernel dimensions 4 and 3. What are all possible values of $\dim \ker(AB)$?

- (A) 1, 2, 3, 4 (B) 3, 4, 5, 6, 7 (C) 4, 5, 6, 7 (D) 7 only (E) 0, 1, ..., 7

Problem 302.2

[See solution](#)

$U, V \subseteq \mathbb{R}^9$ have dimensions 6 and 5, and $U + V = \mathbb{R}^9$. If $T : U \cap V \rightarrow \mathbb{R}$ is nonzero linear, what is $\dim \ker T$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Problem 302.3

[See solution](#)

For linear maps $B : \mathbb{F}^p \rightarrow \mathbb{F}^n$ and $A : \mathbb{F}^n \rightarrow \mathbb{F}^m$, suppose $\text{rank}(AB) = \text{rank } A + \text{rank } B - n$. What must hold?

- (A) $\text{im } B \cap \ker A = \{0\}$. (B) $\ker A \subseteq \text{im } B$.
 (C) $\text{im } B = \{0\}$. (D) A is injective.
 (E) AB is surjective.

Problem 302.4

[See solution](#)

Three subspaces of $\mathbb{R}^7$ each have dimension 5. What is the smallest possible dimension of their common intersection?

- (A) 0 (B) 4 (C) 1 (D) 2 (E) 3

Problem 302.5[See solution](#)

Let $T : V \rightarrow W$ be linear and $U \subset V$ have dimension 7. If $\dim(U \cap \ker T) = 3$, what is $\dim T(U)$?

- (A) 7
- (B) 4
- (C) 3
- (D) 10
- (E) It cannot be determined.

Supplement. Breadth beyond the book

Linear algebra

303 Quadratic forms and positive definiteness

Breadth supplement. [Find in compact glossary](#)

In this entry: quadratic form; positive definite matrix; positive semidefinite matrix; Sylvester criterion; Sylvester law of inertia.

Definition and key ideas

A real quadratic form has the shape $q(x) = x^T A x$, where A may be taken symmetric because the skew-symmetric part contributes zero. It is positive definite if $q(x) > 0$ for every nonzero x , and positive semidefinite if $q(x) \geq 0$ for all x . By the spectral theorem, these conditions mean respectively that all eigenvalues of A are positive or nonnegative.

Sylvester's criterion tests positive definiteness of a real symmetric matrix by positivity of all leading principal minors. For semidefiniteness one must check all principal minors, not merely the leading ones. Orthogonal diagonalization turns the quadratic form into a sum of eigenvalues times squared coordinates. On the unit sphere its extrema are the extreme eigenvalues; constraints may change the relevant subspace.

Sylvester's law of inertia says that an invertible real linear change of variables preserves the numbers of positive, negative, and zero squares in a diagonalized real quadratic form. Such a change acts by congruence $A \mapsto P^T A P$; it need not preserve the individual eigenvalues. A principal minor uses the same subset of row and column indices; a leading principal minor uses the first k indices.

Problem 303.1

[See solution](#)

For which real t is $x^2 + 2txy + 2y^2 + 2tyz + z^2$ positive for every nonzero (x, y, z) ?

- (A) $|t| < 1$ (B) $|t| < \sqrt{2}$ (C) $t < 1$ (D) $|t| \leq 1$ (E) All real t

Problem 303.2

[See solution](#)

The maximum of $3x^2 + 4xy + 3y^2$ on $x^2 + y^2 = 1$ is M , and the minimum is m . What is M/m ?

- (A) 1 (B) $3/2$ (C) 2 (D) 3 (E) 5

Problem 303.3

[See solution](#)

A real symmetric matrix has all its leading principal minors nonnegative. Which conclusion is justified?

- (A) It is positive definite.
(B) It is positive semidefinite.
(C) It is nonsingular.
(D) No semidefiniteness conclusion follows; $\text{diag}(0, -1)$ is a counterexample.
(E) It has no negative eigenvalues.

Problem 303.4[See solution](#)

How many positive and negative squares occur in a diagonalization by real invertible change of variables of $x^2 + 4xy + y^2 - z^2$?

- (A) Two positive and one negative
- (B) One positive and two negative
- (C) Three positive and none negative
- (D) None positive and three negative
- (E) One positive, one negative, and one zero

Problem 303.5[See solution](#)

What is the maximum of $x^2 + y^2 + 4z^2$ subject to $x^2 + y^2 + z^2 = 1$ and $x + y + z = 0$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Supplement. Breadth beyond the book

Linear algebra

304 Minimal polynomials and nilpotent Jordan structure

Breadth supplement. [Find in compact glossary](#)

In this entry: annihilating polynomial; minimal polynomial; Jordan block; Jordan normal form; nilpotency index.

Definition and key ideas

The minimal polynomial m_A is the monic polynomial of least degree satisfying $m_A(A) = 0$. It divides every annihilating polynomial, including the characteristic polynomial. Over a field where it splits, A is diagonalizable exactly when m_A has no repeated roots. Over $\mathbb{C}$, Jordan normal form expresses A as a direct sum of blocks $\lambda I + N$, where N has ones above the diagonal and is nilpotent.

The exponent of $t - \lambda$ in m_A is the size of the largest block for λ ; its exponent in the characteristic polynomial is the total size of those blocks. For a nilpotent matrix, $\dim \ker A^k - \dim \ker A^{k-1}$ counts blocks of size at least k . These increments are nonincreasing and their sum eventually reaches the whole dimension.

A polynomial p with $p(A) = 0$ is an annihilating polynomial. A nilpotent matrix satisfies $A^r = 0$ for some positive integer; its nilpotency index is the least such integer, and is the size of its largest Jordan block. Repeated-root tests must be performed in the actual scalar field, whose characteristic can change a polynomial's factorization.

Problem 304.1

[See solution](#)

A nilpotent 6×6 matrix has $\dim \ker A = 3$, $\dim \ker A^2 = 5$, and $\dim \ker A^3 = 6$. What is its minimal polynomial?

- (A) t (B) t^2 (C) t^3 (D) t^5 (E) t^6

Problem 304.2

[See solution](#)

A complex matrix has characteristic polynomial $(t - 1)^4(t + 2)^2$ and minimal polynomial $(t - 1)^2(t + 2)$. What is the largest possible dimension of its 1-eigenspace?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Problem 304.3

[See solution](#)

A matrix A has minimal polynomial $t^3 - 2t + 1$. Which expression equals A^{-1} ?

- (A) $A^2 - 2I$ (B) $2I - A^2$ (C) $A^2 + 2I$ (D) $2A - I$ (E) $I - A$

Problem 304.4[See solution](#)

Over the field $\mathbb{F}_2$, let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Which statement is correct?

- (A) $A^2 = I$ and A is diagonalizable.
- (B) $A^2 \neq I$.
- (C) $A^2 = I$, but A is not diagonalizable because its minimal polynomial is $(t - 1)^2$.
- (D) The minimal polynomial is $t - 1$.
- (E) A has two distinct eigenvalues.

Problem 304.5[See solution](#)

If a matrix has minimal polynomial $t^2(t - 1)$, what is the least positive integer k such that A^k is idempotent?

- (A) 1
- (B) 4
- (C) 2
- (D) 3
- (E) No such integer exists.

Supplement. Breadth beyond the book

Abstract algebra

305 Quotient groups and the first isomorphism theorem

Breadth supplement. [Find in compact glossary](#)

In this entry: quotient group; coset multiplication; first isomorphism theorem; factorization through a quotient.

Definition and key ideas

For a normal subgroup $N \trianglelefteq G$, the quotient G/N consists of cosets, with $(gN)(hN) = ghN$. Normality makes this independent of representatives. If G is finite, $|G/N| = |G|/|N|$. A homomorphism $\varphi : G \rightarrow H$ induces an isomorphism $G/\ker \varphi \cong \text{im } \varphi$. Fibers of φ are cosets of its kernel, so all nonempty fibers have the same size.

A homomorphism $G \rightarrow H$ factors through G/N exactly when it kills N . The quotient records what remains after identifying elements whose ratio lies in N . A quotient of an abelian group is abelian; a quotient being abelian does not force the original group to be abelian.

Problem 305.1

[See solution](#)

Let $G = (\mathbb{Z}/12\mathbb{Z}) \times (\mathbb{Z}/18\mathbb{Z})$ and $N = \langle (4, 6) \rangle$. What is the order of $(1, 1) + N$ in G/N ?

- (A) 3 (B) 6 (C) 12 (D) 18 (E) 36

Problem 305.2

[See solution](#)

How many homomorphisms $\varphi : \mathbb{Z}/24\mathbb{Z} \rightarrow \mathbb{Z}/18\mathbb{Z}$ have kernel of order 8?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 6

Problem 305.3

[See solution](#)

Let $\varphi : \mathbb{Z}^2 \rightarrow \mathbb{Z}$ be $\varphi(a, b) = 6a + 9b$. To which group is $\mathbb{Z}^2/\ker \varphi$ isomorphic?

- (A) $\mathbb{Z}$ (B) $\mathbb{Z}/6\mathbb{Z}$ (C) The trivial group (D) $\mathbb{Z}^2$ (E) $\mathbb{Z}/3\mathbb{Z}$

Problem 305.4

[See solution](#)

In S_3 let $H = \{e, (12)\}$. Does $(gH)(kH) = gkH$ define a group operation on the left cosets?

- (A) Yes, since every subgroup permits coset multiplication.
 (B) Yes, since the index of H is three.
 (C) No: using e or (12) to represent H gives different products with $(13)H$.
 (D) No, since left cosets do not partition a group.
 (E) No, since H contains no identity.

Problem 305.5[See solution](#)

Let $\varphi : G \rightarrow H$ be a homomorphism and N a normal subgroup of G contained in $\ker \varphi$. The induced map $\bar{\varphi} : G/N \rightarrow H$ is injective exactly when

- (A) φ is surjective. (B) G/N is finite. (C) H is abelian. (D) $N = \ker \varphi$. (E) $N = \{e\}$.

Problem 305.6[See solution](#)

If G/N is abelian for a normal subgroup N , which statement must hold?

- (A) G is abelian.
(B) N is contained in the center of G .
(C) $aba^{-1}b^{-1} \in N$ for all $a, b \in G$.
(D) N is trivial.
(E) Every element of G/N has order at most two.

Supplement. Breadth beyond the book

Abstract algebra

306 Ideals, quotient rings, and polynomial quotients

Breadth supplement. [Find in compact glossary](#)

In this entry: ideal; quotient ring; prime ideal; maximal ideal; irreducible polynomial; comaximal ideals.

Definition and key ideas

An ideal I in a commutative ring R with identity is an additive subgroup with $rI \subseteq I$ for every $r \in R$. The quotient ring R/I has cosets as elements and inherited addition and multiplication. A proper ideal is prime precisely when R/I is an integral domain, and maximal precisely when R/I is a field. Every maximal ideal is prime; the converse need not hold.

Over a field F , the quotient $F[x]/(p)$ for nonconstant p has vector-space dimension $\deg p$, with representatives of smaller degree. It is a field exactly when p is irreducible. Arithmetic reduces powers using $p(x) = 0$ in the quotient. A polynomial quotient may have zero divisors or nilpotents even though $F[x]$ itself is a domain.

A proper ideal is prime when $ab \in I$ forces $a \in I$ or $b \in I$, and maximal when no proper ideal lies strictly between it and R . A nonconstant polynomial is irreducible over a field when it has no factorization into two positive-degree polynomials. In $F[x]/(p)$, a class f is a unit exactly when $\gcd(f, p) = 1$. Ideals I, J are comaximal if $I + J = R$; the Chinese remainder theorem then identifies $R/(I \cap J)$ with $R/I \times R/J$. A nilpotent element has some positive power equal to zero.

Problem 306.1

[See solution](#)

How many units are there in $\mathbb{F}_2[x]/(x^3 + x)$?

- (A) 2 (B) 3 (C) 4 (D) 6 (E) 7

Problem 306.2

[See solution](#)

In $R = \mathbb{Q}[x]/(x^2 - 2)$, what is the inverse of the class of $1 + x$?

- (A) $1 - x$ (B) $x - 1$ (C) $(1 - x)/3$ (D) $(1 + x)/3$ (E) It has no inverse.

Problem 306.3

[See solution](#)

Which ideal of $\mathbb{Z}[x]$ is prime but not maximal?

- (A) (x) (B) $(2, x)$ (C) $(3, x)$ (D) (1) (E) (x^2)

Problem 306.4

[See solution](#)

Which ring is isomorphic to $\mathbb{R}[x]/(x^2 - 1)$?

- (A) $\mathbb{R}[x]/(x^2)$ (B) $\mathbb{R}$ (C) $\mathbb{C}$ (D) $\mathbb{Z} \times \mathbb{Z}$ (E) $\mathbb{R} \times \mathbb{R}$

Problem 306.5[See solution](#)

How many nonzero nilpotent elements are there in $\mathbb{F}_3[x]/(x^3)$?

- (A) 2 (B) 3 (C) 6 (D) 8 (E) 26

Problem 306.6[See solution](#)

Which polynomial p makes $\mathbb{F}_2[x]/(p)$ a field with sixteen elements?

- (A) $x^4 + x^2 + 1$ (B) $x^4 + 1$ (C) $x^4 + x$ (D) $x^4 + x^3 + x^2 + 1$ (E) $x^4 + x + 1$

Supplement. Breadth beyond the book

Abstract algebra

307 Modules, free modules, and annihilators

Breadth supplement. [Find in compact glossary](#)

In this entry: module; submodule; module homomorphism; free module; torsion module; torsion-free module; annihilator.

Definition and key ideas

A module over a ring with identity generalizes a vector space: it has an abelian addition law and scalar multiplication satisfying distributivity, associativity, and the identity axiom. Scalars need not have inverses. Modules over a field are vector spaces, while modules over $\mathbb{Z}$ are abelian groups. A submodule is closed under addition and scalar multiplication; a module homomorphism preserves both. A free module has a basis with unique finite scalar expressions.

Over an integral domain, a torsion element is killed by some nonzero scalar, a module is a torsion module when every element is torsion, and it is torsion-free when only zero is torsion. The annihilator of an element m is the ideal $\{r : rm = 0\}$. Torsion-free does not imply free over an arbitrary domain, though a finitely generated torsion-free module over a principal ideal domain is free. Quotient modules can be studied through the relations imposed on generators.

Problem 307.1

[See solution](#)

For the $\mathbb{Z}$ -module $M = (\mathbb{Z}/12\mathbb{Z}) \oplus (\mathbb{Z}/18\mathbb{Z})$, what is the annihilator ideal of $m = (2, 3)$?

- (A) $2\mathbb{Z}$ (B) $3\mathbb{Z}$ (C) $6\mathbb{Z}$ (D) $12\mathbb{Z}$ (E) $36\mathbb{Z}$

Problem 307.2

[See solution](#)

Let $R = \mathbb{R}[x]$. What is the real dimension of $\text{Hom}_R(R/(x^2), R/(x^3))$?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 6

Problem 307.3

[See solution](#)

As a module over $\mathbb{Z}$, the additive group $\mathbb{Q}$ is

- (A) free of rank one.
 (B) a nonzero torsion module.
 (C) torsion-free but not free.
 (D) free of rank two.
 (E) neither torsion-free nor divisible.

Problem 307.4

[See solution](#)

Let $M = \{(a, b) \in \mathbb{Z}^2 : a \equiv b \pmod{2}\}$. Which list is a $\mathbb{Z}$ -module basis for M ?

- (A) $(1, 0), (0, 1)$ (B) $(2, 0), (0, 2)$ (C) $(1, 1), (0, 2)$ (D) $(1, 1)$ (E) $(1, 1), (2, 2)$

Problem 307.5[See solution](#)

For $M = \mathbb{Z}^2 / \langle (2, 4) \rangle$, which statement is correct?

- (A) M is finite.
- (B) M is free.
- (C) Every element of M is torsion.
- (D) M has a nonzero element of order two and also an element of infinite order.
- (E) M is the zero module.

Supplement. [Breadth beyond the book](#)

Abstract algebra

308 Group actions, conjugacy, and orbit counting

Breadth supplement. [Find in compact glossary](#)

In this entry: group action; orbit; stabilizer; orbit-stabilizer theorem; conjugacy class; centralizer; Burnside counting lemma.

Definition and key ideas

A group action is a map $(g, x) \mapsto gx$ with $ex = x$ and $(gh)x = g(hx)$. The orbit Gx contains points reachable from x , and the stabilizer G_x consists of elements fixing x . For finite G , $|Gx| = |G|/|G_x|$. Conjugation $g \cdot a = gag^{-1}$ has conjugacy classes as orbits and centralizers as stabilizers.

Burnside's counting lemma says the number of orbits on a finite set X is $|G|^{-1} \sum_{g \in G} |\text{Fix}(g)|$. Count pairs (g, x) with $gx = x$ in two orders to prove it. State which symmetries are allowed: necklaces under rotation and bracelets under rotation plus reflection have different orbit counts. For permutations, conjugacy classes are determined by cycle type.

An action is transitive when it has one orbit. An element of the acted-on set fixed by every group element has a singleton orbit. The cycle-type characterization of conjugacy applies to the full symmetric group S_n ; in a subgroup such as A_n , a single S_n conjugacy class can split into multiple classes.

Problem 308.1

[See solution](#)

How many binary colorings of the vertices of a regular hexagon are distinct up to rotation, when reflections are not identified?

- (A) 10 (B) 12 (C) 13 (D) 14 (E) 16

Problem 308.2

[See solution](#)

What is the size of the conjugacy class of $(1\ 2\ 3)(4\ 5)$ in S_5 ?

- (A) 10 (B) 15 (C) 20 (D) 30 (E) 60

Problem 308.3

[See solution](#)

A group of order 27 acts on a set of 20 elements. What is the smallest possible number of elements fixed by every group element?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 9

Problem 308.4

[See solution](#)

What is the size of the conjugacy class of $(1\ 2\ 3\ 4\ 5)$ in A_5 ?

- (A) 5 (B) 10 (C) 12 (D) 20 (E) 24

Problem 308.5[See solution](#)

A group of order 60 acts transitively on a set of 12 points. What is the order of the stabilizer of any point?

- (A) 12 (B) 48 (C) 5 (D) 720 (E) 15

Supplement. Breadth beyond the book

Abstract algebra

309 Sylow subgroup counts and normality

Breadth supplement. [Find in compact glossary](#)

In this entry: Sylow subgroup; Sylow conjugacy theorem; Sylow counting theorem.

Definition and key ideas

If $|G| = p^a m$ with p prime and $p \nmid m$, a Sylow p -subgroup has order p^a . Sylow's theorems assert existence, conjugacy of all such subgroups, and that their number n_p divides m and satisfies $n_p \equiv 1 \pmod{p}$. A Sylow subgroup is normal exactly when it is unique. Every p -subgroup is contained in a Sylow subgroup.

Combine congruence and divisibility before doing detailed group calculations. Distinct subgroups of prime order intersect only at the identity, enabling element counts. The analogous claim for subgroups of order p^a with $a > 1$ is false; their intersections can be nontrivial. Uniqueness of one Sylow subgroup does not automatically make the whole group abelian.

Problem 309.1

[See solution](#)

Every group of order 45 has how many Sylow 5-subgroups?

- (A) 1
- (B) 3
- (C) 5
- (D) 9
- (E) The number cannot be determined.

Problem 309.2

[See solution](#)

For a group G of order 30, which assertion must hold?

- (A) G is cyclic.
- (B) G has a normal Sylow 2-subgroup.
- (C) At least one of its Sylow 3- and Sylow 5-subgroups is normal.
- (D) Every Sylow subgroup is normal.
- (E) G has an element of order 30.

Problem 309.3

[See solution](#)

A group of order 72 has a unique Sylow 3-subgroup. How many elements g satisfy $g^9 = e$?

- (A) 1 (B) 3 (C) 8 (D) 9 (E) 24

Problem 309.4[See solution](#)

If P and Q are distinct Sylow 2-subgroups of S_4 , what is $|P \cap Q|$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 8

Problem 309.5[See solution](#)

If a finite group has exactly one Sylow p -subgroup P , what must hold?

- (A) G is abelian.
- (B) Every element of G has order a power of p .
- (C) P is contained in the center.
- (D) P is cyclic.
- (E) P is normal.

Supplement. Breadth beyond the book

Abstract algebra

310 Polynomial irreducibility and finite field extensions

Breadth supplement. [Find in compact glossary](#)

In this entry: irreducible polynomial; minimal polynomial of an algebraic element; field extension; extension degree; tower law; finite field; algebraic element; algebraic closure; compositum of fields; Eisenstein criterion; Gauss lemma.

Definition and key ideas

A nonconstant polynomial over a field is irreducible if it cannot be factored into two polynomials of positive degree over that field. For degrees 2 and 3, this is equivalent to having no root in the field; degree 4 can factor into two quadratics without a root. Gauss's lemma says that a primitive integer polynomial factors over $\mathbb{Q}$ exactly when it factors into positive-degree integer polynomials; primitive means that the coefficients have gcd 1. Irreducibility after reduction modulo a prime implies irreducibility over $\mathbb{Q}$ when the degree is preserved. Eisenstein's criterion applies when one prime divides all nonleading coefficients, not the leading coefficient, and its square does not divide the constant coefficient.

A field extension K/F is a field K containing a specified copy of F ; its degree $[K : F]$ is its dimension as an F -vector space. An algebraic element α satisfies a nonzero polynomial over F , and its monic polynomial of least positive degree, the minimal polynomial, is irreducible. If its degree is d , then $[F(\alpha) : F] = d$, with basis $1, \alpha, \dots, \alpha^{d-1}$. For finite degrees the tower law is $[L : F] = [L : K][K : F]$.

Every finite field has p^n elements for a prime p and integer $n \geq 1$; for each such size there is exactly one up to isomorphism. Its nonzero multiplicative group is cyclic of order $p^n - 1$. The field with p^n elements has a subfield of size p^d exactly when $d \mid n$, and that subfield is unique. An algebraic closure is an algebraic extension in which every nonconstant polynomial splits into linear factors; within one algebraic closure, finite fields can be compared through these subfield inclusions. In a common algebraic closure, the intersection of $\mathbb{F}_{p^m}$ and $\mathbb{F}_{p^n}$ has size $p^{\gcd(m,n)}$, while their compositum (smallest containing field) has size $p^{\text{lcm}(m,n)}$.

Problem 310.1

[See solution](#)

Let $K = \mathbb{F}_2[x]/(x^4 + x + 1)$, and let α be the class of x . What is the multiplicative order of α ?

(A) 3 (B) 4 (C) 5 (D) 7 (E) 15

Problem 310.2

[See solution](#)

Inside a fixed algebraic closure of $\mathbb{F}_2$, let $K = \mathbb{F}_{2^6}$ and $L = \mathbb{F}_{2^4}$. What is $[KL : K \cap L]$, where KL is the smallest field containing both?

(A) 2 (B) 3 (C) 4 (D) 6 (E) 12

Problem 310.3[See solution](#)

Which polynomial is irreducible over $\mathbb{Q}$ by Eisenstein's criterion with prime 3?

- (A) $x^5 + 6x^3 + 9$ (B) $x^5 + 6x^3 + 1$ (C) $3x^5 + x + 3$ (D) $x^5 + 6x^3 + 3$ (E) $x^5 + 3x^3 + 6x$

Problem 310.4[See solution](#)

How many elements of $\mathbb{F}_{64}$ satisfy $x^8 = x$?

- (A) 2 (B) 8 (C) 4 (D) 64 (E) 16

Problem 310.5[See solution](#)

What is the degree of the minimal polynomial over $\mathbb{Q}$ of $\alpha = \sqrt{2} + \sqrt{3}$?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 8

Problem 310.6[See solution](#)

The primitive polynomial $p(x) = 2x^2 + 3x + 1$ reduces to the irreducible polynomial $x + 1$ over $\mathbb{F}_2$. Which statement is correct?

- (A) p is irreducible over $\mathbb{Q}$ by reduction modulo two.
(B) p is reducible over $\mathbb{Q}$; the reduction test does not apply because its degree drops.
(C) p is not primitive.
(D) p has no rational root.
(E) Every primitive polynomial of even leading coefficient is irreducible.

Supplement. Breadth beyond the book

Real analysis

311 Pointwise and uniform convergence of functions

Breadth supplement. [Find in compact glossary](#)

In this entry: pointwise convergence; uniform convergence; supremum norm; uniform limit theorem; Weierstrass M-test.

Definition and key ideas

Pointwise convergence $f_n \rightarrow f$ means that for each fixed x the values approach $f(x)$, with the required index allowed to depend on x . Uniform convergence means that one index works for every x : $\sup_{x \in E} |f_n(x) - f(x)| \rightarrow 0$. The supremum need not be attained. Uniform limits of continuous functions are continuous. On a compact interval, uniform convergence of Riemann-integrable functions permits passing the limit through the integral.

For a series of functions, the Weierstrass M-test gives uniform absolute convergence if $|u_n(x)| \leq M_n$ for all x and $\sum M_n < \infty$. Pointwise convergence alone does not preserve continuity or integrals. The domain matters: convergence may be uniform on every compact subinterval and fail on the whole open interval.

Problem 311.1

[See solution](#)

For $f_n(x) = nx/(1 + n^2x^2)$ on $[0, 1]$, which statement is true?

- (A) It converges uniformly to 0.
- (B) It converges pointwise to 0, but not uniformly.
- (C) It converges pointwise to 1.
- (D) It fails to converge at $x = 0$.
- (E) It converges uniformly to a discontinuous function.

Problem 311.2

[See solution](#)

For which of the following domains does $\sum_{n=1}^{\infty} x^n/n^2$ converge uniformly? I. $[-1, 1]$; II. $(-1, 1)$; III. $[0, 2]$.

- (A) I only (B) II only (C) I and II only (D) II and III only (E) I, II, and III

Problem 311.3[See solution](#)

For $f_n(x) = x^n(1 - x)$ on $[0, 1]$, which statement is correct?

- (A) It diverges at $x = 1$.
- (B) It has no pointwise limit.
- (C) It converges uniformly to one.
- (D) It converges pointwise but not uniformly to zero.
- (E) The sequence converges uniformly to zero.

Problem 311.4[See solution](#)

For $f_n(x) = x/(n + x)$ on $[0, \infty)$, which statement is correct?

- (A) It converges pointwise to one.
- (B) It converges pointwise to zero but not uniformly.
- (C) It has no pointwise limit.
- (D) It converges uniformly to zero.
- (E) It is unbounded for each n .

Problem 311.5[See solution](#)

Let $h(x) = 0$ for $-1 \leq x < 0$ and $h(x) = 1$ for $0 \leq x \leq 1$. What is $\inf_g \sup_{[-1,1]} |g(x) - h(x)|$, where the infimum is over all continuous real functions g ?

- (A) 0
- (B) $1/4$
- (C) $1/2$
- (D) 1
- (E) No finite value exists.

Supplement. Breadth beyond the book

Real analysis

312 Interchanging limits, derivatives, and integrals

Breadth supplement. [Find in compact glossary](#)

In this entry: termwise differentiation; uniform convergence of derivatives; differentiation under the integral sign.

Definition and key ideas

Uniform convergence of functions alone does not justify differentiating the limit. A useful theorem is: if f_n are C^1 on $[a, b]$, $f_n(x_0)$ converges at one point, and f'_n converges uniformly to g , then f_n converges uniformly to a differentiable f with $f' = g$. Its proof writes $f_n(x) = f_n(x_0) + \int_{x_0}^x f'_n$.

For an integral depending on a parameter, differentiating under the integral is valid under appropriate domination; on a compact rectangle, continuity of the integrand and its parameter derivative is a convenient sufficient condition. Improper domains require a bound integrable in the integration variable. Check these conditions, not merely the visual similarity of the formulas.

Problem 312.1

[See solution](#)

Let $f_n(x) = \sin(nx)/n$ on $[0, 2\pi]$. Which assertion holds?

- (A) f_n fails to converge pointwise.
- (B) $f_n \rightarrow 0$ uniformly, but f'_n does not converge pointwise everywhere.
- (C) $f'_n \rightarrow 0$ uniformly.
- (D) $f_n \rightarrow 1$ uniformly.
- (E) The derivative of the limit is 1.

Problem 312.2

[See solution](#)

For $a > 0$, let $F(a) = \int_0^1 (x^a - 1)/\log x \, dx$, with the removable value at $x = 1$ understood. What is $F(a)$?

- (A) $\log a$ (B) $a/(a+1)$ (C) $\log(a+1)$ (D) $1/(a+1)$ (E) $a \log(a+1)$

Problem 312.3[See solution](#)

Suppose $f_n \in C^1([0, 1])$, $f_n(0) \rightarrow a$, and f'_n converges uniformly to g . Which conclusion follows?

- (A) Only pointwise convergence can be concluded.
- (B) g must be zero.
- (C) f_n must converge to the constant a .
- (D) f_n converges uniformly to $a + \int_0^x g(t) dt$.
- (E) No convergence follows without a bound on $f_n(1)$.

Problem 312.4[See solution](#)

For $a \geq 0$, let $F(a) = \int_0^\infty e^{-x} \sin(ax)/x dx$, with its limiting value used at $x = 0$. What is $F(a)$?

- (A) $\arctan a$ (B) $\log(1 + a)$ (C) $a/(1 + a^2)$ (D) $\frac{1}{2} \log(1 + a^2)$ (E) $\sin a$

Problem 312.5[See solution](#)

A sequence of continuous functions converges uniformly to f on $[0, 2]$. If $\int_0^2 f_n = 3 + 1/n$, what is $\int_0^2 f$?

- (A) 3
- (B) 0
- (C) 2
- (D) 4
- (E) It cannot be determined.

Problem 312.6[See solution](#)

If $f'_n \rightarrow 0$ uniformly on $[0, 1]$, which additional condition guarantees that f_n converges uniformly on this interval?

- (A) Each f_n is continuously differentiable.
- (B) Each f_n is bounded.
- (C) The sequence $f_n(0)$ converges.
- (D) Each f_n is increasing.
- (E) Each f_n is positive.

Supplement. Breadth beyond the book

Probability and statistics

313 Sample means, standard error, and the central limit theorem

Breadth supplement. [Find in compact glossary](#)**In this entry:** sample mean; standard error; central limit theorem; sampling distribution.**Definition and key ideas**

For independent identically distributed variables with mean μ and finite variance σ^2 , the sample mean $\bar{X}_n$ has mean μ and variance σ^2/n . Its standard deviation $\sigma/\sqrt{n}$ is the standard error of the mean. Independence, or appropriate covariance information, is essential to the variance computation. Overlapping samples produce correlated means.

The central limit theorem gives $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ converging in distribution to a standard normal when $0 < \sigma^2 < \infty$. It is an approximation at finite n , not exact normality. If the original variables are normal, the sample mean is exactly normal. The law of large numbers concerns convergence of the average to its mean; it is not a claim that the unscaled sum becomes constant.

A sampling distribution is the probability law of a statistic, such as the sample mean, under repeated random sampling. The law of large numbers here means $\bar{X}_n \rightarrow \mu$ in probability: for every $\varepsilon > 0$, $P(|\bar{X}_n - \mu| > \varepsilon) \rightarrow 0$. With finite variance and independence this also follows from Chebyshev's bound $\sigma^2/(n\varepsilon^2)$.

Problem 313.1[See solution](#)

$X_1, \dots, X_{15}$ are independent with common variance 1. Let $A = (X_1 + \dots + X_{10})/10$ and $B = (X_6 + \dots + X_{15})/10$. What is $\text{Var}(A - B)$?

- (A) $1/20$ (B) $1/10$ (C) $3/20$ (D) $1/5$ (E) $1/2$

Problem 313.2[See solution](#)

Independent variables $X_1, \dots, X_{100}$ each have exponential density e^{-x} for $x > 0$. Using the central limit theorem, $P(\sum X_i > 120)$ is approximately

- (A) $1 - \Phi(20)$ (B) $1 - \Phi(2)$ (C) $1 - \Phi(1/5)$ (D) $\Phi(2)$ (E) $1/2$

Problem 313.3[See solution](#)

Independent observations have common variance 9. What sample size makes the standard deviation of their sample mean equal to $1/5$?

- (A) 45 (B) 225 (C) 81 (D) 135 (E) 405

Problem 313.4[See solution](#)

Let Y equal zero or one with probability $1/2$ each, and set $X_i = Y$ for every i . Which pair gives $\text{Var}(\bar{X}_n)$ and $P(|\bar{X}_n - 1/2| > 1/4)$?

- (A) $(1/(4n), 0)$ (B) $(1/4, 1)$ (C) $(1/(4n), 1/2)$ (D) $(1/4, 1/2)$ (E) $(0, 0)$

Problem 313.5[See solution](#)

If $X_1, \dots, X_{100}$ are independent Bernoulli variables with parameter $1/2$, what is the exact probability that their sample mean equals $1/2$?

- (A) 0, because a normal random variable has no atoms.
(B) $1/2$
(C) $2^{-100} \binom{100}{50}$
(D) $1/\sqrt{50\pi}$
(E) 2^{-50}

Supplement. Breadth beyond the book

Probability and statistics

314 Geometric, Poisson, and exponential distributions

Breadth supplement. [Find in compact glossary](#)

In this entry: geometric distribution; Poisson distribution; exponential distribution; memoryless property; Poisson thinning; rate parameter; homogeneous Poisson process.

Definition and key ideas

A geometric variable G counting trials through the first success with success probability p has $P(G = k) = p(1 - p)^{k-1}$ for $k = 1, 2, \dots$, mean $1/p$, and variance $(1 - p)/p^2$. The alternative convention counts failures before success and subtracts 1 from G . A Poisson variable N with parameter $\lambda > 0$ has $P(N = k) = e^{-\lambda}\lambda^k/k!$ for $k \geq 0$, with mean and variance both λ . Independent Poisson variables add their parameters.

An exponential waiting time T with rate $\lambda > 0$ has density $\lambda e^{-\lambda t}$ for $t \geq 0$, survival probability $P(T > t) = e^{-\lambda t}$, mean $1/\lambda$, and variance $1/\lambda^2$. Exponential and geometric waiting times are memoryless: after a period without success, the remaining waiting time has the same law. Independently retaining each event of a Poisson count with probability p gives a Poisson count with parameter $p\lambda$; retained and discarded counts are independent. A homogeneous Poisson process connects Poisson counts to independent exponential gaps.

For a homogeneous Poisson process of rate λ , disjoint time intervals have independent counts, and an interval of length t has Poisson mean λt . Successive event gaps are independent exponentials of that rate. Thus the waiting time to the k th event exceeds t exactly when the count by t is less than k .

Problem 314.1

[See solution](#)

Let N have a Poisson distribution with mean 6. Independently of everything else, each of the N objects is labeled red with probability $1/3$. Given that no object is red, what is $E[N]$?

- (A) 2 (B) 3 (C) 4 (D) 6 (E) 9

Problem 314.2

[See solution](#)

G is the number of independent trials through the first success, each trial succeeding with probability $1/3$. Independently, T is exponential with rate $\log 2$. What is $P(T > G)$?

- (A) $1/6$ (B) $1/4$ (C) $1/3$ (D) $1/2$ (E) $2/3$

Problem 314.3

[See solution](#)

Independent exponential variables have rates 2 and 3. What is the probability that the first is smaller than the second?

- (A) $2/3$ (B) $1/5$ (C) $3/5$ (D) $1/2$ (E) $2/5$

Problem 314.4[See solution](#)

Independent Poisson variables X, Y have means 2, 3. What is $P(X = 1 \mid X + Y = 4)$?

- (A) $108/3125$ (B) $24/125$ (C) $216/625$ (D) $48/625$ (E) $2/5$

Problem 314.5[See solution](#)

A geometric variable G counts trials through the first success, with success probability $1/4$. What is $P(G > 8 \mid G > 5)$?

- (A) $81/256$ (B) $9/16$ (C) $3/4$ (D) $27/64$ (E) $1/64$

Problem 314.6[See solution](#)

Events follow a homogeneous Poisson process of rate $\lambda > 0$. If T_2 is the time of the second event, what is $P(T_2 > t)$ for $t \geq 0$?

- (A) $e^{-2\lambda t}$ (B) $1 - e^{-\lambda t}$ (C) $(1 + \lambda t)e^{-\lambda t}$ (D) $\lambda t e^{-\lambda t}$ (E) $e^{-\lambda t}/(1 + \lambda t)$

Supplement. Breadth beyond the book

Probability and statistics

315 Conditional expectation and total expectation and variance

Breadth supplement. [Find in compact glossary](#)

In this entry: conditional expectation; law of total expectation; law of total variance; conditional variance; random sum.

Definition and key ideas

Conditional expectation $E[X | Y]$ is the average of X given the information in Y , so it is itself a random variable determined by Y . For discrete Y , its value at $Y = y$ is $\sum_x xP(X = x | Y = y)$; use a conditional-density integral in the continuous case. For integrable X , the tower identity is $E[E[X | Y]] = E[X]$. If X and Y are independent, then $E[X | Y] = E[X]$, but a constant conditional mean alone need not imply independence.

With finite second moments, $\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y])$. The two terms distinguish variability within conditional groups from variability among their means. For a nonnegative integer count N independent of independent identically distributed X_i with mean μ and variance σ^2 , the random sum $S = \sum_{i=1}^N X_i$ has $E[S] = \mu E[N]$ and $\text{Var}(S) = \sigma^2 E[N] + \mu^2 \text{Var}(N)$ when these quantities are finite. Forgetting the second term ignores randomness of the count.

Problem 315.1

[See solution](#)

A fair coin is tossed once. Conditional on heads, X is binomial with parameters $(4, 1/4)$; conditional on tails, it is binomial with parameters $(4, 3/4)$. What is $\text{Var}(X)$?

- (A) $3/4$ (B) 1 (C) $3/2$ (D) $7/4$ (E) 2

Problem 315.2

[See solution](#)

Let N be Poisson with mean 2. Independently of N , the variables X_i are independent and identically distributed with mean 3 and variance 4. Put $S = \sum_{i=1}^N X_i$, taking the empty sum as zero. What is $(E[S], \text{Var}(S))$?

- (A) (6, 8) (B) (6, 12) (C) (6, 18) (D) (6, 26) (E) (3, 26)

Problem 315.3

[See solution](#)

The joint density of (X, Y) is $f(x, y) = 6x$ for $0 < x < y < 1$ and zero otherwise. What is $E[X | Y = y]$ for $0 < y < 1$?

- (A) $y/3$ (B) $y/2$ (C) $2y/3$ (D) $3y/4$ (E) $1/2$

Problem 315.4[See solution](#)

Y takes zero and one with equal probabilities. Conditional on $Y = 0$, the probabilities that $X = -1, 0, 1$ are $1/4, 1/2, 1/4$; conditional on $Y = 1$, they are $1/3, 1/3, 1/3$. Which statement is correct?

- (A) X and Y are independent because $E[X | Y] = 0$.
- (B) $E[X | Y] = 0$, but X and Y are dependent.
- (C) $E[X | Y] = Y$.
- (D) $E[X] > 0$.
- (E) Knowing Y determines X exactly.

Problem 315.5[See solution](#)

An integrable random variable satisfies $E[X | Y] = Y^2$ and Y is standard normal. What is $E[X]$?

- (A) It cannot be determined.
- (B) 0
- (C) 2
- (D) 3
- (E) 1

Supplement. Breadth beyond the book

Geometry

316 Triangle geometry and circle power

Breadth supplement. [Find in compact glossary](#)**In this entry:** law of cosines; law of sines; Heron formula; inradius; circumradius; power of a point.**Definition and key ideas**

For a triangle with opposite sides a, b, c and angles A, B, C , the cosine law is $c^2 = a^2 + b^2 - 2ab \cos C$, and the sine law is $a/\sin A = b/\sin B = c/\sin C = 2R$, where R is the circumradius. If $s = (a + b + c)/2$, its area is $\Delta = \sqrt{s(s-a)(s-b)(s-c)} = rs = abc/(4R)$, where r is the inradius.

For a circle of center O and radius R , a point P has power $PO^2 - R^2$. The products of directed distances along secants through P agree. For an exterior point, a tangent segment of length t satisfies $t^2 = PA \cdot PB$ for the near and far intersections of a secant. Specify directed lengths or the exterior-point setting to avoid sign mistakes.

The incircle is tangent to all three sides; its center is the incenter and its radius is r . The circumcircle passes through all three vertices; its center is the circumcenter and its radius is R . If their centers are I and O , Euler's triangle identity is $OI^2 = R(R - 2r)$. With side-side-angle data, the sine law can give two supplementary choices for an unknown angle, and each must be checked against the positive remaining-angle condition. For intersecting chords through an interior point, the ordinary positive segment products equal $R^2 - PO^2$, the negative of the directed power.

Problem 316.1[See solution](#)

A triangle has side lengths 13, 14, 15. What is the ratio of its circumradius to its inradius?

- (A) 13/8 (B) 15/8 (C) 65/32 (D) 35/16 (E) 5/2

Problem 316.2[See solution](#)

From an exterior point P , a tangent to a circle has length 6. A secant has near intersection A and far intersection B , with $AB = 5$. What is PA ?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 9

Problem 316.3[See solution](#)

A triangle has side $a = \sqrt{2}$ opposite angle $A = 30^\circ$, and side $b = 2$. How many possible values of its third side c are there, and what is their product?

- (A) One value; product 2.
- (B) Two values; product 2.
- (C) Two values; product 4.
- (D) Two values; product $\sqrt{3}$.
- (E) No such triangle exists.

Problem 316.4[See solution](#)

Two chords intersect inside a circle. One is split into lengths 3 and 8, and the other into lengths 4 and x . What is x ?

- (A) 5 (B) 12 (C) 6 (D) 8 (E) 4

Problem 316.5[See solution](#)

A right triangle has legs 8 and 15. What is the distance between its circumcenter and incenter?

- (A) $5/2$ (B) $\sqrt{17}$ (C) $17/2$ (D) $\sqrt{34}/2$ (E) $\sqrt{85}/2$

Appendix A. Authentic ETS practice

The links below open selected questions in ETS's released [2024 practice book](#), Form GR3768. Question numbers identify the authentic items; page numbers refer to that PDF. They supplement the original exercises in this companion. Browser support for jumping to a PDF page varies; the question number remains a reliable locator. The official answer key is on PDF page 34.

Question	PDF page	Topic
11	10	Inverse functions
16	12	Images of sets
35	19	Riemann sums
36	19	Connectedness and boundedness
38	20	Composition and nullity
40	21	Counting plane regions
43	22	Equivalence relations
45	23	Inverse Jacobians
46	24	Linear differential equations
48	25	Partial derivatives
49	25	Complex contour integration
52	26	Conjugacy classes
55	28	Sample means
56	28	Idempotent rings
58	29	Distributing identical objects
60	30	Triangular factorization
61	31	Primitive roots of unity
62	31	Metric completeness
64	32	Continuity versus differentiability
65	33	Bayes's theorem
66	33	Closure in a topology

Official sources, checked 6 October 2026: [ETS test content and structure](#); [ETS Mathematics practice book](#). Preserve an unattempted full test for timed practice if possible.

Appendix B. Corrections and qualifications

These notes distinguish mathematical facts from errors or missing hypotheses in the uploaded fourth edition. Page references are the book's printed numbers. This companion uses the corrected statements. The list is selective: it highlights issues that could otherwise teach an incorrect general rule.

Page(s)	Correct interpretation
5	For $f : A \rightarrow B$, B is the codomain. The image or range is $f(A) \subseteq B$, which need not equal B . Surjectivity is the equality $f(A) = B$.
81	Convergence of $f(x)$ as $x \rightarrow \infty$ implies convergence of the sampled sequence $f(n)$. The converse fails: $f(x) = \sin(2\pi x)$ has $f(n) = 0$ but no limit as $x \rightarrow \infty$.
90–94	Power-series differentiation and integration are justified in the open interval of convergence. Endpoints must be checked separately; convergence of the original series there need not survive differentiation.
126	A differentiable graph $z = f(x, y)$ has tangent plane even when $f_x = f_y = 0$: the plane is then horizontal. The normal $(-f_x, -f_y, 1)$ is never zero.
213	Cayley-Hamilton reduces matrix powers to a polynomial of degree less than the matrix size n . Reduction to a linear expression is specific to a quadratic relation, including the 2×2 case.
240	A homomorphism sends an element of finite order m to an element whose order divides m ; it need not preserve the order. The zero homomorphism is the simplest counterexample.
267	A finite computation of a Collatz trajectory verifies that starting value only. It is not a proof that every positive starting integer reaches 1.
297	A bounded function on a compact interval is Riemann integrable exactly when its discontinuity set has Lebesgue measure zero, not exactly when that set is countable. The indicator of the Cantor set is Riemann integrable despite having uncountably many discontinuities.
302	The principal logarithm of a nonzero complex number is $\log z + i \operatorname{Arg} z$ with the chosen principal argument convention. It is the imaginary part of the logarithm that is constrained by that convention, not the argument of the logarithm itself. The usual principal branch is analytic on the plane slit along the nonpositive real axis.
313	The function $1/(1 - z)$ has a simple pole at $z = 1$, not an essential singularity. In a Laurent expansion at that point its principal part is the single term $-1/(z - 1)$.
341	Differentiability does not imply continuity of the derivative. A limit that is a derivative quotient should be handled by the definition unless the stronger hypotheses of a different method have been checked.
357,359	The ratio test with limit below 1 is sufficient for convergence, not necessary. At limit 1 it is inconclusive: $\sum n^{-2}$ converges and $\sum n^{-1}$ diverges.

Page(s)	Correct interpretation
360	An error bound larger than a tolerance does not prove that the actual error is too large. To prove a Taylor degree is minimal, separately show that a smaller degree fails, for example by a positive omitted term lower bound.
389	A cyclic group has exactly one subgroup of each order dividing its order. An arbitrary finite abelian group need not: $\mathbb{Z}_2 \times \mathbb{Z}_2$ has three subgroups of order 2.
390	The tentative map $\mathbb{Z}_3 \rightarrow \mathbb{Z}_6$ specified by $0 \mapsto 0, 1 \mapsto 3, 2 \mapsto 0$ has image $\{0, 3\}$, not $\{0, 4, 5\}$. It still is not a homomorphism, since $\varphi(1+2) = 0 \neq 3 = \varphi(1) + \varphi(2)$.
395	A subgraph may omit vertices as well as edges. A subgraph retaining all vertices is a spanning subgraph. Arguments about spanning trees must retain this distinction.
396	For $(x-2)(x-3)/x > 0$, a sign chart gives $(0, 2) \cup (3, \infty)$. Zeros and the undefined point 0 are excluded by the strict inequality.
325,399	The displayed candidate $f(x) = x/2 + c$ on $[0, 8]$ cannot be a probability density for any c . Normalization forces $16 + 8c = 1$, hence $c = -15/8$, which makes f negative near 0. A density requires nonnegativity as well as total integral 1.
408	Practice Q11 specifies $\ b\ = 3, \ c\ = 7, b \cdot c = 5$, and $b \times c = 8j + 4k$. These data are inconsistent: the identity $\ b \times c\ ^2 = \ b\ ^2\ c\ ^2 - (b \cdot c)^2$ gives 416, whereas the specified cross product has squared norm 80.
408,425	For $y'' = (y')^2$, setting $u = y'$ and dividing by u^2 loses the solutions $u = 0$. Every constant y solves the equation, including nonzero constants. Nonconstant solutions have form $y = -\log x+C + D$ on intervals avoiding $-C$.
414,433	Derivatives of every order, even with bounds on derivatives at one point, do not imply equality to the Taylor series. The smooth function $h(x) = e^{-1/x^2}$ for $x \neq 0$, $h(0) = 0$, has every derivative at 0 equal to zero. Adding Mh preserves those derivatives and changes the value at 1 by M/e . The proposed upper bound in practice Q41 requires an additional analytic or remainder hypothesis.
430	A nonzero gradient of a defining function is normal to its level surface and the tangent plane, rather than tangent to that plane.
437	Solving a recurrence's fixed-point equation does not prove convergence. For $a_{n+1} = (a_n + 2/a_n)/2$ with $a_1 > 0$, AM-GM gives $a_2 \geq \sqrt{2}$, after which the iterates decrease and stay above $\sqrt{2}$. That establishes convergence before selecting the positive fixed point.
441	In an infinite cofinite space, the closed sets are the finite sets <i>and the whole space</i> . The whole space must be closed in every topology.

About this guide

Start with the compact glossary. Each blue-green entry jumps to its explanation and practice problems. Page references in this guide use the same continuous numbering as the PDF pages; the cover is page 1. Entry numbers identify concepts and are separate from page numbers. Each detailed entry supplies the corresponding *printed book pages*; for the uploaded scan, printed page p is PDF page $p + 10$ through the main text. PDF bookmarks provide chapter, section, and entry navigation. The footer links to Contents, Glossary, and A–Z on every reading page. Each entry also links to its chapter list and adjacent entries.

Coverage and order. The source is Steven A. Leduc, *Cracking the GRE Mathematics Subject Test*, fourth edition (2010), supplied as a 468-page scan. The reading pass included the explanatory chapters, examples, review questions, review solutions, full practice test, and its explanations, as well as the front and back matter. Repeated concepts are consolidated at their first substantive teaching location; passing mentions are not treated as definitions. Tightly related terms share a concept entry. The terms line explicitly identifies the definitions covered. Entries first developed in solutions or the practice test follow the seven content chapters. A separately marked breadth supplement covers useful material not developed adequately in the book.

What mastery means. For every entry, be able to state the definition with its hypotheses, produce an example and a counterexample, and solve an unfamiliar application. Definitions and basic ideas alone cannot guarantee a perfect score. This is a broad book-based preparation resource, not a claim that a finite glossary predicts every possible exam question. The practice problems emphasize application, counterexamples, and common traps. Difficulty varies with your background; problems are intended to challenge within undergraduate scope, not to imitate graduate qualifying exams.

Problem provenance. All problems written out in this companion are newly authored GRE-style exercises, not authenticated past GRE questions or copied Princeton Review questions. Each has five choices and one intended answer. Worked reasoning appears in the [separate solutions section at the end](#), reached through each problem’s “See solution” link. Every solution has a “Back to problem” link. Appendix A links selected authentic questions in ETS’s released practice test, with exact question numbers and PDF page locations; these can be attempted alongside the corresponding topics. No unverified question is described as an actual exam item.

Study sequence. Read one subsection, attempt the questions in each entry before opening the solutions, then revisit every error without notes. Mix subjects when reviewing. A correct answer reached by an invalid argument is still a topic to revisit. Use the linked ETS test for a complete timed session.

Notation. Unless stated otherwise, vector spaces have the indicated field (usually $\mathbb{R}$ or $\mathbb{C}$), logarithms in calculus are natural, angles are in radians, and rings with identity are nonzero when that matters. $\mathbb{N}$ denotes positive integers unless a problem explicitly includes zero. An improper integral’s convergence and a Cauchy principal value are different notions. Hypotheses printed in an entry are part of the fact to learn.

Current exam note. ETS currently describes a computer-delivered Mathematics test with

approximately 66 multiple-choice questions in 2 hours 50 minutes. Approximate weights are 50% calculus, 25% algebra, and 25% additional topics. The uploaded book's paper-era logistics should not be used for present arrangements. Current ETS guidance imposes no deduction for incorrect answers, so its old guessing-penalty advice is obsolete. See the official sources linked in Appendix A (checked 6 October 2026).

Alphabetical term finder

Click a term to open its main explanation. Additional references appear when the term belongs to more than one entry.

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Worked solutions

All 1,647 solutions appear here in the same order as the problems. Use “Back to problem” to return to the original question. Entry titles link to the corresponding definitions and problem statements.

Each problem has a matching solution here, grouped by entry and problem number.

1. Precalculus

Introduction ▸ Prerequisite algebra

001 Exponents, radicals, and complex arithmetic

Solution 001.1 ■ Answer C[Back to problem](#)

The left side is $|x - 1| + |x + 2|$. It equals $-2x - 1$ for $x < -2$, 3 for $-2 \leq x \leq 1$, and $2x + 1$ for $x > 1$. Hence the solutions are $x = -3$ and $x = 2$.

Solution 001.2 ■ Answer C[Back to problem](#)

Write $z = a + bi$. Then $3a - bi = 6 + 3i$, so $z = 2 - 3i$. Consequently $z^2 + 1 = -4 - 12i$, whose modulus is $\sqrt{16 + 144} = 4\sqrt{10}$.

Solution 001.3 ■ Answer A[Back to problem](#)

The two radicals are $\sqrt{3} + \sqrt{2}$ and $\sqrt{3} - \sqrt{2}$. Thus $u = 2\sqrt{2}$, so $u^4 = 64$.

Solution 001.4 ■ Answer D[Back to problem](#)

Since $z^2 = i|z|^2 = 10i$, the required modulus is $\sqrt{36 + 100} = 2\sqrt{34}$; the sign of z is irrelevant.

Solution 001.5 ■ Answer A[Back to problem](#)

The expression is $\sqrt{6} - \sqrt{2/3}$. Since $\sqrt{2/3} = \sqrt{6}/3$, it equals $2\sqrt{6}/3$.

Solution 001.6 ■ Answer C[Back to problem](#)

An even root denotes the nonnegative root, so $f(x) = |x|$ for every real x . The inner square root in g requires $x \geq 0$, after which squaring returns x . The rule $(a^r)^s = a^{rs}$ cannot be used here to erase either a sign or a domain restriction.

Functions ▸ Definition and domain

002 Function, domain, codomain, range, and fixed point

Solution 002.1 ■ **Answer B**[Back to problem](#)

The radicand is nonnegative exactly on $(-\infty, -2) \cup [1, \infty)$. The logarithm requires $x \neq 0$, and its appearance in the denominator also excludes $x = \pm 1$. Only 1 lies in the preliminary domain, yielding the stated union.

Solution 002.2 ■ **Answer B**[Back to problem](#)

The function f is strictly increasing. If $f(x) > x$, then $f(f(x)) > f(x) > x$; if $f(x) < x$, then $f(f(x)) < f(x) < x$. A fixed point of $f \circ f$ therefore fixes f . Solving $x = \sqrt{3x+4}$ gives $x = 4, -1$ after squaring, but only $x = 4$ is nonnegative.

Solution 002.3 ■ **Answer D**[Back to problem](#)

Put $t = x^2 \geq 0$, $t \neq 1$. The function $1 + 2/(t-1)$ gives $(-\infty, -1]$ for $0 \leq t < 1$ and $(1, \infty)$ for $t > 1$.

Solution 002.4 ■ **Answer B**[Back to problem](#)

Both logarithms require $-2 < \log(4 - x^2) < 2$. The upper bound is automatic since $4 < e^2$; the lower gives $x^2 < 4 - e^{-2}$.

Solution 002.5 ■ **Answer A**[Back to problem](#)

Fixed points solve $x^2 - x + a = 0$. There is exactly one distinct real root precisely when $1 - 4a = 0$, so only $a = 1/4$ works.

Solution 002.6 ■ **Answer C**[Back to problem](#)

The square is nonnegative and vanishes at $x = \pm 1$, so every output is at least 2. For each $y \geq 2$, choose $x = \sqrt{1 + \sqrt{y-2}}$ to obtain $f(x) = y$. Thus the actual range is $[2, \infty)$, while the originally stated codomain is larger.

Functions ▸ Composition of Functions

003 Composition and iteration of functions

Solution 003.1 ■ **Answer B**[Back to problem](#)

Compute $f^{\circ 2}(x) = (x - 1)/x$, which requires $x \neq 1$ and $x \neq 0$. Its value is never 1, so the third application adds no exclusions and equals $1/[1 - (x - 1)/x] = x$.

Solution 003.2 ■ **Answer D**[Back to problem](#)

The condition is $2 - (2 - t^2)^2 = t$, or $t^4 - 4t^2 + t + 2 = 0$. This factors as $(t - 1)(t + 2)(t^2 - t - 1)$. All four roots are real and distinct; their product is the constant coefficient 2. The correct choice is therefore 2.

Solution 003.3 ■ **Answer E**[Back to problem](#)

The translation $u = x + 3$ changes iteration to $u \mapsto 2u$. Hence $f^{\circ n}(x) = 2^n(x + 3) - 3$, giving 4093.

Solution 003.4 ■ **Answer D**[Back to problem](#)

The equation says $f(x^2) = |x|$, hence $f(t) = \sqrt{t}$ for $t \geq 0$ and in particular $f(9) = 3$. It imposes no restriction on $f(t)$ for $t < 0$, because those inputs are outside the range of g . Each other expression involves an unrestricted negative input.

Solution 003.5 ■ **Answer D**[Back to problem](#)

Set $t = 2x - 1$ to obtain $f(t) = (t + 1)^2/4 + 1$. Then $f(1) = 2$ and $f(2) = 13/4$.

Functions ▸ Inverse Functions

004 Injectivity, surjectivity, bijectivity, and inverse functions

Solution 004.1 ■ **Answer B**[Back to problem](#)

Solve $4x = 3(1 - x^2)$, obtaining $3x^2 + 4x - 3 = 0$. The roots are $(-2 \pm \sqrt{13})/3$, and only the plus sign lies in $(-1, 1)$. Strict positivity of $f'(x) = 2(1 + x^2)/(1 - x^2)^2$ and the endpoint limits justify bijectivity.

Solution 004.2 ■ **Answer C**[Back to problem](#)

If $f(a) = f(a')$, applying g gives $a = a'$, so f is injective. Every $a \in A$ equals $g(f(a))$, so g is onto. Inclusion of a one-element set into a two-element set with a left inverse shows the other two properties need not hold.

Solution 004.3 ■ Answer B[Back to problem](#)

The opposite infinite limits and continuity imply surjectivity. But $f(-\sqrt{3}) = f(0) = f(\sqrt{3}) = 0$, so injectivity fails.

Solution 004.4 ■ Answer B[Back to problem](#)

To recover an input, undo g first and then undo f . Indeed $(f^{-1} \circ g^{-1}) \circ (g \circ f) = \text{id}_A$, and the composition in the other order is id_C . The domains also force this order.

Solution 004.5 ■ Answer B[Back to problem](#)

Complete the square: $y = (x - 2)^2 + 3$. The negative branch $x = 2 - \sqrt{y - 3}$ selects exactly $x \leq 2$, on which f is one-to-one and has range $[3, \infty)$. The unrestricted parabola is not injective.

Graphs in the xy-Plane ▸ Graphs and symmetry

005 Graphs, symmetry, coordinates, and intercepts

Solution 005.1 ■ Answer B[Back to problem](#)

At $x = -1$, the relation gives $f(1) = -f(-1)$. Evenness gives $f(-1) = f(1)$, so $f(1) = 0$. The function $\cos(\pi x/2)$ meets the hypotheses and makes the other listed values nonzero.

Solution 005.2 ■ Answer A[Back to problem](#)

Completing the square gives $(y + 1)^2 = x^2 + a + 1$. For any fixed a , sufficiently large $|x|$ makes the right side positive, yielding two distinct values of y . Thus the vertical-line test always fails.

Solution 005.3 ■ Answer A[Back to problem](#)

The symmetries give $f(2 - x) = f(x)$ and $f(8 - x) = f(x)$. Composing the two reflections yields $f(x + 6) = f(x)$. For example $\cos(\pi(x - 1)/3)$ rules out the smaller listed periods.

Solution 005.4 ■ Answer E[Back to problem](#)

For any h , $g(2 + h) + g(2 - h) = f(h) + f(-h) + 6 = 6$. Thus opposite points have midpoint $(2, 3)$.

Solution 005.5 ■ Answer E[Back to problem](#)

The graph is $y = x + 1$ with the point at $x = 1$ removed. Its intercepts are $(-1, 0)$ and $(0, 1)$; the canceled root $x = 1$ is not an intercept.

Solution 005.6 ■ **Answer C**[Back to problem](#)

If $f(x) = y$, oddness gives $f(-x) = -y$. Inverting gives $f^{-1}(-y) = -f^{-1}(y)$, so the inverse is odd. Its graph is the reflection of the original graph across $y = x$, but that does not make the inverse graph itself symmetric about that line. No differentiability is assumed.

Graphs in the xy -Plane ▸ Piecewise graphs

006 Piecewise functions, absolute value, floor, and graph transformations

Solution 006.1 ■ **Answer C**[Back to problem](#)

Write $x = n + t$ with integer n and $0 \leq t < 1$. The equation is $3n + \lfloor 2t \rfloor = 7$. Since $\lfloor 2t \rfloor$ is 0 or 1, only $n = 2$ and $1/2 \leq t < 1$ work. The solution interval is $[5/2, 3)$, of length $1/2$.

Solution 006.2 ■ **Answer C**[Back to problem](#)

For $x < -1$, the equation is $-3x = 7$, giving $x = -7/3$. For $-1 \leq x \leq 2$, it is $x + 4 = 7$, whose candidate 3 is outside the interval. For $x > 2$, it is $3x = 7$, giving $7/3$. Their sum is 0.

Solution 006.3 ■ **Answer D**[Back to problem](#)

The terms are 0, 1, 2, 3, 3, 4, 5, 6, 6, 7, whose sum is 37. Grouping four consecutive indices also exposes the pattern.

Solution 006.4 ■ **Answer C**[Back to problem](#)

The median $x = 4$ minimizes the sum. On $[1, 4]$ the slope is -1 , and on $[4, 9]$ it is 1; outside this interval the sum is larger. Its value at 4 is 8.

Solution 006.5 ■ **Answer B**[Back to problem](#)

Multiplication by -3 converts the maximum into a minimum. Solve $4 - 2x = -2$ to obtain $x = 3$; the output is $7 - 3(5) = -8$. Because $4 - 2x$ maps $\mathbb{R}$ bijectively onto $\mathbb{R}$, the minimum remains unique.

Solution 006.6 ■ **Answer B**[Back to problem](#)

First solve $f(y) = 2$. The negative branch would give $y = 0$, which is excluded from that branch; the nonnegative branch gives $y = \sqrt{2}$. Now $f(x) = \sqrt{2}$ yields $x = \sqrt{2} - 2 < 0$ on the first branch and $x = 2^{1/4} > 0$ on the second. There are exactly two solutions.

Analytic Geometry ▸ General conic equation

007 Conic sections, completing squares, rotations, and degeneracy

Solution 007.1 ■ Answer B

[Back to problem](#)

The quadratic matrix has eigenvalues 8 and 2, with orthogonal eigenvectors $(1, 1)$ and $(1, -1)$. In rotated orthonormal coordinates the equation is $8u^2 + 2v^2 = 8$, or $u^2 + v^2/4 = 1$.

Solution 007.2 ■ Answer D

[Back to problem](#)

Complete squares to get $(x - 3)^2 + 4(y + 1)^2 = 13 - k$. The locus is one point exactly when the right side is zero, so $k = 13$. Positive values give ellipses and negative values an empty locus.

Solution 007.3 ■ Answer A

[Back to problem](#)

The first equation factors as $(x + y - 2)(x + y + 2) = 0$, giving two parallel lines. The second has a rank-one quadratic part but a nonzero linear term in the null direction, so it is a parabola; the last has no real points. A discriminant of zero alone does not exclude degeneracy.

Solution 007.4 ■ Answer C

[Back to problem](#)

Invert the point map: $y = v$ and $x = u - v$. Substitution gives $(u - v)^2 + 4v^2 = 4$, or $u^2 - 2uv + 5v^2 = 4$. Substituting the forward map into the old equation would describe the wrong transformation.

Solution 007.5 ■ Answer A

[Back to problem](#)

The coefficient of uv is $(6 - 3)\sin(2\theta) + 4\cos(2\theta)$. Setting it to zero gives $3\tan(2\theta) + 4 = 0$, so $\tan(2\theta) = -4/3$. The stated coordinate convention determines the sign.

Analytic Geometry ▸ Lines

008 Lines, slopes, distances, and perpendicular bisectors

Solution 008.1 ■ Answer C

[Back to problem](#)

Equality of squared distances yields $x^2 + y^2 = (x - 4)^2 + (y - 2)^2$, hence $2x + y = 5$. Solving with $x + 2y = 8$ gives $x = 2/3$, $y = 11/3$.

Solution 008.2 ■ Answer A

[Back to problem](#)

Write the line as $2x - y + b = 0$. Its distance from the origin is $|b|/\sqrt{5}$, which must equal the radius $\sqrt{5}$. Thus $b = \pm 5$, with product -25 .

Solution 008.3 ■ Answer B[Back to problem](#)

Solving gives $P = (9/5, 13/5)$. The distance to the line is $|3(9/5) + 4(13/5)|/5 = 79/25$.

Solution 008.4 ■ Answer E[Back to problem](#)

The vertices are $(0, 6), (0, -2), (4, 2)$. The vertical base has length 8 and the horizontal altitude is 4, giving area 16.

Solution 008.5 ■ Answer C[Back to problem](#)

Reflect B to $B^* = (5, -4)$. For P on the axis, $|PB| = |PB^*|$, so the triangle inequality gives a minimum of $|AB^*| = \sqrt{4^2 + 6^2} = 2\sqrt{13}$. The segment AB^* crosses the axis, so equality is attained.

Analytic Geometry ▸ Parabolas

009 Parabola, focus, directrix, vertex, axis, and latus rectum

Solution 009.1 ■ Answer C[Back to problem](#)

The vertex lies halfway between the focus and directrix at $(2, 1)$, with $p = 2$. Thus $(x - 2)^2 = 8(y - 1)$. At $y = 9$ the two roots are $x = 2 \pm 8$, whose sum is 4.

Solution 009.2 ■ Answer B[Back to problem](#)

Completing the square only translates the parabola, so $a = 1/(4p)$. Its latus rectum length is $4|p| = 1/|a|$. Setting this equal to 3 gives $|a| = 1/3$.

Solution 009.3 ■ Answer D[Back to problem](#)

Completing the square gives $(y - 3)^2 = 8(x + 1)$. The vertex is $(-1, 3)$ and $p = 2$, so the focus is $(1, 3)$ and the sum is 4.

Solution 009.4 ■ Answer A[Back to problem](#)

Squared distance to the focus is $(x - 1)^2 + (y - 1)^2$. Squared distance to the line is $(x + y + 2)^2/(1^2 + 1^2)$. Equating and multiplying by 2 gives the first equation; this works without aligning the parabola with a coordinate axis.

Solution 009.5 ■ Answer D[Back to problem](#)

Substitute $x = y + 1$ to obtain $y^2 - 4y - 4 = 0$, whose roots differ by $\sqrt{32} = 4\sqrt{2}$. The corresponding x coordinates have the same difference. Thus the chord length is $\sqrt{2(4\sqrt{2})^2} = 8$.

Analytic Geometry ▸ Circles

010 Circle, center, radius, diameter, chord, and tangent

Solution 010.1 ■ **Answer C**
[Back to problem](#)

The center is $(1, 3)$. The radius to $(5, 5)$ has direction $(4, 2)$, so the tangent is $4(x - 5) + 2(y - 5) = 0$, or $2x + y = 15$. Setting $y = 0$ gives $x = 15/2$.

Solution 010.2 ■ **Answer C**
[Back to problem](#)

Subtracting the circle equations gives $x = 3$. Substituting yields $y^2 = 16$, so the chord endpoints are $(3, 4)$ and $(3, -4)$, a distance 8 apart.

Solution 010.3 ■ **Answer A**
[Back to problem](#)

Its radius is the perpendicular distance $|6 - 4 - 12|/5 = 2$. Therefore the area is 4π .

Solution 010.4 ■ **Answer D**
[Back to problem](#)

The chord lies distance 4 from the center. Put its endpoints at $(4, \pm 3)$; their tangents are $4x \pm 3y = 25$, meeting at $(25/4, 0)$.

Solution 010.5 ■ **Answer A**
[Back to problem](#)

Substitution gives $2x^2 - 14x = 0$, so the other intersection is $(7, 7)$. Its distance from the origin is $\sqrt{98} = 7\sqrt{2}$.

Analytic Geometry ▸ Ellipses

011 Ellipse, foci, axes, eccentricity, and area scaling

Solution 011.1 ■ **Answer D**
[Back to problem](#)

Here $c = 3$ and $c/a = 3/5$, so $a = 5$. Then $b^2 = a^2 - c^2 = 16$ and $b = 4$, giving area $\pi ab = 20\pi$.

Solution 011.2 ■ **Answer C**
[Back to problem](#)

The original semiaxes are 3 and 2. After the map they are 6 and $2/3$. Thus $e = \sqrt{1 - ((2/3)/6)^2} = \sqrt{80}/9$.

Solution 011.3 ■ Answer C[Back to problem](#)

Here $a = 5$, $c = 4$, $b^2 = 9$. At $x = c$, $y^2 = b^2(1 - c^2/a^2) = b^4/a^2$, so the chord length is $2b^2/a = 18/5$.

Solution 011.4 ■ Answer A[Back to problem](#)

Set $x = 3 \cos t$, $y = 2 \sin t$. Then $x + y = 3 \cos t + 2 \sin t$ has amplitude $\sqrt{3^2 + 2^2} = \sqrt{13}$, which is attained.

Solution 011.5 ■ Answer C[Back to problem](#)

The focal-distance sum is 10. Writing the distances as $2t$, t gives $t = 10/3$, so their product is $200/9$. Such a point exists because the distance difference $10/3$ is below the focal separation 8, and the distances satisfy the strict triangle inequalities.

Analytic Geometry ▸ Hyperbolas

012 Hyperbola, branches, transverse axis, vertices, and asymptotes

Solution 012.1 ■ Answer B[Back to problem](#)

The vertex parameter is $a = 2$, and $b/a = 3/2$ gives $b = 3$. Hence $c = \sqrt{a^2 + b^2} = \sqrt{13}$, so the focus separation is $2\sqrt{13}$.

Solution 012.2 ■ Answer A[Back to problem](#)

The center is $(1, 2)$, $c = 5$, and the transverse semiaxis is $a = 3$. Thus $b = \sqrt{25 - 9} = 4$. The vertical hyperbola has asymptotes $y - 2 = \pm(a/b)(x - 1) = \pm 3(x - 1)/4$.

Solution 012.3 ■ Answer B[Back to problem](#)

Here $a = 3$, $b = 4$, $c = 5$, so $e = c/a = 5/3$ and the corresponding directrix is $x = a^2/c = 9/5$. Squaring $\sqrt{(x - 5)^2 + y^2} = (5/3)|x - 9/5|$ simplifies to the given hyperbola.

Solution 012.4 ■ Answer E[Back to problem](#)

Substitution gives $x^2(1/4 - m^2/9) = 1$. Exactly two real roots occur when the coefficient is positive, or $|m| < 3/2$; equality gives no intersection.

Solution 012.5 ■ Answer D[Back to problem](#)

With $u = (x + y)/\sqrt{2}$ and $v = (x - y)/\sqrt{2}$, the equation becomes $u^2 - v^2 = 24$. The vertices have $u = \pm\sqrt{24}$, so their separation is $4\sqrt{6}$.

Polynomial Equations ▸ Polynomials and quadratics

013 Polynomial degree, coefficients, quadratic formula, and discriminant

Solution 013.1 ■ Answer B

[Back to problem](#)

For $k \neq 1$ the discriminant is $4k^2 - 4(k-1)(k+1) = 4$, so there are two distinct real roots. At $k = 1$ the equation becomes $2x + 2 = 0$, which has exactly one solution. Thus there is one such parameter value.

Solution 013.2 ■ Answer C

[Back to problem](#)

Its leading coefficient is positive, so strict positivity everywhere is equivalent to negative discriminant. The condition $4t^2 - 4t - 8 < 0$ becomes $(t-2)(t+1) < 0$, or $-1 < t < 2$. Endpoints give a zero and are excluded.

Solution 013.3 ■ Answer C

[Back to problem](#)

If $a \neq 0$ is the leading coefficient and n the degree, the leading terms on the two sides are a^2x^{2n} and ax^{2n} , so $a^2 = a$ and $a = 1$. Examples $p(x) = x$ and $p(x) = x^2$ disprove either degree-parity requirement and the remaining statements.

Solution 013.4 ■ Answer C

[Back to problem](#)

Each composition has degree 28, so the difference has degree at most 28. The bound is attained, for example with $p(x) = 2x^4$ and $q(x) = x^7$, whose leading coefficients differ.

Solution 013.5 ■ Answer E

[Back to problem](#)

Factor $x^3 - 3x - 2 = (x-2)(x+1)^2$. Its monic quadratic divisors are $(x+1)^2 = x^2 + 2x + 1$, which has unequal linear and constant coefficients, and $(x-2)(x+1) = x^2 - x - 2$, which also has unequal coefficients. Thus none has the required form.

Polynomial Equations ▸ Division Algorithm, Remainder Theorem, and Factor Theorem

014 Polynomial division, remainders, and factors

Solution 014.1 ■ Answer C

[Back to problem](#)

The divisor is $(x-1)(x+2)$, so the remainder is $ax+b$. Its values give $a+b = 3$ and $-2a+b = -6$. Thus $a = 3$, $b = 0$, and the remainder is $3x$.

Solution 014.2 ■ Answer B

[Back to problem](#)

Write the remainder as $ax+b$. Since the difference is divisible by $(x-1)^2$, it and its derivative vanish at 1. Thus $a+b = 1$ and $a = 20$, giving $20x - 19$.

Solution 014.3 ■ **Answer B**[Back to problem](#)

Modulo the divisor, $x^3 = 1$. Since $101 \equiv 2 \pmod{3}$, the remainder is $x^2 \equiv -x - 1$.

Solution 014.4 ■ **Answer A**[Back to problem](#)

Reconstruct $p = (x^2 + 1)(x - 2) + 3x + 4 = x^3 - 2x^2 + 4x + 2$. Synthetic division at 1 gives quotient $x^2 - x + 3$ and remainder 5. The remainder from the first division is not a quotient for the second.

Solution 014.5 ■ **Answer B**[Back to problem](#)

The desired remainder has degree below 3 and must be $r(x) = x + 1 + k(x - 1)^2$. Substituting $x = -1$ gives $4k = 2$, hence $k = 1/2$ and $r(x) = x^2/2 + 3/2$. This construction enforces both congruences.

Polynomial Equations ▸ Fundamental Theorem of Algebra and Roots

015 Fundamental Theorem of Algebra, multiplicity, and polynomial identity

Solution 015.1 ■ **Answer D**[Back to problem](#)

The polynomial $p(x) - x^2$ has degree at most 5 and six distinct zeros. It must vanish identically, so $p(6) = 6^2 = 36$.

Solution 015.2 ■ **Answer B**[Back to problem](#)

Recognize $p(x) = (x^2 - 1)^3 = (x - 1)^3(x + 1)^3$. The two distinct roots are 1 and -1 , each with multiplicity 3. The degree counts multiplicity, not distinct roots.

Solution 015.3 ■ **Answer A**[Back to problem](#)

Reality forces $2 - i$ to be a root of multiplicity 2. The specified roots account for $2 + 2 + 2 = 6$ of the seven roots, leaving one; it must be real.

Solution 015.4 ■ **Answer B**[Back to problem](#)

The difference equation is satisfied by x^2 . The polynomial $p(x) - x^2$ is periodic, hence constant; at zero that constant is 3. Thus $p(10) = 103$.

Solution 015.5 ■ **Answer C**[Back to problem](#)

If the distinct roots have multiplicities m_1, m_2, m_3 , then $m_1 + m_2 + m_3 = 7$. Differentiation lowers each multiplicity by exactly one, so the gcd has degree $\sum(m_i - 1) = 7 - 3 = 4$.

Polynomial Equations > Root Location Theorem

016 Root location and sign changes

Solution 016.1 ■ **Answer D**[Back to problem](#)

Each of the four disjoint intervals $(-2, -1)$, $(-1, 0)$, $(0, 1)$, and $(1, 2)$ contains a root.

Degree 4 permits at most four distinct roots. Therefore there are exactly four, each simple.

Solution 016.2 ■ **Answer A**[Back to problem](#)

As $x \rightarrow -\infty$, a monic cubic tends to $-\infty$, whereas $p(-1) > 0$. Continuity forces a root below -1 . The stated values also force roots in $(-1, 0)$ and $(0, 1)$, exhausting the three roots.

Solution 016.3 ■ **Answer B**[Back to problem](#)

The derivative is $3(x^2 - 1)$, with a local maximum at -1 of value $a + 2$ and a local minimum at 1 of value $a - 2$. The three monotonic intervals each contain a root precisely when the maximum is positive and minimum negative: $-2 < a < 2$. At the endpoints a root is repeated.

Solution 016.4 ■ **Answer E**[Back to problem](#)

Set $u = x^2$. Then $u = 2 \pm \sqrt{5}$, only one of which is positive. That value gives exactly one positive and one negative root of p .

Solution 016.5 ■ **Answer C**[Back to problem](#)

Crossing a real root reverses the sign exactly when its multiplicity is odd. Since the endpoint signs differ, the total multiplicity of the roots crossed must be odd. Additional even-multiplicity roots or several odd-multiplicity roots are possible, so uniqueness and simplicity do not follow.

Polynomial Equations > Rational Roots Theorem

017 Rational root theorem and root testing

Solution 017.1 ■ **Answer D**[Back to problem](#)

Grouping gives $x^2(2x - 3) - 4(2x - 3) = (2x - 3)(x - 2)(x + 2)$. The roots $3/2$, 2 , and -2 are distinct and rational.

Solution 017.2 ■ **Answer E**[Back to problem](#)

A rational root of this monic integer polynomial must be an integer dividing 1 , hence ± 1 . Both yield $1 - 5 + 1 = -3$, so neither is a root.

Solution 017.3 ■ Answer D[Back to problem](#)

The rational-root theorem forces both denominators 3 and 5 to divide the leading coefficient, so their least common multiple 15 divides it. The polynomial $(3x - 2)(5x - 3)$ attains leading coefficient 15.

Solution 017.4 ■ Answer C[Back to problem](#)

Because the polynomial is monic with integer coefficients, rational roots must be integers. Their product is 6, so the unordered pairs are $(1, 6)$, $(2, 3)$, $(-1, -6)$, $(-2, -3)$. Their sums give $k = 7, 5, -7, -5$, exactly four integers.

Solution 017.5 ■ Answer D[Back to problem](#)

Substituting $1/2$ gives $(k - 1)/4 = 0$, so $k = 1$. The total root sum is $-1/2$, so the other two sum to -1 .

Polynomial Equations ▸ Conjugate Root Theorems

018 Radical and complex conjugate roots

Solution 018.1 ■ Answer C[Back to problem](#)

Rational coefficients force $1 - \sqrt{2}$, and real coefficients force $2 - i$. These four roots are distinct, so degree at least 4 is required. The product $(x^2 - 2x - 1)(x^2 - 4x + 5)$ achieves degree 4.

Solution 018.2 ■ Answer B[Back to problem](#)

If $r = \sqrt{2} + \sqrt{3}$, then $r^2 = 5 + 2\sqrt{6}$, so $(r^2 - 5)^2 = 24$ and $r^4 - 10r^2 + 1 = 0$. To see minimality, $1/r = \sqrt{3} - \sqrt{2}$, so both $\sqrt{2} = (r - 1/r)/2$ and $\sqrt{3} = (r + 1/r)/2$ lie in $\mathbb{Q}(r)$. This field has degree 4, since $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$; hence no smaller degree suffices.

Solution 018.3 ■ Answer C[Back to problem](#)

The coefficient field lies in $\mathbb{R}$, so complex conjugation forces $\sqrt{2} - i$. It does not force changing the sign of $\sqrt{2}$, because the coefficients themselves need not be fixed by that change. The polynomial $(x - \sqrt{2})^2 + 1$ is a counterexample to every other proposed necessity.

Solution 018.4 ■ Answer C[Back to problem](#)

The conjugates are also roots. Their product is $|1 + i|^2 |2 + 3i|^2 = 2 \cdot 13 = 26$, which is the constant term of a monic quartic.

Solution 018.5 ■ Answer B[Back to problem](#)

Conjugating the identity in $\mathbb{Q}(\sqrt{2})$ fixes the rational coefficients and sends $\sqrt{2}$ to $-\sqrt{2}$, so $p(-\sqrt{2}) = 3 - 2\sqrt{2}$. Their product is $9 - 8 = 1$.

Polynomial Equations ▸ Sum and Product of the Roots

019 Vieta formulas and symmetric expressions in roots

Solution 019.1 ■ **Answer B**[Back to problem](#)

Vieta gives $r + s + t = 4$ and $rs + rt + st = 5$. Thus the sum of squares is $4^2 - 2 \cdot 5 = 6$.

Solution 019.2 ■ **Answer B**[Back to problem](#)

The new product is $(rs)^3 = 1$, while the new sum is $(r + s)^3 - 3rs(r + s) = 27 - 9 = 18$. The required monic quadratic is $x^2 - 18x + 1$.

Solution 019.3 ■ **Answer B**[Back to problem](#)

The identity $(r + s)(s + t)(t + r) = (r + s + t)(rs + st + tr) - rst$ gives $4 \cdot 5 - 7 = 13$. This uses elementary symmetric sums without finding the roots.

Solution 019.4 ■ **Answer B**[Back to problem](#)

Factoring p over $\mathbb{C}$ and logarithmically differentiating gives $p'(z)/p(z) = \sum_j 1/(z - r_j)$ when $p(z) \neq 0$. At $z = 2$, $p(2) = 7$ and $p'(2) = 10$, hence the sum is $10/7$.

Solution 019.5 ■ **Answer B**[Back to problem](#)

The reciprocal sum is $(rs + rt + st)/(rst)$, which is $-2/5$. The numerator is the pairwise-product sum, not the root sum.

Logarithms ▸ Exponential and logarithmic functions

020 Logarithms, exponentials, laws, and change of base

Solution 020.1 ■ **Answer B**[Back to problem](#)

The domain is $x > 0$, $x \neq 1$. Set $u = \log_2 x$, so $\log_x 2 = 1/u$. The equation $2u^2 - 5u + 2 = 0$ gives $u = 2$ or $1/2$, hence $x = 4$ or $\sqrt{2}$.

Solution 020.2 ■ **Answer B**[Back to problem](#)

Set $u = 3^x > 0$. The equation becomes $u^2 - 4u - 5 = (u - 5)(u + 1) = 0$. Only $u = 5$ is positive, so $x = \log_3 5$.

Solution 020.3 ■ **Answer D**[Back to problem](#)

The logarithms require $x > 1$. Combining them yields $x^2 - 1 = 8$, whose roots are ± 3 ; only 3 belongs to the domain.

Solution 020.4 ■ **Answer D**[Back to problem](#)

The relations imply $b = a^2$ and $c = a^6$. Consequently $ac = a^7$ and $b^2 = a^4$, giving the logarithm $4/7$.

Solution 020.5 ■ **Answer C**[Back to problem](#)

The logarithm requires a positive argument. Since its base is below 1, it is decreasing, so the inequality becomes $0 < x^2 - 3x + 2 < 4$. The first condition gives $x < 1$ or $x > 2$; the second puts x strictly between the roots $(3 \pm \sqrt{17})/2$. Intersect these sets.

Trigonometry ▸ Trig Functions of Acute Angles

021 Right-triangle trigonometry and special angles

Solution 021.1 ■ **Answer B**[Back to problem](#)

$\tan \theta + \cot \theta = (\sin^2 \theta + \cos^2 \theta)/(\sin \theta \cos \theta) = 1/(\sin \theta \cos \theta)$. Hence the desired product is $1/5$.

Solution 021.2 ■ **Answer B**[Back to problem](#)

The hypotenuse has length 4. By similarity, the squares of the legs are $4 \cdot 1$ and $4 \cdot 3$, so the shorter leg is 2. The smaller angle is opposite that leg, and its sine is $2/4 = 1/2$.

Solution 021.3 ■ **Answer A**[Back to problem](#)

If the legs are a, b , then $a + b = 13 + 4 = 17$. Since $a^2 + b^2 = 169$, $2ab = 17^2 - 169 = 120$, so the area is $ab/2 = 30$.

Solution 021.4 ■ **Answer C**[Back to problem](#)

If the height is h and the distance from Q to the base is d , then $h = \sqrt{3}d$ and $h = (d + 20)/\sqrt{3}$. Thus $3d = d + 20$, $d = 10$, and $h = 10\sqrt{3}$. This combines two special right triangles rather than evaluating one ratio.

Solution 021.5 ■ **Answer D**[Back to problem](#)

Divide numerator and denominator by $\cos^2 \theta$. The expression becomes $\tan \theta / (1 + \tan \theta)^2 = 2/9$.

Solution 021.6 ■ **Answer B**[Back to problem](#)

The condition gives $\tan \alpha = 3$, so relative leg lengths are 3 and 1 and the hypotenuse is $\sqrt{10}$. Since $\beta = \pi/2 - \alpha$, $\sec \beta = \csc \alpha = \sqrt{10}/3$ and $\csc \beta = \sec \alpha = \sqrt{10}$. Their sum is $4\sqrt{10}/3$.

Trigonometry ▸ Arbitrary Angles and Real Numbers

022 Unit circle, radian measure, quadrants, and directed angles

Solution 022.1 ■ **Answer A**[Back to problem](#)

Here $r = 13$, so $\sec \theta = -13/5$ and $\csc \theta = 13/12$. Their sum is $(-156 + 65)/60 = -91/60$.

Solution 022.2 ■ **Answer A**[Back to problem](#)

The directed angle is $-17\pi/3$. Adding 6π gives the coterminal angle $\pi/3$, so the endpoint is $(3 \cos(\pi/3), 3 \sin(\pi/3)) = (3/2, 3\sqrt{3}/2)$.

Solution 022.3 ■ **Answer A**[Back to problem](#)

Writing $s = \sin \theta$ gives $2s^2 - s - 1 = 0$, so $s = 1$ or $s = -1/2$. These yield one and two angles, respectively.

Solution 022.4 ■ **Answer B**[Back to problem](#)

The radian measure is $\pi/2 + k\pi$, whose terminal side lies on the y -axis. Thus $\cos \theta = 0$ and $\sin \theta = \pm 1$. Exactly tangent and secant divide by zero; cotangent is defined and equals zero.

Solution 022.5 ■ **Answer C**[Back to problem](#)

The identity $\cos 2\theta = 1 - 2\sin^2 \theta$ gives $\sin^2 \theta = 9/25$. The quadrant selects the negative root $-3/5$.

Trigonometry ▸ Trig Identities and Formulas

023 Reciprocal, opposite-angle, and Pythagorean identities

Solution 023.1 ■ **Answer A**[Back to problem](#)

Since $(\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1$, the second factor is $1/3$. Adding and subtracting gives $\sec \theta = 5/3$ and $\tan \theta = -4/3$. Therefore $\sin \theta = \tan \theta / \sec \theta = -4/5$.

Solution 023.2 ■ **Answer C**[Back to problem](#)

Use the common denominator $\sin x(1 - \cos x)$. The numerator is $(1 - \cos x)^2 + \sin^2 x = 2(1 - \cos x)$, giving $2/\sin x$. The simplification is valid on the original domain, where the cancelled factor is nonzero.

Solution 023.3 ■ **Answer C**[Back to problem](#)

Their product is 1, so the sum of their squares is $16 - 2 = 14$. Squaring again gives the fourth-power sum $14^2 - 2 = 194$.

Solution 023.4 ■ **Answer B**[Back to problem](#)

Use $\sin(-x) = -\sin x$ and $\cos(-x) = \cos x$ to obtain $F(x) = -\tan x + \cot x$. Both summands are odd on the symmetric domain excluding multiples of $\pi/2$, so F is odd. The exclusions remain after simplification.

Solution 023.5 ■ **Answer B**[Back to problem](#)

The square root is $|\cos(\pi/2 + x)| = |-\sin x|$. Since x is acute, $\sin x > 0$, and the result is $\sin x$. A squared identity determines only magnitude; the reduction formula and interval determine the sign.

Trigonometry ▸ Trig Identities and Formulas

024 Addition, subtraction, double-angle, and half-angle formulas

Solution 024.1 ■ **Answer A**[Back to problem](#)

The quadrant information gives $\cos \alpha = -4/5$ and $\sin \beta = 5/13$. Then $\sin(\alpha + \beta) = (3/5)(12/13) + (-4/5)(5/13) = 16/65$.

Solution 024.2 ■ **Answer A**[Back to problem](#)

The negative cosine and stated interval place θ in quadrant III, so $\sin \theta = -24/25$. Thus $\tan(\theta/2) = (-24/25)/(1 - 7/25) = -24/18 = -4/3$. This is consistent with $\theta/2$ lying in quadrant II.

Solution 024.3 ■ **Answer C**[Back to problem](#)

Double-angle reduction gives $\cos x(2 \sin x - 1) = 0$. The first factor contributes $\pi/2, 3\pi/2$; the second contributes $\pi/6, 5\pi/6$. Dividing by $\cos x$ would lose two of these four solutions.

Solution 024.4 ■ **Answer D**[Back to problem](#)

Product-to-sum gives half of $\cos(\pi/2) + \cos(-\pi/3)$, which is $(0 + 1/2)/2 = 1/4$.

Solution 024.5 ■ **Answer C**[Back to problem](#)

Writing $u = \tan \alpha$ and $v = \tan \beta$, the addition formula gives $(u + v)/(1 - uv) = 1$, hence $u + v = 1 - uv$. Therefore $(1 + u)(1 + v) = 1 + u + v + uv = 2$.

Trigonometry ▷ Periodicity and Graphs of the Trig Functions

025 Periodicity, fundamental period, and trigonometric domains and ranges

Solution 025.1 ■ Answer D

[Back to problem](#)

The frequencies 2 and 3 must each acquire phase an integer multiple of 2π under a period; this follows from linear independence of the distinct exponential frequencies. Thus $2T, 3T \in 2\pi\mathbb{Z}$, and their common smallest positive solution is $T = 2\pi$.

Solution 025.2 ■ Answer D

[Back to problem](#)

Tangent excludes $x = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$, four points. Cosecant excludes $x = k\pi/3$ for $k = 0, \dots, 5$, six points. None coincide: $(2j+1)\pi/4 = k\pi/3$ would give $3(2j+1) = 4k$, impossible by parity. Thus ten points are excluded.

Solution 025.3 ■ Answer D

[Back to problem](#)

The maximum is 5, attained when $\sin(4x - \pi/3) = -1$. Thus $4x - \pi/3 = -\pi/2 + 2k\pi$, giving $x = -\pi/24 + k\pi/2$. The smallest positive value corresponds to $k = 1$ and equals $11\pi/24$.

Solution 025.4 ■ Answer E

[Back to problem](#)

Put $s = \sin x \in [-1, 1]$. The expression is $1 + 2s - 2s^2$, a concave quadratic with maximum $3/2$ at $s = 1/2$ and minimum -3 at $s = -1$.

Solution 025.5 ■ Answer A

[Back to problem](#)

The secant range is $(-\infty, -1] \cup [1, \infty)$. Multiplying by 2 and subtracting 3 yields $(-\infty, -5] \cup [-1, \infty)$. Poles occur where $\cos x = 0$, namely $x = \pi/2 + k\pi$, and these remain vertical asymptotes after the affine output change.

Trigonometry ▷ The Inverse Trig Functions

026 Inverse trigonometric functions and principal branches

Solution 026.1 ■ Answer C

[Back to problem](#)

The first principal value is $-\pi/6$, since the sine is $-1/2$. The second is $\pi/3$, since the cosine is $1/2$. Their sum is $\pi/6$.

Solution 026.2 ■ Answer C

[Back to problem](#)

Let $A = \arctan 2 + \arctan 3 = 3\pi/4$, since its tangent is -1 and $A \in (\pi/2, \pi)$. Then $\tan(A - \arctan(1/5)) = (-1 - 1/5)/(1 - 1/5) = -3/2$. The angle remains in quadrant II, so its principal expression is $\pi - \arctan(3/2)$.

Solution 026.3 ■ Answer C[Back to problem](#)

The principal secant inverse lies in $[0, \pi] \setminus \{\pi/2\}$, so $\operatorname{arcsec}(-2) = 2\pi/3$. The cosecant inverse lies in $[-\pi/2, \pi/2] \setminus \{0\}$, so $\operatorname{arccsc}(-2) = -\pi/6$. Their sum is $\pi/2$.

Solution 026.4 ■ Answer B[Back to problem](#)

If $u = \arctan x \in (-\pi/2, \pi/2)$, then $\pi/2 - u \in (0, \pi)$ and its cotangent is $\tan u = x$. Therefore it is the specified inverse. In particular $\operatorname{arccot} 0 = \pi/2$, and using $\arctan(1/x)$ alone fails for negative x .

Solution 026.5 ■ Answer B[Back to problem](#)

Using $\cos 2u = 1 - 2\sin^2 u$ with $u = \arcsin(2/3)$ gives $1 - 2(4/9) = 1/9$.

Chapter 1 Review Questions ▸ Radical equations

027 Radical equations and extraneous roots

Solution 027.1 ■ Answer C[Back to problem](#)

The right side must be nonnegative, so $x \geq 3$. Squaring gives $x + 3 = x^2 - 6x + 9$, or $(x - 1)(x - 6) = 0$. Only $x = 6$ meets the sign condition, and direct substitution confirms it.

Solution 027.2 ■ Answer C[Back to problem](#)

The domain is $[-1, 9]$. Squaring gives $10 + 2\sqrt{(x+1)(9-x)} = 16$, so $(x+1)(9-x) = 9$. Hence $-x^2 + 8x = 0$, giving $x = 0, 8$. Both lie in the domain and produce the original sum 4.

Solution 027.3 ■ Answer C[Back to problem](#)

The domain is $x \geq -2$. Factor the equation as $x(\sqrt{x+2} - 1) = 0$. This gives $x = 0$ or $x = -1$, both valid. Dividing by x before considering its zero would discard one solution.

Solution 027.4 ■ Answer B[Back to problem](#)

Rewrite as $a = \sqrt{x^2 + 1} - x$, which is strictly positive. Conversely, for $a > 0$, squaring gives $x = (1 - a^2)/(2a)$ and then $x + a = (1 + a^2)/(2a) > 0$, so the squared equation is equivalent and supplies a solution. No solution exists for $a \leq 0$.

Solution 027.5 ■ Answer C[Back to problem](#)

On $[0, 1]$ the sum is symmetric about $1/2$, increasing to its maximum $\sqrt{2}$ there and then decreasing. Each value strictly between 1 and $\sqrt{2}$ occurs twice; 1 also occurs twice.

Chapter 1 Review Questions ▸ Rational inequalities

028 Rational inequalities and sign charts

Solution 028.1 ■ **Answer A**[Back to problem](#)

Away from $x = 1$, the squared factor is positive, so the sign is that of $(x + 2)/(x - 3)$, which is negative on $(-2, 3)$. Include the zero at -2 and the internal zero at 1 , but exclude the pole 3 . This gives $[-2, 3)$.

Solution 028.2 ■ **Answer D**[Back to problem](#)

Subtracting gives $3/[(x - 1)(x + 2)] > 0$, so $x < -2$ or $x > 1$. The allowed integers are $-5, -4, -3, 2, 3, 4, 5$, whose sum is 2 .

Solution 028.3 ■ **Answer B**[Back to problem](#)

The original domain excludes 1 and 3 . After cancellation the sign is that of $(x - 1)(x + 2)/(x - 3)$, which is nonnegative on $[-2, 1)$ and $(3, \infty)$. The cancelled point 1 remains excluded even though it appears as a zero in the reduced numerator.

Solution 028.4 ■ **Answer A**[Back to problem](#)

The denominator is positive on the domain, so the numerator must be positive there. If $|a| < 2$, its discriminant is negative and its leading coefficient positive. At $a = \pm 2$ it vanishes at $x = \mp 1$, a permitted input. If $|a| > 2$, it is negative on an interval, which cannot be removed by excluding the single point 2 .

Solution 028.5 ■ **Answer C**[Back to problem](#)

The numerator changes sign at ± 2 , the denominator at ± 1 . Their signs agree outside $[-2, 2]$ and inside $(-1, 1)$; zeros at ± 2 are included and poles at ± 1 are excluded.

Chapter 1 Review Questions ▸ Hyperbolic functions

029 Hyperbolic sine, cosine, tangent, and their inverses

Solution 029.1 ■ **Answer D**[Back to problem](#)

Let $u = e^{2x} > 0$. Then $\tanh x = (u - 1)/(u + 1) = 3/5$, so $5u - 5 = 3u + 3$ and $u = 4$.

Solution 029.2 ■ **Answer C**[Back to problem](#)

The first term is $\log(3/4 + 5/4) = \log 2$. The second is $\frac{1}{2} \log((8/5)/(2/5)) = \frac{1}{2} \log 4 = \log 2$. Their sum is $\log 4$.

Solution 029.3 ■ Answer B[Back to problem](#)

Here $\cosh x = \sqrt{5}$, so $\sinh 2x = 4\sqrt{5}$ and $\cosh 2x = 1 + 2\sinh^2 x = 9$. Their ratio is $4\sqrt{5}/9$.

Solution 029.4 ■ Answer B[Back to problem](#)

Write the expression as $(e^x + 5e^{-x})/2$. AM-GM gives the lower bound $\sqrt{5}$, attained when $e^{2x} = 5$.

Solution 029.5 ■ Answer D[Back to problem](#)

Hyperbolic sine is a bijection on $\mathbb{R}$, so its inverse composition returns x . The principal inverse of hyperbolic cosine has nonnegative output and returns $|x|$, because cosine hyperbolic is even and increasing on $[0, \infty)$. Their sum is $|x| + x$.

2. Calculus I

Limits of Sequences ▸ Definitions

030 Sequences, convergence, and divergence

Solution 030.1 ■ Answer C

[Back to problem](#)

Rationalization gives $n(\sqrt{n^2 + 3} - n) = 3n/(\sqrt{n^2 + 3} + n) = 3/(\sqrt{1 + 3/n^2} + 1) \rightarrow 3/2$. Keeping the outside factor n is essential.

Solution 030.2 ■ Answer E

[Back to problem](#)

$a_{2k} = 0$, while $a_{4k+1} = 4k + 1$ and $a_{4k+3} = -(4k + 3)$. Thus there is a constant zero subsequence and subsequences diverging in opposite directions. Unboundedness does not imply divergence to one signed infinity.

Solution 030.3 ■ Answer C

[Back to problem](#)

The error is exactly $1/n$, independent of parity. The strict inequality holds precisely when $n > 100$. Thus $N = 101$ works for the entire tail, whereas $N = 100$ fails at its first index.

Solution 030.4 ■ Answer A

[Back to problem](#)

Even and odd subsequences tend to 1 and -1 . Every convergent subsequence must eventually approach one of these two values, since its distance to $\{-1, 1\}$ is $1/n$.

Solution 030.5 ■ Answer A

[Back to problem](#)

The difference of squares tends to $4 - 4 = 0$. The other expressions may be undefined, unbounded, or oscillatory; their denominators or extra factors are uncontrolled.

Limits of Sequences ▸ Monotone convergence

031 Boundedness, monotonicity, and recursive limits

Solution 031.1 ■ Answer B

[Back to problem](#)

For positive a_n , AM–GM gives $a_{n+1} \geq \sqrt{6}$. If $a_n > \sqrt{6}$, then $a_{n+1} - a_n = (6 - a_n^2)/(2a_n) < 0$. Starting from 3, the sequence is decreasing and bounded below. Its positive limit solves $2L = L + 6/L$, hence $L = \sqrt{6}$.

Solution 031.2 ■ Answer C

[Back to problem](#)

Taking reciprocals gives $1/a_{n+1} = 1/a_n + 1$. Consequently $1/a_n = 1/2 + (n - 1) = n - 1/2$, so $na_n = n/(n - 1/2) \rightarrow 1$. Merely solving $L = L/(1 + L)$ finds $a_n \rightarrow 0$ but does not determine the requested scaled limit.

Solution 031.3 ■ Answer D[Back to problem](#)

The terms are $0, 1, 0, 1, \dots$. Its odd and even subsequences have limits 0 and 1, so the full sequence does not converge. The fixed-point equation gives a necessary condition on a limit if one exists; it does not prove existence.

Solution 031.4 ■ Answer C[Back to problem](#)

Every factor lies in $(0, 1)$, so the positive sequence decreases and has a limit. Factoring each term gives $\frac{k(k+2)}{(k+1)^2}$, and telescoping yields $b_n = (n+2)/(2(n+1)) \rightarrow 1/2$. Monotonicity establishes existence; the product structure identifies the value.

Solution 031.5 ■ Answer B[Back to problem](#)

Subtract the fixed point $2/3$ to get $a_{n+1} - 2/3 = -(a_n - 2/3)/2$. Thus the errors form a geometric sequence with initial value $-2/3$. Their signs alternate while their magnitudes tend to zero, so convergence does not require eventual monotonicity.

Limits of Sequences ▸ Algebra and estimates

032 Limit laws and the squeeze theorem

Solution 032.1 ■ Answer C[Back to problem](#)

The first summand equals $2/(\sqrt{1+2/n}+1)$ and tends to 1. The absolute value of the second is at most $1/\sqrt{n}$, so it tends to zero by squeezing.

Solution 032.2 ■ Answer D[Back to problem](#)

Eventually $|a_n b_n| > 1/2$. For any $M > 0$, eventually $|a_n| < 1/(2M)$, hence $|b_n| > M$. The sign need not stabilize: $a_n = (-1)^n/n$ and $b_n = (-1)^n n$ satisfy the hypotheses.

Solution 032.3 ■ Answer D[Back to problem](#)

The expression lies between 0 and $x^2/|x| = |x|$. Both bounds tend to zero, so the squeeze theorem applies despite the jumps of the floor function.

Solution 032.4 ■ Answer D[Back to problem](#)

The sampled sequence is identically zero. However, $f(x) = x^2 \sin(\pi x)$ is continuous and vanishes at every integer while being unbounded, lacking a limit at infinity, and having $f(x)/x = x \sin(\pi x)$ fail to tend to zero. Discrete samples do not control intervening values.

Solution 032.5 ■ Answer B[Back to problem](#)

The expression lies between 3 and $3 \cdot 2^{1/n}$, since $3^n \leq 2^n + 3^n \leq 2 \cdot 3^n$. The two bounds tend to 3.

Solution 032.6 ■ **Answer A**[Back to problem](#)

The ratio of successive positive terms is $(1 + 1/n)^m/2 \rightarrow 1/2$. It is therefore at most $3/4$ from some index onward. The terms are bounded by a fixed constant times $(3/4)^n$, which tends to zero. This proves exponential growth dominates every fixed power.

Limits of Functions ▸ Definitions and calculations

033 Function limits, one-sided limits, and epsilon–delta reasoning

Solution 033.1 ■ **Answer B**[Back to problem](#)

Rationalizing gives $f(x) = |x|/[x(\sqrt{1+|x|}+1)]$. The factor $|x|/x$ is -1 on the left and 1 on the right, yielding the stated pair.

Solution 033.2 ■ **Answer D**[Back to problem](#)

For noninteger x , $\lfloor -x \rfloor = -\lfloor x \rfloor - 1$, so $f(x) = -1$. At integer m the sum is 0 . A sufficiently small punctured neighborhood contains no other integer, so both one-sided limits equal -1 .

Solution 033.3 ■ **Answer C**[Back to problem](#)

Cancel $x - 2$ to obtain $(x^2 + 2x + 4)/(x + 2)$ for $x \neq -2$. Evaluation at -2 gives $12/4 = 3$.

Solution 033.4 ■ **Answer D**[Back to problem](#)

The proposed choice is below 1 , so $|x+2| < 5$. Consequently $|x^2 - 4| = |x-2||x+2| < 5\delta = \varepsilon$. Each other choice fails for some permitted ε, x .

Solution 033.5 ■ **Answer C**[Back to problem](#)

I is the sequential criterion. If the limit fails, some $\varepsilon > 0$ admits bad points arbitrarily close to a . Infinitely many lie on one side; choose them successively closer on that side to form a monotone bad sequence, disproving III. II is insufficient: $\sin(\pi/(x-a))$ vanishes on $a \pm 1/n$ but has no limit.

Limits of Functions as $x \rightarrow \pm\infty$ ▷ End behavior

034 Limits at infinity and asymptotes

Solution 034.1 ■ Answer A

[Back to problem](#)

Long division gives $f(x) = x + 2 - 1/(x + 1)$. The remainder tends to zero at either infinity, while the numerator is nonzero at $x = -1$, producing a genuine pole there.

Solution 034.2 ■ Answer B

[Back to problem](#)

Rationalization gives $(4x + 5)/(\sqrt{x^2 + 4x + 5} - x)$. Divide numerator and denominator by $-x > 0$ to obtain $(-4 - 5/x)/(\sqrt{1 + 4/x + 5/x^2} + 1) \rightarrow -2$.

Solution 034.3 ■ Answer B

[Back to problem](#)

Divide by $|x|$, obtaining $\text{sgn}(x)/\sqrt{1 + 1/x^2}$. The denominator tends to 1, while the sign is positive at one end and negative at the other. The asymptotes are $y = 1$ and $y = -1$.

Solution 034.4 ■ Answer B

[Back to problem](#)

The difference from x has absolute value at most $1/x$, so it tends to zero. But $f'(x) = 1 + 2\cos(x^2) - \sin(x^2)/x^2$ approaches 3 along $x = \sqrt{2\pi n}$ and -1 along $x = \sqrt{(2n + 1)\pi}$. An asymptote does not force convergence of tangent slopes.

Solution 034.5 ■ Answer C

[Back to problem](#)

Away from the exclusions the function equals $(x - 1)/(x + 2)$. At 1 it has finite limit zero, allowing a continuous extension. At -2 the numerator tends to -3 and the denominator to zero, giving a vertical asymptote. At either infinite end the limit is 1.

Continuous Functions ▷ Definition and closure properties

035 Continuity and types of discontinuity

Solution 035.1 ■ Answer D

[Back to problem](#)

$x^3 - 3x + 2 = (x - 1)^2(x + 2)$, so the punctured formula is $x + 2$. Its limit at 1 is 3, uniquely determining the extension.

Solution 035.2 ■ Answer C

[Back to problem](#)

Both one-sided limits equal zero for every fixed real a ; the logarithm is defined near zero because $1 + ax > 0$ there. Continuity therefore requires and is guaranteed by $c = 0$. Matching slopes would be an additional differentiability condition.

Solution 035.3 ■ Answer B[Back to problem](#)

At zero the left and right limits are -2 and 0 , so no assigned value can remove the jump. At 1 , the reciprocal term has infinite one-sided limits. At 2 , both limits equal 2 , so defining $f(2) = 2$ fills the hole. These are classifications of the excluded locations, not failures of continuity relative to the existing domain.

Solution 035.4 ■ Answer D[Back to problem](#)

A rational function is continuous at every point of its domain. The excluded points $0, 1$ are not domain points, so relative-domain continuity fails nowhere.

Solution 035.5 ■ Answer B[Back to problem](#)

The bound $|f(x)| \leq |x|$ proves continuity. But $f(x)/x = \sin(1/x)$ has no limit, so the derivative does not exist.

Solution 035.6 ■ Answer B[Back to problem](#)

Continuity is relative to $[0, 1]$. The right-hand limit at 0 is zero and the left-hand limit at 1 is $1/2$. Assigning these endpoint values gives continuity everywhere; values outside the domain are irrelevant.

Continuous Functions ▸ A nontrivial continuity example

036 Rational–irrational constructions and Thomae’s function

Solution 036.1 ■ Answer D[Back to problem](#)

At each nonzero point the rational and irrational limits differ, so continuity and differentiability fail. At zero, $g(0) = 0$ and $|g(h)/h| = |h| \rightarrow 0$, proving $g'(0) = 0$.

Solution 036.2 ■ Answer A[Back to problem](#)

Given $\varepsilon > 0$, exclude all reduced rationals with denominator at most $1/\varepsilon$ from a punctured neighborhood of $2/5$; there are finitely many. Every remaining value of T is less than ε . Thus the limit is zero even though $T(2/5) = 1/5$.

Solution 036.3 ■ Answer B[Back to problem](#)

Both dense branches must approach the same value, requiring $x = 1 - x$. At $1/2$ both branches tend to $1/2$, proving continuity there and nowhere else.

Solution 036.4 ■ Answer D[Back to problem](#)

Both possible outputs of D are rational, so $D(D(x)) = 1$ identically. The other functions have differing rational and irrational limiting values at some points (and $D^2 = D$). Continuity of a composition does not require that both component functions be continuous.

Solution 036.5 ■ **Answer C**[Back to problem](#)

Its discontinuities are precisely the rational points, a measure-zero set. Every lower sum is zero because irrationals are dense; integrability then gives integral zero.

Theorems Concerning Continuous Functions ▸ Extreme value theorem

037 Extreme value theorem and attainment

Solution 037.1 ■ **Answer C**[Back to problem](#)

The minimum is attained at some c . Since every value is positive, $m = f(c) > 0$, and $1/f(x) \leq 1/m$. Positivity without compactness would not ensure this uniform bound.

Solution 037.2 ■ **Answer B**[Back to problem](#)

For $x \geq 0$, $x/(1+x) < 1$ and tends to 1, so its supremum is 1 without attainment. Choice C is unbounded above; A, D, and E attain their maxima.

Solution 037.3 ■ **Answer A**[Back to problem](#)

Outside a sufficiently large compact interval, $f(x) > f(0) + 1$. The minimum on that interval exists by the extreme value theorem and is consequently a global minimum.

Solution 037.4 ■ **Answer B**[Back to problem](#)

The expression is $1/4 - (x - 1/2)^2$. Its supremum $1/4$ is attained at $1/2$; its infimum 0 is approached at the excluded endpoints.

Solution 037.5 ■ **Answer E**[Back to problem](#)

Closed and bounded subsets of $\mathbb{R}$ are compact, so the extreme value theorem applies. Bounded open intervals and closed unbounded sets show why either property alone is insufficient.

Theorems Concerning Continuous Functions ▸ Bolzano and intermediate value theorems

038 Intermediate values, zeros, and fixed points

Solution 038.1 ■ **Answer B**[Back to problem](#)

The polynomial is negative at 0 and positive at 1, so continuity gives at least one root. Its derivative $3x^2 + 1$ is positive everywhere, so it is strictly increasing and has at most one root.

Solution 038.2 ■ **Answer A**[Back to problem](#)

For $g(x) = f(x) - x^2$, $g(0) = f(0) \geq 0$ and $g(1) = f(1) - 1 \leq 0$. The intermediate value theorem supplies a zero. Constant maps refute B, C, or D as appropriate, and the identity map refutes uniqueness in E.

Solution 038.3 ■ Answer D[Back to problem](#)

For every integer n , induction in either direction gives $f(n) = f(0) + 2n$. Given any real target, choose two integers whose function values lie below and above it. The intermediate value theorem on the interval between them gives an input attaining that target.

Solution 038.4 ■ Answer A[Back to problem](#)

The first function is continuous away from zero. Every interval with zero as an interior point or endpoint contains values of $\sin(1/x)$ filling $[-1, 1]$, so it has the intermediate value property there too. It has no limit at zero. The last function is continuous at zero; the other three skip intermediate values on suitable intervals.

Solution 038.5 ■ Answer A[Back to problem](#)

Let $g(x) = f(x + 1/2) - f(x)$. Then $g(1/2) = -g(0)$, so either an endpoint is zero or continuity gives a zero between them.

The Derivative ▸ Definition and geometry

039 Derivative, differentiability, tangent, and normal

Solution 039.1 ■ Answer C[Back to problem](#)

Write $f(a + t) = f(a) + f'(a)t + o(t)$. Substitution of $t = 5h^2$ and $t = -2h^2$ leaves $7f'(a)h^2 + o(h^2)$ in the numerator. This uses differentiability at a and does not assume continuity of f' .

Solution 039.2 ■ Answer B[Back to problem](#)

The positive intercept is $x = \sqrt{3}$. Since $f'(x) = 3x^2 - 3$, the tangent slope there is 6 and the normal slope is $-1/6$. The normal passes through $(\sqrt{3}, 0)$.

Solution 039.3 ■ Answer C[Back to problem](#)

The difference quotient is $x \sin(1/x)$, which tends to zero, so $f'(0) = 0$. For $x \neq 0$, $f'(x) = 2x \sin(1/x) - \cos(1/x)$, whose cosine term has distinct subsequential limits. Existence of a derivative at a point does not make the derivative continuous there.

Solution 039.4 ■ Answer E[Back to problem](#)

At $x = a$, the tangent is $y = 3a^2x - 2a^3$. Thus $a^3 = 8$, so $a = 2$ and the slope is $3a^2 = 12$.

Solution 039.5 ■ Answer C[Back to problem](#)

At the simple zeros ± 1 of $x^2 - 1$, taking absolute value reverses the sign of a nonzero slope, producing corners. Away from these zeros the expression is locally a polynomial.

Solution 039.6 ■ Answer C[Back to problem](#)

The secant slope is $x^{1/3}/x = x^{-2/3} \rightarrow +\infty$ from both sides, so there is no finite derivative but the tangent line is vertical. Equivalently, the inverse parametrization $x = y^3$ has derivative zero at zero. Its perpendicular normal is horizontal.

The Derivative ▸ Rules of differentiation

040 Differentiation rules and the chain rule

Solution 040.1 ■ Answer D[Back to problem](#)

Let $u = f(x^2)$ and $v = xf(x^3)$. At 1, $u = v = 2$, $u' = -6$, and $v' = 2 - 9 = -7$. The quotient derivative is $[(-6)(2) - 2(-7)]/4 = 1/2$.

Solution 040.2 ■ Answer D[Back to problem](#)

The product rule gives $F' = e^{\sin x} \cos x \log(2 + \cos x) - e^{\sin x} \sin x / (2 + \cos x)$. At zero the second term vanishes and the first is $\log 3$.

Solution 040.3 ■ Answer B[Back to problem](#)

Fix x and differentiate the identity with respect to y at $y = 0$. The left derivative is $f'(x)$, while the right derivative is $f(x)f'(0) = 3f(x)$. The parameter held fixed contributes no derivative factor.

Solution 040.4 ■ Answer B[Back to problem](#)

The chain rule gives $f'(g(f(1)))g'(f(1))f'(1) = f'(1)g'(2)f'(1) = 3 \cdot 4 \cdot 3 = 36$.

Solution 040.5 ■ Answer C[Back to problem](#)

The product simplifies exactly to x^3 , so its derivative at zero is 0. The usual product rule is sufficient when both factors are differentiable, but $u'(0)$ does not exist here. Algebra or the definition must establish the derivative instead.

The Derivative ▸ Inverse-function rule

041 Derivative of an inverse function

Solution 041.1 ■ Answer B[Back to problem](#)

First solve $f(a) = 2$: $a = 1$. The derivative is $f'(1) = 4$, so $g'(2) = 1/4$. Substituting 2 into f' would use the wrong corresponding point.

Solution 041.2 ■ Answer A[Back to problem](#)

Because $f(0) = 1$, use $a = 0$. Here $f'(0) = 2$ and $f''(0) = 1$. Twice differentiating $f(g(y)) = y$ yields $g''(1) = -1/2^3 = -1/8$.

Solution 041.3 ■ Answer D[Back to problem](#)

The quotient is $h^{1/3}/h = |h|^{-2/3}$, which tends to $+\infty$ on both sides. Thus g has no finite derivative at zero. The reciprocal-slope formula requires $f'(0) \neq 0$, a hypothesis that fails.

Solution 041.4 ■ Answer E[Back to problem](#)

The derivative is $2g(y)g'(y)$. At 7, $g(7) = 2$ and $g'(7) = 1/3$, giving $4/3$.

Solution 041.5 ■ Answer E[Back to problem](#)

The two branches use different preimages. Since $f'(0) = -3$ and $f'(\sqrt{3}) = 6$, the inverse derivatives are $-1/3$ and $1/6$. Their difference is $1/2$. A local inverse must be identified by its branch.

The Derivative ▸ Table of differentials

042 Elementary derivatives and differentials

Solution 042.1 ■ Answer B[Back to problem](#)

The derivative is $1/[x\sqrt{1-(\log x)^2}]$. At $x = \sqrt{e}$, $\log x = 1/2$, giving $1/[\sqrt{e}(\sqrt{3}/2)] = 2/\sqrt{3e}$.

Solution 042.2 ■ Answer C[Back to problem](#)

The derivative of the first logarithm is $\sec x$. For the second, $(-\csc x \cot x + \csc^2 x)/(\csc x - \cot x) = \csc x$. Thus $F' = \sec x - \csc x$, which vanishes at $\pi/4$.

Solution 042.3 ■ Answer B[Back to problem](#)

The derivative is $1/[2\sqrt{x}(1+x)]$. At $x = 3$ this is $1/(8\sqrt{3})$.

Solution 042.4 ■ Answer D[Back to problem](#)

The derivative is $2x \cot(x^2)$. Here $x^2 = \pi/4$, so the cotangent is 1 and $2x = \sqrt{\pi}$.

Solution 042.5 ■ Answer E[Back to problem](#)

The quotient rule for the outer logarithm gives $(1 + x/\sqrt{x^2 - 1})/(x + \sqrt{x^2 - 1})$. The common factor cancels, leaving $1/\sqrt{x^2 - 1}$.

Solution 042.6 ■ Answer A[Back to problem](#)

The chain rule gives $dy = 2^{\log_3 x} \log 2 \, dx / (x \log 3)$. At $x = 27$, the exponential factor is $2^3 = 8$. Both the exponential base and the logarithm base contribute a factor.

Solution 042.7 ■ **Answer B**[Back to problem](#)

The derivative is $-1/\sqrt{1-x^2} + 1/x$. Both terms are defined near $-1/2$, because $|x| > 0$ there. Substitution gives $-2/\sqrt{3} - 2$; the absolute value permits negative inputs and does not change the logarithmic derivative $1/x$.

Linear Approximations Using Differentials ▸ Local approximation

043 Linearization, differentials, and approximation error

Solution 043.1 ■ **Answer C**[Back to problem](#)

$dy/y = (-3/2)dx/x$. Multiplying 0.004 by $-3/2$ gives -0.006 , or a 0.6% underestimate. The negative exponent reverses the direction.

Solution 043.2 ■ **Answer A**[Back to problem](#)

At 16, $f = 4$ and $f' = 1/8$, so $L(16.8) = 4 + 0.8/8 = 4.1$. Since $f''(x) = -1/(4x^{3/2}) < 0$, the graph is concave down and lies below this tangent.

Solution 043.3 ■ **Answer B**[Back to problem](#)

The first-order differential is zero because $f'(0) = 0$. Nevertheless $\cos h - 1 = -2\sin^2(h/2)$, so dividing by h^2 gives a limit of $-1/2$. A zero linear term does not make the actual change zero.

Solution 043.4 ■ **Answer A**[Back to problem](#)

On $[1, 1.1]$, $|f''(x)| = 1/x^2 \leq 1$. Taylor's bound is $(1/2)(0.1)^2 = 1/200$; concavity also shows the tangent approximation lies above the logarithm.

Solution 043.5 ■ **Answer E**[Back to problem](#)

The linearization is $1/2 - h/4$. Direct subtraction gives $1/(2+h) - 1/2 + h/4 = h^2/[4(2+h)]$, wherever $h \neq -2$.

Implicit Differentiation ▸ Local implicit functions

044 Implicit differentiation and the implicit function theorem

Solution 044.1 ■ Answer B

[Back to problem](#)

First differentiation yields $2x + y + (x + 2y)y' = 0$, so $y' = -1$ at $(1, 1)$. Differentiating again gives $2 + 2y' + 2(y')^2 + (x + 2y)y'' = 0$. Substitution gives $2 + 3y'' = 0$.

Solution 044.2 ■ Answer C

[Back to problem](#)

Here $F_x = 2x$ and $F_y = 2y$. At $(1, 0)$, $F_x \neq 0$ but $F_y = 0$, so the theorem supplies $x = \sqrt{1 - y^2}$ locally. A differentiable $y(x)$ on an open neighborhood of $x = 1$ cannot parametrize this branch.

Solution 044.3 ■ Answer C

[Back to problem](#)

For real numbers, cubing is injective, so $y^3 = x^3$ is equivalent to $y = x$. This is a unique differentiable branch through the origin. The implicit function theorem gives sufficient conditions; their failure alone proves neither nonexistence nor nonuniqueness.

Solution 044.4 ■ Answer B

[Back to problem](#)

Implicit differentiation gives $e^{xy}(y + xy') + y' = 0$. At $(0, 1)$ this becomes $1 + y' = 0$, and the coefficient of y' is nonzero.

Solution 044.5 ■ Answer B

[Back to problem](#)

Implicit differentiation gives $y' = (2y - x^2)/(y^2 - 2x)$ when the denominator is nonzero. A horizontal tangent requires $y = x^2/2$. Substitution gives $x^3(x^3 - 16) = 0$. Apart from the origin, there is one real solution $x^3 = 16$; there $y^2 = 4x$, so the denominator is $2x \neq 0$ and the implicit function theorem applies.

Higher-Order Derivatives ▸ Repeated differentiation

045 Higher-order derivatives

Solution 045.1 ■ Answer B

[Back to problem](#)

Repeated differentiation gives $f^{(n)}(x) = n!3^n(2 - 3x)^{-n-1}$. Each negative exponent derivative and inner derivative -3 combine to a positive factor. Substitution of zero gives the result.

Solution 045.2 ■ Answer D

[Back to problem](#)

In Leibniz's formula, at zero only the term differentiating x^2 exactly twice survives. It is $\binom{101}{2} \cdot 2 \cdot e^0 = 101 \cdot 100 = 10100$.

Solution 045.3 ■ **Answer B**[Back to problem](#)

The first difference quotient is $|h| \rightarrow 0$, so $f'(0) = 0$. Away from zero, $f'(x) = 2|x|$. The next difference quotient is $2|h|/h$, with one-sided limits -2 and 2 , so the second derivative fails. Existing lower derivatives do not guarantee higher ones.

Solution 045.4 ■ **Answer A**[Back to problem](#)

Derivatives repeat every four orders. Since $101 \equiv 1 \pmod{4}$, this derivative is $\cos x$, whose value at $\pi/3$ is $1/2$.

Solution 045.5 ■ **Answer C**[Back to problem](#)

Differentiating $f'(x) = (1+x)^{-1}$ another $n-1$ times gives $(-1)^{n-1}(n-1)!(1+x)^{-n}$. Evaluation at zero yields the stated value.

Curve Sketching ▷ Properties of the first derivative

046 Critical points, stationary points, and monotonicity

Solution 046.1 ■ **Answer D**[Back to problem](#)

The sign of f' is positive on $(-\infty, -2)$, negative on $(-2, 0)$ and $(0, 1)$, and positive on $(1, \infty)$. Thus -2 is a local maximum and 1 a local minimum, while the double zero at 0 is stationary without an extremum.

Solution 046.2 ■ **Answer C**[Back to problem](#)

$f(0) = 0$ and $f(x) < 0$ for every sufficiently small nonzero x , so zero is a strict local maximum. The quotient $f(h)/h = |h|^{2/3}(h-2)/h$ has unbounded one-sided values, so no finite derivative exists.

Solution 046.3 ■ **Answer B**[Back to problem](#)

The mean value theorem makes f nondecreasing. If $f(a) = f(b)$ for some $a < b$, monotonicity would force it to be constant throughout $[a, b]$, making its derivative zero at every point of the uncountable interval (a, b) . This contradicts the stated zero set. Strict increase does not require a strictly positive derivative at every point.

Solution 046.4 ■ **Answer D**[Back to problem](#)

The derivative is $1 - 1/x^2$, which vanishes at $x = \pm 1$. The excluded point zero is not a stationary point.

Solution 046.5 ■ **Answer C**[Back to problem](#)

Critical points are $\pm\sqrt{a}$. Their values are $2a^{3/2}$ and $-2a^{3/2}$, giving difference $4a^{3/2}$.

Curve Sketching ▸ Properties of the second derivative

047 Concavity, convexity, and inflection points

Solution 047.1 ■ Answer B

[Back to problem](#)

$f''(x) = 12x(x - 2)$ changes sign at 0 and 2. At the nonzero inflection point, $f'(2) = 4(8) - 12(4) = -16$.

Solution 047.2 ■ Answer D

[Back to problem](#)

For $x \neq 0$, $(x^{1/3})'' = -2/(9x^{5/3})$, positive for $x < 0$ and negative for $x > 0$. The graph is continuous at zero and changes concavity there, although even its first derivative is not finite there. The other listed functions do not change concavity at zero.

Solution 047.3 ■ Answer B

[Back to problem](#)

We have $f' = e^{-x}(1 - x) > 0$ for $x < 1$ and $f'' = e^{-x}(x - 2) < 0$ for $x < 2$. Their intersection is $(-\infty, 1)$.

Solution 047.4 ■ Answer B

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The chord from 0 to 1 gives $f(1/2) \leq 1$. Monotonicity of secant slopes gives $[f(0) - f(-1)]/(0 + 1) \leq [f(1/2) - f(0)]/(1/2)$, so $f(1/2) \geq -1$. Both bounds, and every intermediate value, are attained by the piecewise linear interpolant through the four stated points; its successive slopes are $-2, 2t, 4 - 2t$, nondecreasing exactly when $-1 \leq t \leq 1$.

Solution 047.5 ■ Answer C

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The mean value theorem writes the secant slope as $f'(c)$ for some $x < c < y$. Since $f'' > 0$, f' is strictly increasing, giving the strict inequalities.

Curve Sketching ▸ Classifying critical points

048 Second-derivative and higher-derivative tests

Solution 048.1 ■ Answer B

[Back to problem](#)

Taylor's local formula gives $f(x) - f(0) = -12x^6/6! + o(x^6)$, which is negative for all sufficiently small nonzero x . The first nonzero order is even with negative coefficient.

Solution 048.2 ■ Answer C

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$\sin x - x = -x^3/6 + O(x^5)$ changes sign across zero, so zero is not an extremum. A and B have positive fourth-order leading terms; D is positive away from zero; E is negative away from zero.

Solution 048.3 ■ **Answer E**[Back to problem](#)

Extend e^{-1/x^2} by zero at the origin. It is smooth with every derivative zero there and has a strict minimum; its negative has a strict maximum. The function xe^{-1/x^2} , also extended by zero, has the same vanishing derivatives and changes sign. Finite-order derivative tests cannot classify a point when there is no first nonzero derivative.

Solution 048.4 ■ **Answer B**[Back to problem](#)

For $x \neq 0$, $x^4 \leq f(x) \leq 3x^4$, so zero is a strict minimum. The first derivative at zero is zero. Away from zero, $f'(x) = 4x^3(2 + \sin(1/x)) - x^2 \cos(1/x)$; dividing by x shows $f''(0) = 0$. The positive second-derivative test is sufficient, not necessary.

Solution 048.5 ■ **Answer E**[Back to problem](#)

If $a > 0$, the leading fourth-power term is positive; if $a = 0$, $x^6 > 0$ away from zero. For $a < 0$, the leading term is negative, producing a local maximum instead.

Theorems Concerning Differentiable Functions ▸ Rolle's theorem

049 Rolle's theorem and repeated zeros

Solution 049.1 ■ **Answer C**[Back to problem](#)

Four distinct zeros would force three zeros of f' , two of f'' , and a zero of $f^{(3)}$, contradicting strict positivity. The bound is attained by $f(x) = x^3 - x$, whose third derivative is 6 and which has three distinct zeros.

Solution 049.2 ■ **Answer D**[Back to problem](#)

The function is continuous on the closed interval and takes value 1 at both endpoints, but it has no derivative at zero. Its derivatives away from zero are -1 and 1 , so no interior point has derivative zero.

Solution 049.3 ■ **Answer C**[Back to problem](#)

The function is continuous on $[0, 1]$, differentiable on $(0, 1)$, and zero at both endpoints. Endpoint differentiability is unnecessary. On the interior, $f'(x) = x^{-1/3}(2 - 5x)/3$, so the unique zero is $c = 2/5$.

Solution 049.4 ■ **Answer C**[Back to problem](#)

Equal endpoint values give $1 - k(e - 1) = 0$, so $k = 1/(e - 1)$. Then $f'(x) = 1/x - 1/(e - 1)$ vanishes at $x = e - 1$, which lies strictly between 1 and e . The construction chooses the secant slope and then identifies the parallel tangent.

Solution 049.5 ■ Answer A[Back to problem](#)

The function $e^x f(x)$ vanishes at all three points. Rolle's theorem gives a derivative zero in each intervening interval; its derivative is $e^x(f' + f)$.

Theorems Concerning Differentiable Functions ▸ Mean value theorem

050 Mean value theorem and derivative bounds

Solution 050.1 ■ Answer D[Back to problem](#)

The mean value theorem gives $f(3) - 2 = 3f'(c)$ for some $c \in (0, 3)$. Multiplying the derivative bounds by 3 and adding 2 gives $5 \leq f(3) \leq 14$.

Solution 050.2 ■ Answer C[Back to problem](#)

The bound gives $|f(x)| \leq |f(0)| + |x|/3$. Hence $f(x) - x$ is positive for sufficiently negative x and negative for sufficiently positive x , giving a fixed point by continuity. If $f(a) = a$ and $f(b) = b$, the mean value bound gives $|a - b| \leq |a - b|/3$, so $a = b$.

Solution 050.3 ■ Answer E[Back to problem](#)

Derivative bounds give $2x \leq f(x) \leq 5x$ on $[0, 1]$. Integrating yields $1 \leq \int f \leq 5/2$. The linear functions $2x$ and $5x$ attain the endpoints.

Solution 050.4 ■ Answer E[Back to problem](#)

The zero-derivative conclusion applies separately to the intervals $(-\infty, 0)$ and $(0, \infty)$. The first constant is fixed at 3, but the second is arbitrary. The mean value theorem cannot be applied across zero, since the function is not defined on the intervening closed interval.

Solution 050.5 ■ Answer B[Back to problem](#)

Apply Rolle's theorem to $g(x) = f(x) - f(0) - [f(1) - f(0)]x^2$. Its endpoint values are both zero, so $g'(c) = f'(c) - 2c[f(1) - f(0)] = 0$. This is the mean value argument with a quadratic comparison function instead of a secant line.

Max/Min Problems ▸ Finding extrema

051 Optimization and constrained extrema

Solution 051.1 ■ Answer C

[Back to problem](#)

Substitute $y = (12 - x)/2$ on $0 \leq x \leq 12$. The objective is $6x^2 - x^3/2$, with derivative $12x - 3x^2/2 = (3x/2)(8 - x)$. The positive interior maximizer is $x = 8$, $y = 2$, giving 128; endpoints give zero.

Solution 051.2 ■ Answer A

[Back to problem](#)

On $(0, 1)$ logarithmic differentiation gives derivative sign equal to that of $a/x - b/(1 - x)$. It changes from positive to negative at $x = a/(a + b)$. Substitution yields the displayed value, and both endpoint values are zero.

Solution 051.3 ■ Answer A

[Back to problem](#)

In the first quadrant let $u = x/3$, $v = y/2$, so $u^2 + v^2 = 1$. The rectangle area is $24uv \leq 12$, with equality at $u = v = 1/\sqrt{2}$.

Solution 051.4 ■ Answer D

[Back to problem](#)

AM-GM gives $x + 4y \geq 2\sqrt{4xy} = 16$. Equality requires $x = 4y$, achieved by $(x, y) = (8, 2)$.

Solution 051.5 ■ Answer E

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With height h , the squared radius is $9 - h^2/4$. The volume is $\pi h(9 - h^2/4)$, whose derivative vanishes at $h^2 = 12$. Endpoint volumes are zero, so $h = 2\sqrt{3}$ maximizes it.

Solution 051.6 ■ Answer D

[Back to problem](#)

The interior critical points are -1 and 1 . Compare all four candidates: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, and $f(3/2) = -9/8$. The same global minimum occurs at an endpoint and an interior stationary point; a local classification alone misses the tie.

Related Rates ▸ Differentiating a geometric constraint

052 Related rates and time differentiation

Solution 052.1 ■ Answer D

[Back to problem](#)

With $r = h/2$, $V = \pi r^2 h/3 = \pi h^3/12$. Thus $dV/dt = (\pi h^2/4)dh/dt$. At $h = 3$, $3\pi = (9\pi/4)dh/dt$, so $dh/dt = 4/3$.

Solution 052.2 ■ Answer D

[Back to problem](#)

At that instant the height is 12. From $x^2 + y^2 = 169$, $y' = -xx'/y = -5/6$. Since $A = xy/2$, $A' = (2 \cdot 12 + 5(-5/6))/2 = 119/12$. The area can increase even while the top descends.

Solution 052.3 ■ **Answer D**[Back to problem](#)

For distance r , $r^2 = x^2 + x^4$. Differentiating gives $2rr' = (2x + 4x^3)x'$. At $x = 1$, $r = \sqrt{2}$, so $r' = 18/(2\sqrt{2})$.

Solution 052.4 ■ **Answer D**[Back to problem](#)

Since $V' = 4\pi r^2 r'$, the radial rate is $3/4$. Then $A' = 8\pi r r' = 8\pi \cdot 2 \cdot 3/4 = 12\pi$.

Solution 052.5 ■ **Answer D**[Back to problem](#)

If x is the spot distance, $x = 30 \tan \theta$, so $dx/dt = 30 \sec^2 \theta d\theta/dt$. At $x = 40$, $\tan \theta = 4/3$ and $\sec^2 \theta = 25/9$. Thus the speed is $30(25/9)(1/10) = 25/3$ meters per second. An angular rate must be converted using radians and the geometric derivative.

Solution 052.6 ■ **Answer A**[Back to problem](#)

Differentiate the constraint and divide by $PV = kT$: $P'/P + V'/V = T'/T$. Hence $V'/V = -0.01 - 0.03 = -0.04$ per second. Both pressure and temperature vary; neither may be treated as fixed.

Indefinite Integration (Antidifferentiation) ▷ Definitions and elementary formulas

053 Antiderivatives and indefinite integration

Solution 053.1 ■ **Answer C**[Back to problem](#)

$\log |x|$ differentiates to $1/x$ on each component. The zero-derivative theorem forces constancy on each interval separately, not one common constant across the missing point zero.

Solution 053.2 ■ **Answer B**[Back to problem](#)

Integration gives $F(x) = x^3 - 2 \log x + C$. The condition $F(1) = 4$ makes $C = 3$, and $F(e) = e^3 - 2 + 3 = e^3 + 1$.

Solution 053.3 ■ **Answer B**[Back to problem](#)

Differentiating the stated expression gives $(2f + 3) - 2f - 3 = 0$. A differentiable function with zero derivative on an interval is constant.

Solution 053.4 ■ **Answer B**[Back to problem](#)

For $a \neq 1$, integrate and subtract the value at zero to get $(a^x - 1)/\log a$. For $a = 1$, the integrand is the constant 1, so the normalized antiderivative is x . The apparent division by zero signals a separate parameter case, not nonexistence of an antiderivative.

Solution 053.5 ■ **Answer D**[Back to problem](#)

Any candidate must be constant on $(-1, 0)$ and equal to x plus a constant on $(0, 1)$. Differentiability requires continuity at zero, forcing the constants to match, but the resulting left and right derivatives are 0 and 1. The function is bounded and Riemann integrable, yet has no antiderivative across its jump.

Techniques of Integration ▸ Integration by Substitution

054 Integration by substitution

Solution 054.1 ■ **Answer B**[Back to problem](#)

Let $u = 1 + x^2$, so $x^3 dx = (u - 1)du/2$ and the limits become 1, 2. The integral is $\frac{1}{2} \int_1^2 (1 - 1/u) du = (1 - \log 2)/2$.

Solution 054.2 ■ **Answer B**[Back to problem](#)

Use $u = \sin^2 x$, $du = 2 \sin x \cos x dx$. The integral is $\frac{1}{2} \int_0^1 du/(1 + u^2) = \frac{1}{2} \arctan(1) = \pi/8$.

Solution 054.3 ■ **Answer A**[Back to problem](#)

The chain-rule antiderivative is $\sin(x^2)$, with equal endpoint values. Hence the integral is zero. Substitution sends both oriented endpoints to $u = 1$, correctly giving zero; injectivity is not required for this reverse-chain-rule identity. Oddness independently verifies the result.

Solution 054.4 ■ **Answer A**[Back to problem](#)

Set $u = 1 + 3x^2$, so $du = 6x dx$ and the endpoints become 1 and 4. The desired integral is one sixth of the given integral, hence 1. No formula for f or its antiderivative is needed.

Solution 054.5 ■ **Answer C**[Back to problem](#)

Absolute convergence follows from comparison with $|\log x|$ near zero and $(\log x)/x^2$ near infinity. Under $x = 1/u$, the measure factor becomes $-du/(1 + u^2)$ and $\log x = -\log u$. Reversing the transformed endpoints shows the integral equals its negative, so it is zero.

Techniques of Integration ▸ Integration by Parts

055 Integration by parts

Solution 055.1 ■ Answer B

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Integrate by parts with $u = (\log x)^2$ and $dv = x dx$. Since $x^2(\log x)^2 \rightarrow 0$ at zero, the result is $-\int_0^1 x \log x dx$. A second integration by parts gives $\int_0^1 x \log x dx = -1/4$, so the answer is $1/4$.

Solution 055.2 ■ Answer B

[Back to problem](#)

Two integrations by parts, or differentiation of a candidate, give an antiderivative $e^x(\sin x - \cos x)/2$. At π this is $e^\pi/2$, and at zero it is $-1/2$, yielding $(e^\pi + 1)/2$.

Solution 055.3 ■ Answer E

[Back to problem](#)

Integrate by parts with $u = x$ and $dv = \sin x dx$. An antiderivative is $-x \cos(x) + \sin(x)$. Evaluating at the endpoints gives π .

Solution 055.4 ■ Answer B

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Use $u = x^n$ and $dv = e^x dx$. The boundary term is $[x^n e^x]_0^1 = e$ for $n \geq 1$, leaving $I_n = e - n \int_0^1 x^{n-1} e^x dx = e - nI_{n-1}$. Together with $I_0 = e - 1$, this determines the entire family.

Solution 055.5 ■ Answer C

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Take $u = \arctan x$ and $dv = dx$. The boundary contribution is $\pi/4$, and the remaining integral is $\int_0^1 x/(1+x^2) dx = \frac{1}{2} \log 2$. Subtracting gives the result. The inverse trigonometric factor becomes simpler when differentiated.

Techniques of Integration ▸ Trig Substitutions

056 Trigonometric substitutions and radical integrals

Solution 056.1 ■ Answer C

[Back to problem](#)

Let $x = a \sin \theta$ with $0 \leq \theta < \pi/2$. The transformed integrand is $a^2 \sin^2 \theta$, and its integral to $\pi/2$ is $a^2 \pi/4$. The singular upper endpoint is handled as an improper limit.

Solution 056.2 ■ Answer B

[Back to problem](#)

With $x = \tan \theta$, $dx = \sec^2 \theta d\theta$ and the denominator is $\sec^3 \theta$ on $[0, \pi/3]$. The integral becomes $\int_0^{\pi/3} \cos \theta d\theta = \sqrt{3}/2$.

Solution 056.3 ■ Answer A[Back to problem](#)

Use $x = 2 \sec \theta$ with $0 \leq \theta \leq \pi/3$. The integrand becomes $2 \tan^2 \theta d\theta = 2(\sec^2 \theta - 1)d\theta$. Thus the value is $[2 \tan \theta - 2\theta]_0^{\pi/3} = 2\sqrt{3} - 2\pi/3$.

Solution 056.4 ■ Answer E[Back to problem](#)

Use $x = 2 \sinh t$ (equivalently a trigonometric substitution), then integrate the resulting square. An antiderivative is $\frac{x\sqrt{x^2+4}}{2} - 2 \operatorname{asinh}\left(\frac{x}{2}\right)$. Evaluating at the endpoints gives $-2 \operatorname{asinh}\left(\frac{\sqrt{5}}{2}\right) + \frac{3\sqrt{5}}{2}$.

Solution 056.5 ■ Answer B[Back to problem](#)

Choose $x = -2 \sec \theta$, with θ decreasing from $\pi/3$ to $\pi/4$. On this range $\sqrt{x^2 - 4} = 2 \tan \theta$ and $dx = -2 \sec \theta \tan \theta d\theta$, so the transformed integrand is $d\theta/2$. The answer is $(\pi/4 - \pi/3)/2 = -\pi/24$, consistent with a negative integrand over increasing x .

Techniques of Integration ▸ The Method of Partial Fractions

057 Partial fractions and rational integration

Solution 057.1 ■ Answer C[Back to problem](#)

Clearing denominators gives $1 = A(x - 1)^2 + Bx(x - 1) + Cx$. Setting $x = 0, 1$ gives $A = 1, C = 1$; comparison of x^2 gives $B = -1$. Their sum is 1.

Solution 057.2 ■ Answer B[Back to problem](#)

Decompose $1/[x(x^2 + 1)] = 1/x - x/(x^2 + 1)$. The antiderivative is $\log x - \frac{1}{2} \log(x^2 + 1)$. Evaluation gives $\log 2 - \frac{1}{2} \log 5 + \frac{1}{2} \log 2 = \frac{1}{2} \log(8/5)$.

Solution 057.3 ■ Answer E[Back to problem](#)

Split the integrand as $1/(x + 1) - 1/(x + 2)$. An antiderivative is $\log\left(\frac{x+1}{x+2}\right)$. Evaluating at the endpoints gives $\log\left(\frac{4}{3}\right)$.

Solution 057.4 ■ Answer D[Back to problem](#)

The linear factor contributes $A_1/(x - 1) + A_2/(x - 1)^2 + A_3/(x - 1)^3$, requiring three slots. The irreducible quadratic contributes $(B_1x + C_1)/(x^2 + x + 1) + (B_2x + C_2)/(x^2 + x + 1)^2$, requiring four more. The rational function is proper, so there is no polynomial quotient: seven slots total.

Solution 057.5 ■ Answer E[Back to problem](#)

Partial fractions give $\frac{1}{2}(1/x - 1/(x + 2))$. The integral from the left to zero diverges to $-\infty$, while the one from zero to the right diverges to $+\infty$. They cannot be cancelled in an ordinary improper integral. The finite value $-\frac{1}{2}\log 3$ belongs only to the symmetric principal value.

Solution 057.6 ■ Answer C[Back to problem](#)

Polynomial division gives $(x + 1) + 1/(x^2 + 1)$. Integrating the polynomial part gives $3/2$, and the irreducible quadratic part gives $\arctan 1 = \pi/4$. Starting with a proper-fraction template before division would omit the polynomial contribution.

Definite Integration ▸ Area as a limit of sums

058 Riemann sums and definite integrals

Solution 058.1 ■ Answer C[Back to problem](#)

Each summand is $(1/n)/(1 + (k/n)^2)$. This is a right-endpoint Riemann sum for $\int_0^1 dx/(1 + x^2) = \arctan 1 = \pi/4$.

Solution 058.2 ■ Answer C[Back to problem](#)

Take partition points $x_k = k^2/n^2$. Then $\Delta x_k = (2k - 1)/n^2$ and the sine is evaluated at the right endpoint. The mesh is $(2n - 1)/n^2 \rightarrow 0$, so the limit is $\int_0^1 \sin x \, dx = 1 - \cos 1$.

Solution 058.3 ■ Answer D[Back to problem](#)

Both choices of tags are valid and both partition meshes tend to zero. Riemann integrability requires a common limit for every such choice. The two distinct limits disprove integrability; boundedness alone is insufficient.

Solution 058.4 ■ Answer C[Back to problem](#)

The small intervals cover $[0, 1/2]$ and their sums approach $\int_0^{1/2} x^2 dx = 1/24$. The final interval never shrinks and contributes $(1/2)(1/2)^2 = 1/8$. The limit is $1/6$, not the full integral $1/3$, because the mesh remains $1/2$ despite the growing number of intervals.

Solution 058.5 ■ Answer D[Back to problem](#)

The sum-of-squares identity gives $S_n = n(n + 1)(2n + 1)/(6n^3) = 1/3 + 1/(2n) + 1/(6n^2)$. Subtracting the integral and multiplying by n yields $1/2 + 1/(6n) \rightarrow 1/2$. This asks for the discretization error, not merely the integral limit.

The Fundamental Theorem of Calculus ▸ Integration and differentiation

059 Fundamental theorem of calculus and variable limits

Solution 059.1 ■ Answer B

[Back to problem](#)

The endpoint formula gives $F'(x) = e^x \sin(e^{2x}) - 2x \sin(x^4)$. At zero the lower-endpoint term is zero, leaving $\sin 1$.

Solution 059.2 ■ Answer E

[Back to problem](#)

First, $F'(x) = 2xe^{x^4}$. A second differentiation gives $F''(x) = 2e^{x^4} + 8x^4e^{x^4}$. At 1 this is $10e$.

Solution 059.3 ■ Answer D

[Back to problem](#)

For positive x , $A(x) = x$. For negative x , reversing the limits gives $A(x) = -x$. Thus $A(x) = |x|$, continuous with unequal one-sided derivatives. The pointwise derivative conclusion of the fundamental theorem needs continuity of the integrand at the point.

Solution 059.4 ■ Answer C

[Back to problem](#)

Differentiate both the moving endpoint and the explicit x in the integrand. The endpoint contribution is $(x-x)e^{x^2} = 0$ and the remaining derivative is $F'(x) = \int_0^x e^{t^2} dt$. Differentiating once more gives $F''(x) = e^{x^2}$, hence e .

Solution 059.5 ■ Answer E

[Back to problem](#)

Differentiate for $0 < x < 1$ to get $f(x) = f(x) + xf'(x) - 2x$. Hence $f'(x) = 2$ on the open interval. The constant-derivative theorem and continuity at zero give $f(x) = 2x + 3$, so $f(1) = 5$.

The Fundamental Theorem of Calculus ▸ Properties of definite integrals

060 Integral properties, signed area, and piecewise integration

Solution 060.1 ■ Answer D

[Back to problem](#)

On $[-1, 1]$, the graph is below the axis, giving $\int_{-1}^1 (1-x^2)dx = 4/3$. On $[1, 2]$, it is above, giving $\int_1^2 (x^2-1)dx = 4/3$. Add the positive areas to get $8/3$.

Solution 060.2 ■ Answer B

[Back to problem](#)

The values are -2 on $[-3/2, -1)$, -1 on $[-1, 0)$, 0 on $[0, 1)$, and 1 on $[1, 3/2]$, ignoring endpoints. Their contributions are $-1, -1, 0, 1/2$, totaling $-3/2$.

Solution 060.3 ■ Answer A[Back to problem](#)

I holds: a nonzero value of f gives, by continuity, an interval where f^2 is bounded below by a positive constant, making its integral positive. For II use $f(x) = x - 1/2$. For III use a function equal to 1 at one point and zero elsewhere; changing one value does not change the integral.

Solution 060.4 ■ Answer B[Back to problem](#)

The odd part integrates to zero on the symmetric interval. The remaining integral is $2 \int_0^3 x^2 dx = 18$; the given value 5 is unnecessary.

Solution 060.5 ■ Answer D[Back to problem](#)

The curves cross at 0 and 1. Use x on $[0, 1]$ and x^2 on $[1, 2]$, giving $1/2 + 7/3 = 17/6$.

Solution 060.6 ■ Answer C[Back to problem](#)

Add adjacent intervals to get $\int_0^5 f = 2$. Linearity gives $\int_0^5 (2f - 1) = 4 - 5 = -1$. Reversing the endpoints changes its sign, so the requested value is 1.

The Average Value of a Function ▸ Integral averages

061 Average value and the integral mean value theorem

Solution 061.1 ■ Answer C[Back to problem](#)

$\int_1^e \log x \, dx = [x \log x - x]_1^e = 1$, so the average is $1/(e - 1)$. Solving $\log c = 1/(e - 1)$ gives the answer; strict increase makes it unique.

Solution 061.2 ■ Answer B[Back to problem](#)

The total integral of f is $2 \cdot 3 = 6$, so $\int_1^2 f(t) dt = 4$. Substituting $t = x^2 + 1$ gives $\int_0^1 x f(x^2 + 1) dx = \frac{1}{2} \int_1^2 f(t) dt = 2$. Since the interval has length one, this is also the average.

Solution 061.3 ■ Answer A[Back to problem](#)

The integrals over the two pieces are $2 \cdot 3 = 6$ and $3 \cdot 8 = 24$. Divide their sum 30 by the total length 5 to get 6.

Solution 061.4 ■ Answer D[Back to problem](#)

The step function has integral and average $1/2$, but its only values are 0 and 1. It is Riemann integrable, so an average exists; its jump prevents the continuity hypothesis from guaranteeing attainment. Each other function is continuous on the closed interval and therefore attains its average.

Solution 061.5 ■ **Answer C**[Back to problem](#)

The nonnegative weight $2x$ has integral 1, so the weighted average lies between the minimum and maximum of f . Continuity supplies an intermediate point attaining it. Constant functions disprove the other choices.

Solution 061.6 ■ **Answer B**[Back to problem](#)

Average velocity is $[s(2) - s(0)]/2 = (2 - 0)/2 = 1$. Velocity $s'(t) = 3t^2 - 3$ changes sign at $t = 1$, where $s(1) = -2$. Total distance is $|-2 - 0| + |2 - (-2)| = 6$, so average speed is $6/2 = 3$. The averaging variable is time, and speed uses absolute velocity.

Finding the Area Between Two Curves ▸ Choosing the integration variable

062 Area between curves

Solution 062.1 ■ **Answer B**[Back to problem](#)

The intersections are $-1, 0, 1$. On $(0, 1)$, $x > x^3$, giving area $\int_0^1 (x - x^3)dx = 1/4$. Symmetry gives an identical region on $(-1, 0)$, so the total is $1/2$.

Solution 062.2 ■ **Answer D**[Back to problem](#)

Horizontal slices run from $x = y/4$ to $x = \sqrt{y}$ for $0 \leq y \leq 4$. Hence the area is $\int_0^4 (\sqrt{y} - y/4)dy = 16/3 - 2 = 10/3$. This avoids the vertical split at $x = 1$.

Solution 062.3 ■ **Answer D**[Back to problem](#)

The graph partitions the 2 by 4 rectangle into two regions. Vertical slices give area 3 below the graph; horizontal slices give the complementary area $\int_0^4 f^{-1}(y)dy$. Their sum is 8, so the desired area is 5. No explicit inverse is needed.

Solution 062.4 ■ **Answer A**[Back to problem](#)

The square-root curve is above x^2 in this interval. Thus the area is $\int_0^1 (x^{1/2} - x^2)dx = 2/3 - 1/3 = 1/3$.

Solution 062.5 ■ **Answer D**[Back to problem](#)

The intersection points are $\pm\sqrt{a}$. The area is $\int_{-\sqrt{a}}^{\sqrt{a}} (a - x^2)dx = 4a^{3/2}/3$. Equating this to $32/3$ gives $a^{3/2} = 8$, hence $a = 4$.

Polar Coordinates ▸ Coordinates and curves

063 Polar coordinates and standard polar curves

Solution 063.1 ■ Answer A

[Back to problem](#)

Multiply by r to get $x^2 + y^2 = 4x - 6y$. Completing squares gives $(x-2)^2 + (y+3)^2 = 4 + 9 = 13$.

Solution 063.2 ■ Answer C

[Back to problem](#)

For $r = -2$ and $\theta = -\pi/6$, $x = -2(\sqrt{3}/2) = -\sqrt{3}$ and $y = -2(-1/2) = 1$. Negative radius reverses the ray; the positive-radius representation is $(2, 5\pi/6)$.

Solution 063.3 ■ Answer B

[Back to problem](#)

The radius is nonnegative and ranges from 0 to 4, attaining its maximum when $\cos \theta = -1$, or $\theta = \pi$ modulo 2π . This gives the point $(-4, 0)$. The cusp is at the origin when $\theta = 0$; the minus sign orients the large lobe leftward.

Solution 063.4 ■ Answer E

[Back to problem](#)

The polar-to-Cartesian identity $x = r \cos \theta$ gives exactly $x = 2$, regardless of the polar coordinates used to represent its points.

Solution 063.5 ■ Answer D

[Back to problem](#)

Since $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$, multiplication by r^2 gives $x^2 - y^2 = 4$, a hyperbola.

Solution 063.6 ■ Answer C

[Back to problem](#)

Cartesian parametrization gives $x = 2 \cos^2 \theta = 1 + \cos 2\theta$ and $y = 2 \sin \theta \cos \theta = \sin 2\theta$. The angle 2θ sweeps 4π , tracing the circle twice. A negative radius does not remove the second traversal; it represents the same points from opposite rays.

Polar Coordinates ▸ Area in polar coordinates

064 Polar area and overlap regions

Solution 064.1 ■ Answer C

[Back to problem](#)

The cardioid is traced once over $[0, 2\pi]$. Its area is $\frac{9}{2} \int_0^{2\pi} (1 + 2 \cos \theta + \cos^2 \theta) d\theta = \frac{9}{2} (2\pi + \pi) = 27\pi/2$.

Solution 064.2 ■ Answer A

[Back to problem](#)

The overlap is in the first quadrant. Its radius is $2 \sin \theta$ on $[0, \pi/4]$ and $2 \cos \theta$ on $[\pi/4, \pi/2]$. Symmetry gives area $4 \int_0^{\pi/4} \sin^2 \theta d\theta = 4(\pi/8 - 1/4) = \pi/2 - 1$.

Solution 064.3 ■ Answer A[Back to problem](#)

A single petal is traced for $0 \leq \theta \leq \pi/3$. Its area is $\frac{1}{2} \int_0^{\pi/3} 4 \sin^2(3\theta) d\theta = \pi/3$.

Solution 064.4 ■ Answer C[Back to problem](#)

One loop lies on $[-\pi/4, \pi/4]$ and has area $\frac{1}{2} \int_{-\pi/4}^{\pi/4} 4 \cos 2\theta d\theta = 2$. The second congruent loop gives total area 4.

Solution 064.5 ■ Answer D[Back to problem](#)

The outer radius exceeds 1 for $|\theta| < \pi/3$. The area is $\frac{1}{2} \int_{-\pi/3}^{\pi/3} (4 \cos^2 \theta - 1) d\theta = \pi/3 + \sqrt{3}/2$.

Solution 064.6 ■ Answer C[Back to problem](#)

For a nonzero common point the radii are positive and its angle is unique modulo 2π . Equating radii gives $\cos \theta = 0$, producing $(0, 1)$ and $(0, -1)$. Both curves also pass through the origin, at different angles (π and 0). Thus there are three intersections; same-angle equality alone misses the origin.

Volumes of Solids of Revolution ▷ Disks and washers

065 Volumes of revolution: disks and washers

Solution 065.1 ■ Answer D[Back to problem](#)

The outer radius is $2 - x^2$ and inner radius $2 - x$. Thus $V = \pi \int_0^1 [(2 - x^2)^2 - (2 - x)^2] dx = \pi \int_0^1 (4x - 5x^2 + x^4) dx = \pi(2 - 5/3 + 1/5) = 8\pi/15$.

Solution 065.2 ■ Answer B[Back to problem](#)

The curves meet at $x = \pm 1, y = 1$. Horizontal disks have radius $\sqrt{y}$ for $0 \leq y \leq 1$ and $\sqrt{2 - y}$ for $1 \leq y \leq 2$. Thus $V = \pi \int_0^1 y dy + \pi \int_1^2 (2 - y) dy = \pi$. Counting both left and right halves separately would double-count the same disks.

Solution 065.3 ■ Answer B[Back to problem](#)

Each vertical segment crosses the axis and generates one disk, not two additive pieces or a washer with a hole. Its radius is $\max\{1, x\}$. Split at $x = 1$: $V = \pi \int_0^1 1 dx + \pi \int_1^2 x^2 dx = \pi(1 + 7/3) = 10\pi/3$.

Solution 065.4 ■ Answer B[Back to problem](#)

Horizontal washers have outer radius 4 and inner radius y^2 , for $0 \leq y \leq 2$. Thus $V = \pi \int_0^2 (16 - y^4) dy = 128\pi/5$.

Solution 065.5 ■ **Answer B**[Back to problem](#)

For an axis $y = c > 1$, washers give $V(c) = \pi \int_a^b [(c - g)^2 - (c - f)^2] dx$. Subtracting the formulas for $c = 3$ and $c = 2$ cancels the quadratic boundary terms and gives $V(3) - V(2) = 2\pi \int_a^b (f - g) dx$. Thus $8\pi = 2\pi A$ and $A = 4$. The question recovers a planar area from two revolutions rather than evaluating another given-radius integral.

Arc Length ▸ Length by integration

066 Arc length

Solution 066.1 ■ **Answer B**[Back to problem](#)

$y' = (3/2)\sqrt{x}$, so $L = \int_0^{4/3} \sqrt{1 + 9x/4} dx = (8/27)[(1 + 9x/4)^{3/2}]_0^{4/3} = (8/27)(8 - 1) = 56/27$.

Solution 066.2 ■ **Answer B**[Back to problem](#)

Use $x = 8 \cos^3 t$, $y = 8 \sin^3 t$ for $0 \leq t \leq \pi/2$. The speed is $24 \sin t \cos t$, so the length is $24 \int_0^{\pi/2} \sin t \cos t dt = 12$. The parametrization avoids infinite graph slopes at an endpoint.

Solution 066.3 ■ **Answer C**[Back to problem](#)

Since $y' = -\tan x$, the speed is $\sqrt{1 + \tan^2 x} = \sec x$ on this interval. Integrate to get $[\log(\sec x + \tan x)]_0^{\pi/3}$.

Solution 066.4 ■ **Answer D**[Back to problem](#)

The squared speed is $9 \sin^2 t + 9 \cos^2 t + 16 = 25$. The speed is therefore constantly 5, giving length $5(2\pi) = 10\pi$.

Solution 066.5 ■ **Answer C**[Back to problem](#)

The speed is $|2t|$, so the parametrized length is $\int_{-1}^1 2|t| dt = 2$. The geometric image is the unit segment from $(0, 0)$ to $(1, 0)$, of length 1. The path traverses it once in each direction; zero net displacement does not mean zero distance traveled.

Solution 066.6 ■ **Answer D**[Back to problem](#)

For the fourth graph, choose $x_k = (\pi/2 + k\pi)^{-1/2}$. Consecutive values have opposite signs and magnitudes x_k, x_{k+1} . Polygonal lengths dominate the sums of vertical changes $\sum (x_k + x_{k+1})$, which diverge like $\sum k^{-1/2}$. Thus the continuous graph is not rectifiable. The first, second, and fifth graphs are monotone and have bounded total variation; the third has only one interior turning point, so all four have finite length.

The Natural Exponential and Logarithm Functions ▸ Definitions and calculus

067 The natural exponential, logarithm, and the number e

Solution 067.1 ■ Answer C

[Back to problem](#)

The base is positive eventually. Its logarithm after exponentiation is $bn \log(1 + a/n) \rightarrow ab$, using $\log(1 + t)/t \rightarrow 1$; the case $a = 0$ is immediate. Exponentiating gives e^{ab} .

Solution 067.2 ■ Answer D

[Back to problem](#)

The first integral gives $A = \log a$. The requested integral is $\log(a^3) - \log(a^{-2}) = 3\log a + 2\log a = 5A$.

Solution 067.3 ■ Answer A

[Back to problem](#)

The rational-coefficient polynomial $q(x) = p(x) - p(1)$ vanishes at the transcendental number e . If it were nonzero, multiplying by a common denominator would give a nonzero integer-coefficient polynomial with root e , a contradiction. Thus q is identically zero and p is constant.

Solution 067.4 ■ Answer B

[Back to problem](#)

Subtract the two derivative quotients $(e^{5x} - 1)/x$ and $(e^{2x} - 1)/x$. Their limits are 5 and 2, giving 3.

Solution 067.5 ■ Answer C

[Back to problem](#)

For $x > 1$, dividing by $x - 1$ and letting $x \downarrow 1$ forces $c \geq 1$. For $0 < x < 1$, division reverses the inequality and $x \uparrow 1$ forces $c \leq 1$. The value $c = 1$ works because $x - 1 - \log x$ has derivative $1 - 1/x$, decreases to zero at 1, and then increases.

The Natural Exponential and Logarithm Functions ▸ Variable exponents

068 Logarithmic differentiation and variable powers

Solution 068.1 ■ Answer B

[Back to problem](#)

The derivative is $x^{\sin x}[\cos x \log x + \sin x/x]$. At $\pi/2$, the first bracket term is zero and the second is $2/\pi$, while the outside factor is $\pi/2$, yielding 1.

Solution 068.2 ■ Answer C

[Back to problem](#)

Logarithmic differentiation gives $f'/f = 2x \log(x + 1) + x^2/(x + 1)$. At 1, $f = 2$, so $f'(1) = 2(2 \log 2 + 1/2) = 1 + 4 \log 2$.

Solution 068.3 ■ **Answer D**[Back to problem](#)

Logarithmic differentiation gives $f'/f = 3/x + 2/(x-1) - 4/(x+1)$. At 2, this is $3/2 + 2 - 4/3 = 13/6$. The requested logarithmic derivative avoids expanding the product or computing $f(2)$.

Solution 068.4 ■ **Answer D**[Back to problem](#)

The logarithm is $(\log x)/x$, with derivative $(1 - \log x)/x^2$. It increases for $x < e$ and decreases for $x > e$. Since exponentiation preserves order, the original function has its unique global maximum at e , of value $e^{1/e}$.

Solution 068.5 ■ **Answer C**[Back to problem](#)

Taking logarithms gives $x \log y = y \log x$. Differentiate to obtain $\log y + (x/y)y' = y' \log x + y/x$. Thus $y' = (y/x - \log y)/(x/y - \log x)$. At $(2, 4)$ this is the stated ratio; the denominator $1/2 - \log 2$ is nonzero, so the local implicit branch is regular.

L'Hopital's Rule ▸ Limits by differentiation

069 L'Hopital's rule and indeterminate forms

Solution 069.1 ■ **Answer C**[Back to problem](#)

The quotient is $0/0$. Applying the rule once gives $(e^x - 1)/\sin x$, again $0/0$. Applying it again gives $e^x/\cos x \rightarrow 1$. All denominator derivatives are nonzero on sufficiently small punctured intervals when required.

Solution 069.2 ■ **Answer D**[Back to problem](#)

Take logarithms: $\cot x \log(1 + 3x) = \log(1 + 3x)/\tan x$. This is $0/0$, and the derivative quotient tends to $[3/(1 + 3x)]/\sec^2 x \rightarrow 3$. Exponentiating yields e^3 .

Solution 069.3 ■ **Answer B**[Back to problem](#)

The identity $F(x) = 1 + \sin x/x$ and a squeeze give limit 1. The quotient of derivatives is $1 + \cos x$, which has no limit. Existence of the derivative-quotient limit is a sufficient condition in the rule, not a necessary condition for the original quotient to converge.

Solution 069.4 ■ **Answer C**[Back to problem](#)

The logarithm is $x \log x = \log x/(1/x)$. L'Hopital gives a limit of zero for the logarithm, hence the original limit is $e^0 = 1$.

Solution 069.5 ■ **Answer B**[Back to problem](#)

Combine the difference into $[\log(1+x) - x]/[x \log(1+x)]$. The numerator is $-x^2/2 + o(x^2)$ and the denominator is $x^2 + o(x^2)$, so the limit is $-1/2$. Equivalently, two justified differentiations of the combined $0/0$ quotient give numerator limit -1 and denominator limit 2 .

Solution 069.6 ■ **Answer B**[Back to problem](#)

This is an infinity-to-zero indeterminate form. Its logarithm is $(\log x)/\sqrt{x}$, an infinity-over-infinity quotient. The derivative quotient is $(1/x)/(1/(2\sqrt{x})) = 2/\sqrt{x} \rightarrow 0$. Exponentiating gives the limit $e^0 = 1$.

Improper Integrals ▸ Definition and evaluation

070 Improper integrals and independent limiting operations

Solution 070.1 ■ **Answer C**[Back to problem](#)

The integrals on $(-1, 0)$ and $(0, 1)$ diverge separately to opposite infinities. They cannot be added as an ordinary improper integral. However, $\int_{-1}^{-\varepsilon} dx/x + \int_{\varepsilon}^1 dx/x = 0$ for every $\varepsilon > 0$, giving principal value zero.

Solution 070.2 ■ **Answer C**[Back to problem](#)

Split at any positive point to handle the singular lower endpoint and infinite upper endpoint separately. With $u = \sqrt{x}$, $dx = 2u du$, the transformed integrand is $2e^{-u}$. Both limits converge, and the value is $2 \int_0^{\infty} e^{-u} du = 2$.

Solution 070.3 ■ **Answer B**[Back to problem](#)

Integration by parts gives $\int_1^R \sin x/x dx = [-\cos x/x]_1^R - \int_1^R \cos x/x^2 dx$. The boundary term has a limit and the remaining integral converges absolutely by comparison with x^{-2} . For the absolute integral, each interval $[k\pi + \pi/6, k\pi + 5\pi/6]$ has $|\sin x| \geq 1/2$ and contributes at least $1/[3(k+1)]$; these contributions diverge harmonically.

Solution 070.4 ■ **Answer A**[Back to problem](#)

Integrate the difference $[1/(x+a) - 1/(x+b)]/(b-a)$ over $[0, R]$. The upper boundary is $\log[(R+a)/(R+b)]/(b-a) \rightarrow 0$; the lower boundary is $\log(a/b)/(b-a)$. Their difference gives the answer. The two partial fractions must be combined before taking the infinite limit; their separate divergent integrals cannot be subtracted.

Solution 070.5 ■ **Answer A**[Back to problem](#)

Use the antiderivative $x \log x - x$. Since $x \log x \rightarrow 0$ as $x \rightarrow 0^+$, the lower endpoint contributes zero and the upper endpoint contributes -1 .

Improper Integrals ▸ Convergence criteria

071 Improper p-integrals and comparison near singularities

Solution 071.1 ■ Answer C

[Back to problem](#)

Near zero the integrand behaves like x^{-p} , requiring $p < 1$. At infinity it behaves like x^{-p-1} , requiring $p + 1 > 1$, or $p > 0$. Both conditions are necessary.

Solution 071.2 ■ Answer D

[Back to problem](#)

Let $u = \log \log x$, so $du = dx/(x \log x)$ and the lower limit is 1. The integral becomes $\int_1^\infty u^{-q} du$, which converges exactly when $q > 1$.

Solution 071.3 ■ Answer D

[Back to problem](#)

With $x = e^{-t}$ the integral becomes $\int_0^\infty t e^{-(1-p)t} dt$. It converges exactly when $1 - p > 0$; at $p = 1$ even the linear integrand diverges.

Solution 071.4 ■ Answer B

[Back to problem](#)

The numerator lies between 1 and 3. The comparison integral is $\int_e^\infty dx/[x(\log x)^2] = 1$, using $u = \log x$. Integrating the pointwise bounds gives $1 \leq I \leq 3$ and proves convergence despite the oscillation. Symmetry does not justify assigning the exact value 2.

Solution 071.5 ■ Answer D

[Back to problem](#)

Being smaller than a divergent comparison function gives no conclusion. With $g(x) = x^{-2}$, the choice $f(x) = 1$ has a convergent integral and ratio $x^2 \rightarrow 0$, while $f(x) = x^{-1}$ has a divergent integral and ratio $x \rightarrow 0$. The positive finite limit required for two-way limit comparison cannot be replaced by zero.

Infinite Series ▸ Definitions and exact sums

072 Infinite series, partial sums, geometric and telescoping sums

Solution 072.1 ■ Answer C

[Back to problem](#)

Decompose $1/[n(n+2)] = \frac{1}{2}(1/n - 1/(n+2))$. The partial sum is $\frac{1}{2}(1 + 1/2 - 1/(N+1) - 1/(N+2))$, whose limit is $3/4$.

Solution 072.2 ■ Answer C

[Back to problem](#)

The first term is $1/3$ and common ratio is $-2/3$. Since its magnitude is less than one, the sum is $(1/3)/(1 + 2/3) = 1/5$.

Solution 072.3 ■ **Answer C**[Back to problem](#)

The finite numerator is $2^{n+1} - 1$. Thus the series is $\sum_{n \geq 0} (2/3)^{n+1} - \sum_{n \geq 0} (1/3)^{n+1} = 2 - 1/2 = 3/2$. Both geometric series converge, so their subtraction is legitimate.

Solution 072.4 ■ **Answer D**[Back to problem](#)

Differentiating the geometric series gives $\sum nr^n = r/(1-r)^2$ for $|r| < 1$. At $r = 1/3$, the result is $(1/3)/(2/3)^2 = 3/4$.

Solution 072.5 ■ **Answer B**[Back to problem](#)

The first term is $a_1 = s_1 = 4/3$, while the series sum is $\lim_{N \rightarrow \infty} s_N = 3$. The formula is given only for $N \geq 1$; substituting $N = 0$ and treating the resulting $1/2$ as an empty partial sum would be invalid, because the empty sum is zero.

Infinite Series ▸ Basic convergence tests

073 The term test, harmonic series, and p-series

Solution 073.1 ■ **Answer D**[Back to problem](#)

For $p > 0$, the term is asymptotic to n^{1-p} , so the benchmark exponent is $p - 1$, which must exceed 1. For $p \leq 0$, the terms do not even tend to zero. At $p = 2$ the series is harmonic in size and diverges.

Solution 073.2 ■ **Answer D**[Back to problem](#)

Rationalization yields $a_n = 1/(\sqrt{n^2 + 1} + n) \rightarrow 0$. Also $na_n \rightarrow 1/2$, so the series behaves like half the harmonic series and diverges. The term test alone is inconclusive here.

Solution 073.3 ■ **Answer E**[Back to problem](#)

The displayed terms tend to 1. The terms of all four other series tend to zero, even though the harmonic series still diverges for a different reason.

Solution 073.4 ■ **Answer C**[Back to problem](#)

For each integer $k \geq 1$, the denominator base equals k at the $2k+1$ integers $k^2 \leq n < (k+1)^2$. Grouping these nonnegative terms gives $\sum_{k \geq 1} (2k+1)/k^p$, whose leading behavior is $2k^{1-p}$. The p-series criterion requires $p - 1 > 1$, or $p > 2$.

Solution 073.5 ■ **Answer C**[Back to problem](#)

Terms of a convergent series tend to zero. Sparse spikes can invalidate the assertion about na_n , and $a_n = 1/n^2$ already makes the square-root series divergent.

Infinite Series ▸ The comparison test

074 Direct and limit comparison for series

Solution 074.1 ■ **Answer D**[Back to problem](#)

I is bounded termwise by $1/n^2$. In II, the ratio to $1/n$ tends to 1, so it diverges. For III, rationalization gives a term at most $1/(4n^2)$, hence convergence.

Solution 074.2 ■ **Answer D**[Back to problem](#)

Eventually $a_n \leq 1$, so $a_n^2 \leq a_n$; the other non-square-root expressions are also at most a_n . But with $a_n = 1/n^2$, the square-root series is harmonic and diverges.

Solution 074.3 ■ **Answer C**[Back to problem](#)

For $t > 0$, $t - \log(1+t) = \int_0^t u/(1+u)du$, which lies strictly between 0 and $t^2/2$. Therefore the series is positive and bounded termwise by the convergent series $\sum 1/(2n^2)$. One cannot evaluate it by separately subtracting two divergent series; comparison applies to their combined positive terms.

Solution 074.4 ■ **Answer D**[Back to problem](#)

Partial sums of the larger nonnegative series dominate those of the divergent smaller one, so they are unbounded. No termwise growth or ratio conclusion follows.

Solution 074.5 ■ **Answer D**[Back to problem](#)

Since $\log(1+t)/t \rightarrow 1$ as $t \rightarrow 0$, the ratio to $1/n^2$ tends to 1. The positive p -series with exponent 2 converges.

Infinite Series ▸ The ratio test

075 Ratio test

Solution 075.1 ■ **Answer C**[Back to problem](#)

The adjacent ratio is $\frac{1}{a}(n/(n+1))^n \rightarrow 1/(ae)$. The ratio test proves convergence when this is below one, namely $a > 1/e$. This question asks about the test's strict convergence conclusion, so it does not classify the boundary.

Solution 075.2 ■ **Answer A**[Back to problem](#)

The ratio is $2(n+1)^3/[n^3(n+1)] = 2(n+1)^2/n^3 \rightarrow 0$. Factorial growth dominates the polynomial and fixed exponential, giving convergence.

Solution 075.3 ■ Answer C[Back to problem](#)

The exact ratio is $a_{n+1}/a_n = e/(1 + 1/n)^n > 1$. Thus every a_n is at least $a_1 = e$, and the terms do not tend to zero. The limiting ratio 1 is inconclusive; the additional inequality settles the boundary through a different test.

Solution 075.4 ■ Answer A[Back to problem](#)

The absolute ratio tends to $|x|/4$. The ratio test proves convergence when this is strictly below 1 and is inconclusive at the two endpoints.

Solution 075.5 ■ Answer C[Back to problem](#)

Ratios alternate between $1/8$ and 2 , so their limit does not exist. Pairing adjacent positive terms gives $\sum_{n \geq 1} 9 \cdot 4^{-n} = 9(1/4)/(1 - 1/4) = 3$. Infinitely many ratios above one do not replace the ratio test's requirement of a limiting ratio above one.

Infinite Series ▸ The root test

076 Root test and limsup formulation

Solution 076.1 ■ Answer B[Back to problem](#)

The roots are $1/2$ at even indices and $1/3$ at odd indices, so their limsup is $1/2 < 1$. The series converges by the limsup root test, or directly from $0 \leq a_n \leq 2^{-n}$.

Solution 076.2 ■ Answer C[Back to problem](#)

The n th root of the term is $(1 + 1/n)^n/c \rightarrow e/c$. Strictly less than one means $c > e$. The boundary $c = e$ is not settled by the root test alone.

Solution 076.3 ■ Answer C[Back to problem](#)

The n th roots equal 1 at powers of two and $1/2$ elsewhere, so their limsup is 1 . The root test is inconclusive at this boundary. Nevertheless infinitely many terms equal 1 , so the terms fail to tend to zero and the series diverges.

Solution 076.4 ■ Answer B[Back to problem](#)

The n th root is $n^{3/n}/c$, tending to $1/c$. A limit strictly below 1 requires $c > 1$.

Solution 076.5 ■ Answer A[Back to problem](#)

Here $a_n^{1/n} = n^{-2/n} \rightarrow 1$. A root-test limit of 1 is inconclusive, although this particular series converges by the p -series test.

Infinite Series ▸ The integral test

077 Integral test and remainder bounds

Solution 077.1 ■ **Answer C**[Back to problem](#)

The associated function is eventually positive and decreasing. With $u = \log x$, its improper integral becomes $\int_{\log 2}^{\infty} u^{-p} du$, convergent exactly when $p > 1$.

Solution 077.2 ■ **Answer B**[Back to problem](#)

Apply the tail bounds with $f(x) = x^{-2}$ and $N = 100$. The lower integral from 101 to infinity is $1/101$, and the upper integral from 100 is $1/100$.

Solution 077.3 ■ **Answer B**[Back to problem](#)

The positive continuous function has integral $e^{-1} + \sum_{n \geq 2} 2^{-n-1} = e^{-1} + 1/4$, since each triangular area is half its base times height. But $f(n) \geq 1$ at every integer $n \geq 2$, so the sampled series diverges. The missing eventual-decrease hypothesis prevents application of the integral test.

Solution 077.4 ■ **Answer D**[Back to problem](#)

The geometric series equals $1/(e-1)$, while the improper integral equals $1/e$. Their difference is $1/[e(e-1)] > 0$. The integral test equates convergence behavior under its hypotheses, not the numerical values of the sum and the integral.

Solution 077.5 ■ **Answer E**[Back to problem](#)

The terms are eventually decreasing, so the integral test applies. Substitutions $u = \log x$ and then $v = \log u$ reduce the tail integral to $\int v^{-p} dv$, which converges exactly for $p > 1$.

Alternating Series ▸ Alternating-series test

078 Alternating series and their error estimates

Solution 078.1 ■ **Answer B**[Back to problem](#)

The stated bound is at most 10^{-6} precisely when $(N+1)^3 \geq 10^6$, or $N+1 \geq 100$. Thus $N = 99$ is the smallest index certified by this bound. This is a question about the bound, not the smallest index for the actual error.

Solution 078.2 ■ **Answer B**[Back to problem](#)

The magnitudes decrease strictly to zero. Even partial sums omit a positive next term and lie below the sum, while odd partial sums omit a negative next term and lie above it. Strict decrease makes the displayed inequalities strict.

Solution 078.3 ■ **Answer C**[Back to problem](#)

The partial sums are the sum of a convergent alternating square-root series and the divergent harmonic partial sums, so they tend to $+\infty$. Yet the individual terms tend to zero and, apart from any initial exception, alternate in sign. Their magnitudes do not satisfy the necessary monotonic control for the alternating-series test.

Solution 078.4 ■ **Answer C**[Back to problem](#)

The adjacent ratio is $(n+1)^2/(2n^2)$, at most one exactly when $n \geq 1 + \sqrt{2}$, or $n \geq 3$ for integers. Therefore the decreasing tail starts with b_3 , so $N = 2$ is the first permitted cutoff. Earlier terms need not be decreasing when the bound is applied only to this tail.

Solution 078.5 ■ **Answer A**[Back to problem](#)

The finite geometric identity is $1/(1+t) = \sum_{k=0}^{N-1} (-t)^k + (-t)^N/(1+t)$. Integrating from zero to one gives the exact remainder. Its sign is the first omitted sign, and its magnitude is strictly smaller than $\int_0^1 t^N dt = 1/(N+1)$, illustrating why the usual next-term bound is not generally an equality.

Alternating Series ▷ Absolute versus conditional convergence

079 Absolute and conditional convergence

Solution 079.1 ■ **Answer C**[Back to problem](#)

The alternating test gives convergence for $p > 0$. The absolute series is a p -series, which converges only for $p > 1$. Thus conditional convergence occurs exactly for $0 < p \leq 1$; at $p \leq 0$ the terms fail to tend to zero.

Solution 079.2 ■ **Answer B**[Back to problem](#)

$|a_n b_n| \leq 7|a_n|$, and the majorant series converges. No convergence of b_n is needed. Quotients may be undefined or large, and square roots can turn $1/n^2$ into the divergent harmonic series.

Solution 079.3 ■ **Answer C**[Back to problem](#)

For every fixed p , the magnitudes eventually decrease to zero, so the alternating series converges. Its absolute series converges exactly for $p > 1$ by the logarithmic integral test.

Solution 079.4 ■ **Answer C**[Back to problem](#)

Let s_m denote the original m th partial sum. After N full blocks, the rearranged sum is $T_N = s_{4N} + \frac{1}{2}s_{2N}$, as one verifies by separating odd and even denominators. Both partial sums tend to S , so $T_N \rightarrow 3S/2$. At most two unpaired terms remain between block endpoints, and they tend to zero. Conditional convergence permits a rearrangement to change the sum.

Solution 079.5 ■ Answer D[Back to problem](#)

Their difference is the convergent signed partial sum, while their sum is the divergent nonnegative absolute partial sum. Hence $P_N = (\sum |a_n| + \sum a_n)/2 \rightarrow \infty$ and $Q_N = (\sum |a_n| - \sum a_n)/2 \rightarrow \infty$, with the sums interpreted through the same finite index N . Conditional convergence balances two unbounded monotone accumulations.

Power Series ▷ Convergence sets

080 Power series, radius, and interval of convergence

Solution 080.1 ■ Answer B[Back to problem](#)

The radius is 3. At $x = -1$, the series becomes $\sum (-1)^n/n$, which converges. At $x = 5$, it becomes the harmonic series and diverges. Hence the interval is $[-1, 5)$.

Solution 080.2 ■ Answer D[Back to problem](#)

The ratio of consecutive coefficients is $(n+1)(2n+1)(2n+2)/[(3n+1)(3n+2)(3n+3)] \rightarrow 4/27$. The radius is its reciprocal, $27/4$.

Solution 080.3 ■ Answer C[Back to problem](#)

Convergence at distance 3 from the center forces $R \geq 3$, and divergence at the same distance forces $R \leq 3$. Thus $R = 3$. Every interior point converges absolutely, and the two endpoint outcomes are given explicitly.

Solution 080.4 ■ Answer A[Back to problem](#)

Treat the ratio as $(x+2)^2/9$. Interior convergence requires $|x+2| < 3$; at either endpoint every term is $1/n$, so both endpoints diverge.

Solution 080.5 ■ Answer C[Back to problem](#)

The even-index roots are $|c_{2n}|^{1/(2n)} = \sqrt{2}$, and the odd-index roots are zero. Hence the limsup is $\sqrt{2}$ and $R = 1/\sqrt{2}$. An adjacent-coefficient ratio is not available because every other coefficient vanishes. Equivalently, sum the geometric series $\sum (2x^2)^n$.

Solution 080.6 ■ Answer B[Back to problem](#)

For the first series the term ratio is $(n+1)|x|$, so every nonzero x gives divergence; its radius is zero. For the second the ratio is $|x|/(n+1)^2 \rightarrow 0$ for every fixed x , so its radius is infinite. A power series need not have a finite positive radius.

Power Series ▷ Functions Defined by Power Series

081 Termwise differentiation and integration of power series

Solution 081.1 ■ Answer B

[Back to problem](#)

From $\sum x^n = 1/(1-x)$ obtain $\sum nx^n = x/(1-x)^2$ by differentiation and multiplication by x . Apply the same operation again: $x d[x/(1-x)^2]/dx = x(1+x)/(1-x)^3$.

Solution 081.2 ■ Answer C

[Back to problem](#)

The derivative series is $\sum_{n=1}^{\infty} x^{n-1}/n$. At $x = 1$ it is harmonic and diverges. Its radius remains 1, illustrating that the radius is preserved while endpoint convergence need not be.

Solution 081.3 ■ Answer D

[Back to problem](#)

The radius remains 1. At both endpoints the absolute terms are $1/[(n+1)(n+2)]$, a convergent telescoping series. Both endpoints are gained even though the original geometric series diverges at each. Integration changes endpoint behavior without changing the radius.

Solution 081.4 ■ Answer E

[Back to problem](#)

Integrating $1/(1-t^2) = \sum t^{2n}$ from zero to x gives $\sum x^{2n+1}/(2n+1) = \operatorname{artanh} x$. Divide by x and set $x = 1/2$ to obtain $2 \operatorname{artanh}(1/2) = \log 3$.

Solution 081.5 ■ Answer A

[Back to problem](#)

Differentiation leaves radius 2. Writing $3x+1 = 3(x+1/3)$ makes the new center $-1/3$; the condition $|3x+1| < 2$ becomes $|x+1/3| < 2/3$. The exact radius is therefore $2/3$.

Power Series ▷ Operations on series

082 Power-series algebra, substitution, and coefficient extraction

Solution 082.1 ■ Answer C

[Back to problem](#)

Write $e^{x+x^2} = e^x e^{x^2}$. Contributions to x^4 come from degrees $(4,0)$, $(2,2)$, and $(0,4)$, giving $1/24 + 1/2 + 1/2 = 25/24$.

Solution 082.2 ■ Answer B

[Back to problem](#)

Multiply $x - x^2/2 + x^3/3 - x^4/4 + x^5/5 + \cdots$ by $1 + x + x^2 + \cdots$. The coefficient is $1 - 1/2 + 1/3 - 1/4 + 1/5 = 47/60$.

Solution 082.3 ■ **Answer C**[Back to problem](#)

The substituted geometric ratio must satisfy $|x^2 - 1| < 1$, or $0 < x^2 < 2$. At zero the terms alternate between 1 and -1 and do not tend to zero. Although the rational expression $1/(2 - x^2)$ is defined there, this particular series does not converge there.

Solution 082.4 ■ **Answer C**[Back to problem](#)

Write the series as $\sum a_n x^n$ and multiply by $1 - x - x^2$. Coefficient matching gives $a_0 = a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$. Thus $a_2, \dots, a_6 = 2, 3, 5, 8, 13$. Division by a series with nonzero constant term produces a recurrence rather than an infinite long division.

Solution 082.5 ■ **Answer C**[Back to problem](#)

All products within c_n have sign $(-1)^n$. Also $\sqrt{(k+1)(n-k+1)} \leq (n+2)/2$, so $|c_n| \geq 2(n+1)/(n+2)$. The terms cannot tend to zero. Each original series converges conditionally, but this does not guarantee convergence of its Cauchy product; absolute convergence is a sufficient safeguard.

Taylor Series ▷ Coefficients and centers

083 Taylor coefficients, Taylor series, and analyticity

Solution 083.1 ■ **Answer C**[Back to problem](#)

$f^{(3)}(x) = 2/x^3$, so the coefficient is $f^{(3)}(1)/3! = 2/6 = 1/3$. Equivalently, substitute $u = x - 1$ in the series for $\log(1 + u)$.

Solution 083.2 ■ **Answer D**[Back to problem](#)

Termwise differentiation of any power-series representation forces the coefficient formula and hence uniqueness. Smoothness alone does not imply a convergent or faithful Taylor representation; the standard flat function e^{-1/x^2} extended by zero refutes B and C.

Solution 083.3 ■ **Answer B**[Back to problem](#)

Continuity first gives $f(0) = 0$. If its first nonzero Taylor coefficient had index m , then $f(x) = x^m(c + o(1))$ with $c \neq 0$, which cannot have nonzero zeros arbitrarily close to zero. Thus every Taylor coefficient vanishes. Analyticity makes f zero near zero, but no assumption controls values outside the given neighborhood.

Solution 083.4 ■ **Answer A**[Back to problem](#)

The Taylor coefficient is $f^{(n)}(0)/n! = n!$. The ratio of successive absolute terms is $(n+1)|x|$, so the Taylor series diverges for every $x \neq 0$. Smoothness does not guarantee any positive radius of convergence; this is distinct from a flat function whose Taylor series converges but does not represent the function.

Solution 083.5 ■ **Answer B**[Back to problem](#)

Analyticity represents f by its Taylor series. Only even powers occur, so changing x to $-x$ leaves the sum unchanged; smoothness alone would not suffice.

Taylor Series > Standard expansions

084 Standard Maclaurin expansions and leading-order limits

Solution 084.1 ■ **Answer E**[Back to problem](#)

$\sin x = x - x^3/6 + O(x^5)$ and $x \cos x = x - x^3/2 + O(x^5)$. Their difference is $x^3/3 + O(x^5)$, giving limit $1/3$.

Solution 084.2 ■ **Answer C**[Back to problem](#)

The first factor is $x^4/2 + O(x^6)$ and the second is $-x^2/2 + O(x^3)$. Their product begins with $-x^6/4$. Thus exactly $k = 6$ gives a finite nonzero limit, namely $-1/4$.

Solution 084.3 ■ **Answer B**[Back to problem](#)

The two convergent arctangent series give $\arctan 1 - \arctan(1/3)$. The tangent subtraction formula gives tangent $(1 - 1/3)/(1 + 1/3) = 1/2$. The angle is between zero and $\pi/2$, so the sum is $\arctan(1/2)$. This uses the valid endpoint of the arctangent series at 1.

Solution 084.4 ■ **Answer B**[Back to problem](#)

The numerator is $2x + x^3/3 + O(x^5)$ and the denominator is $2 + x^2 + O(x^4)$. Their quotient is $x - x^3/3 + O(x^5)$. The third derivative is $3!$ times the cubic coefficient, namely -2 ; it is not the coefficient itself.

Solution 084.5 ■ **Answer B**[Back to problem](#)

Factor $e^x[e^{\sin x - x} - 1]$. Since $\sin x - x = -x^3/6 + O(x^5)$ and $e^u - 1 \sim u$, the quotient tends to $-1/6$.

Taylor Polynomials ▸ Approximation and error

085 Taylor polynomials and rigorous remainder bounds

Solution 085.1 ■ **Answer C**[Back to problem](#)

Here $R = e^\xi(1/2)^4/4! = e^\xi/384$ for $0 < \xi < 1/2$. Since $1 < e^\xi < 2$, the remainder is positive and less than $2/384 = 1/192$. The approximation is an underestimate.

Solution 085.2 ■ **Answer C**[Back to problem](#)

Three terms give $x - x^3/6 + x^5/120$ and alternating error at most $(1/2)^7/7! = 1/645120 < 10^{-4}$. With two terms, the positive remainder is greater than $(1/2)^5/120 - (1/2)^7/5040 = 1/3840 - 1/645120 > 10^{-4}$. One term has still larger error. Thus three is minimal, not merely sufficient.

Solution 085.3 ■ **Answer C**[Back to problem](#)

Expand both $f(h)$ and $f(-h)$ through degree three at zero. Odd terms cancel, the quadratic terms sum to $h^2 f''(0)$, and the two remainders each have magnitude at most $Mh^4/24$. Their sum has magnitude at most $Mh^4/12$; division by h^2 gives the bound.

Solution 085.4 ■ **Answer C**[Back to problem](#)

The function is smooth on $[1, 3]$, so the finite Taylor theorem applies even outside the convergence interval of its infinite Taylor series. Since $f^{(3)}(x) = 2/x^3$, the Lagrange remainder is $(2/\xi^3)2^3/3! = 8/(3\xi^3)$. Directly, $P_2(3) = 0$, so the remainder is $\log 3 > 0$.

Solution 085.5 ■ **Answer E**[Back to problem](#)

The quadratic Taylor polynomial is zero. For any real K , the smooth function $f(x) = Kx^4$ satisfies every stated datum, yet $|f(1)| = |K|$ can be arbitrarily large. A derivative value at the center cannot replace a derivative bound on the whole interval.

3. Calculus II

Analytic Geometry of $\mathbb{R}^3$ ▸ Coordinates and Vectors

086 Euclidean space, vectors, norm, and distance

Solution 086.1 ■ Answer C

[Back to problem](#)

Write $R = P + t(Q - P)$ with $t > 1$. Then $t = 3(t - 1)$, hence $t = 3/2$ and $R = (4, 5/2, 9)$. Its squared norm is $16 + 25/4 + 81 = 413/4$.

Solution 086.2 ■ Answer D

[Back to problem](#)

Expanding $\|u+v\|^2$ gives $16 = 9+4+2u \cdot v$, so $u \cdot v = 3/2$. Thus $\|2u-v\|^2 = 4(9)+4-4(3/2) = 34$.

Solution 086.3 ■ Answer B

[Back to problem](#)

The octant condition selects $u = (1, 2, 3)/\sqrt{14}$ rather than its negative. Expanding componentwise gives $\|u - \mathbf{i}\|^2 = \|u\|^2 + 1 - 2u_1 = 2 - 2/\sqrt{14}$. Normalization uses the square root of the sum of the squared components.

Solution 086.4 ■ Answer C

[Back to problem](#)

A point of the segment is $(1, 2t, 0)$ with $0 \leq t \leq 1$. Its squared distance is $5 + (3 - 2t)^2$. The unrestricted minimizer $t = 3/2$ lies outside the segment, so the segment minimum occurs at $t = 1$ and equals 6. The containing line would instead give squared distance 5.

Solution 086.5 ■ Answer B

[Back to problem](#)

Equating squared distances cancels the quadratic terms. The result is $2(Q - P) \cdot (x, y, z) = \|Q\|^2 - \|P\|^2 = 13 - 5 = 8$. Since $Q - P = (2, -2, 2)$, division by four gives $x - y + z = 2$, the perpendicular bisector plane.

Analytic Geometry of $\mathbb{R}^3$ ▸ The Dot Product

087 Dot product, angle, and orthogonality

Solution 087.1 ■ Answer A

[Back to problem](#)

The perpendicularity gives $\|u\|^2 - u \cdot v - 2\|v\|^2 = 0$, hence $u \cdot v = 2\|v\|^2 = \|u\|\|v\|$. Thus $\cos \theta = 1$.

Solution 087.2 ■ Answer B

[Back to problem](#)

From $\|u + v\|^2 = \|w\|^2$ and the cyclic versions, all pairwise dot products are $-1/2$. The desired expression is $u \cdot v - u \cdot w - v \cdot v + v \cdot w = -3/2$.

Solution 087.3 ■ Answer E[Back to problem](#)

The dot product is -4 , and the norm product is 9 . Dividing gives $-\frac{4}{9}$; using a raw dot product would omit normalization.

Solution 087.4 ■ Answer C[Back to problem](#)

Put $x = y = t$ and set $a = \sqrt{2}t$. Then $a^2 + z^2 = 1$ and the expression is $(3/\sqrt{2})a + 2z$. Cauchy–Schwarz gives the maximum $\sqrt{9/2 + 4} = \sqrt{34}/2$, attained when (a, z) is the positive unit multiple of $(3/\sqrt{2}, 2)$.

Solution 087.5 ■ Answer C[Back to problem](#)

Bilinearity gives $u \cdot u = u \cdot (v + w) = 0$. Positive definiteness of the Euclidean norm forces $u = 0$, so $v = -w$. They need not vanish individually. A zero dot product with a zero vector does not define an angle, so the last assertion is not valid.

Analytic Geometry of $\mathbb{R}^3$ ▸ The Dot Product

088 Scalar and vector projection

Solution 088.1 ■ Answer C[Back to problem](#)

The dot product is 6 , and the direction has squared norm 9 . The minimum is $14 - 6^2/9 = 10$, attained at $t = 2/3$.

Solution 088.2 ■ Answer D[Back to problem](#)

The scalar projection gives $a \cdot b = -2$. The component of a parallel to b has length $2/3$. Pythagoras gives $\sqrt{4 - 4/9} = 4\sqrt{2}/3$.

Solution 088.3 ■ Answer B[Back to problem](#)

Subtract the base point first: $b - P = (3, 1, 2)$. Its projection parameter along $a = (1, 1, 0)$ is $[(b - P) \cdot a]/(a \cdot a) = 4/2 = 2$. Adding $2a$ to P gives $(3, 2, 1)$. Projecting the raw position vector would use the wrong line through the origin.

Solution 088.4 ■ Answer C[Back to problem](#)

The directed unit vector changes from $a/\|a\|$ to $-a/\|a\|$, so the scalar projection changes sign to 2 . In the vector formula, the sign and scale cancel: $[(-3a) \cdot b](-3a)/(9\|a\|^2) = p$. The underlying projection line is unchanged.

Solution 088.5 ■ Answer A[Back to problem](#)

The parallel component is $p = [3/9](1, 2, 2) = (1/3, 2/3, 2/3)$. Reflection sends $b = p + (b - p)$ to $p - (b - p) = 2p - b = (-7/3, 4/3, 4/3)$. Its squared norm remains 9 , providing a check.

Analytic Geometry of $\mathbb{R}^3$ ▸ The Cross Product**089** Cross product, orientation, and area**Solution 089.1** ■ **Answer B**[Back to problem](#)

$(u + v) \times (u - v) = -2u \times v$. The triangle area is therefore $\|u \times v\| = \sqrt{144 - 36} = 6\sqrt{3}$.

Solution 089.2 ■ **Answer C**[Back to problem](#)

$u \times v = (1, -1, 1)$. Reverse its sign to make the second component positive, and divide by its norm $\sqrt{3}$.

Solution 089.3 ■ **Answer B**[Back to problem](#)

The vector triple-product identity gives $u \times (u \times v) = u(u \cdot v) - v(u \cdot u) = u/2 - v$. Dotting with v gives $1/4 - 1 = -3/4$. Replacing the nested cross product by $(u \times u) \times v = 0$ would incorrectly assume associativity.

Solution 089.4 ■ **Answer B**[Back to problem](#)

The cross-product equations give $v_z = 1$ and $v_y - v_x = 2$. Thus $v = (t, t+2, 1)$, whose squared norm is $2(t+1)^2 + 3$. The minimum occurs at $t = -1$, giving $(-1, 1, 1)$. The free component parallel to $(1, 1, 0)$ is removed at the minimum.

Solution 089.5 ■ **Answer C**[Back to problem](#)

Writing the cross product in components shows that reflection of the inputs preserves its first component and reverses the other two. Equivalently, an orientation-reversing orthogonal map satisfies $(Ru) \times (Rv) = -R(u \times v)$. The result is $(1, -2, -3)$.

Analytic Geometry of $\mathbb{R}^3$ ▸ The Triple Scalar Product**090** Scalar triple product and volume**Solution 090.1** ■ **Answer C**[Back to problem](#)

The coefficient matrix relative to u, v, w is $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$, with determinant 2. Multilinearity gives $2[u, v, w] = 10$.

Solution 090.2 ■ **Answer C**[Back to problem](#)

The displacement vectors are $(0, 2, 0)$, $(0, 0, 3)$, $(4, 1, 1)$. Their scalar triple product is $(6, 0, 0) \cdot (4, 1, 1) = 24$. Divide by 6 to obtain 4.

Solution 090.3 ■ **Answer B**[Back to problem](#)

The first three points determine $x + y + z = 1$. The fourth lies in that plane exactly when $g(t) = t^3 + t^2 + t - 1 = 0$. Its derivative $3(t + 1/3)^2 + 2/3$ is positive everywhere. Since $g(0) < 0 < g(1)$, exactly one real root exists. Coplanarity is the zero-volume condition.

Solution 090.4 ■ **Answer B**[Back to problem](#)

The base parallelogram has area $\|u \times v\| = 2 \cdot 3 \sin(\pi/6) = 3$. Its perpendicular height is at most $\|w\| = 4$, with equality when w is parallel to $u \times v$. Thus the maximum volume is 12.

Solution 090.5 ■ **Answer B**[Back to problem](#)

The squared scalar triple product is the determinant of the Gram matrix G =

$$\begin{pmatrix} 1 & 1/2 & 1/2 \\ 1/2 & 1 & 1/2 \\ 1/2 & 1/2 & 1 \end{pmatrix}.$$

Its determinant is $1/2$, so the parallelepiped volume is $1/\sqrt{2}$. Dividing by six gives $\sqrt{2}/12$. Dot products determine the volume even without coordinates or orientation.

Analytic Geometry of R3 ▸ Lines in 3-Space

091 Parametric and symmetric equations of lines

Solution 091.1 ■ **Answer C**[Back to problem](#)

The first two coordinate equations give $t = 1 + s$ and $1 + t = 2 - s$, so $s = 0, t = 1$. The third coordinates are then 1 and a , requiring $a = 1$.

Solution 091.2 ■ **Answer C**[Back to problem](#)

Substitution gives $2 + (-1 + 2t) + (3 - t) = 8$, hence $t = 4$. The displacement is $4(0, 2, -1)$, whose norm is $4\sqrt{5}$.

Solution 091.3 ■ **Answer B**[Back to problem](#)

A common perpendicular is $u \times v = (1, -1, 1)$. Projecting $q - p$ onto its unit direction gives distance $|(q - p) \cdot (u \times v)| / \|u \times v\| = \frac{\sqrt{3}}{3}$.

Solution 091.4 ■ **Answer A**[Back to problem](#)

The zero first direction component means x stays exactly 2. Eliminating t from the other coordinates gives the stated ratio equality. Omitting $x = 2$ would describe a plane, not the original line; division by the zero component is not allowed.

Solution 091.5 ■ **Answer D**[Back to problem](#)

The second direction is -2 times the first, and its base point is the first line's point at $t = 2$. Indeed the second parametrization equals the first with $t = 2 - 2s$, which ranges over every real value. Parallel direction vectors alone would not distinguish coincident from distinct parallel lines.

Analytic Geometry of $\mathbb{R}^3$ ▸ Planes in 3-Space

092 Planes and normal equations

Solution 092.1 ■ **Answer A**[Back to problem](#)

A second direction in the plane is $(-1, 1, 1)$. Crossing $(1, 2, 0)$ with it gives $(2, -1, 3)$. Evaluating at $(1, 0, 1)$ gives constant 5.

Solution 092.2 ■ **Answer B**[Back to problem](#)

Cross the normals: $(1, 1, 1) \times (1, -1, 2) = (3, -1, -2)$. Its dot product with each normal is zero, so it gives the intersection direction.

Solution 092.3 ■ **Answer C**[Back to problem](#)

Normals are $(1, 0, 0)$ and $(1, k, 1)$. The acute-angle cosine is $1/\sqrt{k^2 + 2}$. Setting it equal to $1/2$ gives $k^2 + 2 = 4$, hence $k = \pm\sqrt{2}$. The plane constants do not affect the angle.

Solution 092.4 ■ **Answer C**[Back to problem](#)

The normals are proportional only when $a = 4$, with ratio 2. In that case twice the first plane equation has right side 6, inconsistent with 12, so the planes are distinct and parallel. For every other a the normals are independent and the planes intersect in a line.

Solution 092.5 ■ **Answer C**[Back to problem](#)

Rearrange as $\lambda(x + y - 2) + (-y + z - 1) = 0$. For this to hold for every λ , both coefficients must vanish: $x + y = 2$ and $z - y = 1$. Set $y = t$ to obtain the displayed line. It satisfies every member of the plane family.

Analytic Geometry of $\mathbb{R}^3$ ▸ Cylinders

093 Cylindrical surfaces and generators

Solution 093.1 ■ **Answer B**[Back to problem](#)

Points are $(s + t, s^2 - t, 2t)$. Since $t = z/2$, the base coordinates are $s = x - z/2$ and $s^2 = y + z/2$, giving the claimed equation.

Solution 093.2 ■ **Answer B**[Back to problem](#)

A generator preserves $x - z$ and $y + 2z$. Its direction (a, b, c) therefore satisfies $a - c = 0$ and $b + 2c = 0$, hence is proportional to $(1, -2, 1)$.

Solution 093.3 ■ **Answer C**[Back to problem](#)

Use orthonormal coordinates in the cutting plane: $(x, y, z) = s(1, 0, 1)/\sqrt{2} + t(0, 1, 0)$. The cylinder becomes $s^2/2 + t^2 = 1$, with semiaxes $\sqrt{2}$ and 1. Its area in the plane is $\sqrt{2}\pi$; projecting onto the xy -plane would incorrectly give π .

Solution 093.4 ■ **Answer B**[Back to problem](#)

Every translated point is (s, t, s^2) with independent s, t . Eliminating parameters gives $z = x^2$, with no restriction on y .

Solution 093.5 ■ **Answer D**[Back to problem](#)

The selected equation is independent of x and has circular yz cross-sections centered at $(1, -2)$ with radius 3. Thus varying x traces its generators.

Analytic Geometry of $\mathbb{R}^3$ ▸ Surfaces of Revolution

094 Surfaces of revolution

Solution 094.1 ■ **Answer B**[Back to problem](#)

The distance to the z -axis equals the original nonnegative $x = 2 - z$, and the original segment implies $0 \leq z \leq 2$. Thus $\sqrt{x^2 + y^2} = 2 - z$.

Solution 094.2 ■ **Answer D**[Back to problem](#)

The radial distance from the y -axis is $\sqrt{x^2 + z^2} = y^2$. At $y = 2$, the radius is 4, so the enclosed disk has area 16π .

Solution 094.3 ■ **Answer A**[Back to problem](#)

The original distance to the y -axis is $x = \sqrt{y}$. Rotation replaces its square by $x^2 + z^2$, giving $y = x^2 + z^2$ with $y \geq 0$ automatically.

Solution 094.4 ■ Answer B[Back to problem](#)

The rotation axis is parallel to the z -axis, so height $z = t$ is unchanged. The generating point's distance to the axis is $|t^2 - 1|$. Thus the new distance $\sqrt{(x - 1)^2 + y^2}$ must equal $|z^2 - 1|$. Squaring gives the stated equation, valid at all heights.

Solution 094.5 ■ Answer A[Back to problem](#)

Rotation preserves z and replaces the square of the original distance $|x|$ from the axis by $x^2 + y^2$. The two equatorial semiaxes are 3, and the polar semiaxis is 2. Equal semiaxes perpendicular to the axis make this an ellipsoid of revolution.

Analytic Geometry of $\mathbb{R}^3$ ▸ Level Curves and Level Surfaces

095 Level sets, contours, and quadric surfaces

Solution 095.1 ■ Answer B[Back to problem](#)

At $z = h$, the semiaxes are $2\sqrt{1 + h^2}$ and $\sqrt{1 + h^2}$. Their enclosed area is $2\pi(1 + h^2)$. Doubling the area 2π gives $h^2 = 1$.

Solution 095.2 ■ Answer D[Back to problem](#)

The equation $x^2 - y^2 = (x - y)(x + y) = 0$ is the union of $y = x$ and $y = -x$. Nonzero levels are hyperbolas, but the zero level is degenerate.

Solution 095.3 ■ Answer C[Back to problem](#)

At level -1 , $z^2 - x^2 - y^2 = 1$ gives two sheets. At zero, $z^2 = x^2 + y^2$ gives the two nappes of a circular cone meeting at the origin. At level 1, horizontal sections have radius $\sqrt{1 + z^2}$ at every height, giving one connected sheet.

Solution 095.4 ■ Answer A[Back to problem](#)

The level curve is the set of inputs where $f(x, y) = 9$, namely an ellipse with semiaxes 3 and $3/2$. The contour consists of graph points $(x, y, 9)$ above that ellipse. A level curve lives in the input plane; its graph contour retains the output height.

Solution 095.5 ■ Answer D[Back to problem](#)

Completing squares gives $f = (x - 2)^2 + (y + 3)^2 - 13$. The minimum level -13 consists solely of $(2, -3)$; lower levels are empty and higher levels are circles.

Solution 095.6 ■ Answer C[Back to problem](#)

On $z = 0$ the equation becomes $(x - 1)^2 + (y + 2)^2 = c - 9$. It is empty for $c < 9$, a single point for $c = 9$, and a circle for $c > 9$. This includes the distinction between a spherical level surface and its potentially degenerate planar section.

Analytic Geometry of R3 ▸ Cylindrical Coordinates

096 Cylindrical coordinates

Solution 096.1 ■ **Answer B**[Back to problem](#)

$r^2 \sin 2\theta = 2(r \cos \theta)(r \sin \theta) = 2xy$. Substituting $y = x$ gives $z = 2x^2$.

Solution 096.2 ■ **Answer D**[Back to problem](#)

Its Cartesian coordinates are $(-\sqrt{3}, 1, -3)$. The squared distance is $(-\sqrt{3}-1)^2 + 1^2 + (-4)^2 = 21 + 2\sqrt{3}$.

Solution 096.3 ■ **Answer B**[Back to problem](#)

Expanding gives $r^2 \leq 2r \cos \theta$. For $r > 0$ this means $r \leq 2 \cos \theta$, so only directions with nonnegative cosine enter the disk. The given angular interval covers it once except for the harmless multiple descriptions of the axis at $r = 0$.

Solution 096.4 ■ **Answer B**[Back to problem](#)

The coordinate equations give $x = y = r/\sqrt{2} \geq 0$. Every point of that vertical half-plane has such coordinates, including its boundary on the z -axis. The other half of the plane has azimuth $5\pi/4$, not $\pi/4$.

Solution 096.5 ■ **Answer B**[Back to problem](#)

The Cartesian coordinates are $(t^2 \cos(1/t), t^2 \sin(1/t), t)$. Their absolute values are bounded by t^2, t^2, t , respectively, so they tend to zero. The angle continues rotating; at the limiting point $r = 0$, azimuth is not uniquely defined.

Analytic Geometry of R3 ▸ Spherical Coordinates

097 Spherical coordinates

Solution 097.1 ■ **Answer B**[Back to problem](#)

Expand to $x^2 + y^2 + z^2 = 4z$. Thus $\rho^2 = 4\rho \cos \phi$, so for $\rho > 0$, $\rho = 4 \cos \phi$. Nonnegative radius restricts the sphere to polar angles at most $\pi/2$.

Solution 097.2 ■ **Answer B**[Back to problem](#)

Here $\rho = 5$, $z = 5/2$, and the azimuth makes x and y equal in magnitude with opposite signs. Their sum vanishes, leaving $5/2$.

Solution 097.3 ■ **Answer C**[Back to problem](#)

The equation becomes $\rho \cos \phi = \rho \sin \phi$. For $\rho > 0$ and nonnegative z , this forces $\phi = \pi/4$.

Solution 097.4 ■ Answer A[Back to problem](#)

At $\phi = 0$, $x = y = 0$ and $z = \rho \geq 0$. At $\phi = \pi/2$, $z = 0$ and arbitrary ρ, θ cover the xy -plane. At $\theta = 0$, $y = 0$ and $x = \rho \sin \phi \geq 0$, with z unrestricted. These are the exceptional coordinate surfaces, including their boundaries.

Solution 097.5 ■ Answer A[Back to problem](#)

The radial equation is $\rho^2 - 6\rho \cos \phi + 8 = 0$, with roots $3 \cos \phi \pm \sqrt{1 - 9 \sin^2 \phi}$. Two distinct intersections require a positive discriminant and positive radii. In the northern directions this gives $0 \leq \phi < \arcsin(1/3)$. At equality the ray is tangent; southern directions produce negative roots and no points on the ray.

Partial Derivatives ▸ Definition and Geometric Interpretation

098 Partial derivatives and coordinate traces

Solution 098.1 ■ Answer B[Back to problem](#)

With x held fixed, $z_y = x/(1 + x^2 y^2) + x^2 = 3/2$ at $(1, 1)$. A tangent direction is $(0, 1, 3/2)$, equivalently $(0, 2, 3)$.

Solution 098.2 ■ Answer C[Back to problem](#)

On either coordinate axis, f is zero, so both partial derivatives at the origin are zero. Along $y = x \neq 0$, its value is $1/2$, so it is not continuous there.

Solution 098.3 ■ Answer C[Back to problem](#)

For the partial derivative, hold the independent variable y fixed: $f_x = 1/x$, giving 1. Along the specified path, $g(x) = \log(1/x) = -\log x$, giving derivative -1 . Changing a dependent input during differentiation computes a path derivative, not the same partial derivative.

Solution 098.4 ■ Answer D[Back to problem](#)

The stated piecewise constant function is smooth on D because the excluded line separates its two components. The mean value theorem forces constancy along each connected horizontal interval in D , but cannot cross the missing point $x = 0$. Thus zero x -partial does not identify values on disconnected slices.

Solution 098.5 ■ Answer C[Back to problem](#)

Integrating with respect to x gives $f(x, y) = x^2 y + e^x + C(y)$. The boundary data yield $C(y) = \sin y - 1$. Thus $f_y = x^2 + \cos y$, and $f_y(1, 0) = 2$. The integration constant may depend on the variable held fixed.

Partial Derivatives ▸ Higher-Order Partial Derivatives

099 Mixed partials and equality of differentiation orders

Solution 099.1 ■ Answer A

[Back to problem](#)

$f_{xx} = y^2 e^{xy}$, so $f_{xxy} = (2y + xy^2)e^{xy}$. At $(1, 0)$ this is zero. Differentiating before substituting is essential.

Solution 099.2 ■ Answer A

[Back to problem](#)

For fixed y , $f_x(0, y) = -2y$, including at zero. Hence $f_{xy}(0, 0) = -2$. Likewise $f_y(x, 0) = x$, so $f_{yx}(0, 0) = 1$. Their difference is -3 ; the continuity hypothesis for commuting partials fails.

Solution 099.3 ■ Answer D

[Back to problem](#)

The series is $\sum_{n \geq 0} x^{2n} y^n / n!$. The coefficient of $x^6 y^3$ is $1/3!$, so the indicated derivative at zero is $6!3!(1/3!) = 720$. All other monomials vanish after the differentiations and evaluation. Smoothness permits either derivative order.

Solution 099.4 ■ Answer D

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Equality of mixed partials requires $2x + 6y = 2x + ky$ for all (x, y) , so $k = 6$. This condition is also achievable: $f = x^2 y + 3xy^2 - \cos y$ has exactly these two first partial derivatives when $k = 6$.

Solution 099.5 ■ Answer A

[Back to problem](#)

The first summand is quadratic in x , so three x -derivatives annihilate it. The second is quadratic in y , so four y -derivatives annihilate it. All derivatives are continuous on the domain, allowing the convenient order for each summand; no expansion of the other factor is needed.

Partial Derivatives ▸ The Tangent Plane to a Surface

100 Tangent planes and implicit partial differentiation

Solution 100.1 ■ Answer B

[Back to problem](#)

For $F = xz + yz^2$, $(F_x, F_y, F_z) = (1, 1, 3)$ at the point. Hence $f_x = f_y = -1/3$ and the requested sum is -1 .

Solution 100.2 ■ Answer B

[Back to problem](#)

The first partials at $(1, -1)$ are $1, -1$. Thus $z - 1 = (x - 1) - (y + 1)$, or $z = x - y - 1$. Only $(2, 1, 0)$ satisfies this equation.

Solution 100.3 ■ **Answer A**[Back to problem](#)

Since $|xy| \leq (x^2 + y^2)/2$, the error after subtracting the zero linear map divided by $\sqrt{x^2 + y^2}$ tends to zero. Thus the function is differentiable at the origin with both partials zero, and its tangent plane is horizontal. Nonzero graph partials are not a prerequisite for a tangent plane.

Solution 100.4 ■ **Answer C**[Back to problem](#)

The full gradient is $(2, 0, 0)$, which is nonzero, but its z -component vanishes. Solving for the x -coordinate gives $x = \sqrt{1 - y^2 - z^2}$ near the point. Its two partials there vanish, and the tangent plane is $x = 1$. The failure of one coordinate condition does not make the surface singular.

Solution 100.5 ■ **Answer B**[Back to problem](#)

The zero set is exactly the smooth graph $z = x + y^2$, whose tangent plane at the origin is $z = x$. Cubing its defining equation makes all first partials of F vanish on the surface, without changing the surface. A nonzero gradient is a sufficient regularity test for the chosen equation, not a necessary property of every equation describing a smooth surface.

Partial Derivatives ▸ Linear Approximations

101 Total differential and linear approximation

Solution 101.1 ■ **Answer D**[Back to problem](#)

At $(2, 0)$, $f_x = 4$ and $f_y = 4$. The first-order change is $4(3t) + 4(-t) = 8t$, so $f \approx 4 + 8t$.

Solution 101.2 ■ **Answer B**[Back to problem](#)

The linear relative change $2(0.01) - 0.02$ is zero, but the exact ratio is $(1.01)^2(0.98) = 1.0201(0.98) = 0.999698$. This tests the distinction between a differential and an exact change.

Solution 101.3 ■ **Answer B**[Back to problem](#)

The bound $|f(x, y)| \leq |y| \leq \sqrt{x^2 + y^2}$ proves continuity. Its coordinate-axis restrictions are zero, so both partials vanish. Differentiability would require $f(x, y)/\sqrt{x^2 + y^2} \rightarrow 0$, but along $x = y = t > 0$ this ratio is $1/(2\sqrt{2})$. Continuity and partials together still do not establish a linear approximation.

Solution 101.4 ■ **Answer D**[Back to problem](#)

Logarithmic differentiation gives $dQ/Q = 2dx/x - dy/y - 3dz/z$. The triangle inequality gives the first-order bound $2(1\%) + 2\% + 3(0.5\%) = 5.5\%$. Taking the x -error positive and the other two negative attains that linear bound. It is a first-order estimate, not an exact finite-error guarantee.

Solution 101.5 ■ **Answer E**[Back to problem](#)

For any real c , the smooth function $f(x, y) = 2x - y + (c + 1)x^2$ has the stated value and gradient. Substitution makes the quotient identically c . First-order differentiability controls an $o(\|(x, y)\|)$ remainder, which need not be $o(t^2)$ along this path.

Partial Derivatives ▸ The Chain Rule for Partial Derivatives

102 Multivariable chain rule and held-fixed variables

Solution 102.1 ■ **Answer B**[Back to problem](#)

Here $u = 5, v = 2, u_x = 1, v_x = 2$. The chain rule gives $z_x = 2uu_x + xv_x + v = 10 + 2 + 2 = 14$.

Solution 102.2 ■ **Answer A**[Back to problem](#)

Implicit differentiation gives the three factors $-y/x$, $-z/y$, and $-x/z$. Their product is -1 . Positivity ensures that none of the denominators is zero.

Solution 102.3 ■ **Answer A**[Back to problem](#)

The chain rule gives $F_x x' + F_y y' = 4t^3 + 2t + 2$. Evaluation at $t = 1$ gives 8.

Solution 102.4 ■ **Answer E**[Back to problem](#)

There are three dependency paths from z to y : the direct path, the path through v , and the path through u then v . Thus $z_y = F_y + F_v g_y + F_u f_v g_y = 5 + 33 + 154 = 192$. The direct partial F_y is not the derivative of the whole composite.

Solution 102.5 ■ **Answer D**[Back to problem](#)

Differentiating the first chain rule again gives $r'^T H r' + \nabla f \cdot r''$. The quadratic term is $2 - 2 + 4 = 4$, and the acceleration term is 6, for a total of 10. Omitting the second term is valid only when that contribution vanishes.

Directional Derivatives and the Gradient ▷ Rates of Change

103 Directional derivative and steepest ascent

Solution 103.1 ■ **Answer C**[Back to problem](#)

Write $\nabla f(P) = (3, b)$. Normalizing $(3, 4)$ gives $(3/5, 4/5)$, so $(9 + 4b)/5 = 4$, hence $b = 11/4$. The maximum is $\sqrt{9 + 121/16} = \sqrt{265}/4$.

Solution 103.2 ■ **Answer D**[Back to problem](#)

The gradient is $(y + z, x + z, x + y) = (1, 0, 3)$. The unit direction to the origin is $(-1, -2, 1)/\sqrt{6}$, so the dot product is $2/\sqrt{6}$.

Solution 103.3 ■ **Answer B**[Back to problem](#)

The gradient has length 5, so each direction makes an angle α with it satisfying $\cos \alpha = 3/5$. The two directions lie symmetrically on opposite sides. Their dot product is $\cos(2\alpha) = 2(3/5)^2 - 1 = -7/25$. This solves for directions with a prescribed rate rather than inserting one given direction.

Solution 103.4 ■ **Answer E**[Back to problem](#)

The gradient at the origin is zero, so every first-order directional rate is zero. Nevertheless, on $y = 0$ the values are x^3 , positive and negative arbitrarily close to zero. Vanishing first-order rates do not classify higher-order behavior.

Solution 103.5 ■ **Answer B**[Back to problem](#)

Along $(x, y) = (ta, tb)$ with $b \neq 0$, $f(ta, tb) = t^2 a^3 b / (t^4 a^6 + b^2)$, so $f(ta, tb)/t \rightarrow 0$. When $b = 0$ the restriction is identically zero. Yet along $y = x^3$ the value is $1/2$ for $x \neq 0$. Thus even all zero directional derivatives do not imply continuity or differentiability.

Directional Derivatives and the Gradient ▷ Normals to Level Sets

104 Gradient normals and tangent directions

Solution 104.1 ■ **Answer B**[Back to problem](#)

The normals are $(2, 2, 2)$ and $(1, 2, 3)$. Their cross product is $(2, -4, 2)$, so $(1, -2, 1)$ is a tangent direction.

Solution 104.2 ■ **Answer C**[Back to problem](#)

The tangent directions are orthogonal to $n = (1, 1, 1)$. Project $\nabla f = (1, -1, 2)$ onto that plane: subtract $(2/3)n$. The squared projected norm is $6 - 4/3 = 14/3$, which is the maximum squared rate.

Solution 104.3 ■ Answer C[Back to problem](#)

The gradient is $(y, x, 2z) = (1, 1, 4)$. The normal line is $(1 + t, 1 + t, 2 + 4t)$, so $t = -1/2$ at $z = 0$. It meets that plane at $(1/2, 1/2, 0)$, giving sum 1.

Solution 104.4 ■ Answer D[Back to problem](#)

The gradient vanishes, so its normal equation gives only $0 = 0$. The surface contains lines through the origin in the three independent directions $(1, 0, 1)$, $(0, 1, 1)$, and $(1, 0, -1)$. A tangent plane would have to contain all three, which no plane can do. Direct geometry, not the vacuous gradient equation, settles the singularity.

Solution 104.5 ■ Answer A[Back to problem](#)

Differentiate $r \cdot r = 3$ twice: $2r' \cdot r' + 2r \cdot r'' = 0$. The given velocity has squared length 6, so $r(0) \cdot r''(0) = -6$. A straight line with that tangent velocity has zero acceleration and therefore leaves the sphere; tangency is only a first-order condition.

Max/Min Problems ▷ Critical Points and the Second-Derivative Test

105 Critical points, Hessian, and saddle points

Solution 105.1 ■ Answer C[Back to problem](#)

The Hessian is $\begin{pmatrix} 2 & 2a \\ 2a & 2 \end{pmatrix}$, whose determinant is $4(1 - a^2)$. For $|a| < 1$ it is positive definite. At $a = \pm 1$, a whole line gives value zero, so the minimum is not strict; outside that interval it is a saddle.

Solution 105.2 ■ Answer C[Back to problem](#)

Along $y = 0$, $f = x^4 > 0$ off the origin. Along $y = x$, $f = -x^4 < 0$. The origin is stationary and has both larger and smaller nearby values, hence is a saddle although its Hessian is zero.

Solution 105.3 ■ Answer D[Back to problem](#)

The gradient at the origin is $(2, 0)$, so the stationarity prerequisite fails. Along $y = 0$, the values $x^2 + 2x$ have both signs near zero. The Hessian is positive definite everywhere, but the actual minimum occurs at $(-1, 0)$, not at every point.

Solution 105.4 ■ Answer A[Back to problem](#)

Writing $r = \sqrt{x^2 + y^2}$ gives $f \geq r - |x|/2 \geq r/2$, strictly positive away from the origin. Thus zero is the unique global minimum. On the x-axis the one-sided slopes of $|x| - x/2$ are $1/2$ and $-3/2$, so even the x-partial fails to exist there.

Solution 105.5 ■ **Answer C**[Back to problem](#)

The nonnegative fourth powers make f strictly positive off the origin. For g , the curve $y = x^2$ gives $-x^6 < 0$, while $x = 0$ gives $y^2 > 0$. In fact every fixed line through the origin gives positive g sufficiently near zero away from zero, so checking only straight lines would miss the saddle. A zero Hessian determinant does not determine either outcome.

Solution 105.6 ■ **Answer B**[Back to problem](#)

The gradient equations are $3x^2 - 3 = 0$ and $-2y = 0$, giving $(\pm 1, 0)$. The Hessian is diagonal with entries $6x, -2$. At $x = 1$ its determinant is negative, so the point is a saddle. At $x = -1$ both entries are negative, giving a strict local maximum. There are no other stationary points.

Max/Min Problems ▸ Absolute Extrema on a Closed Region

106 Absolute extrema and boundary analysis

Solution 106.1 ■ **Answer C**[Back to problem](#)

On the axes the value is zero. On the sloping edge, $x = 6 - 2y$, so $xy = 6y - 2y^2$, with maximum at $y = 3/2$, $x = 3$. The maximum is $9/2$.

Solution 106.2 ■ **Answer B**[Back to problem](#)

Complete squares: $f = (x - 1)^2 + (y - 2)^2 - 5$. The point of the square closest to $(1, 2)$ is $(1, 1)$, giving $f = -4$. The unconstrained stationary point $(1, 2)$ is outside the region.

Solution 106.3 ■ **Answer B**[Back to problem](#)

This is squared distance to $(3, 4)$, which lies outside the disk. The nearest point is on the radial boundary, at distance $\sqrt{25} - 2$. Squaring gives 9.

Solution 106.4 ■ **Answer B**[Back to problem](#)

The punctured disk omits the origin, so every value is positive but values tend to zero along points approaching that omission. The infimum is zero and no minimum exists. Boundary points with radius one attain the maximum one. Boundedness without closedness does not guarantee both extrema.

Solution 106.5 ■ **Answer B**[Back to problem](#)

Set $u = |x| \in [0, 1]$. For fixed u , the largest permitted y^2 is $1 - u^2$, so the objective is at most $1 + u - u^2 = 5/4 - (u - 1/2)^2$. Equality occurs at $x = \pm 1/2$, $y = \pm\sqrt{3}/2$. The maximizing boundary points are not the coordinate-axis endpoints.

Max/Min Problems ▸ Max/Min with a Constraint; Lagrange Multipliers

107 Constrained extrema and Lagrange multipliers

Solution 107.1 ■ Answer C

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Cauchy–Schwarz gives $(1, 2, 2) \cdot (x, y, z) \leq 3 \cdot 3 = 9$, attained at $(1, 2, 2)$. Equivalently Lagrange multipliers make the point parallel to $(1, 2, 2)$.

Solution 107.2 ■ Answer B

[Back to problem](#)

Write $2x^2 + 2xy + 2y^2 \leq 3(x^2 + y^2)$ because $2xy \leq x^2 + y^2$. Thus $x^2 + y^2 \geq 2$, with equality when $x = y = \pm 1$.

Solution 107.3 ■ Answer C

[Back to problem](#)

The only feasible point is $(0, 0)$, so it is both a constrained maximum and minimum, of value zero. There $\nabla f = (1, 0)$ and $\nabla g = (0, 0)$, so the multiplier equation cannot hold. The regular-constraint hypothesis fails; the point must be checked separately.

Solution 107.4 ■ Answer C

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Near the top of the circle, $y = \sqrt{1 - x^2}$ and $f = x^2\sqrt{1 - x^2} > 0$ for nearby nonzero x , so the top is a strict local minimum of value zero. Near the bottom the sign is reversed, giving a strict local maximum. Elsewhere on the full circle both positive and negative values occur, so neither is absolute. Dividing the multiplier equations by x would lose these candidates.

Solution 107.5 ■ Answer B

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The constraint gradients are independent on the feasible circle, which is compact. Multiplier equations give $yz = \lambda + 2\mu x$ and cyclic versions. Subtraction gives $(y - x)(z + 2\mu) = 0$ and its cyclic versions. If all coordinates differed, all would equal -2μ , a contradiction; thus two are equal. Writing them as $a, a, -2a$ gives $6a^2 = 6$, so $a = \pm 1$. The products are ∓ 2 , and the maximum 2 occurs at permutations of $(2, -1, -1)$.

Line Integrals ▸ Line Integrals with Respect to Arc Length

108 Smooth curves, arc length, and scalar line integrals

Solution 108.1 ■ Answer B

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The speed is $\sqrt{9\sin^2 t + 9\cos^2 t + 16} = 5$ and the density is $4t$. Thus the mass is $\int_0^\pi 20t \, dt = 10\pi^2$.

Solution 108.2 ■ Answer B

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Cauchy–Schwarz for arc-length integration gives $(\int_C \rho \, ds)^2 \leq (\int_C 1 \, ds)(\int_C \rho^2 \, ds) = L \int_C \rho^2 \, ds$. Thus the lower bound is M^2/L , attained by the constant density $\rho = M/L$. The geometry enters through length and the measure ds , and the equality case verifies sharpness.

Solution 108.3 ■ Answer A

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The speed is $\sqrt{1 + 4t^2}$, matching the density. Their product is $1 + 4t^2$, so integration on $[0, 1]$ gives $1 + 4/3 = 7/3$.

Solution 108.4 ■ Answer D

[Back to problem](#)

Each side has unit speed under its coordinate parametrization. The four density integrals are $\int_0^1 x \, dx$, $\int_0^1 (1 - y) \, dy$, $\int_0^1 (1 - x) \, dx$, and $\int_0^1 y \, dy$, each equal to $1/2$. Sum the four smooth pieces to obtain mass 2; orientation does not change it.

Solution 108.5 ■ Answer E

[Back to problem](#)

The speed is one, so the integral is $\int_0^{3\pi} (2 + \cos t) \, dt = 6\pi$. The path goes clockwise through one full turn and an extra half-turn. Reversal does not introduce a minus sign for ds , and replacing the traversal by its geometric image would omit the repeated half-circle.

Line Integrals ▸ The Line Integral of a Vector Field

109 Vector fields, work, and oriented line integrals

Solution 109.1 ■ Answer B

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On the parabola take $(x, y) = (t, t^2)$: the work is $\int_0^1 (t^2 + 2t^3) \, dt = 5/6$. On the line take (t, t) : the work is $\int_0^1 (t + t^2) \, dt = 5/6$. Equality on these two particular paths does not prove path independence.

Solution 109.2 ■ Answer C

[Back to problem](#)

The field along the path is $(-\sin t, \cos t, t)$ and $r' = (-\sin t, \cos t, 1)$. Their dot product is $1 + t$, whose integral is $2\pi + 2\pi^2$.

Solution 109.3 ■ Answer D[Back to problem](#)

The first segment has $dy = 0$, the second contributes $\int_0^1 dy = 1$, and the return diagonal is $(x, y) = (1 - t, 1 - t)$, contributing $-\int_0^1 (1 - t)dt = -1/2$. The sum is $1/2$. A closed path need not have zero work for a general vector field.

Solution 109.4 ■ Answer B[Back to problem](#)

On the unit circle $\|F\| = 1$, so the scalar line integral is the circumference 2π . The field points counterclockwise, opposite to the velocity everywhere in this traversal, so its work is -2π . A vector work integral pairs with signed displacement; a scalar arc-length integral does not.

Solution 109.5 ■ Answer C[Back to problem](#)

Write the work as $\int_C F \cdot T ds \leq \int_C 3 ds = 15$. Equality means the continuous nonnegative function $3 - F \cdot T$ integrates to zero, so it vanishes everywhere. Equality in $F \cdot T \leq \|F\| \leq 3$ forces both magnitude 3 and alignment with T . No global path-independence conclusion follows.

Line Integrals ▸ The Fundamental Theorem of Calculus for Line Integrals

110 Potential functions and conservative fields

Solution 110.1 ■ Answer A[Back to problem](#)

A potential is $\phi = x^2y + \sin x + y^3$. Its values are $2\pi^2 + 8$ and 1, so the integral is $2\pi^2 + 7$.

Solution 110.2 ■ Answer A[Back to problem](#)

The compatibility condition is $2ay = 2by$ for every y , so $a = b$. When this holds, $\phi = axy^2 + x^2 + \sin y$ is a global potential, proving sufficiency directly.

Solution 110.3 ■ Answer D[Back to problem](#)

The sign function is locally constant on D , so its derivatives there are zero. Both gradients equal $(1/x, 0)$. Their difference is 1 on $x > 0$ and -1 on $x < 0$. The usual uniqueness-up-to-one-constant statement requires a connected domain.

Solution 110.4 ■ Answer C[Back to problem](#)

The chain rule shows $F = \nabla(\phi^3/3)$. Therefore the integral is $[\phi(B)^3 - \phi(A)^3]/3 = [8 - (-1)]/3 = 3$. A scalar function of a potential times its gradient can have a new potential without any coordinate integration.

Solution 110.5 ■ Answer A[Back to problem](#)

The globally defined function $\phi = 1/(x^2 + y^2)$ has gradient F everywhere on this domain. The endpoint difference is zero for every closed path, even one enclosing the missing point. A hole permits curl-free fields without potentials, but does not prevent a particular field from having a potential.

Double Integrals ▸ Iterated Integrals and Reversing Order

111 Double integrals, Fubini, and region bounds

Solution 111.1 ■ Answer B[Back to problem](#)

The region is $0 \leq x \leq y \leq 1$. Reverse order to obtain $\int_0^1 \int_0^y e^{y^2} dx dy = \int_0^1 ye^{y^2} dy = (e - 1)/2$.

Solution 111.2 ■ Answer C[Back to problem](#)

The integral is $\int_0^2 \int_0^{2-x} (4-x-y) dy dx$. Equivalently the affine height has its mean value equal to its centroid value $4 - 2/3 - 2/3 = 8/3$; multiplying by area 2 gives $16/3$.

Solution 111.3 ■ Answer D[Back to problem](#)

Integrating first in y gives $\frac{x(x-1)^2}{2}$. Its integral over $0 \leq x \leq 1$ is $\frac{1}{24}$; the sloping boundary must be retained.

Solution 111.4 ■ Answer B[Back to problem](#)

The region is the triangle with vertices $(0, 0)$, $(1, 1)$, $(2, 0)$. For $0 \leq x \leq 1$, its upper boundary is $y = x$; for $1 \leq x \leq 2$, it is $y = 2 - x$. The lower boundary is zero throughout. Reversal requires splitting where the active upper boundary changes.

Solution 111.5 ■ Answer B[Back to problem](#)

The signed integral is $\int_0^1 (1/2 - x^2) dx = 1/6$. Physical volume integrates the absolute value, splitting at $y = x^2$. For each x , the result is $x^4/2 + (1/2 - x^2 + x^4/2) = 1/2 - x^2 + x^4$. Integrating gives $1/2 - 1/3 + 1/5 = 11/30$.

Solution 111.6 ■ Answer C[Back to problem](#)

For fixed positive x , an antiderivative in y is $y/(x^2 + y^2)$, giving inner integral $1/(1 + x^2)$ and $I = \pi/4$. For fixed positive y , an antiderivative in x is $-x/(x^2 + y^2)$, giving $J = -\pi/4$. Absolute integrability fails near the origin: in polar coordinates the absolute integrand times area element is $|\cos 2\theta| dr d\theta/r$. Therefore the hypotheses for unrestricted order interchange do not hold.

Double Integrals ▸ Double Integrals in Polar Coordinates

112 Polar area element and radial integration

Solution 112.1 ■ Answer B

[Back to problem](#)

The region is $-\pi/2 \leq \theta \leq \pi/2$, $0 \leq r \leq 2\cos\theta$. Thus the integral is $4 \int_{-\pi/2}^{\pi/2} \cos^4\theta \, d\theta = 4(3\pi/8) = 3\pi/2$.

Solution 112.2 ■ Answer B

[Back to problem](#)

The polar bounds are $1 \leq r \leq 3$ and $0 \leq \theta \leq \pi$. The transformed integrand is $(1/r^2)r = 1/r$, giving $\pi \int_1^3 dr/r = \pi \log 3$.

Solution 112.3 ■ Answer C

[Back to problem](#)

In polar coordinates the Jacobian adds a factor r . Thus the integral is $2\pi \int_0^1 r e^{-r^2} dr = -\frac{\pi}{e} + \pi$.

Solution 112.4 ■ Answer C

[Back to problem](#)

The integrand becomes $r^2 \cos 2\theta$ and the area element adds another r . The product integral is $(\int_0^2 r^3 dr)(\int_0^{\pi/4} \cos 2\theta \, d\theta) = 4(1/2) = 2$. Cancellation over a full disk cannot be transferred to a sector that lacks the needed symmetry.

Solution 112.5 ■ Answer B

[Back to problem](#)

The radial integral is $2\pi \int_0^{e^{-1}} dr/[r(\log(1/r))^p]$. With $u = \log(1/r)$ it becomes $2\pi \int_1^\infty u^{-p} du$, convergent exactly when $p > 1$, with value $2\pi/(p-1)$. The Jacobian changes the singularity threshold from the one-dimensional expression one might initially see.

Solution 112.6 ■ Answer C

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One petal has $-\pi/6 \leq \theta \leq \pi/6$ and $0 \leq r \leq \cos 3\theta$. Its area is $\frac{1}{2} \int_{-\pi/6}^{\pi/6} \cos^2(3\theta) d\theta = \pi/12$. Three disjoint petals give $\pi/4$. The full parametrization from zero to 2π traces this odd-petal rose twice, so blindly using that interval doubles the area.

Green's Theorem ▸ Circulation and Area

113 Green's theorem, positive orientation, and area

Solution 113.1 ■ **Answer A**[Back to problem](#)

The scalar curl is $5 - (-3) = 8$. The ellipse has area 2π . Counterclockwise circulation is 16π , and reversal gives -16π .

Solution 113.2 ■ **Answer B**[Back to problem](#)

Green gives $\iint 3(x^2 + y^2)dA = 3 \int_0^{2\pi} \int_1^2 r^3 dr d\theta = 45\pi/2$. The inner boundary positive orientation is clockwise.

Solution 113.3 ■ **Answer C**[Back to problem](#)

The area formula is $\frac{1}{2} \int_0^{2\pi} (xy' - yx')dt$. Expansion and $\cos t \cos 2t + \sin t \sin 2t = \cos t$ reduce the integrand to $6 - 6\cos t$. Half its integral is 6π . The isolated cusp at the matching endpoints is compatible with a piecewise smooth boundary.

Solution 113.4 ■ **Answer B**[Back to problem](#)

Positive orientation of the entire annular boundary uses the outer circle counterclockwise and the inner circle clockwise. Thus Green gives $10\pi - I = 2(4\pi - \pi) = 6\pi$, so $I = 4\pi$. Smoothness has only been assumed near the annulus; applying the theorem to the filled disk would require information not given.

Solution 113.5 ■ **Answer B**[Back to problem](#)

The connector contributions cancel because their orientations are opposite. The two circle contributions are their signed areas, π and -4π , for a total of -3π . The walk is not a simple positively oriented boundary of the union, whose ordinary area is 5π ; apply the area identity to the separate simple loops.

Green's Theorem ▸ Path Independence and Gradient Fields

114 Convexity, simple connectivity, and the hole obstruction

Solution 114.1 ■ Answer D

[Back to problem](#)

Along the path $F = (-\sin t/2, \cos t/2)$ and $r' = (-2\sin t, 2\cos t)$, so the dot product is 1. The twice-traversed circle has integral 4π .

Solution 114.2 ■ Answer B

[Back to problem](#)

Vanishing of every closed-path integral makes integrals between endpoints independent of path. Integrating from a fixed base point then defines a potential. The annulus is not simply connected, so equality of cross-partials alone is insufficient.

Solution 114.3 ■ Answer B

[Back to problem](#)

The disk enclosed by this circle avoids the origin. The field is smooth there and has zero scalar curl, so Green's theorem gives zero integral.

Solution 114.4 ■ Answer B

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The segment joining $(-1, 1)$ to $(-1, -1)$ crosses the removed ray, so D is not convex. However polar coordinates with $r > 0$ and $-\pi < \theta < \pi$ identify it continuously with a strip (use $\log r$), so every loop contracts. The single-valued angle $\theta = \text{Arg}(x + iy)$ on this slit plane has gradient F and is a potential.

Solution 114.5 ■ Answer C

[Back to problem](#)

A convex open set is simply connected, so the curl-free field has a potential. The fundamental theorem for line integrals then makes every closed-curve integral zero.

Chapter 3 Review Questions ▸ Vector Identities

115 Vector triple product and cross-product equations

Solution 115.1 ■ Answer C

[Back to problem](#)

The second equation forces $x \perp a$. Substituting the first into it gives $x \times (a \times x) = a\|x\|^2 - x(x \cdot a) = a$, so $\|x\| = 1$. Consequently $\|b\| = \|a \times x\| = \|a\|$. Conversely, if the lengths agree, $x = (b \times a)/\|a\|^2$ is perpendicular to a , has norm one, and satisfies both equations by the triple-product identity.

Solution 115.2 ■ Answer B

[Back to problem](#)

Apply the triple-product identity with $c = a$: $a \times (b \times a) = b(a \cdot a) - a(a \cdot b) = b - a/2$.

Solution 115.3 ■ Answer C[Back to problem](#)

The triple-product identity gives $T^2(v) = a(a \cdot v) - \|a\|^2 v$. Crossing once more with a annihilates the parallel first term, leaving $T^3(v) = -\|a\|^2 T(v)$. The simpler square identity only holds on the subspace perpendicular to a , not for every vector.

Solution 115.4 ■ Answer C[Back to problem](#)

Every cross product with a is perpendicular to a . Conversely, if $a \cdot b = 0$, then $x = -(a \times b)/\|a\|^2$ is a solution by the vector triple-product identity.

Solution 115.5 ■ Answer A[Back to problem](#)

Apply BAC-CAB with first vector $a \times b$, second vector a , and third vector c . The result is $a[(a \times b) \cdot c] - c[(a \times b) \cdot a] = 3a$. The zero second scalar follows from perpendicularity. This connects vector and scalar triple products without coordinates.

Chapter 3 Review Questions ▸ Distance to a Plane

116 Perpendicular distance to a plane

Solution 116.1 ■ Answer B[Back to problem](#)

Divide the second equation by 2 to obtain the same normal: $2x - y + 2z = 9$. The separation is $(9 - 3)/\sqrt{4 + 1 + 4} = 6/3 = 2$.

Solution 116.2 ■ Answer A[Back to problem](#)

The normal is $n = (1, 2, 2)$, with $\|n\|^2 = 9$, and $n \cdot P - 1 = 3$. Reflection subtracts twice the normal displacement: $P - 2(3/9)n = (4/3, -1/3, -4/3)$.

Solution 116.3 ■ Answer C[Back to problem](#)

The projection is $Q = P - [(n \cdot P - 3)/\|n\|^2]n = (\frac{1}{3}, \frac{7}{3}, \frac{7}{3})$. Squaring and summing its coordinates gives 11.

Solution 116.4 ■ Answer D[Back to problem](#)

The distances are $|x|$ and $|x + y|/\sqrt{2}$. Equality gives $(x + y)^2 = 2x^2$, the union of the two angle-bisector planes $y = (-1 \pm \sqrt{2})x$. Omitting normal normalization would give the wrong bisectors.

Solution 116.5 ■ Answer B[Back to problem](#)

The center is at distance $|5 - 11|/3 = 2$ from the plane. Signed normal displacements from points of the sphere range from $-2 - 3 = -5$ to $-2 + 3 = 1$. Since this interval includes zero, the sphere meets the plane and the minimum distance is zero; the maximum is five. Taking $|2 - 3|$ as the minimum would miss the intersection.

Chapter 3 Review Questions ▸ Continuity at the Origin

117 Multivariable limits and continuity

Solution 117.1 ■ **Answer C**[Back to problem](#)

Along $y = mx$ with $m \neq 0$, the expression is $mx/(x^2 + m^2) \rightarrow 0$; on $y = 0$ it is zero. Along $y = x^2$ it is $1/2$. Also $2|x^2y| \leq x^4 + y^2$, so unboundedness is false.

Solution 117.2 ■ **Answer C**[Back to problem](#)

Since $|xy| \leq (x^2 + y^2)/2$, $|f| \leq r^{2-2p}/2$, proving continuity for $p < 1$. Along $y = x$, the magnitude is $2^{-p}|x|^{2-2p}$, which does not tend to zero when $p \geq 1$.

Solution 117.3 ■ **Answer C**[Back to problem](#)

Put $s = x^2 + y^2 > 0$. The expression depends only on s , so every approach reduces to $s \downarrow 0$. Split it as $(\sqrt{1+s} - 1)/s + (1 - \cos \sqrt{s})/s$. Rationalization gives limit $1/2$ for the first term; setting $u = \sqrt{s}$ gives $(1 - \cos u)/u^2 \rightarrow 1/2$ for the second. Their sum is one.

Solution 117.4 ■ **Answer E**[Back to problem](#)

The absolute value is bounded by $x^2 + y^2$, which tends to zero along every path. The oscillating sine factor is controlled by this radial bound.

Solution 117.5 ■ **Answer A**[Back to problem](#)

Within the stated domain, $0 \leq x^2/(x^2 + y^2) \leq x^2/y^2 \leq y^2 \rightarrow 0$. The x -axis path that would give value one in the full punctured plane is not in D . A multivariable limit is tested only along approaches allowed by its domain.

4. Differential Equations

Introduction ▸ Differential equations and initial conditions

118 Differential equations, order, solutions, integral curves, and IVPs

Solution 118.1 ■ **Answer D**[Back to problem](#)

Integrate to get $y' = 3x^2 + C_1$ and $y = x^3 + C_1x + C_2$. The initial data give $C_1 = 2$ and $C_2 = -1$. Thus $y(2) = 8 + 4 - 1 = 11$.

Solution 118.2 ■ **Answer C**[Back to problem](#)

Differentiation yields $(3y^2 + x)y' + y = 0$, so $y'(1) = -1/4$. Differentiating again gives $(3y^2 + x)y'' + 6y(y')^2 + 2y' = 0$. At $(1, 1)$ this is $4y'' + 3/8 - 1/2 = 0$, hence $y'' = 1/32$.

Solution 118.3 ■ **Answer C**[Back to problem](#)

The unknown depends on two independent variables and partial derivatives appear, so it is a PDE. The highest differentiation order is four. Squaring a third derivative does not turn it into a sixth derivative, but it does make the equation nonlinear.

Solution 118.4 ■ **Answer E**[Back to problem](#)

For every $a \geq 0$, define $y(x) = 0$ for $x \leq a$ and $y(x) = (x - a)^3$ for $x \geq a$. The values and first derivatives match at a , and direct substitution verifies the equation everywhere. All satisfy the initial condition and are distinct. The right side is continuous but not locally Lipschitz in y at zero, so standard uniqueness does not apply.

Solution 118.5 ■ **Answer C**[Back to problem](#)

The equation is $(y' - x)^2 = 0$, which forces $y' = x$. Integration gives $y = x^2/2 + C$, and the initial condition gives $C = 1$. The square is algebraic in the first derivative; it does not require a second-order solution method.

Separable Equations ▸ Separation of variables

119 Separable ODEs, equilibrium solutions, and maximal intervals

Solution 119.1 ■ **Answer B**[Back to problem](#)

Separation gives $-1/y = x + C$, and $y(0) = 2$ gives $y = 2/(1 - 2x)$. Its pole is at $x = 1/2$. The largest open interval containing 0 and avoiding the pole is $(-\infty, 1/2)$.

Solution 119.2 ■ **Answer C**[Back to problem](#)

Integrating $dy/[y(1 - y)] = dx$ gives $\log |y/(1 - y)| = x + C$. The initial ratio is $1/2$, hence $y/(1 - y) = e^x/2$ and $y = e^x/(2 + e^x)$. At $e^x = 4$ this is $4/6 = 2/3$.

Solution 119.3 ■ **Answer A**[Back to problem](#)

Multiply by e^y and integrate: $(e^y)' = -1$, so $e^y = 1 - x$. Positivity of the exponential forces $x < 1$ and the solution is $\log(1 - x)$. An absolute value would add an invalid branch for $x > 1$, and a disconnected set would not be a maximal solution interval through zero.

Solution 119.4 ■ **Answer B**[Back to problem](#)

Separation gives $x = H(y) = \int_0^y e^{s^2} ds$. The function H is strictly increasing, with limits $-\infty$ and $+\infty$, so it has a differentiable inverse on all of $\mathbb{R}$. That inverse is the solution; it increases without a finite upper limit. An elementary antiderivative is unnecessary to establish the global behavior.

Solution 119.5 ■ **Answer D**[Back to problem](#)

A constant solution has derivative zero, so its value must be a root of the right side. The distinct roots are $-1, 0, 2$; the repeated factor at zero contributes only one constant solution.

Homogeneous Equations ▸ Scaling and the substitution $y=xv$

120 Homogeneous functions and first-order homogeneous equations

Solution 120.1 ■ **Answer A**[Back to problem](#)

Set $v = y/x$. Then $v + xv' = v + v^2$, so $dv/v^2 = dx/x$. Integration gives $-1/v = \log x + C$. At $x = 1$, $v = 1$, so $C = -1$ and $y = x/(1 - \log x)$. Its maximal positive interval containing 1 is $(0, e)$.

Solution 120.2 ■ **Answer B**[Back to problem](#)

With $y = xv$, $v + xv' = (1 + v^2)/(2v)$, giving $2v dv/(1 - v^2) = dx/x$. Integration yields $-\log |1 - v^2| = \log |x| + C$, so $x(1 - v^2)$ is constant. Rearrangement gives $y^2 = x^2 + Cx$, after renaming the constant. Direct differentiation verifies also the allowed constant-ratio cases.

Solution 120.3 ■ **Answer C**[Back to problem](#)

Euler's identity is $xF_x + yF_y = 3F$. At $(1, 2)$ this gives $1 + 2F_y = 15$, so $F_y = 7$.

Solution 120.4 ■ **Answer A**[Back to problem](#)

Substitute $y = kx$ directly: $k = k^2 + 2k - 2$, so $(k - 1)(k + 2) = 0$. These are precisely the constant-ratio solutions. In the substitution $v = y/x$, dividing the equation $xv' = (v - 1)(v + 2)$ by either factor would discard one of them.

Solution 120.5 ■ **Answer C**[Back to problem](#)

Under $(x, y) \mapsto (tx, ty)$ with $t > 0$, the ratio y/x is unchanged and both quadratic factors acquire t^2 . Thus the third function is multiplied by t^2 . A homogeneous function need not be a polynomial; logarithms of scale-invariant ratios are allowed.

Exact Equations ▸ Total differentials and potentials

121 Exact equations, total differentials, and potential functions

Solution 121.1 ■ Answer B

[Back to problem](#)

The cross derivatives are both $2x$, so the equation is exact. Integrating M gives $F = x^2y + x + h(y)$, and $F_y = x^2 + h'(y) = x^2 + 3y^2$ gives $h = y^3$. Its value at $(1, 1)$ is 3.

Solution 121.2 ■ Answer A

[Back to problem](#)

The given dx coefficient is the x derivative of $x \cos(xy)$. Thus a potential has form $F = x \cos(xy) + h(y)$, giving $N = -x^2 \sin(xy) + h'(y)$. The condition at $x = 0$ forces $h'(y) = y^2$.

Solution 121.3 ■ Answer A

[Back to problem](#)

The potential is $y^2 - x^2$, and the initial value fixes its level at one. The two branches are disconnected; the positive branch alone contains $(0, 1)$. It is smooth for every real x and never reaches $y = 0$, giving the maximal graph solution. A level equation can contain more than the initial curve.

Solution 121.4 ■ Answer B

[Back to problem](#)

The potential is $x^2 + y^2$, so the level through the point is a smooth circle with a vertical tangent. A C^1 graph through $(1, 0)$ would have to satisfy $2x + 2yy' = 0$, which at that point would say $2 = 0$. It can instead be written locally as $x = \sqrt{1 - y^2}$.

Solution 121.5 ■ Answer C

[Back to problem](#)

The exact relation integrates to $y^2 - x^2 = C$, and the initial condition gives $C = 0$. Away from zero each connected side must choose $y = x$ or $y = -x$. Differentiability at zero requires the same slope on both sides, leaving exactly $y = x$ and $y = -x$. The singular initial point has $M = N = 0$, so the regular local-graph theorem does not assert uniqueness there.

Solution 121.6 ■ Answer C

[Back to problem](#)

The form splits as $(x dx + y dy)/r^2 + (-y dx + x dy)/r^2 = dr/r + d\theta$. Thus a local potential is $\log r + \theta$ and its levels are spirals $r = e^{C-\theta}$. Around one counterclockwise circle the form integrates to 2π , so no single-valued global potential exists on the punctured plane despite equal cross-partials.

Nonexact Equations and Integrating Factors ▸ Factors depending on one variable

122 Integrating factors for nonexact equations

Solution 122.1 ■ Answer D

[Back to problem](#)

Here $M_y = 3$ and $N_x = 1$, so $(M_y - N_x)/N = 2/x$. Hence $\mu = e^{\int 2/x dx} = x^2$. The resulting potential is $x^3y + x^4/4$.

Solution 122.2 ■ Answer B

[Back to problem](#)

The ratio $(N_x - M_y)/M = (2 - 1)/y = 1/y$, so $\mu(y) = y$. The multiplied form is $y^2 dx + (2xy + y^2) dy = d(xy^2 + y^3/3)$. Thus the second expression is constant.

Solution 122.3 ■ Answer D

[Back to problem](#)

The function $y = 0$ makes the multiplied equation zero identically, but its derivative is zero rather than one. The first three functions solve both equations, while $y = 2x$ fails the multiplied equation except at one point. Multiplication by a factor that vanishes can add solutions.

Solution 122.4 ■ Answer A

[Back to problem](#)

Multiplication by $x + y$ gives $y dx + x dy = d(xy)$. This nonzero factor depends on both variables. In fact the x-only criterion gives $(x - y)/[x(x + y)]$ and the y-only criterion gives $(y - x)/[y(x + y)]$, neither a function of the required single variable. Failure of those shortcuts does not mean no integrating factor exists.

Solution 122.5 ■ Answer B

[Back to problem](#)

Multiplication by e^{3x} changes the left side to $(e^{3x}y)'$. Its derivative contributes the required $3e^{3x}y$ term.

First-Order Linear Equations ▸ Standard form and integrating factor

123 First-order linear ODEs and the product-derivative method

Solution 123.1 ■ Answer C

[Back to problem](#)

The integrating factor is x^2 , so $(x^2y)' = x^4$ and $y = x^3/5 + C/x^2$. From $y(1) = 1$ we get $C = 4/5$. Thus $y(2) = 8/5 + 1/5 = 9/5$.

Solution 123.2 ■ Answer B

[Back to problem](#)

Multiplication by x gives $(xy)' = x \cos x$. Integrating yields $xy = x \sin x + \cos x + C$. A finite limit at zero requires $C = -1$, canceling the constant numerator. The remaining expression tends to 0.

Solution 123.3 ■ Answer A[Back to problem](#)

After multiplication by $1 + x$, the left side becomes $(1 + x)^2 y' + 2(1 + x)y = [(1 + x)^2 y]'$. The standard normalized equation has integrating factor $(1 + x)^2$, but normalization first divides the whole equation by $1 + x$. The net multiplier for the original unnormalized equation is therefore $1 + x$.

Solution 123.4 ■ Answer B[Back to problem](#)

Reverse the variables to get the linear equation $dx/dy - 2x = y$. Its solution is $x = Ce^{2y} - y/2 - 1/4$. The condition $x(1) = 0$ gives $C = 3e^{-2}/4$, so $x(2) = (3e^2 - 5)/4$. Along $1 \leq y \leq 2$, $dx/dy = [3e^{2(y-1)} - 1]/2 > 0$, so the inverse graph remains valid up to the requested point.

Solution 123.5 ■ Answer B[Back to problem](#)

A particular solution is $(2 \sin x - \cos x)/5$, as direct substitution confirms. The general solution adds Ce^{-2x} . Periodicity requires $C(e^{-4\pi} - 1) = 0$, hence $C = 0$. Therefore $y(0) = -1/5$. The transient exponential cannot be part of a nonzero periodic component.

Higher-Order Linear Equations with Constant Coefficients ▸ Linear structure and general solutions

124 Linear differential operators, superposition, and solution spaces

Solution 124.1 ■ Answer B[Back to problem](#)

Linearity gives $L[2u - v] = 2f - 2f = 0$. No product rule or square rule implies analogous cancellation for the nonlinear expressions. The other linear combinations give nonzero multiples of f .

Solution 124.2 ■ Answer B[Back to problem](#)

The homogeneous solution space is precisely the span of e^x, e^{-x} . Since $\cosh x = (e^x + e^{-x})/2$ lies in that span, adding it to x^2 preserves the forcing term. The other added terms do not lie in the stated homogeneous space.

Solution 124.3 ■ Answer D[Back to problem](#)

Relative to the independent basis e^x, e^{-x} , the coefficient matrix is $\begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}$, with determinant $1 - a^2$. It is invertible precisely when $a \neq \pm 1$. At either exceptional value the two functions are proportional, so two listed solutions do not span the full two-dimensional space.

Solution 124.4 ■ Answer A[Back to problem](#)

Expanding the first equation gives $e^x y'' + x^2 y' + 2xy = 0$: every appearance of the unknown is linear, and the forcing is zero. The second and fifth equations are nonlinear; the third is nonhomogeneous; the fourth has constant coefficients. Coefficients may depend on the independent variable in a linear equation.

Solution 124.5 ■ Answer C[Back to problem](#)

On each side of zero, separation gives $y = C_- x^2$ for $x < 0$ and $y = C_+ x^2$ for $x > 0$. The equation at zero requires $y(0) = 0$, and any two constants give matching value and first derivative zero there. Thus the space has two independent parameters. The usual dimension-equals-order theorem does not apply across a zero of the leading coefficient.

Higher-Order Linear Equations with Constant Coefficients ▷ Auxiliary equation: distinct real roots

125 Characteristic polynomial and distinct exponential modes

Solution 125.1 ■ Answer A[Back to problem](#)

The characteristic polynomial is $(r - 3)(r + 2)$, so $y = Ae^{3x} + Be^{-2x}$. Boundedness forces $A = 0$, and the initial condition gives $B = 4$. At $x = \log 2$, $y = 4 \cdot 2^{-2} = 1$.

Solution 125.2 ■ Answer B[Back to problem](#)

The characteristic polynomial is $r(r - 2)(r + 2)$, so solutions are $A + Be^{2x} + Ce^{-2x}$. Boundedness at $+\infty$ forces $B = 0$ and at $-\infty$ forces $C = 0$. Only the one-dimensional constant subspace remains.

Solution 125.3 ■ Answer C[Back to problem](#)

The characteristic roots are $1, -2$, so $y = Ae^x + Be^{-2x}$. The initial equations $A + B = 2$ and $A - 2B = 1$ give $A = 5/3$, $B = 1/3$. Hence $y(1) = (5e + e^{-2})/3$.

Solution 125.4 ■ Answer B[Back to problem](#)

The monic characteristic polynomial has roots $1, 2$ and a third root r . Its constant term is $-2r = -6$, so $r = 3$. Expanding $(t - 1)(t - 2)(t - 3) = t^3 - 6t^2 + 11t - 6$ gives $a = -6$, $b = 11$. This reconstructs the operator from its modes.

Solution 125.5 ■ Answer B[Back to problem](#)

The solution is $Ae^x + Be^{2x} = e^x(A + Be^x)$. The first factor never vanishes. If $B \neq 0$, the second is strictly monotone and has at most one zero; if $B = 0$ a nonzero solution has none. The example $e^x - e^{2x}$ attains the bound, with its only zero at zero.

Higher-Order Linear Equations with Constant Coefficients ▸ Auxiliary equation: repeated roots

126 Repeated characteristic roots and polynomial factors

Solution 126.1 ■ **Answer A**[Back to problem](#)

The characteristic root 2 is double, so $y = (A + Bx)e^{2x}$. The data give $A = 1$ and $B + 2A = 0$, hence $B = -2$. Therefore $y(1) = -e^2$.

Solution 126.2 ■ **Answer C**[Back to problem](#)

The characteristic polynomial is $r^3(r - 1)^2$. Its general solution is $A + Bx + Cx^2 + (D + Ex)e^x$. The polynomial solutions are precisely the span of $1, x, x^2$, a three-dimensional space.

Solution 126.3 ■ **Answer D**[Back to problem](#)

The x^3e^{2x} term requires characteristic root 2 of multiplicity at least four. The xe^{-x} term requires root -1 of multiplicity at least two. Distinct exponential factors cannot cancel each other under a constant-coefficient operator. Thus the minimal characteristic polynomial is $(r - 2)^4(r + 1)^2$, of degree six.

Solution 126.4 ■ **Answer C**[Back to problem](#)

Every solution is $(A + Bx + Cx^2)e^{-x}$. All three modes decay at positive infinity and are bounded on the nonnegative half-line. At negative infinity, every nonzero polynomial times e^{-x} is unbounded, so only the zero solution is bounded on all of $\mathbb{R}$. The two subspace dimensions are three and zero.

Solution 126.5 ■ **Answer C**[Back to problem](#)

The operator factors as $(D - 1)^2(D + 2)$. Acting on e^xv shifts each D to $D + 1$, giving $e^xD^2(D + 3)v$. With $v = x^2$, this is $e^xD^2(2x + 3x^2) = 6e^x$. The double root at one annihilates e^x and xe^x , but not the next polynomial multiple.

Higher-Order Linear Equations with Constant Coefficients ▸ Auxiliary equation: nonreal roots

127 Complex characteristic roots, oscillation, and damping

Solution 127.1 ■ **Answer B**[Back to problem](#)

The roots are $-2 \pm 3i$, so $y = e^{-2x}(A \cos 3x + B \sin 3x)$. The initial data give $A = 0$, $B = 1$. Since the exponential never vanishes, the first positive zero is where $3x = \pi$, namely $\pi/3$.

Solution 127.2 ■ **Answer C**[Back to problem](#)

The characteristic polynomial is $(r^2 + 1)^2$, so i and $-i$ each have multiplicity 2. Thus both sine and cosine are accompanied by their x multiples, giving the stated four-parameter family.

Solution 127.3 ■ **Answer B**[Back to problem](#)

The solution is $e^{-x} \sin 2x$. Its derivative is $e^{-x}(2 \cos 2x - \sin 2x)$, which first vanishes when $2x = \arctan 2$. It changes from positive to negative there, giving a local maximum. Ignoring the exponential derivative would incorrectly put the maximum at $\pi/4$.

Solution 127.4 ■ **Answer C**[Back to problem](#)

The solution is $e^{-x}(A \cos 2x + B \sin 2x)$ with $(A, B) \neq (0, 0)$. The sinusoidal factor has infinitely many zeros, and the exponential never vanishes. Yet the whole solution tends to zero at positive infinity. A periodic function with this limit would have to be identically zero, so no nonzero solution is periodic.

Solution 127.5 ■ **Answer B**[Back to problem](#)

Write the roots as $\alpha \pm i\beta$. Consecutive zeros are separated by π/β , so $\beta = 2$. Consecutive extrema have the same spacing and equal absolute trigonometric factors, so their amplitude ratio is $e^{\alpha\pi/2} = e^{-\pi}$, giving $\alpha = -2$. The polynomial is $(r - \alpha)^2 + \beta^2 = r^2 + 4r + 8$.

Higher-Order Linear Equations with Constant Coefficients ▸ Particular solutions of nonhomogeneous equations

128 Particular-solution guessing, undetermined coefficients, and resonance

Solution 128.1 ■ Answer C

[Back to problem](#)

The operator factors as $(D - 1)(D - 2)$, so e^x is a homogeneous mode. Put $y = Axe^x$. Direct substitution gives $L[Axe^x] = -Ae^x$, so $A = -1$.

Solution 128.2 ■ Answer C

[Back to problem](#)

Resonance suggests $Ax \sin x$. Its second derivative plus itself is $2A \cos x$, so $A = 1/2$. This function and its first derivative vanish at 0, so no homogeneous correction is needed.

Solution 128.3 ■ Answer B

[Back to problem](#)

Use the full quadratic trial $Ax^2 + Bx + C$. Substitution gives $-Ax^2 - Bx + (2A - C) = x^2$, so $A = -1$, $B = 0$, and $C = -2$. A trial containing only the leading quadratic term misses the constant produced by differentiation.

Solution 128.4 ■ Answer C

[Back to problem](#)

The operator is $(D - 2)^2$. Writing $y = e^{2x}v$ reduces the equation to $v'' = 1$, so one may take $v = x^2/2$. Both e^{2x} and xe^{2x} are homogeneous modes; the double root requires two powers of x , not one.

Solution 128.5 ■ Answer A

[Back to problem](#)

The general solution is $A \cos x + B \sin x - \frac{1}{2}x \cos x$. At $x = 2\pi n$ its value is $A - \pi n$, which is unbounded. No bounded homogeneous correction can cancel the growing resonant term. A bounded forcing term need not produce a bounded response.

Solution 128.6 ■ Answer B

[Back to problem](#)

The operator sends x to x and sends $\sin 2x$ to $-3 \sin 2x$. Linearity therefore makes $x - (1/3) \sin 2x$ a particular solution for the sum. The frequency two forcing is nonresonant for the homogeneous frequency one equation.

Chapter 4 Review Questions ▸ Growth proportional to population

129 Exponential growth, decay, doubling time, and half-life

Solution 129.1 ■ **Answer B**[Back to problem](#)

The three-hour multiplier is $3/2$, so $k = \log(3/2)/3$. Set $e^{kt} = 4$ to obtain $t = \log 4/k = 3 \log 4 / \log(3/2)$.

Solution 129.2 ■ **Answer B**[Back to problem](#)

The rate is $k = \frac{1}{2} \log(4/5)$. Therefore $N(5)/N(0) = e^{5k} = (4/5)^{5/2}$. A fixed percentage loss per two days is multiplicative, not a linear loss of ten percentage points per day.

Solution 129.3 ■ **Answer D**[Back to problem](#)

The ratio is $2^{-t/2}/2^{-t/3} = 2^{-t/6}$. Set this equal to $1/4 = 2^{-2}$ to get $t = 12$. Comparing two amounts uses the difference of their exponential rates, not either sample's individual half-life.

Solution 129.4 ■ **Answer B**[Back to problem](#)

The remaining fraction is $[2^{-t} + 2^{-t/2}]/2$. Put $z = 2^{-t/2} > 0$; the half-total equation becomes $z^2 + z = 1$, so $z = (\sqrt{5} - 1)/2$. Hence $t = -2 \log_2 z = 2 \log_2((1 + \sqrt{5})/2)$. Averaging the two half-lives does not give the mixture's half-time.

Solution 129.5 ■ **Answer C**[Back to problem](#)

The periodic factor repeats after one unit, so the third function has the required multiplier. Its logarithmic derivative is $\log 3 + 2\pi \cos 2\pi t / (2 + \sin 2\pi t)$, which is not constant. The first two functions do have constant proportional rates, and the last two have the wrong unit-time multiplier.

Chapter 4 Review Questions ▸ Area conditions and differential modeling

130 Converting integral or geometric conditions into ODEs

Solution 130.1 ■ **Answer D**[Back to problem](#)

Differentiating gives $f = \frac{1}{4}(f + xf')$, so $xf' = 3f$ and $f = Cx^3$. The value at 1 gives $C = 2$. This function satisfies the original integral identity, and $f(2) = 16$.

Solution 130.2 ■ **Answer C**[Back to problem](#)

Differentiating once yields $f'(x) = \int_0^x f(t) dt$ and again gives $f'' = f$. The original identity and its first derivative imply $f(0) = 2$ and $f'(0) = 0$. Thus $f = 2 \cosh x$, which substitution or integration verifies.

Solution 130.3 ■ **Answer E**[Back to problem](#)

Differentiation gives $f' = 2f$ and evaluation at zero gives $f(0) = 1$. Thus $f = e^{2x}$, so $f(1) = e^2$.

Solution 130.4 ■ **Answer C**[Back to problem](#)

The tangent equation gives x-intercept $x - f(x)/f'(x)$. Equating this to $2x$ yields $xf' = -f$. Separation gives $f = C/x$, and the given point gives $C = 2$. Substitution verifies the required intercept; here the geometric condition supplies the differential equation.

Solution 130.5 ■ **Answer B**[Back to problem](#)

Differentiate to get $f = f + xf'$, so $f' = 0$ and $f = C$ on the interval. Substitution in the original relation gives $C(x - 1) = Cx$, forcing $C = 0$. Differentiation alone leaves an arbitrary constant that the original base-point condition eliminates.

Chapter 4 Review Questions ▷ Power-product integrating factors

131 Monomial integrating factors and coefficient matching

Solution 131.1 ■ Answer A

[Back to problem](#)

Exactness gives $2(n+2) = 2(m+1)$ for the $x^m y^{n+1}$ terms and $3(n+1) = 4m$ for the $x^{m-1} y^n$ terms. The first gives $m = n+1$; substituting into the second gives $n = -1$, $m = 0$. Thus $m+n = -1$, and the integrating factor is $1/y$.

Solution 131.2 ■ Answer A

[Back to problem](#)

The multiplied form is $(2xy + 1)dx + x^2 dy = d(x^2 y + x)$. Thus the stated expression is constant. The factor x is nonzero on the specified region, so the transformed equation is equivalent there.

Solution 131.3 ■ Answer C

[Back to problem](#)

The cross derivatives after multiplication are $(n+1)x^m y^n$ and $(m+1)x^m y^n$, so exactness is equivalent to $m = n$. Coefficient matching need not give a unique pair. When $m = n = -1$, a potential is $\log(xy)$; otherwise it is $(xy)^{m+1}/(m+1)$.

Solution 131.4 ■ Answer E

[Back to problem](#)

Dividing the exactness condition by the nonzero factor gives $1 + n + n/y = (m+2)x + m/x$. Independence of y on the right forces $n = 0$. Then $1 = (m+2)x + m/x$ for every $x > 0$, or $x = (m+2)x^2 + m$, an impossible polynomial identity. Failure of this suggested monomial form does not exclude other forms of integrating factor.

Solution 131.5 ■ Answer A

[Back to problem](#)

On the horizontal axis $y = 0$, the differential relation is satisfied because $dy = 0$. The factor y^{-2} is undefined there, so its transformed level family cannot include that solution. Each of the other lines has constant x/y away from the origin and is represented by the transformed family.

Chapter 4 Review Questions ▸ Orthogonal trajectories

132 Orthogonal trajectories and eliminating a family parameter

Solution 132.1 ■ **Answer B**[Back to problem](#)

Differentiate $y = cx^2$ and eliminate c to get $y' = 2y/x$. The orthogonal slope is $-x/(2y)$. Thus $2y \, dy = -x \, dx$, which integrates to $y^2 = -x^2/2 + C$, or $x^2 + 2y^2 = C$.

Solution 132.2 ■ **Answer A**[Back to problem](#)

Differentiate the given family: $x + yy' = cy'$. Eliminating $c = (x^2 + y^2)/(2y)$ gives $y' = 2xy/(x^2 - y^2)$. The orthogonal slope is $(y^2 - x^2)/(2xy)$. Differentiating $x^2 + y^2 = 2kx$ and eliminating k produces exactly this slope.

Solution 132.3 ■ **Answer B**[Back to problem](#)

The normal to the first family is $(2x, -2y)$, and the normal to $xy = K$ is (y, x) . Their dot product is $2xy - 2xy = 0$. Rotating both normals by a right angle preserves their dot product, so the tangent directions are also perpendicular. This checks orthogonality directly without solving a slope ODE.

Solution 132.4 ■ **Answer C**[Back to problem](#)

Circle tangents are perpendicular to the radial vector (x, y) , so an orthogonal trajectory has radial tangent direction and constant polar angle. Its components are rays along lines through the origin. The vertical line is essential; expressing everything as $y(x)$ can omit it when taking reciprocal slopes.

Solution 132.5 ■ **Answer E**[Back to problem](#)

At $(0, 1)$ the parameter must satisfy $c^2 = 1$, so the two members have slopes 1 and -1 . A perpendicular direction to one has slope -1 , while a perpendicular direction to the other has slope 1. No single nonzero direction satisfies both. Eliminating a family parameter can give multiple slope branches rather than one direction field.

Chapter 4 Review Questions ▸ First-order partial differential equations

133 Linear first-order PDEs, characteristic directions, and level curves

Solution 133.1 ■ **Answer E**[Back to problem](#)

The invariant along direction $(3, -2)$ is $2x + 3y$, so write $u = F(2x + 3y)$. The data give $F(3y) = e^y$, hence $F(s) = e^{s/3}$. Therefore $u(3, 1) = e^{(6+3)/3} = e^3$.

Solution 133.2 ■ **Answer C**[Back to problem](#)

Along the characteristic through $(1, 2)$, trace back by $(2, 2)$ to $(-1, 0)$. The initial value there is 1, and advancing parameter time 2 adds $2 \cdot 2 = 4$. Thus the value is 5. Equivalently $u(x, y) = (x - y)^2 + 2y$.

Solution 133.3 ■ **Answer B**[Back to problem](#)

The diagonal is a characteristic. Along it, the chain rule gives $d[u(t, t)]/dt = u_x + u_y = 0$, contradicting data I. Data II are compatible but constrain only one characteristic: every $u = F(x - y)$ with $F(0) = 1$ works. Characteristic data can be inconsistent or insufficient for uniqueness.

Solution 133.4 ■ **Answer A**[Back to problem](#)

Characteristics satisfy $x' = x$ and $y' = -y$, so xy stays constant and $u' = xu_x - yu_y = 0$. Each positive hyperbola $xy = k$ is connected and traversed by a characteristic, so u depends only on $k = xy$. Direct differentiation also verifies that every $F(xy)$ solves the PDE.

Solution 133.5 ■ **Answer C**[Back to problem](#)

Set $h(y) = e^{1-1/y^2}$ for $y < 0$ and $h(y) = 0$ for $y \geq 0$. Define $u = h(y)$ on the part with $x > 0$ and $u = 0$ elsewhere in D . Flatness of h at zero makes this smooth, and $u_x = 0$ locally everywhere. The two prescribed values hold. At negative heights the missing half-axis splits each horizontal characteristic slice; connectedness of the whole domain alone does not reconnect those slices.

5. Linear Algebra

Solutions of Linear Systems ▸ Consistency and Number of Solutions

134 Linear equations and consistent systems

Solution 134.1 ■ **Answer A**[Back to problem](#)

The second equation is compatible with the first exactly when $a = 2$. The first and third equations are independent and then leave one free variable, for every real b .

Solution 134.2 ■ **Answer C**[Back to problem](#)

$A(3u - 2v) = 3b - 2b = b$. The coefficients sum to one. None of the other choices has this guarantee when $b \neq 0$.

Solution 134.3 ■ **Answer C**[Back to problem](#)

Choose x_2, x_3, x_4 arbitrarily, with three choices each. The equation then determines exactly one residue for x_1 . Thus there are $3^3 = 27$ solutions. The zero-one-infinity rule for real systems does not apply over this finite field.

Solution 134.4 ■ **Answer D**[Back to problem](#)

Inconsistency is possible, for example with equations $x_1 = 0$ and $x_1 = 1$. If the system is consistent, its rank is at most two, so at least two of the four unknowns are free. It therefore has infinitely many solutions, never exactly one.

Solution 134.5 ■ **Answer E**[Back to problem](#)

Solvability for every right side means the columns span all of $\mathbb{R}^m$, so row rank is m . Rank cannot exceed the number of columns, hence $m \leq n$.

Matrices and Matrix Algebra ▸ Dimensions and Matrix Operations

135 Matrices, entries, and triangular form

Solution 135.1 ■ **Answer B**[Back to problem](#)

Only entries strictly above the diagonal remain free. Their number is $4 + 3 + 2 + 1 = 10$. The fixed diagonal does not contribute free choices.

Solution 135.2 ■ **Answer C**[Back to problem](#)

Being both triangular types forces A to be diagonal. Each of its four diagonal entries must square to 1, giving two independent choices per entry and $2^4 = 16$ matrices.

Solution 135.3 ■ **Answer B**[Back to problem](#)

Adding the equations gives $3A = C + D$, and then $3B = C - 2D$. Consequently $A + 2B = (C + D + 2C - 4D)/3 = C - D$. These operations act entrywise and require equal dimensions; rectangular matrices cause no difficulty.

Solution 135.4 ■ **Answer B**[Back to problem](#)

Above the diagonal, L is zero, so U must copy M . On and below the diagonal, U is zero, so L must copy M . This constructs a decomposition and forces every entry, proving uniqueness. Without the zero-diagonal restriction the diagonal could be split in many ways.

Solution 135.5 ■ **Answer C**[Back to problem](#)

Reversing both indices sends a position above the diagonal to one below it. The latter is zero by triangularity, so every off-diagonal entry is zero. On the diagonal the symmetry pairs positions 1, 9; 2, 8; 3, 7; and 4, 6, leaving position 5 alone. There are five independent diagonal values.

Matrices and Matrix Algebra ▷ The Transpose

136 Transpose and row-column duality

Solution 136.1 ■ **Answer C**[Back to problem](#)

$A^T A x = 0$ implies $0 = x^T A^T A x = \|Ax\|^2$, hence $Ax = 0$; the converse is immediate. Both nullspaces have dimension $5 - 2 = 3$.

Solution 136.2 ■ **Answer B**[Back to problem](#)

Apply the reversed-order rule twice: $(ABC)^T = C^T(AB)^T = C^T B^T A^T$. One can also verify the entry sums directly; preserving the original order is generally false.

Solution 136.3 ■ **Answer C**[Back to problem](#)

The trace of $A^T A$ is the sum of the squares of all entries of A , since each diagonal entry is a column squared norm. Adding them gives 16.

Solution 136.4 ■ **Answer C**[Back to problem](#)

Ordinary transposition gives $A^T A = 1 + i^2 = 0$. Conjugate transposition gives $A^* A = 1 + (-i)i = 2$. Thus the nonzero matrix A has $A^T A = 0$; the real nullspace identity fails for ordinary transpose over complex scalars, while conjugation restores positivity.

Solution 136.5 ■ **Answer D**[Back to problem](#)

The expression is a one-by-one matrix, hence equals its transpose. Reversing the factors gives $(x^T A x)^T = x^T A^T x = -x^T A x$. A real scalar equal to its negative is zero. This argument uses transposition rather than entry-by-entry expansion.

Matrices and Matrix Algebra ▸ Matrix Multiplication

137 Matrix multiplication and noncommutativity

Solution 137.1 ■ **Answer C**
[Back to problem](#)

The difference is $AB + BA$. Compute $AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ and $BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$; both traces equal 3, giving 6.

Solution 137.2 ■ **Answer D**
[Back to problem](#)

For $X = (x_{ij})$, equality $DX = XD$ forces $x_{12} = x_{21} = 0$ because the diagonal entries of D differ. Exactly the displayed diagonal choice satisfies this.

Solution 137.3 ■ **Answer A**
[Back to problem](#)

The commutator is $AB - BA = \begin{bmatrix} 4 & 0 \\ 0 & -4 \end{bmatrix}$. Squaring this matrix and taking its trace gives 32. Although a commutator has trace zero, its square need not.

Solution 137.4 ■ **Answer B**
[Back to problem](#)

Every column of B lies in the one-dimensional nullspace of A , so $\text{rank } B \leq 1$. Equality occurs with $A = \text{diag}(1, 1, 0)$ and $B = \text{diag}(0, 0, 1)$. A zero product does not force either matrix to be zero.

Solution 137.5 ■ **Answer B**
[Back to problem](#)

Associativity gives $M^2 = u(v^T u)v^T = -2M$. Induction yields $M^k = (-2)^{k-1}M$, hence $M^5 = 16M$. The middle product is a scalar; reversing the original outer product without checking dimensions would be invalid.

Matrices and Matrix Algebra ▸ Identity Matrices and Inverses

138 Identity matrices and one-sided inverses

Solution 138.1 ■ **Answer B**
[Back to problem](#)

$(BA)^2 = B(AB)A = BA$. Also $\text{rank}(BA) \leq 2 < 3$, so it is singular. In fact its rank is 2; rectangular factors cannot produce I_3 in the reverse order.

Solution 138.2 ■ **Answer D**
[Back to problem](#)

Multiply the original equation on the left by A^{-1} and on the right by B^{-1} . This gives $X = A^{-1}CB^{-1}$; moving these factors across C is not justified.

Solution 138.3 ■ **Answer C**[Back to problem](#)

The equation fixes the first two rows of B as $(1, 0)$ and $(0, 1)$, while its last row is arbitrary, say (s, t) . The sum is $2 + s^2 + t^2$, minimized at $s = t = 0$. A rectangular matrix can have an entire family of right inverses.

Solution 138.4 ■ **Answer A**[Back to problem](#)

Invert a product by reversing its factor order: the result is $C^{-1}(B^{-1})^{-1}(A^T)^{-1} = C^{-1}B(A^{-1})^T$. Direct multiplication in either order cancels adjacent inverse pairs.

Solution 138.5 ■ **Answer C**[Back to problem](#)

The left-inverse equation forces $\text{rank } A = n$, and the right-inverse equation forces $\text{rank } A = m$, so $m = n$. Associativity also gives $L = L(AR) = (LA)R = R$. Having both one-sided inverses forces the unique two-sided inverse.

Matrices and Matrix Algebra ▸ Matrix Powers and Polynomial Identities

139 Matrix powers and polynomial identities

Solution 139.1 ■ **Answer C**[Back to problem](#)

Using $A^2 = 3A - 2I$, obtain $A^3 = 7A - 6I$ and $A^4 = 7(3A - 2I) - 6A = 15A - 14I$.

Solution 139.2 ■ **Answer B**[Back to problem](#)

Rearrange to $A(-A^2 - 2A + I) = 3I$. Dividing by 3 supplies a right inverse, which for a square matrix is its inverse.

Solution 139.3 ■ **Answer D**[Back to problem](#)

Expand the given identity as $A^2 - 3A + 2I = 0$. Then $(2I - A)^2 = 4I - 4A + A^2 = 2I - A$. The matrix $\text{diag}(1, 2)$ disproves the other four claims. A zero product need not make either factor zero.

Solution 139.4 ■ **Answer C**[Back to problem](#)

Multiplication gives $(I - N)(I + N + N^2) = I - N^3 = I$, and the factors commute because both are polynomials in N . The nonzero N^2 term is needed; N itself cannot be invertible. This is a terminating geometric-series identity.

Solution 139.5 ■ **Answer D**[Back to problem](#)

Take $A = I_2$ and $B = \text{diag}(1, 2)$. They commute, but every polynomial in I_2 is a scalar multiple of I_2 , so none equals B . The remaining claims follow by distributing products of powers; for the last, transposition reverses identical factors and leaves each power unchanged.

Gaussian Elimination ▸ Row Operations and Echelon Form

140 Gaussian elimination and echelon form

Solution 140.1 ■ **Answer C**
[Back to problem](#)

Subtract the first two equations from the third: $(a - 4)z = 1$. When $a = 4$ this is impossible. For $a \neq 4$, z is determined and the first two equations determine x, y uniquely.

Solution 140.2 ■ **Answer A**
[Back to problem](#)

The pivot variables are x_1, x_3 . Put $x_2 = s, x_4 = t$. The rows give $x_1 = 3 - 2s + t$ and $x_3 = 2 - 4t$, with both parameters arbitrary.

Solution 140.3 ■ **Answer E**
[Back to problem](#)

The row represents $0 = 2$, which no choice of variables can satisfy. The corresponding homogeneous row would instead be $0 = 0$ and would pose no inconsistency.

Solution 140.4 ■ **Answer B**
[Back to problem](#)

The original row space is the plane of vectors $(s, 2s, t)$. In B both rows lie in it and are independent: their coefficient pairs $(2, 1)$ and $(1, 1)$ have nonzero determinant. Hence their reduced row echelon form is the original matrix. C has rank one; A, D, and E contain a row outside that plane. Equal rank alone is not row equivalence.

Solution 140.5 ■ **Answer D**
[Back to problem](#)

The row transformation has matrix $\begin{pmatrix} 1 & 1 \\ 1 & t \end{pmatrix}$ and determinant $t - 1$. For $t \neq 1$ it is invertible, so the original equations can be recovered. At $t = 1$ both new rows are the sum of the old ones; for example $x = 0, y = 0$ becomes two copies of $x + y = 0$, enlarging the solution set.

Gaussian Elimination ▸ Free Variables and Homogeneous Systems

141 Homogeneous systems and affine solution spaces

Solution 141.1 ■ **Answer D**
[Back to problem](#)

The directions in the solution set form $\ker A$, whose dimension is number of columns minus rank, $7 - 3 = 4$. The number of equations is not the number of independent constraints.

Solution 141.2 ■ **Answer D**
[Back to problem](#)

The squared norm is $(1 + t)^2 + (2 - t)^2 + t^2 = 5 - 2t + 3t^2$, minimized at $t = 1/3$. Equivalently the minimizing solution is perpendicular to the homogeneous direction $(1, -1, 1)$. Minimizing the norm is the same as minimizing its square.

Solution 141.3 ■ Answer A[Back to problem](#)

Subtracting $Ax_0 = b$ from $Ax = b$ gives $A(x - x_0) = 0$. Conversely any nullspace vector can be added to x_0 , yielding precisely the stated affine plane.

Solution 141.4 ■ Answer B[Back to problem](#)

The three specified solutions say that all three columns of A equal b . Therefore $Ax = (x_1 + x_2 + x_3)b$, which equals the nonzero vector b exactly when the coordinate sum is one. This reconstructs the affine plane from known solutions.

Solution 141.5 ■ Answer C[Back to problem](#)

The solution set is $x_p + \ker A$, with nullity two. Its linear span is $\text{span}\{x_p\} + \ker A$: differences of solutions give every kernel vector. Since $Ax_p = b \neq 0$, x_p is outside the kernel, increasing the dimension to three. The affine dimension of the solution set itself is only two.

Solving Matrix Equations Using an Inverse ▸ The Two-by-Two Formula

142 Inverse solution formula and the two-by-two inverse

Solution 142.1 ■ Answer D[Back to problem](#)

The determinant is 1, so $A^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. The product is $\begin{pmatrix} 2 & -2 \\ -3 & 4 \end{pmatrix}$, whose entries sum to 1.

Solution 142.2 ■ Answer D[Back to problem](#)

The coefficient determinant is $t^2 - 1$. It is nonzero exactly for $t \neq \pm 1$. At $t = 1$ there are infinitely many solutions, while at $t = -1$ the two equations contradict each other.

Solution 142.3 ■ Answer C[Back to problem](#)

The first equation gives $x_1 + x_2 = 0$. Subtract it from the second to obtain $\varepsilon x_2 = \varepsilon$, so $x = (-1, 1)^T$ for every positive ε . The right-hand side tends to zero, but the inverse matrices grow as the determinant ε tends to zero.

Solution 142.4 ■ Answer B[Back to problem](#)

Write $(1, 1)^T = \frac{1}{2}(2, 0)^T + \frac{1}{4}(0, 4)^T$. Linearity of the inverse then gives $\frac{1}{2}(1, 1)^T + \frac{1}{4}(1, -1)^T = (3/4, 1/4)^T$. Known images can determine an inverse action without reconstructing all matrix entries.

Solution 142.5 ■ **Answer C**[Back to problem](#)

The two-by-two inverse formula gives $\text{tr}(A^{-1}) = (a + d)/(ad - bc) = \text{tr } A / \det A$. Thus $3 = 3/\det A$, forcing determinant one. Such matrices exist, for example $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

Vector Spaces ▸ Vector Spaces and Subspaces

143 Vector space axioms and the subspace test

Solution 143.1 ■ **Answer D**[Back to problem](#)

Both equations in D are homogeneous linear constraints, so their joint solution set is a subspace. A and B omit zero; C fails closure under addition; E fails closure under negative scalar multiplication.

Solution 143.2 ■ **Answer C**[Back to problem](#)

If neither contains the other, choose $u \in U \setminus V$ and $v \in V \setminus U$. Their sum lies in neither: if $u + v \in U$ then $v \in U$, and similarly for V . This contradicts closure of the union.

Solution 143.3 ■ **Answer B**[Back to problem](#)

The bijection $u \mapsto \log u$ carries $\oplus$ to ordinary addition and $\odot$ to ordinary scalar multiplication on $\mathbb{R}$. Hence all vector-space axioms hold. The additive identity for the defined operation is 1, and $u \oplus (1/u) = 1$. A zero vector need not be the usual number zero when operations change.

Solution 143.4 ■ **Answer D**[Back to problem](#)

The dimension formula subtracts the intersection counted twice: $\dim(U + V) = 4 + 5 - 2 = 7$.

Solution 143.5 ■ **Answer E**[Back to problem](#)

The intersection contains zero and is closed under addition and scalar multiplication because both original subspaces are. A union or arbitrary affine translate need not have these properties.

Solution 143.6 ■ **Answer B**[Back to problem](#)

Real multiples and sums of purely imaginary numbers remain purely imaginary, so W is a real subspace. Complex scalar multiplication fails: $i \cdot i = -1 \notin W$. The set already includes zero; the relevant change is the scalar field.

Vector Spaces ▷ The Nullspace

144 Nullspace and homogeneous directions

Solution 144.1 ■ **Answer A**[Back to problem](#)

The equations are $x + 2y + z = 0$ and $y + z = 0$. Let $z = t$, giving $y = -t$ and $x = t$. Thus the nullspace is spanned by the nonzero vector $(1, -1, 1)$.

Solution 144.2 ■ **Answer C**[Back to problem](#)

Trivial nullspace means the n columns are independent in $\mathbb{R}^m$, so $n \leq m$. Equality and trivial nullspace of the transpose need not hold when the matrix has extra rows.

Solution 144.3 ■ **Answer C**[Back to problem](#)

For $x = x_1e_1 + x_2e_2 + x_3e_3$, $Ax = x_2e_1 + x_3e_2$ and $A^2x = x_3e_1$. Thus $A^2x = 0$ exactly when $x_3 = 0$. The kernel grows from a line to a plane, so $\ker A^2 = \ker A$ is not a general identity.

Solution 144.4 ■ **Answer D**[Back to problem](#)

All rows are identical and nonzero, giving rank one. The kernel consists of vectors whose coordinates sum to zero, of dimension $4 - 1 = 3$.

Solution 144.5 ■ **Answer B**[Back to problem](#)

The kernel of the stacked map is exactly the intersection of the two kernels, which is trivial by hypothesis. Cancellation can prevent $A + B$ or $A - B$ from being injective.

Vector Spaces ▷ Linear Combinations

145 Linear combinations and span

Solution 145.1 ■ **Answer E**[Back to problem](#)

A combination is $(s, s + t, t)$. Matching the first and third entries forces $s = 1, t = 3$, hence the middle entry must be $a = 4$.

Solution 145.2 ■ **Answer D**[Back to problem](#)

The coefficient matrix $\begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}$ is invertible precisely when $1 - a^2 \neq 0$. At $a = \pm 1$ the two new vectors become dependent and span only a line.

Solution 145.3 ■ **Answer B**[Back to problem](#)

The sets share no listed vector, so $S \cap T$ is empty and its span is $\{0\}$. Each pair separately spans the plane, so the intersection of their spans is the plane. Taking spans does not generally commute with intersection.

Solution 145.4 ■ **Answer B**[Back to problem](#)

Membership in U is equivalent to $y = x + z$, while membership in W is equivalent to $x = z$. Together they give $(x, y, z) = t(1, 2, 1)$. Intersecting spans can be done by combining their defining linear constraints.

Solution 145.5 ■ **Answer D**[Back to problem](#)

The family contains $x^2, (x - 1)^2, (x + 1)^2$. Their difference $(x + 1)^2 - (x - 1)^2 = 4x$ gives x , and their sum minus $2x^2$ gives 2, so the span contains $1, x, x^2$. Every generator lies in P_2 , giving equality. Negative coefficients allow combinations that are not nonnegative.

Vector Spaces ▸ Linear Independence

146 Linear independence and dependence

Solution 146.1 ■ **Answer B**[Back to problem](#)

The identity $1 - \sin^2 x - \cos^2 x = 0$ is a nontrivial dependence relation. No pair consists of scalar multiples, as evaluation at 0 and $\pi/2$ shows. Pairwise nonproportionality therefore does not establish independence of three functions.

Solution 146.2 ■ **Answer B**[Back to problem](#)

For B, $a(u + v) + b(u - v) + cw = (a + b)u + (a - b)v + cw = 0$ forces $a = b = c = 0$. In A the third vector is the sum of the first two; the other lists visibly contain proportional pairs.

Solution 146.3 ■ **Answer E**[Back to problem](#)

Dependence is equivalent to determinant zero: $t^2 - t - 2 = 0$. The discriminant is positive, and Vieta gives the product as the constant coefficient divided by the leading coefficient, namely -2 .

Solution 146.4 ■ **Answer C**[Back to problem](#)

Orthogonality eliminates every cross term, leaving $a_j \|v_j\|^2 = 0$. The nonzero-vector assumption makes the squared norm positive, so each coefficient vanishes. Orthogonality alone would not suffice if the list contained zero.

Solution 146.5 ■ **Answer D**[Back to problem](#)

The relation space has dimension one, so the span has dimension three. Omitting any of v_1, v_2, v_3 removes a vector recoverable from the other two and leaves a basis with v_4 . Omitting v_4 leaves the displayed dependent triple. A dependent list does not permit arbitrary deletion.

Vector Spaces ▸ Bases and Dimension

147 Basis, dimension, and minimal spanning sets

Solution 147.1 ■ **Answer A**[Back to problem](#)

One independent equation in $\mathbb{R}^4$ gives dimension 3. The three vectors in A lie in V and are independent, as successively comparing the first, second, and third coordinates shows. The other lists fail membership, independence, or spanning.

Solution 147.2 ■ **Answer C**[Back to problem](#)

Every basis has exactly four vectors. The spanning-set reduction theorem guarantees a four-element basis can be selected from the six, so exactly two are removed.

Solution 147.3 ■ **Answer B**[Back to problem](#)

Every such polynomial is a finite linear combination of positive powers of x , and uniqueness of polynomial coefficients proves their independence. No finite list suffices because degrees are unbounded. A basis in an infinite-dimensional space still represents each vector by a finite sum.

Solution 147.4 ■ **Answer C**[Back to problem](#)

The new list must contain x^3 , since the original polynomials have degree at most two. Within P_2 the original span has coefficient vectors $(a, a + b, b)$, satisfying constant minus linear plus quadratic coefficient equal to zero. Each of $1, x, x^2$ violates this equation, so any one extends that plane to P_2 . There are three valid pairs.

Solution 147.5 ■ **Answer B**[Back to problem](#)

An independent list has at most the dimension many vectors. The other assertions omit independence, spanning, or cardinality requirements and admit elementary counterexamples.

Vector Spaces ▸ The Rank, Column Space, and Row Space

148 Rank, column space, and row space

Solution 148.1 ■ **Answer C**[Back to problem](#)

The second row is twice the first, so every image has second coordinate twice its first. Thus $b = 2$ is necessary. The first and third rows are independent, so this single constraint is also sufficient.

Solution 148.2 ■ **Answer C**[Back to problem](#)

The pivot columns are 1, 2, 4. Those columns of the original matrix are independent and span its column space. Column 3 equals twice column 1 minus column 2.

Solution 148.3 ■ Answer B[Back to problem](#)

The row space of A is the column space of A^T , which consists exactly of vectors $A^T x$ with $x \in \mathbb{R}^4$. Both row and column rank equal three, but the row space lies in $\mathbb{R}^6$ and the column space in $\mathbb{R}^4$. Their ambient dimensions cannot be interchanged.

Solution 148.4 ■ Answer B[Back to problem](#)

The columns of AB are combinations of those of A . If B is invertible, $A = (AB)B^{-1}$ gives the reverse containment, hence the column spaces coincide.

Solution 148.5 ■ Answer C[Back to problem](#)

If $Ax = b$, then $y^T b = y^T Ax = (A^T y)^T x = 0$, contradicting the given value one. A vector orthogonal to every column but not to b certifies that b lies outside the column space.

Vector Spaces ▸ Other Vector Spaces

149 Polynomial, function, and matrix spaces

Solution 149.1 ■ Answer C[Back to problem](#)

These polynomials have the form $x(x-1)q(x)$ with $q \in P_2$. Multiplication by $x(x-1)$ is an injective correspondence from the three-dimensional space P_2 .

Solution 149.2 ■ Answer D[Back to problem](#)

Each row has three entries subject to one independent equation, leaving two free entries per row. Across three rows the dimension is $3 \cdot 2 = 6$.

Solution 149.3 ■ Answer A[Back to problem](#)

The upper-left 3×3 block can be chosen freely. Each remaining entry in the first three rows or columns is then forced, and the final entry is consistent because total row and column sums agree. Thus the dimension is 9.

Solution 149.4 ■ Answer D[Back to problem](#)

Finite sums and scalar multiples preserve boundedness and continuity. The functions $1, \cos x, \cos 2x, \dots$ contain arbitrarily large independent finite lists: integrate a proposed relation against each cosine on $[0, 2\pi]$ and use orthogonality. Thus no finite spanning set exists. Bounded polynomials alone would not establish this on all of $\mathbb{R}$.

Solution 149.5 ■ Answer E[Back to problem](#)

Even polynomials have basis $1, x^2, x^4, x^6, x^8, x^{10}$, of size six. The single nonzero linear evaluation condition removes one dimension, leaving five.

Solution 149.6 ■ **Answer B**[Back to problem](#)

The constrained entries must be $a_{11} = t$ and $a_{23} = -t$, represented by $t(E_{11} - E_{23})$. The other four entries are free and have their own matrix units. These five matrices are independent by comparing entries and span every permitted matrix.

Determinants ▸ Permutations and the Defining Formula

150 Permutation signs and determinant definition

Solution 150.1 ■ **Answer B**[Back to problem](#)

A five-cycle has sign $(-1)^4 = 1$, while one transposition has sign -1 . The product permutation has sign -1 .

Solution 150.2 ■ **Answer B**[Back to problem](#)

The column list $(3, 1, 4, 2)$ has three inversions: $(3, 1)$, $(3, 2)$, $(4, 2)$. Its permutation is odd, so the coefficient is -1 .

Solution 150.3 ■ **Answer C**[Back to problem](#)

Every determinant monomial must select each column exactly once. This product uses column three twice and never uses column two, so it cannot arise from any permutation. Its coefficient is zero, regardless of the sign of any proposed row ordering.

Solution 150.4 ■ **Answer A**[Back to problem](#)

The sign of a product is the product of signs, and a permutation and its inverse have the same sign. Thus the displayed sign is $(\operatorname{sgn} \sigma)^2 (\operatorname{sgn} \tau)^2 = 1$. The permutation itself need not be the identity.

Solution 150.5 ■ **Answer B**[Back to problem](#)

Only two permutations can yield nonzero products: choose every diagonal entry, giving t^4 , or choose all four cyclic off-diagonal entries, giving t^4 . Choosing one off-diagonal entry forces the next choice around the cycle. The four-cycle is odd, so its term has a minus sign.

Determinants ▸ Properties and Elimination

151 Determinant laws and row operations

Solution 151.1 ■ Answer E[Back to problem](#)

The scalar factor contributes $2^3 = 8$. The remaining determinant is $(\det A)^{-1} \det A^T = (-2)^{-1}(-2) = 1$, so the result is 8.

Solution 151.2 ■ Answer C[Back to problem](#)

The net multiplier is $(-1)(-3)(1) = 3$. Thus $12 = 3 \det A$, giving 4. The row addition does not change the determinant.

Solution 151.3 ■ Answer C[Back to problem](#)

The row transformation is $B = CA$ with $C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. Its determinant is two, so

$\det B = 2 \det A = 10$. Expanding the three row sums separately also leaves two nonzero terms of the same sign; simultaneous row sums are not independent row-addition operations.

Solution 151.4 ■ Answer B[Back to problem](#)

Transposition preserves determinant, while multiplying all five rows by -1 reverses its sign. Thus $\det A = \det A^T = \det(-A) = -\det A$, forcing zero. Odd dimension is essential to this argument.

Solution 151.5 ■ Answer D[Back to problem](#)

Exchange the first block of three rows with the last block. This permutation has $3 \cdot 3 = 9$ inversions and reverses determinant sign. The resulting block diagonal matrix has determinant $\det C \det B = -6$. The original determinant is therefore 6.

Determinants ▸ Laplace Expansions

152 Minors, cofactors, and Laplace expansion

Solution 152.1 ■ **Answer B**[Back to problem](#)

Expand along the second column, then use the first column of the remaining minor. The result is $4(5 \cdot 8 - 6 \cdot 7) = 4(-2) = -8$. The entry 2 does not contribute.

Solution 152.2 ■ **Answer A**[Back to problem](#)

The derivative with respect to the $(2, 2)$ entry is its cofactor. The relevant minor is $\begin{pmatrix} 1 & 3 \\ 2 & 0 \end{pmatrix}$, with determinant -6 , and the cofactor sign is positive.

Solution 152.3 ■ **Answer C**[Back to problem](#)

Replace row two by row one. The cofactors used in expansion along row two are unchanged, since their minors delete that row. The expansion is the requested sum, but the modified matrix has two equal rows and hence determinant zero.

Solution 152.4 ■ **Answer C**[Back to problem](#)

Expansion along an end row gives $D_n = 2D_{n-1} - D_{n-2}$. Starting with $D_1 = 2$ and $D_2 = 3$ gives $D_3 = 4$, $D_4 = 5$, $D_5 = 6$. The recurrence, not the product of diagonal entries, accounts for the adjacent off-diagonal terms.

Solution 152.5 ■ **Answer C**[Back to problem](#)

Each updated row is its old row plus a scalar multiple of the same row v^T . Multilinearity expands the determinant into terms. Any term using updates from two or more rows has proportional rows and vanishes. Only the constant and linear terms remain; either or both coefficients can be zero.

Solution 152.6 ■ **Answer B**[Back to problem](#)

Expansion along the first row gives $x(b - d) - y(a - c) + ad - bc = 0$. If the points differ, the coefficients of x, y are not both zero, and the equation defines their line. If the points coincide, every coefficient vanishes and every point satisfies the equation.

Determinants ▸ The Adjugate Matrix

153 Adjugate and the invertibility criterion

Solution 153.1 ■ **Answer D**[Back to problem](#)

$\det(2A) = 2^4 \cdot 3 = 48$. For a 4×4 matrix the adjugate determinant is the cube of its determinant, so the answer is $48^3 = 110592$.

Solution 153.2 ■ **Answer C**[Back to problem](#)

Each cofactor is a signed 3×3 minor. Every such minor is zero because the full matrix has rank only 2. Thus all adjugate entries vanish.

Solution 153.3 ■ **Answer B**[Back to problem](#)

At least one $(n - 1)$ -minor is nonzero, so the adjugate is nonzero. Since $A \operatorname{adj} A = 0$, its columns lie in the one-dimensional kernel of A , forcing rank one.

Solution 153.4 ■ **Answer C**[Back to problem](#)

The $(1, 3)$ entry of the adjugate is cofactor C_{31} , not C_{13} . Deleting row three and column one leaves $\begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$, with determinant six and positive cofactor sign. The transpose in the definition is essential.

Solution 153.5 ■ **Answer B**[Back to problem](#)

Multiply the equation by A . The adjugate identity gives $(\det A)I = 2I$, hence $\det A = 2$.

Determinants ▸ Cramer's Rule

154 Cramer's rule and a selected unknown

Solution 154.1 ■ **Answer A**[Back to problem](#)

The coefficient determinant is -12 . Replacing column two by $(0, 3, 1)^T$ gives determinant 6, so $y = 6/(-12) = -1/2$. Direct substitution gives $(x, y, z) = (3/2, -1/2, -1/2)$.

Solution 154.2 ■ **Answer B**[Back to problem](#)

By multilinearity only the $-3c_2$ term in the second column contributes; the other terms duplicate another column and have zero determinant. The ratio is -3 , matching the second coordinate in Cramer notation.

Solution 154.3 ■ **Answer A**[Back to problem](#)

Every replacement matrix still has a zero column, so all three numerators vanish. Yet every vector Ax has second coordinate zero, whereas b has second coordinate one. Thus the system is inconsistent despite every formal Cramer numerator being zero; division by the zero determinant supplies no solution criterion.

Solution 154.4 ■ **Answer A**[Back to problem](#)

Transpose the equation to obtain $A^T x = b$. Ordinary Cramer column replacement in A^T becomes row replacement when transposed back, and determinants are unchanged by transpose. The orientation of the original matrix equation determines which replacement is appropriate.

Solution 154.5 ■ **Answer C**[Back to problem](#)

Multilinearity leaves the original determinant and three single-column replacements. Every term containing two copies of b vanishes. Cramer gives the total as $(1 + x_1 + x_2 + x_3) \det A$. Dividing by the nonzero determinant yields the sum six.

Linear Transformations ▸ Linearity and Component Vectors

155 Linear transformations and coordinates in a basis

Solution 155.1 ■ **Answer B**[Back to problem](#)

Write $(5, 0) = a(1, 2) + b(2, -1)$. Then $a + 2b = 5$ and $2a - b = 0$, giving $a = 1, b = 2$. Therefore $T(5, 0) = (3, 1) + 2(1, 4) = (5, 9)$.

Solution 155.2 ■ **Answer B**[Back to problem](#)

$3 + x = 2(1 + x) + (1 - x)$, so its coordinate vector is $(2, 1)^T$. Linearity gives $T(3 + x) = 2 \cdot 1 + x = 2 + x$.

Solution 155.3 ■ **Answer D**[Back to problem](#)

The first two input vectors form a basis, so their assigned images define one linear map. Linearity forces $T(1, 1) = (1, 2) + (2, -1) = (3, 1)$. Thus the third assignment is compatible exactly for $a = 3$.

Solution 155.4 ■ **Answer D**[Back to problem](#)

Evaluation and differentiation are linear, and so is their stated linear combination. The other maps violate additivity or scalar homogeneity: the translation does not send zero to zero, squares and products scale quadratically, and absolute value fails for negative scalars.

Solution 155.5 ■ **Answer C**[Back to problem](#)

The third input is the first minus the second, and its prescribed image has the same relation, so it adds no constraint. The first two inputs are independent and can be extended to a basis of $\mathbb{R}^3$. The image of its remaining basis vector is any vector in $\mathbb{R}^2$, giving two free real parameters.

Linear Transformations ▸ Standard Matrix Representative

156 Matrix representation of a linear map

Solution 156.1 ■ **Answer A**[Back to problem](#)

$T(1) = 1$, $T(x) = 1 + 2x$, and $T(x^2) = 2x + 3x^2$. These coordinate columns give A. The row-column convention rules out the transpose.

Solution 156.2 ■ **Answer B**[Back to problem](#)

By linearity this is $T(3, 0, 1) = (3, 1)$. Equivalently multiply by the standard matrix $\begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$.

Solution 156.3 ■ **Answer D**[Back to problem](#)

The monomial x^k is sent to $(k+1)x^k$, so T^2 has diagonal entries 1, 4, 9, 16. Their sum is 30. Squaring the trace instead would incorrectly introduce cross terms.

Solution 156.4 ■ **Answer B**[Back to problem](#)

The domain basis has three vectors and the codomain basis four, giving four rows and three columns. The second input vector is x , whose image is $x^2/2$; its coefficient in the third output basis position is $1/2$.

Solution 156.5 ■ **Answer B**[Back to problem](#)

The three outputs are $0, 1, 2x$. In the specified output basis their coordinates are $(0, 0)^T$, $(1/2, 1/2)^T$, and $(1, -1)^T$. Place these coordinate vectors in columns. Using standard output coefficients would give the wrong representation.

Solution 156.6 ■ **Answer B**[Back to problem](#)

Any matrix product sends the zero vector to zero, but $F(0, 0) = (1, 0)$. The nonzero translation prevents an ordinary matrix representation of the stated dimensions. Reordering a basis changes coordinates, not the image of zero.

Linear Transformations ▸ The Rank Plus Nullity Theorem

157 Kernel, image, and rank-nullity

Solution 157.1 ■ **Answer C**[Back to problem](#)

Rank-nullity gives rank $7 - 4 = 3$. Thus it is neither injective nor surjective, and its image is a three-dimensional subspace of $\mathbb{R}^5$.

Solution 157.2 ■ **Answer D**[Back to problem](#)

For x^k , $T(x^k) = (k - 2)x^k$. Exactly the basis vector x^2 maps to zero, so the kernel is one-dimensional. Since $\dim P_4 = 5$, the rank is 4.

Solution 157.3 ■ **Answer C**[Back to problem](#)

Every polynomial has a polynomial antiderivative, so differentiation is onto. Its kernel consists of constant polynomials and is nontrivial, so it is not one-to-one. Equal domain and codomain do not make injectivity and surjectivity equivalent in an infinite-dimensional space.

Solution 157.4 ■ **Answer E**[Back to problem](#)

The kernel has dimension two. Its intersection with U has dimension at least $4 + 2 - 5 = 1$ and at most two. Rank-nullity for the restriction gives $\dim T(U) = 4 - \dim(U \cap \ker T)$, hence two or three. A basis adapted to the kernel constructs subspaces attaining either value.

Solution 157.5 ■ **Answer A**[Back to problem](#)

Evaluation is linear. If $E(p) = 0$, then a polynomial of degree at most two has three distinct roots, so $p = 0$. The kernel is trivial. Both spaces have dimension three, hence injectivity implies surjectivity and E is an isomorphism.

Linear Transformations ▸ A Note on Inverses and Compositions

158 Composition and inverse transformations

Solution 158.1 ■ **Answer B**[Back to problem](#)

$T(S(x, y)) = (x + y, -x)$. Solving $a = x + y$, $b = -x$ gives $x = -b$, $y = a + b$, so the inverse is $(-b, a + b)$.

Solution 158.2 ■ **Answer B**[Back to problem](#)

The identity factorization implies T is onto $\mathbb{R}^2$, so its rank is 2. Its domain has dimension 3, and rank-nullity gives nullity 1.

Solution 158.3 ■ Answer B[Back to problem](#)

The composition sends (x, y, z) to $(x - z, 0)$. Its kernel is therefore the plane $x = z$, parametrized as (s, t, s) . A vector need only map under S into $\ker T$; it need not belong to $\ker S$.

Solution 158.4 ■ Answer B[Back to problem](#)

The product rule gives $D(xp) = p + xp'$, whereas $M(Dp) = xp'$. Their difference is p for every polynomial, hence the identity. Composition order matters even for two elementary linear operators.

Solution 158.5 ■ Answer B[Back to problem](#)

Write $R = (I_V - TS)^{-1}$. Multiplying $(I_W - ST)(I_W + SRT)$ gives $I_W + S(R - I_V - TSR)T = I_W$, since $(I_V - TS)R = I_V$. Multiplication in the other order uses $R(I_V - TS) = I_V$ and also gives the identity. The formula is valid with different dimensions for V and W .

Eigenvalues and Eigenvectors ▷ Characteristic Equation

159 Eigenvalues, eigenvectors, and characteristic polynomial

Solution 159.1 ■ Answer C[Back to problem](#)

Apply the expression to v : the scalar becomes $2^3 - 3(2) + 4(1/2) = 8 - 6 + 2 = 4$.

Solution 159.2 ■ Answer C[Back to problem](#)

$\det(tI - A) = t^2 + 4$, whose roots are $\pm 2i$. Thus it has no real eigenvectors despite being a nonzero real matrix.

Solution 159.3 ■ Answer B[Back to problem](#)

The characteristic polynomial is a real polynomial of odd degree. Its values have opposite signs for sufficiently large positive and negative arguments, so the intermediate value theorem gives a real root. Nonreal roots can still occur in conjugate pairs; zero, positivity, and distinctness are not forced.

Solution 159.4 ■ Answer C[Back to problem](#)

$A^2 = I$, so $A^2v = v$. But $Av = (1, -1)^T$ is not a scalar multiple of $(1, 1)^T$. The forward rule that an eigenvector of A is an eigenvector of $p(A)$ has no general converse.

Solution 159.5 ■ Answer B[Back to problem](#)

The basis vectors are eigenvectors, so A is diagonal with entries $\lambda_1, \lambda_2, \lambda_3$. Applying the hypothesis to each $e_i + e_j$ forces $\lambda_i = \lambda_j$. Hence $A = \lambda I$ and $3\lambda = 9$, giving $A = 3I$.

Eigenvalues and Eigenvectors ▸ Trace and Determinant

160 Trace and eigenvalue sums and products

Solution 160.1 ■ Answer C

[Back to problem](#)

If the remaining eigenvalues are a, b , then $a + b = 4$ and $ab = 4$. Thus $3^2 + a^2 + b^2 = 9 + (a + b)^2 - 2ab = 9 + 16 - 8 = 17$.

Solution 160.2 ■ Answer C

[Back to problem](#)

For a two-by-two matrix, $\det(A + cI) = c^2 + c \operatorname{tr} A + \det A$. With $c = 2$, the result is $4 + 2 - 2 = 4$. Equivalently its eigenvalues are 2, -1 , which shift to 4, 1.

Solution 160.3 ■ Answer D

[Back to problem](#)

Cyclic trace gives $\operatorname{tr}(AB - BA) = 0$, whereas $\operatorname{tr} I_n = n \neq 0$. Thus the equation is impossible in finite dimension over $\mathbb{R}$. Matrix commutativity is not required for equality of the two traces.

Solution 160.4 ■ Answer D

[Back to problem](#)

Every vector splits as $Ax + (x - Ax)$, with the first term in the image and the second in the kernel. Their intersection is zero. In an adapted basis, A has ones on the image and zeros on the kernel, so its trace equals its rank, three. Rank-nullity gives nullity five.

Solution 160.5 ■ Answer A

[Back to problem](#)

For eigenvalues $\lambda_1, \lambda_2, \lambda_3$ counted with multiplicity, their pair-product sum is $[(\sum \lambda_i)^2 - \sum \lambda_i^2]/2 = (9 - 5)/2 = 2$. The characteristic polynomial is therefore $t^3 - 3t^2 + 2t - 0$. The identities remain valid without diagonalizability.

Eigenvalues and Eigenvectors ▸ Eigenspaces

161 Eigenspaces and geometric multiplicity

Solution 161.1 ■ Answer C

[Back to problem](#)

$(A - 2I)(x, y, z)^T = (y, 0, 0)^T$. Thus $y = 0$, while x, z are free. The eigenspace has dimension 2, although the eigenvalue has algebraic multiplicity 3.

Solution 161.2 ■ Answer A

[Back to problem](#)

The eigenspaces provide four independent eigenvectors, hence a basis. The eigenvalues in that basis are 1, 1, 2, 3, so their product is 6.

Solution 161.3 ■ **Answer D**[Back to problem](#)

Diagonalizability requires four independent eigenvectors. With only one eigenvalue, all eigenvectors lie in its two-dimensional eigenspace, so there cannot be an eigenbasis.

Solution 161.4 ■ **Answer B**[Back to problem](#)

From $Av = 2u + 5w$ and $5v = 5u + 5w$, subtraction gives $5v - Av = 3u$. Distinct eigenvalues allow separate eigenspace components to be isolated by a polynomial in the operator.

Solution 161.5 ■ **Answer C**[Back to problem](#)

Geometric multiplicity is at least one and at most algebraic multiplicity. The values one, two, and three can all occur, according to the Jordan block structure; four cannot.

The Cayley-Hamilton Theorem ▸ Reducing Matrix Powers

162 Cayley–Hamilton theorem

Solution 162.1 ■ **Answer C**[Back to problem](#)

Cayley–Hamilton gives $(A - I)^2 = 0$. Write $A = I + N$ with $N^2 = 0$; the binomial expansion terminates at $A^{100} = I + 100N = 100A - 99I$. Knowing only the repeated eigenvalue does not justify replacing A by I .

Solution 162.2 ■ **Answer B**[Back to problem](#)

The characteristic polynomial is $t(t - 1)(t + 1) = t^3 - t$, so $A^3 = A$. If $A^2 = aA + bI$, the zero diagonal position forces $b = 0$; the positions for eigenvalues 1 and -1 then force $a = 1$ and $a = -1$, a contradiction. In size three a quadratic remainder can be necessary.

Solution 162.3 ■ **Answer C**[Back to problem](#)

Every eigenvalue must be a root of $t^2 - 3t + 2$, hence must be 1 or 2, while a three-by-three characteristic polynomial has degree three. Only C satisfies both requirements, and it is realized by $\text{diag}(1, 1, 2)$. An annihilating polynomial need not itself be the characteristic polynomial.

Solution 162.4 ■ **Answer B**[Back to problem](#)

Cayley–Hamilton gives $A^3 - A = I$. Factoring the left side as $(A - I)(A^2 + A)$ proves the stated inverse; the factors commute because they are polynomials in A . This also proves invertibility of the shifted matrix directly.

Solution 162.5 ■ **Answer C**[Back to problem](#)

Cayley–Hamilton gives $A^3 = 0$. The matrix with ones immediately above the diagonal and zeros elsewhere has rank two, a one-dimensional kernel, and a nonzero square, so none of the stronger claims is necessary.

Chapter 5 Review Questions ▸ Counterexamples to Scalar Rules

163 Nilpotent and involutory matrices

Solution 163.1 ■ **Answer C**[Back to problem](#)

Commutation permits the binomial expansion $(N + M)^3 = N^3 + 3N^2M + 3NM^2 + M^3 = 0$. The square can equal the nonzero matrix $2NM$. For example, on a basis e_1, e_2, e_3, e_4 , let $Ne_3 = e_1$, $Ne_4 = e_2$, $Me_2 = e_1$, and $Me_4 = e_3$, with all other images zero. These maps commute and have square zero, but $NMe_4 = e_1$. Thus square zero and zero product do not follow.

Solution 163.2 ■ **Answer B**[Back to problem](#)

Eigenvalues of an involution satisfy $\lambda^2 = 1$. Trace zero forces one 1 and one -1 , so the determinant is -1 . Alternatively Cayley–Hamilton gives $A^2 + (\det A)I = 0$, hence $\det A = -1$.

Solution 163.3 ■ **Answer E**[Back to problem](#)

The kernel has dimension one. For each k , restriction of N^k maps $\ker N^{k+1}$ into $\ker N$ with kernel $\ker N^k$, so the kernel dimensions can increase by at most one per power. To reach dimension five therefore requires at least five powers. Cayley–Hamilton bounds the index by five, giving exactly five.

Solution 163.4 ■ **Answer B**[Back to problem](#)

Both N^2 and M^2 are zero, but $NM = \text{diag}(1, 0)$ and $MN = \text{diag}(0, 1)$. Hence $(N + M)^2 = NM + MN = I$. A sum of nilpotents need not be nilpotent, because the factors need not commute.

Solution 163.5 ■ **Answer B**[Back to problem](#)

Apply A and use $A^2v = v$: $Av_+ = (Av + v)/2 = v_+$ and $Av_- = (Av - v)/2 = -v_-$. Also $v = v_+ + v_-$, and the two eigenspaces intersect only at zero. An involution splits the space into fixed and sign-reversed components.

Solution 163.6 ■ **Answer C**[Back to problem](#)

The image lies in the kernel, so $r \leq 7 - r$, giving $r \leq 3$. Equality is achieved by three independent chains $e_2 \mapsto e_1 \mapsto 0$, $e_4 \mapsto e_3 \mapsto 0$, and $e_6 \mapsto e_5 \mapsto 0$, with $e_7 \mapsto 0$. Thus the upper bound is sharp.

Chapter 5 Review Questions ▷ Symmetric Matrix Spaces

164 Symmetric and skew-symmetric matrices

Solution 164.1 ■ **Answer D**[Back to problem](#)

The symmetric space has dimension $4 \cdot 5/2 = 10$. The trace condition is one nonzero linear constraint, reducing the dimension to 9.

Solution 164.2 ■ **Answer D**[Back to problem](#)

Write $S = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$. The diagonal errors are minimized at $a = 1, d = 3$. The remaining sum is $(4 - b)^2 + (-2 - b)^2 = 2(b - 1)^2 + 18$, giving minimum eighteen at the symmetric part $(A + A^T)/2$.

Solution 164.3 ■ **Answer A**[Back to problem](#)

Symmetry lets each entry strictly above the diagonal be chosen independently and determines the matching lower entry. There are $\binom{n}{2} = n(n - 1)/2$ such positions.

Solution 164.4 ■ **Answer C**[Back to problem](#)

Since $A^T = -A$, $\|Ax\|^2 = x^T A^T Ax = -x^T A^2 x$. Thus the square of a real skew-symmetric matrix is negative semidefinite. In particular it cannot equal I in positive dimension.

Solution 164.5 ■ **Answer C**[Back to problem](#)

Because $A^T = A$ and $B^T = B$, $(AB)^T = BA$. Symmetry of the product is exactly the equality $AB = BA$.

Solution 164.6 ■ **Answer C**[Back to problem](#)

For nonzero x , the quadratic form of the convex combination is $tx^T Ax + (1 - t)x^T Bx > 0$. Symmetry is also preserved. Subtraction can yield zero, and negative scaling reverses the sign, so this class is not a vector subspace.

Chapter 5 Review Questions ▸ Rotations and Reflections

165 Rotation and reflection operators

Solution 165.1 ■ **Answer C**[Back to problem](#)

The composition rotates by twice the angle from the first reflecting line to the second:
 $2(\pi/3 - \pi/6) = \pi/3$ counterclockwise.

Solution 165.2 ■ **Answer A**[Back to problem](#)

Use $SR = R^{-1}S$ and $S^2 = I$: $SRSR = R^{-1}S^2R = I$. The composition SR is itself a reflection.

Solution 165.3 ■ **Answer B**[Back to problem](#)

The reflecting line is fixed pointwise, so solve $(Q - I)(x, y)^T = 0$. The equation $-2x + 4y = 0$ gives $y = x/2$. The perpendicular direction has eigenvalue -1 ; also $Q^T Q = I$ and $\det Q = -1$ verify the reflection interpretation.

Solution 165.4 ■ **Answer B**[Back to problem](#)

A planar orthogonal matrix with determinant -1 is a reflection and has trace zero. Thus this matrix is a rotation, with $2\cos\theta = 1$. Its angle is $\theta = \pm\pi/3$ modulo 2π ; trace alone does not distinguish the direction of rotation.

Solution 165.5 ■ **Answer C**[Back to problem](#)

The matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has determinant one, so it preserves orientation and area. But it sends $(0, 1)$, of length one, to $(1, 1)$, of length $\sqrt{2}$. A positive determinant specifies orientation, not orthogonality.

Chapter 5 Review Questions ▸ The Cross Product as a Linear Map

166 Cross-product linear operators

Solution 166.1 ■ **Answer A**[Back to problem](#)

The triple-product identity gives $v \times (v \times x) = v(v \cdot x) - \|v\|^2 x$. Orthogonality removes the first term and $\|v\|^2 = 9$.

Solution 166.2 ■ **Answer C**[Back to problem](#)

T^2 is zero on the line spanned by v and equals multiplication by $-\|v\|^2$ on the perpendicular plane. Its image is therefore that plane, of dimension 2.

Solution 166.3 ■ **Answer C**[Back to problem](#)

The eigenvalues of T over $\mathbb{C}$ are 0 and $\pm i\|v\|$. Hence $\det(I + T) = (1 + i\|v\|)(1 - i\|v\|) = 1 + \|v\|^2 = 7$.

Solution 166.4 ■ **Answer B**[Back to problem](#)

The triple-product identities give $u \times (v \times x) - v \times (u \times x) = v(u \cdot x) - u(v \cdot x) = (u \times v) \times x$. The order and minus sign matter: the two cross-product operators generally fail to commute.

Solution 166.5 ■ **Answer C**[Back to problem](#)

Use $T^2 = vv^T - r^2I$ and $T(vv^T) = 0$. Then $(I+T)(I-T+vv^T) = I - T^2 + vv^T = (1+r^2)I$. The reverse product is the same because $(vv^T)T = 0$ as well. The positive denominator ensures invertibility for every real v .

Solution 166.6 ■ **Answer C**[Back to problem](#)

Compare with $\begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}$. The entries give $a = 1, b = -2, c = 3$. For example the first column is $v \times e_1 = (0, 3, 2)^T$, fixing the left-cross-product sign convention.

6. Number Theory and Abstract Algebra

Part A: Number Theory ▸ Divisibility

167 Divisibility, parity, primes, and coprimality

Solution 167.1 ■ Answer C

[Back to problem](#)

The expression is divisible by 2 by parity, by 3 by checking $n \equiv 0, \pm 1$, and by 5 by Fermat. Thus 30 always divides it. At $n = 2$ it equals 30, so no larger universal divisor is possible.

Solution 167.2 ■ Answer C

[Back to problem](#)

Divisibility requires prime powers 2^2 and 3^2 , so $36 \mid n$. The integer 36 itself satisfies both requirements, proving that 36 is the largest forced divisor. Multiplying 12 and 18 would double-count shared prime factors.

Solution 167.3 ■ Answer C

[Back to problem](#)

Since $72 = 8 \cdot 9$, both digit tests must hold. The digit sum gives $a + b = 3$ or 12. The last three digits give $4a + 4 + b \equiv 0 \pmod{8}$. In the first case $3a + 7 \equiv 0$, yielding $(a, b) = (3, 0)$; in the second $3a \equiv 0$, yielding $(8, 4)$. Both digit pairs are valid, giving 4320 and 4824.

Solution 167.4 ■ Answer C

[Back to problem](#)

For any prime, the three exponents have pairwise even sums, so their parities are equal. At least one exponent is zero because the common gcd is one; hence all three exponents are even. This works for every prime. Pairwise coprimality is stronger and is not forced: 36, 100, 225 satisfy the hypotheses but each pair has a nontrivial gcd.

Solution 167.5 ■ Answer C

[Back to problem](#)

If every prime exponent of m is at most one, each prime dividing m and a^2 divides a , so their product divides a . Conversely write $m = \prod p^{e_p}$ and take $a = \prod p^{\lceil e_p/2 \rceil}$. Then $m \mid a^2$, but if any $e_p \geq 2$, its exponent in a is smaller, so $m \nmid a$. The case $m = 1$ also works.

Part A: Number Theory ▸ The Division Algorithm

168 Quotient and remainder

Solution 168.1 ■ Answer C

[Back to problem](#)

The floor quotient is $q = -16$, since $9(-16) = -144$. Consequently $r = 7$, and $q + r = -9$.

Solution 168.2 ■ Answer D

[Back to problem](#)

Write $N = 35k + 17$. Since N is even, k is odd. Thus $N^2 \equiv 35k^2 + 9 \equiv 44 \pmod{70}$.

Solution 168.3 ■ Answer C[Back to problem](#)

Division by seven removes the final base-seven digit, giving quotient $(35)_7 = 3 \cdot 7 + 5 = 26$. Division by $49 = 7^2$ removes the last two digits, giving quotient three. The sum is twenty-nine; the quotient digits must still be interpreted in base seven.

Solution 168.4 ■ Answer A[Back to problem](#)

The condition is exactly $r + s \geq m$, since the sum is less than $2m$. For a fixed r , the permitted s values are $m - r, \dots, m - 1$, giving r choices. Summing $0 + 1 + \dots + (m - 1)$ yields $m(m - 1)/2$. This counts the carry generated by adding remainders.

Solution 168.5 ■ Answer C[Back to problem](#)

Let the divisor be d and the common quotient and remainder q . Then $100 = q(d + 1)$ with $0 \leq q < d$. Thus $k = d + 1$ is a divisor of 100 with $k \geq 2$ and $100/k < k - 1$. Of $k = 2, 4, 5, 10, 20, 25, 50, 100$, exactly 20, 25, 50, 100 satisfy the inequality, yielding four divisors d .

Part A: Number Theory ▸ Primes

169 Prime factorization and the distribution of primes

Solution 169.1 ■ Answer D[Back to problem](#)

Trailing zeros count paired factors $2 \cdot 5$. There are more factors of 2, so the answer is $v_5(250!) = \lfloor 250/5 \rfloor + \lfloor 250/25 \rfloor + \lfloor 250/125 \rfloor = 50 + 10 + 2 = 62$.

Solution 169.2 ■ Answer C[Back to problem](#)

Exponent patterns arise from multiplicative partitions of 18: (17), (8, 1), (5, 2), and (2, 2, 1). Assign larger exponents to smaller primes. The corresponding minima are 2^{17} , $2^8 3$, $2^5 3^2 = 288$, and $2^2 3^2 5 = 180$. The least is 180.

Solution 169.3 ■ Answer B[Back to problem](#)

The prime factorization is $2^4 \cdot 3^3 \cdot 5^2$. A square divisor has an even exponent at each prime, giving $3 \cdot 2 \cdot 2 = 12$ choices.

Solution 169.4 ■ Answer B[Back to problem](#)

For each $j = 2, \dots, k + 1$, the number $(k + 1)! + j$ is divisible by j and is larger than j , hence composite. These are exactly k consecutive integers. This proves arbitrarily long prime-free intervals without contradicting the existence of primes in every interval $(n, 2n)$.

Solution 169.5 ■ Answer B[Back to problem](#)

The theorem gives $\pi(x^2) \sim x^2/\log(x^2) = x^2/(2\log x)$ and $x\pi(x) \sim x^2/\log x$. Taking their ratio gives $1/2$. The approximation is asymptotic; it is not an exact count at finite x .

Solution 169.6 ■ Answer B[Back to problem](#)

Bertrand's postulate gives a prime p with $n < p < 2n$. In $(2n)!$ there is exactly one multiple of p , and $p^2 > 2n$, while $n!$ contains no factor p . Thus the binomial coefficient has prime exponent $v_p = 1$, which rules out a perfect square.

Solution 169.7 ■ Answer C[Back to problem](#)

The reciprocal-prime series is a known divergent series. The squared-reciprocal series is bounded above by the convergent series $\sum_{n=2}^{\infty} 1/n^2$. Infinitely many primes alone would not prove reciprocal divergence; that is a stronger distribution result.

Part A: Number Theory ▸ The Greatest Common Divisor and the Least Common Multiple; The Euclidean Algorithm

170 GCD, LCM, Euclid, and Bezout

Solution 170.1 ■ Answer B[Back to problem](#)

Write $a = 12u, b = 12v$ with $\gcd(u, v) = 1$. Then $uv = 420/12 = 35$. Each prime factor 5, 7 must lie wholly in one of u, v , giving $2^2 = 4$ ordered pairs.

Solution 170.2 ■ Answer E[Back to problem](#)

Euclid gives $101 = 2(37) + 27$, $37 = 27 + 10$, $27 = 2(10) + 7$, $10 = 7 + 3$, and $7 = 2(3) + 1$. Back-substitution yields $1 = 11(101) - 30(37)$. Thus $37^{-1} \equiv -30 \equiv 71 \pmod{101}$.

Solution 170.3 ■ Answer C[Back to problem](#)

Any common divisor divides $4(n^2 + n + 1) - (2n + 1)^2 = 3$. Thus the gcd is one or three. Both arguments are divisible by three precisely when $n \equiv 1 \pmod{3}$, establishing the stated cases without factoring either large expression.

Solution 170.4 ■ Answer D[Back to problem](#)

Divide by $d = \gcd(a, b)$ to get $(a/d)x + (b/d)y = 1$. Every common divisor of x, y divides this integer combination and hence divides one. Therefore $\gcd(x, y) = 1$, even when a, b themselves are not relatively prime.

Solution 170.5 ■ Answer A[Back to problem](#)

For a fixed prime let the exponents in a, b, c be α, β, γ . The left side has exponent $\min(\max(\alpha, \beta), \max(\alpha, \gamma)) = \max(\alpha, \min(\beta, \gamma))$, exactly the exponent on the right in A. Agreement for every prime proves the identity.

Part A: Number Theory ▸ The Diophantine Equation $ax + by = c$

171 Linear Diophantine equations

Solution 171.1 ■ **Answer B**[Back to problem](#)

Divide by 6: $3x + 5y = 42$. Modulo 3, $y \equiv 0$, so $y = 3t$ and $x = 14 - 5t$. Nonnegativity permits exactly $t = 0, 1, 2$, yielding three pairs.

Solution 171.2 ■ **Answer A**[Back to problem](#)

Divide by 13 to get $2x + 3y = 5$. Modulo 3, $2x \equiv 2$, hence $x \equiv 1 \pmod{3}$. The least positive choice is $x = 1$, and then $y = 1$.

Solution 171.3 ■ **Answer B**[Back to problem](#)

Divide by six to obtain $2x + 3y = 0$. Since two and three are coprime, $3 \mid x$; write $x = 3t$, which forces $y = -2t$. The first parametrization gives only every sixth solution and is incomplete. Here t ranges over all integers.

Solution 171.4 ■ **Answer C**[Back to problem](#)

If both rational numbers are nonzero, use a common positive denominator d to write $u = a/d, v = b/d$ with nonzero integers a, b . Then $a^3 + b^3 = d^3$, contradicting Fermat's last theorem. If one is zero, the other is one, giving exactly $(0, 1)$ and $(1, 0)$.

Solution 171.5 ■ **Answer C**[Back to problem](#)

One solution is $(-3, 2)$, and all are $x = -3 + 11t, y = 2 - 7t$ for integer t . The squared norm is $170t^2 - 94t + 13$, whose real minimum occurs at $47/170$. The nearest integer is zero; checking the two adjacent integers confirms the minimum 13 at $t = 0$.

Solution 171.6 ■ **Answer C**[Back to problem](#)

The gcd of the coefficients is fourteen, so solvability requires $k^2 + 3 \equiv 0 \pmod{14}$. Modulo seven this means $k \equiv \pm 2$, and modulo two it requires k odd. Combining gives $k \equiv 5$ or $9 \pmod{14}$. The first ninety-eight integers contain seven complete blocks, giving fourteen values; ninety-nine and one hundred add none.

Part A: Number Theory ▸ Congruences

172 Congruence arithmetic and valid cancellation

Solution 172.1 ■ **Answer A**[Back to problem](#)

Write $x = 2 + 7k$. Then $7k \equiv 3 \pmod{11}$, so $k \equiv 2 \pmod{11}$ because $7^{-1} \equiv 8$. Hence $x \equiv 16 \pmod{77}$.

Solution 172.2 ■ **Answer D**[Back to problem](#)

The gcd of 18 and 84 is 6. Thus $84 \mid 18(a - b)$ is equivalent to $14 \mid 3(a - b)$ and, because 3 is coprime to 14, to $14 \mid a - b$.

Solution 172.3 ■ **Answer D**[Back to problem](#)

The gcd of the moduli is six. Compatibility requires $b \equiv 5 \pmod{6}$, and only eleven among the choices has this residue. The resulting solution class is modulo the lcm thirty-six, not the product two hundred sixteen.

Solution 172.4 ■ **Answer D**[Back to problem](#)

The congruences are compatible modulo their common divisor three. Writing $x = 2 + 6k$ gives $6k \equiv 3 \pmod{9}$, so $k \equiv 2 \pmod{3}$ and $x \equiv 14 \pmod{18}$. Modulo fifty-four, this class lifts to 14, 32, 50, giving three distinct classes.

Solution 172.5 ■ **Answer D**[Back to problem](#)

Modulo each of the primes three and five, x must be 1 or -1 . The two sign choices are independent under CRT, giving four classes modulo fifteen: 1, 4, 11, 14. Over a composite modulus, $(x - 1)(x + 1) \equiv 0$ does not force one whole factor to vanish modulo the full modulus.

Part A: Number Theory ▸ Congruences

173 Fermat, Euler, and modular powers

Solution 173.1 ■ **Answer E**[Back to problem](#)

Fermat reduces 222 modulo 12 to 6. Since $7^2 \equiv 10$, $7^3 \equiv 5$, and $7^6 \equiv 25 \equiv 13 \pmod{12}$, the remainder is 12.

Solution 173.2 ■ **Answer A**[Back to problem](#)

The base is coprime to 40. More directly than Euler, $3^4 = 81 \equiv 1 \pmod{40}$. Since $4 \mid 1000$, the required remainder is 1.

Solution 173.3 ■ Answer C[Back to problem](#)

The totient is $360(1 - 1/2)(1 - 1/3)(1 - 1/5) = 96$, so every unit satisfies the congruence by Euler. A nonunit shares a prime factor with 360; reducing modulo that prime makes the left side zero and the right side one, impossible. Thus exactly the ninety-six units work.

Solution 173.4 ■ Answer E[Back to problem](#)

Modulo eight the power is zero. Modulo nine, Euler gives $2^6 \equiv 1$, so the power is one. Among multiples of eight below seventy-two, only sixty-four is one modulo nine. Reducing $120 \bmod \varphi(72) = 24$ directly would be invalid because two is not a unit modulo seventy-two.

Solution 173.5 ■ Answer C[Back to problem](#)

Raising the given congruence to the thirty-fourth power gives $2^{340} \equiv 1$. But $341 = 11 \cdot 31$ is composite. A Fermat congruence for one base is a necessary condition for primality in the usual coprime setting, not a sufficient condition. Here the order divides ten.

Part A: Number Theory ▸ The Congruence Equation $ax \equiv b \pmod{n}$

174 Linear congruences and solution counts

Solution 174.1 ■ Answer D[Back to problem](#)

Here $d = 42$, and $42 \mid 84$. Dividing gives $x \equiv 2 \pmod{3}$, which has $126/3 = 42$ representatives modulo 126.

Solution 174.2 ■ Answer C[Back to problem](#)

Dividing by 5 gives $7x \equiv 2 \pmod{11}$, so $x \equiv 5 \pmod{11}$. Among the listed values, only $115 = 5 + 11(10)$ has that residue. The interval alone need not make the solution unique; the question asks which listed integer satisfies it.

Solution 174.3 ■ Answer B[Back to problem](#)

The coefficient gcd is twelve. The congruence is solvable exactly for the five multiples of twelve modulo sixty: 0, 12, 24, 36, 48. Each such right-hand side has twelve preimages. The question counts permissible right-hand sides, rather than the solutions for one fixed right-hand side.

Solution 174.4 ■ Answer C[Back to problem](#)

Translation by b is always a permutation, with inverse translation by $-b$. Multiplication by a is a permutation exactly when $\gcd(a, n) = 1$: then every linear congruence has a unique solution; otherwise the congruence $ax \equiv 0$ has more than one solution. There are $\varphi(n)$ choices for a and independently n for b , giving $n\varphi(n)$.

Solution 174.5 ■ Answer B[Back to problem](#)

Divide the congruence by twelve to obtain $x \equiv 1 \pmod{5}$. Such residues already avoid the prime five. CRT pairs this fixed residue modulo five with any residue modulo twelve; exactly $\varphi(12) = 4$ of those are coprime to twelve. Thus four of the twelve original solutions are units modulo sixty: 1, 11, 31, 41.

Part B: Abstract Algebra ▸ Binary Structures and the Definition of a Group

175 Binary operations, semigroups, and monoids

Solution 175.1 ■ Answer D[Back to problem](#)

The identity is 0, and $1 + (x * y) = (1 + x)(1 + y)$ proves associativity. However, $(-1) * y = -1$ for every y , so -1 has no inverse. Thus the structure is a monoid but not a group.

Solution 175.2 ■ Answer A[Back to problem](#)

For an element a , finiteness forces two nonnegative powers to agree, say $a^i = a^j$ with $i < j$. Cancel a^i to obtain $a^{j-i} = e$. Thus a^{j-i-1} is a two-sided inverse. This holds for every element; commutativity is not required.

Solution 175.3 ■ Answer C[Back to problem](#)

Both $(x * y) * z$ and $x * (y * z)$ equal x , so the operation is associative. Also $x * e = x$ for any e , making every element a right identity. A left identity would need $e * x = e = x$ for every x , impossible on a set with at least two elements. Identity uniqueness concerns two-sided identities.

Solution 175.4 ■ Answer C[Back to problem](#)

The row-column property holds because negation is a bijection modulo three. Nevertheless, $(0 * 0) * 1 = 0 * 1 = 2$, while $0 * (0 * 1) = 0 * 2 = 1$. This explicit triple disproves associativity; a Latin-square table alone is insufficient.

Solution 175.5 ■ Answer A[Back to problem](#)

Evaluating $l * r$ by the left-identity property gives r , and by the right-identity property gives l . Hence $l = r$, and the common element has both properties. This proof does not use associativity.

Part B: Abstract Algebra ▸ Binary Structures and the Definition of a Group

176 Groups, inverses, and commutativity

Solution 176.1 ■ **Answer A**
[Back to problem](#)

Multiply on the left by a to obtain $xba = ac$, then on the right by $a^{-1}b^{-1}$. Thus $x = aca^{-1}b^{-1}$. The order of these final factors is essential.

Solution 176.2 ■ **Answer D**
[Back to problem](#)

Every element is its own inverse. Thus $ab = (ab)^{-1} = b^{-1}a^{-1} = ba$. Elementary Abelian 2-groups can be noncyclic and arbitrarily large, so the other assertions are not forced.

Solution 176.3 ■ **Answer A**
[Back to problem](#)

If $ax = xb$, then $a^2x = xb^2$. Since $a^2 = e$, cancellation of x gives $b^2 = e$, contrary to b having order three. This uses the orders of the elements without commuting any factors.

Solution 176.4 ■ **Answer C**
[Back to problem](#)

The identity is $abab = aabb$. Cancel a on the left and b on the right to obtain $ba = ab$ for arbitrary elements.

Solution 176.5 ■ **Answer C**
[Back to problem](#)

Reverse the order and invert each factor: $(aba^{-1})^{-1} = (a^{-1})^{-1}b^{-1}a^{-1} = ab^{-1}a^{-1}$.

Solution 176.6 ■ **Answer C**
[Back to problem](#)

Rectangular real matrices add entrywise, with the zero matrix as identity and $-A$ as inverse; addition is commutative and there are infinitely many matrices. The first set is not closed under addition. Positive integers lack multiplicative inverses in that set, all square matrices include singular ones, and positive reals under addition omit the identity zero.

Part B: Abstract Algebra ▸ Examples of Groups

177 General and special linear groups

Solution 177.1 ■ **Answer C**
[Back to problem](#)

The first column of an invertible matrix has $3^2 - 1 = 8$ choices; the second avoids its three-element span, giving 6 choices. Thus GL has 48 elements. Determinants 1 and 2 occur equally often, so SL has 24.

Solution 177.2 ■ **Answer B**
[Back to problem](#)

An integer matrix has an integer inverse exactly when its determinant is ± 1 . Here $4 - 3k = 1$ gives $k = 1$, whereas $4 - 3k = -1$ gives the noninteger $5/3$. Exactly one integer works.

Solution 177.3 ■ Answer A[Back to problem](#)

Positive determinants contain the identity and stay positive under products and reciprocals, so the subgroup test holds. Each other set omits the identity: its determinant is one, its trace is two, and its off-diagonal entries are zero.

Solution 177.4 ■ Answer C[Back to problem](#)

Both A and A^{-1} have integer entries. Every common divisor of the coordinates of Av divides those of $A^{-1}(Av) = v$, so that divisor is one. More generally the coordinate gcd is preserved in both directions. Such matrices preserve primitive integer vectors, not necessarily their lengths.

Solution 177.5 ■ Answer C[Back to problem](#)

Cayley–Hamilton gives $A^2 - (\text{tr } A)A + I = 0$. Using $A^2 = I$ yields $(\text{tr } A)A = 2I$, so the trace is nonzero and A is scalar. Its square being I then forces the scalar to be 1 or -1 . The determinant-one restriction excludes the nontrivial planar reflections.

Part B: Abstract Algebra ▷ Examples of Groups

178 Roots of unity as a finite group

Solution 178.1 ■ Answer D[Back to problem](#)

The common roots are precisely the $\gcd(84, 126) = 42$ roots of unity of order dividing 42. Bezout also shows the two equations imply $z^{42} = 1$.

Solution 178.2 ■ Answer D[Back to problem](#)

The least positive m with $(\zeta^{12})^m = 1$ satisfies $45 \mid 12m$. Dividing by $\gcd(45, 12) = 3$ gives $15 \mid m$, so the order is 15.

Solution 178.3 ■ Answer D[Back to problem](#)

Expand each sixth power and sum by exponent. The geometric sum $\sum_{k=0}^4 \zeta^{jk}$ is five when $5 \mid j$ and zero otherwise. For $0 \leq j \leq 6$, only $j = 0, 5$ survive. Thus the result is $5\left[\binom{6}{0} + \binom{6}{5}\right] = 5(1 + 6) = 35$.

Solution 178.4 ■ Answer C[Back to problem](#)

Write $z = e^{2\pi ik/60}$. The equation is $14k \equiv 2 \pmod{60}$, equivalent to $7k \equiv 1 \pmod{30}$. Since seven is a unit modulo thirty, this determines one class there and two classes modulo sixty, namely $k = 13, 43$. Both correspond to valid elements of the specified group.

Solution 178.5 ■ Answer D[Back to problem](#)

Write $u = \zeta^a, v = \zeta^b$ for a primitive eighth root ζ . The primitive exponents are the four odd residues 1, 3, 5, 7, in agreement with $\varphi(8) = 4$. Their sum is even, so the product has order dividing four. Taking exponent pairs (1, 7), (1, 3), (1, 1) gives orders one, two, and four respectively.

Part B: Abstract Algebra ▸ Examples of Groups

179 Permutations and cycle decompositions

Solution 179.1 ■ Answer C[Back to problem](#)

Choose the three elements of the 3-cycle in $\binom{7}{3}$ ways and arrange that cycle in $2!$ ways. Choose two of the remaining four for the transposition in $\binom{4}{2}$ ways. The total is $35 \cdot 2 \cdot 6 = 420$.

Solution 179.2 ■ Answer C[Back to problem](#)

The order of σ is $\text{lcm}(4, 3) = 12$. Squaring gives order $12 / \gcd(12, 2) = 6$. Equivalently, the 4-cycle splits into two transpositions while the 3-cycle still has order 3.

Solution 179.3 ■ Answer A[Back to problem](#)

Track the images: $1 \mapsto 2, 2 \mapsto 4, 4 \mapsto 3, \text{ and } 3 \mapsto 1$. These form the four-cycle (1 2 4 3). The opposite composition gives a different cycle; overlapping cycles cannot simply be treated as disjoint.

Solution 179.4 ■ Answer C[Back to problem](#)

The equation is equivalent to $\tau\sigma = \sigma\tau$. Once $\tau(1)$ is chosen, commutation forces $\tau(\sigma^j(1)) = \sigma^j(\tau(1))$ for all j , determining the entire permutation. Each of the five choices produces one power of σ , so exactly five work.

Solution 179.5 ■ Answer B[Back to problem](#)

Squaring an odd cycle leaves one odd cycle of the same length; squaring a cycle of length $2k$ gives two k -cycles. Consequently even-length cycles in a square must occur in pairs of each length. Only B satisfies that requirement, and $(1\,3\,2\,4)^2 = (1\,2)(3\,4)$. All choices are even permutations, so parity alone would not decide the question.

Part B: Abstract Algebra ▸ Examples of Groups

180 Dihedral groups and geometric symmetry

Solution 180.1 ■ **Answer C**
[Back to problem](#)

Use $sr^7 = r^{-7}s$: $(r^4s)(r^7s) = r^{4-7}s^2 = r^{-3} = r^6$.

Solution 180.2 ■ **Answer D**
[Back to problem](#)

There are ten reflections, each of order 2. Among rotations only the half-turn r^5 has order 2. Hence there are eleven such elements.

Solution 180.3 ■ **Answer C**
[Back to problem](#)

All twelve rotations commute with r , and no reflection does, because conjugating r by a reflection gives $r^{-1} \neq r$. A rotation r^k commutes with s exactly when $r^k = r^{-k}$, or $2k \equiv 0 \pmod{12}$; these are $k = 0, 6$. Removing those two leaves ten.

Solution 180.4 ■ **Answer C**
[Back to problem](#)

The two reflections generate the rotation r^6 , of order $15/\gcd(15, 6) = 5$. Every word is either one of its five powers or such a power followed by s . The two sets are disjoint, giving a subgroup of order ten, isomorphic to D_5 .

Solution 180.5 ■ **Answer B**
[Back to problem](#)

Conjugation by r^j gives $r^j s r^{-j} = r^{2j}s$, yielding the five even-exponent reflections. Conjugation by $r^j s$ yields the same forms. No odd-exponent reflection appears, so the ten reflections split into two conjugacy classes, each of size five.

Part B: Abstract Algebra ▸ Examples of Groups

181 Residue groups and multiplication tables

Solution 181.1 ■ **Answer C**
[Back to problem](#)

Factor $84 = 2^2 \cdot 3 \cdot 7$. Then $\varphi(84) = 84(1 - 1/2)(1 - 1/3)(1 - 1/7) = 24$.

Solution 181.2 ■ **Answer C**
[Back to problem](#)

The units are 1, 3, 5, 7. Each nonidentity unit squares to 1 modulo 8, so the group has order four and no element of order four. It is the Klein four-group, not the cyclic group of order four.

Solution 181.3 ■ Answer D[Back to problem](#)

CRT separates modulo forty into modulo eight and modulo five. All four units modulo eight square to one, while modulo five only $1, -1$ do. Hence there are $4 \cdot 2 = 8$ roots of $u^2 = 1$. Every root is a unit, and only the identity has order one, leaving seven elements of exact order two.

Solution 181.4 ■ Answer B[Back to problem](#)

The row and column of b reproduce the headers, so b is the identity. The products $ac = ca = b$ show $a^{-1} = c$. Successive powers of a are a, d, c, b , covering all four elements; hence the group is cyclic. The table is symmetric, also confirming commutativity.

Solution 181.5 ■ Answer C[Back to problem](#)

For a prime modulus every nonzero class is coprime to the modulus and hence invertible. If n is composite, write $n = ab$ with $1 < a, b < n$. The nonzero classes of a, b multiply to zero, so the proposed set is not even closed. The full unit group exists for every modulus, but it can be smaller than the nonzero classes.

Part B: Abstract Algebra ▷ Cyclic Groups

182 Cyclic groups and generators

Solution 182.1 ■ Answer C[Back to problem](#)

The number is $\varphi(90) = 90(1 - 1/2)(1 - 1/3)(1 - 1/5) = 24$.

Solution 182.2 ■ Answer D[Back to problem](#)

The subgroup generated by a^r, a^s consists of powers whose exponents are integer combinations of r, s , hence equals $\langle a^{\gcd(r,s)} \rangle$. Only 15 and 22 are coprime.

Solution 182.3 ■ Answer D[Back to problem](#)

If a subgroup contained any a^k with $k > 0$, it would also contain a^{-k} , which is outside P because a has infinite order. Thus only the identity can occur. Nonnegative powers form a cancellative monoid, but do not form the entire infinite cyclic group.

Solution 182.4 ■ Answer C[Back to problem](#)

The first element must be one of the three nonidentity elements, and the second must be a different nonidentity element, giving two choices. Such a pair generates both elements, their distinct product, and the identity. An identity or repeated element generates at most a subgroup of order two, so the count is $3 \cdot 2 = 6$.

Solution 182.5 ■ Answer A[Back to problem](#)

Zero generates only zero. If $q \neq 0$, its integer multiples cannot include $q/2$, since $kq = q/2$ would require the noninteger $k = 1/2$. Thus no rational number generates all of $\mathbb{Q}$. An infinite Abelian group need not be infinite cyclic.

Part B: Abstract Algebra ▷ Subgroups

183 Subgroups and the center

Solution 183.1 ■ Answer D[Back to problem](#)

The three subgroups of order two in V_4 each contain the identity and one different non-identity element, so their union is all of V_4 . A cyclic group cannot be covered by proper subgroups, since a generator belongs to none of them. The theorem about a union of two subgroups does not extend to a union of three.

Solution 183.2 ■ Answer C[Back to problem](#)

Commuting with every diagonal invertible matrix forces all off-diagonal entries to vanish. Commuting with every permutation matrix then forces all diagonal entries to agree. Conversely, every nonzero scalar matrix commutes with every matrix.

Solution 183.3 ■ Answer A[Back to problem](#)

The set HK has size $|H||K|/|H \cap K| \leq 36$, forcing intersection size at least six. In a cyclic group of order 36, the subgroups of orders 12, 18 intersect in the subgroup of order six.

Solution 183.4 ■ Answer B[Back to problem](#)

Writing elements as $g^a z$ and $g^b w$ with z, w central shows that they commute. Thus $G = Z(G)$, although an abelian group need not be cyclic.

Solution 183.5 ■ Answer D[Back to problem](#)

With index two there is only one coset other than H , so the nontrivial left and right cosets both equal $G \setminus H$. Therefore all left and right cosets agree.

Solution 183.6 ■ Answer C[Back to problem](#)

I contains one, and $(xy^{-1})^2 = x^2/y^2$ remains rational, so the subgroup test applies. II contains two but omits its inverse $1/2$. For III, left multiplication by any member permutes the finite subset, since ambient cancellation makes it injective. Surjectivity first forces the identity into the subset and then supplies each inverse. This proves III without assuming those properties at the start.

Part B: Abstract Algebra ▸ Cyclic Subgroups

184 Element order and generated subgroups

Solution 184.1 ■ **Answer D**[Back to problem](#)

The order n of g is a multiple of both ten and six, hence of thirty. Also $g^{60} = e$, so $n \mid 60$. The possibilities are thirty and sixty. At thirty the power g^6 would have order five, contrary to the hypothesis, so $n = 60$.

Solution 184.2 ■ **Answer E**[Back to problem](#)

Certainly $(ab)^{120} = e$. If $(ab)^k = e$, then $a^k = b^{-k}$ belongs to both $\langle a \rangle$ and $\langle b \rangle$. Their intersection is trivial because its order divides both 8 and 15. Thus both 8 and 15 divide k , forcing $120 \mid k$.

Solution 184.3 ■ **Answer D**[Back to problem](#)

Cycle types show the possible orders are 1, 2, 3, 4. Their least common multiple is twelve. No permutation of four symbols has order twelve, so a group exponent need not be attained as the order of one element.

Solution 184.4 ■ **Answer E**[Back to problem](#)

The composition is $s(t(x)) = x - 2$. Its k th positive power is $x \mapsto x - 2k$, never the identity. Thus two finite-order elements can have a product of infinite order when they do not commute; the Abelian product-order rule cannot be used.

Solution 184.5 ■ **Answer D**[Back to problem](#)

Any finite list of positive rationals uses only finitely many primes in its numerators and denominators. Products of those rationals and their inverses introduce no new prime factors. A prime outside that finite list therefore cannot be generated. The group is Abelian but requires infinitely many generators.

Part B: Abstract Algebra ▸ Generators and Relations

185 Group presentations and word reduction

Solution 185.1 ■ Answer D

[Back to problem](#)

Conjugation preserves powers, so $ab^ma^{-1} = b^{2m}$. Each of the three conjugations by a doubles the exponent, starting from two; the resulting exponent is $2 \cdot 2^3 = 16$. No interchange of a and b is valid.

Solution 185.2 ■ Answer C

[Back to problem](#)

Eliminate b using $b = a^2$. The remaining relations reduce to $a^4 = e$, so at most four elements remain. The cyclic group of order four realizes all relations with a of order four, proving that the order is exactly four.

Solution 185.3 ■ Answer D

[Back to problem](#)

The relation $ba = a^{-1}b$ moves each b to the right. Every word reduces to a^k or a^kb , with $0 \leq k < 4$, so the order is at most eight. A quarter-turn a and a reflection b of a square satisfy all three relations and generate eight distinct symmetries. This realization gives the matching lower bound; the relation does not allow replacing b by a power of a .

Solution 185.4 ■ Answer C

[Back to problem](#)

Commutation reduces every word to a^mb^n for integers m, n . Realize $a = (1, 0)$ and $b = (0, 1)$ in the additive group $\mathbb{Z}^2$; all such pairs are distinct and satisfy the relation. Thus no further identifications occur. Commuting generators do not acquire finite orders without corresponding relations.

Solution 185.5 ■ Answer B

[Back to problem](#)

Conjugating by a three times sends b to b^8 , but $a^3 = e$ makes this equal to b . Hence $b^7 = e$, and together with $b^5 = e$ this forces $b = e$. The presentation reduces to $\langle a \mid a^3 = e \rangle$. A cyclic group of order three realizes the relations, proving the order is exactly three.

Part B: Abstract Algebra ▸ Some Theorems Concerning Subgroups

186 Lagrange, Cauchy, and Sylow existence

Solution 186.1 ■ **Answer E**[Back to problem](#)

Since $180 = 2^2 \cdot 3^2 \cdot 5$, Sylow existence guarantees subgroups of orders 4, 9, and 5, and their prime-power subgroups. Among these choices only 9 is such a guaranteed prime power. The question specifies what follows from Sylow existence alone.

Solution 186.2 ■ **Answer C**[Back to problem](#)

There is exactly one subgroup for every divisor. Since $360 = 2^3 3^2 5$, the divisor count is $(3 + 1)(2 + 1)(1 + 1) = 24$.

Solution 186.3 ■ **Answer B**[Back to problem](#)

Lagrange requires the group order to be divisible by seven. Cauchy excludes any order divisible by three or five, leaving twenty-eight. The cyclic group of order twenty-eight realizes the conditions, since its element orders divide twenty-eight.

Solution 186.4 ■ **Answer A**[Back to problem](#)

Such a subgroup would have index two and hence be normal, giving a quotient of order two. Every three-cycle must map to the quotient identity, because its image has order dividing both three and two. There are eight three-cycles, so the subgroup would contain them and the identity, more than six elements. This contradicts its proposed size and disproves the general converse to Lagrange.

Solution 186.5 ■ **Answer B**[Back to problem](#)

Cauchy gives elements of orders two and three. They commute, so their product has order six, producing the required cyclic subgroup. The group $\mathbb{Z}_2^3 \times \mathbb{Z}_3^2$ has order seventy-two and every element has order dividing six, ruling out the other guarantees. Existence of a subgroup of a divisor order does not imply existence of a cyclic subgroup of that order.

Part B: Abstract Algebra ▸ The Concept of Isomorphism

187 Isomorphism and structural invariants

Solution 187.1 ■ **Answer B**
[Back to problem](#)

The element i has order four and generates the group. Therefore it is cyclic of order four and isomorphic to $\mathbb{Z}_4$; both other groups of order four listed have exponent two.

Solution 187.2 ■ **Answer C**
[Back to problem](#)

In additive notation the first group is divisible by two: $x = a/2$ solves $x + x = a$. The analogous multiplicative equation $x^2 = a$ in positive rationals has no rational solution at $a = 2$. Here a group square root means an element whose product with itself, under the specified group operation, is the given element.

Solution 187.3 ■ **Answer A**
[Back to problem](#)

Because 3 and 4 are coprime, $(1, 1)$ has order twelve in the first product. The other abelian products have no element of order twelve, and the nonabelian groups cannot be cyclic.

Solution 187.4 ■ **Answer D**
[Back to problem](#)

The first group contains an element of order eight, while every element of the second has order dividing four. Isomorphisms preserve element orders.

Solution 187.5 ■ **Answer B**
[Back to problem](#)

The map $q \mapsto 2^q$ is a bijection onto the displayed group and converts addition to multiplication. The other groups differ in divisibility, cardinality, or torsion properties.

Part B: Abstract Algebra ▸ The Classification of Finite Abelian Groups

188 Direct products, direct sums, and element orders

Solution 188.1 ■ **Answer D**
[Back to problem](#)

The component orders are $18/6 = 3$, $25/5 = 5$, and $21/3 = 7$. Their least common multiple is 105.

Solution 188.2 ■ **Answer D**
[Back to problem](#)

The equation $4x = 0$ has four solutions in each factor, giving 16 elements whose order divides four. The equation $2x = 0$ has two solutions per factor, giving four with order dividing two. Subtract to get $16 - 4 = 12$.

Solution 188.3 ■ Answer C[Back to problem](#)

An element of S is a binary sequence with only finitely many ones, specified by a finite subset of the natural numbers. There are countably many such subsets. Elements of P are all binary sequences, an uncountable set by the diagonal argument. Finite direct sums and products agree, but this infinite distinction is substantial.

Solution 188.4 ■ Answer C[Back to problem](#)

The integer multiples of $(1, 1)$ are $(k, k \bmod 2)$. Only $(2, 0)$ among the choices satisfies this relation. The generator has infinite order, yet its subgroup is proper because it omits $(0, 1)$.

Solution 188.5 ■ Answer B[Back to problem](#)

Products and inverses are coordinatewise, so $(g, g)(h, h) = (gh, gh)$ and $(g, g)^{-1} = (g^{-1}, g^{-1})$ stay in D . The map $g \mapsto (g, g)$ is an isomorphism from S_3 to D , so D has six elements and is non-Abelian. A subgroup of a direct product need not be a product of independent coordinate subgroups.

Part B: Abstract Algebra ▸ The Classification of Finite Abelian Groups

189 Elementary divisors and invariant factors

Solution 189.1 ■ Answer D[Back to problem](#)

The partitions of 4 number five, and those of 3 number three. The choices for the two primary components are independent, so the total is $5 \cdot 3 = 15$.

Solution 189.2 ■ Answer A[Back to problem](#)

The 2-power factors are 4, 8 and the 3-power factors are 3, 9. Multiply corresponding sorted entries to obtain 12, 72, and indeed $12 \mid 72$. Their product is the group order 864.

Solution 189.3 ■ Answer C[Back to problem](#)

For a product of r nontrivial cyclic two-power groups, there are $2^r - 1$ elements of exact order two. The count three forces two cyclic factors. Their exponents partition four into two parts: $3 + 1$ or $2 + 2$. The first gives an element of order eight and is excluded; the second yields $\mathbb{Z}_4 \times \mathbb{Z}_4$.

Solution 189.4 ■ Answer C[Back to problem](#)

Projection onto the three two-primary coordinates is a three-dimensional vector space over $\mathbb{F}_2$, requiring at least three generators. Conversely the group is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{18}$, by combining one $\mathbb{Z}_2$ with the coprime $\mathbb{Z}_9$. One generator for each factor gives three.

Solution 189.5 ■ Answer C[Back to problem](#)

An index-three subgroup is the kernel of a surjective homomorphism to $\mathbb{Z}_3$. Such a map is $f(x, y, z) = \alpha x + \beta y \pmod{3}$; the factor of order five must map to zero. There are eight nonzero pairs (α, β) , and two maps, differing by multiplication by -1 , have each kernel. Hence there are four kernels, given by $x = 0, y = 0, x + y = 0, x - y = 0$ modulo three.

Part B: Abstract Algebra ▷ Group Homomorphisms

190 Homomorphisms and maps from cyclic groups

Solution 190.1 ■ Answer C[Back to problem](#)

The image y of 1 must satisfy $18y = 0 \pmod{30}$. This congruence has $\gcd(18, 30) = 6$ solutions, each defining exactly one homomorphism.

Solution 190.2 ■ Answer B[Back to problem](#)

The image of 1 must generate $\mathbb{Z}_8$, hence be 1, 3, 5, or 7. Each works because $8 \mid 24$, so the relation $24 \cdot 1 = 0$ is respected. There are four maps.

Solution 190.3 ■ Answer B[Back to problem](#)

The image of the generator must be killed by six modulo four, which holds for the image two. Equivalently replacing x by $x + 6$ changes $2x$ by twelve, a multiple of four, and additivity holds. A and E depend on the representative; C also fails the identity condition; D is well defined but fails additivity at $1 + 1$.

Solution 190.4 ■ Answer B[Back to problem](#)

Multiplication by two is additive and injective in both groups. In $\mathbb{Z}$ its image omits odd integers, whereas every rational q is the image of $q/2$. Thus the same algebraic rule has different surjectivity depending on the group.

Solution 190.5 ■ Answer B[Back to problem](#)

For any $g \in G$, its order divides twenty-seven, and the order of its image divides that order as well as twenty. The only common divisor is one, so every element maps to the identity. The trivial map exists, giving exactly one homomorphism, without any cyclicity assumption on G .

Solution 190.6 ■ Answer A[Back to problem](#)

Belonging to H means $3a + 4b$ is divisible by six. Modulo two this forces a even, and modulo three it forces b divisible by three. Conversely these two conditions make the expression divisible by six. Thus the inverse image is the displayed subgroup of the domain.

Solution 190.7 ■ Answer D[Back to problem](#)

Always $j(ab) = b^{-1}a^{-1}$, while $j(a)j(b) = a^{-1}b^{-1}$. These are equal for all a, b exactly when all inverse elements commute, equivalently when G is Abelian. Bijection by itself does not make a map an automorphism.

Part B: Abstract Algebra ▸ Group Homomorphisms

191 Kernels and normal subgroups

Solution 191.1 ■ Answer B[Back to problem](#)

The kernel has order $72/12 = 6$. Each nonempty fiber is a coset of the kernel, so every specified image element has exactly six preimages.

Solution 191.2 ■ Answer B[Back to problem](#)

The map is well defined since $24 \mid 10 \cdot 36$. The kernel condition is $24 \mid 10x$, equivalent to $12 \mid x$. In $\mathbb{Z}_{36}$ the possibilities are 0, 12, 24, so the kernel has order three.

Solution 191.3 ■ Answer B[Back to problem](#)

The determinant is multiplicative, with kernel $SL_n(\mathbb{R})$. If $B = \text{diag}(-2, 1, \dots, 1)$, then $f^{-1}(-2) = BSL_n(\mathbb{R})$. This fiber is nonempty but omits the identity, so it is not a subgroup.

Solution 191.4 ■ Answer C[Back to problem](#)

The kernel is a subgroup of a prime-order group, so its order is either 1 or p . The latter would make the entire image the identity, contrary to the hypothesis. Thus the kernel is trivial and f is injective. Its codomain can be larger and non-Abelian.

Solution 191.5 ■ Answer D[Back to problem](#)

Kernels are normal. Conjugating (12) by (123) gives (23) , which is outside $\{e, (12)\}$. The other three occur as kernels of the identity map, the sign map, and the trivial map, respectively.

Solution 191.6 ■ Answer D[Back to problem](#)

The stated codomain is $m\mathbb{Z}$: every element there is mk for a unique integer k . Addition is preserved, so f is an isomorphism. Yet $m\mathbb{Z}$ is a proper subgroup of $\mathbb{Z}$. This can happen for infinite groups; a finite group cannot be isomorphic to a proper subgroup because cardinalities would differ.

Part B: Abstract Algebra ▷ Group Homomorphisms

192 Inner automorphisms and automorphism groups

Solution 192.1 ■ **Answer C**
[Back to problem](#)

Two elements induce the same conjugation map exactly when they differ by a central element. Hence the number of maps is $|G|/|Z(G)| = 60/5 = 12$.

Solution 192.2 ■ **Answer B**
[Back to problem](#)

The multiplier 5 is a unit modulo 24, so the map is an automorphism. Its square is multiplication by $25 \equiv 1 \pmod{24}$, but it is not itself the identity. Its order is therefore two.

Solution 192.3 ■ **Answer C**
[Back to problem](#)

An automorphism permutes the three nonzero elements. Any such permutation preserves addition: each nonzero element added to itself is zero, and the sum of two distinct nonzero elements is the third. Thus all $3! = 6$ permutations work. Since V is Abelian, every inner automorphism is the identity.

Solution 192.4 ■ **Answer B**
[Back to problem](#)

At g , the left side is $\alpha(a\alpha^{-1}(g)a^{-1}) = \alpha(a)g\alpha(a)^{-1}$. This is $c_{\alpha(a)}(g)$, showing in particular that inner automorphisms form a normal subgroup of the full automorphism group.

Solution 192.5 ■ **Answer D**
[Back to problem](#)

Every inner automorphism must be the identity, so G is Abelian. In an Abelian group, inversion $g \mapsto g^{-1}$ is an automorphism. It too must be the identity, giving $g = g^{-1}$. The group of order two satisfies the hypothesis, so triviality is not forced.

Part B: Abstract Algebra ▷ Rings

193 Rings, unity, and subrings

Solution 193.1 ■ **Answer C**
[Back to problem](#)

An identity $e \in S$ must satisfy $3e \equiv 3 \pmod{15}$, so $e \equiv 1 \pmod{5}$. The unique such element of S is 6. Every member is a multiple of 3, and multiplication by 6 fixes all of them modulo 15.

Solution 193.2 ■ **Answer D**
[Back to problem](#)

Upper triangular matrices are closed under subtraction and multiplication and contain zero. The trace-zero and symmetric conditions fail under multiplication; determinant-one and invertible matrices do not contain the additive zero.

Solution 193.3 ■ Answer C[Back to problem](#)

Zero multiplication is associative, commutative, and distributive over addition, so the ring axioms hold. An identity would have to satisfy $e * 1 = 1$, but every product is zero. Cancellation and a unity are not required by the stated ring convention.

Solution 193.4 ■ Answer A[Back to problem](#)

Subtraction gives $ere - ese = e(r - s)e$. A product is $(ere)(ese) = e(res)e$, using $e^2 = e$, so S is a subring. Also $e(ere) = (ere)e = ere$. The ambient identity need not belong to S ; a matrix projection provides a proper corner subring.

Solution 193.5 ■ Answer D[Back to problem](#)

Both sides in I equal a , and both sides in III equal $a + b$. In II the left side is a , while the right side is $2a$, so this law fails, for instance at $a = 1$. Associative multiplication and an Abelian additive group do not suffice for a ring; both distributive laws must hold.

Part B: Abstract Algebra ▸ Rings

194 Ring characteristic

Solution 194.1 ■ Answer D[Back to problem](#)

The matrix ring has characteristic 12, since a scalar multiple of its identity vanishes exactly when that scalar is divisible by 12. The product characteristic is $\text{lcm}(12, 18) = 36$.

Solution 194.2 ■ Answer C[Back to problem](#)

The characteristic is a prime p . The additive subgroup generated by 1 has order p , which divides 125 by Lagrange. The only possible prime is 5.

Solution 194.3 ■ Answer C[Back to problem](#)

The formal monomials $1, x, x^2, \dots$ are distinct, so the ring is infinite. Adding the constant polynomial 1 to itself p times gives zero, with no smaller positive multiple doing so. Positive characteristic therefore does not imply finiteness. The nonconstant polynomial x has no polynomial inverse.

Solution 194.4 ■ Answer A[Back to problem](#)

No positive integer annihilates $(1, 0)$, so the characteristic is zero. But $a \neq 0$ and $a + a = (0, 0)$, giving additive order two. Characteristic zero excludes a common positive annihilator of every element; it does not exclude individual torsion elements.

Solution 194.5 ■ **Answer D**[Back to problem](#)

Twice every matrix is zero, while the identity is nonzero, so the characteristic is two. The nonzero matrix E_{12} has square zero. Thus prime characteristic does not imply that a ring is a domain or field; the implication in the exposition goes in the opposite direction.

Part B: Abstract Algebra ▸ Rings

195 Polynomial, Gaussian-integer, and function rings

Solution 195.1 ■ **Answer B**[Back to problem](#)

If $zw = 1$, norms give $N(z)N(w) = 1$ with nonnegative integers. Thus $N(z) = 1$, whose solutions are $z = 1, -1, i, -i$. All four have Gaussian integer inverses.

Solution 195.2 ■ **Answer C**[Back to problem](#)

If $fg = 1$ pointwise, f cannot vanish anywhere. Conversely, when $f(x) \neq 0$ for every x , the function $g(x) = 1/f(x)$ belongs to this unrestricted function ring and is its inverse.

Solution 195.3 ■ **Answer A**[Back to problem](#)

Products in S remain in S because $(\sqrt{2})^2 = 2$. In contrast, $\sqrt[3]{2} \in T$ but its square is not in T : such a representation would give a polynomial of degree at most two over $\mathbb{Q}$ satisfied by $\sqrt[3]{2}$. Its minimal polynomial is $x^3 - 2$, irreducible by the rational-root test for a cubic. Thus T fails multiplicative closure.

Solution 195.4 ■ **Answer D**[Back to problem](#)

The identity $(1 + 2x)^2 = 1$ holds modulo four. A unit must have unit constant coefficient and nilpotent nonconstant coefficients; $1 + x$ fails the latter requirement.

Solution 195.5 ■ **Answer C**[Back to problem](#)

Both functions are continuous and nonzero. At each x , at least one is zero, so their pointwise product vanishes. Their sum is $|x|$, which vanishes at zero and is not a unit. Moreover $f \circ g = g$, not the zero function, illustrating why composition is not the multiplication in this ring.

Solution 195.6 ■ **Answer B**[Back to problem](#)

The leading coefficient is one, so this formal polynomial is nonzero. Fermat's theorem gives $t^p = t$ for every $t \in \mathbf{F}_p$, including zero, hence its associated function is identically zero. A formal indeterminate is not a variable already restricted to the finite set of field elements.

Part B: Abstract Algebra ▸ Rings

196 Nilpotent elements

Solution 196.1 ■ **Answer C**
[Back to problem](#)

Since $72 = 2^3 3^2$, nilpotence requires divisibility by both 2 and 3, hence by 6. Conversely any multiple of 6 has a sufficiently high power divisible by 72. There are $72/6 = 12$ such residue classes, including zero.

Solution 196.2 ■ **Answer B**
[Back to problem](#)

$U^2 = I$, so U is a unit, and $N^2 = 0$ with $N \neq 0$. But $U + N = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ is singular. The general identity making $1 + N$ invertible does not justify replacing 1 by an arbitrary noncommuting unit. If the unit commutes with N , factoring it out does give an invertible sum.

Solution 196.3 ■ **Answer B**
[Back to problem](#)

Since $288 = 2^5 3^2$ and $12^k = 2^{2k} 3^k$, the power vanishes modulo 288 precisely when $2k \geq 5$ and $k \geq 2$. The least such integer is $k = 3$. This asks for the first vanishing power, not merely whether the element is nilpotent.

Solution 196.4 ■ **Answer C**
[Back to problem](#)

Because a, b commute, the binomial expansion applies. Every term of $(a + b)^4$ contains a^2 or b^3 , so it is zero. But $(a + b)^3 = 3ab^2 \neq 0$ over $\mathbb{R}$. Hence the index is exactly four, and the exponent bound $2 + 3 - 1$ is sharp here.

Solution 196.5 ■ **Answer A**
[Back to problem](#)

Idempotence implies $a^k = a$ for every positive integer k , by induction. In particular $a = a^m = 0$. No unity, commutativity, or characteristic assumption is needed. Nilpotence and idempotence have very different meanings despite their similar names.

Part B: Abstract Algebra ▸ Ring Homomorphisms

197 Ring homomorphisms and evaluation

Solution 197.1 ■ **Answer D**
[Back to problem](#)

Such a map is $f(n) = ne$, and multiplicativity holds precisely when $e^2 = e$ in $\mathbb{Z}_6$. The possible values are 0, 1, 3, 4, giving four maps.

Solution 197.2 ■ **Answer B**
[Back to problem](#)

The factor theorem says $f(3) = 0$ exactly when $x - 3$ divides $f(x)$. Thus the kernel is the ideal of all multiples of $x - 3$. The larger-degree polynomial $x^2 - 9$ does not generate every such multiple.

Solution 197.3 ■ Answer C[Back to problem](#)

A unital map must send the class of k to the class of k , since it preserves addition and 1. The relation $m \cdot 1 = 0$ in the domain therefore requires $m = 0$ modulo n . Conversely, when $n \mid m$, reduction modulo n is well defined and preserves both operations and unity.

Solution 197.4 ■ Answer D[Back to problem](#)

Since $xB = Bx$ for each constant matrix B , a homomorphism fixing coefficients must satisfy $AB = BA$ for every B . Commuting with E_{11} forces A diagonal, and then commuting with E_{12} forces equal diagonal entries. Conversely a scalar matrix commutes with all coefficients, so substitution respects products.

Solution 197.5 ■ Answer C[Back to problem](#)

For I, the images of $1 \cdot 1$ and the product of the images are 2 and 4. For II, $\det(I + I) = 4$ but $\det I + \det I = 2$. Complex conjugation preserves sums and products, so III is a ring automorphism. Preserving just the additive or multiplicative structure is insufficient.

Part B: Abstract Algebra ▷ Ring Homomorphisms

198 Frobenius endomorphism

Solution 198.1 ■ Answer A[Back to problem](#)

Apply Frobenius twice: $(x + 2)^{25} = x^{25} + 2^{25}$. In $\mathbf{F}_5$, $2^4 = 1$, so $2^{25} = 2$. Thus the answer is $x^{25} + 2$.

Solution 198.2 ■ Answer C[Back to problem](#)

If F is injective and $a^m = 0$, choose $p^k \geq m$. Then $F^k(a) = a^{p^k} = 0$, and repeated injectivity forces $a = 0$. Conversely, if $F(a) = F(b)$ then $(a - b)^p = 0$, so absence of nonzero nilpotents gives $a = b$. A product of two copies of $\mathbf{F}_p$ has zero divisors but injective Frobenius, disproving necessity of a domain.

Solution 198.3 ■ Answer C[Back to problem](#)

Every element satisfies $z^{16} = z$: for nonzero z , its multiplicative order divides 15, and zero also satisfies the identity. Thus F^4 is the identity and $F^3(z) = z^8$ is the inverse of F . Both compositions give $z^{16} = z$.

Solution 198.4 ■ Answer D[Back to problem](#)

In this commutative ring of characteristic three, $(a + b\varepsilon)^3 = a^3 + b^3\varepsilon^3 = a$. Hence the fiber over 1 consists of $1, 1 + \varepsilon, 1 + 2\varepsilon$. Frobenius is not injective here: the nilpotent direction disappears.

Solution 198.5 ■ **Answer C**[Back to problem](#)

In $\mathbb{Z}_4$, $(1 + 1)^2 = 0$ while $1^2 + 1^2 = 2$, so additivity fails. In the matrix ring take $A = E_{12}$ and $B = E_{21}$: their squares are zero, but $(A + B)^2 = I$. $\mathbb{H}$ is commutative of characteristic two, so Frobenius applies. Both the characteristic and commutativity hypotheses matter.

Solution 198.6 ■ **Answer B**[Back to problem](#)

A field has no nonzero nilpotents, so $(u - v)^p = 0$ implies $u = v$. If $(a(t)/b(t))^p = t$ for nonzero polynomials a, b , then $p \deg a = 1 + p \deg b$, an impossibility modulo p . Thus t has no preimage. Even for a field, infinitude prevents the finite-set argument for surjectivity.

Part B: Abstract Algebra ▸ Ring Homomorphisms

199 Ideals and absorption

Solution 199.1 ■ **Answer D**[Back to problem](#)

Every integer combination is divisible by 42. Conversely $126 - 84 = 42$, so all multiples of 42 lie in the generated ideal. Hence it is $42\mathbb{Z}$.

Solution 199.2 ■ **Answer D**[Back to problem](#)

Ideals of $\mathbb{Z}_n$ correspond to divisors $d \mid n$ and are generated by the residue of d . Since $30 = 2 \cdot 3 \cdot 5$, it has $(1 + 1)^3 = 8$ positive divisors, including the ideals 0 and the whole ring.

Solution 199.3 ■ **Answer A**[Back to problem](#)

Left multiplication sends a zero column to a zero column, so $AL \subseteq L$ for every A . But $E_{11} \in L$ and $E_{11}E_{12} = E_{12} \notin L$. Thus right absorption fails. Closure under subtraction is immediate.

Solution 199.4 ■ **Answer C**[Back to problem](#)

A two-sided ideal containing I must contain every matrix $A = AI$. Diagonal matrices form a proper subring containing I , so they cannot form an ideal. They are closed under addition and multiplication; commutativity alone is not an obstruction to being an ideal.

Solution 199.5 ■ **Answer D**[Back to problem](#)

For arbitrary k, ℓ , the product $E_{ki}AE_{j\ell} = a_{ij}E_{k\ell}$ belongs to I . Multiplication by $a_{ij}^{-1}I$ shows that every matrix unit $E_{k\ell}$ belongs to I . Absorption by scalar matrices and addition now produce every matrix. This uses both sides of the ideal condition.

Part B: Abstract Algebra ▸ Integral Domains

200 Zero divisors, cancellation, and integral domains

Solution 200.1 ■ **Answer B**[Back to problem](#)

There are $\varphi(45) = 45(1 - 1/3)(1 - 1/5) = 24$ units. Of the 44 nonzero classes, the remaining $44 - 24 = 20$ are zero divisors.

Solution 200.2 ■ **Answer D**[Back to problem](#)

The ring $\mathbb{Z}[\sqrt{3}]$ is a subring of the field $\mathbb{R}$ containing 1, so nonzero products cannot vanish. The other rings have, respectively, the zero-divisor pairs $3, 7$; suitable matrix units; $(1, 0), (0, 1)$; and nonzero functions with disjoint supports.

Solution 200.3 ■ **Answer C**[Back to problem](#)

Commutativity gives $(a - b)(a + b) = 0$. Since there are no nonzero zero divisors, one factor is zero. One cannot simply cancel a factor of a against a factor of b . Both signs are possible, for example in $\mathbb{Z}$, and being a unit is unnecessary.

Solution 200.4 ■ **Answer B**[Back to problem](#)

We have $LR = I$. The nonzero projection $P = I - RL$ sends e_0 to itself and all other basis vectors to zero; $LP = 0$. Thus L is a left zero divisor. If $BL = 0$, then $B = BLR = 0$, so it is not a right zero divisor. A one-sided inverse in an infinite-dimensional setting need not be two-sided.

Solution 200.5 ■ **Answer C**[Back to problem](#)

It is not a unit because it vanishes at zero. If $fg = 0$ with g continuous, then $g(x) = 0$ for every $x > 0$, and continuity also gives $g(0) = 0$. Thus multiplication by f is injective even though f has no inverse in this ring. In infinite rings, a nonunit need not be a zero divisor.

Part B: Abstract Algebra ▸ Fields

201 Units, division rings, fields, and finite domains

Solution 201.1 ■ **Answer A**[Back to problem](#)

The product $(2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$, so the conjugate itself is the inverse.

Solution 201.2 ■ **Answer D**[Back to problem](#)

A finite integral domain is a field: for $a \neq 0$, the map $x \mapsto ax$ is injective, hence hits 1. Thus exactly the $49 - 1 = 48$ nonzero elements are units.

Solution 201.3 ■ Answer A[Back to problem](#)

The inverse of $1 + i$ is $(1 - i)/2$. Multiplying the first equation on the left gives $q = (1 - i)j/2 = (j - k)/2$. Multiplying the second on the right gives $r = j(1 - i)/2 = (j + k)/2$, since $ij = k$ but $ji = -k$. The side on which one divides matters.

Solution 201.4 ■ Answer C[Back to problem](#)

This set is a commutative unital subring. A matrix in it has determinant $a^2 - tb^2$. For $t < 0$, this vanishes only when $a = b = 0$, and the inverse of a nonzero member has the same form. For $t = 0$ take $a = 0, b = 1$; for $t > 0$ take $a = \sqrt{t}, b = 1$. These nonzero singular matrices prevent a field.

Solution 201.5 ■ Answer B[Back to problem](#)

If $a + b\sqrt{2} = c + d\sqrt{3}$ with rational coefficients and $b, d \neq 0$, squaring $b\sqrt{2} - d\sqrt{3} = c - a$ gives $2b^2 + 3d^2 - 2bd\sqrt{6} = (c - a)^2$, forcing $\sqrt{6}$ rational, a contradiction. If either b or d is zero, irrationality forces the other to be zero as well. Hence the common elements are exactly the rationals.

Solution 201.6 ■ Answer C[Back to problem](#)

A tuple is invertible exactly when every coordinate is invertible, with coordinatewise inverse. The three factors have 2, $\varphi(12) = 4$, and $4 - 1 = 3$ units. Thus the count is $2 \cdot 4 \cdot 3 = 24$. The zero coordinate in a field is not allowed.

Solution 201.7 ■ Answer B[Back to problem](#)

The determinant is $|z|^2 + |w|^2 > 0$. The ordinary 2×2 inverse has numerator $\begin{pmatrix} \bar{z} & -w \\ \bar{w} & z \end{pmatrix} = H(\bar{z}, -w)$. Thus every nonzero member has its inverse within the displayed set. In this representation both conjugation and the sign of the off-diagonal parameter matter.

Part B: Abstract Algebra ▷ Fields

202 Idempotents and Boolean rings

Solution 202.1 ■ Answer D[Back to problem](#)

Factor $180 = 4 \cdot 9 \cdot 5$ into pairwise coprime prime powers. In each component, $e(e - 1) = 0$ forces $e \equiv 0$ or 1 , since consecutive integers are coprime. The Chinese remainder theorem combines the $2^3 = 8$ independent choices.

Solution 202.2 ■ Answer B[Back to problem](#)

If a unit u satisfies $u^2 = u$, multiply by u^{-1} to obtain $u = 1$. Conversely 1 is a unit. Boolean rings can have zero divisors and have characteristic two, so the other assertions need not hold.

Solution 202.3 ■ Answer A[Back to problem](#)

Expanding in the given order gives $(e + f)^2 = e^2 + ef + fe + f^2 = e + f$. Orthogonality removes both mixed products, with no commutativity assumption on the entire ring. The sum need not equal 1; two coordinate projections in $M_3(\mathbb{R})$ give a counterexample.

Solution 202.4 ■ Answer D[Back to problem](#)

Applying the hypothesis to $2x$ gives $2x = (2x)^2 = 4x^2 = 4x$, hence $2x = 0$. Expanding $(x + y)^2 = x + y$ then gives $xy + yx = 0$. Since every element is its own additive inverse, $xy = yx$. A nonzero Boolean ring therefore has characteristic two, but it need not be a field.

Solution 202.5 ■ Answer D[Back to problem](#)

Direct multiplication gives $A^2 = A$ and $B^2 = B$. However $AB = \text{diag}(2, 0)$, whose square is $\text{diag}(4, 0)$. The claim that products of idempotents are idempotent requires commutation; this pair does not commute.

7. Additional Topics

Logic ▸ Sentential Calculus

203 Propositions and logical connectives

Solution 203.1 ■ Answer C

[Back to problem](#)

The implications forbid a true statement followed by a false one in the chain P, Q, R . The allowable triples are $(F, F, F), (F, F, T), (F, T, T), (T, T, T)$, giving four.

Solution 203.2 ■ Answer B

[Back to problem](#)

An implication fails exactly when its antecedent is true and its consequent false. Thus its negation is $(P \vee Q) \wedge \neg R$.

Solution 203.3 ■ Answer C

[Back to problem](#)

The necessary condition says $x \in A \Rightarrow x \in B$, giving $A \subseteq B$. The sufficient condition says $x \in C \Rightarrow x \in A$, giving $C \subseteq A$. Neither implication can be reversed without further information.

Solution 203.4 ■ Answer C

[Back to problem](#)

The first two clauses say that exactly one of P, Q is true, so their truth values differ and the biconditional is false. Either individual variable could be true. The final clause forces R to be true. This is exclusive disjunction built from inclusive disjunction, conjunction, and negation.

Solution 203.5 ■ Answer C

[Back to problem](#)

No integer has square two. Thus the implication in I is true for every integer, by its false antecedent. It is also true for at least one integer, so II holds. III demands an actual integer satisfying both equations and is false. An existentially quantified implication does not assert its antecedent.

Logic ▸ Logical Equivalences and Truth Tables

204 Truth tables, tautologies, and contraposition

Solution 204.1 ■ Answer C

[Back to problem](#)

The last implication has contrapositive $R \Rightarrow P$. Together these give the cycle $P \Rightarrow Q \Rightarrow R \Rightarrow P$. Thus if any is true all are true, and otherwise all are false.

Solution 204.2 ■ Answer B

[Back to problem](#)

Reverse each quantifier and negate the final conjunction. The result is that some real x has no larger y with positive $f(y)$, equivalently $\exists x \forall y [y \leq x \vee f(y) \leq 0]$.

Solution 204.3 ■ Answer C[Back to problem](#)

The converse $Q \Rightarrow P$ is false exactly at $Q = T, P = F$. At that assignment the original implication is vacuously true and the inverse $\neg P \Rightarrow \neg Q$ is false. Converse and inverse are equivalent to each other, but not to the original implication.

Solution 204.4 ■ Answer E[Back to problem](#)

The implication equals $(\neg P \wedge \neg Q) \vee R$. Distributing gives $(\neg P \vee R) \wedge (\neg Q \vee R)$, which is the conjunction of the two implications.

Solution 204.5 ■ Answer D[Back to problem](#)

If P and $\neg Q$ are true, the implication in F is false; otherwise another conjunct is false. Thus F is always false. One of $P, \neg P$ is always true, so G is always true. The formula F is a contradiction; this is distinct from replacing an implication by its contrapositive.

Solution 204.6 ■ Answer A[Back to problem](#)

Double negation removes the two negations. Associativity and commutativity collect the repeated Q ; $Q \vee Q \equiv Q$. Distributing $P \wedge (Q \vee R)$ then gives the stated expression. For example, choice D incorrectly allows R to make the formula true when P is false.

Set Theory ▸ Subsets and Complements

205 Sets, subsets, and complements

Solution 205.1 ■ Answer C[Back to problem](#)

The member $\emptyset$ is a subset of every set. The other member $\{\emptyset\}$ is also a subset of A because its sole element $\emptyset$ belongs to A . Thus both members qualify.

Solution 205.2 ■ Answer B[Back to problem](#)

An element of $C \setminus B$ lies in C and not in B . Since $A \subseteq B$, it cannot lie in A , so it lies in $C \setminus A$. Taking complements reverses inclusion.

Solution 205.3 ■ Answer E[Back to problem](#)

The two elements of B are compulsory. Each of the remaining three elements can be included or excluded independently, giving $2^3 = 8$ subsets.

Solution 205.4 ■ Answer C[Back to problem](#)

Transitivity gives $A \subseteq C$, so intersecting A with C leaves exactly A . The inclusions need not be equalities.

Solution 205.5 ■ Answer B[Back to problem](#)

$S = \{-2, -1, 0, 1, 2\}$, whereas $T = \{0, 1, 4\}$: repeated outputs are listed only once in a set. Therefore $S \setminus T = \{-2, -1, 2\}$ and $T \setminus S = \{4\}$. The two uses of set-builder notation describe a condition on inputs and an image set, respectively.

Set Theory ▸ Union and Intersection; Venn Diagrams

206 Unions, intersections, and symmetric difference

Solution 206.1 ■ Answer B[Back to problem](#)

An element must lie in A and fail to lie in $B \cap C^c$. De Morgan gives $A \cap (B^c \cup C)$, which distributes to $(A \setminus B) \cup (A \cap C)$.

Solution 206.2 ■ Answer B[Back to problem](#)

Take symmetric difference with A on both sides. Associativity and $A \triangle A = \emptyset$ give $B = C$. Equivalently, adding the same membership indicator modulo two is cancellative.

Solution 206.3 ■ Answer C[Back to problem](#)

Exactly one-set regions contribute $10 - 4 - 5 + 2 = 3$, $12 - 4 - 6 + 2 = 4$, and $14 - 5 - 6 + 2 = 5$ elements. A threefold symmetric difference also includes the two elements in all three sets. The total is $3 + 4 + 5 + 2 = 14$. Equivalently, use $\sum |A| - 2 \sum |A \cap B| + 4|A \cap B \cap C|$.

Solution 206.4 ■ Answer C[Back to problem](#)

For each fixed n , some $m \geq n$ makes $A_{n,m} = U$, so each inner union is U . For each fixed m , choosing $n > m$ makes $A_{n,m}$ empty, so each inner intersection is empty. The order of an indexed union and intersection cannot in general be exchanged; it reflects $\forall n \exists m$ versus $\exists m \forall n$.

Solution 206.5 ■ Answer A[Back to problem](#)

The left shaded component lies only in A , outside both B and C . The lower-right shaded component lies in both B and C but outside A . Their union is the expression in A. The triple intersection is unshaded, unlike a threefold symmetric difference.

Set Theory ▸ Cartesian Products; Intervals of the Real Line

207 Cartesian products and real intervals

Solution 207.1 ■ **Answer B**[Back to problem](#)

Both 0 and 1 lie in every interval, as does every point between them. If $x < 0$ or $x > 1$, choose n sufficiently large that the corresponding endpoint passes x , excluding it. Thus the intersection is $[0, 1]$.

Solution 207.2 ■ **Answer B**[Back to problem](#)

Intersection of products is $(A \cap B) \times (B \cap C)$. Here $A \cap B = [1, 2]$ and $B \cap C = [2, 3]$, yielding the stated rectangle.

Solution 207.3 ■ **Answer D**[Back to problem](#)

Neither endpoint zero nor one is in any interval. Every point strictly between them lies in one for all sufficiently large n , so the union is $(0, 1)$.

Solution 207.4 ■ **Answer C**[Back to problem](#)

Every finite real x is smaller than some positive integer n , so no real point survives all the intervals. In the extended line, $+\infty$ is explicitly included by the bracket in every interval for J . It is not an element of the ordinary real intervals defining I .

Solution 207.5 ■ **Answer D**[Back to problem](#)

If either set is empty, both products are empty. If both are nonempty, fix an element of each. For any $a \in A$, pairing it with a fixed $b \in B$ and using equality shows $a \in B$; reversing the argument gives $B \subseteq A$. Hence $A = B$. Equal cardinality alone does not identify coordinate sets.

Set Theory ▸ Cardinality

208 Cardinality, countability, and algebraic numbers

Solution 208.1 ■ **Answer D**[Back to problem](#)

Finite subsets and finite coefficient lists form countable unions of finite products of countable sets. Algebraic numbers and rational pairs are countable. Binary sequences are uncountable by the diagonal argument, since any purported listing misses the sequence obtained by flipping its diagonal digits.

Solution 208.2 ■ **Answer C**[Back to problem](#)

There are countably many such finite coefficient strings, and each polynomial has finitely many roots, so the set is countable. It is infinite: for each integer $m \geq 1$, the polynomial $1 + x + x^{2m+1}$ has a root in $(-1, 0)$ by the intermediate value theorem. Roots associated to different m are distinct, since a shared root would give equal distinct powers of a number with absolute value strictly between zero and one.

Solution 208.3 ■ **Answer B**[Back to problem](#)

Fix an enumeration of $\mathbb{Q}$. From each interval select the first rational in that enumeration that lies inside it. Density guarantees such a rational, and disjointness makes the selected rationals distinct. This injects the family into $\mathbb{Q}$. A single interval is already uncountable, so the union need not be countable.

Solution 208.4 ■ **Answer D**[Back to problem](#)

Every positive integer has a unique factorization $2^m u$ with u odd and positive. Writing $u = 2n + 1$ gives a unique pair $(m, n) \in \mathbb{N}_0^2$. In $\mathbb{C}$, integers with other prime factors, such as five, are missed; the other formulas have repeated values.

Solution 208.5 ■ **Answer E**[Back to problem](#)

A subset of an enumerated set can be listed by retaining its members in that enumeration. It may be finite, but cannot be uncountable; its power set may nevertheless be uncountable.

Set Theory ▸ Cardinality

209 Power sets, uncountability, and the continuum

Solution 209.1 ■ **Answer D**[Back to problem](#)

The inner power set has $2^3 = 8$ members, its product with B has $8 \cdot 4 = 32$ members, and the outer power set therefore has 2^{32} members.

Solution 209.2 ■ **Answer E**[Back to problem](#)

Cantor gives $|\mathcal{P}(\mathbb{R})| > |\mathbb{R}|$. The plane, irrationals, power set of naturals, and finite real sequences all have cardinality $\mathfrak{c}$, so they are not larger.

Solution 209.3 ■ **Answer B**[Back to problem](#)

The codomain has $2^3 = 8$ elements. Each of three domain elements has eight independent image choices, giving $8^3 = 512$ functions.

Solution 209.4 ■ **Answer C**[Back to problem](#)

The denominator is nonzero since $t \neq -3$. If $y = (2t + 1)/(t + 3)$ were algebraic, then $y \neq 2$ and $t = (1 - 3y)/(y - 2)$ would be algebraic, since algebraic numbers form a field. This contradicts the hypothesis. The other choices are the distinct algebraic numbers 0, 1, $\sqrt{2}$, and $\sqrt{3} + \sqrt{5}$.

Solution 209.5 ■ **Answer B**[Back to problem](#)

The derivative of $x/(1 - x^2)$ is $(1 + x^2)/(1 - x^2)^2 > 0$, and its limits at the endpoints are $-\infty$ and $+\infty$. Continuity gives surjectivity and strict increase gives injectivity. A and D are bounded, C is not injective, and E is undefined at $\pm 1/2$.

Solution 209.6 ■ **Answer D**[Back to problem](#)

If $D = f(a)$ for some a , then $a \in D$ would be equivalent to $a \notin D$, by the defining rule. This is impossible. The set D is a subset of A , so it belongs to the codomain but is missed by f . The proof works for every set, including the empty set.

Graph Theory ▷ Vertices, Edges, and Degree

210 Simple graphs and vertex degrees

Solution 210.1 ■ **Answer C**[Back to problem](#)

The degree sum is $35 + 5d$, which must be even. Thus d must be odd, excluding 4. The other sequences are graphical; for example the Havel–Hakimi criterion confirms them. The parity obstruction alone uniquely identifies the listed impossible value.

Solution 210.2 ■ **Answer C**[Back to problem](#)

The degree sum is $6(5) + 2(4) = 38$, so the graph has 19 edges. Since K_8 has 28 edges, exactly $28 - 19 = 9$ are missing.

Solution 210.3 ■ **Answer C**[Back to problem](#)

With n vertices, possible degrees are $0, 1, \dots, n - 1$. But degrees zero and $n - 1$ cannot both occur: a universal vertex would be adjacent to an allegedly isolated vertex. Thus n vertices occupy at most $n - 1$ possible degrees, so the pigeonhole principle forces repetition.

Solution 210.4 ■ **Answer D**[Back to problem](#)

The two vertices of degree five would each be adjacent to all other vertices, forcing every other degree to be at least two. That contradicts the degrees one. The other sequences are realized by $K_{3,3}$, a six-cycle, K_6 minus a perfect matching, and a six-vertex star. An even degree sum is necessary but not sufficient.

Solution 210.5 ■ **Answer B**[Back to problem](#)

Each loop contributes two to the degree, each ordinary incident edge contributes one, and parallel edges are counted separately. Thus the degree is $2 \cdot 2 + 3 + 1 = 8$. The distinct other neighbors are only u, w , so there are two. Degree equals the number of neighbors only under the simple-graph assumptions.

Graph Theory ▷ Graph Isomorphism; Review Matrices

211 Graph isomorphisms and incidence matrices

Solution 211.1 ■ **Answer B**[Back to problem](#)

The diagonal of BB^T contains the vertex degrees. Their sum is twice the number of edges, so $2m = 26$ and $m = 13$.

Solution 211.2 ■ **Answer D**[Back to problem](#)

Both graphs have six vertices, six edges, all degrees two, and no odd-degree vertices. The six-cycle is connected, whereas the two triangles form two connected components.

Solution 211.3 ■ Answer B[Back to problem](#)

The diagonal entries of A^3 count closed walks of length three. Every triangle contributes three starting vertices and two directions, hence six walks. Divide 42 by six to obtain seven triangles.

Solution 211.4 ■ Answer D[Back to problem](#)

Here $BB^T = 3I + A$, where A is the adjacency matrix. Since $\text{tr } A = 0$ and $\text{tr } A^2 = 8 \cdot 3$, the trace is $9 \cdot 8 + 24 = 96$.

Solution 211.5 ■ Answer B[Back to problem](#)

Multiplication by two is a bijection modulo five. It sends differences ± 1 exactly to ± 2 , preserving adjacency in both directions. The identity and cube maps send the edge $\{0, 1\}$ to a nonedge of H ; the square and constant maps are not bijections.

Graph Theory ▸ Paths, Trees, and Subgraphs

212 Walks, circuits, trees, and spanning subgraphs

Solution 212.1 ■ Answer C[Back to problem](#)

A spanning tree keeps $17 - 1 = 16$ edges, so $29 - 16 = 13$ can be deleted. Fewer than sixteen remaining edges cannot connect seventeen vertices.

Solution 212.2 ■ Answer C[Back to problem](#)

If there are L leaves, the degree sum is $L + 4(3) + 2(4) = L + 20$. A tree on $L + 6$ vertices has $L + 5$ edges, so $L + 20 = 2(L + 5)$, giving $L = 10$.

Solution 212.3 ■ Answer B[Back to problem](#)

Write the original vertex and edge counts as n, m , so $m = n - c$. After deletion they are $n - 1, m - d$, and the graph is still a forest. Its component count is $(n - 1) - (m - d) = c + d - 1$. This also handles an isolated vertex ($d = 0$), whose deletion removes one component.

Solution 212.4 ■ Answer C[Back to problem](#)

No edge repeats, so the sequence is a trail. Its endpoints agree, making it a closed trail, or circuit. The vertex a also occurs in the middle, so it is not a cycle. Closed walks, circuits, and cycles impose different repetition restrictions.

Solution 212.5 ■ **Answer D**[Back to problem](#)

Without diagonal 13, a spanning tree is obtained by deleting one of the four cycle edges, giving four. With the diagonal included, vertex 2 must attach by exactly one of 12, 23, and vertex 4 by exactly one of 34, 41, giving $2 \cdot 2 = 4$. These cases exhaust the trees, for a total of eight.

Algorithms ▸ Tracing and Termination

213 Algorithms, loops, and invariants

Solution 213.1 ■ **Answer D**[Back to problem](#)

The states after successive iterations are $(a, b) = (3, 7), (1, 17), (-1, 35)$. The loop then stops because $a > 0$ is false, leaving $b = 35$.

Solution 213.2 ■ **Answer C**[Back to problem](#)

The successive states are $(85, 51), (51, 34), (34, 17)$, and $(17, 0)$. Thus there are four replacements, and the final gcd is 17.

Solution 213.3 ■ **Answer B**[Back to problem](#)

Subtracting one integer from the other preserves their gcd. Both integers remain positive, and their sum decreases at every nonterminal step by the smaller positive value. Thus termination follows, separately from the invariant. When $a = b$, their common value is the original gcd.

Solution 213.4 ■ **Answer C**[Back to problem](#)

The printed values are 3, 10, 5, 16, 8, 4, 2, 1, so there are eight. The input six is not printed, and one is printed before the next condition test ends the loop. This finite computation concerns this input only and supplies no general termination proof.

Solution 213.5 ■ **Answer C**[Back to problem](#)

Input zero skips the loop, and input one reaches zero in one update. The update $T(x) = 1 - x$ satisfies $T(T(x)) = x$. For every other integer, the two values $x, 1 - x$ are both nonzero, so the procedure cycles forever. A deterministic update need not terminate.

Combinatorics ▸ The Multiplication Principle

214 Sum and product counting principles

Solution 214.1 ■ **Answer D**
[Back to problem](#)

If the last digit is 0, there are $5 \cdot 4 \cdot 3 = 60$ choices. If it is 2 or 4, there are 4 allowed nonzero leading digits, then 4 and 3 choices, giving $2 \cdot 4 \cdot 4 \cdot 3 = 96$. The total is 156.

Solution 214.2 ■ **Answer C**
[Back to problem](#)

Use inclusion-exclusion: all 4^5 strings minus 3^5 missing A minus 3^5 missing B , plus 2^5 missing both. The result is $1024 - 486 + 32 = 570$.

Solution 214.3 ■ **Answer B**
[Back to problem](#)

The first three specified arguments have 5, 4, 3 ordered image choices. Each of the other five arguments still has all five choices, including images already used. Multiplying yields the answer. The distinguished arguments are fixed, and only their three images must be distinct.

Solution 214.4 ■ **Answer D**
[Back to problem](#)

If the rooks were ordered, there would be 20 choices for the first square and $3 \cdot 4 = 12$ for the second. Each unordered placement is counted twice, so the answer is $20 \cdot 12/2 = 120$. Equivalently choose two rows, two columns, and one of the two matchings: $\binom{4}{2} \binom{5}{2} \cdot 2$.

Solution 214.5 ■ **Answer B**
[Back to problem](#)

For each element of A , the union condition allows three states: in S only, in T only, or in both. This gives 3^6 pairs. If the intersection is empty, only the first two states remain, giving 2^6 pairs to subtract. The ordered roles of S, T are retained.

Combinatorics ▸ Permutations and Combinations

215 Permutations, repeated symbols, and circular orders

Solution 215.1 ■ **Answer C**
[Back to problem](#)

Arrange the non- A letters B,N,N in $3!/2! = 3$ ways. They create four gaps, and choose three of those gaps for the identical A letters: $\binom{4}{3} = 4$. The total is 12.

Solution 215.2 ■ **Answer C**
[Back to problem](#)

There are $5! = 120$ total circular arrangements. If the specified pair is adjacent, merge it into a block: five circular objects give $4!$ orders, with two internal pair orders, totaling 48. Subtract to obtain 72.

Solution 215.3 ■ Answer B[Back to problem](#)

For any fixed positions occupied by 1, 2, 3, their $3!$ possible internal relative orders occur equally often. Independently, the two orders of 4, 5 occur equally often. Exactly one of the $3! \cdot 2!$ patterns meets both conditions, so the count is $7!/(3!2!) = 420$.

Solution 215.4 ■ Answer D[Back to problem](#)

The available multiset has six symbols, so exactly one is unused. Omitting an A leaves counts (2, 2, 1) and gives $5!/(2!2!) = 30$ strings; omitting a B gives $5!/(3!) = 20$; omitting C gives $5!/(3!2!) = 10$. The cases have different multiplicities and are disjoint, giving 60.

Solution 215.5 ■ Answer B[Back to problem](#)

Fix one specified person to remove rotational freedom; the opposite seat is then fixed for the other. The six remaining people can be assigned in $6!$ ways. Reflection pairs these arrangements with no fixed arrangement, because all people are distinct and some seats are exchanged. Dividing by two gives 360.

Combinatorics ▷ Permutations and Combinations

216 Combinations, binomial coefficients, and partitions into teams

Solution 216.1 ■ Answer B[Back to problem](#)

Order all eight people, group consecutive pairs, and divide by $2!$ for each pair and $4!$ for permutation of the four pairs. The result is $8!/[(2!)^4 4!] = 105$.

Solution 216.2 ■ Answer E[Back to problem](#)

Rewrite as $(1 - x^2)^4(1 + x)^6$. Only even exponents 2, 4, 6 from the first factor contribute, giving $-4\binom{6}{5} + 6\binom{6}{3} - 4\binom{6}{1} = -24 + 120 - 24 = 72$.

Solution 216.3 ■ Answer C[Back to problem](#)

Choose which of A, B contains the specified pair (two choices), then its third member (six choices). From the five remaining people choose three for the other labeled three-person team ($\binom{5}{3} = 10$ choices); the final two form C . The result is $2 \cdot 6 \cdot 10 = 120$. There is no extra division by $2!$ because A, B have distinct labels.

Solution 216.4 ■ Answer D[Back to problem](#)

Count four-element subsets of $\{1, \dots, 9\}$ by their largest element. If it is $k + 1$, the other three are chosen from the first k elements, giving $\binom{k}{3}$. Summing gives $\binom{9}{4} = 126$. This hockey-stick identity also follows by telescoping Pascal's identity.

Solution 216.5 ■ **Answer A**[Back to problem](#)

Choose which two of the seven factors supply x , which three supply $-2y$, and which two supply z . There are $7!/(2!3!2!) = 210$ assignments, each contributing $(-2)^3 = -8$. Thus the coefficient is -1680 .

Combinatorics ▸ With Repetitions Allowed

217 Repetition and stars and bars

Solution 217.1 ■ **Answer D**[Back to problem](#)

Substitute $y_1 = x_1 - 2$, $y_2 = x_2 - 1$. The four nonnegative variables now sum to 9, so stars and bars gives $\binom{12}{3} = 220$.

Solution 217.2 ■ **Answer C**[Back to problem](#)

Without upper bounds there are $\binom{12}{2} = 66$ triples. For $x \geq 6$, shifting gives $\binom{6}{2} = 15$ violations, likewise for y, z . Two coordinates cannot both be at least six. Hence the count is $66 - 3(15) = 21$.

Solution 217.3 ■ **Answer C**[Back to problem](#)

The ordered pair in positions one and six has $4 \cdot 3$ choices. Positions two and five share one freely chosen symbol, and positions three and four are independently free, contributing 4^3 . The total is $4 \cdot 3 \cdot 4^3 = 768$; equality of two positions removes one choice, not both positions.

Solution 217.4 ■ **Answer B**[Back to problem](#)

Write eight consecutive ones. Each of the seven gaps either receives a separator or does not. Each separator pattern specifies a unique ordered positive composition of eight, so there are $2^7 = 128$. For example, $3 + 5$ and $5 + 3$ count separately; this is not an integer-partition count.

Solution 217.5 ■ **Answer D**[Back to problem](#)

The first two entries form an unordered selection with repetition from $\{1, 2, 3\}$, giving $\binom{4}{2} = 6$ possibilities. The final two similarly form a multiset from $\{3, 4, 5\}$, also giving six. These choices are independent once the middle entry is fixed, so the count is 36.

Combinatorics ▸ The Pigeonhole Principle

218 Pigeonhole and averaging arguments

Solution 218.1 ■ **Answer D**[Back to problem](#)

Group integers by their odd part: $u, 2u, 4u, \dots$ for the ten odd $u \leq 20$. Eleven choices force two in one group, and the smaller divides the larger. The ten numbers 11 through 20 have no divisibility pair, so eleven is sharp.

Solution 218.2 ■ **Answer C**[Back to problem](#)

There are seventeen residue classes modulo 17, so eighteen integers force a repeated class. Seventeen consecutive integers occupy distinct classes, proving that seventeen is not sufficient.

Solution 218.3 ■ **Answer C**[Back to problem](#)

Partition the square into four half-size squares, assigning shared boundaries consistently. Two of the five points lie in one cell, whose diameter is $\sqrt{2}/2$. This bound is attained by placing points at the four corners and the center: the shortest distance is $\sqrt{2}/2$. Thus a smaller universal bound is impossible.

Solution 218.4 ■ **Answer D**[Back to problem](#)

If color i occurs n_i times, it supplies $\lfloor n_i/4 \rfloor$ disjoint groups. If at most two groups are available in total, write $n_i = 4q_i + r_i$ with $0 \leq r_i \leq 3$; the total is at most $4 \cdot 2 + 4 \cdot 3 = 20$. Hence 21 forces three groups. Counts $(11, 3, 3, 3)$ show that twenty is insufficient.

Solution 218.5 ■ **Answer C**[Back to problem](#)

There are $\binom{7}{3} = 35$ triple sums, and each entry occurs in $\binom{6}{2} = 15$ of them. Their average is $15 \cdot 28/35 = 12$, so at least one is at most twelve. Taking every entry equal to four makes every triple sum twelve, attaining the bound.

Probability and Statistics ▸ Elementary Probability

219 Equally likely outcomes and geometric probability

Solution 219.1 ■ Answer C

[Back to problem](#)

An even sum uses either zero or two odd numbers. There are $\binom{5}{3} + \binom{5}{2}\binom{5}{1} = 10 + 50 = 60$ favorable subsets out of $\binom{10}{3} = 120$, giving $1/2$.

Solution 219.2 ■ Answer C

[Back to problem](#)

For $0 < x \leq 1/4$, every $y \in [0, 1]$ works. For $1/4 < x \leq 1$, the allowable height is $1/(4x)$. The area is $1/4 + \int_{1/4}^1 (4x)^{-1} dx = (1 + \log 4)/4 = (1 + 2 \log 2)/4$.

Solution 219.3 ■ Answer B

[Back to problem](#)

The favorable region is the upper-right triangle in the unit square with legs $1/2$. Its area, and hence probability, is $\frac{1}{2}(1/2)^2 = 1/8$.

Solution 219.4 ■ Answer D

[Back to problem](#)

The excluded region consists of two right triangles, each with legs $2/3$. Subtract their total area from one: $1 - (2/3)^2 = 5/9$.

Solution 219.5 ■ Answer D

[Back to problem](#)

Partition according to the first choice. The three paths to $X = 1$ have probabilities $1/3$, $(1/3)(1/2)$, and $(1/3)(1/3)$. Their sum is $1/3 + 1/6 + 1/9 = 11/18$. Although both steps are individually uniform on their allowed choices, the final values $1, 2, 3$ are not equally likely.

Probability and Statistics ▸ Probability Spaces

220 Sample spaces, events, and probability measures

Solution 220.1 ■ Answer C

[Back to problem](#)

Continuity from above for a probability measure gives the probability of the decreasing intersection as $\lim_n P(A_n) = 1/3$.

Solution 220.2 ■ Answer D

[Back to problem](#)

Each event is determined uniquely by deciding which of the four disjoint nonempty blocks to include. There are $2^4 = 16$ choices, including the empty union and the whole sample space.

Solution 220.3 ■ Answer B[Back to problem](#)

Complements and finite unions preserve finite-or-cofinite status. But the union of all even singletons is neither finite nor cofinite, so the family is not a sigma-algebra. Finite additivity holds: disjoint cofinite sets cannot both occur, and a finite set disjoint from a cofinite one lies in its finite complement. Countable additivity fails even for the union $\mathbb{N} = \bigcup_n \{n\}$, since all singleton values are zero but $P(\mathbb{N}) = 1$.

Solution 220.4 ■ Answer C[Back to problem](#)

Each singleton has uniform probability zero. The rational points form a countable union of such singletons, so countable additivity gives $P(A) = 0$. Its complement has probability one. Both sets are nonempty, showing that probability zero does not mean impossible as a set and probability one does not mean the whole sample space.

Solution 220.5 ■ Answer B[Back to problem](#)

Since $1/[n(n+1)] = 1/n - 1/(n+1)$, the total mass is $c \sum_{n \geq 1} (1/n - 1/(n+1)) = c$. Normalization forces $c = 1$. The tail from $n = 4$ telescopes to $1/4$. Both normalization and event probabilities here use countable sums of disjoint singleton masses.

Probability and Statistics ▸ Probability Spaces

221 Independence, exclusion, and inclusion-exclusion

Solution 221.1 ■ Answer C[Back to problem](#)

The union has probability $2/5 + 3/4 - (2/5)(3/4) = 17/20$. Since $A \cap (A \cup B) = A$, the conditional probability is $(2/5)/(17/20) = 8/17$.

Solution 221.2 ■ Answer C[Back to problem](#)

Pairwise independence gives each pairwise intersection probability $1/4$. Inclusion-exclusion gives $3(1/2) - 3(1/4) + 0 = 3/4$. The triple intersection cannot be replaced by $1/8$ because mutual independence was not assumed.

Solution 221.3 ■ Answer C[Back to problem](#)

The two disjoint possibilities are $A \cap B^c$ and $A^c \cap B$. Their probabilities sum to $p(1-q) + (1-p)q = p + q - 2pq = \frac{8}{15}$.

Solution 221.4 ■ Answer E[Back to problem](#)

The probabilities of two reds given A, B are $\binom{4}{2}/\binom{5}{2} = 3/5$ and $\binom{2}{2}/\binom{5}{2} = 1/10$. Bayes gives $[(1/3)(3/5)]/[(1/3)(3/5) + (2/3)(1/10)] = (1/5)/(4/15) = 3/4$. Both the prior bag probabilities and the changing composition after the first draw matter.

Solution 221.5 ■ Answer D[Back to problem](#)

Independence gives $P(A)P(B) = P(A \cap B) = 0$. Since probabilities are nonnegative, at least one is zero. That event need not be empty: two distinct singleton events in a uniform continuous sample space are disjoint, nonempty, and independent because both have probability zero.

Solution 221.6 ■ Answer B[Back to problem](#)

The intersection satisfies $\max(0, P(A) + P(B) - 1) \leq P(A \cap B) \leq \min(P(A), P(B))$, giving $7/20 \leq P(A \cap B) \leq 3/5$. Dividing by $3/4$ gives $[7/15, 4/5]$. Every intermediate intersection is attainable by assigning nonnegative probabilities to the four membership regions, so this is the full range.

Probability and Statistics ▸ Bernoulli Trials

222 Bernoulli trials and the binomial distribution

Solution 222.1 ■ Answer C[Back to problem](#)

The ratio is $P(X = 4)/P(X = 3) = (5/4)p/(1-p)$. Setting it equal to one gives $5p = 4(1-p)$, hence $p = 4/9$.

Solution 222.2 ■ Answer B[Back to problem](#)

Conditional on three heads, all $\binom{8}{3} = 56$ head-position sets are equally likely. Arrange the five tails, producing six gaps; place one head in each of three selected gaps. Thus $\binom{6}{3} = 20$ outcomes qualify and the probability is $20/56 = 5/14$.

Solution 222.3 ■ Answer C[Back to problem](#)

Adding the expansions of $(q + p)^n$ and $(q - p)^n$ cancels the odd powers of p . Thus $P(X \text{ even}) = [1 + (1 - 2p)^n]/2 = \frac{41}{81}$.

Solution 222.4 ■ Answer B[Back to problem](#)

Matching means gives $p + q = 2r$. Matching the probability of two successes gives $pq = r^2$. Hence $(p - q)^2 = (p + q)^2 - 4pq = 0$, so $p = q = r$. Conversely, equal success probabilities and independence do produce a binomial sum. Independence alone is insufficient.

Solution 222.5 ■ Answer B[Back to problem](#)

The ratio of variance to mean is $1 - p = 3/4$, so $p = 1/4$ and $n = 4/p = 16$. The mass at one is $\binom{16}{1}(1/4)(3/4)^{15} = 4(3/4)^{15}$. The variance here determines p , not the number of trials by itself.

Probability and Statistics ▸ Random Variables

223 Random variables and cumulative distribution functions

Solution 223.1 ■ **Answer C**
[Back to problem](#)

The condition $X \geq 1$ has probability $3/4$. Its intersection with $0 < X < 2$ is $X = 1$, whose probability is the CDF jump $3/4 - 1/4 = 1/2$. Thus the conditional probability is $(1/2)/(3/4) = 2/3$.

Solution 223.2 ■ **Answer C**
[Back to problem](#)

The mass at the left endpoint must be restored: $P(2 \leq X \leq 5) = F(5) - F(2) + P(X = 2) = 9/10 - 7/10 + 1/5 = 2/5$.

Solution 223.3 ■ **Answer B**
[Back to problem](#)

Right continuity at zero forces $c = \lim_{t \downarrow 0} (1+t)/2 = 1/2$. With this value, F is nondecreasing, right-continuous everywhere, and has the correct limits. It describes a mass of $1/2$ at zero plus probability $1/2$ spread uniformly on $(0, 1)$; differentiating away from zero does not recover the atom.

Solution 223.4 ■ **Answer C**
[Back to problem](#)

$P(Y \leq t) = P(X \geq -t) = 1 - P(X < -t) = 1 - F((-t)^-)$. The left limit is essential: using $1 - F(-t)$ would discard any mass at $X = -t$. This is a decreasing transformation, so the inequality reverses.

Solution 223.5 ■ **Answer B**
[Back to problem](#)

A measurable function must be constant on each block $\{1, 2\}$ and $\{3, 4\}$: otherwise a threshold between two values separates members of a block, yielding a set outside $\mathcal{F}$. The tuple in B has this property, and every inverse image of an open real set is a union of the two blocks.

Solution 223.6 ■ **Answer C**
[Back to problem](#)

A point mass is the jump $F(a) - F(a^-)$. Continuity makes every such jump zero. This is the only forced conclusion: continuous CDFs can be singular, such as the Cantor distribution, so a density need not exist. Infinite means, flat CDF intervals, and disconnected supports are also possible.

Probability and Statistics ▸ Random Variables

224 Densities and transformations of continuous variables

Solution 224.1 ■ Answer A

[Back to problem](#)

For $0 < y < 1$, $P(Y \leq y) = P(X \leq \sqrt{y}) = \int_0^{\sqrt{y}} 2x \, dx = y$. Differentiating gives the uniform density 1.

Solution 224.2 ■ Answer C

[Back to problem](#)

By the exponential survival law, the conditional probability is $e^{-\lambda(5-2)} = P(X > 3) = 1/8$. The memoryless property removes the elapsed two time units.

Solution 224.3 ■ Answer C

[Back to problem](#)

The triangle under f has base one and height c , so its area is $c/2$ and normalization gives $c = 2$. Then $P(|X| \leq 1/4) = 4 \int_0^{1/4} (1 - 2x) \, dx = 4(1/4 - 1/16) = 3/4$. The height $f(0) = 2$ is valid; density values are not point probabilities.

Solution 224.4 ■ Answer C

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Each admissible preimage $x = \pm\sqrt{y}$ contributes $(1/3)/(2\sqrt{y})$. Both preimages lie in $(-2, 1)$ for $0 < y < 1$, but only the negative preimage does for $1 < y < 4$. The density must therefore change at one. Its two integrals are $2/3$ and $1/3$, totaling one.

Solution 224.5 ■ Answer B

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The minimum exceeds t exactly when both variables exceed t . Independence gives $P(T > t) = e^{-2t}e^{-3t} = e^{-5t}$. Thus $F_T(t) = 1 - e^{-5t}$ and differentiating yields $5e^{-5t}$. For a minimum the survival probabilities multiply; adding the densities would integrate to two.

Probability and Statistics ▸ Expectation, Variance, and Standard Deviation

225 Expectation, indicators, and moments

Solution 225.1 ■ Answer C

[Back to problem](#)

Let I_j indicate that j is fixed. Every j has probability $1/20$ of being fixed. Therefore $E[\sum I_j] = 20(1/20) = 1$, regardless of dependence among the indicators.

Solution 225.2 ■ Answer C

[Back to problem](#)

Use the density directly: $E[1/X] = \int_0^1 x^{-1}(3x^2) \, dx = 3 \int_0^1 x \, dx = 3/2$. The singularity at zero is integrable against this density.

Solution 225.3 ■ Answer B[Back to problem](#)

For a fixed box, every ball avoids it with probability $(2)/3$, so its empty probability is that ratio to the 5th power. Sum the 3 indicator expectations to get $\frac{32}{81}$; independence between boxes is unnecessary.

Solution 225.4 ■ Answer E[Back to problem](#)

The positive-tail first moment is $\pi^{-1} \int_0^\infty x/(1+x^2) dx = \infty$, and symmetry gives the same divergence for the negative part. Their difference is not defined. Symmetric truncations have integral zero, but that principal value is not a probabilistic expectation; $E|X| = \infty$.

Solution 225.5 ■ Answer D[Back to problem](#)

Use $X^3 = X(X-1)(X-2) + 3X(X-1) + X$. Counting ordered distinct success indices gives expectations $3 \cdot 2 \cdot 1(1/3)^3 = 2/9$ and $3 \cdot 2(1/3)^2 = 2/3$, while $E[X] = 1$. Hence $E[X^3] = 2/9 + 3(2/3) + 1 = 29/9$. In particular it differs from $(E[X])^3 = 1$.

Probability and Statistics ▸ Expectation, Variance, and Standard Deviation

226 Variance, standard deviation, covariance, and correlation

Solution 226.1 ■ Answer C[Back to problem](#)

Use $a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$. This gives $9(4) + 4(9) + 2(3)(-2)(2) = 36 + 36 - 24 = 48$.

Solution 226.2 ■ Answer B[Back to problem](#)

The first moment is $E[X] = 2/3$ and the second moment is $E[X^2] = 1/2$. Thus the variance is $1/2 - (2/3)^2 = 1/2 - 4/9 = 1/18$.

Solution 226.3 ■ Answer D[Back to problem](#)

The original correlation is $-3/(2 \cdot 3) = -1/2$. Translation does not affect it, and multiplying only one variable by a negative coefficient reverses its sign. Explicitly, $\text{Cov}(U, V) = 18$, $\sigma_U = 4$, and $\sigma_V = 9$, so the new correlation is $18/36 = 1/2$.

Solution 226.4 ■ Answer B[Back to problem](#)

We have $E[X] = E[X^3] = 0$, $E[Y] = 2/3$, $\text{Var}(X) = 2/3$, and $\text{Var}(Y) = 2/3 - 4/9 = 2/9$. Thus covariance and correlation are zero. But $P(Y = 0 \mid X = 0) = 1$, whereas $P(Y = 0) = 1/3$, proving dependence. A deterministic nonlinear relation can have zero linear correlation.

Solution 226.5 ■ Answer A[Back to problem](#)

The variance is $9 + 4t^2 - 6t = 4(t - 3/4)^2 + 27/4$. Its unique minimum is attained at $t = 3/4$ and equals $27/4$. The covariance term determines the optimal adjustment; minimizing by discarding it would give the wrong coefficient.

Solution 226.6 ■ Answer B[Back to problem](#)

Standardize $U = (X - 1)/2$ and $V = (Y + 3)/6$. Then $E[U + V] = 0$ and $\text{Var}(U + V) = 1 + 1 + 2(-1) = 0$, so $U + V = 0$ almost surely. Rearrangement gives $Y = -3X$. A constant has standard deviation zero, making the denominator in its correlation coefficient zero; the coefficient is undefined, not zero.

Probability and Statistics ▸ The Normal Distribution

227 Normal distributions and standardization

Solution 227.1 ■ Answer A[Back to problem](#)

The mean of $X + 2Y$ is $2 + 2(-1) = 0$, and its variance is $9 + 4(4) = 25$. Its standard deviation is 5, so the standardized threshold is $5/5 = 1$.

Solution 227.2 ■ Answer C[Back to problem](#)

Strict increase of Φ gives $(4 - \mu)/\sigma = -1$ and $(10 - \mu)/\sigma = 2$. Subtracting yields $6 = 3\sigma$, so $\sigma = 2$ and $\mu = 6$. Hence $\mu + \sigma = 8$.

Solution 227.3 ■ Answer A[Back to problem](#)

Complete the square: $-2x^2 + 8x = -2(x - 2)^2 + 8$. Comparing with a normal density gives mean 2 and variance $1/4$. Its normalizing coefficient in front of $e^{-2(x-2)^2}$ is $\sqrt{2/\pi}$, so $C = e^{-8}\sqrt{2/\pi}$. Finally $E[X^2] = \text{Var}(X) + (E[X])^2 = 1/4 + 4 = 17/4$.

Solution 227.4 ■ Answer C[Back to problem](#)

The event $|Z| > 2$ is contained in $|Z| > 1$, so conditional probability is their probability ratio. Symmetry gives tail probabilities $2[1 - \Phi(2)]$ and $2[1 - \Phi(1)]$; the factors of two cancel. The ratio is approximately $0.0228/0.1587 \approx 0.144$.

Solution 227.5 ■ Answer A[Back to problem](#)

The interval must be centered at the mean 3, so $a = 1$. More explicitly, the derivative of the interval probability is $f(a + 4) - f(a)$, positive for $a < 1$ and negative for $a > 1$ by the symmetric unimodal density. The optimal interval $[1, 5]$ lies one standard deviation on either side of the mean, giving $\Phi(1) - \Phi(-1) = 2\Phi(1) - 1$.

Probability and Statistics ▸ The Normal Approximation to the Binomial Distribution

228 Normal approximation and continuity correction

Solution 228.1 ■ **Answer C**[Back to problem](#)

The mean is 50 and standard deviation 5. The event $X \geq 60$ begins at corrected boundary 59.5, so the standardized threshold is $(59.5 - 50)/5 = 1.9$.

Solution 228.2 ■ **Answer B**[Back to problem](#)

The integer event is $90 \leq X \leq 109$, so use the continuous interval $[89.5, 109.5]$. The mean is 100 and variance $400(1/4)(3/4) = 75$, yielding upper and lower standardized endpoints $9.5/\sqrt{75}$ and $-10.5/\sqrt{75}$.

Solution 228.3 ■ **Answer A**[Back to problem](#)

The expected successes are $n^{1-\alpha}$, which diverge exactly when $\alpha < 1$. For every fixed $\alpha > 0$, expected failures are $n(1 - n^{-\alpha}) \sim n$, which diverge. At $\alpha = 1$, the expected successes remain 1, so large n alone does not justify a normal approximation.

Solution 228.4 ■ **Answer C**[Back to problem](#)

The sample proportion has standard deviation $\sqrt{(1/4)(3/4)/n} = \sqrt{3}/(4\sqrt{n})$. The stated tolerance is twice this amount. Standardizing and applying the normal limit gives $P(|Z| \leq 2) = 2\Phi(2) - 1$. Integer rounding or a half-unit correction changes standardized count endpoints by a quantity tending to zero.

Solution 228.5 ■ **Answer D**[Back to problem](#)

The worst-case estimated margin is $1.96\sqrt{(1/4)/n} = 0.98/\sqrt{n}$. Requiring it to be at most 0.01 gives $\sqrt{n} \geq 98$, or $n \geq 9604$. This inverts an approximation-based design formula; it does not assert exact binomial coverage for every p .

229 Topology, open and closed sets, and neighborhoods

Solution 229.1 ■ **Answer C**[Back to problem](#)

Intersection adds $\{b\}$ and union adds $\{a, b, c\}$. Together with $\emptyset, X$ and the two original sets these give six sets. Their unions and intersections add nothing further.

Solution 229.2 ■ **Answer B**[Back to problem](#)

The topology axioms persist under intersection. On $\{a, b, c\}$, take $\mathcal{T}_1 = \{\emptyset, X, \{a\}\}$ and $\mathcal{T}_2 = \{\emptyset, X, \{b\}\}$. Their union lacks $\{a, b\}$, so fails closure under unions. Unequal comparable topologies have a union that is a topology.

Solution 229.3 ■ **Answer C**[Back to problem](#)

The closed sets are $X, \{b, c\}, \{c\}, \emptyset$. Intersecting those that contain b gives $\{b, c\}$.

Solution 229.4 ■ **Answer C**[Back to problem](#)

A neighborhood must contain an open interval about the point. This holds at every $x \in (-1, 1)$. Every interval about -1 or 1 includes points just outside N , and every interval about the isolated member 2 does too. A neighborhood need not itself be open, but merely belonging to it is insufficient.

Solution 229.5 ■ **Answer A**[Back to problem](#)

Arbitrary intersections of closed sets are closed by taking complements and using arbitrary unions of open sets. II fails for $F_n = \{1/n\}$, whose union omits its limit point zero. III fails for $U_n = (-1, 1/n)$, whose intersection is $(-1, 0]$. The topology axioms distinguish finite intersections from arbitrary ones.

230 Hausdorff separation

Solution 230.1 ■ **Answer E**[Back to problem](#)

For a fixed point x , separate x from each of the other four points and intersect the four neighborhoods of x . This makes $\{x\}$ open. Every singleton is open, hence every subset is open; there are $2^5 = 32$.

Solution 230.2 ■ **Answer C**[Back to problem](#)

The compact subset K is closed because X is Hausdorff, so its complement cannot contain the limit of a sequence lying in K . None of finiteness, isolation, openness, or compactness of the whole space follows.

Solution 230.3 ■ **Answer E**[Back to problem](#)

Singletons are closed, giving T_1 . Any two nonempty open sets intersect because each omits only finitely many points, so distinct points cannot have disjoint neighborhoods.

Solution 230.4 ■ **Answer B**[Back to problem](#)

Two distinct proposed limits have disjoint neighborhoods. Convergence would force every sufficiently late term into both neighborhoods, a contradiction. The sequence need not be eventually constant.

Solution 230.5 ■ **Answer C**[Back to problem](#)

Any two nonempty open sets intersect, because the union of their countable complements cannot exhaust uncountable X . Thus the space is not Hausdorff. If $x_n \rightarrow x$, remove the countable set of all terms different from x ; its complement is an open neighborhood of x . Convergence forces the sequence eventually to equal x , so a second limit is impossible. Uniqueness of sequence limits alone does not characterize Hausdorff spaces.

231 Discrete, indiscrete, finer, and coarser topologies

Solution 231.1 ■ Answer B[Back to problem](#)

Only the preimages of $\emptyset$ and Y need to be checked, so f is continuous. A singleton is open in X but its image is not open in the infinite indiscrete space Y .

Solution 231.2 ■ Answer D[Back to problem](#)

If two points have different images, an open interval containing one image and excluding the other has a nonempty proper preimage, which is not open. Thus exactly the constant functions are continuous, one for each real value.

Solution 231.3 ■ Answer A[Back to problem](#)

In a discrete space, convergence to a point requires eventual membership in its singleton, hence an eventually constant sequence. Alternation fails this. In an indiscrete space, the only open neighborhood of any point is X , which contains every term, so both points are limits. Finer topologies impose stronger convergence requirements.

Solution 231.4 ■ Answer C[Back to problem](#)

The inverse image under the identity of a $\mathcal{T}_1$ -open set is itself, and it belongs to $\mathcal{T}_2$. The reverse implication requires the reverse inclusion.

Solution 231.5 ■ Answer C[Back to problem](#)

A discrete subspace is covered by its singletons, so an infinite one has no finite subcover; finite spaces are always compact. Every subspace of an indiscrete space is indiscrete, and any nonempty open cover must contain the whole subspace, giving a one-set subcover. The empty subspace is compact as well.

Point-Set Topology ▸ The Subspace Topology

232 Subspace or relative topology

Solution 232.1 ■ Answer D

[Back to problem](#)

$[0, 1] = S \cap (-1, 3/2)$ is relatively open, and it is closed in $\mathbb{R}$, hence in S . The first and fifth sets omit relative limit points; the second and third are not relatively open at their nonisolated boundary points.

Solution 232.2 ■ Answer C

[Back to problem](#)

Every $1/n$ is isolated, so their union $S \setminus \{0\}$ is open. It is not closed because 0 is a limit point. The singleton $\{1\}$ and the empty set are both open and closed.

Solution 232.3 ■ Answer B

[Back to problem](#)

If S is open, each relatively open set $S \cap U$ is an intersection of two ambient open sets and is open. Conversely, S is always open in itself, so the stated property applied to S forces S to be open in $\mathbb{R}$. Relative openness alone does not imply ambient openness.

Solution 232.4 ■ Answer E

[Back to problem](#)

The irrational boundary point is absent from the subspace. Both the chosen set and its complement are intersections of $\mathbb{Q}$ with ordinary open rays, so both are relatively open.

Solution 232.5 ■ Answer C

[Back to problem](#)

A point of S is in the relative closure exactly when every relative neighborhood meets A , equivalently every ambient neighborhood meets A . This gives the intersection with S . The analogous interior equality need not hold: for $A = S$ equal to a line in $\mathbb{R}^2$, the relative interior is the line but the ambient interior is empty.

Point-Set Topology ▸ The Interior, Exterior, Boundary, Limit Points, and Closure of a Set

233 Interior, exterior, and boundary

Solution 233.1 ■ Answer B

[Back to problem](#)

The closure is $[0, 1] \cup [2, 3]$ and the interior is $(2, 3)$. Subtracting the interior gives $[0, 1] \cup \{2, 3\}$, since rational and irrational points are both dense in $[0, 1]$.

Solution 233.2 ■ Answer C

[Back to problem](#)

The exterior is the complement of the closure. The only extra limit point of the reciprocal sequence is 0, and $[2, 4)$ adds the limit point 4. This description also includes the many open gaps between consecutive reciprocals.

Solution 233.3 ■ Answer A[Back to problem](#)

If a point lies outside both boundaries, it has neighborhoods on which membership in A and in B is separately constant. Their intersection makes membership in $A \cup B$ constant, so the point is not on its boundary. Equality can fail: $A = \mathbb{Q}$ and $B = \mathbb{R} \setminus \mathbb{Q}$ each have boundary $\mathbb{R}$, while their union has empty boundary.

Solution 233.4 ■ Answer E[Back to problem](#)

The integers form a closed set with empty interior. Hence their boundary, closure minus interior, is the whole set $\mathbb{Z}$.

Solution 233.5 ■ Answer C[Back to problem](#)

Since A is closed, $\partial A = A \setminus \text{int } A$ is closed. It has empty interior: a nonempty open set contained in it would be an open subset of A , hence contained in $\text{int } A$, a contradiction. Its own boundary is therefore itself. Closedness matters; for $A = \mathbb{Q}$, the first boundary is $\mathbb{R}$ and the second is empty.

Point-Set Topology ▷ The Interior, Exterior, Boundary, Limit Points, and Closure of a Set

234 Limit points, derived set, closure, and density

Solution 234.1 ■ Answer A[Back to problem](#)

The only accumulation points are 0 and 2, so $A' = \{0, 2\}$. A finite subset of the real line has no accumulation points; its derived set is empty.

Solution 234.2 ■ Answer B[Back to problem](#)

Both A and B are dense in $(0, 1)$ and accumulate at its endpoints, so their closures are $[0, 1]$. Their intersection is empty, showing why closure need not commute with intersections. Their union is $(0, 1)$.

Solution 234.3 ■ Answer D[Back to problem](#)

The derived set is $\{0\} \cup \{1/n : n \geq 1\}$. Each $1/n$ is approached by holding n fixed and letting $m \rightarrow \infty$; zero is approached by $n = m \rightarrow \infty$. Conversely, take a convergent sequence of distinct values $1/n + 1/m$. If both indices tend to infinity along a subsequence, its limit is zero. Otherwise some index has a fixed value along a subsequence; distinctness then forces a further subsequence of the other index to tend to infinity, giving limit $1/n$ or $1/m$. These exhaust possible limits. This derived set has only zero as an accumulation point, so its derived set is $\{0\}$.

Solution 234.4 ■ **Answer C**[Back to problem](#)

Closure is closed, so taking its closure adds no points. The other assertions fail, for example using disjoint dense sets or nonclosed subsets.

Solution 234.5 ■ **Answer C**[Back to problem](#)

Any nonempty open interval meets A in a nonempty open set. Density of B makes that set meet B as well, proving the intersection meets every open interval.

Point-Set Topology ▷ Basis for a Topology

235 Topological bases and the lower-limit topology

Solution 235.1 ■ **Answer C**[Back to problem](#)

Every basic neighborhood $[0, \varepsilon)$ eventually contains $1/n$. Each of the other sequences has infinitely many negative terms or remains away from zero, and therefore fails the neighborhood criterion.

Solution 235.2 ■ **Answer B**[Back to problem](#)

Every right-hand neighborhood of 0 meets A , but $[1, 1 + \varepsilon)$ misses it. Thus the closure is $[0, 1)$. The interior remains $(0, 1)$, leaving boundary $\{0\}$.

Solution 235.3 ■ **Answer A**[Back to problem](#)

Rational endpoints give intervals refining every ordinary neighborhood, and the local intersection condition follows by choosing rational endpoints within an overlap. II fails that condition when two length-one intervals overlap in an interval shorter than one. III fails where two closed intervals meet only at one endpoint: no positive-length closed interval fits inside that singleton.

Solution 235.4 ■ **Answer B**[Back to problem](#)

The interval is basic open. Its complement $(-\infty, 0) \cup [1, \infty)$ is also a union of basic intervals, so it is closed. But $[0, 1/2)$ and $[1/2, 1)$ form a disjoint, nonempty, relatively open separation. Ordinary interval connectedness does not carry over to this finer topology.

Solution 235.5 ■ **Answer C**[Back to problem](#)

For each real x , a basis must supply B_x with $x \in B_x \subseteq [x, x + 1)$. Such a set has minimum x , so the chosen sets for distinct x are distinct. There must therefore be at least as many basis elements as real numbers. Rational left endpoints cannot refine a basic neighborhood beginning at an irrational point.

Point-Set Topology ▸ The Product Topology

236 Product topology and open rectangles

Solution 236.1 ■ Answer B[Back to problem](#)

The closure is $[0, 1]^2$, since rational first coordinates are dense. The interior is empty because every open rectangle contains an irrational first coordinate. Therefore the entire closed square is the boundary.

Solution 236.2 ■ Answer D[Back to problem](#)

The closed hyperbola $xy = 1$ projects to $\mathbb{R} \setminus \{0\}$, which is not closed. Continuity gives preservation of compactness and connectedness; products of intervals show projections are open.

Solution 236.3 ■ Answer B[Back to problem](#)

A map into a product is continuous exactly when each coordinate map into its stated factor is continuous. The second coordinate of I has singleton preimages, which are not open in the usual line. For III, the preimage of $\{0\}$ in the second factor is $[0, 1)$, again not open. II has a continuous first coordinate and a constant second coordinate.

Solution 236.4 ■ Answer B[Back to problem](#)

S is the inverse image of the closed set $\{0\}$ under the continuous function $(x, y) \mapsto xy$, so it is closed. Every disk contains a point with both coordinates nonzero, so the interior is empty. Its projections are each the whole real line, illustrating that open projections do not imply that the original subset is open. It is unbounded and hence not compact.

Solution 236.5 ■ Answer C[Back to problem](#)

If $x \notin \pi(C)$, the entire compact fiber $\{x\} \times [0, 1]$ lies in the open complement of C . Each fiber point has a product neighborhood avoiding C . Finitely many second-coordinate neighborhoods cover $[0, 1]$; intersect their first-coordinate neighborhoods to obtain a neighborhood of x avoiding $\pi(C)$. Thus the projection is closed. Taking $C = \mathbb{R} \times \{0\}$ shows boundedness need not follow.

237 Connectedness, separations, and connected components

Solution 237.1 ■ **Answer D**[Back to problem](#)

The components are $(-\infty, 0)$, $(1, \infty)$, and the intervals $(1/(n+1), 1/n)$ for $n \geq 1$. Each is an interval, and no connected subset can cross a removed point.

Solution 237.2 ■ **Answer B**[Back to problem](#)

A is the continuous image of the interval $(0, 1]$, hence is connected. Its closure is B , and the closure of a connected set is connected. This argument asserts connectedness without asserting path connectedness.

Solution 237.3 ■ **Answer B**[Back to problem](#)

Every connected subset of $\mathbb{R}$ is an interval. Any interval containing two distinct points contains a rational, so a connected subset of the irrationals has at most one point. Thus all components are singletons.

Solution 237.4 ■ **Answer E**[Back to problem](#)

Continuous images of connected spaces are connected. The only nonempty connected subsets of a discrete space are singletons, so such a map is constant.

Solution 237.5 ■ **Answer D**[Back to problem](#)

A separation of the union would place each connected subset wholly on one side. Their common point forces the same side for both, contradicting a nontrivial separation.

Solution 237.6 ■ **Answer D**[Back to problem](#)

Each coordinate has two connected components, the negative and positive rays. The eight sign choices produce products of three intervals, each connected. A connected subset cannot meet two different sign choices, since a coordinate projection would then be a connected subset of $\mathbb{R} \setminus \{0\}$ containing both signs. Thus these eight products are exactly the components.

238 Paths, path connectedness, and convexity

Solution 238.1 ■ **Answer C**[Back to problem](#)

The closed sine curve is connected because it is the closure of the connected graph over $(0, 1]$. Suppose a path joined its vertical segment to the graph. In the parameter interval where its first coordinate has just become positive, that coordinate is continuous and tends to zero at the left endpoint. It therefore takes arbitrarily small values at which $\sin(1/x) = 1$ and also values at which $\sin(1/x) = -1$, arbitrarily near that endpoint. The second coordinate could not be continuous there. Hence no such path exists. The punctured plane and square are path connected, while the punctured line and rationals are disconnected.

Solution 238.2 ■ **Answer D**[Back to problem](#)

A is the disjoint union of the first and third open quadrants, giving a separation. Every point of B can be joined to the origin by a straight segment lying in B , and two such segments join any pair of points.

Solution 238.3 ■ **Answer D**[Back to problem](#)

Two points can be joined by a polygonal path detouring around the missing origin. But the segment between opposite points passes through the origin, so convexity fails.

Solution 238.4 ■ **Answer B**[Back to problem](#)

Join each of two points to a star center by the segments contained in the set, then concatenate those segments. Star-shapedness is weaker than convexity, which requires every direct segment.

Solution 238.5 ■ **Answer B**[Back to problem](#)

Choose $q_0 \in C$. Points farther from p than q_0 cannot improve the minimum, so restrict to the nonempty closed bounded set $C \cap \overline{B}(p, \|p - q_0\|)$. Compactness and continuity give a minimizer. If distinct u, v both minimized with distance r , their midpoint lies in C and satisfies $\|p - (u + v)/2\|^2 = r^2 - \|u - v\|^2/4 < r^2$, a contradiction. Closedness gives existence; convexity gives uniqueness.

239 Open covers and compactness

Solution 239.1 ■ Answer D[Back to problem](#)

Every positive $x \leq 1$ exceeds $1/n$ for some sufficiently large n . A finite subcollection is contained in its largest member U_N , which misses $1/(N+1)$. Thus it witnesses failure of compactness.

Solution 239.2 ■ Answer B[Back to problem](#)

If the intersection were empty, the complements would form an open cover of K . A finite subcover would imply some finite intersection of the F_n was empty; nestedness makes that intersection one of the nonempty F_n , a contradiction.

Solution 239.3 ■ Answer B[Back to problem](#)

Compact metric spaces are limit-point compact. The limit point may lie outside the subset, as zero does for $\{1/n : n \geq 1\} \subset [0, 1]$.

Solution 239.4 ■ Answer E[Back to problem](#)

The finite product $X \times Y$ is compact. The function $(x, y) \mapsto f(x) - g(y)$ is continuous, so C is the inverse image of the closed set $\{0\}$ and is closed in the product. A closed subset of a compact space is compact, even when the spaces are not Hausdorff. No overlap of the two ranges was assumed, so C may be empty.

Solution 239.5 ■ Answer B[Back to problem](#)

Given an open cover of K , select a member containing x . By convergence it contains all but finitely many of the terms x_n . Choose one cover member for each remaining point; together these form a finite subcover. This argument uses the cover definition and holds without Euclidean boundedness or a Hausdorff assumption. A convergent reciprocal sequence with its limit need not be connected.

Point-Set Topology ▸ Compactness

240 Boundedness, norm, and the Heine–Borel theorem

Solution 240.1 ■ Answer C

[Back to problem](#)

The third set is an intersection of closed sets and lies in the unit disk, so is compact. The hyperbola and reciprocal graph are unbounded. The open disk and reciprocal sequence omit limit points.

Solution 240.2 ■ Answer E

[Back to problem](#)

The set is closed for every a . If $a > 0$, then $|x| \leq 1$ and $|y| \leq 1/\sqrt{a}$. If $a \leq 0$, every point $(0, y)$ satisfies the inequality, so the set is unbounded.

Solution 240.3 ■ Answer C

[Back to problem](#)

Every e_n has norm one, while distinct members have distance $\sqrt{2}$. Any convergent sequence from E must be eventually constant, so all its limits lie in E and E is closed. The sequence $e_1, e_2, \dots$ has no convergent subsequence, so E is not compact. This is an infinite-dimensional counterexample to using closed-and-bounded outside the Euclidean Heine–Borel setting.

Solution 240.4 ■ Answer A

[Back to problem](#)

The punctured closed disk is bounded but omits the limit point $(0, 0)$, so it is not closed. Every other listed set is closed and bounded.

Solution 240.5 ■ Answer E

[Back to problem](#)

This is the Heine–Borel theorem for finite-dimensional Euclidean space. Closedness alone allows unbounded sets, and boundedness alone allows missing limit points.

Point-Set Topology ▸ Metric Spaces

241 Metrics, open balls, and equivalent metric topologies

Solution 241.1 ■ Answer B

[Back to problem](#)

For $0 < p \leq 1$, $(a + b)^p \leq a^p + b^p$ for nonnegative a, b , so the usual triangle inequality implies the new one. For $p > 1$, the triple $0, 1, 2$ would require $2^p \leq 2$, which fails.

Solution 241.2 ■ Answer C

[Back to problem](#)

Each squared coordinate difference is at most $\|x - y\|_\infty^2$, so the sum is at most three times that square. Equality occurs for $x - y = (1, 1, 1)$, proving $C = \sqrt{3}$ is sharp.

Solution 241.3 ■ Answer E[Back to problem](#)

Truncation at one preserves the triangle inequality because $\min(1, a + b) \leq \min(1, a) + \min(1, b)$. The other choices violate the triangle inequality, zero-distance condition, or nonnegativity.

Solution 241.4 ■ Answer C[Back to problem](#)

Set $u = x^n$, which ranges over all of $[0, 1]$. The difference is $u - u^2 = u(1 - u)$, whose maximum is $1/4$ at $u = 1/2$, attained when $x = 2^{-1/n}$. The result is independent of n . In particular, the functions x^n are not Cauchy in this metric, even though their pointwise behavior is simple.

Solution 241.5 ■ Answer A[Back to problem](#)

The strict inequality $|x|/(1 + |x|) < 1/2$ is equivalent to $|x| < 1$. Every distance in this metric is less than one, so the whole space is bounded. More generally, for $0 < r < 1$ the metric ball is an ordinary interval of radius $r/(1 - r)$, so the metric still induces the usual topology; boundedness is not determined by topology alone.

Point-Set Topology ▸ Continuous Functions

242 Topological and metric continuity

Solution 242.1 ■ Answer A[Back to problem](#)

The two fibers must both be open, so the fiber of 0 must be clopen. Only $\emptyset$ and X are clopen in this topology. Thus exactly the two constant maps work.

Solution 242.2 ■ Answer B[Back to problem](#)

The identity from a finer topology to a coarser one is continuous. But $g^{-1}([0, 1)) = (-1, 0]$, which is not lower-limit open at 0 because every basic neighborhood of 0 contains positive points.

Solution 242.3 ■ Answer B[Back to problem](#)

At zero, $|f(x) - f(0)| = |x|$, so $\delta = \varepsilon$ proves continuity. The inverse image of U contains $(-1, 1)$, hence is a neighborhood of zero. But $3/2$ belongs to it and arbitrarily close irrational points greater than one do not, so that inverse image is not open. Continuity at one point requires a neighborhood there, not global openness of every such preimage.

Solution 242.4 ■ Answer D[Back to problem](#)

Metric spaces admit the sequential criterion for continuity. Failure of continuity at a point yields points approaching it whose images stay a fixed positive distance from its image, proving the converse as well.

Solution 242.5 ■ **Answer A**[Back to problem](#)

Take $x_n = 1/n$ and $y_n = 1/(n+1)$. Their distances tend to zero, but the image difference is always one. The other listed functions have bounded derivative or are constant on this interval.

Point-Set Topology ▸ Continuous Functions

243 Continuous images and the extreme value theorem

Solution 243.1 ■ **Answer C**[Back to problem](#)

The image is connected, so it contains every number between -2 and 5 . The stated minimum and maximum rule out any values outside that interval, and both endpoints occur.

Solution 243.2 ■ **Answer B**[Back to problem](#)

The minimum is attained at some $x_0 \in K$. Since f is strictly positive at every point, $c = f(x_0) > 0$. Positivity alone on a noncompact domain would not guarantee a positive lower bound.

Solution 243.3 ■ **Answer A**[Back to problem](#)

Continuous images of compact sets are compact. The open interval $(0, 1)$ is not compact, while each other set is compact and can occur as such an image.

Solution 243.4 ■ **Answer D**[Back to problem](#)

The inverse image is $\{k\pi : k \in \mathbb{Z}\}$, a closed unbounded set of isolated points. It is not compact and is disconnected. The target set K is both compact and connected, so this example shows that continuity preserves those properties for direct images, not generally for inverse images. Closedness is preserved under inverse images.

Solution 243.5 ■ **Answer B**[Back to problem](#)

Continuity on a compact metric space implies uniform continuity. Given $\varepsilon > 0$, choose one $\delta > 0$ valid for every pair in $[0, 1]$; for $1/n < \delta$, every difference in the maximum is less than ε . The maximum exists by continuity on the compact interval. For $f(x) = x$, $nM_n = 1$, so no asserted rate of that kind follows.

Point-Set Topology ▸ Open Maps and Homeomorphisms

244 Open maps, homeomorphisms, and topological invariants

Solution 244.1 ■ Answer B[Back to problem](#)

The map $x \mapsto \tan(\pi(x - 1/2))$ is a homeomorphism from $(0, 1)$ onto $\mathbb{R}$. Compactness rules out the pairs involving a compact and a noncompact space. Removing one point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$.

Solution 244.2 ■ Answer C[Back to problem](#)

For any closed $F \subseteq X$, compactness of X makes F compact. Its image is compact in Hausdorff Y , hence closed. Thus f is a closed bijection and its inverse is continuous.

Solution 244.3 ■ Answer C[Back to problem](#)

The image of any open set is a subset of the discrete codomain, and every such subset is open, so f is an open map. But the open singleton $\{0\}$ has inverse image $(-\infty, 0]$, which is not open in the domain. Thus f is not continuous; being an open map does not require a bijection or surjectivity.

Solution 244.4 ■ Answer C[Back to problem](#)

The map is a continuous bijection. Away from $(1, 0)$, a local angle can be chosen continuously within $(0, 2\pi)$. But $f(2\pi - 1/n) \rightarrow (1, 0) = f(0)$, while the inverse images $2\pi - 1/n$ do not converge to zero. Thus the inverse fails continuity exactly at $(1, 0)$. The half-open domain is not compact, so the compact-to-Hausdorff theorem does not apply.

Solution 244.5 ■ Answer B[Back to problem](#)

Homeomorphisms and their inverses preserve open covers, hence compactness. Boundedness is not topological: the bounded interval $(0, 1)$ is homeomorphic to the unbounded line $\mathbb{R}$.

Real Analysis ▸ The Completeness of the Real Numbers

245 Bounds, supremum, infimum, and order completeness

Solution 245.1 ■ Answer A[Back to problem](#)

Even indices give $1 + 1/n$, maximized at $n = 2$ with value $3/2$. Odd indices give $-1 + 1/n > -1$ and approach -1 , so the infimum is -1 but is not attained.

Solution 245.2 ■ Answer C[Back to problem](#)

Every difference is at most $3 - 2 = 1$. Given $\varepsilon > 0$, choose $a > 3 - \varepsilon/2$ and $b < 2 + \varepsilon/2$ using the definitions of supremum and infimum. Then $a - b > 1 - \varepsilon$, proving the supremum is 1.

Solution 245.3 ■ Answer C[Back to problem](#)

Rational density shows that elements of A approach $\sqrt{2}$ from below, so its supremum in $\mathbb{R}$ is $\sqrt{2}$. Any rational upper bound lies strictly above this irrational number, and a smaller rational number can be found between them. Thus no rational upper bound is least. Being bounded does not establish completeness of the ambient ordered field.

Solution 245.4 ■ Answer D[Back to problem](#)

Choose $a_1 \in A$. Given $a_n < s$, the supremum property supplies $a_{n+1} \in A$ strictly greater than both a_n and $s - 1/(n + 1)$, since their maximum is still below s . This constructs a strictly increasing sequence bounded above by s , with $a_n \rightarrow s$. The argument uses approximation to a supremum without assuming that it is attained.

Solution 245.5 ■ Answer E[Back to problem](#)

Reciprocation reverses order on positive numbers. The reciprocal values are bounded above by $1/2$ and approach it as elements of A approach their infimum; attainment is unnecessary.

Real Analysis ▸ The Completeness of the Real Numbers

246 Cauchy sequences and complete metric spaces

Solution 246.1 ■ Answer C[Back to problem](#)

The map $\arctan$ is injective, so d is a metric. Since $\arctan n \rightarrow \pi/2$, the sequence is Cauchy. A limit $x \in \mathbb{R}$ would require $\arctan x = \pi/2$, which is impossible.

Solution 246.2 ■ Answer D[Back to problem](#)

The fourth set contains its only possible new accumulation point, 0, so is closed in complete $\mathbb{R}$ and is complete. Each other set omits a finite limit of a Cauchy sequence of its own points.

Solution 246.3 ■ Answer C[Back to problem](#)

Consecutive differences are $\log(1 + 1/n) \rightarrow 0$. Nevertheless $|x_{2n} - x_n| = \log 2$ for every n , so the Cauchy condition fails at any fixed $\varepsilon < \log 2$. The condition must control every pair of sufficiently late terms, not just consecutive pairs.

Solution 246.4 ■ Answer D[Back to problem](#)

A tail lies in a ball of radius one about a fixed tail term. The finitely many earlier terms can be enclosed by enlarging that ball, proving boundedness. Convergence additionally requires completeness.

Solution 246.5 ■ Answer C[Back to problem](#)

Total boundedness lets one extract a Cauchy subsequence from every sequence by nested finite covers. Completeness supplies its limit, giving sequential compactness and hence compactness in a metric space.

Solution 246.6 ■ Answer A[Back to problem](#)

Given $\varepsilon > 0$, all sufficiently late terms are within $\varepsilon/2$ of each other. Choose a sufficiently late subsequence term also within $\varepsilon/2$ of x . The triangle inequality puts every sufficiently late term within ε of x . Completeness is not needed once one cluster limit is already present.

247 Outer measure and measurable sets

Solution 247.1 ■ Answer C[Back to problem](#)

The union is open and contains all rationals, hence is dense. Countable subadditivity bounds its measure by the sum of interval lengths, $\sum_{n \geq 1} \varepsilon/2^{n+1} = \varepsilon/2$. Density does not force large measure.

Solution 247.2 ■ Answer C[Back to problem](#)

Countable intersections belong to the sigma-algebra. Every real subset is an uncountable union of measurable singletons, so arbitrary-union closure would incorrectly make all subsets measurable. Arbitrary functions need not preserve measurability in either direction.

Solution 247.3 ■ Answer C[Back to problem](#)

Translating a countable interval cover preserves all lengths; multiplying by -3 multiplies lengths by three. Taking infima gives an upper bound of $3m^*(A)$, and applying the inverse affine transformation gives the reverse inequality. Thus outer measure scales by the absolute factor, yielding $6/5$, whether or not A is measurable.

Solution 247.4 ■ Answer B[Back to problem](#)

If E were measurable, the Caratheodory criterion applied to $A = [0, 1]$ would require $1 = m^*(E) + m^*(A \setminus E) = 3/2$, a contradiction. Outer measure is defined on every set but is only subadditive in general; equality across a measurable cut is the additional requirement.

Solution 247.5 ■ Answer C[Back to problem](#)

Monotonicity gives outer measure zero for each subset. Sets of outer measure zero satisfy the Lebesgue measurability criterion, which is the completeness property of Lebesgue measure.

248 Measure, null sets, and countable additivity

Solution 248.1 ■ **Answer C**[Back to problem](#)

The intervals are pairwise disjoint: the right endpoint for $n+1$ is less than 2^{-n} . Their lengths sum to $\sum_{n \geq 1} 4^{-n} = 1/3$. Adding countably many rational points does not change the measure.

Solution 248.2 ■ **Answer C**[Back to problem](#)

For each positive integer k , at most k disjoint members can have measure at least $1/k$, since their measures sum to at most one. Every positive-measure member belongs to one of these finite families for sufficiently large k , so the whole family is countable. Countably infinitely many disjoint intervals, for example $(2^{-j}, 2^{-j+1})$ for $j \geq 1$, show the bound is attainable.

Solution 248.3 ■ **Answer B**[Back to problem](#)

After n stages, the remaining closed set consists of 2^n intervals of length 3^{-n} , so $m(C) \leq (2/3)^n$ for every n and hence $m(C) = 0$. Each infinite binary sequence specifies ternary digits zero or two, yielding a distinct point of C . There are uncountably many such sequences. The intersection is closed, so it is measurable.

Solution 248.4 ■ **Answer C**[Back to problem](#)

Every E_n has infinite measure, but no finite real number belongs to all of them, so the intersection is empty and has measure zero. Continuity of measure from above requires that some set in the decreasing family have finite measure. Probability measures automatically meet that restriction; general Lebesgue-measure examples need not.

Solution 248.5 ■ **Answer E**[Back to problem](#)

The symmetric difference consists of the disjoint sets $A \setminus B$ and $B \setminus A$. Both have measure zero, so removing $A \setminus B$ from A leaves intersection measure one.

249 Measurable functions and characteristic functions

Solution 249.1 ■ Answer D

[Back to problem](#)

At 0, $|f(x)| \leq x \rightarrow 0$. At any $a > 0$, rational inputs tending to a give values tending to a , while irrational inputs give zero. The incompatible limits show discontinuity at every positive point.

Solution 249.2 ■ Answer D

[Back to problem](#)

Measurability follows from measurability of E . Every neighborhood meets both sets, so indicator values 0 and 1 occur arbitrarily close to every point. No continuity point exists; density alone does not imply almost-everywhere constancy.

Solution 249.3 ■ Answer E

[Back to problem](#)

If an open target set omits zero, its inverse image under f is a subset of the null set N and is measurable. If it contains zero, the complement of the inverse image is such a subset. Thus f is measurable regardless of the assigned values on N . The constant sequence of zero functions converges to f almost everywhere, although not necessarily at points of N .

Solution 249.4 ■ Answer D

[Back to problem](#)

The preimage of $(1/2, \infty)$ is E , proving necessity. Conversely, the preimage of any open real set under an indicator is one of $\emptyset, E, E^c, \mathbb{R}$, all measurable when E is measurable. This proves sufficiency directly from the stated open-set definition.

Solution 249.5 ■ Answer E

[Back to problem](#)

For each real a , the threshold set can be written $\{f > a\} = \bigcup_{k \geq 1} \bigcup_{N \geq 1} \bigcap_{n \geq N} \{f_n > a + 1/k\}$. If $f(x) > a$, choose $1/k < f(x) - a$ and use eventual convergence; conversely eventual values above $a + 1/k$ force $f(x) \geq a + 1/k > a$. The expression uses only countable measurable-set operations. Thus f is measurable; continuity, boundedness, integrability, and uniform convergence require extra assumptions.

Real Analysis ▸ Lebesgue Measurable Functions

250 Simple functions and approximation from below

Solution 250.1 ■ Answer D[Back to problem](#)

Apart from endpoints, s is 2 on $(0, 1)$, -1 on $(1, 2)$, and -3 on $(2, 3)$. Therefore the integral of its absolute value is $2 + 1 + 3 = 6$; the signed integral would be -2 .

Solution 250.2 ■ Answer B[Back to problem](#)

There are 2^n subintervals of length 2^{-n} with values $k2^{-n}$ for $k = 0, \dots, 2^n - 1$. The integral is $2^{-2n} \sum k = (1 - 2^{-n})/2$, approaching the integral of x from below.

Solution 250.3 ■ Answer A[Back to problem](#)

Every finite-valued simple function is bounded. If $s_n \rightarrow f$ uniformly, choose one s_n with $\sup |s_n - f| < 1$; then f is bounded. Conversely, for bounded measurable f , rounding its values down to multiples of $1/n$ produces a finite-valued measurable function with error less than $1/n$. The range bound, rather than the size of the underlying domain, is decisive.

Solution 250.4 ■ Answer E[Back to problem](#)

Increasing the truncation level cannot decrease the rounded value, and refinement of the dyadic grid cannot decrease it either. At each point where f is finite, eventually $0 \leq f - s_n < 2^{-n}$; at a point where $f = \infty$, $s_n = n \rightarrow \infty$. Each s_n is simple.

Solution 250.5 ■ Answer C[Back to problem](#)

The actual function is $\chi_{[0,1)}$, so it is integrable with integral one. The displayed representation uses overlapping sets of infinite measure, and subtracting their separate integrals would produce the invalid expression $\infty - \infty$. Rewrite the function using its disjoint level sets before integrating.

Real Analysis ▸ Lebesgue Integrable Functions

251 Lebesgue integral and positive and negative parts

Solution 251.1 ■ **Answer C**[Back to problem](#)

For $p < 1$, $\int_0^1 x^{-p} dx = 1/(1 - p)$. At $p = 1$ the logarithmic integral diverges, and for $p > 1$ the divergence is stronger. Positivity makes ordinary convergence and absolute convergence the same here.

Solution 251.2 ■ **Answer B**[Back to problem](#)

If $A = \int f^+$ and $B = \int f^-$, then $A - B = 3$ and $A + B = 11$. Subtraction gives $2B = 8$, hence $B = 4$. The negative part itself is nonnegative.

Solution 251.3 ■ **Answer A**[Back to problem](#)

With $u = \log x$, the integral becomes $\int_1^\infty u^{-p} du$, which is finite exactly when $p > 1$. Since f is positive, absolute and ordinary Lebesgue integrability coincide.

Solution 251.4 ■ **Answer C**[Back to problem](#)

For each positive integer k , nonnegativity gives $0 = \int f \geq k^{-1}m(\{f \geq 1/k\})$, so each such level set is null. Their countable union is $\{f > 0\}$ and is null. A function can nevertheless be nonzero on an uncountable null set, or have arbitrarily large finite values at a countable set of points.

Solution 251.5 ■ **Answer B**[Back to problem](#)

Each simple function is bounded, so its integral on an interval of length one is finite. But on $E_k = (2^{-k}, 2^{-k+1}]$, $f \geq 2^{k-1}$. Thus the simple minorant $s_N = \sum_{k=1}^N 2^{k-1} \chi_{E_k}$ has integral $N/2$. Their integrals have infinite supremum. This distinguishes individually finite lower approximations from a finite integral of the limit.

Real Analysis ▸ Lebesgue Integrable Functions

252 Riemann versus Lebesgue integration

Solution 252.1 ■ Answer B[Back to problem](#)

C is closed with empty interior, so the discontinuity set of its indicator is exactly C . This null set satisfies the Riemann criterion despite being uncountable. Both integrals equal the measure of C , namely zero.

Solution 252.2 ■ Answer C[Back to problem](#)

Integrating over full unit intervals gives the alternating harmonic series, which converges to $\log 2$. An unfinished last interval contributes at most $1/k$, tending to zero, so arbitrary real cutoffs have the same limit. The positive intervals contribute the divergent sum over odd reciprocals, and the negative intervals contribute the divergent sum over even reciprocals. Thus the improper integral converges conditionally while the Lebesgue integral is undefined.

Solution 252.3 ■ Answer B[Back to problem](#)

Every nondegenerate subinterval contains both rational and irrational points, so its infimum is zero and supremum is one. Hence every lower sum is zero and every upper sum is the total interval length, one. The rational set is null, so the Lebesgue integral is zero. No partition can make the Darboux gap small.

Solution 252.4 ■ Answer D[Back to problem](#)

The integral over $[\varepsilon, 1]$ is $2(1 - \sqrt{\varepsilon})$ and tends to two. Positivity makes this also the Lebesgue integral. Proper Riemann integrability on a compact interval requires boundedness; the function is unbounded near zero regardless of its assigned value at zero. Thus having only one discontinuity does not suffice without the boundedness hypothesis.

Solution 252.5 ■ Answer A[Back to problem](#)

A bounded Riemann-integrable function has a null discontinuity set and is Lebesgue integrable with the same integral. The discontinuity set can be uncountable. Altering a function on the rationals can destroy Riemann integrability while leaving its Lebesgue integral unchanged.

Real Analysis ▸ Lebesgue Integrable Functions

253 Monotone and dominated convergence

Solution 253.1 ■ Answer C

[Back to problem](#)

Each integral is $n(1/n) = 1$, but for every fixed $x > 0$ the function is eventually zero. Thus the pointwise limit has integral zero. No single integrable function can dominate all f_n almost everywhere, or dominated convergence would contradict this calculation.

Solution 253.2 ■ Answer C

[Back to problem](#)

The nonnegative functions increase pointwise to $x^{-1/2}$. Monotone convergence gives the limit $\int_0^1 x^{-1/2} dx = 2$. Directly, split at $x = n^{-2}$ to get $1/n + 2(1 - 1/n) = 2 - 1/n$.

Solution 253.3 ■ Answer B

[Back to problem](#)

Apply monotone convergence to the nonnegative increasing functions $f_n + g$. Subtract the finite number $\int g$ from the resulting integral identity. Every negative part is bounded by g , so the signed integrals are well defined, although a positive part and the limit integral may be infinite. An integrable lower bound is enough; literal nonnegativity can be obtained by shifting.

Solution 253.4 ■ Answer A

[Back to problem](#)

The integrand tends to zero at every point and satisfies $0 \leq nx/(1 + n^2x^2) \leq 1/2$. Dominated convergence gives zero. Direct integration yields $\log(1 + n^2)/(2n)$, with the same limit.

Solution 253.5 ■ Answer B

[Back to problem](#)

For $n \geq 2$ the integrands lie between zero and one and converge pointwise to e^{-x} . Dominated convergence on the finite interval gives $\int_0^2 e^{-x} dx = 1 - e^{-2}$.

Real Analysis ▸ Lebesgue Integrable Functions

254 Lebesgue spaces and completeness

Solution 254.1 ■ Answer B

[Back to problem](#)

The relevant integral is $\int_0^1 x^{-p/3} dx$, finite exactly when $p/3 < 1$. Thus the admissible range is $1 \leq p < 3$, with logarithmic divergence at $p = 3$.

Solution 254.2 ■ Answer C

[Back to problem](#)

$\int_1^\infty x^{-3/2} dx$ is finite, but $\int_1^\infty x^{-3/4} dx$ diverges. The first two choices belong to both spaces, and the last two belong to neither.

Solution 254.3 ■ Answer D[Back to problem](#)

For $p < 4$, the power $x^{-p/4}$ is locally integrable, even with the logarithmic factor. At $p = 4$, substitution $u = \log(1/x)$ gives $\int_1^\infty u^{-2} du < \infty$. For $p > 4$, the power singularity dominates the logarithmic decay and the integral diverges.

Solution 254.4 ■ Answer E[Back to problem](#)

Cauchy–Schwarz gives $\int |f| \leq (\int |f|^2)^{1/2} (\int 1)^{1/2} = 2\sqrt{3}$. Equality holds for a constant function of absolute value $2/\sqrt{3}$, so the bound is optimal.

Solution 254.5 ■ Answer E[Back to problem](#)

The exponents 3 and $3/2$ are conjugate, so Holder’s inequality gives $\int |fg| \leq \|f\|_3 \|g\|_{3/2}$. Without a finite-measure assumption or further bounds, the other integrals need not be finite.

Solution 254.6 ■ Answer C[Back to problem](#)

Completeness gives an L^p norm limit. If both f and g are norm limits, the triangle inequality implies $\|f - g\|_p = 0$, so $f = g$ almost everywhere. Values on null sets do not specify distinct elements of L^p . Completeness alone does not supply pointwise convergence of the full sequence or uniform convergence.

Solution 254.7 ■ Answer D[Back to problem](#)

The support at level k has measure 2^{-k} , so $\|f_n\|_p = 2^{-k/p} \rightarrow 0$. For every fixed x , exactly one interval at each level contains x , giving infinitely many values one; the other intervals give infinitely many values zero. Thus the full sequence has no pointwise limit anywhere. This does not contradict the existence of a pointwise almost-everywhere convergent subsequence.

Complex Variables ▸ Introduction

255 Complex arithmetic and conjugation

Solution 255.1 ■ Answer C[Back to problem](#)

Writing $z = x + iy$ gives $2x = a + ib$. The left side is real, so $b = 0$ is necessary. In that case $x = a/2$ and y is arbitrary, giving an entire vertical line of solutions. This real-linear equation can be singular; complex-looking notation does not guarantee a unique solution.

Solution 255.2 ■ Answer C[Back to problem](#)

Taking moduli gives $|z|^2 = |z|$, so either $z = 0$ or $|z| = 1$. In the latter case $\bar{z} = 1/z$, and the equation becomes $z^3 = 1$, giving three distinct additional solutions.

Solution 255.3 ■ **Answer D**[Back to problem](#)

In an ordered field, $1 = 1^2 > 0$, hence $-1 < 0$. Whether a nonzero element u is positive or negative, its square is positive, since in the latter case $u^2 = (-u)^2$. Taking $u = i$ would make $-1 = i^2 > 0$, a contradiction. The ordinary complex operations otherwise satisfy the field axioms.

Solution 255.4 ■ **Answer D**[Back to problem](#)

Since $z\bar{z} = 1$, $z^2 + \bar{z}^2 = (z + \bar{z})^2 - 2 = -1$. Squaring again gives $z^4 + \bar{z}^4 = (-1)^2 - 2 = -1$.

Solution 255.5 ■ **Answer B**[Back to problem](#)

Numerator and denominator have equal modulus, so every value lies on the unit circle. Solving $w = (t - i)/(t + i)$ gives $t = i(1 + w)/(1 - w)$, which is real for every unit $w \neq 1$. The equation $w = 1$ would require $-i = i$, so that point is omitted.

Complex Variables ▸ The Polar Form

256 Modulus, arguments, and polar geometry

Solution 256.1 ■ **Answer B**[Back to problem](#)

The triangle inequality gives $|z| + |2 - z| \geq 2$. Equality requires the two vectors z and $2 - z$ to point in the same direction, allowing a zero vector. Since they sum to the positive real number two, both must be nonnegative real. Thus precisely $0 \leq z \leq 2$ works. This is the degenerate ellipse case.

Solution 256.2 ■ **Answer B**[Back to problem](#)

Writing $z = x + iy$ and squaring gives $(x - 1)^2 + y^2 = 4((x + 1)^2 + y^2)$. This simplifies to $(x + 5/3)^2 + y^2 = 16/9$, a circle centered at the real point $-5/3$.

Solution 256.3 ■ **Answer C**[Back to problem](#)

For $z = x + iy$, equality of squared distances reduces to $x = y$, the perpendicular bisector of the segment from one to i . On this line, $|z - 2|^2 = (x - 2)^2 + x^2 = 2(x - 1)^2 + 2$. The minimum is two for the squared distance, hence $\sqrt{2}$, attained at $z = 1 + i$.

Solution 256.4 ■ **Answer C**[Back to problem](#)

The squared distance is $|z - 3|^2 = |z|^2 + 9 - 6 \operatorname{Re} z = 13 - 6 \operatorname{Re} z$. It is largest when the real part is zero, attained at $z = \pm 2i$, giving $\sqrt{13}$.

Solution 256.5 ■ **Answer C**[Back to problem](#)

Conjugation negates the argument, cubing multiplies it by three, and multiplication by -1 adds π . Thus the resulting argument is $-3\pi/5 + \pi = 2\pi/5$, already in the principal range.

Complex Variables ▸ The Exponential Form

257 Euler formula, complex exponential, and de Moivre formula

Solution 257.1 ■ **Answer B**
[Back to problem](#)

$1 + i = \sqrt{2}e^{i\pi/4}$, hence the twelfth power is $2^6e^{3\pi i} = -64$. Tracking both modulus and angle avoids expansion.

Solution 257.2 ■ **Answer B**
[Back to problem](#)

The exponent must equal $2\pi ik$. Dividing by $1 + i$ gives $z = 2\pi ik(1 - i)/2 = \pi k(1 + i)$ for every integer k .

Solution 257.3 ■ **Answer C**
[Back to problem](#)

The sum is the real part of $(1 + i)^8$. Since $1 + i = \sqrt{2}e^{i\pi/4}$, its eighth power is $16e^{2\pi i} = 16$.

Solution 257.4 ■ **Answer B**
[Back to problem](#)

The modulus equation gives $\operatorname{Re} z = 1$, and the argument equation gives $\operatorname{Im} z = (2k + 1)\pi$. The smallest absolute imaginary part is π , so the minimum modulus is $\sqrt{1 + \pi^2}$.

Solution 257.5 ■ **Answer B**
[Back to problem](#)

The modulus $e^{\operatorname{Re} z}$ ranges over all positive numbers, while the angle ranges strictly between zero and 2π . This gives exactly the plane with the nonnegative real ray removed. Equal exponentials have exponents differing by $2\pi ik$; two imaginary parts in this open interval cannot differ by such a nonzero multiple, so the map is injective.

Complex Variables ▸ Complex Roots

258 Complex roots and roots of unity

Solution 258.1 ■ **Answer B**
[Back to problem](#)

The roots have modulus 2 and arguments $\pi/6, \pi/2, 5\pi/6, 7\pi/6, 3\pi/2, 11\pi/6$. Only the first and last have positive real part; the two purely imaginary roots do not count.

Solution 258.2 ■ **Answer C**
[Back to problem](#)

Factor $(z^7 - 1)/(z - 1) = \prod_{k=1}^6 (z - \omega^k) = 1 + z + \cdots + z^6$. Evaluating at $z = 1$ yields the product 7.

Solution 258.3 ■ **Answer B**
[Back to problem](#)

Pair k with $9 - k$. For $a \neq 1$, $1/(1 - a) + 1/(1 - a^{-1}) = 1$. There are four such pairs, giving 4.

Solution 258.4 ■ Answer B[Back to problem](#)

The roots of $z^6 = -1$ have arguments $k\pi/6$ with k odd. The primitive twelfth roots correspond to $k = 1, 5, 7, 11$; the remaining two, $k = 3, 9$, are $\pm i$. Divide $z^6 + 1$ by $z^2 + 1$ to obtain $z^4 - z^2 + 1$. Its four distinct roots are exactly the desired ones.

Solution 258.5 ■ Answer D[Back to problem](#)

The roots have modulus two and principal arguments $\pi/3, \pi, -\pi/3$. Their principal square roots are $\sqrt{2}e^{i\pi/6}, i\sqrt{2}$, and $\sqrt{2}e^{-i\pi/6}$. The conjugate pair sums to $2\sqrt{2}\cos(\pi/6) = \sqrt{6}$, leaving the stated total. Taking principal roots does not preserve the original rotational symmetry.

Complex Variables ▸ Complex Logarithms

259 Multivalued logarithm and principal logarithm

Solution 259.1 ■ Answer D[Back to problem](#)

The first two logarithms sum to $3\pi i/2$. The product has principal argument $-\pi/2$, so its principal logarithm is $-\pi i/2$. The difference is $2\pi i$.

Solution 259.2 ■ Answer B[Back to problem](#)

The real parts always agree. The imaginary parts agree exactly when 2θ already lies in the principal interval $(-\pi, \pi]$. The asymmetric endpoint convention excludes $-\pi/2$ and includes $\pi/2$.

Solution 259.3 ■ Answer E[Back to problem](#)

The exponential has modulus e^2 and argument $7\pi/3$ modulo 2π . Its principal argument is $\pi/3$, so its principal logarithm is $2 + \pi i/3$.

Solution 259.4 ■ Answer E[Back to problem](#)

If L existed, the continuous function $[L(e^{it}) - it]/(2\pi i)$ on $[0, 2\pi]$ would be integer-valued, hence constant. It would follow that $L(e^{2\pi i}) - L(1) = 2\pi i$, although both arguments are the same point. Choosing a principal value introduces a jump; it cannot give a continuous logarithm on the whole circle.

Solution 259.5 ■ Answer E[Back to problem](#)

Any analytic logarithm on this connected domain differs from the principal analytic branch by a constant integer multiple of $2\pi i$. The normalization gives $L = \text{Log} + 4\pi i$, so $L(i) = \pi i/2 + 4\pi i = 9\pi i/2$.

Complex Variables ▸ Complex Powers

260 Complex powers and branch dependence

Solution 260.1 ■ **Answer B**
[Back to problem](#)

All logarithms of i equal $i(\pi/2 + 2\pi k)$. Multiplying by the exponent i gives $-\pi/2 - 2\pi k$, hence the stated positive real values.

Solution 260.2 ■ **Answer D**
[Back to problem](#)

Changing logarithm by $2\pi i k$ multiplies the value by $e^{2\pi i(3k/8)}$. Since 3 and 8 are relatively prime, these factors assume exactly eight distinct values, each multiplying a nonzero number.

Solution 260.3 ■ **Answer A**
[Back to problem](#)

On this consistent branch, differentiation gives $F'(z) = iz^{i-1}$ and $F''(z) = i(i-1)z^{i-2}$. At one, the principal logarithm is zero, so the last power equals one. Therefore $F''(1) = i(i-1) = -1 - i$. Branch differentiation applies locally on its analytic domain.

Solution 260.4 ■ **Answer B**
[Back to problem](#)

Changing the logarithm by $2\pi i k$ multiplies a value of z^w by $e^{2\pi i k w}$. For $w = 2$ this is always one. For $w = \sqrt{2}$, equality at two different indices would make a nonzero integer multiple of $\sqrt{2}$ an integer. For $w = 1 + i$, the factors are $e^{-2\pi k}$, which have distinct positive moduli. This proves all three counts.

Solution 260.5 ■ **Answer C**
[Back to problem](#)

I compares $i^2 = -1$ with $\sqrt{1} = 1$, so it fails. II has left side $\sqrt{1} = 1$, so it fails. III is valid here: the left side is $e^{2\pi i/3}$, and the principal cube root is $e^{\pi i/3}$, whose ordinary square is that same value. Familiar power rules must be checked using the chosen arguments.

Complex Variables ▸ The Trigonometric Functions

261 Complex trigonometric functions and equations

Solution 261.1 ■ **Answer A**
[Back to problem](#)

With $w = e^{iz}$ the equation is $w^2 - 4w + 1 = 0$, so $w = 2 \pm \sqrt{3}$. These are positive reciprocals. Therefore $iz = \pm \log(2 + \sqrt{3}) + 2\pi i k$, yielding the stated set after absorbing the sign.

Solution 261.2 ■ **Answer A**
[Back to problem](#)

$\sin z = 0$ implies $e^{iz} = e^{-iz}$, hence $e^{2iz} = 1$. Thus $2iz = 2\pi i k$, giving $z = k\pi$. No additional nonreal zeros occur.

Solution 261.3 ■ **Answer C**[Back to problem](#)

For $z = x + iy$, $\sin z = \sin x \cosh y + i \cos x \sinh y$. The real part forces $x = k\pi$. Then $(-1)^k \sinh y = 3$, giving $y = L$ for even k and $y = -L$ for odd k .

Solution 261.4 ■ **Answer A**[Back to problem](#)

At a point where tangent is defined, set $w = e^{2iz} \neq 0$. Then $\tan z = (w - 1)/(i(w + 1))$, with $w \neq -1$. Equating this to i gives $w - 1 = -(w + 1)$ and hence $w = 0$, impossible for an exponential. Thus complex tangent can omit a value even though complex sine and cosine are unbounded.

Solution 261.5 ■ **Answer C**[Back to problem](#)

The zeros of $\cos z$ are $\pi/2 + k\pi$. Its derivative $-\sin z$ is nonzero there, while the numerator $\sin z$ is also nonzero. Thus each is a simple pole of the quotient.

Solution 261.6 ■ **Answer A**[Back to problem](#)

On their domains, $F = (1 - \sin^2 z)/\cos^2 z = 1$ and $G = (1 - \cos^2 z)/\sin^2 z = 1$. Thus each omitted point can be filled with value one to give a holomorphic function everywhere. Poles in individual summands can cancel exactly in a difference.

Complex Variables ▸ The Hyperbolic Functions

262 Hyperbolic functions and complex trigonometric modulus

Solution 262.1 ■ **Answer C**[Back to problem](#)

The squared modulus is $\sin^2 x + \sinh^2(\log 2)$, minimized when $\sin x = 0$. Since $\sinh(\log 2) = (2 - 1/2)/2 = 3/4$, the minimum modulus is $3/4$.

Solution 262.2 ■ **Answer C**[Back to problem](#)

Adding the two modulus formulas gives $1 + 2 \sinh^2 y$, which is $\cosh(2y)$. The identity $\sin^2 z + \cos^2 z = 1$ does not imply the corresponding sum of squared moduli equals one.

Solution 262.3 ■ **Answer C**[Back to problem](#)

The equation $(e^z + e^{-z})/(e^z - e^{-z}) = 2$ gives $e^{2z} + 1 = 2(e^{2z} - 1)$, hence $e^{2z} = 3$. Thus $2z = \log 3 + 2\pi ik$. At these points $e^{2z} \neq 1$, so $\sinh z \neq 0$ and no inadmissible solution was introduced.

Solution 262.4 ■ **Answer B**[Back to problem](#)

The real and imaginary parts are $u = \sin x \cosh y$ and $v = \cos x \sinh y$. Divide by the positive semiaxes and add squares to obtain the ellipse equation. Conversely, the sine-cosine parametrization traverses the whole ellipse. This is an image-locus problem, not a bound obtained by maximizing one term.

Solution 262.5 ■ **Answer B**[Back to problem](#)

The exponential definitions give $\sin(iz) = i \sinh z$ and $\cos(iz) = \cosh z$. Dividing gives $\tan(iz) = i \tanh z$ wherever the denominator is nonzero.

Complex Variables ▸ The Derivative of a Function of a Complex Variable

263 Complex differentiability

Solution 263.1 ■ **Answer D**[Back to problem](#)

Writing $f = x^2 + y^2 + i0$, the Cauchy–Riemann equations require $2x = 0$ and $2y = 0$. At zero the quotient $|h|^2/h = \bar{h} \rightarrow 0$, confirming derivative zero there.

Solution 263.2 ■ **Answer C**[Back to problem](#)

The difference quotient of F is the conjugate of $[f(\bar{a} + \bar{h}) - f(\bar{a})]/\bar{h}$. As $h \rightarrow 0$, also $\bar{h} \rightarrow 0$ through every direction. Taking the conjugate of the limit proves the formula. Conjugating both the input and the output is different from conjugating just one.

Solution 263.3 ■ **Answer B**[Back to problem](#)

The derivative of the inner path at zero is $i(1 + i) = -1 + i$. The chain rule gives $(3 + i) = f'(1 + i)(-1 + i)$. Dividing and rationalizing yields $(3 + i)/(-1 + i) = -1 - 2i$. A derivative along a parametrized path includes the complex velocity of that path.

Solution 263.4 ■ **Answer D**[Back to problem](#)

Since $|f(z)| = |z|$, it is continuous at zero. Along the real axis $f(t) = t$, and along the imaginary axis $f(it) = it$, so the partial derivatives even satisfy the Cauchy–Riemann equations at zero. But $f(z)/z = (\bar{z}/z)^2$ equals 1 on the real axis and -1 on the line $z = t(1 + i)$, so no complex derivative exists.

Solution 263.5 ■ **Answer B**[Back to problem](#)

The polynomial z^2 is complex differentiable everywhere. If their sum were differentiable at any point, subtracting z^2 would make $\bar{z}$ differentiable there, but its difference quotients along real and imaginary increments are 1 and -1 .

Complex Variables ▸ The Cauchy–Riemann Equations

264 Cauchy–Riemann equations

Solution 264.1 ■ **Answer B**
[Back to problem](#)

The equations are $2x = bx$ and $2ay = -by$ for all x, y . Thus $b = 2$ and $a = -1$, producing exactly $f(z) = z^2$.

Solution 264.2 ■ **Answer D**
[Back to problem](#)

Here $u_x = y$, $v_y = -2y$, $u_y = x$, and $v_x = 2x$. The equations give $3y = 0$ and $3x = 0$. Continuous polynomial partials make those conditions sufficient at the origin.

Solution 264.3 ■ **Answer B**
[Back to problem](#)

The Cauchy–Riemann equations are $ae^{ax} \cos by = be^{ax} \cos by$ and $-be^{ax} \sin by = -ae^{ax} \sin by$. They hold everywhere precisely when $a = b$, in which case $f(z) = e^{az}$.

Solution 264.4 ■ **Answer D**
[Back to problem](#)

The Cauchy–Riemann equations put J in the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. Thus $J^T J = (a^2 + b^2)I$ and $\det J = a^2 + b^2 = |f'|^2 > 0$. Locally the derivative scales all directions equally and preserves orientation. Symmetry and zero trace are not forced.

Solution 264.5 ■ **Answer A**
[Back to problem](#)

For an analytic function, $f' = u_x + iv_x = u_x - iv_y$ by the Cauchy–Riemann equations. Thus $f'(1 + 2i) = 3 + 4i$.

Complex Variables ▸ The Cauchy–Riemann Equations

265 Harmonic functions and harmonic conjugates

Solution 265.1 ■ **Answer D**
[Back to problem](#)

The given real part is $\operatorname{Re}(z^3 + 2z)$. Hence $v = 3x^2y - y^3 + 2y + C$. The initial value gives $C = 5$, and at $(1, 2)$ this is $6 - 8 + 4 + 5 = 7$.

Solution 265.2 ■ **Answer D**
[Back to problem](#)

The logarithmic radius is the real part of a local logarithm, whose conjugate is the angle. The angle increases by 2π around the origin and cannot be globally single-valued. The other functions are real parts of globally entire functions.

Solution 265.3 ■ Answer B[Back to problem](#)

Differentiation gives $\Delta(u^2) = 2(u_x^2 + u_y^2) + 2u\Delta u = 2(u_x^2 + u_y^2)$. If both functions are harmonic, this vanishes, forcing both first derivatives to be zero. Connectedness then makes u constant. Every real constant meets the assumptions, including zero.

Solution 265.4 ■ Answer C[Back to problem](#)

Its Laplacian is $u_{xx} + u_{yy} = 2(a + c)$, so harmonicity requires $a + c = 0$. When this holds, u is the real part of $(a - ib/2)z^2$, which supplies a global harmonic conjugate.

Solution 265.5 ■ Answer D[Back to problem](#)

Since $1/z = (x - iy)/(x^2 + y^2)$ is analytic on the punctured plane, its imaginary part $-y/(x^2 + y^2)$ is a global harmonic conjugate. A punctured domain does not prevent every harmonic function from having one.

Complex Variables ▸ Analytic Functions

266 Analytic, holomorphic, and entire functions

Solution 266.1 ■ Answer B[Back to problem](#)

With $u = 0$, the Cauchy–Riemann equations give $v_x = v_y = 0$. Since the plane is connected, v is constant. The constant need not be zero.

Solution 266.2 ■ Answer C[Back to problem](#)

The polynomial components have continuous partial derivatives. The equation $u_x = v_y$ is $2y = 2y$, while $u_y = -v_x$ requires $2x = -2x$, so exactly $x = 0$ gives differentiability. There $f' = u_x + iv_x = 2y$. No point of this line has a plane neighborhood contained in the line, so the function is analytic nowhere.

Solution 266.3 ■ Answer B[Back to problem](#)

The entire function $f(z) - z^2$ has zeros $1/n$ accumulating at zero, a point of its domain. By the identity theorem it vanishes identically, so $f(i) = i^2 = -1$.

Solution 266.4 ■ Answer C[Back to problem](#)

Put $h = z - 1$. The numerator is $(1 + h)^5 - 5(1 + h) + 4 = 10h^2 + 10h^3 + 5h^4 + h^5$. After division, the extension is $10 + 10h + 5h^2 + h^3$. Its second derivative at $h = 0$ is $2! \cdot 5 = 10$. Cancellation produces a holomorphic extension with derivatives of every order.

Solution 266.5 ■ Answer B[Back to problem](#)

If $f(a) \neq 0$, continuity makes f nonzero on a neighborhood of a . The identity then forces $f(-z) = 0$ throughout that neighborhood, so f vanishes on an open set and hence everywhere, a contradiction. Therefore no such a exists.

Complex Variables ▸ Complex Line Integrals

267 Complex line integrals and primitives

Solution 267.1 ■ **Answer D**
[Back to problem](#)

Set $z = e^{it}$ for $0 \leq t \leq 2\pi$. Then $\bar{z} = e^{-it}$ and $dz = ie^{it}dt$, so the integrand is $i dt$ and the integral is $2\pi i$. Nonanalyticity prevents applying Cauchy's theorem.

Solution 267.2 ■ **Answer A**
[Back to problem](#)

The entire primitive is $z^3/3$, so the integral is $(1+i)^3/3 = (-2+2i)/3$. No information about the intermediate path is needed.

Solution 267.3 ■ **Answer E**
[Back to problem](#)

Writing $z = x + iy$ gives $\bar{z} dz = x dx + y dy + i(x dy - y dx)$. The real part integrates to zero, and Green's theorem makes the imaginary part twice the enclosed area. The triangle has area 2, so the result is $4i$.

Solution 267.4 ■ **Answer C**
[Back to problem](#)

On C_1 , write $z = t(1+i)$, giving integral $\int_0^1 t(1+i)dt = (1+i)/2$. On the two segments of C_2 , the integrals are $\int_0^1 t dt = 1/2$ and $\int_0^1 i dt = i$. Subtracting yields $i/2$. This nonholomorphic integrand has genuine path dependence.

Solution 267.5 ■ **Answer C**
[Back to problem](#)

The partial fraction decomposition is $1/(z-i) + 1/(z+i)$. Both simple poles are inside C , each contributing $2\pi i$, so the integral is $4\pi i$.

Complex Variables ▸ Theorems Concerning Analytic Functions

268 Cauchy theorem and simply connected domains

Solution 268.1 ■ **Answer C**
[Back to problem](#)

The function $1/z^2$ has primitive $-1/z$. Every other listed function has Laurent coefficient 1 at z^{-1} , so its integral around the unit circle is $2\pi i$; a global primitive would force that integral to vanish.

Solution 268.2 ■ **Answer D**
[Back to problem](#)

Cauchy's theorem applied to the region between the circles makes the two counterclockwise integrals equal. Reversing the outer orientation changes its sign, giving $-6\pi i$.

Solution 268.3 ■ **Answer C**[Back to problem](#)

On this slit plane choose the continuous argument in $(-\pi/2, 3\pi/2)$. The logarithm $\log|z| + i \arg z$ is a primitive of $1/z$. Its values at -1 and 1 are πi and zero, so the integral is $-\pi i$. The specified domain prevents paths from differing by a full turn around the origin.

Solution 268.4 ■ **Answer D**[Back to problem](#)

The two contributions cancel on the large circle, but a sufficiently small circle centered at one encloses only that pole and has integral $2\pi i$. A global primitive would make every closed-contour integral zero. The function is holomorphic on D ; the obstruction comes from the separate holes and their nonzero periods.

Solution 268.5 ■ **Answer B**[Back to problem](#)

Cauchy's theorem on a simply connected domain makes every such closed-curve integral zero, equivalently giving an analytic primitive. Boundedness, polynomial form, and absence of zeros do not follow from analyticity on an arbitrary domain.

Complex Variables ▸ Theorems Concerning Analytic Functions

269 Morera theorem

Solution 269.1 ■ **Answer C**[Back to problem](#)

On the unit circle, $|z|^2 = 1$, so the integral is $\oint dz = 0$. But $|z|^2$ is complex differentiable only at zero and hence is not entire. The other four choices are entire. Morera requires zero integrals for every qualifying triangle, not just one chosen closed curve.

Solution 269.2 ■ **Answer B**[Back to problem](#)

Split any triangle crossing the real axis into pieces above and below it. Cauchy's theorem, continuity at the line, and cancellation of the shared segments show the boundary integral is zero. Morera then gives holomorphy on all of $\mathbb{C}$.

Solution 269.3 ■ **Answer A**[Back to problem](#)

These are precisely the local hypotheses of Morera's theorem, which yields analyticity. They impose no polynomial or constant restriction, nor do they guarantee extension beyond the disk.

Solution 269.4 ■ **Answer B**[Back to problem](#)

Local uniform convergence gives continuity of f and convergence of integrals along each fixed triangle boundary. Each integral of f_n is zero, so the integral of f is zero. Morera's theorem now gives analyticity.

Solution 269.5 ■ **Answer D**[Back to problem](#)

The function is discontinuous at zero. Changing an integrand at a single point does not affect these ordinary contour integrals, but Morera's theorem requires continuity on the domain. Thus the integral condition alone does not suffice.

Complex Variables ▸ Theorems Concerning Analytic Functions

270 Cauchy integral and derivative formulas

Solution 270.1 ■ **Answer C**[Back to problem](#)

Use $f(z) = z^2 e^z$ and $n = 2$. Since $f''(z) = e^z(z^2 + 4z + 2)$, its value at 1 is $7e$. Multiplying by $2\pi i/2!$ gives $7\pi i e$.

Solution 270.2 ■ **Answer B**[Back to problem](#)

The fourth derivative of e^{2z} at zero is 16. Positive orientation would give $2\pi i \cdot 16/4! = 4\pi i/3$; clockwise orientation negates it.

Solution 270.3 ■ **Answer C**[Back to problem](#)

For a inside, the value formula applied to the analytic numerator z^2 gives $2\pi i a^2$. For a outside, the entire integrand is holomorphic on and inside the contour, so Cauchy's theorem gives zero. The same algebraic integrand produces different parameter regions; the contour-location condition cannot be dropped.

Solution 270.4 ■ **Answer E**[Back to problem](#)

The coefficient of z^4 in $e^{z^2} = 1 + z^2 + z^4/2 + \dots$ is $1/2$. After division by z^5 , this is the residue, so the integral is $2\pi i(1/2) = \pi i$.

Solution 270.5 ■ **Answer D**[Back to problem](#)

The point one lies on the path. With $z = e^{it}$ near $t = 0$, the parametrized integrand $ie^{it}/(e^{it} - 1)$ behaves like $1/t$, so the corresponding one-sided improper integral does not converge. A separately specified principal value would be a different operation. Cauchy's formula requires analyticity on the contour itself.

Complex Variables ▸ Theorems Concerning Analytic Functions

271 Cauchy derivative estimates and growth bounds

Solution 271.1 ■ Answer B

[Back to problem](#)

Cauchy's estimate gives $2! \cdot 9/3^2 = 2$. The function $f(z) = z^2$ attains this bound and satisfies the boundary condition, so it cannot be improved.

Solution 271.2 ■ Answer B

[Back to problem](#)

For $n \geq 2$, the n th Taylor coefficient has modulus at most $10(1 + R^{3/2})/R^n$, which tends to zero. Thus only constant and linear terms remain. A nonzero linear function satisfies such a bound.

Solution 271.3 ■ Answer D

[Back to problem](#)

The estimate is $2e^R/R^2$. Its logarithmic derivative is $1 - 2/R$, negative before $R = 2$ and positive after, so the minimum is $e^2/2$. This value is even attainable: $f(z) = (e^2/4)z^2$ obeys the growth bound because e^r/r^2 has minimum $e^2/4$.

Solution 271.4 ■ Answer C

[Back to problem](#)

The coefficient of z^2 is one. Cauchy's formula writes it as the average of $f(e^{it})e^{-2it}$, whose real part is at most one. Its average real part equals one, so continuity forces that real part to equal one everywhere. The modulus bound then forces $f(e^{it})e^{-2it} = 1$ throughout the boundary. The maximum principle applied to $f - z^2$ gives $f = z^2$ inside.

Solution 271.5 ■ Answer B

[Back to problem](#)

Cauchy gives $|a_n| \leq 4/2^n$. On $|z| \leq 1$, bound each tail term by $4/2^n$ and sum: $\sum_{n=3}^{\infty} 4/2^n = 4(1/8)/(1 - 1/2) = 1$. The question asks for the resulting summed estimate, not whether every coefficient bound can be attained simultaneously by one function.

Complex Variables ▸ Theorems Concerning Analytic Functions

272 Liouville theorem

Solution 272.1 ■ Answer A

[Back to problem](#)

The reciprocal $1/f$ is entire and bounded by $1/2$, so is constant by Liouville. It is nonzero, hence f itself is constant; its value need not equal 2.

Solution 272.2 ■ Answer A

[Back to problem](#)

The entire function $e^{if(z)}$ has modulus $e^{-\operatorname{Im} f(z)} < 1$, so is constant. Differentiating gives $if'(z)e^{if(z)} = 0$; the exponential never vanishes, hence $f' = 0$ and f is constant.

Solution 272.3 ■ Answer B[Back to problem](#)

The leading term gives $|P(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$, so $1/P$ is bounded outside a sufficiently large disk. The no-zero assumption makes $1/P$ continuous on the remaining compact disk, where it has a finite maximum modulus. It is also entire. Liouville would make it constant, contradicting the positive degree of P .

Solution 272.4 ■ Answer C[Back to problem](#)

I is Liouville's theorem. For II, boundedness near the isolated missing point makes it removable; extending there gives a bounded entire function, again constant. For III, $f(z) = z$ is bounded and nonconstant. The puncture case needs the removable-singularity step, not an unsupported application of Liouville to a proper domain.

Solution 272.5 ■ Answer C[Back to problem](#)

Both exponentials are entire, nonconstant, and nowhere zero, with product one. In each other pair the reciprocal has a pole, so it is not entire. Liouville requires boundedness, not merely absence of zeros.

Complex Variables ▸ Theorems Concerning Analytic Functions

273 Maximum modulus principle

Solution 273.1 ■ Answer D[Back to problem](#)

The triangle inequality gives $|z^2 + 2z| \leq |z|^2 + 2|z| \leq 3$. Equality occurs at $z = 1$. The maximum principle also ensures the maximum lies on the boundary because the polynomial is nonconstant.

Solution 273.2 ■ Answer B[Back to problem](#)

Apply the maximum principle to f to obtain $|f| \leq 1$ inside. Apply it to the holomorphic reciprocal to obtain $1/|f| \leq 1$. Thus $|f| = 1$ throughout the disk, and an interior maximum forces constancy on this connected disk.

Solution 273.3 ■ Answer D[Back to problem](#)

Writing $w = (1 + z)/(1 - z)$ gives $\operatorname{Re} w = (1 - |z|^2)/|1 - z|^2 > 0$, hence $|f| < 1$. Along points corresponding to $w = n$, which tend to $z = 1$, the values tend to zero. Along points corresponding to $w = 1 + 2\pi in$, which also tend to one from within the disk, the values are e^{-1} . The two limits disagree. Boundedness inside a bounded domain does not supply boundary continuity.

Solution 273.4 ■ Answer E[Back to problem](#)

The maximum modulus principle first gives $|f| \leq 1$ throughout the disk. Equality at an interior point forces f to be constant, and the stated value makes the constant one.

Solution 273.5 ■ **Answer E**[Back to problem](#)

The reciprocal $1/f$ is analytic on the disk and has modulus at most $1/2$ on the boundary. The maximum modulus principle gives the same bound inside, hence $|f| \geq 2$. The zero-free hypothesis is essential.

Complex Variables ▸ Taylor Series for Functions of a Complex Variable

274 Complex Taylor series and disks of convergence

Solution 274.1 ■ **Answer C**[Back to problem](#)

The only singularities are i and $-i$, both at distance $\sqrt{2^2 + 1^2} = \sqrt{5}$ from the center 2. Neither is removable, so that distance is the exact Taylor radius.

Solution 274.2 ■ **Answer B**[Back to problem](#)

This is geometric in $4z^2$ and converges exactly when $|4z^2| < 1$, or $|z| < 1/2$. The zero coefficients at odd indices illustrate why the general root formula uses limsup rather than assuming a limit exists.

Solution 274.3 ■ **Answer D**[Back to problem](#)

For $z \neq 1$ on the unit circle, the partial sums of z^n are bounded by the finite geometric-sum formula. Dirichlet's test with $1/n \downarrow 0$ gives convergence. At $z = 1$ the series is harmonic and diverges. At every boundary point its absolute-value series is the harmonic series, so none of this boundary convergence is absolute.

Solution 274.4 ■ **Answer B**[Back to problem](#)

The analytic extension is $1/(z - 3)$. The apparent singularity at 1 is removable and does not limit its Taylor series; the actual pole at 3 gives radius 3.

Solution 274.5 ■ **Answer A**[Back to problem](#)

Multiplication by z^2 shifts degrees by two. The desired coefficient is the coefficient of z^5 in $e^{3z} = \sum_{n \geq 0} 3^n z^n / n!$, namely $3^5/5!$.

Complex Variables ▸ Singularities, Poles, and Laurent Series

275 Isolated singularities and their classification

Solution 275.1 ■ Answer C

[Back to problem](#)

$\sin z - z = -z^3/6 + z^5/120 - \dots$. Dividing by z^5 gives $-1/(6z^2) + 1/120 - \dots$, so the most negative power is z^{-2} with nonzero coefficient.

Solution 275.2 ■ Answer B

[Back to problem](#)

The cosine series gives $f(z) = 1/2 - z^2/24 + \dots$, hence a removable singularity with extension value $1/2$. The series $g(z) = \sum_{n \geq 0} z^{-n}/n!$ has infinitely many negative powers and an essential singularity.

Solution 275.3 ■ Answer E

[Back to problem](#)

For every nonzero integer k , the point $z = 1/(k\pi)$ makes $\sin(1/z) = 0$. These are poles, since the derivative of the denominator there is nonzero, and they accumulate at zero. There is therefore no punctured disk about zero on which f is holomorphic. The removable/pole/essential trichotomy for isolated singularities does not apply.

Solution 275.4 ■ Answer C

[Back to problem](#)

The bound makes the isolated singularity removable and forces the extension value to be zero. Then $|F(z)/z| \leq |z|^{1/2} \rightarrow 0$, proving $F'(0) = 0$. The example $F(z) = z^2$ meets the bound on the unit disk and has second derivative two, so neither identically zero nor a vanishing second derivative is forced.

Solution 275.5 ■ Answer D

[Back to problem](#)

The numerator has a simple zero at $1/2$, since its derivative there is $-\pi \sin(\pi/2) = -\pi \neq 0$. One of the denominator's two zero factors cancels, leaving a simple pole.

Complex Variables ▸ Singularities, Poles, and Laurent Series

276 Laurent series, annuli, and principal parts

Solution 276.1 ■ Answer B

[Back to problem](#)

Partial fractions give $-1/[2(z-1)] + 1/[2(z-3)]$. For $|z| > 1$, the first term is $-\frac{1}{2} \sum_{n \geq 0} z^{-n-1}$. For $|z| < 3$, the second term has only nonnegative powers. Thus the coefficient is $-1/2$.

Solution 276.2 ■ Answer C

[Back to problem](#)

Put $w = z - 2$. Then $1/(z-1) = 1/(w+1) = w^{-1}/(1+w^{-1})$. The geometric series in $-w^{-1}$ converges for $|w| > 1$ and yields the third choice.

Solution 276.3 ■ Answer B[Back to problem](#)

The expansion is $1 + z^{-1} + z^{-2}/2! + \cdots + z^2$. The nonnegative powers form the analytic part $1 + z^2$, while all the negative powers form the principal part. The analytic part exists even though the original function has an essential singularity at the center; its coefficients are not derivatives of that undefined original value.

Solution 276.4 ■ Answer C[Back to problem](#)

The rational function is holomorphic at zero and has a simple pole at one. The displayed Laurent series is centered at zero but valid only outside the unit circle, so it is not a local punctured-disk expansion at either candidate singularity. Infinitely many negative powers in such an exterior expansion do not establish an essential singularity at one. This is precisely where the center and annulus matter.

Solution 276.5 ■ Answer E[Back to problem](#)

The singularity moduli are 1, 2, 4. They give the four maximal annuli $0 < |z| < 1$, $1 < |z| < 2$, $2 < |z| < 4$, and $|z| > 4$. The innermost expansion has zero principal part and extends to the Taylor series on the full disk $|z| < 1$, including its center.

Complex Variables ▸ The Residue Theorem

277 Residues and residue computation

Solution 277.1 ■ Answer B[Back to problem](#)

The coefficient of z^2 in e^{2z} is $2^2/2! = 2$. Dividing by z^3 turns that into the coefficient of z^{-1} , so the residue is 2.

Solution 277.2 ■ Answer D[Back to problem](#)

Expand $z^2 e^{1/z} = \sum_{n \geq 0} z^{2-n}/n!$. The reciprocal power occurs at $2 - n = -1$, hence $n = 3$, with coefficient $1/3! = 1/6$.

Solution 277.3 ■ Answer C[Back to problem](#)

Differentiate the Laurent series term by term: $c_n(z - a)^n$ becomes $nc_n(z - a)^{n-1}$. A reciprocal term would require $n = 0$, whose derivative coefficient is zero. This works for any isolated singularity, including an essential one. A derivative can still have a genuine pole, as $(-1/z)' = 1/z^2$ shows.

Solution 277.4 ■ Answer D[Back to problem](#)

Logarithmic differentiation gives $F'/F = 4/(z - a) + h'/h$. The second term is holomorphic near a because h is nonzero there. Thus the residue is the zero multiplicity four, independently of the value or derivative of h .

Solution 277.5 ■ **Answer B**[Back to problem](#)

At π , the denominator $\sin z$ has a simple zero with derivative $\cos \pi = -1$. The numerator is also -1 , so the simple-pole residue is $\cos \pi / (\sin z)'|_{z=\pi} = 1$.

Complex Variables ▸ The Residue Theorem

278 Residue theorem and contour evaluation

Solution 278.1 ■ **Answer A**[Back to problem](#)

The enclosed poles are 0 and 1. Their residues are $1/3$ and $-1/2$, summing to $-1/6$. Multiplication by $2\pi i$ gives $-\pi i/3$; the pole at 3 lies outside.

Solution 278.2 ■ **Answer B**[Back to problem](#)

The weighted residue sum is $2 \cdot 3 + (-1) \cdot 5 = 1$. The generalized residue theorem therefore gives $2\pi i$, with the negative winding number accounting for clockwise turning about 2.

Solution 278.3 ■ **Answer C**[Back to problem](#)

Put $z = e^{i\theta}$, so $\cos \theta = (z + z^{-1})/2$ and $d\theta = dz/(iz)$. The integral becomes $\oint_{|z|=1} dz/[i(-2z^2 + 5z - 2)]$. The roots of the quadratic are $1/2$ and two; only the first lies inside. Its residue is $1/[i(-4(1/2) + 5)] = 1/(3i)$. Multiplication by $2\pi i$ gives $2\pi/3$.

Solution 278.4 ■ **Answer B**[Back to problem](#)

The essential singularity at zero does not obstruct the residue theorem. The Laurent expansion $e^{1/z} = 1 + z^{-1} + z^{-2}/2! + \dots$ has residue one, giving $2\pi i$.

Solution 278.5 ■ **Answer C**[Back to problem](#)

The logarithmic derivative has residue three at the zero $z = 1$ and residue minus two at the pole $z = -1$, both inside. The zero at four is outside. The total enclosed residue is $3 - 2 = 1$, so the integral is $2\pi i$. Counting multiplicities avoids expanding the derivative of the rational quotient.

Numerical Analysis ▸ Iterative Methods

279 Iterations, error, tolerance, and convergence rate

Solution 279.1 ■ **Answer C**
[Back to problem](#)

$e_1 = 3(1/10)^2 = 3/100$. Squaring again gives $e_2 = 3(3/100)^2 = 27/10000$. This is quadratic error reduction, not multiplication by a fixed ratio.

Solution 279.2 ■ **Answer C**
[Back to problem](#)

After n iterations the error factor is 4^{-n} . Since $4^4 = 256 < 1000$ and $4^5 = 1024 \geq 1000$, five iterations suffice and four do not.

Solution 279.3 ■ **Answer D**
[Back to problem](#)

The successive error ratio is $2^{-(n+1)^2+n^2} = 2^{-2n-1} \rightarrow 0$. But the ratio to the square of the previous error is $2^{-(n+1)^2+2n^2} = 2^{n^2-2n-1} \rightarrow \infty$. Thus convergence is superlinear in the ratio sense but is not quadratic. Faster than a fixed linear ratio need not mean quadratic.

Solution 279.4 ■ **Answer B**
[Back to problem](#)

Telescope the remaining changes and bound their absolute values by a geometric series: $|\alpha - x_n| \leq (1/1000) \sum_{j=1}^{\infty} 4^{-j} = 1/3000$. Equality is possible when all future changes have the same sign and each is exactly one fourth of its predecessor. Small changes become a certified stopping rule only because a bound on all later changes is supplied.

Solution 279.5 ■ **Answer D**
[Back to problem](#)

Relative error divides the absolute error by the exact magnitude: $(\sqrt{2} - 7/5)/\sqrt{2} = 1 - 7/(5\sqrt{2}) = 1 - 7\sqrt{2}/10$. Dividing by the approximation instead would use a different convention.

Numerical Analysis ▸ Bisection Method

280 Bisection and certified root brackets

Solution 280.1 ■ **Answer C**
[Back to problem](#)

The midpoint error bound is $3/2^{n+1}$. The requirement is $2^{n+1} \geq 3000$. Since $2^{11} = 2048$ and $2^{12} = 4096$, the least n is 11.

Solution 280.2 ■ **Answer D**
[Back to problem](#)

Testing $3/2$ gives a positive value, leaving $[1, 3/2]$. Testing $5/4$ gives $125/64 - 2 < 0$, leaving $[5/4, 3/2]$. Testing $11/8$ gives a positive value, leaving $[5/4, 11/8]$. Its midpoint is $21/16$.

Solution 280.3 ■ Answer A[Back to problem](#)

Every retained interval has its negative endpoint below $\sqrt{2}$ and its positive endpoint at or above it. Its width halves, so the endpoints and midpoints converge to $\sqrt{2}$. But f takes only the values minus one and one. The missing continuity hypothesis, rather than any error in halving the interval, invalidates the root guarantee.

Solution 280.4 ■ Answer C[Back to problem](#)

The initial endpoint signs are negative and positive. At the first midpoint $5/2$, the product is negative, so the retained bracket is $[5/2, 5]$. Its only root is three. Every later retained sign-changing bracket contains that root, and shrinking widths force convergence to it. An initial bracket containing several roots does not make bisection find the smallest one or all of them.

Solution 280.5 ■ Answer C[Back to problem](#)

Continuity and opposite endpoint signs give a root by the intermediate value theorem. At each nonzero midpoint, one half retains opposite endpoint signs and hence a root. Without continuity, a jump can change sign without any zero.

Solution 280.6 ■ Answer D[Back to problem](#)

Choose an unsampled point c in the interval and set $P(x) = \prod_j (x - x_j)$. The polynomial $[1 - P(x)/P(c)]^2$ is nonnegative, equals one at every sample point, and vanishes at c . In contrast, the constant function one has identical sampled values and no zero. A zero can be touched without a sign change, so the bisection criterion is sufficient, not necessary.

Numerical Analysis > Newton Method

281 Newton iteration and multiple roots

Solution 281.1 ■ Answer C[Back to problem](#)

The iteration is $x_{n+1} = (x_n + 3/x_n)/2$. Thus $x_1 = 7/4$, and $x_2 = (7/4 + 12/7)/2 = 97/56$.

Solution 281.2 ■ Answer C[Back to problem](#)

For $x_n \neq 2$, $f/f' = (x_n - 2)/4$, so $x_{n+1} - 2 = (3/4)(x_n - 2)$. The initial error is one, hence the error after n steps is $(3/4)^n$, demonstrating linear convergence at a multiple root.

Solution 281.3 ■ Answer C[Back to problem](#)

Newton's update is $x_{n+1} = x_n - x_n \log x_n$. Thus $x_1 = e - e = 0$, outside the domain of the logarithm. This example shows that Newton's method need not remain in the function's domain.

Solution 281.4 ■ Answer B[Back to problem](#)

The update is $(x_n + a/x_n)/2$. Subtracting $\sqrt{a}$ and putting over the denominator $2x_n$ gives $(x_n^2 - 2x_n\sqrt{a} + a)/(2x_n)$, the stated square. This also shows that the new iterate is at least $\sqrt{a}$.

Solution 281.5 ■ Answer E[Back to problem](#)

Since $f'(x) = 3(x - 1)^2$, the correction is $3(x - 1)^3/[3(x - 1)^2] = x - 1$. Hence $x_{n+1} = 1$. The multiplicity factor eliminates the linear slowdown of ordinary Newton iteration for this polynomial.

Solution 281.6 ■ Answer B[Back to problem](#)

For f , $f(0) = 2$ and $f'(0) = -2$ give the next iterate one; $f(1) = 1$ and $f'(1) = 1$ send it back to zero, repeating a two-cycle. For g , $g(0) = -1$ but $g'(0) = 0$, so division by the derivative is impossible at the initial step. Neither failure contradicts a local theorem near a simple root with a sufficiently close start.

8. Solutions to the Chapter Review Questions

Chapter 5 Review Questions ▸ Question 20: complex matrices

282 Hermitian matrices, conjugate transpose, and the spectral theorem

Solution 282.1 ■ Answer C

[Back to problem](#)

If the eigenvalues are $\lambda_1, \lambda_2, \lambda_3$, their pairwise-product sum is $(6^2 - 14)/2 = 11$. Their characteristic polynomial is $t^3 - 6t^2 + 11t - 6 = (t - 1)(t - 2)(t - 3)$. Thus $H + I$ has eigenvalues 2, 3, 4, and the requested trace is $1/2 + 1/3 + 1/4 = 13/12$.

Solution 282.2 ■ Answer C

[Back to problem](#)

Choose an orthonormal basis whose first vector is $(1, 1, 1)^T/\sqrt{3}$. In that basis the first column of H is zero, and Hermitian symmetry makes the first row zero. The remaining block is $\begin{pmatrix} a & z \\ \bar{z} & b \end{pmatrix}$ with $a, b \in \mathbb{R}$ and $z \in \mathbb{C}$. The trace condition gives $b = -a$, leaving the three real parameters $a, \operatorname{Re} z, \operatorname{Im} z$.

Solution 282.3 ■ Answer B

[Back to problem](#)

In two-by-two block form, the equation forces both diagonal blocks of H to vanish. The upper-right block is an arbitrary complex 2×2 matrix, and the lower-left block is its conjugate transpose. Four complex parameters give eight real dimensions.

Solution 282.4 ■ Answer C

[Back to problem](#)

Directly, $\det(H + iI) = (1 + i)(3 + i) - i(-i) = 1 + 4i$, whose squared modulus is 17. Equivalently, Hermitian eigenvalues are real and the answer is $\det(H^2 + I)$.

Solution 282.5 ■ Answer D

[Back to problem](#)

Conjugate transpose reverses products, so $(HK)^* = KH$. Thus HK is Hermitian only if the factors commute. In contrast, $(HK + KH)^* = HK + KH$, and conjugating the scalar i supplies the minus sign that makes $[i(HK - KH)]^* = i(HK - KH)$. Noncommuting Hermitian matrices provide counterexamples to I.

Solution 282.6 ■ Answer D

[Back to problem](#)

The characteristic polynomial is $(2 - \lambda)(3 - \lambda) - 1 = \lambda^2 - 5\lambda + 5$, so the real eigenvalues are $(5 \pm \sqrt{5})/2$. In an orthonormal eigenbasis the quotient is a weighted average of these two numbers. Its maximum is the larger one, attained at a corresponding eigenvector.

Chapter 6 Review Questions ▸ Question 8: a subgroup-order counterexample

283 Alternating groups and permutation parity

Solution 283.1 ■ **Answer D**[Back to problem](#)

The only possible cycle types are $(2, 4)$ and $(3, 3)$. Both are even because their signs are $(-1)^{6-2} = 1$. Type $(2, 4)$ has $6!/(2 \cdot 4) = 90$ permutations, and type $(3, 3)$ has $6!/(3^2 \cdot 2!) = 40$. The total is 130.

Solution 283.2 ■ **Answer C**[Back to problem](#)

Every element of order 2 must map to 0, because its image has order dividing both 2 and 3. Thus every map kills the normal Klein four subgroup $V = \{e, (12)(34), (13)(24), (14)(23)\}$ and factors through A_4/V , a cyclic group of order 3. A map from this quotient is surjective exactly when a generator maps to 1 or 2, giving two maps.

Solution 283.3 ■ **Answer C**[Back to problem](#)

A subgroup of index two is normal, so A_4/H would have order two. The image of a 3-cycle has order dividing both three and two, hence is the identity. There are $\binom{4}{3} \cdot 2 = 8$ distinct 3-cycles, so H would contain them all, as well as the identity, contradicting $|H| = 6$. Lagrange's divisibility condition is not an existence theorem.

Solution 283.4 ■ **Answer B**[Back to problem](#)

Conjugating relabels a cycle, so $aba^{-1} = (234)$ and $b^{-1} = (142)$. Applying the product $(234)(142)$ sends $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$, giving $(12)(34)$. It is a nonidentity even permutation and lies in the normal Klein four subgroup. Thus A_4 is not Abelian even though its quotient by that subgroup is cyclic.

Solution 283.5 ■ **Answer A**[Back to problem](#)

A 7-cycle is even and has order seven. The partitions that could give larger orders in S_7 are $5 + 2$ (order ten) and $4 + 3$ (order twelve), but each has odd sign. The other even cycle types have order at most seven.

9. Practice Test

Practice Test ▸ Question 26: surfaces of revolution

284 Torus and circular cross sections

Solution 284.1 ■ Answer D

[Back to problem](#)

The torus equation is $(x^2 + y^2 + z^2 + 3)^2 = 16(x^2 + y^2)$. On the sphere it reduces to $64 = 16(x^2 + y^2)$, hence $x^2 + y^2 = 4$ and $z^2 = 1$. The intersection consists of two circles, each of radius 2, so the total length is $2(2\pi \cdot 2) = 8\pi$.

Solution 284.2 ■ Answer B

[Back to problem](#)

The horizontal area is $12\pi\sqrt{4 - z^2}$. Therefore the volume is $12\pi \int_{-1}^1 \sqrt{4 - z^2} dz$. Using the antiderivative $\frac{1}{2}z\sqrt{4 - z^2} + 2\arcsin(z/2)$ gives the integral $\sqrt{3} + 2\pi/3$, and the volume is $12\pi\sqrt{3} + 8\pi^2$.

Solution 284.3 ■ Answer E

[Back to problem](#)

At height one, the meridional circle gives radial distances $5 \pm \sqrt{2^2 - 1^2} = 5 \pm \sqrt{3}$. The annular area is $\pi[(5 + \sqrt{3})^2 - (5 - \sqrt{3})^2] = 20\pi\sqrt{3}$.

Solution 284.4 ■ Answer C

[Back to problem](#)

The generating circumference is $2\pi r = 4\pi$, and its centroid travels a circle of length $2\pi R = 10\pi$. Pappus's surface theorem gives their product, $40\pi^2$. The axis does not intersect the generating circle.

Solution 284.5 ■ Answer B

[Back to problem](#)

Parametrize the surface by $((3 + \cos v)\cos u, (3 + \cos v)\sin u, \sin v)$. The area element is $(3 + \cos v) du dv$, obtained from the magnitude of the cross product of the parameter derivatives. The requested half has $-\pi/2 \leq v \leq \pi/2$. Its area is $\int_0^{2\pi} \int_{-\pi/2}^{\pi/2} (3 + \cos v) dv du = 2\pi(3\pi + 2)$. The outer half has more area than the inner half.

Practice Test ▸ Question 38: a quantitative continuity condition

285 Lipschitz and Holder conditions; constancy from a strong difference bound

Solution 285.1 ■ Answer B

[Back to problem](#)

The exponent-2 bound forces constancy on each interval, so $f = 0$ on $[0, 1]$ and $f = c$ on $[2, 3]$. Taking $x = 1, y = 2$ gives $|c| \leq 1$. The value $c = 1$ is attainable because points in different intervals have distance at least 1; within each interval the difference is zero.

Solution 285.2 ■ Answer B

[Back to problem](#)

For $x \geq y \geq 0$, $(\sqrt{x} - \sqrt{y})^2 \leq x - y$, so $\alpha = 1/2$ works with $C = 1$. Taking $y = 0$ would require $x^{1/2-\alpha} \leq C$ for all positive $x \leq 4$. If $\alpha > 1/2$, the left side diverges as $x \downarrow 0$, so no finite C can work.

Solution 285.3 ■ Answer D

[Back to problem](#)

Divide any interval $[x, y]$ into n equal pieces. The triangle inequality bounds the total change by $(y - x)/\log(en/(y - x))$, which tends to zero. Therefore $f(x) = f(y)$ for every pair of points.

Solution 285.4 ■ Answer A

[Back to problem](#)

The derivative is $f'(x) = 3\sqrt{x}/2$, whose maximum on the interval is 3. The mean value theorem makes 3 a valid Lipschitz constant; difference quotients approaching $x = 4$ tend to 3, so no smaller constant works.

Solution 285.5 ■ Answer E

[Back to problem](#)

The inequalities give $|f(g(x)) - f(g(y))| \leq 2|g(x) - g(y)| \leq 6|x - y|$. Equality is attained by $f(t) = 2t$ and $g(t) = 3t$ on $\mathbb{R}$, so six is optimal.

Solution 285.6 ■ Answer B

[Back to problem](#)

The inequality $|f(x)| \leq x$ gives continuity at zero. For $x > 0$, $f'(x) = \sin(\log x) + \cos(\log x)$ has absolute value at most $\sqrt{2}$. The mean-value theorem on any subinterval, including one with endpoint zero, yields a Lipschitz bound. But $[f(x) - f(0)]/x = \sin(\log x)$ oscillates without a limit as $x \downarrow 0$, so no right derivative exists.

Practice Test ▸ Question 57: Cauchy mean-value formula

286 Cauchy mean-value theorem and simultaneous derivative ratios

Solution 286.1 ■ **Answer A**[Back to problem](#)

The ordinary equations are $2c_f = 1$ and $3c_g^2 = 1$, giving $c_f = 1/2$ and $c_g = 1/\sqrt{3}$. The Cauchy equation is $3c_C^2 \cdot 1 = 2c_C \cdot 1$. Its only solution in $(0, 1)$ is $c_C = 2/3$. The distinct points show why separate ordinary applications cannot justify a shared point.

Solution 286.2 ■ **Answer C**[Back to problem](#)

The quotient of endpoint differences is $\log 4/(1/4 - 1) = -4 \log 4/3$. The derivative quotient is $(1/c)/(-1/c^2) = -c$, so $c = 4 \log 4/3$. It lies in $(1, 4)$; the theorem guarantees this and the elementary inequalities $3/4 < \log 4 < 3$ also show it.

Solution 286.3 ■ **Answer B**[Back to problem](#)

The endpoint differences are $\Delta f = -2$ and $\Delta g = 0$. The valid cross-multiplied equation is $(-2)g'(c) = 0$, so $\sin(2c) = 0$. Its only solution in $(0, \pi)$ is $c = \pi/2$. At that point $g'(c) = 0$ and $f'(c) = -1$, so division by $g'(c)$ is invalid. Equal endpoint values of g do not invalidate the cross-multiplied theorem.

Solution 286.4 ■ **Answer A**[Back to problem](#)

For each $x > 0$, Cauchy's theorem gives $f(x)/g(x) = f'(c)/g'(c)$ for some $0 < c < x$. Strict increase makes this less than $f'(x)/g'(x)$. Also $g(x) > 0$. Differentiating the quotient yields $(f/g)' = (g'/g)[f'/g' - f/g] > 0$, proving the claim.

Solution 286.5 ■ **Answer A**[Back to problem](#)

Since $g' > 0$, its secant denominator is positive. The Cauchy mean-value theorem identifies the secant ratio with $f'(c)/g'(c)$ at some interior point, so it obeys the same two bounds.

Practice Test ▸ Question 58: a topology with finite complements

287 Cofinite topology and nonunique limits

Solution 287.1 ■ **Answer E**[Back to problem](#)

For any $x \in \mathbb{R}$, a neighborhood contains all but finitely many real numbers and therefore all but finitely many terms of the distinct sequence $1, 2, 3, \dots$. Thus $x_n \rightarrow x$ for every real x . The set of limits is all of $\mathbb{R}$, which is uncountable.

Solution 287.2 ■ **Answer C**[Back to problem](#)

A nonconstant map is continuous here exactly when each point inverse is finite: inverse images of finite closed sets are then finite unions of finite sets, and the converse follows from closed singletons. Fibers of x^2 have at most two points and those of e^x at most one. But $\sin^{-1}(\{0\}) = \pi\mathbb{Z}$ is infinite and proper, hence is not closed in the domain.

Solution 287.3 ■ **Answer E**[Back to problem](#)

A cofinite neighborhood of zero excludes only finitely many nonzero values, so eventually it contains both the even terms and all odd terms. For any $a \neq 0$, the neighborhood $\mathbb{R} \setminus \{0\}$ contains a but excludes infinitely many even terms, preventing convergence to a .

Solution 287.4 ■ **Answer E**[Back to problem](#)

Every subset inherits the cofinite topology (which is discrete when the subset is finite). In an open cover of a nonempty subset, one nonempty member leaves only finitely many uncovered points, each covered by one further member. Thus a finite subcover always exists.

Solution 287.5 ■ **Answer E**[Back to problem](#)

Any two nonempty open subsets of an infinite cofinite space intersect. If the map took two distinct values, disjoint Hausdorff neighborhoods of those values would have disjoint nonempty open preimages, a contradiction. Hence the map is constant.

10. Practice Test Answers and Explanations

Practice Test Answers and Explanations ▸ Question 46: Pascal triangle

288 Pascal triangle and binomial sum identities

Solution 288.1 ■ Answer D

[Back to problem](#)

Use $\binom{6}{k} = \binom{6}{6-k}$. The sum is the coefficient of x^6 in $(1+x)^6(1+x)^6 = (1+x)^{12}$, namely $\binom{12}{6} = 924$. Combinatorially, choose six people from two groups of six and sum over how many are chosen from the first group.

Solution 288.2 ■ Answer C

[Back to problem](#)

Write $k^2 = k(k-1) + k$. The two weighted binomial identities give $8 \cdot 7 \cdot 2^6 + 8 \cdot 2^7 = 3584 + 1024 = 4608$. Equivalently the general formula is $n(n+1)2^{n-2}$, which at $n = 8$ gives $8 \cdot 9 \cdot 64$.

Solution 288.3 ■ Answer D

[Back to problem](#)

The sum is the coefficient of x^8 in $(1-x)^8(1+x)^8 = (1-x^2)^8$. That coefficient is $(-1)^4 \binom{8}{4} = 70$.

Solution 288.4 ■ Answer B

[Back to problem](#)

Work modulo two. Repeated squaring gives $(1+x)^8 = 1+x^8$ and $(1+x)^2 = 1+x^2$, so $(1+x)^{10} = (1+x^8)(1+x^2) = 1+x^2+x^8+x^{10}$ modulo two. Exactly four coefficients are odd. This uses the binary decomposition $10 = 8 + 2$, rather than differentiating a row sum.

Solution 288.5 ■ Answer C

[Back to problem](#)

Integrate $(1+x)^6 = \sum_{k=0}^6 \binom{6}{k} x^k$ from zero to one. Termwise integration gives the desired reciprocal weights, while the left side integrates to $[(1+x)^7/7]_0^1 = (2^7 - 1)/7 = 127/7$. This complements differentiation for polynomial weights.

Solution 288.6 ■ Answer D

[Back to problem](#)

Write the weight $10 - k$ as the number of integers j from k to nine. Interchange the two finite sums to obtain $\sum_{j=3}^9 \sum_{k=3}^j \binom{k}{3}$. The inner hockey-stick sum is $\binom{j+1}{4}$; applying the identity again gives $\binom{11}{5} = 462$. The extra weight requires a second counting or summation layer.

Supplement. Breadth beyond the book

Multivariable calculus

289 Jacobians and multivariable change of variables

Solution 289.1 ■ Answer B

[Back to problem](#)

Put $u = x + y$, $v = x - y$. Then $x^2 - y^2 = uv$ and $|\partial(x, y)/\partial(u, v)| = 1/2$. The integral is $\frac{1}{2}(\int_0^2 u \, du)(\int_0^1 v \, dv) = 1/2$.

Solution 289.2 ■ Answer C

[Back to problem](#)

The determinant is $4(u^2 + v^2) = 4r^2$. Polar integration gives $\int_0^\pi \int_0^1 4r^3 \, dr \, d\theta = \pi$. On the half disk the angle doubles through a full turn exactly once; using the full disk would count the image twice.

Solution 289.3 ■ Answer C

[Back to problem](#)

Use $u = xy$ and $v = y/x$. The inverse is $x = \sqrt{u/v}$, $y = \sqrt{uv}$, with absolute Jacobian $1/(2v)$. Thus the area is $\int_1^4 \int_1^9 (2v)^{-1} \, dv \, du = (3/2) \log 9 = 3 \log 3$.

Solution 289.4 ■ Answer D

[Back to problem](#)

Set $x = 2r \cos \theta$, $y = 3r \sin \theta$. The integrand becomes r^2 and the area Jacobian is $6r$. The integral is $6 \int_0^{2\pi} \int_0^1 r^3 \, dr \, d\theta = 3\pi$.

Solution 289.5 ■ Answer B

[Back to problem](#)

The Jacobian determinant is $e^{2u}(\cos^2 v + \sin^2 v) = e^{2u} > 0$, so the inverse function theorem applies at every point. But $T(u, v + 2\pi) = T(u, v)$, so global injectivity fails. Its image is the punctured plane. Nonzero Jacobian is a local criterion.

Multivariable calculus

290 Triple integrals and solid coordinate systems

Solution 290.1 ■ Answer A

[Back to problem](#)

The cone requires $0 \leq \phi \leq \pi/4$. Thus the integral is $\int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^4 \sin \phi \, d\rho \, d\phi \, d\theta = \frac{2\pi}{5}(1 - 1/\sqrt{2})$.

Solution 290.2 ■ Answer C

[Back to problem](#)

Integrating first in z gives $\frac{1}{2} \int_0^1 \int_0^{1-x} xy(1-x-y)^2 \, dy \, dx$. The inner integral is $x(1-x)^4/24$. Finally $\int_0^1 x(1-x)^4 \, dx = 1/30$, so the result is $1/720$.

Solution 290.3 ■ Answer C[Back to problem](#)

By rotational symmetry, the integral is one third of $\iiint r^2 dV$. In spherical coordinates this is $(1/3)4\pi \int_0^2 r^4 dr = 128\pi/15$.

Solution 290.4 ■ Answer D[Back to problem](#)

In cylindrical coordinates $0 \leq r \leq 2$, and the vertical height is $4 - r^2$. Thus the volume is $2\pi \int_0^2 (4 - r^2)r dr = 8\pi$.

Solution 290.5 ■ Answer C[Back to problem](#)

This tetrahedron has coordinate intercepts 2, 3, 4, so its volume is one sixth of their product: $2 \cdot 3 \cdot 4/6 = 4$. Equivalently, scaling the standard simplex contributes Jacobian 24 to its volume $1/6$.

Multivariable calculus

291 Surface area and flux integrals

Solution 291.1 ■ Answer B[Back to problem](#)

The upward vector area element is $(-2x, -2y, 1) dx dy$. Its dot product with $\mathbf{F}$ on the surface is $-x^2 - y^2$. Thus the flux is $-\int_0^{2\pi} \int_0^1 r^3 dr d\theta = -\pi/2$.

Solution 291.2 ■ Answer C[Back to problem](#)

The area integral is $2\pi \int_0^1 r\sqrt{1+4r^2} dr$. With $u = 1 + 4r^2$ it becomes $\frac{\pi}{4} \int_1^5 u^{1/2} du = \frac{\pi}{6}(5\sqrt{5} - 1)$.

Solution 291.3 ■ Answer C[Back to problem](#)

The parameter derivatives are $(\cos v, \sin v, 0)$ and $(-u \sin v, u \cos v, 1)$. Their cross product is $(\sin v, -\cos v, u)$, with norm $\sqrt{1+u^2}$. Thus the area is $2\pi \int_0^1 \sqrt{1+u^2} du = \pi(\sqrt{2} + \log(1 + \sqrt{2}))$. The nonzero norm also verifies regularity, including along $u = 0$.

Solution 291.4 ■ Answer C[Back to problem](#)

In spherical coordinates the polar angle runs from zero to $\pi/3$, and $dS = 4 \sin \phi d\phi d\theta$. Integrating gives area $8\pi(1 - 1/2) = 4\pi$. The outward unit normal has vertical component $\cos \phi$, so the flux is $8\pi \int_0^{\pi/3} \cos \phi \sin \phi d\phi = 3\pi$. Flux uses the direction as well as the scalar area element.

Solution 291.5 ■ Answer C[Back to problem](#)

Use orthonormal tangent vectors $u = (1, -1, 0)/\sqrt{2}$ and $v = (1, 1, -2)/\sqrt{6}$ on the original plane. Their transformed vectors have cross product $(24, 16, 12)/\sqrt{12}$, whose norm is $\sqrt{976/12} = 2\sqrt{61/3}$. This constant area factor applies to the whole patch. The determinant 24 is a three-dimensional volume factor, not this two-dimensional area factor.

Multivariable calculus

292 Divergence, curl, and differential identities

Solution 292.1 ■ **Answer D**
[Back to problem](#)

The divergence is $ay + 2y + x = (a + 2)y + x$, independent of b . Thus $a = -2$ and b is arbitrary; a condition on curl was not requested.

Solution 292.2 ■ **Answer A**
[Back to problem](#)

For $\mathbf{F} = (P, Q, R)$, the first component of the left side is $Q_{xy} + R_{xz} - P_{yy} - P_{zz}$. The first component of $\nabla(\nabla \cdot \mathbf{F}) - \Delta \mathbf{F}$ is the same after the P_{xx} terms cancel. The other components follow cyclically. Continuous second partials justify the interchanges.

Solution 292.3 ■ **Answer B**
[Back to problem](#)

Each component is independent of its own differentiation variable, so divergence is zero. Moreover $\mathbf{F} = \nabla(xyz)$, so its curl is zero; direct cross-partial differentiation gives the same result.

Solution 292.4 ■ **Answer A**
[Back to problem](#)

The divergence of a curl is zero because mixed partial derivatives cancel. The other fields can have nonzero divergence; for example, $\nabla \cdot \nabla f = \Delta f$ need not vanish.

Solution 292.5 ■ **Answer B**
[Back to problem](#)

Here $\phi \mathbf{F} = (-yz, xz, 0)$. Its curl is $(0 - \partial_z(xz), \partial_z(-yz) - 0, \partial_x(xz) - \partial_y(-yz)) = (-x, -y, 2z)$.

Multivariable calculus

293 The divergence theorem

Solution 293.1 ■ **Answer C**
[Back to problem](#)

The divergence is $3(x^2 + y^2 + z^2)$. Integrating over the ball gives $3 \cdot 4\pi \int_0^1 \rho^4 d\rho = 12\pi/5$.

Solution 293.2 ■ **Answer C**
[Back to problem](#)

Close the hemisphere with the disk at $z = 0$. Divergence is 3, so total flux is $3(2\pi/3) = 2\pi$. The outward normal of the disk is downward and its flux is $-\pi$. Hence the curved hemisphere contributes $2\pi - (-\pi) = 3\pi$.

Solution 293.3 ■ Answer C[Back to problem](#)

On the unit sphere the field equals the outward unit normal, so the integrand is one and the flux is the area 4π . The field is smooth on the surface, so the surface integral exists. Its singularity at the origin prevents applying the divergence theorem directly to the full ball; removing the origin requires an inner boundary contribution.

Solution 293.4 ■ Answer D[Back to problem](#)

The divergence theorem makes the integral of the divergence over each such ball zero. If the continuous divergence were positive at a point, it would remain positive on a sufficiently small ball, giving a positive integral; the negative case is analogous. Thus it vanishes pointwise. A rotational field such as $(-y, x, 0)$ shows that curl need not vanish.

Solution 293.5 ■ Answer D[Back to problem](#)

The field is smooth and has divergence zero throughout the shell, so its total boundary flux is zero. Directly, the outer sphere contributes 4π and the inner sphere, whose shell-outward normal points toward the origin, contributes -4π .

Multivariable calculus

294 Stokes's theorem

Solution 294.1 ■ Answer E[Back to problem](#)

The curl is $(-1, -1, 1)$. Use the disk in the plane $z = x + y$ with upward vector area element $(-1, -1, 1) dx dy$. Their dot product is 3, giving 3π .

Solution 294.2 ■ Answer A[Back to problem](#)

Replace S by the upward unit disk. The curl is $(-3z^2, -3x^2, -3y^2)$, so its flux is $-3 \iint_D y^2 dA = -3\pi/4$. Both surfaces induce the same counterclockwise boundary orientation.

Solution 294.3 ■ Answer C[Back to problem](#)

If such $\mathbf{A}$ existed, it would be defined on a neighborhood of the sphere. Stokes's theorem on this closed surface, whose boundary is empty, would give zero curl flux, contradicting the given value one. Divergence-free is a local necessary condition for a vector potential, but this domain has a global flux obstruction. Applying the divergence theorem to the missing-origin ball would be invalid.

Solution 294.4 ■ Answer B[Back to problem](#)

The curl is $(0, 0, 2)$. Clockwise boundary orientation corresponds to a downward disk normal, so the integral is -2 times the disk area 4π , yielding -8π .

Solution 294.5 ■ Answer D[Back to problem](#)

A closed surface has empty boundary, so Stokes's theorem gives zero. Equivalently, the divergence theorem and the identity $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ yield the same conclusion.

Differential equations

295 Existence, uniqueness, and maximal solution intervals

Solution 295.1 ■ Answer C[Back to problem](#)

The equation implies $|y'| \geq 1$. Continuity of y' prevents its sign from changing on a connected interval, so there are two branches: $y' = \sqrt{1 + y^2}$ or $y' = -\sqrt{1 + y^2}$. Each branch has a smooth right side and a unique local solution, namely $y = \sinh t$ or $y = -\sinh t$. Uniqueness for one explicit normal form does not choose between two implicit derivative branches.

Solution 295.2 ■ Answer B[Back to problem](#)

Separation gives $y = (1 - 2t)^{-1/2}$ on the branch fixed by the positive initial value. It blows up at $t = 1/2$, so the maximal interval through 0 is $(-\infty, 1/2)$. A smooth polynomial right side guarantees local uniqueness, but does not prevent finite-time blowup.

Solution 295.3 ■ Answer D[Back to problem](#)

Local uniqueness forces the zero solution on the connected time interval $(-1, 1)$. Although the function zero is bounded and can be written at every real time, the specified differential equation is undefined at $t = \pm 1$. A solution interval containing zero cannot cross those times. A finite maximal endpoint need not result from blowup of the solution itself.

Solution 295.4 ■ Answer C[Back to problem](#)

For $f(t, y) = \sin(t + y^2)$, both f and $f_y = 2y \cos(t + y^2)$ are continuous near $(0, 0)$. The fractional powers lack a continuous y derivative there, and the reciprocal fields are not even defined at the initial point.

Solution 295.5 ■ Answer D[Back to problem](#)

Define $G(y) = \int_0^y [1 + \sqrt{|s|}]^{-1} ds$. It is continuously differentiable with positive derivative, so has a differentiable local inverse. Every solution satisfies $\frac{d}{dt}G(y(t)) = 1$ and the initial condition gives $G(y(t)) = t$. Hence every solution must be the same inverse function $G^{-1}(t)$, which also directly supplies existence. No waiting at zero is possible because the right side is one there.

Differential equations

296 Autonomous equations and phase-line stability

Solution 296.1 ■ **Answer B**[Back to problem](#)

The sign alternates across the simple roots: negative below 0, positive on $(0, 1)$, negative on $(1, 2)$, and positive above 2. Arrows point toward 1 from both sides and away from 0 and 2.

Solution 296.2 ■ **Answer C**[Back to problem](#)

For $y < 0$, the derivative is positive. Uniqueness prevents crossing the equilibrium 0, so the solution increases and is bounded above. It exists for all future time while remaining in $[-1, 0]$. Its limit must be an equilibrium, hence is 0. The equilibrium is attracting from the left but repelling from the right.

Solution 296.3 ■ **Answer B**[Back to problem](#)

For the first equation, the derivative points toward zero on either side. Solutions remain between their initial value and zero and converge to zero, so the equilibrium is asymptotically stable. For the second, the arrows point away on both sides; any nonzero initial value eventually leaves a fixed small neighborhood. Both linearized derivatives vanish, so the sign analysis supplies information that linearization does not.

Solution 296.4 ■ **Answer C**[Back to problem](#)

Every initial value sufficiently near zero lies in $[-1, 1]$, and uniqueness makes its solution constant. Such solutions remain arbitrarily close to zero, giving stability. A nonzero constant solution, however close its initial value, never converges to zero, so asymptotic stability fails. The behavior outside this interval is irrelevant.

Solution 296.5 ■ **Answer D**[Back to problem](#)

A periodic differentiable solution attains a maximum at some time t_0 , where $y'(t_0) = 0$. Thus its value $a = y(t_0)$ satisfies $f(a) = 0$. The constant solution $y = a$ passes through the same initial value. Local uniqueness, continued along the connected time domain, forces the proposed periodic solution to be that constant. This is the scalar autonomous no-crossing principle in a global form.

Differential equations

297 Euler-Cauchy differential equations

Solution 297.1 ■ **Answer B**
[Back to problem](#)

The indicial polynomial is $m(m-1) - 3m + 4 = (m-2)^2$. Thus $y = x^2(C_1 + C_2 \log x)$. The initial value gives $C_1 = 0$, and the derivative at 1 gives $C_2 = 1$.

Solution 297.2 ■ **Answer B**
[Back to problem](#)

Try $y = Ax$. The three terms give $0 + Ax + Ax = 2Ax$, so $A = 1/2$. Under $t = \log x$, the equation becomes $Y'' + Y = e^t$, which gives the same particular solution. No resonance occurs because 1 is not a root of $m^2 + 1$.

Solution 297.3 ■ **Answer E**
[Back to problem](#)

The indicial equation is $m(m-1) - 2 = (m-2)(m+1) = 0$. In particular, $y = Cx^2$ is a twice continuously differentiable solution on the whole interval for every real C , and each satisfies the two initial conditions. Thus there are infinitely many. Dividing by x^2 produces a coefficient singular at the initial point, so the ordinary uniqueness theorem cannot be applied there.

Solution 297.4 ■ **Answer E**
[Back to problem](#)

The indicial equation is $r^2 + 4 = 0$, so the real solution is $C_1 \cos(2 \log x) + C_2 \sin(2 \log x)$. At $x = 1$ the conditions give $C_1 = 1$ and $2C_2 = 0$.

Solution 297.5 ■ **Answer B**
[Back to problem](#)

The homogeneous indicial polynomial is $(r-1)(r-2)$, so x^2 is resonant. For $y = x^2 \log x$, $y' = 2x \log x + x$ and $y'' = 2 \log x + 3$. Substitution gives exactly x^2 .

Differential equations

298 Linear differential systems and matrix exponentials

Solution 298.1 ■ **Answer B**
[Back to problem](#)

First $y = e^{2t}$. Then $(e^{-2t}x)' = 1$, and $x(0) = 0$ gives $x = te^{2t}$. The factor t reflects the non-trivial repeated-eigenvalue block.

Solution 298.2 ■ **Answer B**
[Back to problem](#)

The characteristic polynomial is $\lambda^2 + 2\lambda + 5$, with roots $-1 \pm 2i$. Both real parts are negative, so every solution is an exponentially decaying linear combination of oscillatory terms. The oscillations do not make the trajectories periodic because their amplitudes decay.

Solution 298.3 ■ Answer B[Back to problem](#)

Since $A^2 = -4I$, its exponential is $I \cos(2t) + (A/2) \sin(2t)$. Acting on $(1, 0)^T$ gives $(\cos 2t, \sin 2t)^T$, with the stated first component.

Solution 298.4 ■ Answer C[Back to problem](#)

For an arbitrary fundamental matrix, write $v(t) = \Phi(t)c$ and solve $\Phi(0)c = v(0)$. Here $\Phi(0) = \text{diag}(1, 2)$, so $c = (1, 1)^T$. The normalized propagator is $\Phi(t)\Phi(0)^{-1}$; multiplying directly by the initial vector would be valid only if $\Phi(0) = I$.

Solution 298.5 ■ Answer C[Back to problem](#)

The characteristic polynomial is $\lambda^2 - 4$, so the distinct real eigenvalues are 2 and -2 . Only initial vectors in the one-dimensional -2 eigenspace have decaying solutions; any component in the 2 eigenspace grows.

Solution 298.6 ■ Answer C[Back to problem](#)

Since $N^2 = 0$, $e^{tN} = I + tN$. Initial data $(0, \varepsilon)^T$ yield $(t\varepsilon, \varepsilon)^T$. No matter how small nonzero ε is, this trajectory eventually leaves a fixed neighborhood of zero. The nontrivial Jordan block on the imaginary axis creates polynomial growth; eigenvalue real parts alone do not settle this boundary case.

Linear algebra

299 Inner products, orthogonality, and Gram-Schmidt

Solution 299.1 ■ Answer C[Back to problem](#)

Write $p = x^2 + ax + b$. Orthogonality to x gives $a \int_{-1}^1 x^2 = 0$, so $a = 0$. Orthogonality to 1 gives $2/3 + 2b = 0$, hence $b = -1/3$.

Solution 299.2 ■ Answer D[Back to problem](#)

A normal to $\text{span}(v_1, v_2)$ is $(1, -1, -1)$, of squared norm 3. The residual of v_3 is its projection onto that normal, with squared norm $((-2)^2)/3 = 4/3$. This avoids computing the first two normalized vectors.

Solution 299.3 ■ Answer C[Back to problem](#)

The zero-mean condition gives $\int x f = \int (x - 1/2)f$. Cauchy-Schwarz bounds this by $\sqrt{\int_0^1 (x - 1/2)^2} \sqrt{\int_0^1 f^2} = 1/\sqrt{12}$. Equality is attained by $f(x) = \sqrt{12}(x - 1/2)$, which meets both constraints. Centering first improves the bound obtained from x alone.

Solution 299.4 ■ **Answer B**[Back to problem](#)

The coefficient is $\langle v, w \rangle / \langle v, v \rangle = (1 - i)/2$. Multiplication by $(1, i)$ gives the stated vector. The residual is $((1 + i)/2, (1 - i)/2)$, whose inner product with v is zero. Conjugation belongs in the first argument under the specified convention.

Solution 299.5 ■ **Answer B**[Back to problem](#)

The matrix is symmetric. Positive definiteness requires its leading principal minors to be positive: $1 > 0$ and $4 - t^2 > 0$. Thus exactly $|t| < 2$ gives an inner product; equality leaves a nonzero vector of zero squared norm.

Solution 299.6 ■ **Answer D**[Back to problem](#)

The first two vectors are independent. The third is their sum, so its residual is zero and adds no dimension. Their span consists of $(a, b, a + b)$, which does not contain $(1, 1, 0)$. The fourth therefore has a nonzero residual and adds a third orthonormal vector. A dependent intermediate vector does not prevent later vectors from enlarging the span.

Linear algebra

300 Orthogonal projections and least squares

Solution 300.1 ■ **Answer C**[Back to problem](#)

The normal equations are $3a + 3b = 3$ and $3a + 5b = 4$. Subtraction gives $2b = 1$. Therefore the slope is $1/2$ and the intercept is $1/2$.

Solution 300.2 ■ **Answer C**[Back to problem](#)

The space has dimension 3, so the projection has three eigenvalues 1 and one 0, giving trace 3. Since $Pe_1 = e_1 - \frac{1}{4}(1, 1, 1, 1)$, its squared norm is $9/16 + 3/16 = 3/4$.

Solution 300.3 ■ **Answer B**[Back to problem](#)

Put $s = x_1 + x_2$. The squared residual is $(s - 1)^2 + 4s^2 + 1 = 5(s - 1/5)^2 + 9/5$. Thus the minimizing sum is uniquely $1/5$, but its two coefficients are not unique. Every minimizer gives the same fitted vector $(1/5, 2/5, 0)^T$. Singular normal equations can have many solutions rather than no solution.

Solution 300.4 ■ **Answer D**[Back to problem](#)

An orthogonal projection is contractive by the orthogonal decomposition and the Pythagorean theorem. Conversely, let $u \in \text{im } P$ and $v \in \text{ker } P$. Contractivity applied to $u + tv$ gives $\|u\|^2 \leq \|u + tv\|^2$ for every real t . The linear coefficient $2\langle u, v \rangle$ must vanish, so range and kernel are orthogonal. The other listed spectral properties do not distinguish oblique idempotents.

Solution 300.5 ■ **Answer D**[Back to problem](#)

At the minimum, the residual is orthogonal to every column of A , so $A^T(Ax - b) = 0$. Independent columns make $A^T A$ invertible, yielding $x = (A^T A)^{-1} A^T b$. The other formulas either omit the Gram matrix or generally have incompatible dimensions.

Linear algebra

301 Change of basis, similarity, and diagonalization

Solution 301.1 ■ **Answer B**[Back to problem](#)

The first basis vector is a 2-eigenvector and the second is a 3-eigenvector: $T(1, 1) = (3, 3)$. The columns in the new coordinates are therefore $(2, 0)$ and $(0, 3)$.

Solution 301.2 ■ **Answer B**[Back to problem](#)

The annihilating polynomial has distinct roots, so A is diagonalizable with eigenvalues in $\{1, 2\}$. If k eigenvalues equal 2, its trace is $4 + k = 6$, hence $k = 2$. The determinant is $1^2 \cdot 2^2 = 4$.

Solution 301.3 ■ **Answer D**[Back to problem](#)

The images of the basis vectors are $(2, 3)$ and $(2, -3)$. Their coordinates in this basis are $(5/2, -1/2)$ and $(-1/2, 5/2)$, respectively. These coordinate columns form the stated matrix.

Solution 301.4 ■ **Answer C**[Back to problem](#)

Rank-nullity gives $\dim \ker(A - I) = 5 - 3 = 2$, while the algebraic multiplicity of one is three. This shortfall prevents an eigenbasis, regardless of the eigenspace dimension for two. The determinant is $1^3 2^2 = 4$, so singularity is not the issue.

Solution 301.5 ■ **Answer E**[Back to problem](#)

Similarity preserves polynomial identities and hence the minimal polynomial. Entries, eigenvectors, and coordinate bases can change under similarity. Different coordinate matrices of the same operator are precisely related by a basis-change similarity.

Linear algebra

302 Rank inequalities and dimensions of intersections

Solution 302.1 ■ **Answer C**[Back to problem](#)

$\text{im } B$ has dimension 7. Its intersection with the 4-dimensional kernel of A can have any dimension from 1 to 4. Thus the rank of AB ranges from 6 down to 3, giving nullities 4 through 7. Coordinate subspaces realize each intersection.

Solution 302.2 ■ **Answer B**[Back to problem](#)

The intersection has dimension $6 + 5 - 9 = 2$. A nonzero linear functional has rank 1, so rank-nullity gives kernel dimension 1.

Solution 302.3 ■ **Answer B**[Back to problem](#)

The precise rank formula gives $\dim(\text{im } B \cap \ker A) = n - \text{rank } A = \dim \ker A$. Since the intersection is a subspace of $\ker A$ with its full dimension, it equals $\ker A$. Thus the image of B contains every direction annihilated by A . The other conclusions need not hold; for example, take $A = \text{diag}(1, 0)$ and $B = \text{diag}(0, 1)$.

Solution 302.4 ■ **Answer C**[Back to problem](#)

Each subspace has codimension two, so the common intersection has codimension at most six and dimension at least one. Equality is attained by imposing disjoint pairs of coordinate-zero conditions on the first six coordinates, leaving only the seventh coordinate free.

Solution 302.5 ■ **Answer B**[Back to problem](#)

Restrict T to U . Its kernel is exactly $U \cap \ker T$, so rank-nullity on the seven-dimensional domain gives $\dim T(U) = 7 - 3 = 4$.

Linear algebra

303 Quadratic forms and positive definiteness

Solution 303.1 ■ **Answer A**[Back to problem](#)

The symmetric matrix is $\begin{pmatrix} 1 & t & 0 \\ t & 2 & t \\ 0 & t & 1 \end{pmatrix}$. Its leading minors are 1, $2 - t^2$, and $2 - 2t^2$. All are positive exactly when $|t| < 1$. At $|t| = 1$ there is a nonzero null vector.

Solution 303.2 ■ **Answer E**[Back to problem](#)

The symmetric matrix $\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$ has eigenvalues 5 and 1, with orthogonal eigenvectors $(1, 1)$ and $(1, -1)$. The unit-sphere extrema are therefore 5 and 1, giving ratio 5.

Solution 303.3 ■ **Answer D**[Back to problem](#)

The leading minors of $\text{diag}(0, -1)$ are both zero, but its quadratic form is $-y^2$, which is negative for $(0, 1)$. For positive semidefiniteness, all principal minors must be nonnegative, including the lower-right one-by-one minor. The strictly positive leading-minor criterion for positive definiteness cannot simply be weakened to nonnegative leading minors.

Solution 303.4 ■ **Answer B**[Back to problem](#)

The x, y block has eigenvalues 3 and -1 , while the z coefficient is -1 . Sylvester's law of inertia preserves these sign counts under invertible changes of variables, giving one positive and two negative squares.

Solution 303.5 ■ **Answer C**[Back to problem](#)

On the unit sphere the form equals $1 + 3z^2$. The additional constraint gives $z^2 = (x + y)^2 \leq 2(x^2 + y^2) = 2(1 - z^2)$, so $z^2 \leq 2/3$ and the maximum is three. Equality is attained at $(x, y, z) = (1, 1, -2)/\sqrt{6}$. The full-space largest eigenvalue four is not attainable within the required plane.

Linear algebra

304 Minimal polynomials and nilpotent Jordan structure

Solution 304.1 ■ **Answer C**[Back to problem](#)

The nullities show $A^2 \neq 0$ but $A^3 = 0$, so the nilpotency index is 3 and $m_A = t^3$. The successive increments 3, 2, 1 also give block sizes 3, 2, 1.

Solution 304.2 ■ **Answer C**[Back to problem](#)

The 1-blocks have total size 4, with largest size exactly 2. To maximize their number, choose sizes 2, 1, 1, giving three independent 1-eigenvectors. Four size-1 blocks would contradict the square factor in the minimal polynomial. The -2 blocks do not affect this eigenspace.

Solution 304.3 ■ **Answer B**[Back to problem](#)

The identity $A^3 - 2A + I = 0$ gives $A(2I - A^2) = I = (2I - A^2)A$. Thus A is invertible and its inverse is $2I - A^2$. The nonzero constant term permits recovering an inverse as a polynomial in the matrix.

Solution 304.4 ■ **Answer C**[Back to problem](#)

Write $A = I + N$ with $N \neq 0$ and $N^2 = 0$. In characteristic two, $A^2 = I + 2N + N^2 = I$. Nevertheless, the minimal polynomial is exactly $(t - 1)^2$, so there is a nontrivial Jordan block. Here $t^2 - 1 = (t - 1)^2$ has a repeated root; the distinct-root argument valid in characteristic zero does not apply.

Solution 304.5 ■ **Answer C**[Back to problem](#)

The polynomial identity gives $A^3 = A^2$, hence $A^4 = A^2$, so A^2 is idempotent. The matrix A itself cannot be idempotent, because then its minimal polynomial would divide $t(t - 1)$, contrary to the given polynomial.

Abstract algebra

305 Quotient groups and the first isomorphism theorem

Solution 305.1 ■ Answer E

[Back to problem](#)

The subgroup is $\{(0, 0), (4, 6), (8, 12)\}$. Membership of (k, k) requires $k \equiv 4t \pmod{12}$ and $k \equiv 6t \pmod{18}$ for one $t \in \{0, 1, 2\}$. Compatibility modulo 6 requires $4t \equiv 0 \pmod{6}$, forcing $t = 0$. Thus k must be a multiple of both 12 and 18, whose least positive value is 36.

Solution 305.2 ■ Answer C

[Back to problem](#)

The image must have order $24/8 = 3$. A map from a cyclic group is determined by the image of 1, which must have order exactly 3. The two elements of order 3 in $\mathbb{Z}/18\mathbb{Z}$ are 6 and 12, and both satisfy $24a = 0$.

Solution 305.3 ■ Answer A

[Back to problem](#)

The image is $3\mathbb{Z}$, since the greatest common divisor of 6, 9 is three. The first isomorphism theorem gives the quotient isomorphic to $3\mathbb{Z}$, which as an abstract additive group is isomorphic to $\mathbb{Z}$, not to $\mathbb{Z}/3\mathbb{Z}$.

Solution 305.4 ■ Answer C

[Back to problem](#)

If the proposed product were independent of representatives, $(13)H$ would equal $(12)(13)H$. Equality would require $(13)^{-1}(12)(13) = (23)$ to belong to H , which it does not. The subgroup is not normal. Its cosets still partition S_3 , but cannot inherit this multiplication.

Solution 305.5 ■ Answer D

[Back to problem](#)

The map is well-defined by $\bar{\varphi}(gN) = \varphi(g)$ because φ kills N . Its kernel consists of the cosets in $(\ker \varphi)/N$. This kernel is trivial exactly when $\ker \varphi = N$. Its image remains in φ , so injectivity of the induced map does not require surjectivity onto H .

Solution 305.6 ■ Answer C

[Back to problem](#)

Commutativity in the quotient says $(aN)(bN) = (bN)(aN)$, which is equivalent to the commutator $aba^{-1}b^{-1}$ lying in N . This need not make the commutator trivial in G : for example, S_3/A_3 is abelian whereas S_3 is not.

Abstract algebra

306 Ideals, quotient rings, and polynomial quotients

Solution 306.1 ■ **Answer A**[Back to problem](#)

Factor $x^3 + x = x(x + 1)^2$ over $\mathbb{F}_2$. A representative of degree below 3 is invertible iff it is divisible by neither x nor $x + 1$, that is, its values at both 0 and 1 are 1. Write $ax^2 + bx + c$: this requires $c = 1$ and $a + b = 0$, leaving two possibilities.

Solution 306.2 ■ **Answer B**[Back to problem](#)

In this quotient $x^2 = 2$. Thus $(1 + x)(x - 1) = x^2 - 1 = 1$. Equivalently rationalize with the conjugate: $(1 + x)(1 - x) = -1$, so the inverse is $x - 1$.

Solution 306.3 ■ **Answer A**[Back to problem](#)

Evaluation at zero gives $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$, which is an integral domain but not a field. Thus (x) is prime but not maximal. The next two quotients are the fields $\mathbb{F}_2$ and $\mathbb{F}_3$; (1) is not proper; and the class of x is a nonzero nilpotent modulo x^2 .

Solution 306.4 ■ **Answer E**[Back to problem](#)

The ideals $(x - 1)$ and $(x + 1)$ are comaximal, so the Chinese remainder theorem gives a product of their quotients, each isomorphic to $\mathbb{R}$. Explicitly, send a polynomial class to $(p(1), p(-1))$.

Solution 306.5 ■ **Answer D**[Back to problem](#)

An element has representative $a + bx + cx^2$. If it is nilpotent, evaluation at zero shows a is nilpotent in the field $\mathbb{F}_3$, hence $a = 0$. Conversely, every multiple of x has cube zero modulo x^3 . There are $3^2 = 9$ such elements, of which eight are nonzero.

Solution 306.6 ■ **Answer E**[Back to problem](#)

A degree-four polynomial over $\mathbb{F}_2$ is irreducible iff it has no linear factor and is not divisible by the only monic irreducible quadratic $x^2 + x + 1$. For $x^4 + x + 1$, neither zero nor one is a root, and reduction modulo $x^2 + x + 1$ gives remainder one. The first choice is $(x^2 + x + 1)^2$; the others have a root zero or one. The irreducible quotient has dimension four over $\mathbb{F}_2$, hence sixteen elements.

Abstract algebra

307 Modules, free modules, and annihilators

Solution 307.1 ■ **Answer C**[Back to problem](#)

The condition $km = 0$ requires $12 \mid 2k$ and $18 \mid 3k$. Both are equivalent to $6 \mid k$. Thus the annihilator is exactly $6\mathbb{Z}$, not the ideal determined by the orders of the entire summands.

Solution 307.2 ■ **Answer C**[Back to problem](#)

A module map is determined by the image v of the class of 1, and must satisfy $x^2v = 0$ modulo x^3 . Write $v = a + bx + cx^2$. Then $x^2v = ax^2$ modulo x^3 , so $a = 0$. The free real parameters b, c give dimension 2.

Solution 307.3 ■ **Answer C**[Back to problem](#)

If a nonzero integer kills a rational number, that number is zero, so $\mathbb{Q}$ is torsion-free. Every rational number is divisible by two within $\mathbb{Q}$. In a nonzero free $\mathbb{Z}$ -module, a basis vector cannot equal twice another vector, because its coefficient one would have to be twice an integer. Thus $\mathbb{Q}$ is not free. It is not finitely generated, so the finitely generated PID theorem is not contradicted.

Solution 307.4 ■ **Answer C**[Back to problem](#)

Every $(a, b) \in M$ has the unique expression $a(1, 1) + ((b - a)/2)(0, 2)$ with integer coefficients. Thus the listed vectors span M and are independent over $\mathbb{Z}$. The ambient standard basis is not contained in M , while the two even coordinate vectors fail to generate $(1, 1)$.

Solution 307.5 ■ **Answer D**[Back to problem](#)

The class of $(1, 2)$ is nonzero because it is not an integer multiple of $(2, 4)$, and doubling it gives zero. The map $(a, b) \mapsto b - 2a$ kills the relation and induces a surjection $M \rightarrow \mathbb{Z}$, so M also contains an infinite-order element. In fact, using the basis $(1, 2), (0, 1)$ shows $M \cong (\mathbb{Z}/2\mathbb{Z}) \oplus \mathbb{Z}$.

Abstract algebra

308 Group actions, conjugacy, and orbit counting

Solution 308.1 ■ **Answer D**
[Back to problem](#)

A rotation by k steps has $\gcd(6, k)$ cycles, hence fixes $2^{\gcd(6, k)}$ colorings. The six counts are 64, 2, 4, 8, 4, 2. Their average is $84/6 = 14$.

Solution 308.2 ■ **Answer C**
[Back to problem](#)

A commuting permutation must preserve the two cycle supports since their sizes differ. It acts by one of three rotations on the 3-cycle and one of two rotations on the 2-cycle, so the centralizer has order 6. The class has size $5!/6 = 20$.

Solution 308.3 ■ **Answer C**
[Back to problem](#)

Orbit sizes divide 27, so every nontrivial orbit has size divisible by three. The number of singleton orbits is therefore congruent to 20, or two, modulo three, and is at least two. This is attainable: let the cyclic group of order 27 act through a quotient of order nine on two disjoint nine-point orbits, fixing two further points.

Solution 308.4 ■ **Answer C**
[Back to problem](#)

Its centralizer in S_5 consists of its five powers: a commuting permutation is determined by the image of one symbol in this single cycle. All five powers are even, so its centralizer in A_5 also has order five. The conjugacy class has size $60/5 = 12$. Although all twenty-four five-cycles are conjugate in S_5 , they split into two classes in A_5 .

Solution 308.5 ■ **Answer C**
[Back to problem](#)

Transitivity makes the orbit size twelve. Orbit–stabilizer gives $|G| = |\text{Orb}(x)| |\text{Stab}(x)|$, so each stabilizer has order $60/12 = 5$.

Abstract algebra

309 Sylow subgroup counts and normality

Solution 309.1 ■ **Answer A**
[Back to problem](#)

The number divides 9 and is congruent to 1 modulo 5. Among the divisors 1, 3, 9, only 1 satisfies the congruence. Thus this subgroup is unique and normal.

Solution 309.2 ■ **Answer C**
[Back to problem](#)

Sylow gives $n_5 \in \{1, 6\}$ and $n_3 \in \{1, 10\}$. If both exceeded 1, the distinct prime-order subgroups would contribute $6(5 - 1) + 10(3 - 1) = 44$ nonidentity elements, more than 29. Thus one count is 1. The dihedral group of order 30 refutes the stronger alternatives.

Solution 309.3 ■ Answer D[Back to problem](#)

If $g^9 = e$, its order divides nine, so the cyclic subgroup it generates is a 3-subgroup and is contained in the unique Sylow 3-subgroup P . Conversely, $|P| = 9$, so every element of P satisfies $g^9 = e$ by Lagrange. Exactly its nine elements qualify, whether P is cyclic or not.

Solution 309.4 ■ Answer D[Back to problem](#)

The Klein four-group $V = \{e, (12)(34), (13)(24), (14)(23)\}$ is normal in S_4 . For any Sylow 2-subgroup P , the product PV is a subgroup of power-of-two order containing P ; its order cannot exceed eight, the largest power of two dividing 24. Hence $V \subseteq P$, and likewise $V \subseteq Q$. Their intersection has order at least four and divides eight. Order eight would give $P = Q$, so it is four.

Solution 309.5 ■ Answer E[Back to problem](#)

Conjugating a Sylow p -subgroup gives another Sylow p -subgroup. Uniqueness therefore makes every conjugate equal to P , which is normality. The subgroup need not be cyclic or central.

Abstract algebra

310 Polynomial irreducibility and finite field extensions

Solution 310.1 ■ Answer E[Back to problem](#)

The polynomial has no root in $\mathbb{F}_2$. The only monic irreducible quadratic there is $x^2 + x + 1$; reducing $x^4 + x + 1$ modulo it gives 1, so the quartic is irreducible. Thus K is a field with 16 elements and α has order dividing 15. The orders 1, 3, 5 are impossible: $\alpha \neq 1$, $\alpha^3 \neq 1$, and $\alpha^5 = \alpha^2 + \alpha \neq 1$ by independence of $1, \alpha, \alpha^2, \alpha^3$. The order is 15.

Solution 310.2 ■ Answer D[Back to problem](#)

The intersection is $\mathbb{F}_{2^{\gcd(6,4)}} = \mathbb{F}_{2^2}$, while the compositum is $\mathbb{F}_{2^{\text{lcm}(6,4)}} = \mathbb{F}_{2^{12}}$. This follows from the subfield divisibility criterion. The tower law gives $[KL : K \cap L] = 12/2 = 6$.

Solution 310.3 ■ Answer D[Back to problem](#)

For $x^5 + 6x^3 + 3$, three divides every nonleading coefficient, does not divide the leading coefficient, and nine does not divide the constant term three. Zero coefficients are divisible by three. The other choices fail at least one required condition: their constant term is divisible by nine, a nonleading coefficient is not divisible by three, or the leading coefficient is divisible by three.

Solution 310.4 ■ Answer B[Back to problem](#)

The roots of $x^{2^3} - x$ in $\mathbb{F}_{2^6}$ form its unique subfield of size $2^{\gcd(3,6)} = 8$. The polynomial has distinct roots since its derivative in characteristic two is -1 .

Solution 310.5 ■ Answer D[Back to problem](#)

Since $\alpha^{-1} = \sqrt{3} - \sqrt{2}$, the numbers $(\alpha - \alpha^{-1})/2 = \sqrt{2}$ and $(\alpha + \alpha^{-1})/2 = \sqrt{3}$ both lie in $\mathbb{Q}(\alpha)$. Hence this field equals $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and has degree four: $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, as squaring a proposed expression $a + b\sqrt{2}$ with rational a, b shows. Squaring $\alpha^2 = 5 + 2\sqrt{6}$ yields $\alpha^4 - 10\alpha^2 + 1 = 0$, its degree-four minimal polynomial.

Solution 310.6 ■ Answer B[Back to problem](#)

Over $\mathbb{Z}$, $p = (2x + 1)(x + 1)$, so it is reducible over $\mathbb{Q}$. Modulo two, the first factor becomes the constant one and the degree drops from two to one. The degree-preservation hypothesis is essential in the reduction criterion. The coefficients have gcd one, so primitivity does hold and is not the failed assumption.

Real analysis

311 Pointwise and uniform convergence of functions

Solution 311.1 ■ Answer B[Back to problem](#)

At $x = 0$ every term is 0, and at fixed $x > 0$ the denominator grows quadratically, so the limit is 0. But $f_n(1/n) = 1/2$, and in fact that is its maximum. Thus the supremum of the error never approaches zero.

Solution 311.2 ■ Answer C[Back to problem](#)

On either of the first two domains, $|x^n/n^2| \leq 1/n^2$, so the M-test applies. At $x = 2$ the terms do not approach zero; there is not even pointwise convergence on the third domain.

Solution 311.3 ■ Answer E[Back to problem](#)

The maximum occurs at $x = n/(n + 1)$ and equals $n^n/(n + 1)^{n+1} \leq 1/(n + 1)$. Thus the supremum norm tends to zero, proving uniform convergence, including the endpoint $x = 1$ where every value is zero.

Solution 311.4 ■ Answer B[Back to problem](#)

For every fixed x , the denominator tends to infinity, so the limit is zero. For each fixed n , however, $f_n(x) \rightarrow 1$ as $x \rightarrow \infty$, so $\sup_{x \geq 0} |f_n(x)| = 1$. Uniform convergence fails.

Solution 311.5 ■ **Answer C**[Back to problem](#)

If the supremum error is E , then approaching zero from the left and using continuity gives $|g(0)| \leq E$, while the value at zero gives $|g(0) - 1| \leq E$. The triangle inequality implies $1 \leq 2E$. The constant function $g = 1/2$ attains error $1/2$. Thus this discontinuous function cannot even be approximated arbitrarily closely in supremum norm by continuous functions.

Real analysis

312 Interchanging limits, derivatives, and integrals

Solution 312.1 ■ **Answer B**[Back to problem](#)

$|f_n| \leq 1/n$, so convergence to zero is uniform. Yet $f'_n(\pi) = \cos(n\pi) = (-1)^n$ does not converge. The limit function has derivative zero; differentiating the sequence does not commute with taking this limit.

Solution 312.2 ■ **Answer C**[Back to problem](#)

The identity $(x^a - 1)/\log x = \int_0^a x^t dt$ has a nonnegative integrand. Tonelli allows integration in the other order: $F(a) = \int_0^a \int_0^1 x^t dx dt = \int_0^a (1+t)^{-1} dt = \log(a+1)$. It also yields $F'(a) = 1/(a+1)$.

Solution 312.3 ■ **Answer D**[Back to problem](#)

By the fundamental theorem, $f_n(x) = f_n(0) + \int_0^x f'_n(t) dt$. The uniform difference from $a + \int_0^x g$ is at most $|f_n(0) - a| + \|f'_n - g\|_\infty$, which tends to zero. Also g is continuous as a uniform limit of continuous derivatives.

Solution 312.4 ■ **Answer A**[Back to problem](#)

The parameter derivative of the integrand is $e^{-x} \cos(ax)$, bounded in absolute value by the integrable function e^{-x} . Differentiation is therefore justified, giving $F'(a) = \int_0^\infty e^{-x} \cos(ax) dx = 1/(1+a^2)$, by elementary integration of the damped cosine. Since $F(0) = 0$, integrating in the parameter yields $\arctan a$.

Solution 312.5 ■ **Answer A**[Back to problem](#)

Uniform convergence on a finite interval permits exchanging limit and integral. The error in the integral is bounded by $2\|f_n - f\|_\infty$, so $\int f = \lim_n (3 + 1/n) = 3$.

Solution 312.6 ■ **Answer C**[Back to problem](#)

For differentiable f_n , the mean value theorem bounds $|f_n(x) - f_n(0)|$ by $\sup |f'_n|$, which tends to zero. Thus convergence of the constants $f_n(0)$ yields uniform convergence to that constant limit. Without an anchor, $f_n(x) = n + x/n$ has derivatives tending uniformly to zero but diverges at every point; it also meets all the other listed conditions.

Probability and statistics

313 Sample means, standard error, and the central limit theorem

Solution 313.1 ■ **Answer B**[Back to problem](#)

The five shared variables cancel. The difference is $1/10$ times five independent variables minus five other independent variables. Hence its variance is $10/100 = 1/10$. Equivalently $\text{Cov}(A, B) = 5/100$.

Solution 313.2 ■ **Answer B**[Back to problem](#)

Each variable has mean 1 and variance 1, so the sum has mean 100 and standard deviation 10. The threshold is $(120 - 100)/10 = 2$ standard deviations above the mean. The approximation is $1 - \Phi(2)$, where Φ is the standard normal cumulative distribution function.

Solution 313.3 ■ **Answer B**[Back to problem](#)

The standard deviation of the mean is $\sigma/\sqrt{n} = 3/\sqrt{n}$. Setting this to $1/5$ gives $\sqrt{n} = 15$ and $n = 225$.

Solution 313.4 ■ **Answer B**[Back to problem](#)

The observations are identical, not independent, so their average is exactly Y for every sample size. Its variance stays $1/4$, and its distance from $1/2$ is always $1/2$, giving probability one. Averaging alone does not force concentration or justify a central-limit approximation. Covariance terms exactly offset the usual independent-sample variance reduction.

Solution 313.5 ■ **Answer C**[Back to problem](#)

The sample mean equals one half exactly when fifty of the hundred observations equal one. Independence gives the binomial probability $\binom{100}{50}(1/2)^{100}$, which is positive. The central limit theorem approximates the sampling distribution after scaling, but it does not turn a finite discrete distribution into an exactly continuous normal distribution.

Probability and statistics

314 Geometric, Poisson, and exponential distributions

Solution 314.1 ■ Answer C

[Back to problem](#)

The unnormalized conditional probability at $N = n$ is $e^{-6}6^n(2/3)^n/n! = e^{-6}4^n/n!$. Dividing by $P(\text{no red}) = e^{-2}$ gives $e^{-4}4^n/n!$. Thus the conditional distribution is Poisson with mean 4. Equivalently the nonred count is independent of the red count.

Solution 314.2 ■ Answer B

[Back to problem](#)

Condition on $G = k$. The exponential tail is $P(T > k) = 2^{-k}$. Hence $P(T > G) = \sum_{k=1}^{\infty} \frac{1}{3}(\frac{2}{3})^{k-1}2^{-k} = \frac{1}{6} \sum_{j=0}^{\infty} (\frac{1}{3})^j = 1/4$. Using the failures-before-success convention would give a different answer; the question specifies the convention.

Solution 314.3 ■ Answer E

[Back to problem](#)

If the variables are X, Y , then $P(X < Y) = \int_0^{\infty} 2e^{-2t}P(Y > t) dt = \int_0^{\infty} 2e^{-5t} dt = 2/5$. The faster rate is more likely to produce the smaller waiting time.

Solution 314.4 ■ Answer C

[Back to problem](#)

Conditionally on a sum of four, X is binomial with parameters 4, $2/5$. Thus the probability is $\binom{4}{1}(2/5)(3/5)^3 = 216/625$.

Solution 314.5 ■ Answer D

[Back to problem](#)

Given five failures, the additional trials retain the same independent success probability. The event requires three more failures, so the conditional probability is $(3/4)^3 = 27/64$.

Solution 314.6 ■ Answer C

[Back to problem](#)

The second event has not occurred by time t exactly when the count by that time is zero or one. Since that count is Poisson with parameter λt , the probability is $e^{-\lambda t} + (\lambda t)e^{-\lambda t}$. Although each gap is exponential, the time to the second event is a sum of two gaps and is not itself exponentially distributed.

Probability and statistics

315 Conditional expectation and total expectation and variance

Solution 315.1 ■ Answer D

[Back to problem](#)

The two conditional means are 1 and 3, each with probability $1/2$, so their variance is 1. Each conditional variance is $4(1/4)(3/4) = 3/4$. Total variance therefore gives $\text{Var}(X) = 3/4 + 1 = 7/4$.

Solution 315.2 ■ Answer D

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Given N , the sum has conditional mean $3N$ and variance $4N$. Thus $E[S] = 3E[N] = 6$, and total variance gives $E[4N] + \text{Var}(3N) = 4 \cdot 2 + 9 \cdot 2 = 26$.

Solution 315.3 ■ Answer C

[Back to problem](#)

The marginal density is $f_Y(y) = \int_0^y 6x \, dx = 3y^2$. Dividing gives the conditional density $2x/y^2$ on $(0, y)$. Its mean is $\int_0^y x(2x/y^2) \, dx = 2y/3$. Both the support and the normalizing factor depend on the conditioned value.

Solution 315.4 ■ Answer B

[Back to problem](#)

Symmetry gives conditional mean zero for either value of Y . But the conditional probability of $X = 0$ is $1/2$ in one case and $1/3$ in the other, so the conditional distributions differ and independence fails. A conditional mean summarizes only the first moment. Neither variable is a deterministic function of the other in this example.

Solution 315.5 ■ Answer E

[Back to problem](#)

By the tower property, $E[X] = E[E[X | Y]] = E[Y^2]$. A standard normal has mean zero and variance one, so its second moment is one.

Geometry

316 Triangle geometry and circle power

Solution 316.1 ■ Answer C

[Back to problem](#)

The semiperimeter is 21 and Heron gives area $\sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$. Thus $r = 84/21 = 4$ and $R = 13 \cdot 14 \cdot 15 / (4 \cdot 84) = 65/8$. Their ratio is $65/32$.

Solution 316.2 ■ Answer B

[Back to problem](#)

Write $PA = x > 0$. Then $PB = x + 5$ and the power identity gives $x(x + 5) = 36$. Factoring $x^2 + 5x - 36 = (x + 9)(x - 4)$ selects the positive solution $x = 4$.

Solution 316.3 ■ Answer B[Back to problem](#)

The cosine law at angle A gives $2 = 4 + c^2 - 2\sqrt{3}c$, whose roots are $c = \sqrt{3} + 1$ and $c = \sqrt{3} - 1$, both valid positive triangle sides. Their product is two. Equivalently, the sine law gives $\sin B = 1/\sqrt{2}$, allowing $B = 45^\circ$ or 135° ; either leaves a positive third angle. This is the ambiguous side-side-angle case.

Solution 316.4 ■ Answer C[Back to problem](#)

The intersecting-chords theorem equates products of the two segment lengths: $3 \cdot 8 = 4x$. Therefore $x = 6$. These are products of segments measured from the interior intersection point.

Solution 316.5 ■ Answer E[Back to problem](#)

The hypotenuse is 17, so the circumradius is $R = 17/2$. The inradius is $(8 + 15 - 17)/2 = 3$. Euler's center formula gives $OI^2 = R(R - 2r) = (17/2)(5/2) = 85/4$, hence $OI = \sqrt{85}/2$.